GE Medical Systems

Technical Publications

Direction 2295400

Revision 4

GEMSO 3.0T/940 Active Shield Magnet and Cryogens Subsystem

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete **"Damage Loss Claim Form"** after damage is found. Form is available via MS Exchange Mail at **Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/Damage Loss Claim Form.** Send completed form to e-mail address listed on form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address: **ftp://3.87.40.2/ globepro/qualsys/Docs/190016MF.PDF**

REV 0, 11/17/2000

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WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
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注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

SAFETY ALERT USAGE

CAUTION

“A ’CAUTION’” Safety Alert is “used to identify conditions or actions for which a potential hazard may exist which will or can cause minor personal injury or property damage if the instructions are ignored.”*

WARNING!

“A ’WARNING’” SAFETY ALERT IS “USED TO IDENTIFY CONDITIONS OR ACTIONS FOR WHICH A SPECIFIC HAZARD IS KNOWN TO EXIST WHICH MAY CAUSE SEVERE PERSONAL INJURY, DEATH OR SUBSTANTIAL PROPERTY DAMAGE IF THE INSTRUCTIONS ARE IGNORED.”*

DANGER!!

“A ’DANGER’” SAFETY ALERT IS “USED TO IDENTIFY CONDITIONS OR ACTIONS FOR WHICH A SPECIFIC HAZARD IS KNOWN TO EXIST WHICH WILL CAUSE SEVERE PERSONAL INJURY OR SUBSTANTIAL PROPERTY DAMAGE IF THE INSTRUCTIONS ARE IGNORED.”*

* Direction 46-015099, MR Style Guide.

DIRECTION 295400 APPLICATION

Direction 295400 covers the set-up, commissioning and service of GE Superconducting Magnet Model 3T/940.

This magnet has superconducting coils immersed in a liquid helium vessel which is surrounded by an insulating cryostat. The coils are ramped to a field strength of 3.0 Tesla at a nominal, continuous coil current of 253.5 amperes. Once the magnet is ramped, no external power is required to maintain the field.

The magnet is used in Magnetic Resonance Imaging (MRI) systems and is designated as Class I equipment per IEC (International Electrical Code) Standard 60601. MR system components and connections are covered in the MR Systems (Signa) Pre-Installation manual.

REVISION HISTORY

REV	DATE	PCN	PRIMARY REASON(S) FOR CHANGE
A	Feb. 28, 2001		
B	Mar. 16, 2001		
C	April 19, 2001		
D	June 6, 2001		
E	July 10, 2001		
F	Sept. 25, 2001		
0	Oct. 25, 2001		
1	Dec. 12, 2001	188928	Procedural / equipment / measurement refinements / corrections to the Detransiting, Magnet Evacuation, Shield Cooler Installation, Nitrogen Precool, Ramping, and Shimming sections; added GEMSO telephone number; closure of Service Turret Ball Valve before Magnet Venting; removal of N2 drain Dewars / cylinders before He Purge; additional cable connection illustrations in Ramping, Ramp Down and Shimming.
2	Jan. 21, 2002	188942	More conservative overnight / 24 hour ramp parking and final ramping with additional Warnings and Cautions and the use of Current Control to set parking field; clarify Shimming procedure (units of measure, plotting radius, shim polarity) and software usage; new table in Ramping, Shimming and Ramp Down giving Shim Power Supply Channel / Shim Order correspondances; new RT Shim Tube removal Caution in Warm Up section; clarify Field Plot Data and Shim Plot Data DATA SHEETS. Inclusion of 2301164PRE, revision 1.
3	March 18, 2002	188972	SET-UP AND CALIBRATION: Revised leak detection plumbing in Detransiting, Magnet Evacuation and Purge sections and pump-out plumbing in Detransiting; more comprehensive Detransiting procedure; new Service Turret Top Plate Leak Test sections in Precool and Purge; refined Precool, Purge, Fill / Top-Off, Ramping and Shimming sections; new table of ramp current designs by magnet serial number in Ramping section; Fill section renamed “Liquid Helium Fill / Top-Off”; all Plotting Rig / Teslameter set-up moved to new Section 10, Preparations for Field Plotting. FUNCTIONAL CHECKS: Two additional shim DSV specifications in Section 1, Magnet Specifications; 1000 amp ESS7.5-1000-2-D-1236 is now the preferred Main Power Supply in Section 6; Section 7, Temperature Sensor Measurement revised to use a four-wire resistance measurement method. REPLACEMENT / MAINTENANCE: Revised Rampdown and Coldhead Replacement sections; new Section 11, Preparation for Site Removal. SCHEMATICS / INTERCONNECTS: Added Magnet Service Wiring subsection. RENEWAL PARTS: Section 1, Magnet System: reorganized to reflect shipping packaging; Section 4, Service Tools & Equipment: 1000 amp P/S and 1000 amp Ramp Kit 2180589 added; content of Passive Shim Kit 2292493 and Plotting Rig 2292542 listed. DATA SHEETS: Added “Percent Full of Helium” and “Comments” columns to Volumetric Conversion and resistors measured to Temperature Sensor Log.

REVISION HISTORY (continued)

REV	DATE	PCN	PRIMARY REASON(S) FOR CHANGE
4	July 1, 2002	229015	Added Safety Alert Usage and Direction 2295400 Application pages at front of manual. INTRODUCTION: New Magnet Siting, Installation & Commissioning Flow Chart & Check List in Section 1. SET-UP AND CALIBRATION: Procedural refinements / clarifications and additional tools / part number information throughout Tab; vacuum / helium pump designation / preparation added to Section1; revised helium flexline installation (Section 6); 10-Pin and 19-Pin Breakout Boxes and temperature monitoring refinements added to Section 7; new LN2 Drain Plumbing illustration and revised N2 Purge and He Vessel Leak Check procedures in Section 8; Tools / Equipment subsection, Probe Scale illustration and numerous enhancements added to Section 10; new Ramp Cable Strain Relief Clamp illustration, updated telephone numbers and new B0 Coil 'ON' Warning in Section 11; Section 12: updated web address, telephone numbers, version of SHIMCALC.XLS and instructions for using MULTI.EXE plus passive shim options reflecting 2321737. FUNCTIONAL CHECKS: Added bandwidth spec. to Section 1; added revised fringe field illustration and Shim Adapter Box usage to Section 2; usage of 10-Pin and 19-Pin Breakout Boxes, rewiring Level Probe, and table for 37-pin RT Shim pigtail and 10-pin Demountable Level Probe added to Section 3; revised Section 4, including setting B0 Clock; illustration and usage of 10-Pin and 19-Pin Breakout Boxes added to Section 7. REPLACEMENT / MAINTENANCE: Procedural refinements / clarifications and additional tools / part number information throughout Tab; new Ramp Cable Strain Relief Clamp illustration, updated telephone numbers in Section 1; new Section 11, Helium Flexline Connections / Replacement; adjusted periodicities in Section 12. RENEWAL PARTS: Added longer flexlines as special order; added Ramp Cable Strain Relief parts to Ramp / Energize Kit 2296056; added 3T/940 O-Ring Kit 2347635.

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RENEWAL PARTS

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DATA SHEETS

MR MAGNET — SAFETY REQUIREMENTS

1 MAGNET SAFETY

1-1 Magnetic Field Considerations

The Magnetic Resonance Imaging (MRI) System utilizes a magnet, which can have a field strength several thousand times greater than that of the earth's magnetic field. The magnetic field surrounding the magnet is called the fringe field, and it extends from the magnet's iso-center in all dimensions. The fringe field may present a hazard to personnel and equipment within the immediate area. Therefore, magnetic field precautions must be applied to the floors above and below the magnet, as well as to the surrounding space on the same level.

To minimize the risk to personnel and equipment when the magnet is at field, follow the precautions listed below:

1. Post "WARNING" signs, part number 46-255326P1, outside the 5 gauss zone alerting personnel with cardiac pacemakers, neurostimulators and other biostimulation devices of the effect of the magnetic field on these devices. See Illustration 1. Post these signs two days prior to the activation of the magnet for maximum impact.
2. Post "SECURITY ZONE" signs, part number 46-255325P1, outside the exam room to alert personnel of the high magnetic field and not to bring ferromagnetic objects into the exam room. See Illustration 2.
3. Set up the "AUTHORIZED PERSONNEL" signs, part number 2289812, at the entrance(s) to the magnet room prior to the start of magnet service procedures. See Illustration 3.
4. Notify responsible personnel two days prior to the activation of the magnet to allow for preparatory actions to be accomplished.
5. Do not bring ferromagnetic objects (including TOOLS, pens, tape measures, and vacuum pumps) into the exam room when the magnet is at field unless otherwise called for in the service documentation. Do not bring large metal objects near the outside walls of the exam room.
6. Use only nonmagnetic cylinders, nonmagnetic cylinder carts, and Dewars when transferring cryogens into an energized superconducting magnet.
7. Do not take self-winding watches, magnetically coded credit cards, magnetic recording heads, magnetic tapes or cameras near the magnet when it is at field.

1-1 Magnetic Field Considerations (continued)

Note

The Local GE Field Service Operation will provide the signs in the primary local languages. The highly visible (orange, black & white) security and warning signs are available from GE under the following catalog numbers:

- E8819AA MR Warning Sign Kit (2 security signs, 10 exclusion signs) in English
- E8819A One MR Security Warning Sign, in English.
- E8819B One MR Exclusion Warning Sign, in English.
- E8819C One MR Security Warning Sign, in Spanish.
- E8819D One MR security Warning Sign, in French.
- E8819BA MR Warning Label Kit (Peel-Off Back)
- 2289812 Authorized Personnel Warning Sign, in English.

1-1 Magnetic Field Considerations (continued)

MAGNETIC FIELD WARNING SIGN (46-255326P1)
ILLUSTRATION 1

1-1 Magnetic Field Considerations (continued)

MAGNETIC FIELD SECURITY ZONE SIGN (46-255325P1)
ILLUSTRATION 2

1-1 Magnetic Field Considerations (continued)

MAGNETIC FIELD SECURITY ZONE SIGN (2289812)
ILLUSTRATION 3

1-2 Emergency Situations

Description

The Magnet technology used on MR Systems may vary from one product design to another. There are 3 basic magnet technologies used by suppliers today: 1) Superconducting Magnets, 2) Permanent Magnets, and 3) Resistive Magnets. Each magnet technology will have emergency response processes unique to that design. For example, If a person is pinned to the magnet by a ferromagnetic object or that object has become attached to the magnet then the Magnet technology will dictate the appropriate action to be taken. Therefore, go to the appropriate Magnet section in this document and refer to the First Aid / Emergency Situations subsection for specific actions. Revision 1 of this document addresses designs relating to Superconducting Magnets. The sections for Permanent Magnets and Resistive Magnets will be included in a future update of this document.

2 SUPERCONDUCTING MAGNETS

2-1 Introduction

Superconducting magnets contain cryogenic liquids (liquid helium = –452°F, –269°C; liquid nitrogen = –320°F, –196°C) and generate a three–dimensional magnetic field several times the strength of the earth's magnetic field. These conditions require safety precautions to be taken to prevent serious injury (cryogenic burns, asphyxiation, explosion, fire, shock, ferromagnetic projectiles or other magnetic field effects).

The safety precautions / requirements contained in the following sections must be reviewed, understood and implemented prior to performing any service on a Superconducting Magnet. Make sure that all magnet service is performed by trained / authorized personnel and all safety equipment is in place.

All persons working with cryogenic liquids should understand the following:

- Nature and properties of liquid and gaseous helium and nitrogen
- Specific instructions on the equipment and clothing
- Use and care of protective equipment and clothing
- Safety and first aid
- Handling emergency situations such as leaks, spills and fires

2-2 Cryogen Safety Requirements

2-2-1 Site Precautions

The Field Service Engineer is responsible to make sure the following items are in compliance:

- Review all service processes to be performed on the magnet with the facility safety officer or with the local authorities (i.e. fire department, safety council, etc.) and obtain any required permits prior to magnet commissioning and servicing.
- Magnet room venting shall be installed, tested, and operating in conformance with site planning requirements.
- A portable floor fan and exhaust duct shall be used to remove the cold nitrogen gas that stratifies at floor level, from the room to the outside, during nitrogen pre–cool.
- The magnet shall be vented in conformance with the site planning requirements. Magnet plumbing and vent system shall be inspected for leaks during magnet installation.
- The door into the magnet room must be secured in the open position prior to any service action that will result in the handling or release of cryogens from the magnet. The ceiling hatch and rear door shall be secured in the open position if in a mobile van.
- Any service actions which release large quantities of cryogenic gas shall have provisions for exhausting the gas through the magnet vent system.

2-2-1 Site Precautions (continued)

- Make sure a second person (GE, contractor, or hospital personnel trained in these safety requirements) is present in the area during magnet service in case of emergency.
- A working phone line accessing an outside line shall be available in case of an emergency.
- Set up “Authorized Personnel Only” signs outside the magnet room door(s) prior to any magnet servicing.
- Make sure an Oxygen Monitoring device is calibrated and operating properly before beginning any magnet service.
- Always maintain a clear exit path no less than 28 inches wide.

2-2-2 Equipment Handling Precautions

- Do not bring any ferromagnetic equipment, Dewars or cylinders into the magnet room when the magnet is ramped. They will become dangerous projectiles in the presence of the magnetic field.
- Helium gas cylinders are pressurized to 2400 psi. Secure each helium cylinder before removing the protective cap and open the main valve very slowly to prevent any possibility of a fatal release of explosive gas. Make sure gas cylinders are stored in an upright secured position.
- Firmly hold the unattached end of the gas hose during a purge to prevent hose’s “whipping“ motion.
- Keep Dewars in the vertical position at all times. Dewars should have wheels mounted at the base for transport. If wheels are not present, use a low platform dolly, which fully encompasses the Dewar’s base for moving. Do not slide or roll Dewars. Use a suitable hand truck to move gas cylinders.
- Store Dewars in a well ventilated area as outlined in the room ventilation pre-installation manual.
- Make sure that safety relief valves and regulators are operating properly.
- Check all equipment, Dewars, and gas cylinders for leaks.
- Make sure the transportation route for Dewars and gas cylinders is clear of obstacles and restrictions.

2-2-3 Cryogen Handling Precautions

- Contact of liquid cryogens or their vapors with the eyes can cause severe frostbite even when contact is too brief to affect the skin. Always wear safety glasses and face shield when handling cryogens.
- Never allow any unprotected part of the body to touch cold magnet plumbing.
- Protective clothing (long sleeve shirt, long pants, protective apron / jacket) and dry, non-absorbent insulated gloves shall be worn when handling or being exposed to cryogens, to prevent cold burns from contact with the cryogenic liquid or gas.
- Make sure a calibrated, functioning oxygen monitor is installed in the magnet room in conformance with site requirements or place a functioning portable oxygen monitor in the magnet room and cryogen storage area. The position of the O_2 monitor sensor should be at a level that corresponds to the cryogen product being stored or dispensed (i.e., for liquid helium the sensor should be near the ceiling; for liquid nitrogen the sensor should be near the floor).
- Smoking is prohibited in the magnet room and around cryogens. Liquid cryogens can liquefy atmospheric oxygen producing a highly enriched oxygen liquid.
- When pluming a stinger, always point the stinger toward the ceiling and away from the face at a 45-degree angle.
- Do not bring more than 1000 liters or 2 Dewars (whichever is less) of cryogens into the magnet room.

2-2-4 Magnet Servicing Precautions

- **RAMP MAGNET DOWN TO ZERO FIELD** prior to any service requiring opening / exposure to the helium vessel and / or cryogenic gas / liquid, except for the removal of fill and ramp lead port caps, to prevent the possibility of a magnet quench and rapid expulsion of cryogenic helium gas and liquid.
- Make sure non-ferromagnetic fiber or composition safety shoes are worn in the magnet room.
- Observe helium vessel pressure gauge and vent the magnet down to < 0.5 psi before removing ramp / fill port caps or loosening component resulting in the release of cryogenic helium gas and liquid.
- Never allow a helium Dewar to empty during a magnet fill, resulting in a magnet quench from the introduction of warm helium gas.
- Wear proper personal protective equipment (PPE).
- Use caution when inserting ramp leads or a shim lead into a ramped magnet to prevent a quench.

2-2-5 First Aid / Emergency Situations

- Notify proper emergency responders.
- Do not make any rescue attempts until the environment has been verified as being safe for normal occupancy, then move persons suffering from a lack of oxygen immediately to an area with normal atmosphere and seek medical assistance.
- Use large volumes of tepid water, within the temperature range of 105°F (41°C) to 115°F (46°C), to flush frostbitten or cold burn areas and get medical attention immediately. Notify the physician that this is a cold burn.
- In case of a magnet cryogenic vent failure during a quench stay near the floor where the oxygen will be and immediately exit the magnet room.
- If a ferromagnetic object has become attached to the magnet and cannot be safely removed by two people, the magnet will have to be ramped down.

2-3 Magnet Field Safety Requirements

2-3-1 Site Precautions

The magnet will create a fringe field that may present a hazard to other personnel working or visiting near the MR Suite. Make sure these precautions are followed to minimize risk to personnel.

- Post warning signs outside the 5-gauss zone, including areas above and below the magnet room, prior to ramping the magnet to warn personnel with cardiac pacemakers, neurostimulators, steel plates or other conditions affected by a magnetic field not to proceed into the area.
- Post “Ramped Magnet” warning sign at magnet room entrance prior to ramping the magnet.
- Notify site administration before ramping the magnet.
- Remove all ferromagnetic material from the magnet room that is not properly installed on the system.
- Prior to ramping the magnet make sure that the Magnet Rundown Unit (MRU) is operating properly.
- Do not bring any ferromagnetic tools / objects into the magnet room when the magnet is ramped unless specifically called for in the service documentation.
- Do not loosen any ferromagnetic components on a ramped magnet unless specifically called for in the service documentation.
- Wear leather gloves when opening or closing the coldhead motor shield.
- Use caution when opening or closing the top half of the coldhead motor shield. Never put your hand or fingers between the motor shield and mounting bracket.

2-3-2 Ramping Precautions

A superconducting magnet at field is a high-energy storage device capable of discharging rapidly (quenching), creating a high voltage across the main leads. Make sure the following precautions are observed when ramping a superconducting magnet.

- When working with the main lead connections that are installed into a ramped magnet do not touch both main lead extensions at the same time or allow them to come in contact with one another.
- Allow main lead extensions to cool before fully inserting them into a ramped magnet to prevent any possibility of a quench.
- Make sure the power supply has passed all functional checks and the input power cable is disconnected before connecting it to the main power leads.
- Make sure the final magnet “parking” current and voltage polarity has been recorded and will be available if a rampdown is required. An incorrect polarity connection will result in a magnet quench.
- Use the appropriate hold-down tool to properly secure ramp leads to the magnet.

2-3-3 Emergency Rampdown of Magnet

- Make sure the Magnet Rundown Unit (MRU) has been installed in conformance with the GE magnet service manual, the batteries have been charged for 24 hours and the MRU has passed the functional checks in conformance with the supplier manual before ramping the magnet.
- Make sure the rampdown methods covered in the introduction of the GE magnet service manual are fully understood by field service engineers involved with the MR equipment.
- Hospital personal involved with the MR equipment should be familiar with the equipment operator’s manual and Magnet Rundown Unit operation.

2-4 Magnet & Cryogens Service Safety Requirements

TABLE 1
SERVICE PROCEDURE SAFETY REQUIREMENTS

	Service Procedure (X = Required Item For Procedure)													
Safety Requirement	**Detransitin Magnet & Vacuum**	**Conversion to Operating Configuration**	**Venting Installation**	**Nitrogen Precool / Purge**	**Helium Precool / Warm-Up**	**Helium Fill / Top-Off**	**Magnet Ramping**	**Magnet S/C Shim**	**Magnet P-Shim**	**Burst Disk / Plumbing Component Replacement**	**Magnet / Recondenser De-Icing**	**Shim Lead Replacement**	**Baffle / Instrumentation Lead Replacement**	**Cryocooler Service**
1. Cryogenic vent system installed per requirements.	–	–	–	X	X	X	X	X	–	X	X	X	X	X
2. Special permits obtained, if required by local authorities, for use of inert gasses / cryogens during magnet commissioning and service procedures.	X	X	–	X	X	X	X	–	X	X	X	X	X	X
3. Mobile van ceiling hatch secured open.	X	X	X	X	X	X	X	X	–	X	X	X	X	X
4. Room ventilation fan tested and running (room exhaust at ceiling).	–	X	X	–	X	X	X	X	–	X	X	X	X	X
5. Room ventilation fan tested and running (room exhaust at floor).	–	–	–	X	–	–	–	–	–	–	–	–	–	–
6. Vent magnet and inspect exhaust.	X	X	X	X	X	X	X	X	–	–	X	X	X	X
7. Second person on site and trained in safety. practices for procedure being performed.	X	X	X	X	X	X	X	X	X	X	X	X	X	X
8. Fully functional land-line telephone (with emergency response numbers clearly posted beside it) in known and immediately available location near Magnet Room.	X	X	X	X	X	X	X	X	X	X	X	X	X	X
9. Magnet Room doors secured open to assist in room ventilation. Ensure cross ventilation to outside or other large volume non-workspace.	X	X	X	X	X	X	X	X	–	X	X	X	X	X
10. O_2 Monitor on site, tested and functional for both visual and audible alarms, whenever cryogens are being stored or dispensed. Portable device 2287000 for magnets up to and including 3.0T.	X	X	X	X	X	X	X	X	–	X	X	X	X	X
11. Cryogen Safety Kit (46-271137G1) on site for use. Kit includes cryogen gloves, face shield and goggles.	X	X	X	X	X	X	X	X	–	X	X	X	X	X
12. Nonferrous safety shoes.	X	X	X	X	X	X	X	X	X	X	X	X	X	X

2-4 Magnet & Cryogens Service Safety Requirements (continued)

TABLE 1 (CONTINUED)
SERVICE PROCEDURE SAFETY REQUIREMENTS

	Service Procedure (X = Required Item For Procedure)													
Safety Requirement	**Detransitin Magnet & Vacuum**	**Conversion to Operating Configuration**	**Venting Installation**	**Nitrogen Precool / Purge**	**Helium Precool / Warm-Up**	**Helium Fill / Top-Off**	**Magnet Ramping**	**Magnet S/C Shim**	**Magnet P-Shim**	**Burst Disk / Plumbing Component Replacement**	**Magnet / Recondenser De-Icing**	**Shim Lead Replacement**	**Baffle / Instrumentation Lead Replacement**	**Cryocooler Service**
13. Secure area and display signs alerting of intended process.	X	X	X	X	X	X	X	X	X	X	X	X	X	X
14. Verify dewar transportation path is clear.	-	-	-	X	X	X	X	-	-	-	-	-	-	-
15. Dewars stored in a ventilated area.	-	-	-	X	X	X	X	X	-	-	-	-	-	-
16. Not-in-use gas cylinders secured to wall with protective cap in place.	X	-	-	X	X	X	X	X	-	X	X	X	X	X
17. Before removing protective cap on gas cylinder, secure to wall or nonmagnetic cart.	X	-	-	X	X	X	X	X	-	X	X	X	X	X
18. Move gas cylinders / Dewars without wheels using hand trucks designed for purpose. (Nonmagnetic gas cylinder cart: 46-306717G1)	X	-	-	X	X	X	X	X	-	X	X	X	X	X
19. Keep gas cylinders / Dewars in upright position at all times.	X	-	-	X	X	X	X	X	X	-	-	-	-	-
20. Keep ferromagnetic equipment / objects out of magnet room when magnet is ramped.	-	-	-	-	-	X	X	X	X	X	X	X	X	X
21. Use only nonmagnetic tools in the magnet room when the magnet is ramped (46-320273G4).	-	-	-	-	-	X	X	X	X	X	X	X	X	X
22. Wear long-sleeve shirt and apron designed for cryogenic servicing.	-	-	-	X	X	X	X	X	-	X	X	X	X	X
23. Inspect all pressure regulators for proper functioning.	X	-	-	X	X	X	X	X	-	X	X	X	X	X
24. Inspect all pressure relief valves and ensure proper configuration to achieve recommended pressure (e.g., 20 psi).	X	-	-	X	X	X	X	X	-	X	X	X	X	X
25. Inspect all valves and fittings for leaks.	X	-	-	X	X	X	X	X	-	X	X	X	X	X
26. MRU installed and tested prior to magnet ramp.	-	-	-	-	-	-	X	X	X	X	X	X	X	X
27. Post "Ramped Magnet" warning signs included in kit 46-258770G4.	-	-	-	-	-	-	X	-	-	-	-	-	-	-

2-4 Magnet & Cryogens Service Safety Requirements (continued)

TABLE 1 (CONTINUED)
SERVICE PROCEDURE SAFETY REQUIREMENTS

	Service Procedure (X = Required Item For Procedure)													
Safety Requirement	**Detransitin Magnet & Vacuum**	**Conversion to Operating Configuration**	**Venting Installation**	**Nitrogen Precool / Purge**	**Helium Precool / Warm-Up**	**Helium Fill / Top-Off**	**Magnet Ramping**	**Magnet S/C Shim**	**Magnet P-Shim**	**Burst Disk / Plumbing Component Replacement**	**Magnet / Recondenser De-Icing**	**Shim Lead Replacement**	**Baffle / Instrumentation Lead Replacement**	**Cryocooler Service**
28. Post “Authorized Personnel Only” signs (2289812) at entrance to MR suite.	X	X	X	X	X	X	X	X	X	X	X	X	X	X
29. Notify Hospital Administrator when magnet is ramping.	-	-	-	-	-	-	X	-	-	-	-	-	-	-
30. Remove all ferromagnetic material from Magnet Room before ramping or shimming magnet.	-	-	-	-	-	X	X	X	X	-	-	-	-	-
31. Use only single Dewar connection (i.e., no daisy-chain connections)	-	-	-	X	X	X	X	X	-	-	-	-	-	-
32. Identify and label Dewars as “full” or “empty”; label Dewar storage area.	-	-	-	X	X	X	X	-	-	-	-	-	-	-
33. Remove frost from valve operators on magnet after each Dewar change.	-	-	-	X	X	X	X	-	-	-	-	-	-	-
34. Notify hospital engineering personnel prior to use of any equipment (i.e., heat guns, propane torches, welding equipment, etc.) that could set off the fire alarm system.	X	X	X	X	X	X	X	X	X	X	X	X	X	X
35. Identify and understand the properties and operation of the site’s fire suppression system (i.e., water sprinkler, halon, CO_2, etc.) and how to respond to alarms.	X	X	X	X	X	X	X	X	X	X	X	X	X	X
36. Reduce magnet helium vessel pressure to 0.2 psig before performing service.	-	X	-	-	-	X	X	X	-	X	X	X	X	-
37. Use product-defined tools and personal protective equipment (such as gloves, goggles and protective clothing) when removing or installing passive shim material / assemblies.	-	-	-	-	-	-	-	-	X	-	-	-	-	-

2-5 Buddy System Requirements & Certification

The Buddy System (second person on site) is required whenever service is performed on the superconducting magnet subsystem. It is critical that a ”Buddy” is always present to ensure your safety. A ”Buddy” is more than just a second person that is ”around” while you are working. A ”Buddy” is someone who will maintain communication with you at all times and sound the alarm in case of an emergency and may even assist in performing the service procedure, if qualified.

All individuals selected to perform the task of either “Work Assistant” or “Work Observer” must comply with the prerequisite training and complete the certification form located in Section 2-5-1, Worker Certification / Agreement Form. A signed copy of the certification must remain on file at the customer site.

2-5-1 Worker Certification / Agreement Form

IMPORTANT !!!

All individuals selected to perform the task of either "Work Assistant" or "Work Observer" must comply with the prerequisite training and complete the certification. Do not allow a person to function as a buddy (second person) until ALL the MANDATORY requirements are completed for the category being performed.

Instructions: Table 2 is to be completed by the individual that is being certified to function as either a Work Assistant or Work Observer.

TABLE 2
SECOND PERSON – BUDDY REQUIREMENTS

Safety Requirment	Work Assistant	Work Observer	Initialed by Second Person When Complete	Comments
1. Must have completed all required EHS training.	MANDATORY			
2. Completed and have on file the medical clearance form "Magnetic Resonance Employee Questionnaire For Metallic Foreign Body and Electronic Devices," available from the GEMS Medical department.	MANDATORY	MANDATORY		
3. Must have completed the magnet and cryogen safety training portion of MR 508, Principles of MRI, or MR 512, Magnet & Cryogen Safety (CD-ROM).	MANDATORY	MANDATORY		
4. Must know how to wear the proper PPE for the procedure being performed as described in the service documentation and/or demonstrated by the Site FE performing the qualification process.	MANDATORY			
5. Needs to be a FE, GE employee, or magnet-knowledgeable person.	MANDATORY			
6. Know how to evacuate the area (Emergency escape route) as demonstrated by the primary FE	MANDATORY	MANDATORY		
7. Know how to notify emergency responders as per site emergency process.	MANDATORY	MANDATORY		
8. Will not attempt a rescue until the atmosphere is determined safe by use of a calibrated instrument designed for measuring atmospheric air quality.	MANDATORY	MANDATORY		
9. Know how to control and operate the valves on the gas cylinders / regulators and cryogenic Dewars.	MANDATORY			

2-5-1 Worker Certification / Agreement Form (continued)

TABLE 2 (CONTINUED)
SECOND PERSON – BUDDY REQUIREMENTS

Safety Requirment	Work Assistant	Work Observer	Initialed by Second Person When Complete	Comments
10. Know how to terminate the cryogen transfil operation on both the Dewars and the magnet in the event of a problem.	MANDATORY			
11. Know how and when to perform an emergency run down procedure as defined in the service documentation.	MANDATORY	MANDATORY		
12. Know how and when to initiate ’Emergency OFF’ as defined in the service documentation.	MANDATORY	MANDATORY		
13. Know how to respond to an oxygen monitor alarm.	MANDATORY	MANDATORY		
14. Know what to do during a quench.	MANDATORY	MANDATORY		
15. Know the physical changes or indicators of working in an oxygen deficient atmosphere.	MANDATORY	MANDATORY		**Your buddy MUST:** • Listen for changes in your voice; it will be higher than usual when working with helium. • Watch for physical indicators; you may begin staggering or slur your speech when working with nitrogen. • Notify the emergency response team when necessary.
16. Identify and understand the properties and operation of the site’s fire suppression system (i.e., water sprinkler, halon, CO_2, etc.) and how to respond to alarms.	MANDATORY	MANDATORY		

2-5-1 Worker Certification / Agreement Form (continued)

Instructions: Table 3 is to be completed by the Field Engineer responsible to perform any of the listed service procedures. He or she must certify the capabilities of the "Second Person," either by interview or training, and initial the appropriate "Assistant" or "Observer" column to show they have been certified.

TABLE 3
SECOND PERSON – SERVICE PROCEDURE CERTIFICATION

Service Procedure	**Assistant** (FE initials required if meets certification)	**Observer** (FE initials required if meets certification)
1. Detransit Magnet		
2. Conversion to Operating Configuration		
3. Venting Installation		
4. Nitrogen Precool / Purge		
5. Helium Precool / Warm Up		
6. Helium Fill / Top-Off		
7. Magnet Ramping		
8. Magnet P-Shim		
9. Magnet S/C Shim		
10. Burst Disk / Plumbing Component Replacement		
11. Magnet / Recondensor De-Icing		
12. Shim Lead Replacement		
13. Baffle / Instrumentation Lead Replacement		
14. Cryocooler Service		
15. Other [ ]		
16. Other [ ]		

2-5-1 Worker Certification / Agreement Form (continued)

Certification Agreement

I understand and will comply with the MR Buddy System Requirements.

(Print) Name of Buddy	Signature	Date

I certify the person above meets or exceeds the MR Buddy System Requirements for the selected category / procedures.

(Print) Name of FE	Signature	Date

The primary Field Engineer must keep the completed forms on site.

REVISION HISTORY

REV	DATE	PCN #	AUTHOR	PRIMARY REASONS FOR CHANGE
0	May 10, 2001		P. Senski	Initial Document Release
1	Oct. 23, 2001	224297	P. senski	Added clarification of O_2 sensor position under fourth bullet in Sec. 2-3. Made the following revisions to Table 1: 1. Reversed order of column headers for Magnet S/C Shim and Magnet P-Shim only. 2. Corrected required item errors for line item 1. 3. Corrected required item errors for column 1. 4. Changed content of safety requirement line item 10. 5. Added new requirement line items 36 and 37.

INTRODUCTION

TABLE OF CONTENTS

SECTION 1 – MANUAL CONTENT & WARNINGS

IMPORTANT !!!

The most recent version of this manual is available at the MR Service Engineering web site:

`http://3.87.118.28/optec3/common/mangets/magnets.htm#magnet%20 manuals`

1-1 MANUAL CONTENT

Direction 2295400, *GEMSO 3.0T/940 Active Shield Magnet and Cryogens Subsystem* covers the GEMSO 3T/940 Magnet System. The major Tabs of Direction 2295400 and the topics covered in each are:

SAFETY REQUIREMENTS:
Contained in 2301164PRE, *MR Magnet – Safety Requirements*.

INTRODUCTION:
Documentation organization and system and component identification are in this section.

SET-UP AND CALIBRATION PROCEDURES:
Magnet system installation, commissioning, adjustment, and calibration procedures are in this section. Procedures from supplier manuals are referenced where applicable.

Note
Manual sequence for Magnet Commissioning is shown in Illustration 1-4. An overview of the entire Siting, Installation and Commissioning process is shown in Table 1-1.

FUNCTIONAL CHECKS:
Procedures for performing subsystem checks are in this section, such as procedures done for diagnostics and periodic maintenance. Magnet resistance values and guideline tables are provided in this section.

REPLACEMENT / MAINTENANCE:
Procedures and illustrations to aid in subsystem maintenance and component replacement are in this section.

SCHEMATICS / INTERCONNECTS:
A cable Interconnect diagram for the system, schematics for all nonsupplier subsystem circuits and power supply controls, meters and indicators are in this section.

RENEWAL PARTS:
Renewal part identification and exploded views for the Magnet / Cryogen Subsystem and GE Part Number Reference Tables for supplier renewal parts are in this section.

DATA SHEETS:
Contains logs, charts and tables for Helium Fill, Ramping and Shimming.

Note
All schematics / circuit diagrams, component parts lists / descriptions, adjustments / calibrations and other information necessary for the field service of this magnet system are contained within *DIRECTION 2295400.*

1-1 MANUAL CONTENT (continued)

TABLE 1-1
MAGNET SITING, INSTALLATION & COMMISSIONING
FLOW CHART & CHECK LIST

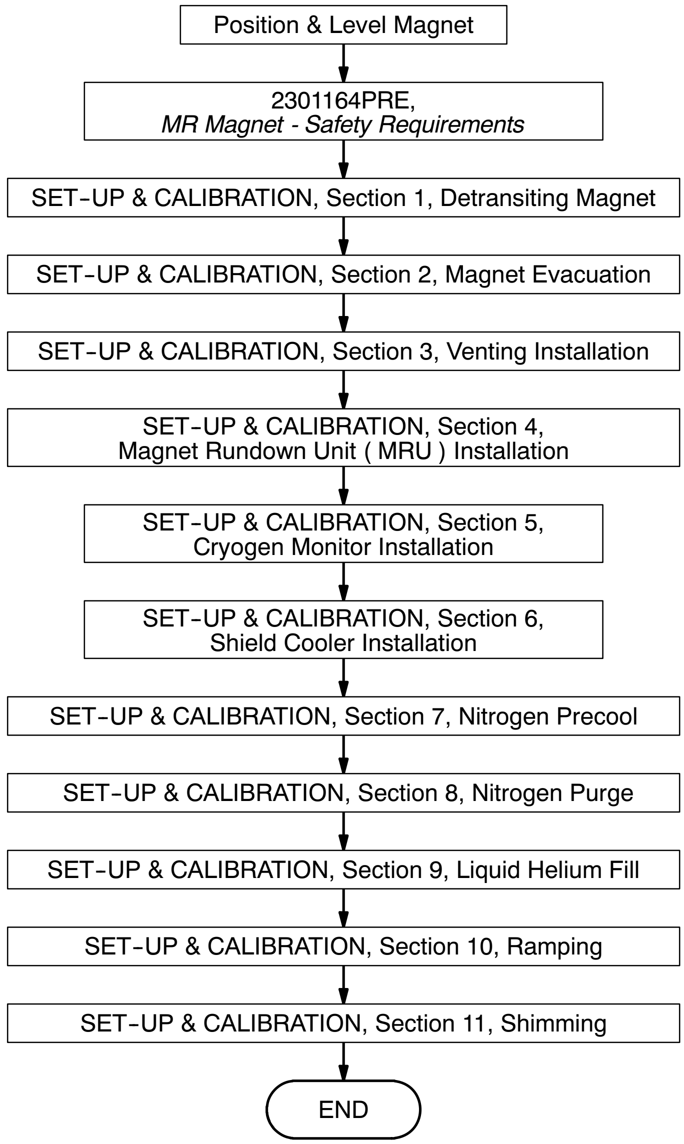
SITING & INSTALLATION			
Responsibility	**Function**	**Reference Document**	**Complete**
GE Installation Specialist & Shield Room Vendor	Magnet Room Requirements: • Safety (Exhaust & Venting) • Magnet Anchoring • Power & cooling	Direction 2221775, Signa 3T/94 & VH3 Pre-Installation	☐
GE Installation Specialist	Preinstallation Check List Completed	Direction 2221775, Signa 3T/94 & VH3 Pre-Installation	☐
GE Field Engineer & Rigger	Site Delivery Review: • Access & Route • Clearances	Direction 2341078, GEMSO 3.0T/940 Delivery & Installation	☐
Rigger	Magnet Delivery	Direction 2341078, GEMSO 3.0T/940 Delivery & Installation	☐
GE Field Engineer	Magnet Component Checks: • Physical & Electrical	Direction 2341078, GEMSO 3.0T/940 Delivery & Installation	☐
Rigger	Move Magnet to MR Suite	Direction 2341078, GEMSO 3.0T/940 Delivery & Installation	☐
Rigger	Leveling & Bolting Down Magnet	Direction 2341078, GEMSO 3.0T/940 Delivery & Installation	☐
	Hand-off to GE Field Engineer for “Commissioning”		

1-1 MANUAL CONTENT (continued)

TABLE 1-1 (CONTINUED)
MAGNET SITING, INSTALLATION & COMMISSIONING
FLOW CHART & CHECK LIST

COMMISSIONING			
Responsibility	**Function**	**Reference Document**	**Complete**
	From “Siting & Installation” Phase		
GE Field Engineer	MR Magnet Safety Document 2301164PRE Review & Compliance	Direction 2295400, SAFETY REQUIREMENTS Tab	☐
GE Field Engineer	Magnet Conversion to Operating Configuration AND Immediate Vent Installation	Direction 2295400, SET-UP AND CALIBRATION Tab	☐
GE Field Engineer	Magnet System Components Installation & Functional Checks: • Cryocooler • MRU • Magnet Monitor	Direction 2295400, SET-UP AND CALIBRATION Tab	☐
GE Field Engineer	Magnet Commissioning Checks	Direction 2295400, SET-UP AND CALIBRATION Tab	☐
GE Field Engineer	Connect Magnet Monitor	Direction 2230681, GEMS Magnet Monitoring Hardware (SHEM) Installation	☐
GE Field Engineer	Magnet Precool & Fill	Direction 2295400, SET-UP AND CALIBRATION Tab	☐
GE Field Engineer	Ramp Magnet	Direction 2295400, SET-UP AND CALIBRATION Tab	☐
GE Field Engineer	Shim Magnet	Direction 2295400, SET-UP AND CALIBRATION Tab	☐
GE Field Engineer	Main Field Adjustment A/R	Direction 2295400, SET-UP AND CALIBRATION Tab	☐
	Hand Off to Systems Calibration		

1-1 MANUAL CONTENT (continued)



MAGNET COMMISSIONING FLOW DIAGRAM
ILLUSTRATION 1-4

1-2 WARNINGS

CAUTION

In accordance with International Standard IEC 601-1 the manufacturer is not responsible for any consequences caused by unauthorized modification of this type B equipment.

All procedures covered in this manual, other than the indicator lamp checks on the Magnet Rundown Unit, should be performed by a General Electric authorized service representative.

Note

All electrical installations preliminary to positioning equipment at the site prepared for the equipment shall be performed by licensed contractors. Additionally, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between electrical equipment, calibrations and testing shall be performed by qualified GE Medical Systems personnel. The products involved and their accompanying electrical installations are highly sophisticated. Special engineering competence is required.

GE uses its own specially-trained field engineers to perform all electrical work on these products. All GE electrical work on these products will comply with the requirements of applicable electrical codes. The purchaser of GE Medical Systems equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel from third-party service companies with equivalent training or licensed electricians) to perform electrical servicing on the equipment.

WARNING!

- **DURING INSTALLATION ALWAYS HAVE MAGNET CONNECTED TO A GOOD ELECTRICAL GROUND VIA THE DESIGNATED GROUNDING POINTS.**
- **DO NOT ENERGIZE MAGNET UNTIL COOLING IS COMPLETED. WHILE ENERGIZING MAGNET, LIMIT POWER SUPPLIED TO MAGNET CURRENT PLUS 2%.**
- **AN ENERGIZED MAGNET PRODUCES A VERY STRONG MAGNETIC FIELD THAT REMAINS STRONG SIGNIFICANTLY BEYOND THE MAGNET ITSELF. POST WARNING SIGNS OUTSIDE THE 5 GAUSS ZONE TO ALERT PERSONNEL AND RESTRICT PUBLIC ACCESS.**
  - **A. DO NOT ALLOW ANYONE WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR OTHER BIOSTIMULATION DEVICES WITHIN THE 5 GAUSS ZONE.**
  - **B. DO NOT ALLOW ANYONE WITH STEEL IMPLANTS SUCH AS ANEURYSM OR SURGICAL CLIPS TO APPROACH THE MAGNET.**

1-2 WARNINGS (continued)

CAUTION

A magnet at field can interfere with sensitive electrical devices such as oscilloscopes and photomultiplier tubes, can damage cameras and analog watches, and can erase magnetic storage media such as floppy / hard disks, digital tape and credit / bank cards.

WARNING!

- **DO NOT WEAR SHOES, CLOTHING OR PERSONAL ITEMS CONTAINING MAGNETIC MATERIALS NEAR A RAMPED MAGNET.**
- **ONLY USE NONMAGNETIC TOOLS / EQUIPMENT WITHIN THE 5 GAUSS ZONE WHILE THE MAGNET IS ENERGIZED (AT FIELD). MAGNETIC OBJECTS CAN BECOME DANGEROUS PROJECTILES WITHIN THE FIELD.**
  - **A. MAGNETIC METALS: COBALT ALLOYS, IRON, NICKEL ALLOYS, STEEL (EXCEPT STAINLESS STEEL)**
  - **B. NONMAGNETIC METALS: ALUMINUM, BRASS, BRONZE, COPPER, GOLD, STAINLESS STEEL, STERLING SILVER**

DANGER!!

DO NOT ATTEMPT TO MOVE ANY LARGE MAGNETIC OBJECT "STUCK" TO A MAGNET AT FIELD. THE OBJECT MAY VIOLENTLY REALIGN ITSELF, CAUSING A SERIOUS OR FATAL INJURY TO ANYONE STRUCK.

WARNING!

SMOKING IS PROHIBITED IN THE EXAM AND CRYOGEN STORAGE ROOMS. LIQUID CRYOGENS CAN LIQUIFY ATMOSPHERIC OXYGEN, PRODUCING A HIGHLY COMBUSTION SUPPORTING FLUID.

1-2 WARNINGS (continued)

A superconducting magnet at field can rapidly de-energize, releasing large amounts of helium. This is called a magnet quench. A magnet will quench under certain fault conditions. A quench can also be initiated by the operator to handle emergencies. All quenches need to be reported to GEMSO.

WARNING!

THE HELIUM GAS RELEASED DURING A QUENCH IS EXTREMELY COLD AND WILL CAUSE COLD BURNS. THE VENTED CRYOGENIC GASES WILL DISPLACE OXYGEN IN THE AIR, CAUSING ASPHYXIATION. BEFORE COMMENCING ANY PROCEDURE INVOLVING CRYOGENS OR HAVING QUENCH POTENTIAL:

- **MAKE SURE ALL REQUIRED PROTECTIVE CLOTHING IS WORN TO PREVENT CRYOGEN EXPOSURE / BURNS: GOGGLES / FACE SHIELD, NONABSORBENT GLOVES, LONG-SLEEVE SHIRT AND LONG PANTS, PROTECTIVE APRON / JACKET.**
- **MAKE SURE ROOM EXHAUST SYSTEM, VENT SYSTEM AND SUPPLEMENTAL FAN ARE INSTALLED AND RUNNING IN CONFORMANCE WITH SITE AND PROCEDURE REQUIREMENTS.**
- **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH SITE REQUIREMENTS OR THAT PROPERLY FUNCTIONING OXYGEN MONITORS ARE WORN ON ALL PERSONNEL. WEAR BREATHING APPARATUS / RESPIRATOR IF REQUIRED TO BE IN AREAS WITH LESS THAN 19.5% OXYGEN.**

The magnet user is responsible to take all necessary precautions to prevent accident or loss from use of the magnet. GEMSO accepts no responsibility for any loss or damage caused either directly or indirectly from use of magnet.

TYPE B EQUIPMENT (International Standard IEC 601-1)	
Environmental conditions for normal operation:	
a. Ambient temperature	+10°C to -40°C
b. Relative humidity	30% to 75%
c. Atmospheric pressure	700 hPa to 1060 hPa
d. Inlet temperature of compressor cooling water	no higher than 25C
Environmental conditions during trasport and storage:	
a. Ambient temperature	-20°C to +40°C
b. Relative humidity	30% to 75%
c. Atmospheric pressure	700 hPa to 1060 hPa

1-3 EMERGENCY DISCHARGE

WARNING!

MAKE SURE MAGNET VENTING IS INSTALLED, EXHAUST FANS ARE OPERATING AND MAGNET ROOM DOOR IS SECURED IN OPEN POSITION BEFORE ENGAGING MRU TO PREVENT POSSIBILITY OF HELIUM LEAKING INTO ROOM AND POSSIBLE ASPHYXIATION.

The magnet system includes a Magnet Rundown Unit (MRU), also called the Emergency Discharge Unit (EDU), connected to an internal heater incorporated in the magnet coil windings. The the main MRU is located in the electronics cabinet, with remote switches inside and outside the magnet room. If the magnet needs to be discharged quickly in an emergency, lift the cover over the Emergency Discharge Button and press and hold the button for 10 seconds or until the magnet quench begins. The magnet will discharge in about a minute.

SECTION 2 – VENDOR MANUAL MATRIX

Equipment	Vendor	GE and Vendor Manual Numbers	GE and Vendor Model Numbers
Liquid Helium Level Monitor	GEMSO (Magnex Scientific Ltd.)	(E5011MAN)	2255532 (E5011)
Shield Cooler Coldhead	Leybold	(GA 12.112/7)	2255517 (RGD 5/100–1)
Shield Cooler Compressor	Leybold	(GA 12.129/4.02)	2255524 (CP6000)
Main Power Supply Cabinet	Electronics Measurements Inc. (EMI)	750 Amp 46–294439P6 (83–452–010 Rev.4 04/03/92)	750 Amp 46–260776G3 (EMI Model 452–62–1)
Superconducting Shim Power Supply	GEMSO (Magnex Scientific Ltd.)	(E1030MAN)	(E1030)
	Electronics Measurements Inc. (EMI)	46–294439P7 (83–452–011)	46–260777G3 (EMI Model 452–62–2)
Magnet Rundown Unit (Emergency Discharge Unit)	GEMSO (Magnex Scientific Ltd.)	Installation: (E7006OPE)	2295525–8 (E7006)
		Operation: (E7006OPE)	2295525–8 (E7006)

SECTION 3 – MAGNET SYSTEM COMPONENT PARTS

3–1 EVACUATION PORT AND DROP–OFF PLATE

Cryostat Valve

The cryostat is evacuated through the cryostat valve. When the correct vacuum level has been reached, the valve can be closed, maintaining the vacuum in the cryostat after pumps have been removed.

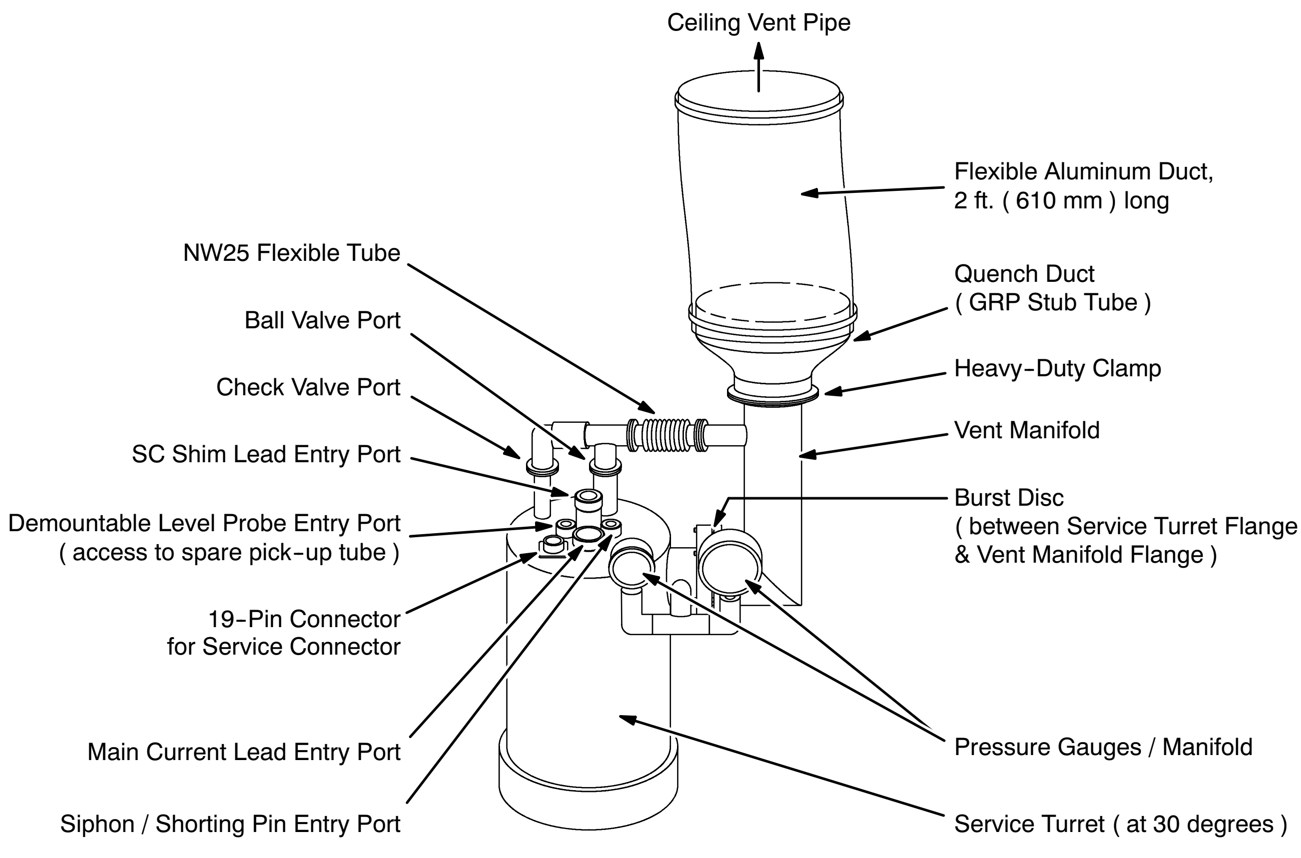
Drop–Off Plate

This plate is supplied with some systems to prevent a dangerous pressure build–up in the vacuum space.

Ten–Pin Seals

The ten–pin seals located on the cryostat near the cryostat valve provide electrical access to the vacuum space. The cryostat shield temperature sensors are accessed through these seals.

3–2 SERVICE TURRET



SERVICE TURRET
ILLUSTRATION 3–1

3-2 SERVICE TURRET (continued)

Burst Disc

The burst disc, shown in illustration 3-1 with respect to the service turret, breaks when pressure exceeds its designed pressure capacity, normally 5 psi.

Non-Return (Helium Exhaust) Valve

During normal operation the non-return valve allows evaporated helium gas to exit without allowing air to enter the cryostat. Typically the valve is set to open at 0.05 bar pressure. The valve should be bypassed during helium fills.

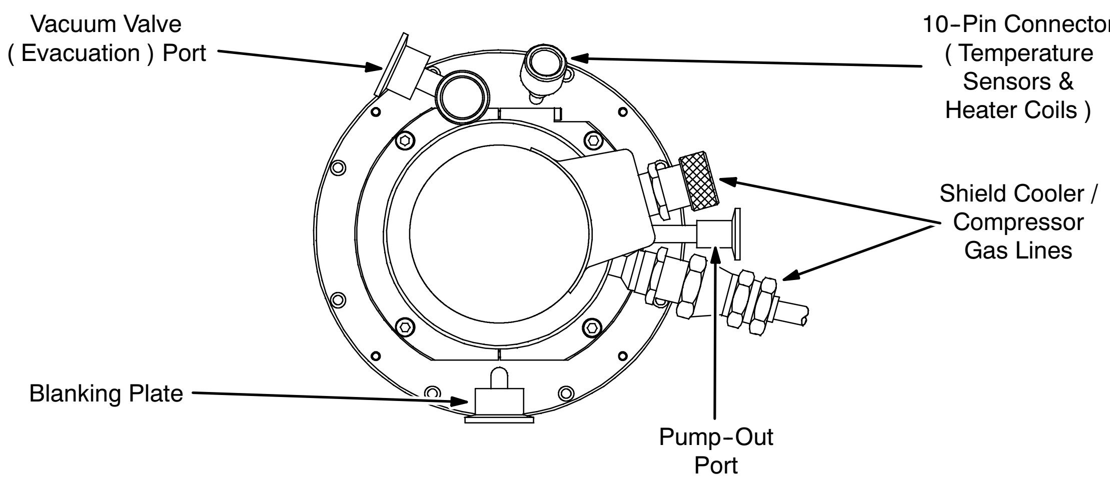
Siphon / Shorting Pin Entry

This entry point allows blow-out tube, siphon and shorting pin insertion.

Demountable Helium Level Probe Entry

The access point where this probe can enter the helium vessel.

3-3 COLDHEAD TURRET



COLDHEAD TURRET
ILLUSTRATION 3-2

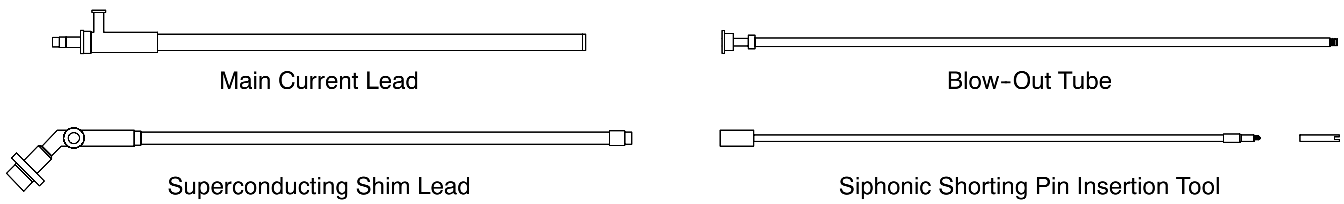
Shield Cooler Evacuation Valve

This valve is for evacuating the shield cooler.

Shield Cooler Electrical Feed-Through / Ten-Pin Seal

This is an electrical connection access point for temperature sensors and heaters.

3-4 AUXILIARY ITEMS



AUXILIARY ITEMS
ILLUSTRATION 3-3

Main Current Lead

The main current lead shown in Illustration 3-3 engages a connector inside the magnet to deliver power to the magnet. At its top it connects to the main power cables via two copper terminal blocks. The power cables in turn connect to the magnet power supply. When the magnet has reached a stable field the main current leads may be removed from the cryostat. The power cables have two different sized lugs to ensure correct polarity when connected.

Superconducting Shim Lead

The superconducting shim lead shown in Illustration 3-3 engages a connector inside the cryostat to energize the shims. Outside the magnet the shim lead connects to the power supply via the shim cable. Once the shims have been energized the shim lead can be removed.

Blow-Out Tube

The blow-out tube shown in Illustration 3-3 is used to fill and remove liquid nitrogen from the helium vessel after the pre-cool stage. The tube enters the cryostat through the siphon entry and screws into a connector inside.

Shorting Pin Tool

The shorting pin tool shown in Illustration 3-3 enters the cryostat through the siphon entry to insert / remove the shorting pin. The shorting pin plugs into a connector inside the magnet and can be inserted when the magnet field is stable.

3-5 HELIUM LEVEL MONITOR

The helium level monitor measures the amount of helium in the helium vessel, displaying the results in millimeters. The volume of helium remaining is a non-linear function of depth because of the annular nature of the helium vessel and the monitor probe's arc shape inside the vessel.

The helium level monitor probe inside the magnet is a fine superconducting wire with a resistive matrix. During the instrument's read cycle a pulse of current is passed along the wire. This heats the top of the wire and creates a "normal zone" on the wire. The normal zone propagates along the wire. If the current pulse is chosen correctly, the normal zone propagates only to where the wire meets the liquid helium's surface. Thus, the resistance of the wire is proportional to the wire's length above the liquid level. This is measured and converted to a reading in millimeters. This is covered further in FUNCTIONAL CHECKS, Section 9, Liquid Helium Level Measurement.

The level monitor can trigger a low helium warning.

3-6 MAGNET RUNDOWN UNIT

The Magnet Rundown Unit (MRU), also called the Emergency Discharge Unit (EDU), is a small power supply with battery back-up. The MRU is used in an emergency to energize a heater embedded in the magnet coil windings and initiate a quench. GEMSO 3T/940 magnets also provide for manually dumping the B0 shield coil.

The MRU's power supply is backed-up by lead acid batteries. These batteries are constantly being trickle-charged while the main power is connected. Check the battery level regularly. If both the battery and heater lights illuminate when the battery's green "Test" button is pressed, the battery is charged and the heater connected.

SET-UP AND CALIBRATION

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SECTION 1 - DETRANSITING MAGNET

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

1-1 INTRODUCTION

Upon magnet arrival check what arrived against the packing list. Also, check the shock sensors on the shipping crate and magnet.

If fitted, the small yellow Tiny Tag shock logger box mounted on two studs at the top of the magnet should be removed and sent to GE Medical Systems Oxford immediately, along with the date and time of its removal. This box records in-transit shocks to the magnet.

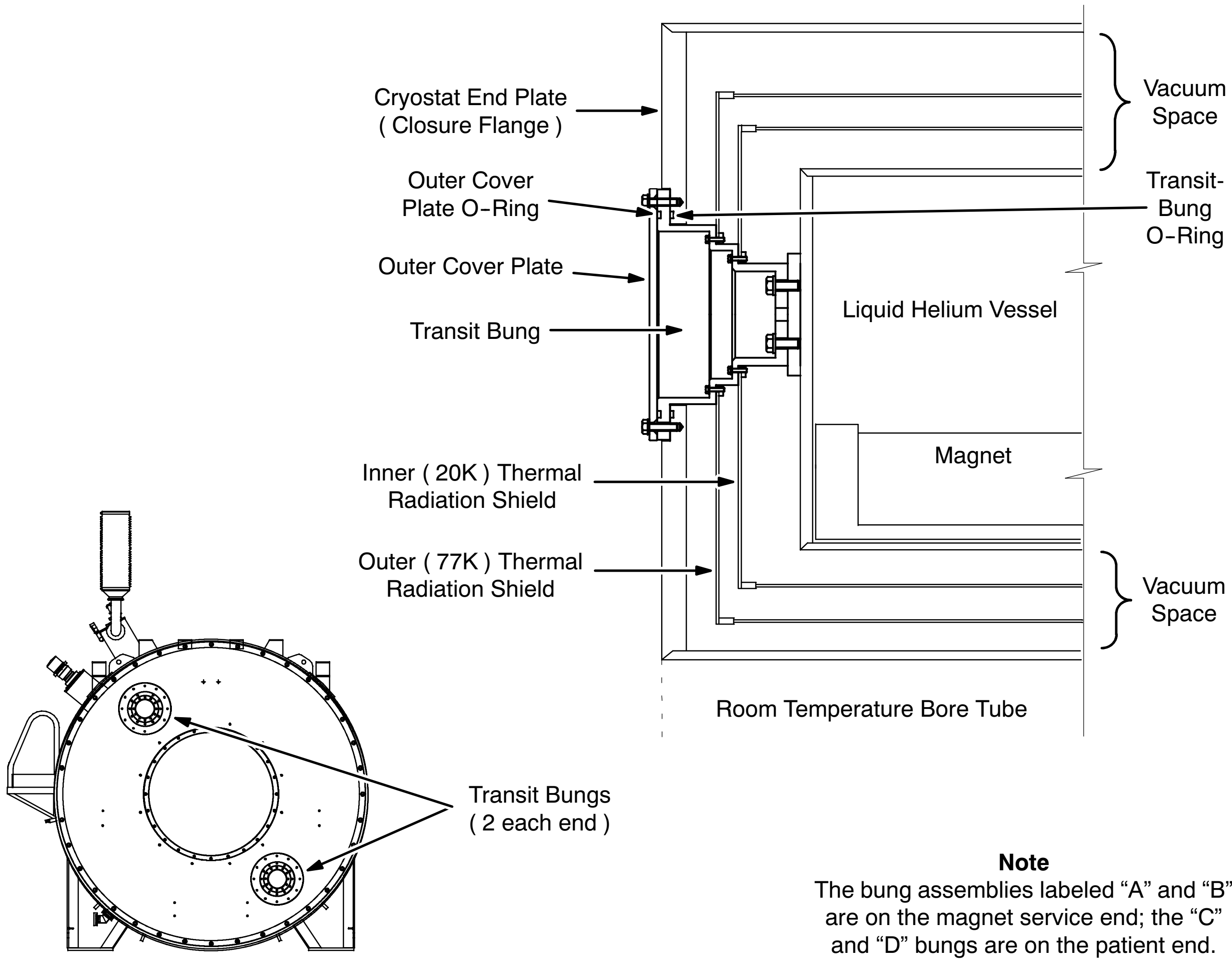
If any shock sensors have tripped during shipping, immediately contact GE Medical Systems Oxford at 8-011-44-1235-5490-00.

IMPORTANT !!!

Perform the resistance checks given in FUNCTIONAL CHECKS, Section 3, Magnet Electrical Checks, before detransiting the magnet. Contact your MAC team representative if the magnet fails to pass these checks.

On warm-shipped magnets the internal cryostat vessel is secured using four bungs inserted into the cryostat via holes in the cryostat end plate. A typical transit bung is shown in Illustration 1-1. These bungs need to be removed and replaced with inner vessel cover plates before the system can be cooled down.

1-1 INTRODUCTION (continued)



TRANSIT BUNGS
ILLUSTRATION 1-1

The cryostat is shipped under vacuum. Before the bungs can be removed the vacuum space must be purged with nitrogen gas.

The cryostat is sealed with a 47 mm (1.85 inch) diameter brass bung. A Pump-Out Port Actuator Tool (included with the regionalized Fittings and Electronics Kits) is used to remove the bung and allow connection to vacuum pumps. The tool is fitted to the cryostat with a NW50 o-ring and clamp arrangement, and a gaseous nitrogen source is connected to the Pump-Out Tool’s NW50 port. Once the tool is screwed into the M8 tapped hole in the bung, the bung can be pulled out of the cryostat using the tool’s handle. The bung fits in the cryostat tightly, so considerable effort may be required.

1-2 TOOLS / EQUIPMENT

1-2-1 Safety Equipment

- Portable Oxygen Monitor (2106236 or 2106237) if Oxygen Monitor Kit 2107184 not installed and functional in room.
- Cryogen Warning Signs specified in *2301164PRE, MR Magnet - Safety Requirements*.
- Floor exhaust fan. Fan to be specified by Installation Specialist.
- Leather gloves
- Safety glasses (46-271135P12; included in Safety Wear Kit 46-271137G1).
- Nonmagnetic safety shoes

1-2-2 Tools

- M8 Allen key
- M10 socket
- Fittings and Electronics Kit (U.S. 2297067; Japan 2297071; Euro 2297712):
  — M24 socket with extension fitting
  — Large side-cutting wire cutters (46-294167P58)

1-2-3 Equipment

Pump / Precool Kit Components*

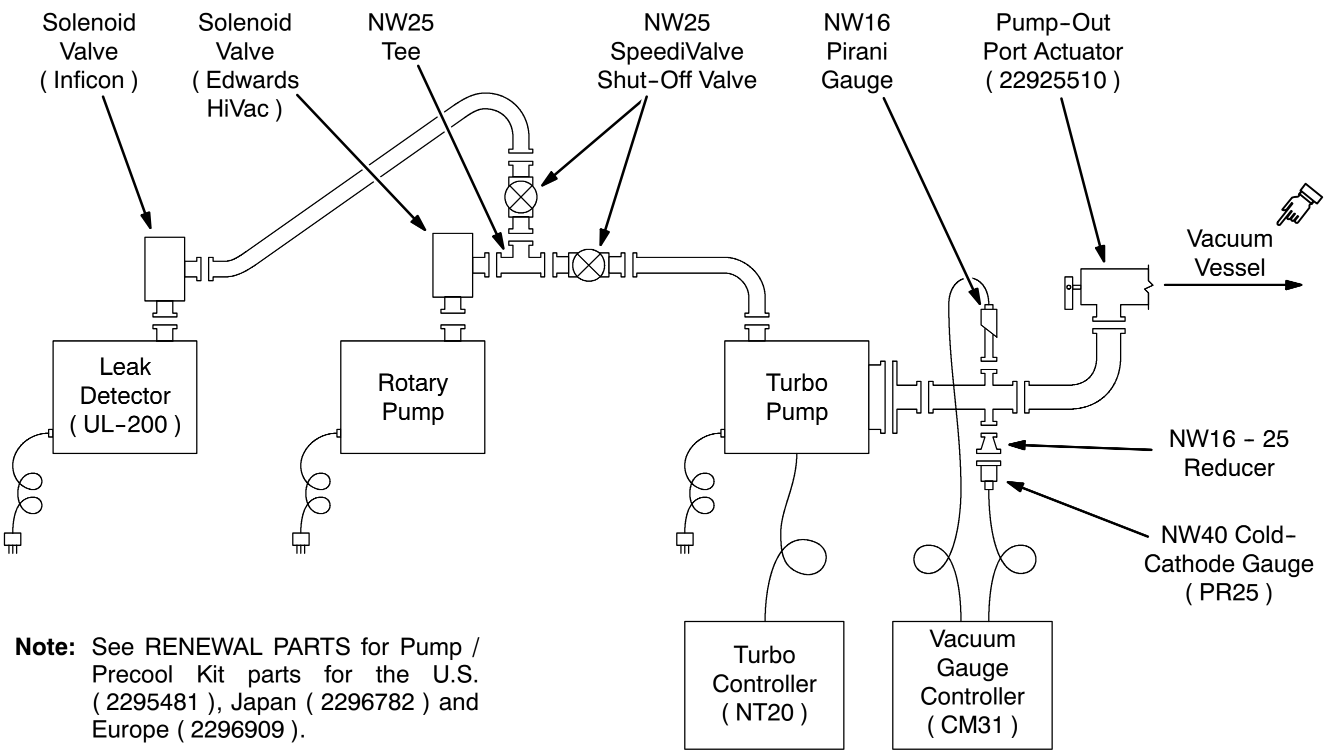
- Leak Detector Kit (U.S.: 2296796; Japan: 2296797; Europe: 2296920)
- Turbomolecular Pump (U.S. / Euro.: 2294584; Japan: 2295480)
- Vacuum Pump Kit (U.S.: 2296794; Japan: 2296798; Europe: 2296912)
- Pump-Out Port Actuator Tool (2292510) with NW50 clamp, o-ring and carrier
- Vacuum lines and fittings
- Gas regulator and lines
- 0 - 1 bar pressure gauge with NW50-25 flange adapter, NW25 tee and NW25 hand-wheel valve

* These items are included in the kits contained within the regional Pump / Precool Kits (U.S.: 2295481; Japan: 2296782; Europe: 2296909).

1-2-3 Equipment (continued)

Other Equipment

- GEMSO Detransiting Kits (one kit for each Transit Bung)
- Liquid nitrogen Dewar with “Gas Use” valve
- Bottle of nitrogen gas with pressure regulator
- Aluminized adhesive tape
- Isopropyl alcohol (46-183000P164)
- Anti-seize lubricant (2119594)
- Vacuum grease (2294083-9)



Note: See RENEWAL PARTS for Pump / Precool Kit parts for the U.S. (2295481), Japan (2296782) and Europe (2296909).

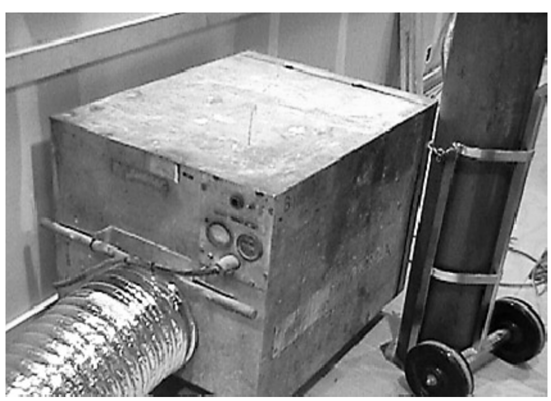
EVACUATION AND LEAK DETECTION PLUMBING
ILLUSTRATION 1-2

1-3 PREPARATION

WARNING!

- **NITROGEN GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLORLESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:**
  - **A. MAKE SURE THAT THE MAGNET ROOM DOOR IS PROPPED OPEN,**
  - **B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND**
  - **C. THE PORTABLE FLOOR FAN IS INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THIS PROCEDURE.**
- **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT A PROPERLY FUNCTIONING OXYGEN MONITORS IS USED IN THE MAGNET AND/OR STORAGE ROOM(S) BY SERVICE PERSONNEL.**
- **TRANSIT BUNG COVER PLATES ARE HEAVY. TO PREVENT INJURY WHILE HANDLING, WEAR LEATHER GLOVES AND EXERCISE CAUTION.**

1. Obtain floor exhaust fan specifications from Installation Specialist. Make arrangements to procure fan which meets specifications. A typical unit is shown in Illustration 7-3.



TYPICAL FLOOR EXHAUST FAN
ILLUSTRATION 1-3

2. Install floor exhaust fan in magnet room near magnet. Route exhaust ducting to the building exterior. Make sure the exit area is ventilated at least as well as that required of a cryogens storage area.

1-3 PREPARATION (continued)

3. Complete all items marked “X” in the “Detransiting” column of SAFETY REQUIREMENTS, *2301164PRE, MR Magnet - Safety Requirements*, Table 1, Service Procedure Safety Requirements.

IMPORTANT !!!

Make sure all identified Detransit defects are corrected before continuing with this procedure.

IMPORTANT !!!

Make sure no component part or tool is dropped inside the cryostat as they can not be retreived.

1-4 DETRANSITING PROCEDURE

The GEMSO 3T/940 Magnet is shipped with four internal bung assemblies for stability purposes. The bungs labeled “A” and “B” are in the magnet’s service end. Those labeled “C” and “D” are in the magnet’s patient end. These internal bung asemblies must be removed and replaced one at a time with internal cover plates from the GEMSO Detransiting Kit. The shipping vacuum must first be broken.

1-4-1 Breaking Vacuum and Purging Vacuum Vessel

1. Check the contents of the regional Pump / Precool Kits (U.S.: 2295481; Japan: 2296782; Europe: 2296909). There should be two rotary vacuum pumps and one turbomolecular pump. Label one rotary pump “FOR HELIUM USE ONLY.”

2. On all pumps **NOT** designated for helium use:

   a. Blank off the input, attach the exhaust oil mist filter and turn on (“I” position) the pump’s gas ballast facility.

   Note

   The pump’s gas ballast facility is activated by a small switch next to the oil site glass’ bottom. When on, small amounts of air are bled into the pump to sweep out condensates and clean the oil.

   b. Turn on power to this pump and allow it to pump on itself for six hours.

   c. Turn off (“O” position) the gas ballast.

   d. Label this clean pump as “FOR VACUUM USE ONLY.”

1-4-1 Breaking Vacuum and Purging Vacuum Vessel (continued)

3. Use a M8 Allen wrench to slightly loosen the twelve M12 bolts in the outer cover plates. See Illustrations 1-1 and 1-5. Loosen only until an air leak is heard. Then retighten the bolts only enough to stop the air leak.

4. Connect a the Pump-Out Port Actuator Tool (2292510) to the Pump-Out Port on the Cryostat. See Illustration 1-4.

5. Attach a pressure gauge and a NW25 SpeediValve as shown in Illustration 1-4.

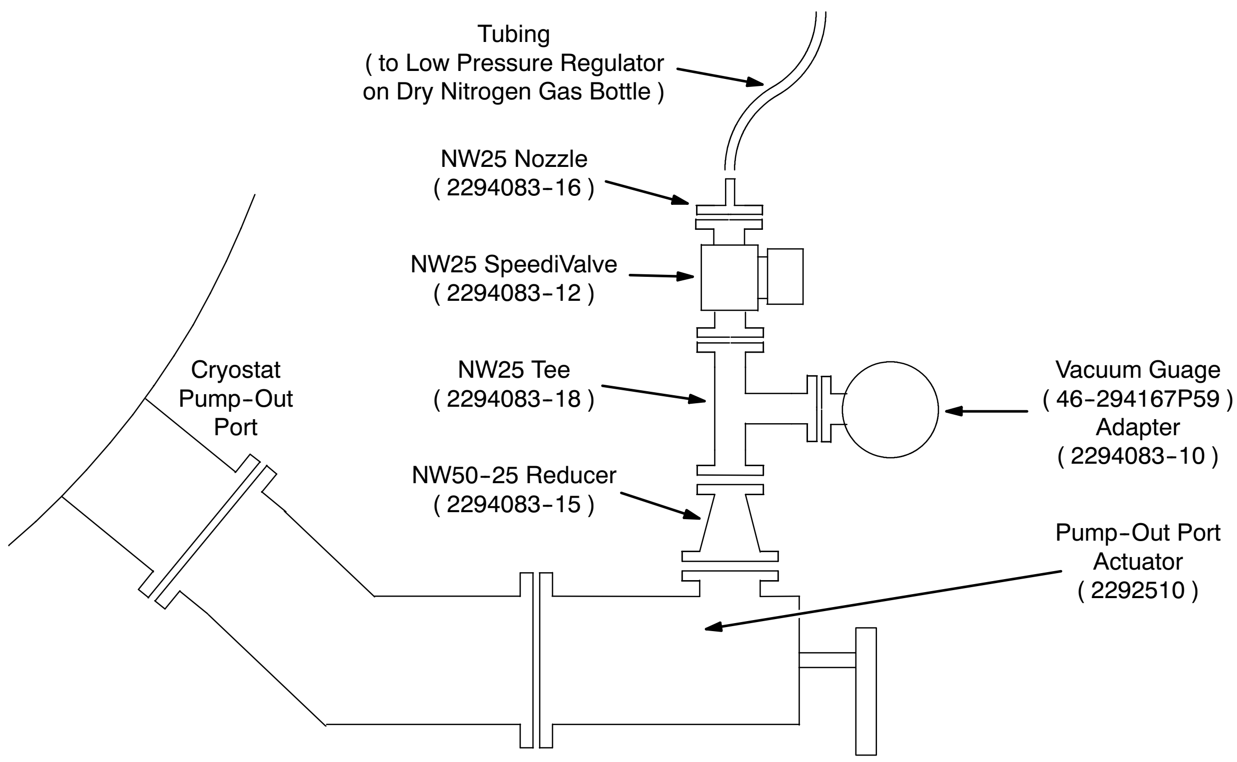
6. Connect the vacuum pump. Open the SpeediVavle and evacuate the Pump-Out Port Actuator Tool.

7. Pull out the Pump-Out Tool's Plunger.

8. Close the SpeediValve, disconnect the vacuum pump and connect a supply of dry nitrogen gas.

9. Put on your safety glasses.

10. Turn on the dry nitrogen gas supply, letting gas into the Cryostat. Adjust the pressure to 0.5 psig, then slowly open the SpeediValve.



BREAKING MAGNET VACUUM WITH THE PUMP-OUT TOOL
ILLUSTRATION 1-4

1-4-2 Detransiting the Bungs

The bung assemblies (like the one shown in Illustration 1-1) need to be removed one at a time to prevent convective air circulation inside the vacuum space and consequent loss of purge gas cover. Each hole must be covered using Detransiting Kit contents before progressing to the next bung assembly.

IMPORTANT !!!

Remove bung assemblies and replace with Detransiting Kit contents one at a time.

1-4-2-1 Bung Assembly Removal

CAUTION

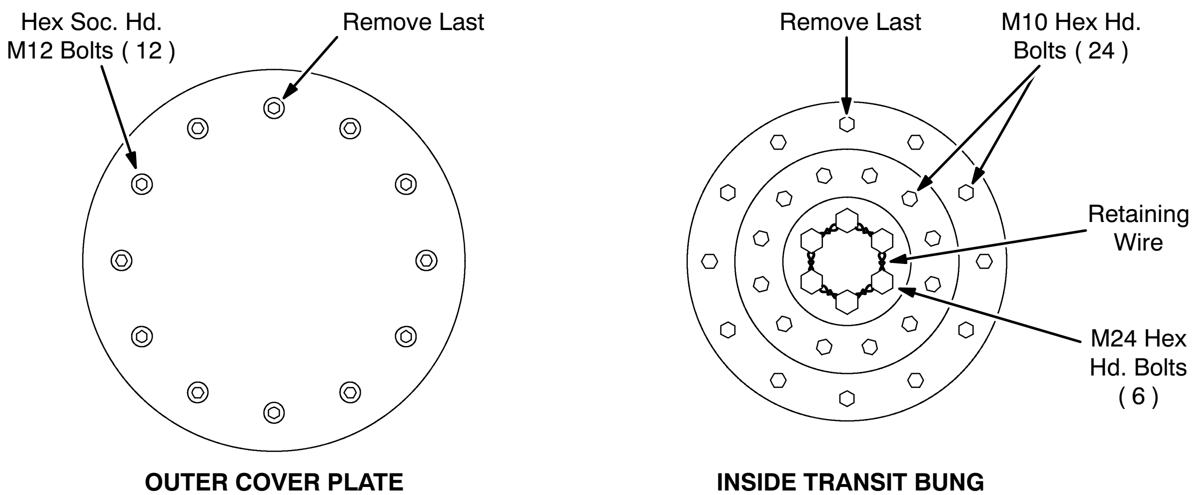
The outer cover plates covering the bungs are heavy. To avoid injury:

- **Wear leather gloves.**
- **Use each cover plate's top bolt for support while removing the other eleven bolts, then hold the plate with a "free" hand while removing the top bolt. Use both hands when lowering each plate.**

1. While the magnet vacuum space is under N_2 purge with positive pressure, use a M8 Allen wrench to remove (one at a time) the eleven lowest M12 bolts in one of the outer cover plates. Slightly loosen but do not remove the top bolt. See Illustration 1-5.
2. Remove the top bolt while securely holding the cover plate with a "free" hand. Carefully lower the plate with both hands. Rest plate with o-ring seal surface facing up.

Note

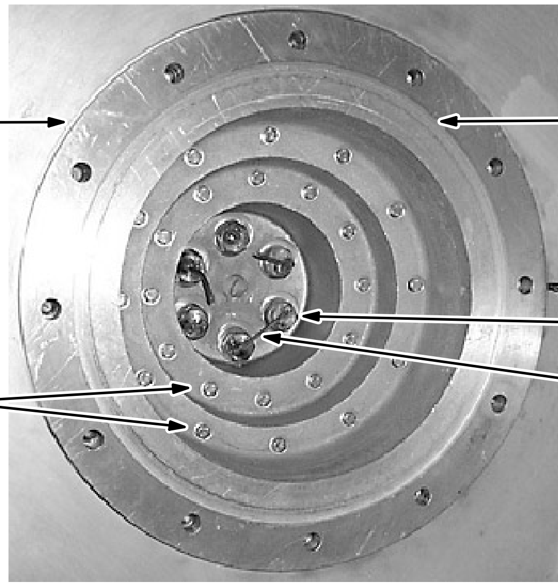
Nitrogen gas will be excaping through "weep holes" around the inside radius of the internal bung assembly.



BUNG BOLT ARRANGEMENTS
ILLUSTRATION 1-5

1-4-2-1 Bung Assembly Removal (continued)

3. Use large side-cutting wire cutters to cut the bolt retaining wires connecting the six M24 bolts inside the bung. See Illustrations 1-5 and 1-6. Remove the wire.



BOLT RETAINING WIRES CUT
ILLUSTRATION 1-6

4. Loosen all six M24 bolts to spread forces among bolts evenly. Then remove the six M24 bolts. Save these bolts in a bag marked “3T/940 Transit Bolts” for future use.

CAUTION

The transit bung weighs approximately 35 pounds (15.88 kg). To avoid injury use each bung’s top bolt for support while removing the other 23 bolts, then have a second technician hold the bung while removing the top bolt. Use both hands when lowering each bung.

5. Remove the 23 lowest M10 bolts securing the transit bung, leaving the top M10 bolt in place. See Illustration 1-5. Save all M10 bolts in the “3T/940 Transit Bolts” bag.

1-4-2-1 Bung Assembly Removal (continued)

6. Remove the top bolt while a second technician securely holds the bung. Carefully lower the bung with both hands.

Note

Save all bung assemblies. They are specifically machined to match each magnet.

7. Remove one of the bung assembly's identical inner / outer o-rings and place in a plastic bag. The other o-ring will be cleaned and used during Detransiting Kit installation.

IMPORTANT !!!

Do not remove the next bung assembly until the present hole is closed in accordance with Sections 1-4-2-2 and 1-4-2-3 below.

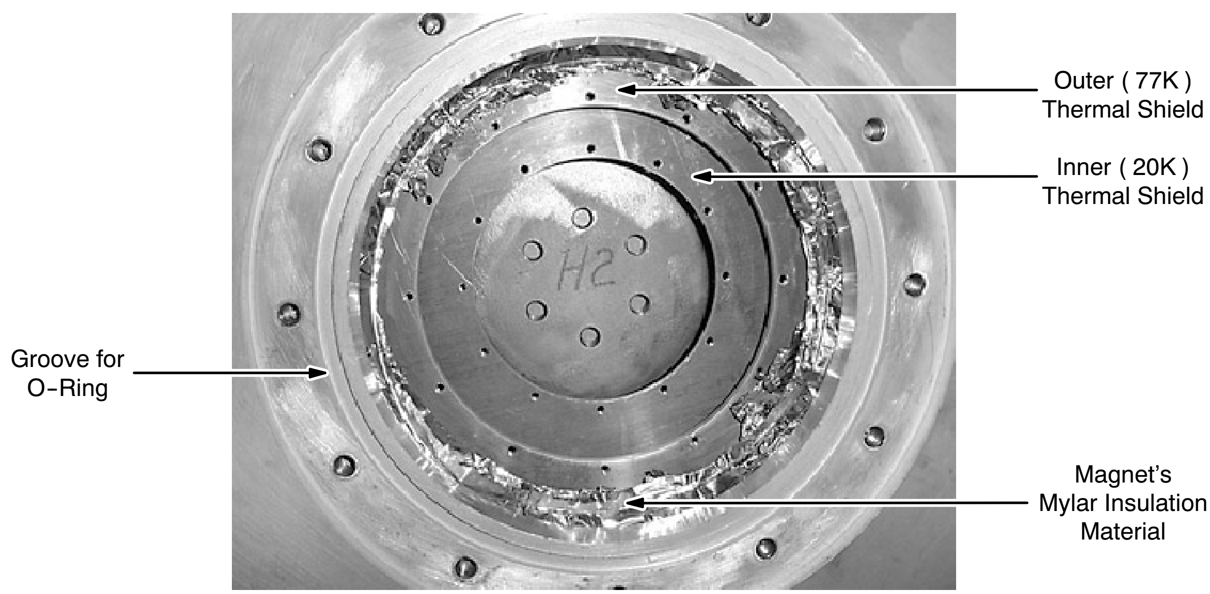
1-4-2-2 Detransiting Kit Installation

There are four Detransiting Kits per magnet, each containing the inner vessel cover plates ("cat flaps"), mylar insulation material and appropriate fasteners needed to close one hole that shipped plugged by a bung assembly.

1. Clean the surface of the inner (20K) thermal radiation shield (Illustration 1-7) and the mating surface of the smaller inner vessel cover plate with isopropyl alcohol.

IMPORTANT !!!

Do not drop inner vessel cover plates or bolts inside the cryostat as they can not be retreived.

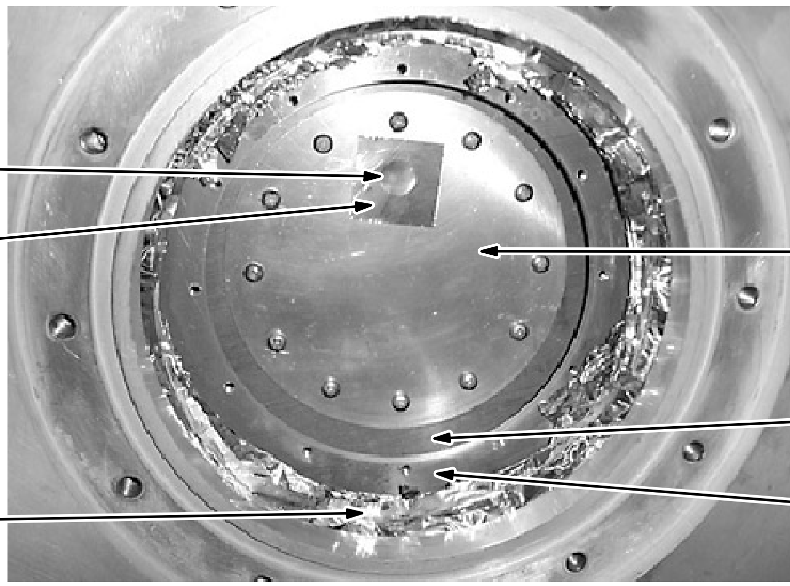


BUNG HOLE WITH TRANSIT BUNG REMOVED

ILLUSTRATION 1-7

1-4-2-2 Detransiting Kit Installation (continued)

2. With one finger through the plate's "finger hole," hold the smaller inner vessel cover plate over the opening in the inner (20K) thermal shield. Make sure the bolt holes align. See Illustration 1-8.
3. Insert one slotted head bolt to secure the inner vessel cover plate in place.
4. Cover the "finger hole" with aluminized adhesive tape. See Illustration 1-8.



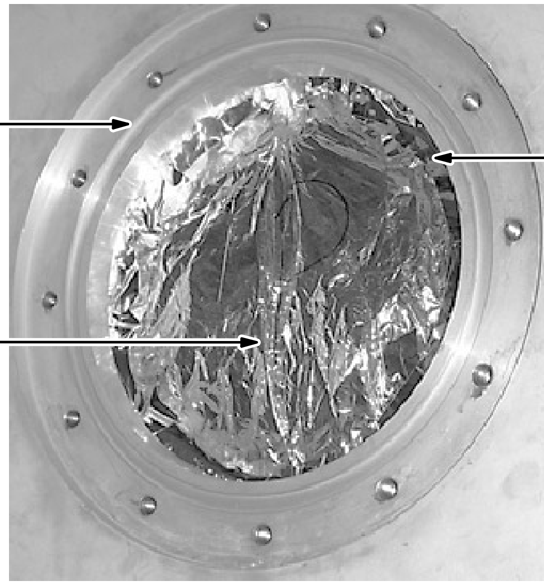
SMALL INNER VESSEL COVER PLATE INSTALLATION
ILLUSTRATION 1-8

5. Install the remaining slotted head bolts. Firmly tighten all bolts in a cross-pattern sequence.
6. Repeat Steps 1 through 5 to install the larger inner vessel plate, but misalign the "finger hole" locations.
7. Check one of the mylar insulation patches. The layers should be held together only by a plastic stud. Cut any tape holding the layers' edges together since the tape is conductive.
8. Place a mylar insulation patch on the larger inner vessel plate's surface. Make sure the patch's shiney side faces out.
9. Cover the entire gap between the magnet's mylar insulation material and the patch with overlapping strips of aluminized mylar tape. See Illustration 1-9.

IMPORTANT !!!

Do not remove next bung assembly until present hole is closed in accordance with Section 1-4-2-3.

1-4-2-2 Detransiting Kit Installation (continued)



MYLAR INSULATION PATCH INSTALLATION
ILLUSTRATION 1-9

1-4-2-3 Outer Cover Plate Reinstallation

1. Clean the surface of the o-ring groove surrounding the hole in the cryostat end plate and the mating surface of the outer cover plate with isopropyl alcohol.
2. Clean and apply a thin coat of vacuum grease to the o-ring not stored in Section 1-4-2-1, Step 7. Insert this o-ring into the cleaned groove.
3. Reinstall the outer cover plate using new hex head bolts from the Detransiting Kit. Apply anti-sieze lubrication to the bolts before insertion.
4. Firmly tighten the countersunk socket-head bolts in a cross-pattern sequence.

Note:
The outer cover plate should now be flush with the magnet's outer surface.

5. Repeat Sections 1-4-2-1, 1-4-2-2 and 1-4-2-3, in order, to detransit each of the other bung holes.

1-4-2-3 Outer Cover Plate Reinstallation (continued)

6. Inspect the gap between the closure flange and the room temperature bore (seen in Illustration 1-1) on both ends of the magnet. If the gaps are clearly unequal, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715) immediately. Elsewise continue to Step 7.

Note

If the closure flange moves during venting, its o-ring can get trapped and damaged. A damaged flange o-ring will prevent the system from being pumped down.

7. Close the magnet's vacuum valve apparatus. Close the dewar's "Gas Use" port valve and shut off the regulator. Remove the plumbing connecting the dewar and magnet.

SECTION 2 - MAGNET EVACUATION

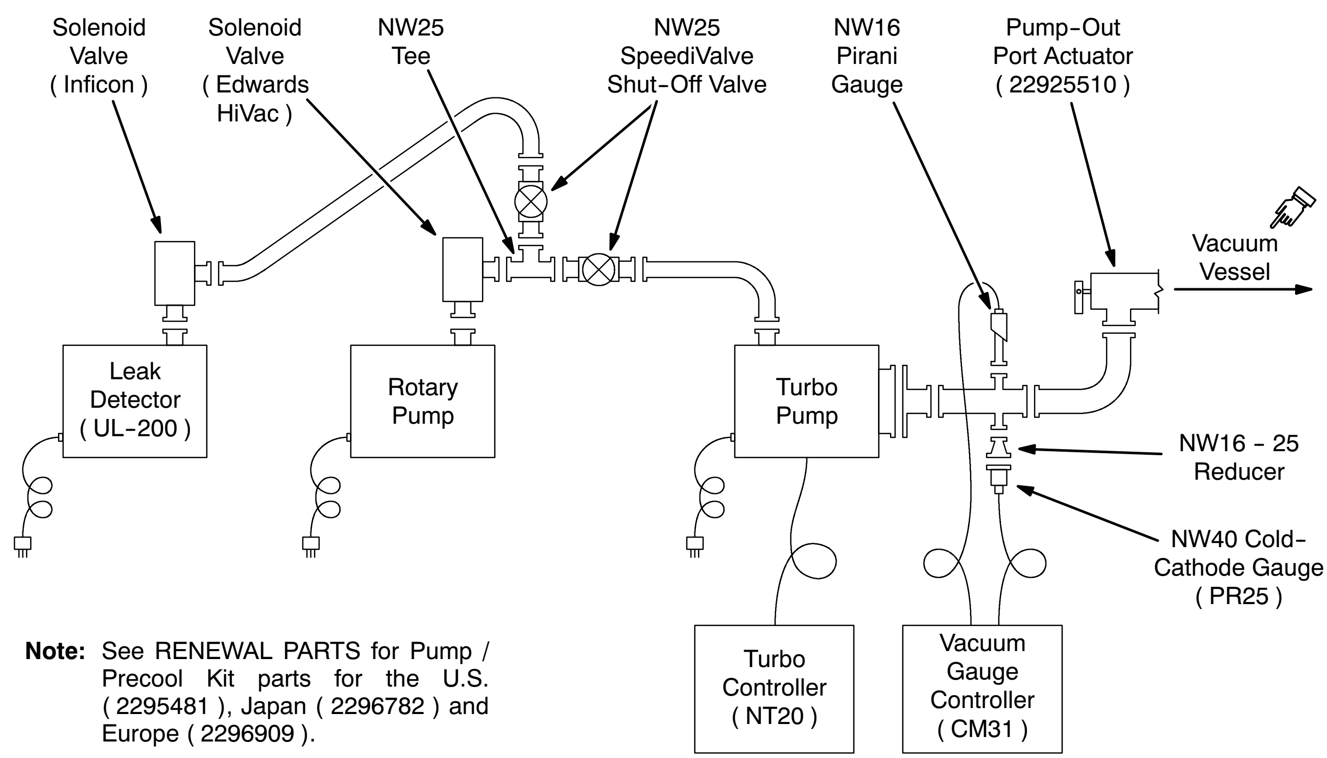
2-1 EVACUATION AND HELIUM LEAK DETECTION

1. Temporarily clamp the drop-off plate (situated next to the pump-out port):
   a. Remove the drop-off plate's four M8 bolts.
   b. Screw M8 nuts onto each drop-off plate bolt.
   c. Reinsert the four bolts with a washer between the nut and the drop-off plate.
2. Remove the pressure gauge, NW25 tee and NW25-50 reducer.
3. Connect the Turbomolecular Pump to the Pump-out Port Actuator. See Illustration 2-1.
4. Make sure the oil mist filter from the regional Pump / Precool Kit is installed on the Rotary Pump.

Note

If possible, duct the exhaust from the Rotary Pump's oil mist filter to a well-ventilated area outside the room.

5. Connect the Rotary Pump, Leak Detector, solenoid valves and SpeediValve shut-off valves. Illustration 2-1.



Note: See RENEWAL PARTS for Pump / Precool Kit parts for the U.S. (2295481), Japan (2296782) and Europe (2296909).

EVACUATION AND LEAK DETECTION PLUMBING
ILLUSTRATION 2-1

2-1 EVACUATION AND HELIUM LEAK DETECTION (continued)

6. Turn on the Leak Detector, Turbomolecular Pump and Rotary Pump.
7. Open both sets of solenoid and shut-off valves to pump the evacuation pumping system.
8. Leak test the evacuation plumbing system using the Vacuum Gauge Controller. Repair any leaks found and repump the system, then turn off both the Turbo Pump and the Rotary Pump.
9. Close the shut-off valve to the Rotary Pump and Leak Detector. Leave the Leak Detector running while it comes up to operating temperature.
10. Wait until the Turbo Pump stops rotating, then make sure the shut-off valve to the Rotary Pump is closed and turn on the Rotary Pump.
11. Open the Pump-out Port Actuator.
12. Make sure the solenoid valve at the Rotary Pump is open and the shut-off valves are closed.
13. Turn on the Rotary Pump and gradually open the shut-off valve to the Rotary Pump to "rough" pump the magnet vacuum.
14. When the vacuum reading reaches .5 mbar (0.376 Torr), remove the drop-off plate's bolts and reinsert without washers or nuts.
15. When the vacuum reading reaches .1 mbar (0.075 Torr), which will take about 24 hours of pumping, turn on the high vacuum turbomolecular pump.
16. Continue pumping until the vacuum reading reaches 0.0005 mbar (0.375 mTorr). This often takes 48 hours.
17. Open the shut-off valve to the Leak Detector, then close the Pump-Out Port Actuator.
18. Calibrate the Leak Detector.
19. Open the Pump-Out Port Actuator and leak test the cryostat using the Leak Detector.
20. Leak test the helium vessel by pressurizing the vessel to 4 psig using 99.999% Helium gas. Monitor the leak rate on the Leak Detector. Continue pumping until the leak rate is < 3.75×10^{-8} Torr l s^{-1} (0.2×10^{-8} mbar l s^{-1}) (0.2×10^{-8} cc/s^{-1}) with helium background < 0.15×10^{-8} Torr l s^{-1} (5.0×10^{-8} mbar l s^{-1}) (5.0×10^{-8} cc/s^{-1}).
21. Verify that the outer cover plate bolts are hand tight and not tighter.
22. Keep pumping and monitoring the cryostat with this assembly throughout Section 7, Nitrogen Precool. A resulting vacuum < 1.5×10^{-6} millibar (0.113 milliTorr) can be expected.

SECTION 3 – VENTING INSTALLATION

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

WARNING!

EXHAUST GAS EXITING THE VENTILATION EXHAUST DUCT IS EXTREMELY COLD. TO PREVENT COLD BURNS RESTRICT ACCESS WITHIN 25 FEET (7.62 METERS) OF THE EXTERIOR EXIT OPENING.

PROTECT THE VENTILATION EXHAUST EXIT OPENING FROM ENTRY OF RAIN, SNOW OR DEBRIS THAT COULD BLOCK THE EXHAUST SYSTEM.

THE MAXIMUM ALLOWABLE PRESSURE DROP IN THE VENT DUCT BETWEEN THE MAGNET AND THE BUILDING EXTERIOR OPENING IS 17 PSI (117.2 KPA).

CAUTION

Never place weight / objects on the Coldhead or Sleeve as irreparable damage may result.

SECTION 3 – VENTING INSTALLATION (continued)

1. Make sure the vent ducting is fully attached, contains no blockages and exits the building.
2. Make sure the pressure drop in the vent system does not exceed 17 psi (117.2 KPa) from the stub tube at the magnet to the exit outside the building. Use Table 3–1 to compute the pressure drop in the system.

TABLE 3–1
HELIUM VENT LINE PRESSURE DROP MATRIX*

Inside Diameter of Vent Pipe	Cryogenic Pressure Drop, 3T/940 Magnet		Pressure Drop Per Elbow			
	Distance, Vent Component to Magnet	Pressure Drop, Straight Vent Pipe with Smooth Inside Surface	Standard Sweep**		Long Sweep**	
			45° Elbow***	90° Elbow***	45° Elbow***	90° Elbow***
in. (mm)	ft. (m)	psi/ft. (Kpa/m)	psi (Kpa)	psi (Kpa)	psi (Kpa)	psi (Kpa)
8 (200)	0.0–16.4 (0– 5)	0.024 (0.55)	0.47 (3.26)	1.13 (7.79)	0.36 (2.46)	0.55 (3.80)
	16.4–32.8 (5–10)	0.049 (1.10)	0.94 (6.50)	2.25 (15.51)	0.71 (4.90)	1.01 (7.56)
	32.8–49.2 (10–15)	0.073 (1.65)	1.42 (9.76)	3.38 (23.30)	1.07 (7.37)	1.65 (11.35)
	49.2–65.6 (15–20)	0.098 (2.21)	1.89 (13.02)	4.51 (31.10)	1.43 (9.83)	2.20 (15.15)
	65.6–82.0 (20–25)	0.125 (2.83)	2.36 (16.25)	5.63 (38.82)	1.78 (12.27)	2.74 (18.91)
	> 82.0 (> 25)	0.149 (3.38)	2.83 (19.52)	6.76 (46.61)	2.14 (14.73)	3.29 (22.70)
10 (250)	0.0–16.4 (0– 5)	0.006 (0.14)	0.15 (1.07)	0.37 (2.55)	0.12 (0.81)	0.18 (1.24)
	16.4–32.8 (5–10)	0.015 (0.34)	0.31 (2.14)	0.74 (5.10)	0.23 (1.61)	0.36 (2.49)
	32.8–49.2 (10–15)	0.021 (0.48)	0.46 (3.20)	1.11 (7.65)	0.35 (2.42)	0.54 (3.73)
	49.2–65.6 (15–20)	0.027 (0.62)	0.62 (4.27)	1.48 (10.20)	0.47 (3.25)	0.72 (4.97)
	65.6–82.0 (20–25)	0.037 (0.83)	0.77 (5.34)	1.85 (12.76)	0.58 (4.03)	0.90 (6.21)
	> 82.0 (> 25)	0.043 (0.97)	0.93 (6.41)	2.22 (15.31)	0.70 (4.84)	1.08 (7.46)
12 (300)	0.0–16.4 (0– 5)	0.003 (0.07)	0.06 (0.43)	0.15 (1.03)	0.05 (0.33)	0.07 (0.50)
	16.4–32.8 (5–10)	0.006 (0.14)	0.13 (0.87)	0.30 (2.07)	0.09 (0.65)	0.15 (1.01)
	32.8–49.2 (10–15)	0.009 (0.21)	0.19 (1.30)	0.45 (3.10)	0.14 (0.98)	0.22 (1.51)
	49.2–65.6 (15–20)	0.009 (0.21)	0.25 (1.70)	0.59 (4.07)	0.19 (1.29)	0.29 (1.98)
	65.6–82.0 (20–25)	0.012 (0.28)	0.31 (2.14)	0.74 (5.10)	0.23 (1.61)	0.36 (2.49)
	> 82.0 (> 25)	0.015 (0.34)	0.37 (2.57)	0.89 (6.14)	0.28 (1.94)	0.43 (2.99)

* Maximum pressure drop must not exceed 17 psi (117.2 KPa) through entire vent system (magnet stub tube to exit outside building).
** “Standard sweep” elbows have a 0.5 ratio of the radius of the pipe bend (R) to the vent pipe’s inside diameter (D). “Long sweep” elbows have a 1.5 R/D ratio.
*** Elbows with angles greater than 90° must not be used.

Note

If the total pressure drop calculated exceeds the maximum specified pressure drop off 17 psi (117.22 KPa), then larger diameters for some vent line sections must be selected and the total pressure drop recalculated until it is less than 17 psi (117.22 KPa).

SECTION 3 - VENTING INSTALLATION (continued)

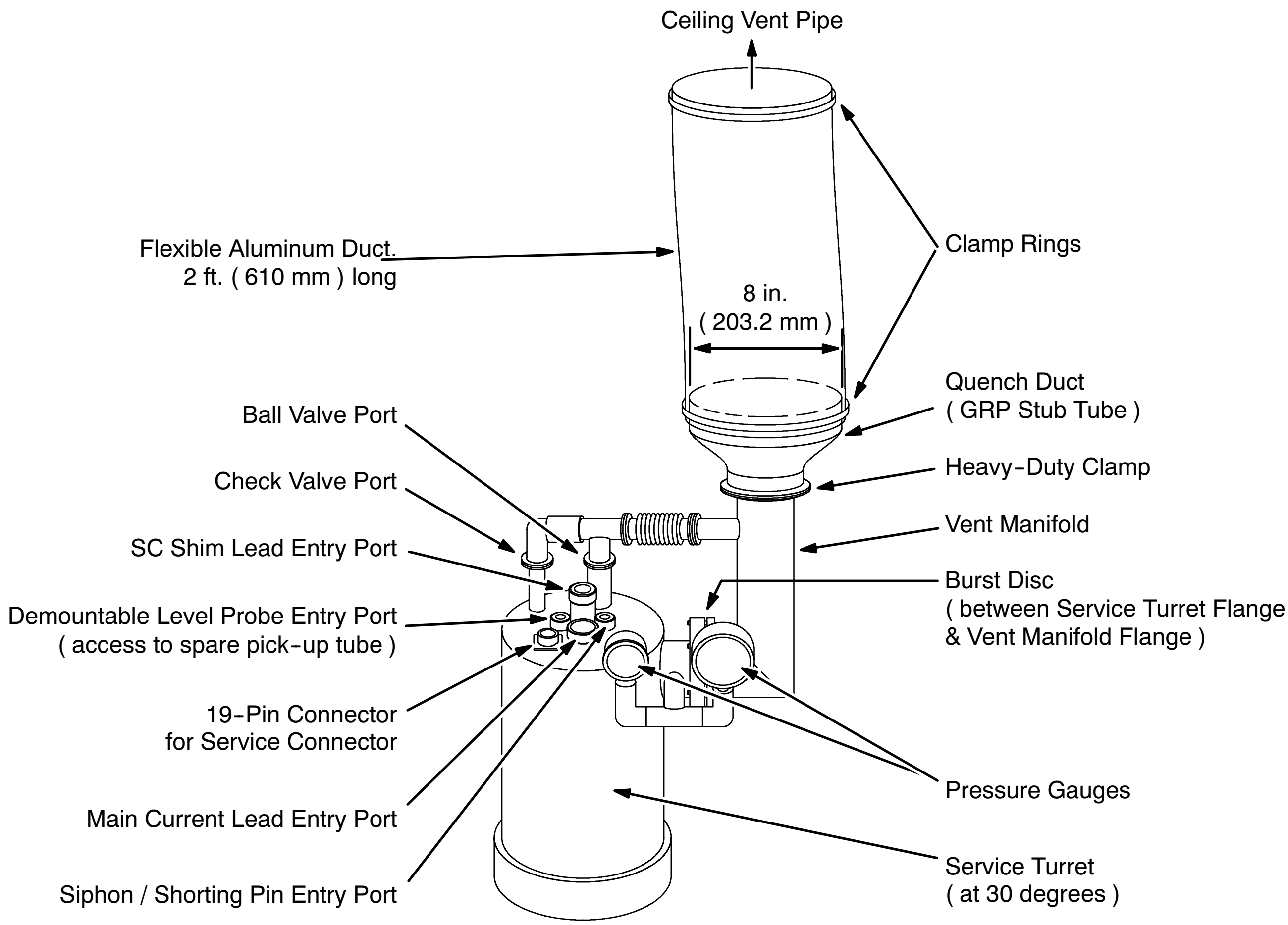
CAUTION

Insulate the vent duct in the magnet room if there are horizontal sections that could collect condensation and drip on personnel.

3. Measure and cut the 2 foot (610 mm) long flexible aluminum vent duct for proper fit at the ceiling vent pipe. See Illustration 3-1.

Note

Do not cut the duct too short to fully fit over both the GRP Stub Tube and the ceiling vent pipe nor so long as to create excess bend in the installed duct.



EXHAUST VENT CONNECTION
ILLUSTRATION 3-1

SECTION 3 – VENTING INSTALLATION (continued)

4. Install a clamp ring onto the each end of the flexible vent duct.

Note

Make sure both clamp rings are on the flexible vent duct before sliding the duct onto the GRP Stub Tube or the deiling vent pipe.

5. Slide one end of the flexible duct onto the GRP Stub Tube at the magnet and the other end onto the ceiling vent pipe, as in Illustration 3–1.

6. Slide the lower clamp ring over the overlapping flexible duct and stub tube and tighten the ring's screw.

7. Slide the upper clamp ring over the overlapping flexible duct and ceiling vent pipe, adjust the duct's position as needed and tighten the ring's screw.

SECTION 4
MAGNET RUNDOWN UNIT (MRU) INSTALLATION

Note
Use this section in conjunction with *Operating Manual for Magnex Model E7006 Emergency Discharge Unit*, available at the GE MR Engineering web site.

4-1 DESCRIPTION

The Magnet Rundown Unit (MRU) (2295525-8), also called the Emergency Discharge Unit (EDU), is a small power supply with battery back-up. It provides a means to quickly discharge the magnet's field in cases of emergency. This is done by passing current through a heater embedded in the magnet coil's windings when the MRU's red Emergency Discharge Button is pressed. The 30 seconds (± 20%) of current flow will cause the magnet to quench in about one minute, thus collapsing the magnet's field.

The MRU includes a function testing facility covered in FUNCTIONAL CHECKS, Section 4, Emergency Discharge Unit (MRU) Checks.

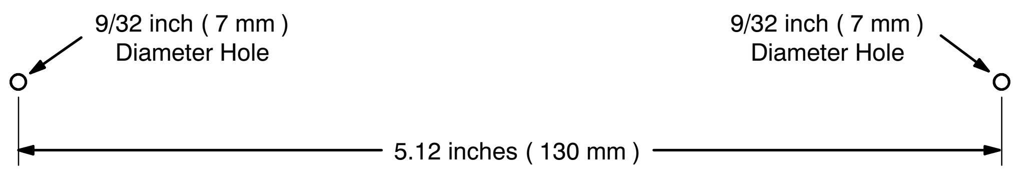
The MRU also houses a timer and manual controls to heat and release the field in the magnet's B0 shield coil. The B0 coil picks up current from the magnet's main coil over time. The B0 coil should be manually discharged during ramping and shimming procedures and at regular intervals via the clock thereafter.

4-2 MRU INSTALLATION

The E7006 MRU has a wall-mounted main housing that normally should be located in the magnet room and two remote quench buttons, one to be mounted elsewhere within the magnet room and one outside the magnet room. The main housing should be located near a power outlet.

1. Mark and drill two 9/32 inch (7 mm) diameter holes in the magnet room wall 5.12 inches (130 mm) apart horizonatlly. See Illustration 4-1.

2. Secure one of the wall anchors included with the MRU in each hole.



HOLE TEMPLATE, MOUNTING MRU TO WALL
ILLUSTRATION 4-1

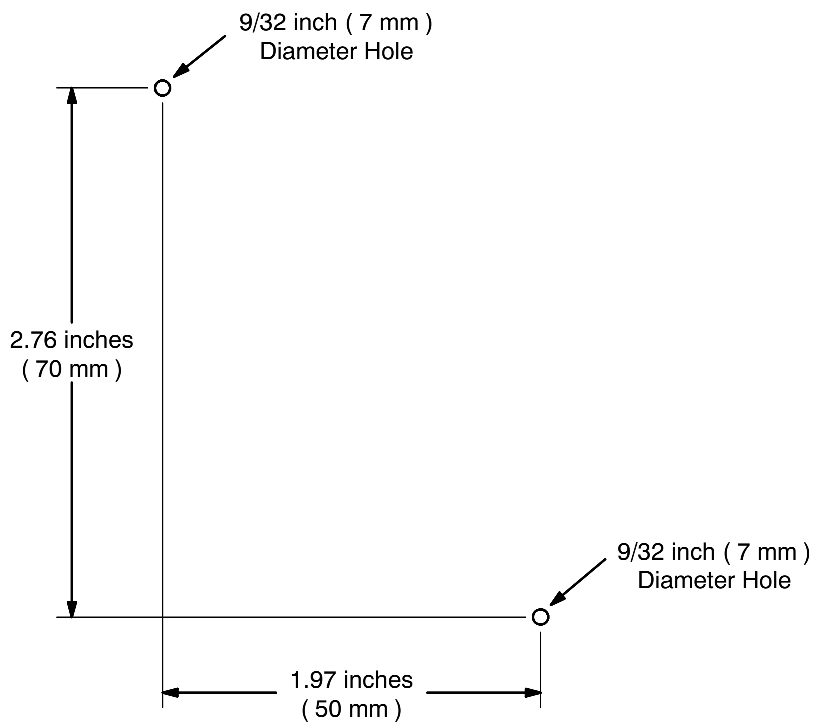
4-2 MRU INSTALLATION (continued)

3. Remove the MRU cover and place the MRU housing on the wall with the housing's mounting holes aligned to the wall anchors.

4. Insert and tighten the screws included with the MRU.

5. Locate where each of the two remote MRU buttons will be mounted. One button should be elsewhere inside the magnet room and the other outside the magnet room.

6. At each remote MRU button location mark and drill two 9/32 inch (7 mm) diameter holes in the wall in the arrangement shown in Illustration 4-2.



HOLE TEMPLATE, MOUNTING MRU TO WALL
ILLUSTRATION 4-2

7. Secure one of the wall anchors included with the MRU in each hole.

8. Remove the cover of each remote MRU button and place its housing on the wall with the housing's mounting holes aligned to the wall anchors.

9. Insert and tighten the screws included with the MRU.

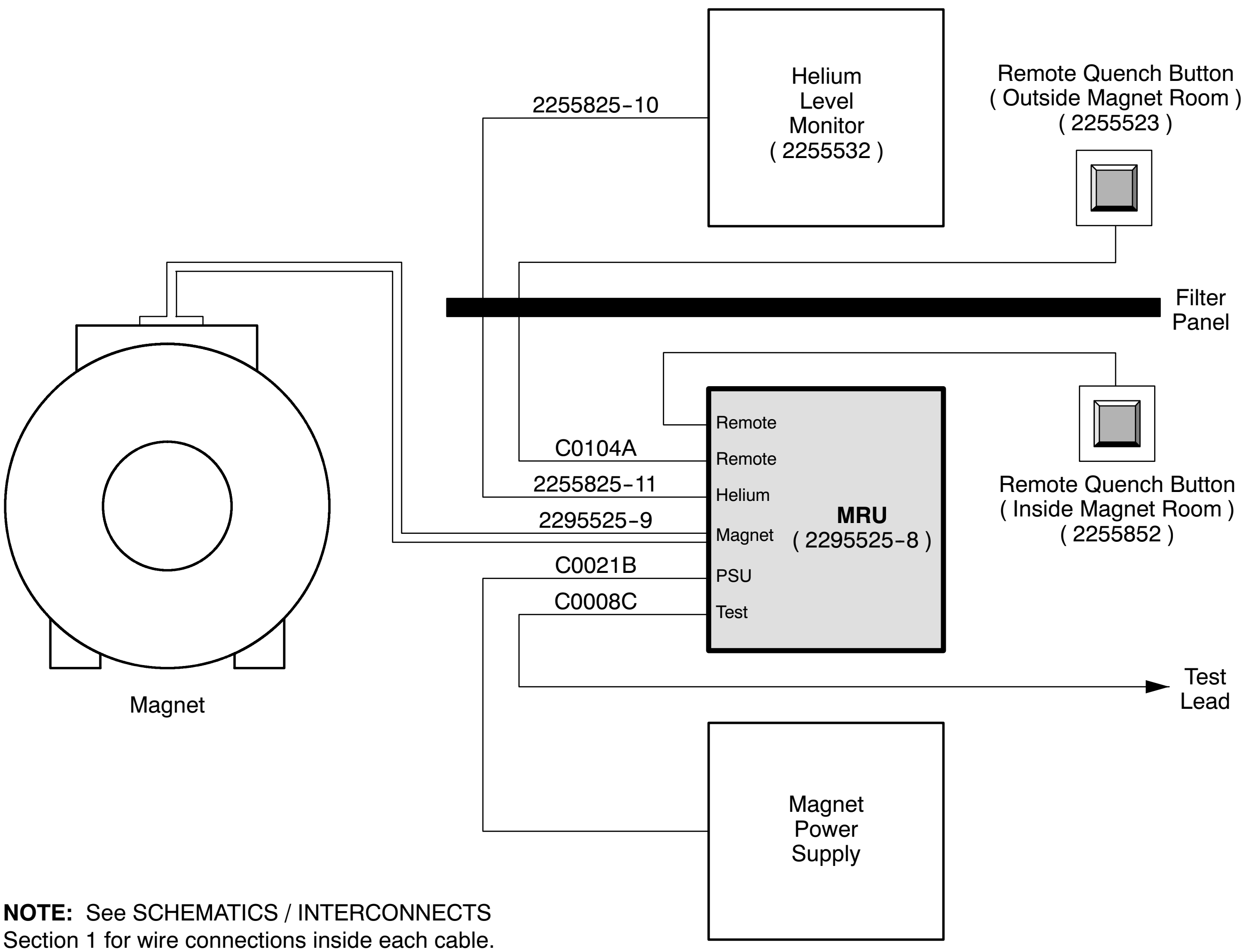
10. Make sure the main input voltage is correct, then connect the MRU to facility power. Make sure the "Power" LED illuminates.

Note

The MRU is shipped set for a 115VAC 50-60 Hz main power supply, but can be set to 220VAC.

4-2 MRU INSTALLATION (continued)

11. Attach a cable ferrite to cable 2295525-9 3/4 inch (20 mm) from its 25-pin D-plug, if a cable ferrule is not already attached, and secure with a cable tie.
12. Route cable 2295525-9 far from the gradient and other cables to prevent induced current from effecting the switch heaters and causing the magnet to quench.
13. Make the cable connections with the MRU shown in Illustration 4-3 and in SCHEMATICS / INTERCONNECTS, Section 1.
14. Replace the cover of the MRU and each remote MRU button.



MRU CABLE CONNECTIONS
ILLUSTRATION 4-3

4-2 MRU INSTALLATION (continued)

CAUTION

Battery power to the MRU is required for emergency discharge and manual / timed B0 Coil discharges during main power supply outages.

15. Unlock and open the MRU's access panel. Make sure the battery is connected to terminal 3.

CAUTION

Failure to set the B0 clock may cause the B0 coil heater to remain energized.

16. Set the B0 clock in conformance with FUNCTIONAL CHECKS, Section 4-3, Setting the B0 Dump Clock.
17. Press the "Reset" button. Make sure the LED goes out.
18. Relock the MRU door and remove the key.
19. Press the green "Test" button. Make sure both the "Battery OK" and "Heater OK" LED's illuminate. Recharge or change batteries if the "Battery OK" LED goes not light.

Note
MRU functional checks are also covered in FUNCTIONAL CHECKS, Section 4.

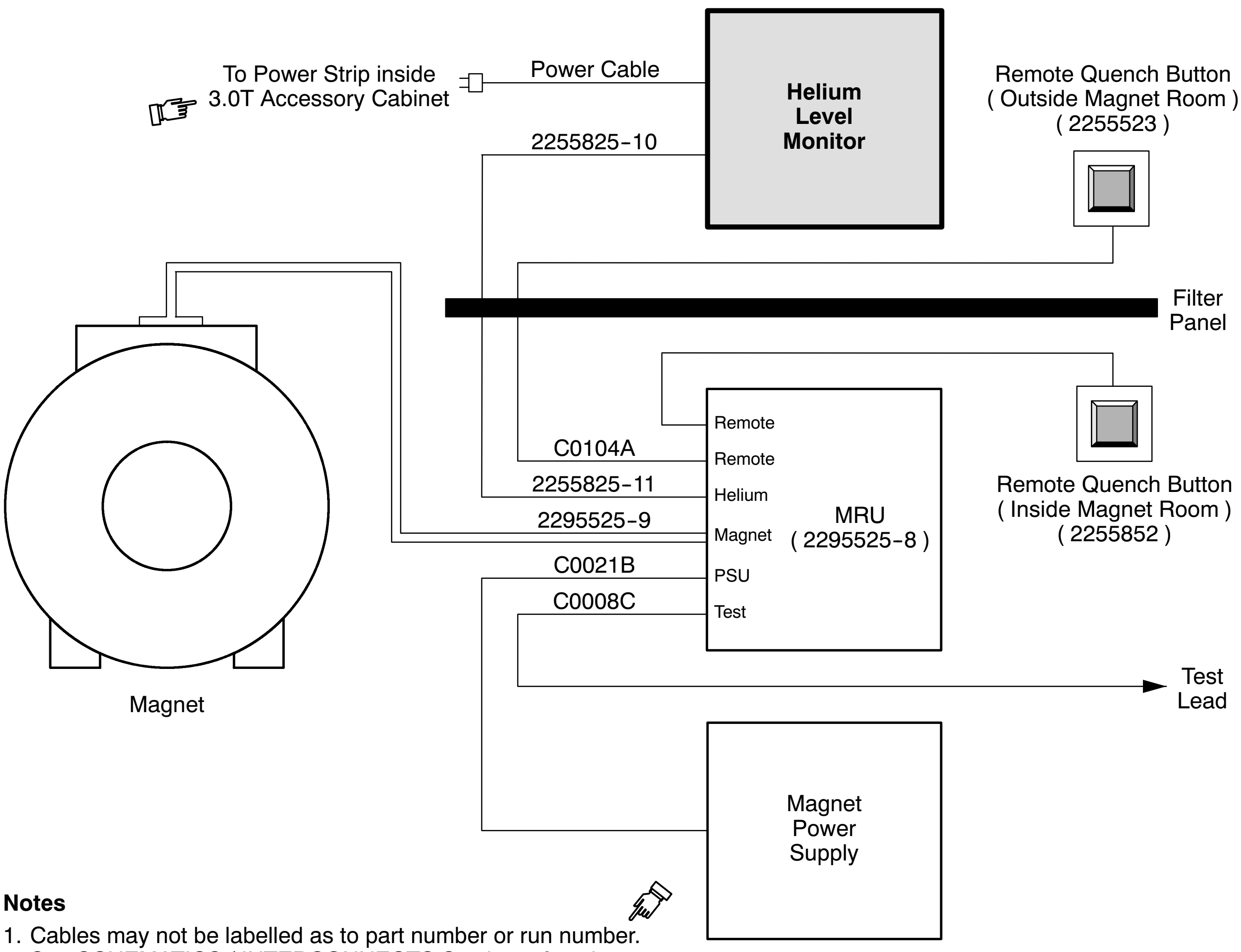
SECTION 5
HELIUM LEVEL MONITOR INSTALLATION

Note
Use this section in conjunction with *Operating Instructions and Specifications for Magnex Model E5011 Liquid Helium Level Monitor*, available at the GE MR Engineering web site.

5-1 INSTALLATION

1. Unpack the Helium Level Monitor (2255532) unit.
2. Install the Helium Level Monitor into the top position inside the 3.0T Accessory Cabinet.
3. Make the cable connections with the Helium Level Monitor shown in Illustration 5-1 and in SCHEMATICS / INTERCONNECTS, Section 1.
4. Verify that the voltage setting on the back of the Helium Level Monitor matches site power.
5. Connect the monitor to the main power supply and turn the monitor on. Make sure the “Power” LED illuminates.
6. Check probe circuit continuity. Table 5-1 shows the potential cause of certain the front panel display conditions.

5-1 INSTALLATION (continued)



Notes

1. Cables may not be labelled as to part number or run number.
2. See SCHEMATICS / INTERCONNECTS Section 1 for wire connections inside each cable.

HELIUM LEVEL MONITOR CABLE CONNECTIONS

ILLUSTRATION 5-1

5-1 INSTALLATION (continued)

TABLE 5-1
TROUBLESHOOTING PROBE CIRCUIT CONTINUITY

Front Panel Display Condition				
"Power" LED	**LCD Level Display**	**"Alarm" LED**	**"Read" LED**	**Potential Cause***
None	None	—	—	Monitor not connected to power supply or power not turned on.
				Monitor fuse blown. See Section 5-2.
				Loss of F1, F2 (at on rear panel) or F3 (at printed circuit board)
On	Full	Flashing	None	Loss of I+ / I- connections or F3 (at printed circuit board)
On	Continuously Full	Off	Normal	Loss of V+ connection
On	—	Won't Change	Continuously On	Loss of V- connection
* Check continuity / resistance values at connection(s) mentioned first, then check all combinations at cryostat connector.				

5-2 CHANGING THE FUSE

In the event the Helium Level Monitor is known to have power and all connections are correct, but it doesn't function, try changing fuses.

1. Disconnect power from the monitor.
2. Remove the monitor's top cover and then the fuse installation cover.
3. Remove the suspect fuse(s) and replace with fuse(s) of the rating shown in Table 5-2.
4. Replace the fuse installation cover and the monitor top cover.
5. Reconnect power, then check probe circuit continuity according to Section 5-1, Step 6.

TABLE 5-2
FUSE RATINGS

Fuse	Rating
F1, F2	220VAC power supply: T100mA 250VAC
	110VAC power supply: T250mA 250VAC
F3	T800mA 250VAC
F5	T200mA 250VAC

SECTION 6 - SHIELD COOLER INSTALLATION

6-1 INTRODUCTION

The Shield Cooler System comes with a Coldhead already installed on the magnet and a separate Compressor unit, which will be located in the Equipment Room. A power cable and gas supply and return lines connect the two units. Locate and read the supplier manual supplied with your system to become thoroughly familiar with the configuration, site requirements and procedures before installation. Compressor installation instructions are covered in the supplier's manual. The Shield Cooler Interconnect Diagram is shown in SCHEMATICS / INTERCONNECTS, Section 2.

6-2 COMPRESSOR INSTALLATION

1. Unpack and install the Compressor in conformance with the supplier service manual. Inspect the unit for visible damage and notify the shipper if any is found.
2. Make sure input power and the water supply are in conformance with the supplier manual specifications.
3. Position the Compressor in the equipment room as required in the site plan. Refer to site planning manual.
4. Check the Compressor voltage setting on the attached tag. If the line voltage differs from the value stated on the tag, it is essential to reconnect the terminals on the power supply. See supplier manual.
5. Connect the input power cable in conformance with the supplier manual (Leybold p/n GA 12.129) and local electrical code.

CAUTION

Do NOT connect flexlines until system gas pressure check is performed in Section 6-3, Gas Pressure Adjustment, to prevent line contamination.

6-3 GAS PRESSURE ADJUSTMENT

Refer to the supplier's manual for Compressor gas pressure requirements. If pressure is out of specified range, use the following procedures to make the required adjustments.

6-3-1 Preparation For Increasing Shield Cooler Gas Pressure

CAUTION

The following procedure purges air out of the regulator and connecting lines before the line is connected to a new cylinder of certified 99.999% helium gas.

1. Obtain a cylinder of 99.999% helium gas.

6-3-1 Preparation For Increasing Shield Cooler Gas Pressure (continued)

2. Loosely attach the high pressure regulator (46-294009P1) to the helium cylinder as follows:

   a. Thread in the screw connecting the regulator and helium cylinder about 2 turns.

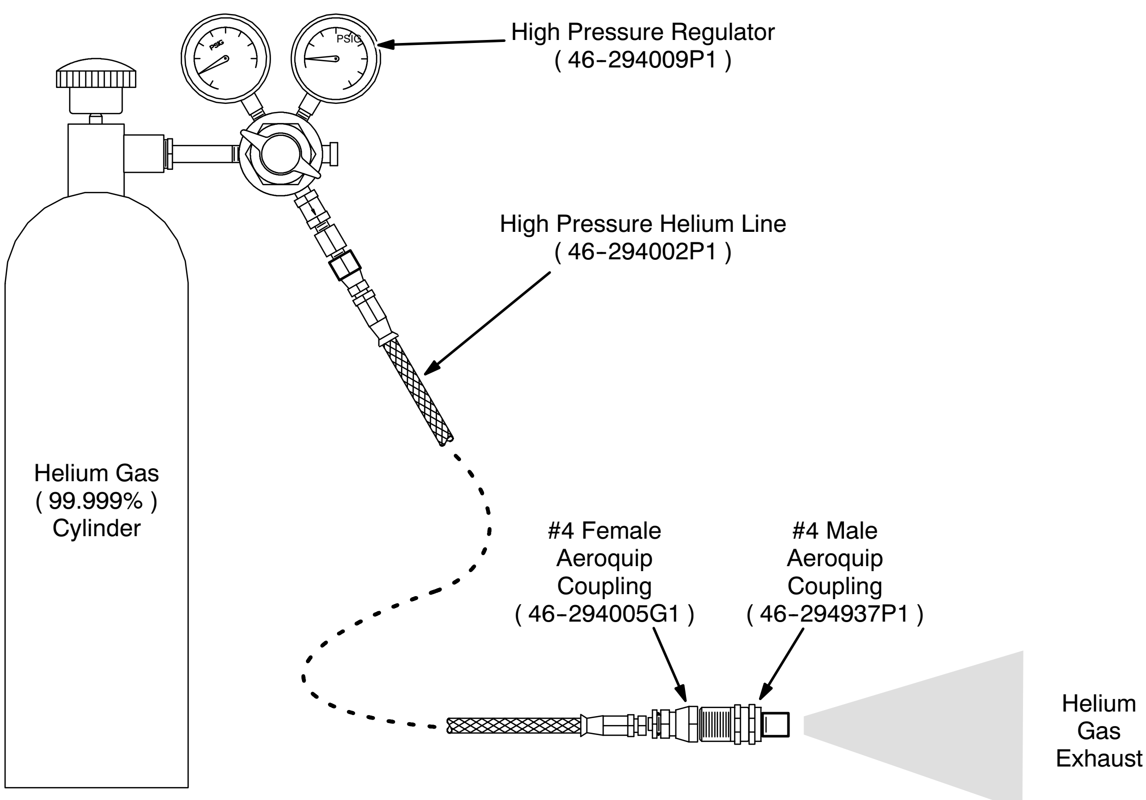
   b. Turn the regulator handle fully clockwise to open the regulator.

   c. Open the helium cylinder and immediately tighten the regulator to the helium cylinder.

   d. Close the regulator valve by turning the handle counterclockwise.

3. Attach the high pressure helium line (46-294002P1) to the regulator shut-off valve. See Illustration 6-1.

4. Attach the #4 female Aeroquip coupling (46-294005G1) to the end of the high pressure charging line. See Illustration 6-1.



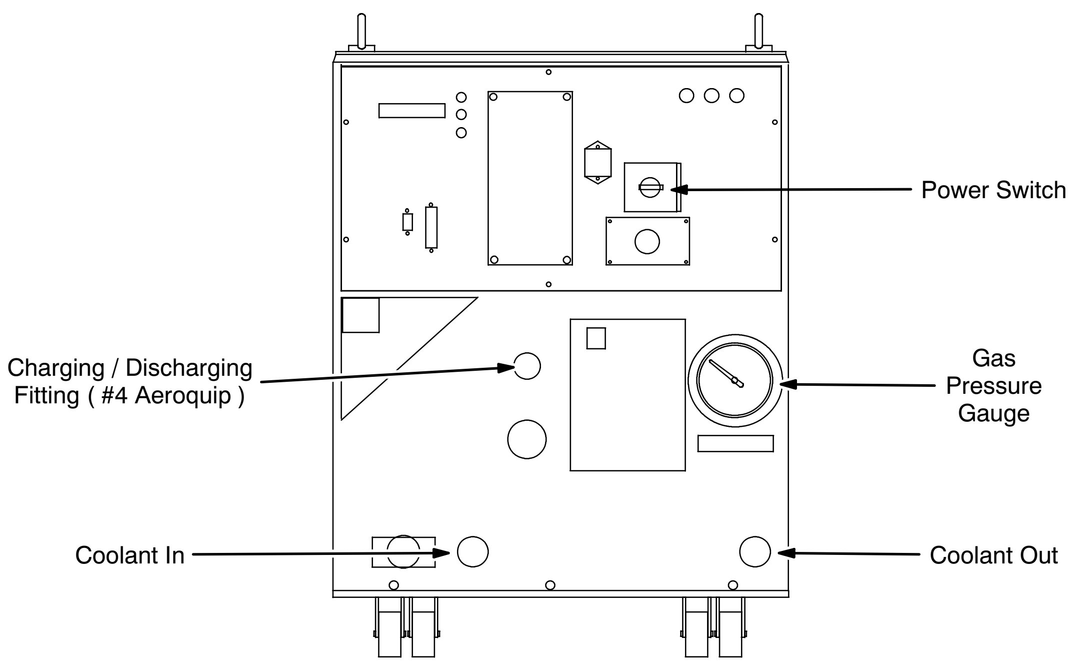
SET-UP FOR COMPRESSOR CHARGING
ILLUSTRATION 6-1

6-3-1 Preparation For Increasing Shield Cooler Gas Pressure (continued)

DANGER!!

FATAL EXPLOSIVE HAZARD!! TO PREVENT POSSIBLE FATAL EXPLOSIVE RELEASE OF GAS, OPEN MAIN VALVE ON GAS CYLINDER VERY SLOWLY. GAS IS AT 2400 PSI.

5. Attach the #4 male Aeroquip coupling (46-294937P1) to the #4 female Aeroquip coupling. Hand tighten the fittings together. This will open the helium circuit to allow the charging assembly to be purged.
6. Establish gas flow through the helium line and fittings by slowly opening the main valve on the helium cylinder and tightening the male coupling into the female Aeroquip coupling.
7. Allow helium to purge out the assembly for 2 minutes.
8. Remove the male Aeroquip fitting coupling from the female coupling.
9. Close the regulator. Shut off Compressor power to let supply and return pressures equalize in the compressor.
10. Fully open the valve on the helium cylinder.
11. Adjust the regulator control valve to achieve a pressure of approximately 200 psig.
12. Attach the female Aeroquip coupling of the purged Charging Line Assembly to the “charging” fitting on the front of the Compressor. See Illustration 6-2.



CP6000 COMPRESSOR CHARGING FITTING LOCATION
ILLUSTRATION 6-2

6-3-2 Increasing Gas Pressure

1. Increase the Compressor helium pressure by adjusting the regulator until the Compressor's gauge reads static pressure within the specification range.
2. If too much helium gas has been added, refer to Section 6-3-4, Decreasing Shield Cooler Gas Pressure, to lower the helium pressure.

6-3-3 Decreasing Gas Pressure

1. Remove the protective cap from the Compressor front panel fitting.
2. Connect the Oil-Charging Hose And Fitting (46-294003P1) to the small fitting on the front of the Compressor. See Illustration 6-2.
3. Slowly tighten the fitting until you hear gas escaping.
4. Once pressure has decreased to within specification range, immediately unscrew the fitting and hose to prevent further gas removal.
5. Replace the protective cap on the front panel fitting.

6-3-4 Disconnection and Stowage of Hoses and Regulator

1. Remove the Aeroquip coupling from Compressor. Restore Compressor power.
2. Close the regulator valve.
3. Attach the high pressure hose and #4 female Aeroquip coupling to the #4 male Aeroquip coupling until the helium circuit is open and gas is flowing.
4. Allow the high pressure line and Aeroquip couplings to depressurize.
5. Remove the male and female Aeroquip couplings from the high pressure charging line.
6. Remove the high pressure charging line from the regulator.
7. Close the helium cylinder valve. Bleed off pressure from the regulator.
8. Remove the regulator from the helium cylinder.
9. Store all equipment in the Shield Cooler Installation / Maintenance Kit (46-281088G3) carrying case.

6-4 GAS LINE INSTALLATION

6-4-1 Gas Line Routing and Shielding

CAUTION

When routing 1 inch (25.4 mm) copper gas lines make sure all slack is coiled within the equipment room to prevent white pixel generation in images and that all curvatures / coils in the gas lines have radii greater than 17 inches (432 mm) to prevent bending force on Aeroquip fittings and strain / "chirping" noise on the line.

1. Feed the gas supply line through either port in the Penetration Panel. Position one end at the Compressor and the other at the Coldhead on the Cryostat.
2. Feed the gas return line through the other port in the Penetration Panel and route parallel with the gas supply line.
3. Adjust the slack on the Helium supply and return flex lines for the most suitable length on both sides of the Penetration Panel. Loosely coil excess line length in the equipment room.

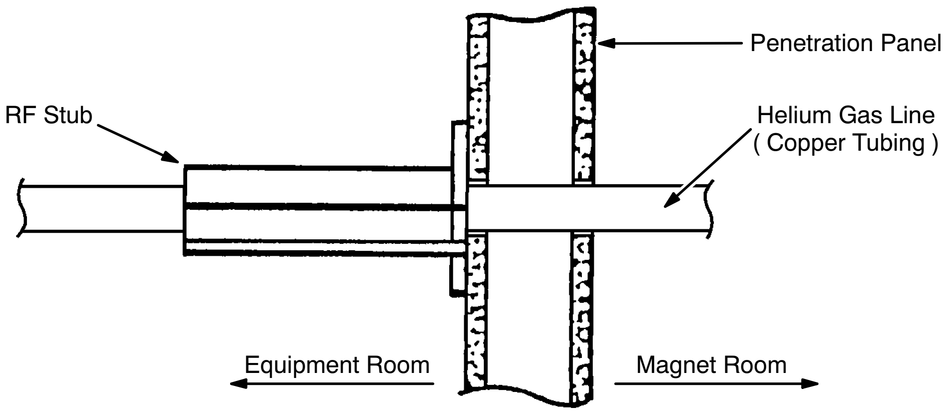
Note
RF shielding is performed from the equipment room side of the Penetration Panel using the following procedure and is applicable for both the helium supply and return lines.

4. Install RF shielding around each gas line as it passes through the Penetration Panel:
   a. Position half of the RF Stub Assembly underneath the helium supply line, with the flange end oriented toward the Penetration Panel. See Illustration 6-3.

   Note
   Make sure the RF Stub Assembly is clean (i.e., bright copper appearance). Wire brush the surface as required to insure a clean surface.

   b. Secure the flange of the RF Stub Assembly to the Penetration Panel with four screws, aligning the flange holes with the holes in the Penetration Panel.
   c. Insert copper wool around the helium supply line (top and bottom) over the length of the RF Penetration Stub where it passes through the waveguide on the Penetration Panel.
   d. Position the other half of the RF Stub Assembly over the helium supply line, with the flange oriented toward the Penetration Panel and align it over the lower half.
   e. Secure the flange of the top half of the RF Penetration Stub Assembly to the Penetration Panel with four screws, aligning the flange holes with holes in the Penetration Panel.
   f. Repeat Steps 4a and 4e for the Helium return line.

6-4-1 Gas Line Routing and Shielding (continued)

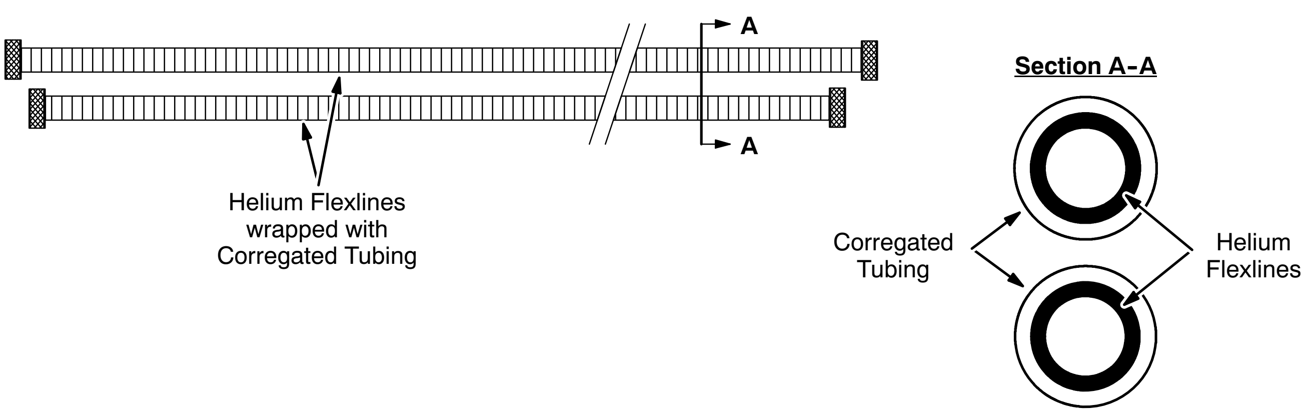


RF STUB ASSEMBLY MOUNTING
ILLUSTRATION 6-3

5. Cover both lines completely with Corregated Plastic Tubing (2251611) from the Penetration Panel where the lines exit the magnet room to the helium line fittings at the Coldhead. See Illustration 6-4.

Note

Corregated plastic tubing is used to isolate helium gas lines from each other and from ground to prevent spike noise.



HELIUM LINE INSULATION
ILLUSTRATION 6-4

6-4-2 Gas Line Connections

1. Connect the gas lines to the Compressor and Coldhead Aeroquip fittings in the sequence and by the procedure shown in the supplier manual.

Note

Refer to the supplier manual for the procedure and sequence on the connecting gas line and Aeroquip couplings.

2. Observe the Compressor pressure gauge reading after connecting the flexlines. The reading should remain constant.

Note

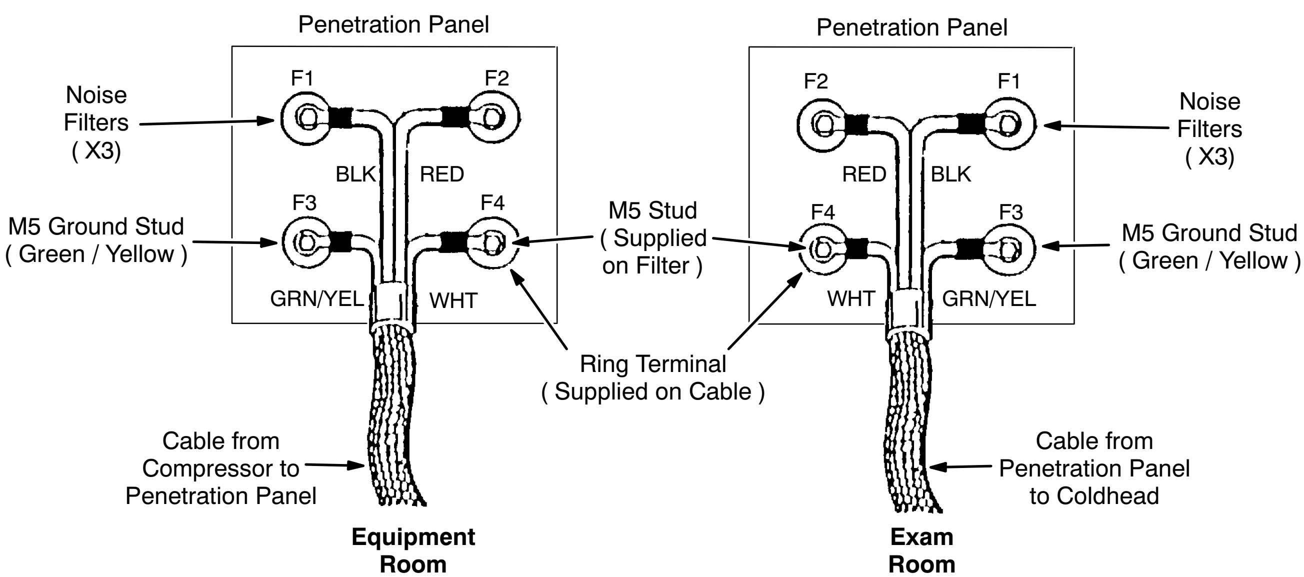
If the reading is lower, this indicates the flexlines were not fully charged when received or that a leak may exist. Refer to the supplier manual for troubleshooting instructions.

6-5 SHIELD COOLER ELECTRICAL CABLE CONNECTIONS

Note

If the Shield Cooler is installed prior to Penetration Panel installation, use a terminal block to connect the four ring terminals on the two Coldhead power cables. Make sure the same-color wires are connected.

1. Connect the four ring terminals on the end of the Coldhead electrical cable to the four studs of the noise filter inside the Penetration Panel. See Illustration 6-5.



SHIELD COOLER ELECTRICAL CABLE CONNECTIONS
ILLUSTRATION 6-5

6-5 SHIELD COOLER ELECTRICAL CABLE CONNECTIONS (continued)

2. Ty-wrap the cable to the motor shield for strain relief.
3. Connect the other end of the cable to the mating connector on the Coldhead.
4. Connect the four ring terminals on the end of the Compressor electrical cable to the four studs of the noise filter outside the Penetration Panel (equipment room side).

CAUTION

Make sure the wire colors and numbers properly match on the noise filter studs on both sides of the Penetration Panel to prevent improper operation or damage.

5. Connect the other end of this cable to the Coldhead connector on the Compressor.

SECTION 7 – NITROGEN PRECOOL

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

7–1 INTRODUCTION

Twenty to twenty–four 230–liter Dewars of liquid nitrogen (LN_2) are used to precool a warm (room temperature) cryostat over a four–day period in preparation for liquid helium (LHe) fill. The latent heat capacity of liquid helium is much lower than liquid nitrogen (21 vs. 198 KJ/KG). Therefore precooling with liquid nitrogen is more effective, economical and less time consuming.

Purging of air and water vapor with dry nitrogen before nitrogen precool is required to prevent water vapor inside the helium vessel from freezing. See SET–UP AND CALIBRATION, Section 1–4–1, Breaking Vacuum and Purging Vacuum Vessel.

IMPORTANT !!!

Liquid nitrogen is used in this procedure. Make sure of the following prior to starting the precool:

- **A MINIMUM of two fully trained and authorized persons are required to throughout this procedure.**
- **All required equipment is in place and functional.**
- **The area is secured to restrict access by unauthorized personnel.**

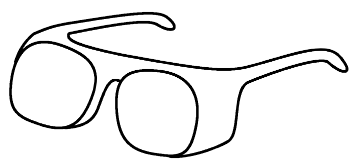
7-2 TOOLS / EQUIPMENT

7-2-1 Safety Equipment

- Goggles or safety glasses, a face shield and insulated nonabsorbant gloves from the Cryogen Safety Kit 46-271137G1. Two sets required, one per person. See Illustration 7-1.
- Protective apron / jacket.
- Nonmagnetic safety shoes.
- Portable Oxygen Monitor (2106236 or 2106237) if oxygen monitor 2107184 is not installed and functional in room.
- Cryogen Warning Signs specified in 2301164PRE, *MR Magnet - Safety Requirements*.
- Floor exhaust fan. Fan to be specified by Installation Specialist and obtained locally.

Face Shield*

Leather Gloves*



Goggles*

Protective Apron / Jacket

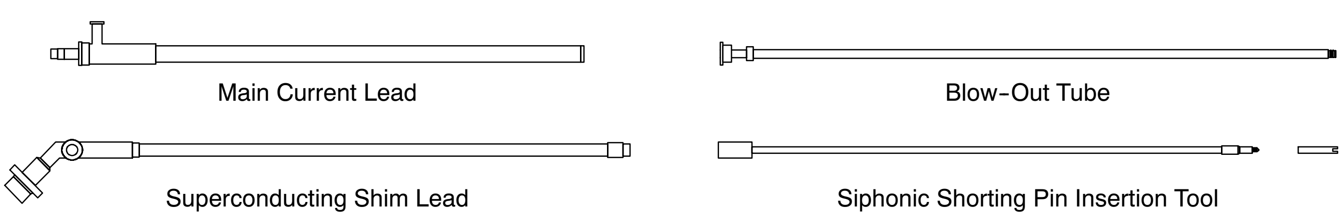
* Contained in Cryogen Safety Kit (46-271137G1).

SAFETY EQUIPMENT
ILLUSTRATION 7-1

7-2-2 Auxiliary Equipment

- Main Current Lead (2255510; for 2.9M ceiling: 2294705)
- Superconducting Shim Lead (Long: 2255512; Short: 2255511)
- Blow-Out Tube (2255509; for 2.9M ceiling: 2294703)
- Siphonic Shorting Pin & Insertion Tool (2255522; hinged for 2.9M ceiling: 2294710)
- Exhaust Gas Vent Assembly (2292536)
- 10-Pin Breakout Box (2295140)
- 19-Pin Breakout Box (2295141)
- Digital Volt / Ohm Meter (DVOM)
- Fittings and Electronics Kit (U.S. 2297067; Japan 2297071; Euro 2297712):
  — NW25 x 1 meter (39.37 inch) Flexible Stainless Steel Hose (2294083-11)
  — NW25 clamp (2294083-16)
  — Centering Ring & O-Ring (2294083-17)
  — Pressure Gauge, +/- psig (294167P63)
  — NW25 x 500mm (19.7 inches) Flexible Pipe (2294083-28)
  — NW25 to .25 NPTM Pipe Adapter (2294083-10)
- 90° Stainless Steel (or aluminum) Elbow (________)
- Flow Seals (________)
- Blanking Caps (________)

The Main Current Lead (2255510), Superconducting Shim Lead (2255512), Blow-Out Tube (2255509) and Siphonic Shorting Pin & Insertion Tool (2255522) are shown in Illustration 7-2. Lower ceiling height versions exist for each.



AUXILIARY ITEMS
ILLUSTRATION 7-2

7-2-3 Cryogen Transfer / Monitoring Equipment

- Helium Fill Kit (2295507)
- LN_2 Line Cut-Off Valve (_______)

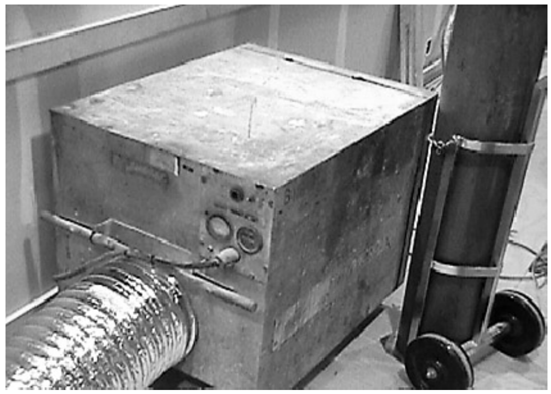
7-3 PREPARATION

WARNING!

- **NITROGEN GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLOR-LESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:**
  - **A. MAKE SURE THE MAGNET ROOM DOOR IS PROPPED OPEN,**
  - **B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND**
  - **C. THE PORTABLE FLOOR FAN IS INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THIS PROCEDURE.**
- **MAKE SURE ALL REQUIRED PERSONAL PROTECTIVE EQUIPMENT IS WORN TO PREVENT CRYOGEN EXPOSURE / BURNS: GOGGLES / SAFETY GLASSES AND FACE SHIELD; INSULATED NONABSORBENT GLOVES, LONG-SLEEVE SHIRT AND LONG PANTS; AND PROTECTIVE APRON / JACKET.**
- **SMOKING IS PROHIBITED AROUND CRYOGENS. CRYOGENS CAN CONDENSE ATMOSPHERIC OXYGEN PRODUCING A HIGHLY COMBUSTION SUPPORTING FLUID. CLOTHING WHICH HAS ABSORBED LIQUID OXYGEN CAN REMAIN FLAMMABLE FOR LONG PERIODS OF TIME.**
- **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT A PROPERLY FUNCTIONING OXYGEN MONITOR IS USED IN THE MAGNET AND/OR CRYOGEN STORAGE ROOM(S) BY SERVICE PERSONNEL.**

7-3 PREPARATION (continued)

1. Obtain floor exhaust fan specifications from Installation Specialist. Make arrangements to procure fan which meets those specifications. A typical unit is shown in Illustration 7-3.



TYPICAL FLOOR EXHAUST FAN
ILLUSTRATION 7-3

2. Install floor exhaust fan in magnet room. Route exhaust ducting to the building exterior. Make sure the exit area is ventilated at least as well as that required of a cryogens storage area.

3. Complete all items marked "X" in the "Nitrogen Precool / Purge" column of SAFETY REQUIREMENTS, *2301164PRE, MR Magnet - Safety Requirements*, Table 1, Service Procedure Safety Requirements.

IMPORTANT !!!

Make sure all identified Nitrogen Precool defects are corrected before continuing with magnet service.

7-4 MAGNET VENTING INSTRUCTIONS

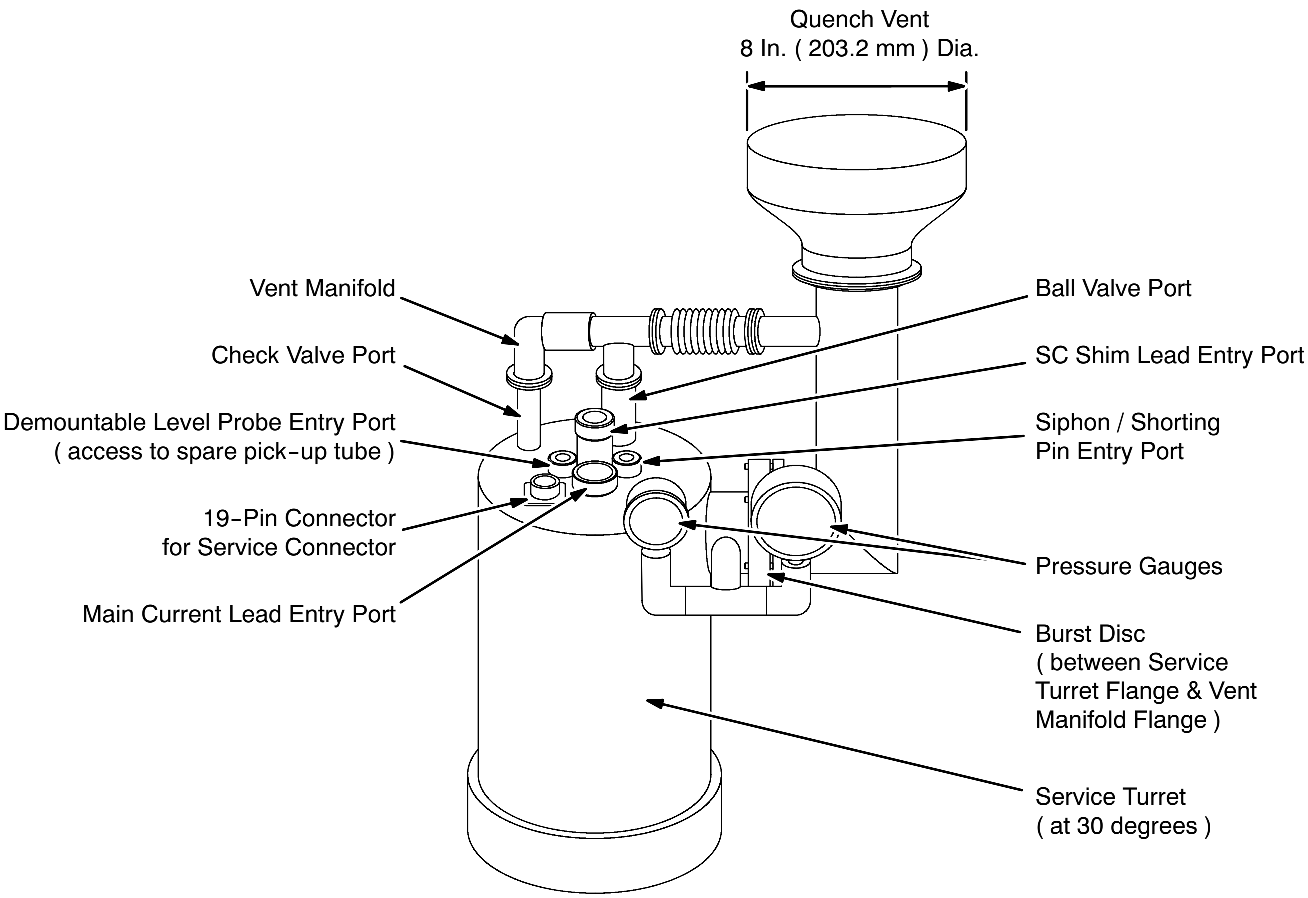
IMPORTANT !!!

Cryogens potentially released during LN_2 Fill MUST be exhausted safely outside the room to the building exterior.

Prepare the room exhaust system as follows:

1. Open the Ball Valve shown in Illustration 7-4 to vent the magnet to < 0.5 psig, then close the valve for Steps 2 and 3.

2. Remove the Vent Manifold and Pressure Gauge Assembly. See Illustration 7-4.

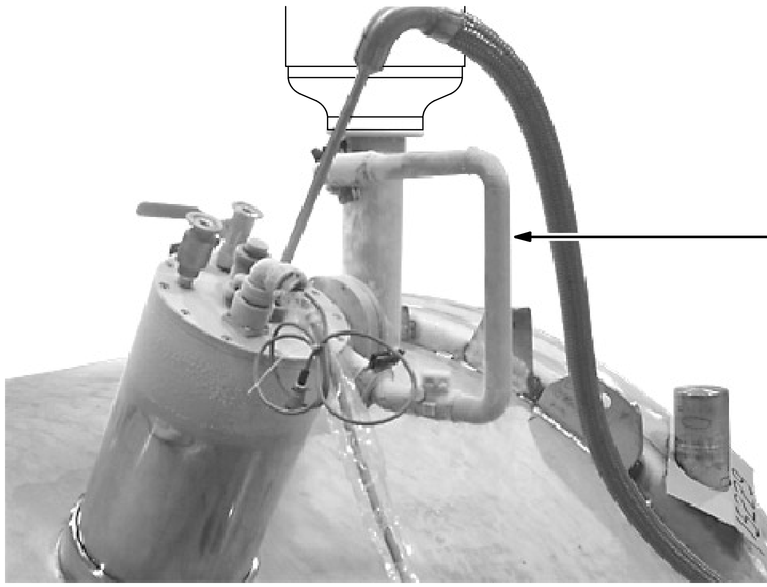


TYPICAL FACTORY-INSTALLED 3T SERVICE TURRET PLUMBING
ILLUSTRATION 7-4

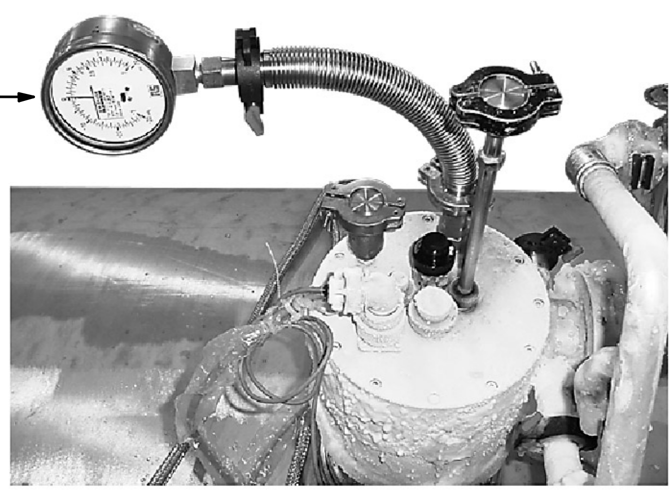
7-4 MAGNET VENTING INSTRUCTIONS (continued)

3. Connect the Exhaust Gas Vent Assembly (2292536) between the NW25 port on the quench exhaust line and the NW25 port where the pressure gauge assembly was removed. See Illustration 7-5.

4. Connect a pressure gauge to the ball valve port using a 2 foot (600 mm) flexible KF25 hose to keep the gauges from freezing. Open the ball valve. See Illustration 7-6.



EXHAUST GAS VENT CONNECTION
ILLUSTRATION 7-5



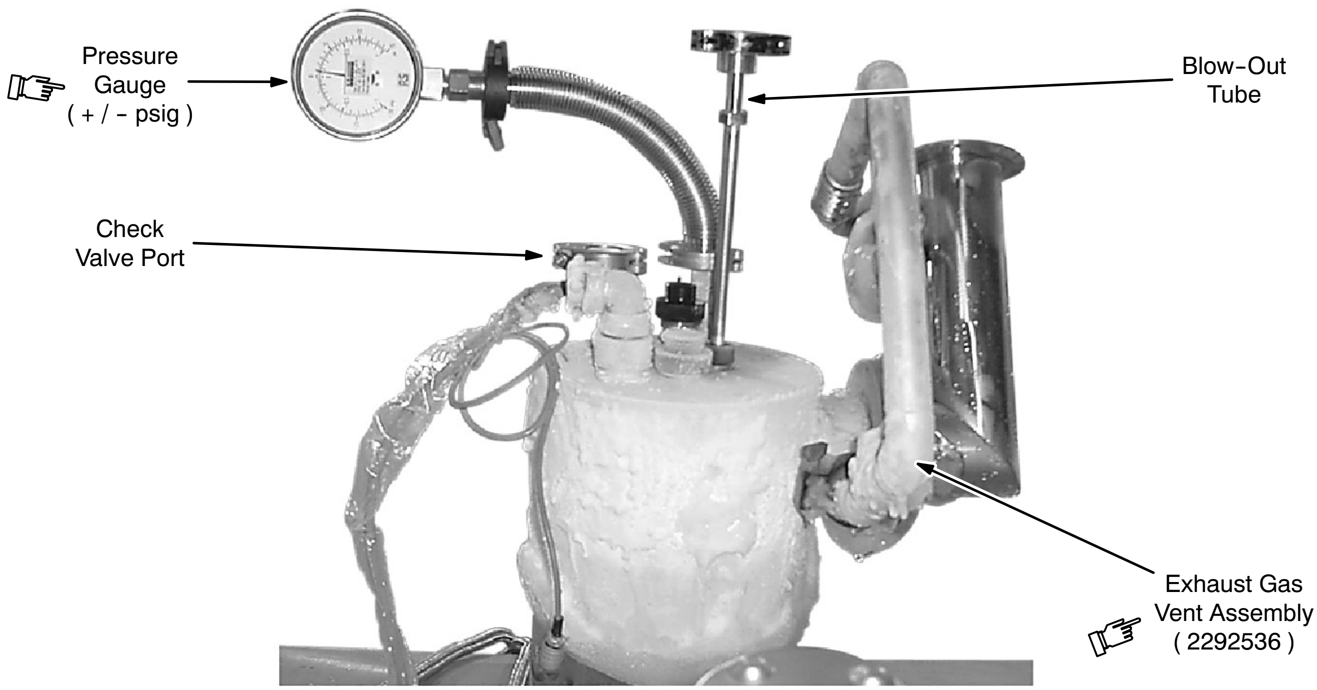
PRESSURE GUAGE CONNECTION
ILLUSTRATION 7-6

7-4 MAGNET VENTING INSTRUCTIONS (continued)

5. Temporarily cap off the check valve port. See Illustration 7-7.

Note

All cryogen operations are performed with the Burst Disc in place to prevent cryogens from being discharged into the magnet room.



CHECK VALVE PORT & BLOW-OUT TUBE
ILLUSTRATION 7-7

6. Remove plug on siphon port compression fitting. See Illustration 7-4.

7. Assemble collar nut, washer and o-ring onto Blow-Out Tube.

CAUTION

At least two turns of thread engagement are required in Step 8. Be careful not to cross-thread the Blow-out Tube. See Illustration 7-7.

8. Insert Blow-Out Tube into siphon port and lightly screw into cup. Tighten the collar nut. See Illustration 7-7.

7-5 LIQUID NITROGEN (LN_2) PRECOOL

WARNING!

MAKE SURE BOTH THE MAGNET ROOM VENT EXHAUST FAN AND THE FLOOR FAN ARE TURNED ON AND FUNCTIONING BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE NITROGEN GAS GENERATED DURING THIS PROCEDURE AND TO PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.

SKIN CONTACT WITH LIQUID CRYOGENS WILL CAUSE BURNS. WEAR PROTECTIVE CLOTHING (LONG-SLEEVE SHIRT AND APRON), GLOVES (INSULATED NON-ABSORBENT MATERIAL), GOGGLES / SAFETY GLASSES AND FACESHIELD WHEN TRANSFERRING CRYOGENS.

SMOKING IS PROHIBITED IN THE MAGNET AND CRYOGEN STORAGE ROOMS. LIQUID CRYOGENS CAN LIQUIFY ATMOSPHERIC OXYGEN, PRODUCING A HIGHLY COMBUSTIBLE SUPPORTING FLUID THAT RENDERS CLOTHING HIGHLY FLAMABLE FOR AS MUCH AS 15 MINUTES AFTER EXPOSURE

1. Obtain a full liquid nitrogen Dewar. Verify that all valves are in the closed position.

Note
Other sources of gaseous and liquid nitrogen may be used with appropriate setup apparatus.

2. Open the Dewar's pressure-building valve; verify that the Dewar pressure gauge reading does not exceed 20 psig. See Illustration 7-8.

3. Connect a nitrogen transfer line to the "Liquid" valve on the Dewar. See Illustrations 7-8 and 7-9.

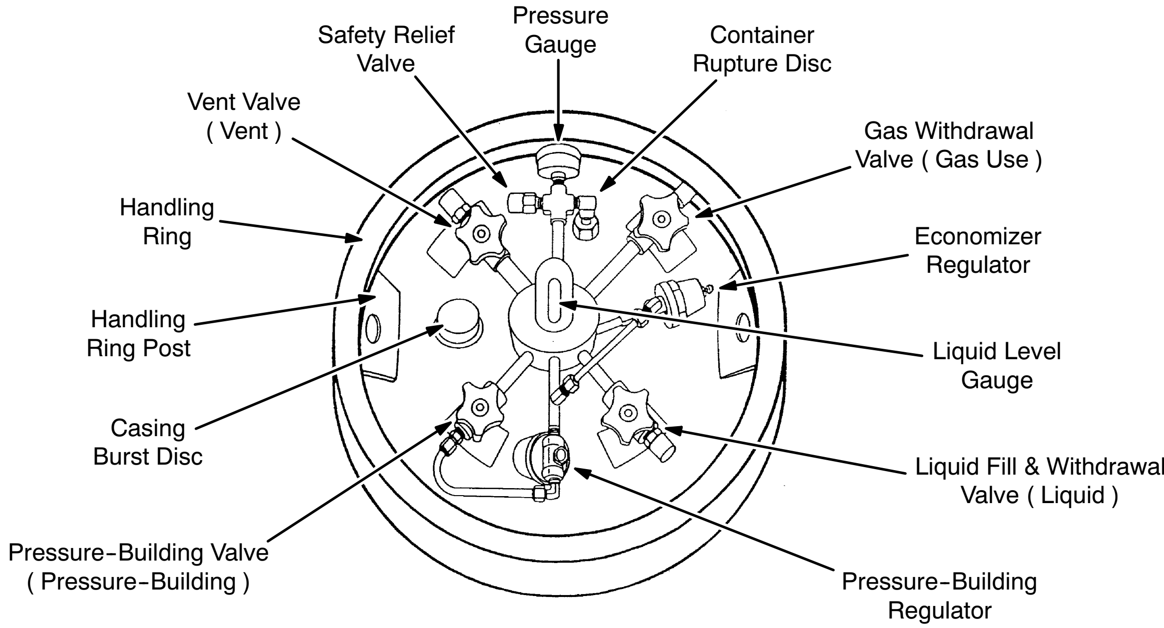
Note
Use foam pipe insulation on the Nitrogen Transfer Line to control frost and increase efficiency on longer line lengths.

4. Make sure all personal protective equipment required for this procedure is being worn before continuing.

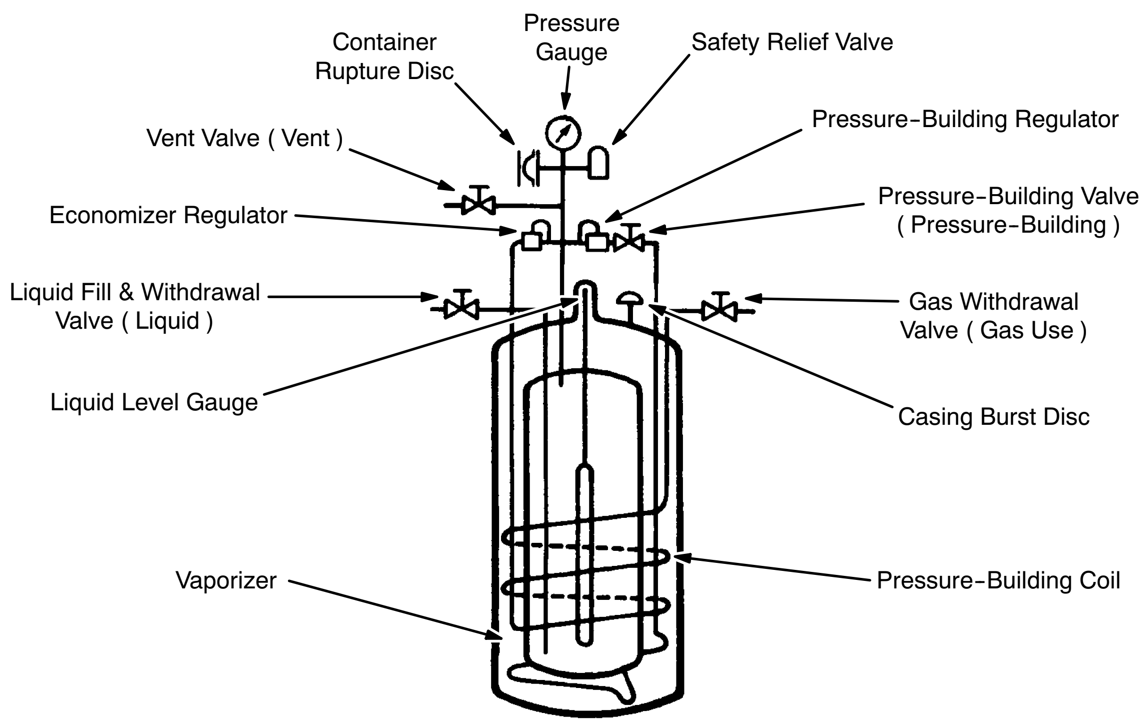
5. Purge the Nitrogen Transfer Line Assembly by partially opening the "Liquid" valve on the Dewar for 10 seconds, then close the valve.

6. Connect the other end of the nitrogen transfer line to the Blow-Out Tube.

7-5 LIQUID NITROGEN (LN_2) PRECOOL (continued)



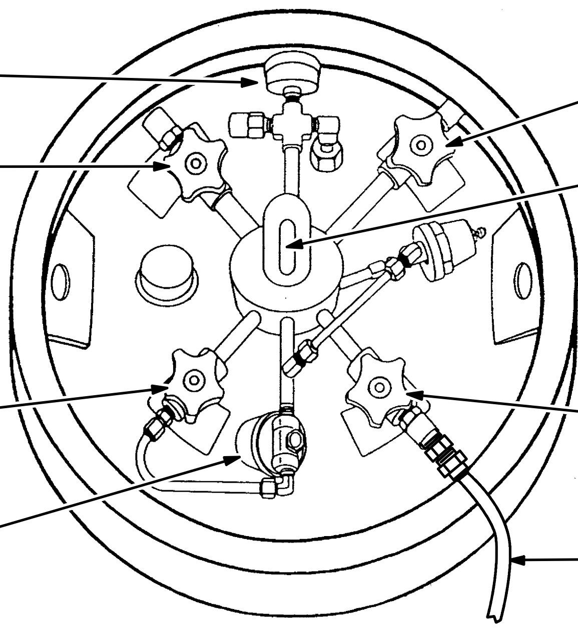
Top View of Nitrogen Dewar



Text in parenthesis ( ) refers to the identification labels on the dewars.

NITROGEN GAS / LIQUID DEWAR
ILLUSTRATION 7-8

7-5 LIQUID NITROGEN (LN_2) PRECOOL (continued)



Top View of Nitrogen Dewar

NITROGEN PURGE / PRECOOL ADAPTER
ILLUSTRATION 7-9

7. Open the "Liquid" valve 1/4 to 1/2 turn and monitor the pressure gauge on the cryostat. Adjust the valve as required to maintain a 2 to 3 psig reading on the cryostat pressure gauge.

Note

Continued pressure rise above 3 psig indicates a blocked vent. This requires correction.

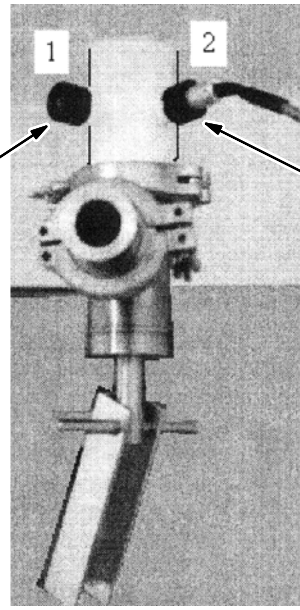
8. Advance the liquid valve 1/4 turn each half hour until the Blow-Out Tube develops frost and a downward mist plume. Keep the magnet pressure below 3 psig.

CAUTION

Make sure the cryostat pressure stays < 5 psig to prevent any possible damage to the Burst Disc.

9. Connect a DVOM and 19-Pin Breakout Box 2295141 to the 19-pin connector on the magnet's Service Turret. See Illustration 7-4. Connect 10-Pin Breakout Box 2295140 to the right-hand pump-out port 10-pin connector. See Illustration 7-10.

7-5 LIQUID NITROGEN (LN_2) PRECOOL (continued)



PUMP-OUT PORT SENSOR CONNECTIONS
ILLUSTRATION 7-10

10. With a DVOM connected to the 19-Pin Breakout Box measure and record the PT100 (magnet middle) resistance readings in comformance with FUNCTIONAL CHECKS, Section 7, Temperature Sensor Measurement, at 30-minute intervals.
11. Keep these valve settings for the first 150 liters of liquid nitrogen, typically emptying the first Dewar.
12. Turn on the Shield Cooler. Perform Shield Cooler checks per FUNCTIONAL CHECKS, Section 5, Shield Cooler Compressor Pressure Checks.
13. Turn off the pressure-building valve and the “Liquid” valve when the Dewar is empty. The first Dewar may take as long as 4 hours to empty and show negligible temperature change in the cryostat.
14. Shut off the LN_2 line cut-off valve and disconnect the nitrogen transfer line from the empty Dewar.
15. Exchange Dewars.
16. Connect the nitrogen transfer line to the “Liquid” port on the new Dewar, open the LN_2 line cut-off valve and then open the Dewar’s “Liquid” valve 1/2 to 1 turn and open the pressure-building valve.
17. Again look for frost and a downward plume on the Blow-Out Tube, opening the valve subject to the 2-3 psig limit.

7-5 LIQUID NITROGEN (LN_2) PRECOOL (continued)

18. Repeat this process, increasing the valve opening 1/2 turn each Dewar change but limiting the valve opening to keep pressure < 3 psig in the cryostat.

Note

The time required to empty a Dewar decreases as Dewars are transferred into the magnet, finally approaching 30 minutes.

Remove frost build-up from the Service Turret at each Dewar change.

19. When approximately 4500 liters (twenty 230-liter Dewars) of liquid nitrogen have been transferred, check the Allen-Bradley resistance (bottom magnet temperature) using the DVOM connected to the 19-Pin Breakout Box in conformance with FUNCTIONAL CHECKS, Section 7, Temperature Sensor Measurement.
____________________Bottom Magnet Temperature: ________________

20. After each additional Dewar is transfered, check the Allen-Bradley resistance again. If the resistance as changed < 0.4 ohm since the previous reading, indicating a temperature change of approximately 1K, discontinue liquid nitrogen transfer.

Note

Precool will require a total of twenty to twenty-four 230-liter Dewars of liquid nitrogen.

21. Record:
    - Vacuum Level: ________________Torr
    - Leak Rate: ________________cc / sec
22. Close the vacuum valve on the Cryostat by engaging the plunger.
23. Turn off — but do not disconnect — the vacuum equipment during thermal soak.

7-6 PREPARE THE CRYOSTAT FOR THERMAL SOAK

Note

In the soak interval the thermal radiation shields inside the cryostat cool down to minimize liquid helium (LHe) losses during and after LHe fill.

1. Maintain liquid nitrogen in vessel until the temperature of the 20K Shield is ≤ 100K.
2. Monitor vessel pressure for decrease to < 0.25 psig.
3. Disconnect the nitrogen transfer line from the blow-out tube.
4. Remove the blow-out tube from the siphon port and replace the siphon port plug.

Note

The cryostat is now able to thermally soak unattended for one to three days while the shields cool down and approach their steady-state operating temperature.

7-7 SERVICE TURRET TOP PLATE LEAK TEST (DURING THERMAL SOAK)

Note

The Service Turret Top Plate should be leak tested after the cryostat has thermally soaked for a couple hours. Leak testing reduces the chances of icing later on.

1. Once the cryostat has been thermally soaking for a couple hours, verify that the Ball Valve on the Service Turret is closed.
2. Warm and dry the Service Turret Top Plate using a heat gun.
3. Loosen both clamps securing the Exhaust Gas Vent Assembly (2292536) in the position shown in Illustration 7-5, then remove the bottom clamp and rotate the assembly out of the way. Attach a NW25 SpeediValve (2294083-2) to the flange where the Exhaust Gas Vent Assembly had been attached.
4. Connect the pressure gauge and flexible hose shown in Illustration 7-6 to the SpeediValve and open the SpeediValve
5. Connect a source of helium gas to the Ball Valve, set the pressure to 2 psig and open the Ball Valve until the cryostat pressurizes to 2 psig, then close the Ball Valve.
6. Bubble test all joints on the Service Turret Top Plate, the 19-pin connector and other electrical connectors, seal nuts and welds. There should be no more than one 1 mm diameter bubble every 2.6 seconds.

   Note

   The maximum permitted leak under a vacuum leak test (vacuum on one side and 100 percent helium on the other) is .0015 mbar/l/sec. This translates to ice building up at $\leq$ 1 cm^3 per year under normal working conditions (30 mbar maximum pressure differential).
7. Check the bungs on the Service Turret Top Plate. If they are in place, warm, dry and their o-rings are free to rotate, the seal nuts need only be finger tight.
8. Close the SpeediValve and remove the pressure gauge from the end of the flexible hose.
9. Connect the loose end of the flexible hose to the bottom of the exhaust fitting.
10. Open the SpeediValve to exhaust the overpressure from the cryostat into the vent line.
11. Remove the SpeediValve and reconnect the Exhaust Gas Vent Assembly as shown in Illustration 7-5.
12. Reattach the flexible hose and pressure gauge to the Ball Valve as shown in Illustration 7-6.
13. Open the Ball Valve and continue thermally soaking the cryostat.

SECTION 8 – NITROGEN PURGE

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

8–1 INTRODUCTION

The 3T helium vessel has completed its thermal soak, and the inner (20K) thermal shield should be at a temperature of ≤ 100K. The helium vessel is more than half full of liquid nitrogen (LN_2). The magnet is now ready to prepare for helium fill. All traces of N_2 must be removed from the helium vessel to prevent malfunction of the level gauges. Again, we seek to release no cryogens or cryogen residuals into the magnet room by returning the liquid to empty Dewars, which are to be sent back to the supplier.

IMPORTANT !!!

Liquid nitrogen is used in this procedure. Make sure of the following prior to starting the precool:

- **TWO fully trained and authorized persons are required to throughout this procedure.**
- **All required equipment is in place and functional.**

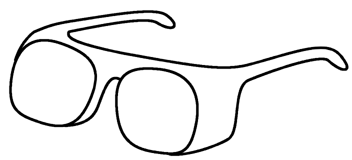
8-2 TOOLS / EQUIPMENT

8-2-1 Safety Equipment

- Goggles or safety glasses, a face shield and insulated nonabsorbant gloves from the Cryogen Safety Kit 46-271137G1. Two sets required, one per person. See Illustration 7-1.
- Protective Apron / Jacket.
- Nonmagnetic safety shoes.
- Portable Oxygen Monitor (2106236 or 2106237) if Oxygen Monitor Kit 2107184 is not installed and functional in room.
- Cryogen Warning Sign specified in 2301164PRE, *MR Magnet - Safety Requirements*.
- Floor exhaust fan.

Face Shield*

Leather Gloves*



Goggles*

Protective Apron / Jacket

* Contained in Cryogen Safety Kit (46-271137G1).

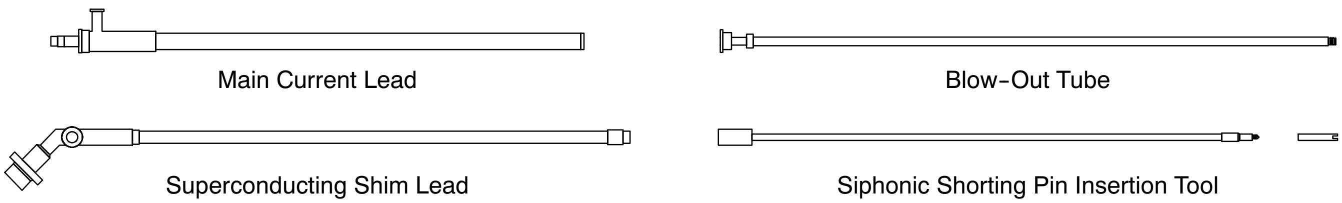
SAFETY EQUIPMENT
ILLUSTRATION 8-1

8-2-2 Auxiliary Equipment

- Main Current Lead (2255510; for 2.9M ceiling: 2294705)
- Superconducting Shim Lead (Long: 2255512; Short: 2255511)
- Blow-Out Tube (2255509; for 2.9M ceiling: 2294703)
- Siphonic Shorting Pin & Insertion Tool (2255522; hinged for 2.9M ceiling: 2294710)

Versions of these for ceilings higher that 2.9M are shown in Illustration 8-2. Lower ceiling height versions exist for each.

8-2-2 Auxiliary Equipment (continued)



AUXILIARY ITEMS
ILLUSTRATION 8-2

8-2-3 Other Equipment

- Twelve empty 150-liter or eight empty 230-liter liquid nitrogen Dewars (at least 1800 liters total capacity) with level gauges.
- Four to six helium gas bottles
- Pressure gauge with a range of ± 15 psig
- Leak Detector Kit (U.S.: 2296796; Japan: 2296797; Europe: 2296920)*
- Turbomolecular Pump (U.S. / Euro.: 2294584; Japan: 2295480)*
- Vacuum Pump Kit (U.S.: 2296794; Japan: 2296798; Europe: 2296912)*
- Pump-Out Port Actuator Tool (2292510) with NW50 clamp, o-ring and carrier*
- NW25 hose (19.7 inch (500 mm); 36.4 inch (1 M): 2294083-28;)*
- 19-Pin Breakout Box (2295141)
- Digital Volt-Ohm Meter
- Tee fitting with valve

* These items are included in the kits contained within the regional Pump / Precool Kits (U.S.: 2295481; Japan: 2296782; Europe: 2296909).

8-2-4 Cryogen Transfer / Monitoring Equipment

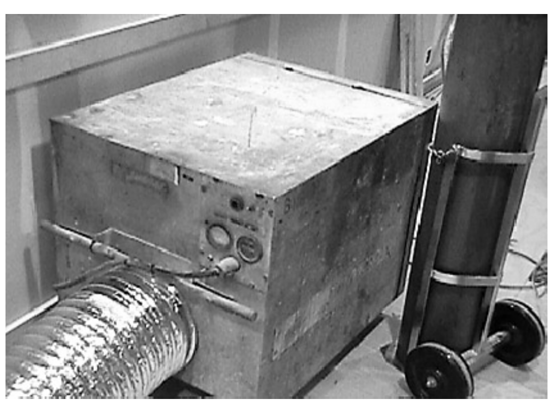
- Helium Fill Kit, 3T (2295507)
- LN_2 Line Cut-Off Valve with pressure safety.

8-3 PREPARATION

WARNING!

- **HELIUM GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLORLESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:**
  - **A. MAKE SURE THE MAGNET ROOM DOOR IS PROPPED OPEN,**
  - **B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND**
  - **C. THE PORTABLE FLOOR FAN IS INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THIS PROCEDURE.**
- **MAKE SURE ALL REQUIRED PERSONAL PROTECTIVE EQUIPMENT IS WORN TO PREVENT CRYOGEN EXPOSURE / BURNS: GOGGLES / SAFETY GLASSES AND FACE SHIELD; INSULATED NONABSORBENT GLOVES, LONG-SLEEVE SHIRT AND LONG PANTS; AND PROTECTIVE APRON / JACKET.**
- **SMOKING IS PROHIBITED AROUND CRYOGENS. CRYOGENS CAN CONDENSE ATMOSPHERIC OXYGEN PRODUCING A HIGHLY COMBUSTION SUPPORTING FLUID. CLOTHING WHICH HAS ABSORBED LIQUID OXYGEN CAN REMAIN FLAMMABLE FOR LONG PERIODS OF TIME.**
- **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT A PROPERLY FUNCTIONING OXYGEN MONITOR IS USED IN THE MAGNET AND/OR CRYOGEN STORAGE ROOM(S) BY SERVICE PERSONNEL.**

1. Install floor exhaust fan in magnet room. Route exhaust ducting outside building. Make sure the exit area is ventilated at least as well as that required of a cryogens storage area. See Illustration 8-3.



TYPICAL FLOOR EXHAUST FAN
ILLUSTRATION 8-3

8-4 LIQUID NITROGEN (LN_2) DRAIN PROCEDURE

1. Identify a LN_2 Dewar whose pressure checks at < 20 psig. Bring that Dewar into the magnet room for use in pressurizing the magnet. Position it in front of the magnet.

Note

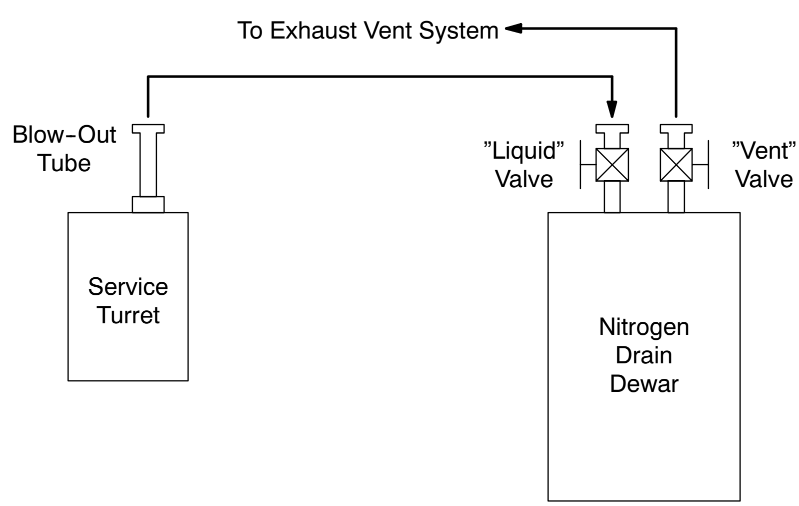
Other pressurizing apparatus may be used with appropriate plumbing and pressure regulation.

2. Connect a flexible stainless LN_2 transfer line to the Dewar's service port using fittings from the Universal Fill Kit for S-I and Oxford magnets.
3. Remove the Exhaust Gas Vent Assembly and connect the LN_2 transfer line to the pressure gauge flange where the Exhaust Gas Vent Assembly was connected.

Note

The Dewar vent valve should remain closed for now but will be used to start / stop pressurizing gas flow into the helium vessel later.

4. Obtain an empty Dewar. Make sure its pressure is less than 20 psig. This dewar will be used as the drain Dewar in the magnet room. Position it beside and forward of the service turret. Make sure its pressure-building valve is closed.
5. Connect a LN_2 transfer line to an open inline cryogen ball valve and connect the valve to the liquid service fitting on the empty Dewar. See Illustration 8-4.



LN_2 DRAIN PLUMBING
ILLUSTRATION 8-4

8-4 LIQUID NITROGEN (LN_2) DRAIN PROCEDURE (continued)

WARNING!

IT IS IMPORTANT NOT TO TRAP LIQUID OR GAS IN THE SHORT PIECE OF TUBE BETWEEN THESE VALVES; I.E., ONE VALVE SHOULD ALWAYS BE OPEN WHEN THE FITTINGS ARE TIGHT TO PREVENT RUPTURE.

6. Reinstall the Blow-Out Tube in the syphon port and carefully screw it in at least two full turns. Tighten the collar seal on the siphon port.
7. Connect the stainless LN_2 transfer line from the Dewar's "Liquid" service port to the Blow-Out Tube using a NW25 adapter.
8. Place a 0.5 inch (12.7 mm) tube to KF25 speed flange adapter fitting on the drain Dewar's "Vent" port. This adapter mates to a flexible steel transfer line having speed flanges.
9. Vent the drain Dewar down to < 1 psig pressure.
10. Connect the drain Dewar's vent port to the quench line's NW25 port where the Vent Manifold was connected.
11. Check the drain Dewar pressure to be less than 1 psig. Open the vent port on the drain Dewar.
12. Open the liquid service port on the drain Dewar.

WARNING!

MAKE SURE MAGNET PRESSURE STAYS BELOW 5 PSIG TO PREVENT BURST DISK RUPTURE.

Note

Magnet pressure can be monitored on the pressure gauge attached to the ball valve during SET-UP AND CALIBRATION, Section 7, Nitrogen Precool.

13. Check the pressure in the pressurizing Dewar. Vent as required to maintain pressure below 5 psi.
14. Blank off the Service Turret Check Valve with a KF25 blanking flange.
15. Slowly open the gas service valve on the pressurizing Dewar, watching the magnet pressure to be sure it stays below 5 psig.

8-4 LIQUID NITROGEN (LN_2) DRAIN PROCEDURE (continued)

16. The Blow-Out Tube should develop frost and a downward vapor mist plume very quickly, indicating LN_2 flow from the Dewar through the tube. Liquid oxygen and argon may drip from the tube after a short time.

Note

The drain Dewar is full when the LN_2 liquid level gauge shows full or when the drain Dewar vent line frosts heavily, develops a strong downward vapor mist plume, and perhaps drips liquid oxygen or nitrogen. A full Dewar should be exchanged with an empty one.

17. Shut off the pressure-building valve on the pressurizing Dewar and dial back that Dewar's pressure regulator to zero pressure.
18. Turn off the gas supply valve (vent) on the pressurizing Dewar.
19. Close the ball valve in the stainless steel flex-line leading from the Blow-Out Tube to the drain Dewar. Watch the helium vessel pressure. If it rises, reopen the valve to the drain Dewar's liquid service tube and let the pressure subside before reclosing the valve.
20. Open the fitting on the liquid service port until gas escapes, about one half turn. Then close the liquid service valve on the drain Dewar.
21. Remove the line from the liquid port on the drain Dewar.
22. Remove the line from the vent port of the drain Dewar.
23. Remove the drain Dewar from the magnet room and bring in another empty drain Dewar after confirming its internal pressure is < 20 psig.
24. Reattach the flex lines to the drain Dewar, open the gas vent on the pressurizing Dewar, open the pressure-building valve on the pressurizing Dewar, open the two valves on the drain Dewar liquid line, close the pressure-building valve on the drain Dewar and look for frosting of the Blow-Out Tube.
25. Return full Dewars to the supplier.
26. Repeat this process until the helium vessel is empty as indicated by light frosting on the Blow-Out Tube (about twelve 150-liter or eight 230-liter Dewars if the vessel was allowed to become half full).
27. Make sure all nitrogen Dewars / cylinders are removed from the magnet room.

CAUTION

Do not bring helium (He) Dewars / cylinders into the magnet room until all nitrogen (N_2) Dewars / cylinders are removed from the room.

28. Close the Service Turret's ball valve, remove the pressure gauge and then connect a cylinder of helium gas to it. Set the cylinder's regulator to 2 psig.

8-4 LIQUID NITROGEN (LN_2) DRAIN PROCEDURE (continued)

29. Connect the Blow-Out Tube to the exhaust vent.
30. Open the ball valve to allow helium gas through the helium vessel and drive any nitrogen out.
31. Restart the vacuum equipment. Verify that the foreline pressure is < 10^{-5} Torr and the leak rate is < 10^{-9} cc/sec. If not, troubleshoot the vacuum equipment.
32. Open the Cryostat Vacuum Valve.
33. Observe helium background pressure on the leak detector to check for helium leaking into the vacuum space.

    Background Helium Pressure: ______________
34. Close the ball valve when one half of a cylinder of helium gas has been used. Disconnect the helium cylinder and remove the Blow-Out Tube.
35. Turn on the Helium Level Monitor (2255532) for 20 - 30 seconds to evaporate any residual nitrogen droplets potentially trapped within the Helium Level Probe.

The helium vessel is now ready for helium flushing.

8-5 NITROGEN PURGE PROCEDURE

CAUTION

Do not bring helium (He) Dewars / cylinders into the magnet room until all nitrogen (N_2) Dewars / cylinders are removed from the room.

IMPORTANT !!!

Make sure current is flowing to the helium level sensors to prevent sensor probe icing during purge procedure.

Note

The regional Pump / Precool Kits (U.S.: 2295481; Japan: 2296782; Europe: 2296909) should contain two rotary pumps which were prepared in SET-UP AND CALIBRATION, Section 1, Detransiting Magnet, for either helium or vacuum use (labelled “FOR HELIUM USE ONLY” or “FOR VACUUM USE ONLY” respectively). Only use the rotary pump labelled “FOR HELIUM USE ONLY” to pump the helium vessel.

If only one rotary pump is available, perform Steps 1 through 4. If using two rotary pumps — one already labelled “FOR HELIUM USE ONLY” and one prepared / labelled “FOR VACUUM USE ONLY” — skip Steps 1 through 4.

8-5 NITROGEN PURGE PROCEDURE (continued)

1. Check the main cryostat Vacuum Pump-Out Port Operator to be certain the main vacuum valve is closed.

2. Turn off the vacuum turbo pump and roughing pump.

3. Close the valve between the roughing pump and turbo pump.

4. Disconnect the roughing pump to use it to pump vacuum on the helium vessel.

WARNING!

TO PREVENT NITROGEN GAS FROM FREEZING IN THE VACUUM VESSEL AND TO PREVENT OVERLY STRESSING THE BURST DISC, DO NOT PUMP TO < 500 MILLIBARS.

5. Remove the blanking plate from the gauge assembly port. Connect one end of the pump line — about 8 feet (4.44 M) of 1 inch (25.4 mm) stainless steel flex-hose fitted with NW25 flanges — to the gauge assembly port and to a vacuum valve at the other.

WARNING!

COLD GAS WILL INCREASE THE VISCOSITY OF THE PUMP OIL AND CAUSE VACUUM PUMP MALFUNCTION. USE ENOUGH VACUUM TUBE LENGTH TO KEEP THE FROST ZONE AWAY FROM THE PUMP. HEATING THE TUBE WITH A HEAT GUN IS AN ALTERNATIVE TO LONG TUBES.

6. Connect a vacuum pressure gauge to the ball valve and then open the ball valve.

7. Connect the roughing pump to the stainless steel flex-hose via a vacuum valve.

8. Turn on the helium level monitor to evaporate any residual liquid from its wires.

9. Start the pump and monitor the pressure decrease in the helium vessel.

10. When the pressure approaches 500 millibar (376 Torr), stop the vacuum pump.

11. Close the ball valve and then disconnect the pressure gauge.

8-5 NITROGEN PURGE PROCEDURE (continued)

12. Connect a helium bottle through a low pressure regulator to the normal helium vessel vent port (i.e., the ball valve port on the Service Turret).
13. Check that the regulator is working and backed out to zero pressure.
14. Break vacuum and bring the helium vessel to approximately one atmosphere using the helium bottle and low pressure regulator.
15. Adjust the regulator to 0 psig and close the helium bottle's valve.
16. Close the ball valve and disconnect the vacuum gauge.
17. Remove the blanking plate from the check valve. Connect the check valve to the Exhaust Vent's NW25 flange using NW25 hose.
18. Wait 2 hours.
19. If the helium vessel has been pumped and flushed only once, then disconnect the NW25 hose from the check valve, reattach the blanking plate to the check valve and repeat Steps 9 - 17.
20. When the helium vessel has been pumped and flushed twice, attach 19-Pin Breakout Box 2295141 to the 19-pin Service Turret Connector and measure the bottom magnet temperature (Allen-Bradley resistor connected to pins S and T minus resistance between pins S and R) in conformance with FUNCTIONAL CHECKS, Section 7, Temperature Sensor Measurement.

    First Bottom Magnet Temperature: _______________

21. Verify that the check valve is not blanked off and that a NW25 hose is connected between the check valve and the Exhaust Vent, then let the magnet sit in this configuration for at least twelve hours.
22. At least twelve hours later, measure the bottom magnet temperature again.

    Second Bottom Magnet Temperature: _____________

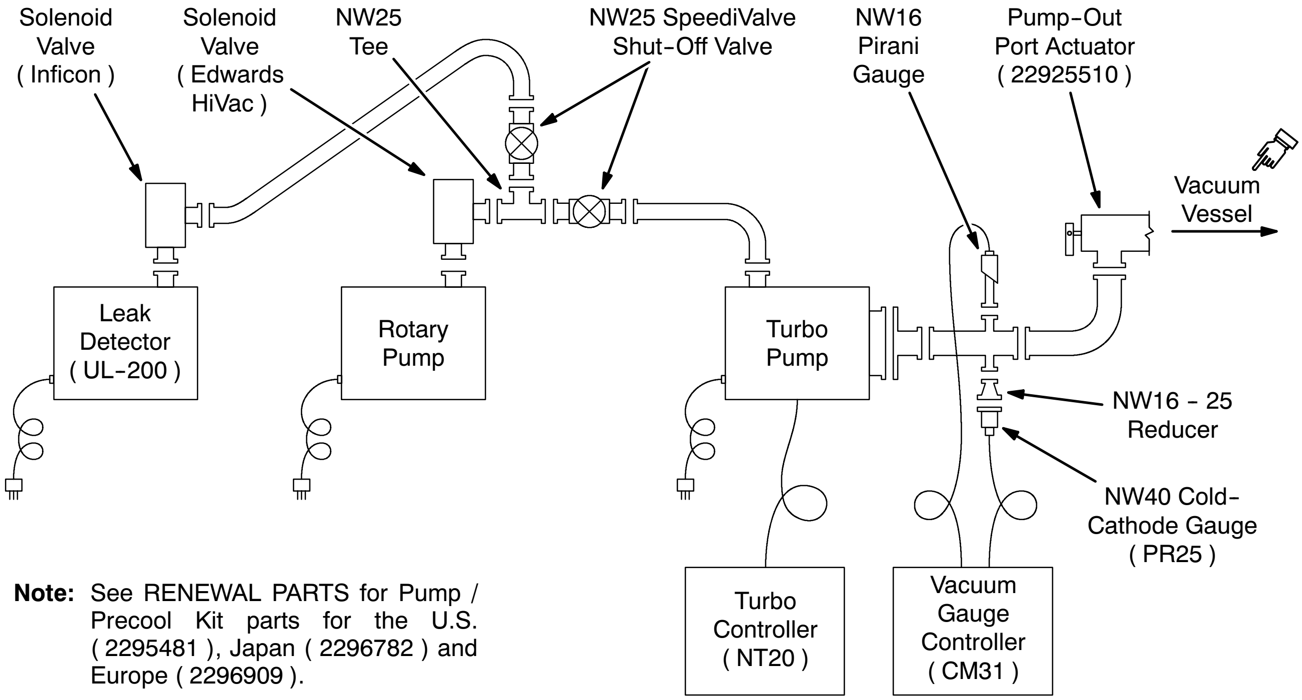
23. If the temperature measured in Step 21 is higher than the temperature measured in Step 20 and Step 21's temperature is ≥ 85K (resistance 6.5 ohms < Allen-Bradley resistance at 77K), then continue to Section 8-6, Helium (He) Vessel Leak Check Procedure. Otherwise, repeat Steps 9 - 21.

8-6 HELIUM (He) VESSEL LEAK CHECK PROCEDURE

At this time it is appropriate to check the helium vessel for leaks.

1. "Clean" the roughing pump in conformance with Step 2 of SET-UP AND CALIBRATION, Section 1-4-1, for one hour.
2. Reconnect the roughing pump to the turbo pump and to a leak detector according to Illustration 8-5.
3. Turn on the rotary pump and turbo pump according to established procedures.

8-6 HELIUM (He) VESSEL LEAK CHECK PROCEDURE (continued)



Note: See RENEWAL PARTS for Pump / Precool Kit parts for the U.S. (2295481), Japan (2296782) and Europe (2296909).

VACUUM PUMPING AND LEAK CHECKING SCHEMATIC DIAGRAM
ILLUSTRATION 8-5

4. Connect a gas line from a helium gas bottle's regulator to the Ball Valve Port on Service Turret.
5. Set the regulator to 1 psig and open the ball valve.
6. When pressure in the helium vessel reaches 0.25 - 0.50 psig as read on the magnet pressure gauge, close the ball valve.
7. If the current helium pressure level is greater than the background helium pressure level detected in Section 8-4, Step 33, and there has been no possibility of helium contamination of the roughing pump oil, contact GE Medical Systems Oxford at 8-011-44-1235-5490-00.
8. If there is no helium contamination of the main vacuum space and the helium vessel remains at 0.25 psig:
   a Close the ball valve and connect the vent manifold from the ball valve to the exhaust vent.
   b Open the ball valve to depressurize the helium vessel
   c Then remove the vacuum pump line from the gauge assembly port and blank off that port with a NW25 blanking plate.
9. Remove the pressurizing line from the ball valve port on the Service Turret and close the port with a blanking plate.

8-7 SERVICE TURRET TOP PLATE LEAK TEST (POST NITROGEN PURGE)

1. Close the ball valve on the service turret and reconnect the pressure gauge assembly.
2. Temporarily blank off the check valve on the service turret.
3. Connect a cylinder of helium gas to the ball valve. Set the cylinder's regulator to 2 psig.
4. Open the ball valve and allow the cryostat to pressurize to 2 psig.
5. Make sure the Service Turret Top Plate is warm and dry.
6. Check the bungs on the Service Turret Top Plate. If they are in place, warm, dry and their o-rings are free to rotate, the seal nuts need only be finger tight.
7. Bubble (Snoop) test all joints on the Service Turret Top Plate, electrical connectors, seal nuts and welds.
8. Close the ball valve and disconnect the helium cylinder.
9. Reconnect the ball valve to the exhaust vent and reopen the ball valve to vent the helium gas from the cryostat.
10. Close the ball valve once the helium is vented.

Note
The magnet is now secure and may be safely left to begin helium fill preparations.

SECTION 9 – LIQUID HELIUM FILL / TOP–OFF

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

9–1 INTRODUCTION

Liquid helium fill of the 3T/940 Magnet is required for the following conditions:

- Initial fill during installation.
- Top–off during magnet operation to keep level above minimum (1878 liters, 61.19%, 500 mm monitor reading).
- Minimum level for magnet ramping. (2466 liters, 80.43%, 920 mm monitor reading)

Helium vessel volume is 3066 liters. Approximatelly twelve to fifteen 500–liter Dewars (6000 – 7500 liters) of liquid helium is required to cool the magnet down from liquid nitrogen temperature (77K) and to fill with helium.

9-1 INTRODUCTION (continued)

IMPORTANT !!!

Liquid helium is used in this procedure. Make sure of the following prior to starting the precool:

- **TWO fully trained and authorized persons are required to throughout this procedure.**
- **All required equipment is in place and functional.**

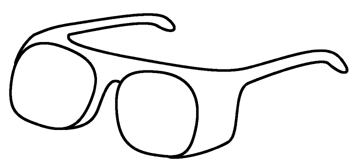
9-2 TOOLS / EQUIPMENT

9-2-1 Safety Equipment

- Goggles or safety glasses, a face shield and insulated nonabsorbant gloves from the Cryogen Safety Kit 46-271137G1. Two sets required, one per person. See Illustration 9-1.
- Protective Apron / Jacket.
- Nonmagnetic safety shoes.
- Portable Oxygen Monitor (2106236 or 2106237) if Oxygen Monitor Kit 2107184 is not installed and functional in room.
- Cryogen Warning Sign specified in 2301164PRE, *MR Magnet - Safety Requirements*.

Face Shield*

Leather Gloves*



Goggles*

Protective Apron / Jacket

* Contained in Cryogen Safety Kit (46-271137G1).

SAFETY EQUIPMENT
ILLUSTRATION 9-1

9-2-2 Cryogen Transfer / Monitoring Equipment

- Required number of liquid helium Dewars
- Required number of helium gas cylinders.
- Helium Fill Kit, 3T (2295507), which includes:
  — 0.5 inch (12.7 mm) diameter Helium Transfer Siphon (2255866; for 2.9M ceilings: 2292330)
  — Pressure gauge, +/- psig, with line and KF25 fitting (2294083-7)
  — Siphon Entry Adapter, NW25 (2294083-7), for Dewars with a NW25 flange

Note
Universal Fill Line Kit 46-294705G1 — plus a 23 inch Stinger — is an alternative to the Helium Fill Kit, 3T (2295507).

- Regulator Kit (46-306734G1).
- Hose Assembly (46-271135P16).
- Heat Gun (U.S.: 46-294167P63; Japan: 46-294167P66; Euro: 46-294167P61).
- Compression fittings 46-318619P1, 46-260272P1 and 46-260342P9 (or local equivalent)
- Liquid Helium Level Monitor (2255532).
- Resistance measuring equipment for monitoring magnet temperature, such as a Digital Volt-Ohm Meter (DVOM).

9-3 MAGNET BYPASS VENTING

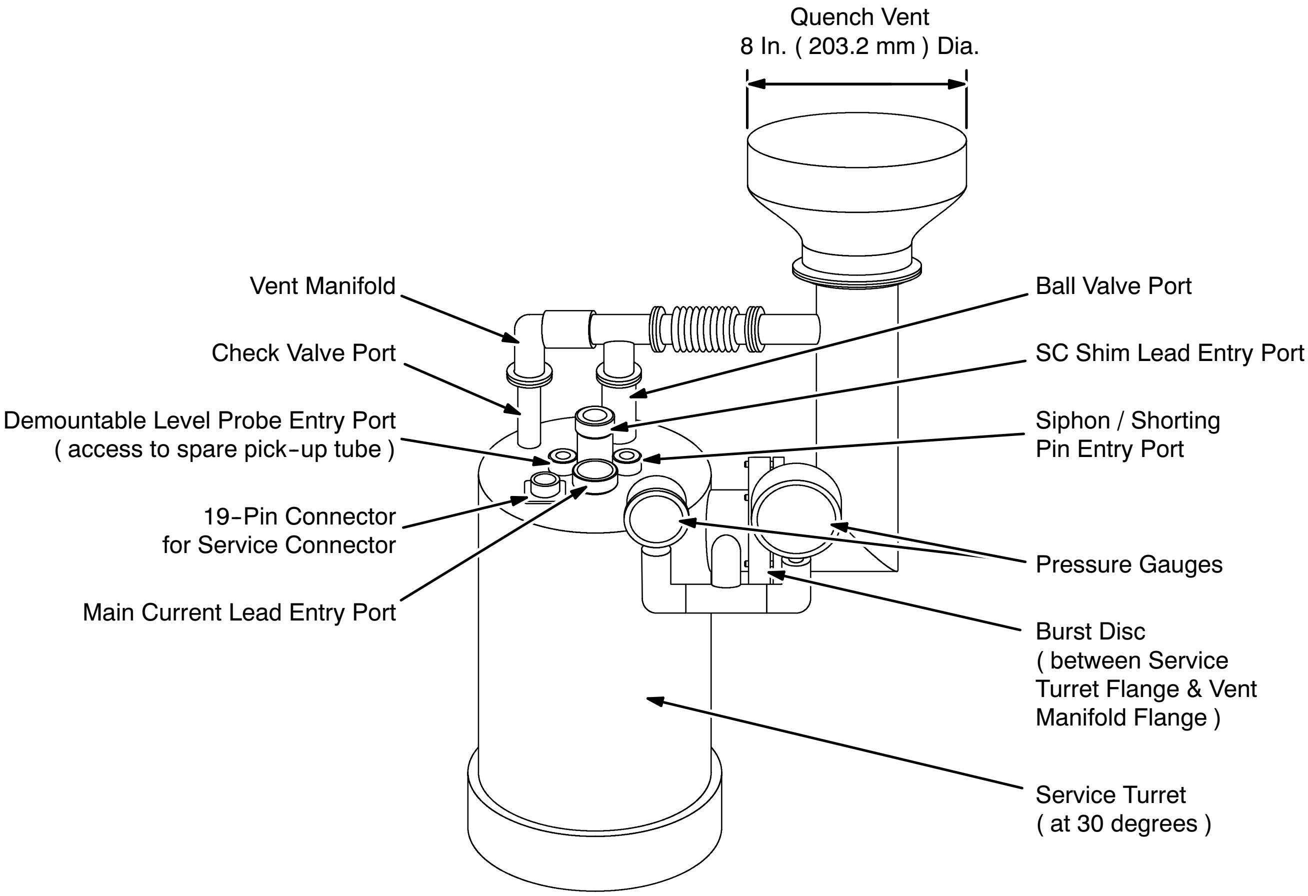
CAUTION

Make sure the magnet is vented down to < 0.5 psig before removing the Vent Manifold.

IMPORTANT !!!
Illustration 9-2 shows the typical 3T / 940 turret plumbing installed by the factory and received at the site. All cryogens must be exhausted outside the room to a safe location outside the building. The following steps are required to prepare the exhaust system for this requirement during liquid helium (LHe) fill if the Exhaust Gas Vent Assembly (2292536) is not already installed.

1. Make sure the magnet is vented to < 0.5 psig (open / close the ball valve as required), then remove Vent Manifold.

9-3 MAGNET BYPASS VENTING (continued)



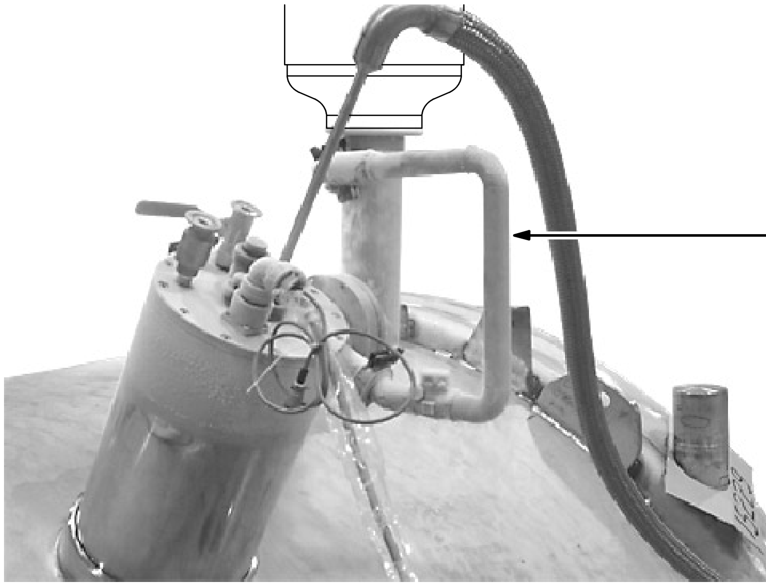
TYPICAL FACTORY-INSTALLED 3T SERVICE TURRET PLUMBING
ILLUSTRATION 9-2

2. Close the ball valve and remove the Pressure Gauge Assembly. See Illustration 9-2.
3. Connect the bottom of the Exhaust Gas Vent Assembly (2292536) to the NW25 flange where the pressure gauges were attached. See Illustrations 9-2 and 9-3.
4. Disconnect the Exhaust Vent Manifold from the Quench Vent.
5. Connect the top of the Exhaust Gas Vent to the flange where the Vent manifold was attached. See Illustrations 9-2 and 9-3.
6. Close the ball valve and remove the Vent Manifold entirely.
7. Connect a pressure gauge to the open turret exhaust port. See Illustration 9-4.
8. Temporarily cap off the check valve port. See Illustration 9-4.

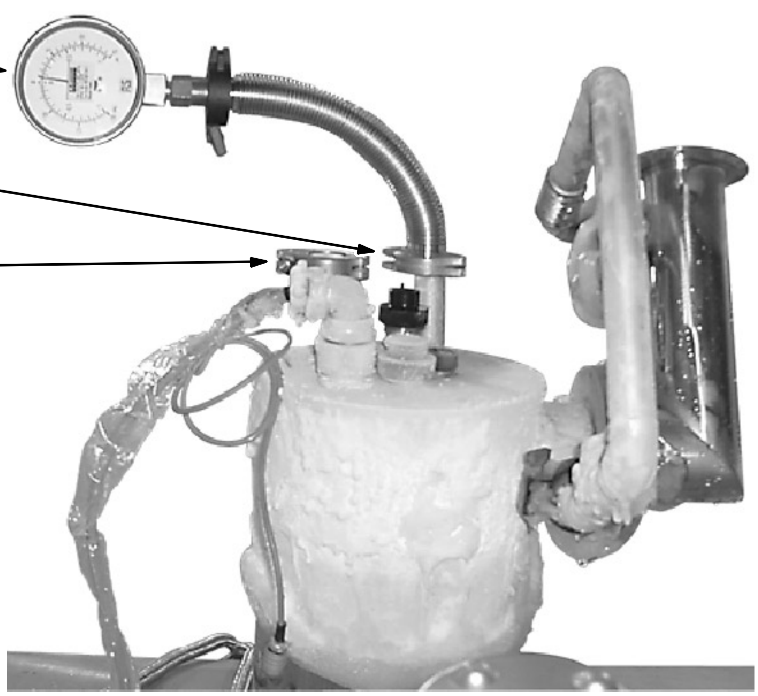
Note

All cryogen operations are performed with the Burst Disc in place to prevent leakage of cryogens into the magnet room.

9-3 MAGNET BYPASS VENTING (continued)



EXHAUST GAS VENT CONNECTION
ILLUSTRATION 9-3



PRESSURE GUAGE CONNECTION
ILLUSTRATION 9-4

9-4 PREPARATION

WARNING!

- **HELIUM GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLORLESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:**
  - **A. MAKE SURE THE MAGNET ROOM DOOR IS PROPPED OPEN,**
  - **B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND**
  - **C. THE PORTABLE FLOOR FAN IS INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THIS PROCEDURE.**
- **MAKE SURE ALL REQUIRED PERSONAL PROTECTIVE EQUIPMENT IS WORN TO PREVENT CRYOGEN EXPOSURE / BURNS: GOGGLES / SAFETY GLASSES AND FACE SHIELD; INSULATED NONABSORBENT GLOVES, LONG-SLEEVE SHIRT AND LONG PANTS; AND PROTECTIVE APRON / JACKET.**
- **SMOKING IS PROHIBITED AROUND CRYOGENS. CRYOGENS CAN CONDENSE ATMOSPHERIC OXYGEN PRODUCING A HIGHLY COMBUSTION SUPPORTING FLUID. CLOTHING WHICH HAS ABSORBED LIQUID OXYGEN CAN REMAIN FLAMMABLE FOR LONG PERIODS OF TIME.**
- **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT A PROPERLY FUNCTIONING OXYGEN MONITOR IS USED IN THE MAGNET AND/OR CRYOGEN STORAGE ROOM(S) BY SERVICE PERSONNEL.**
- **METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD. DO NOT BRING ANY FERROMAGNETIC OBJECTS, EQUIPMENT OR TOOLS INTO THE MAGNET ROOM UNLESS SPECIFICALLY CALLED FOR IN THE SERVICE DOCUMENTATION.**

1. Complete all items marked “X” in the “Helium Fill / Top-Off” column of SAFETY REQUIREMENTS, *2301164PRE, MR Magnet - Safety Requirements*, Table 1, Service Procedure Safety Requirements.

IMPORTANT !!!

Make sure all identified Helium Fill / Top-Off defects are corrected before continuing with magnet service.

9-4 PREPARATION (continued)

2. Make sure the Liquid Helium Level Monitor (2255532) is calibrated and operating to monitor progress of the fill procedure.

   a. Ordinarily the monitor is pre-calibrated for a particular probe. If recalibration is required, recalibrate both probes using the procedure contained in the E5011 Liquid Helium Level Monitor manual.

   b. Perform the helium monitor checks found in FUNCTIONAL CHECKS, Section 9-2, Helium Level Monitor Checks.

   c. Set the helium monitor to initially read from "Probe 1."

   d. Unplug the 19-pin service cable until liquid helium starts to collect.

3. If this is an **<u>initial</u>** fill (no LHe in the vessel):

   a. Make sure the Shield Cooler is operating, then connect the 19-Pin Breakout Box (2295141) from the Electrical Test Kit (2296053 **)** to the Service Turret Top Plate's 19-pin connector.

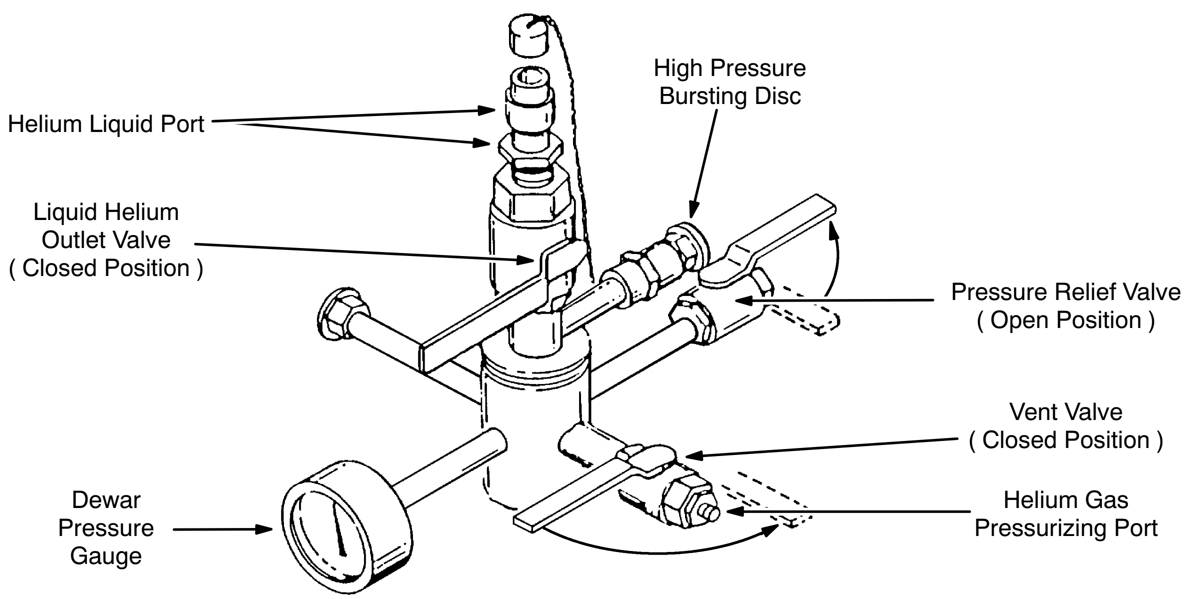
   b. Connect a Digital Volt-Ohm meter to the 19-Pin Breakout Box.

   c. Monitor the magnet cartridge cool down by checking the Allen-Bradley resistor's Resistance S-T minus Resistance R-S reading on the DVOM. When the readings stabilize (no further changes), liquid helium has started accumulating around the magnet cartridge, cooling it to 4.2K.

Note

See FUNCTIONAL CHECKS, Section 7, Temperature Sensor Measurement, for connecting the DVOM and interpreting its display.

4. Obtain a full 250-liter liquid helium (LHe) Dewar. Check the Dewar pressure gauge; if the pressure reading is above 1 psig, slowly open Dewar's vent valve and reduce pressure in the Dewar to 1 psig. See Illustration 9-5.



DEWAR CONNECTIONS
ILLUSTRATION 9-5

9-4 PREPARATION (continued)

WARNING!

IF DEWAR PRESSURE DOES NOT VENT DOWN TO 1 PSIG, VERIFY THAT DEWAR PRESSURE RELIEF VALVE IS LEFT IN OPEN POSITION. CONTACT CRYOGEN SUPPLIER IMMEDIATELY.

Note

The pressure relief valve is normally open during shipping and storage to prevent excessive build up of pressure inside the Dewar. Therefore, always reopen the pressure relief valve after using a Dewar.

Note

If "99.999% Helium Gas" is used (five nines certified gas), the purity of the gas remaining in the cylinder will degrade as a result of this process (i.e., the purity of the remaining gas will be something less than 99.999%).

5. Obtain one full aluminum cylinder (135 SCF) of gaseous helium (GHe) for every two 250-liter liquid helium (LHe) Dewars required.

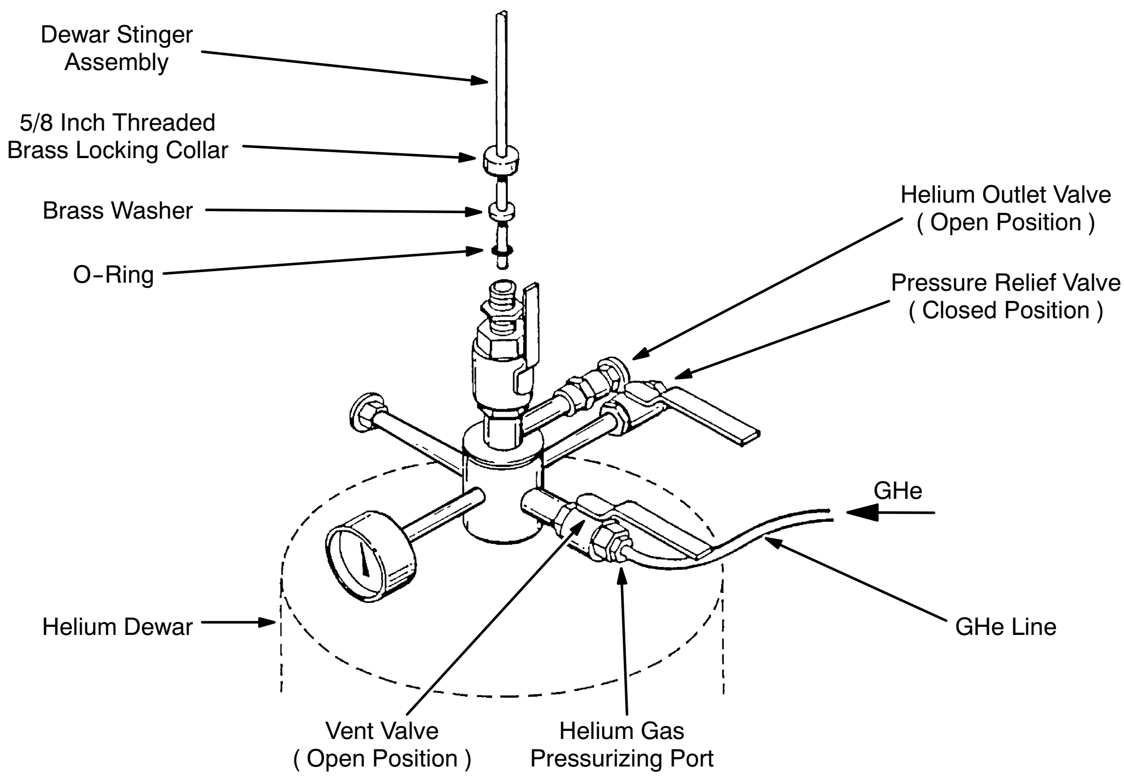
WARNING!

SECURE CYLINDER TO A NONMAGNETIC GAS BOTTLE CART BEFORE REMOVING PROTECTIVE VALVE CAP TO PREVENT CYLINDER FROM FALLING, WHICH COULD RESULT IN SHEARING THE VALVE OUTLET AND CAUSING HAZARDOUS HIGH-PRESSURE GAS RELEASE.

6. Connect a standard GHe cylinder regulator and hose assembly to valve outlet (CGA 580) on the GHe cylinder.
7. Make sure the regulator adjusting handle is fully backed out, then slowly open GHe cylinder valve.
8. Observe the regulator's high pressure gauge. Make sure the pressure reads approximately 2000 psig, indicating a full cylinder.

9-5 LIQUID HELIUM (LHe) FILL / TOP-OFF

1. Verify the Dewar's liquid helium outlet valve is in the closed position. See Illustration 9-5.
2. If the helium Dewar has a NW25 flange on top, connect the Siphon Entry Adapter (2294083-7) from Helium Fill Kit 2295507 to the Dewar-end siphon's stinger. If the Dewar has compression fitting like in Illustration 9-6:
   a. Loosen the 5/8 inch (15.9 mm) locking collar.
   b. Remove 1/2 inch (12.7 mm) cap and adapters, exposing a 5/8 inch (15.9 mm) brass locking collar.
   c. Connect the 5/8 inch (15.9 mm) brass locking collar, brass washer and o-ring to the Dewar-end stinger.
3. Verify that the pressure relief valve is in the open position. See Illustration 9-6.
4. Make sure the valve on the Dewar Stinger Assembly is closed and insert the assembly through the 5/8 inch (15.9 mm) locking collar until the stinger tip contacts the helium outlet valve. See Illustration 9-6.



HELIUM DEWAR TRANSFER LINE EXTENSION CONNECTION
ILLUSTRATION 9-6

9-5 LIQUID HELIUM (LHe) FILL / TOP-OFF (continued)

5. Open the helium outlet valve.
6. Slowly insert the Dewar Stinger Assembly into the Dewar until the stinger tip contacts liquid helium (indicated by a pressure increase on Dewar's pressure gauge and by expulsion of gas from pressure relief valve port).
7. Continue to insert Dewar Stinger Assembly at a rate that maintains a maximum 5 psig reading on the Dewar's pressure gauge.
8. When the Dewar Stinger Assembly contacts the bottom of the Dewar, raise the stinger assembly 1 inch (25 mm) and securely tighten the 5/8 inch (15.9 mm) threaded locking collar.

Note

If ceiling height prohibits insertion of the Dewar Stinger Assembly into the Dewar, the Dewar must be moved to an area with a higher ceiling and transported back into the magnet room.

9. When Dewar pressure stabilizes at or below 5 psig, close the pressure relief valve.
10. For LHe top-offs (LHe already in the vessel) **ONLY**: Attach the Phase Separator to the fill siphon. **Do not** attach the Phase Separator for initial fill (no LHe in the vessel).
11. Attach the selected Fill Siphon Assembly onto the helium transfer line, if not a one-piece assembly.
12. Install the threaded locking collar, retainer ring, and o-ring on the fill siphon.
13. Attach the opposite end of the helium transfer line onto the Dewar Stinger Assembly.

WARNING!

FIRMLY HOLD UNATTACHED END OF HOSE WHILE PURGING REGULATOR AND GAS LINE ASSEMBLY TO PREVENT WHIPPING MOTION.

14. Purge the GHe regulator and the gas line assembly by alternately turning the regulator handle fully in and out 3 times. Upon completion of purge, back the regulator out until a minimal flow is observed exiting the gas line assembly.

Note

Steps 15 and 16 will provide a helium-rich environment for connecting the GHe line to the helium gas pressurizing port.

15. Open helium Dewar's vent valve to allow a small amount of gas to flow.
16. Attach the purged gas line assembly to the liquid helium Dewar's helium gas pressurizing port. See Illustration 9-6.
17. Back out the regulator handle all the way.

9-5 LIQUID HELIUM (LHe) FILL / TOP-OFF (continued)

18. Assemble the o-ring, washer and locking collar onto the siphon.

CAUTION

Make sure the magnet vent is attached and inspected and the cryostat is at or below 0.5 psig.

19. Partially open the Dewar Stinger Assembly valve, allowing liquid helium to purge and precool the transfer line assembly.

CAUTION

Make sure the transfer line assembly is pointing in a direction safely away from all personnel. A plume of liquid helium will exit the assembly.

20. Allow the Dewar pressure to go down to < 5 psig before continuing.
21. With a plume present, uncap the siphon port and fully insert the siphon until it enters the siphon cone.
    - For initial fill (no LHe in the vessel), make sure the siphon is **fully seated**, then securely tighten the fill port compression fitting.
    - For top-offs (LHe already in the vessel) **ONLY**: When the siphon is fully seated, retract the siphon one inch (25 mm) to allow the Phase Separator to distribute LHe outside the cone.

WARNING!

MAKE SURE MAGNET PRESSURE STAYS BELOW 5 PSIG TO PREVENT BURST DISK RUPTURE.

22. Fully open the Dewar Stinger Assembly valve.

Note

If gas is observed escaping from the compression fitting on the fill port or the helium Dewar, use a heat gun to warm the compression fitting and recheck compression fitting tightness.

23. Open the vent valve at the helium gas pressurizing port on the Dewar.
24. Verify that the GHe cylinder's valve is fully open. Adjust the GHe cylinder's regulator to obtain a Dewar pressure of 4.0 psig during the entire fill. Do not set Dewar pressure above 4.0 psig.

9-5 LIQUID HELIUM (LHe) FILL / TOP-OFF (continued)

CAUTION

If frost is detected on the transfer line, stop the fill immediately and change lines. Frosted transfer lines have lost vacuum.

Note

The exhaust stack should be frosting up, indicating there is no restriction in the venting circuit. When liquid helium begins to collect, helium boil-off will decrease, the frost on the exhaust vent will start to melt and a slight decrease in helium vessel pressure will occur.

25. If the cryostat is empty, check the pressure gauge shown in Illustration 9-4 for a pressure increase until liquid helium begins to collect, at which point the pressure drops.
26. Once 1220 liters of helium are in the helium vessel, periodically press the “update” button on the Helium Level Monitor (2255532) and read the “Probe 1” helium level. Reset the monitor’s probe switch to “Probe 2” once 2506 liters of liquid helium are in the helium vessel. Read the “Probe 2” level from 2506 liters until the vessel is full at 3066 liters.

CAUTION

If the Helium Level Monitor (2255532) reading is not increasing, check the magnet and fill equipment for frosting or blockage. If the monitor reading is decreasing, stop the fill immediately, remove the siphon from the magnet, and contact service.

Note

Multiple Dewars may be required to achieve a 100% fill of the magnet cryostat.

27. Monitor Dewar transfer for one or more of the following to prevent blowing helium gas into the helium vessel:
    a. Listen for a whistling sound coming from the transfer line, indicating the Dewar is empty. Depending on equipment and conditions, the whistle may not always be heard.
    b. Monitor the Dewar pressure gauge and watch for a decrease in pressure. (A decrease in cryostat pressure could also be caused by an empty gas cylinder.)
    c. Monitor the percent change on the Liquid Helium Level Monitor. Stop the fill process when there is no positive (increasing) change in its reading.
    d. Make sure the Dewar stinger remains frost free during the fill process. Frost on the stinger is one indicator for passing helium vapor. Stop filling if frost is present.

9-5 LIQUID HELIUM (LHe) FILL / TOP-OFF (continued)

28. Record information for each Dewar in DATA SHEETS, Sections 2, Helium Refill Log, and 4, Helium Fill Data.

29. When the cryostat is full (100%), or when changing helium Dewars, close the Dewar Stinger Assembly's valve, close the GHe cylinder's valve, close the Dewar's vent valve and open the Dewar's pressure relief valve.

CAUTION

Always remove the cryostat siphon from the fill port and start the fill procedure over again when changing helium Dewars with the magnet at field. This is done to avoid the possibility of introducing helium gas from the helium transfer line into the magnet.

Note

If the magnet is NOT at field (unramped), the siphon may be left in the fill port when changing Dewars. In this case both the siphon and transfer line will require purging of helium pressure from the vessel backwards through the line before the Dewar stinger connection is made.

30. Remove the siphon from the fill port and immediately replace the fill port cap.

31. If additional Dewars are required, change helium Dewars in conformance with Section 9-6, Changing Helium Dewars, before continuing this procedure.

Note

A heat gun may be required to remove frost from the Fill Port Assembly before removing the stinger.

32. Tighten fill port compression fitting to prevent a leak from occurring.

33. Disconnect the Helium Transfer Line Assembly from the helium Dewar Stinger Assembly.

34. Make sure the helium Dewar is < 1 psig. Then remove the stinger assembly from helium Dewar.

35. Close the GHe cylinder's valve.

36. Disconnect the helium gas line from the Dewar's helium gas pressurizing port.

37. Back off the pressure regulator adjustment handle on the helium gas cylinder counterclockwise (CCW) until no resistance is felt.

38. Verify following Dewar configuration.
    - Liquid Helium Outlet Valve closed
    - Helium Vent Valve closed
    - Pressure Relief Valve open

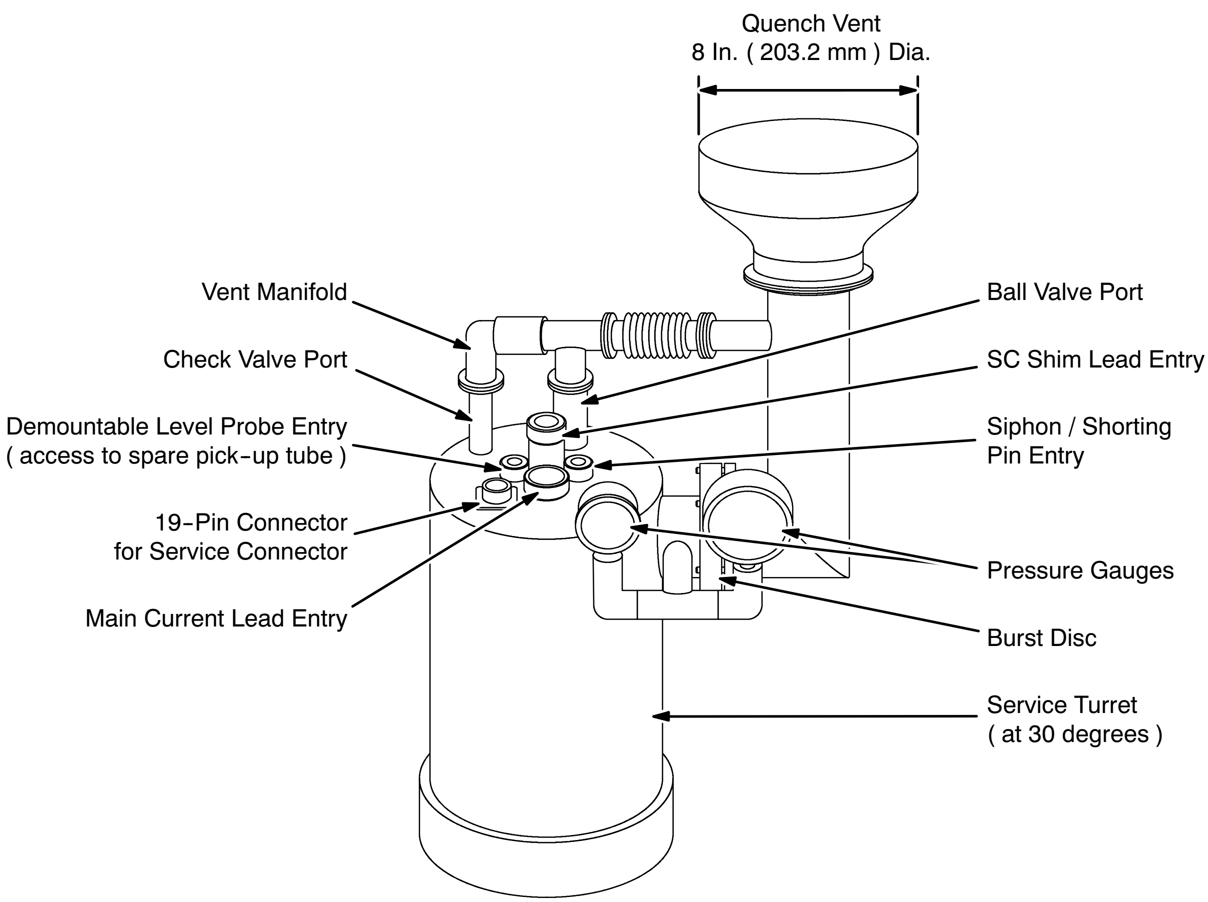
9-5 LIQUID HELIUM (LHe) FILL / TOP-OFF (continued)

39. Replace all adapters on the liquid helium valve outlet.

40. Remove the regulator from helium gas cylinder and reinstall its protective valve cap.

41. Remove the magnet bypass venting, then convert the magnet venting to operating configuration. See Illustration 9-7.

42. Leak test (Snoop) the fill port cap and all plumbing. Repair as necessary.



MAGNET VENTING IN OPERATING CONDITION
ILLUSTRATION 9-7

9-6 CHANGING HELIUM DEWARS

CAUTION

Always remove the cryostat siphon from the fill port and start the fill procedure over again when changing helium Dewars with the magnet at field. This is done to avoid the possibility of introducing helium gas from the helium transfer line into the magnet.

Note

If the magnet is NOT at field (unramped), the siphon may be left in the fill port when changing Dewars. In this case both the siphon and transfer line will require purging of helium pressure from the vessel backwards through the line before the Dewar stinger connection is made in Step 21.

1. Remove the siphon and cap the fill port.

Note

If "99.999% Helium Gas" is used, the purity of the gas remaining in the cylinder will degrade as a result of this process (i.e., the purity of the remaining gas will be something less than 99.999%).

2. Obtain full liquid helium Dewar. Check Dewar pressure gauge. If pressure is above 1 psig, slowly open Dewar vent valve and reduce pressure to 1 psig. See Illustration 9-6.

WARNING!

IF DEWAR PRESSURE DOES NOT VENT DOWN TO 1 PSIG, VERIFY THAT DEWAR PRESSURE RELIEF VALVE IS LEFT IN THE OPEN POSITION. CONTACT CRYOGEN SUPPLIER IMMEDIATELY.

Note

The pressure relief valve is normally open during shipping and storage to prevent excessive build up of pressure in the Dewar. Therefore, always leave the pressure relief valve open after using a Dewar.

3. Observe GHe cylinder regulator's high pressure gauge. Make sure indicated pressure is at least 1000 psig, indicating sufficient gas volume for transferring a full 250 liter helium Dewar.

9-6 CHANGING HELIUM DEWARS (continued)

WARNING!

SECURE CYLINDER TO A NONMAGNETIC GAS BOTTLE CART BEFORE REMOVING PROTECTIVE VALVE CAP TO PREVENT CYLINDER FROM FALLING, WHICH COULD RESULT IN SHEARING THE VALVE OUTLET AND CAUSING HAZARDOUS HIGH-PRESSURE GAS RELEASE.

4. If a new aluminum cylinder of GHe is required, connect a standard GHe regulator and hose assembly to the valve outlet (CGA 580) on the GHe cylinder.
5. Make sure that regulator adjusting handle is fully backed out, then slowly open GHe cylinder valve.
6. Refer to Section 9-2, Tools / Equipment for appropriate equipment.
7. Observe cryostat's pressure gauge and vent, temporarily opening the helium vent valve as required to obtain a pressure of 0.5 psig or lower.
8. Verify that the helium outlet valve is closed on a full Dewar.
9. Loosen the 5/8 inch (15.9 mm) locking collar on the Dewar.
10. Remove the 1/2 inch (12.7 mm) cap and adapters, exposing the 5/8 inch (15.9 mm) brass locking collar.
11. Verify that the pressure relief valve is in the open position.
12. Verify that the Dewar Stinger Assembly's valve is in the closed position. Disconnect the helium transfer line from Dewar Stinger Assembly.
13. Remove the Dewar Stinger Assembly from empty the Dewar.
14. Wipe off frost or moisture on the Dewar stinger and insert Dewar Stinger Assembly thru the 5/8 inch (15.9 mm) locking collar until the stinger tip contacts the helium outlet valve in the full Dewar. See Illustration 9-6.
15. Insert the Dewar Stinger Assembly through the 5/8 inch (15.9 mm) locking collar until the stinger tip contacts the helium outlet valve.
16. Open the helium outlet valve.
17. Slowly insert the Dewar Stinger Assembly into the dewar until the stinger tip contacts liquid helium (indicated by a pressure increase on the Dewar's pressure gauge and by expulsion of gas from the pressure relief valve port).
18. Continue to insert the Dewar Stinger Assembly at a rate that maintains a maximum 5 psig reading on the Dewar pressure gauge.

9-6 CHANGING HELIUM DEWARS (continued)

19. When the Dewar Stinger Assembly contacts the dewar's bottom, raise the stinger assembly 1 inch (25.7 mm) and securely tighten the 5/8 inch (15.9 mm) threaded locking collar.

Note

If ceiling height prohibits insertion of the Dewar Stinger Assembly into the Dewar, the Dewar must be moved to an area with a higher ceiling and transported back into the magnet room.

20. When the Dewar's pressure stabilizes at 5 psig, close the pressure relief valve.
21. Attach the helium transfer line to the Dewar Stinger Assembly.

WARNING!

FIRMLY HOLD UNATTACHED END OF HOSE WHILE PURGING REGULATOR AND GAS LINE ASSEMBLY TO PREVENT WHIPPING MOTION.

22. Disconnect the helium gas line from the empty Dewar. Purge the GHe cylinder's regulator and the gas line assembly by alternately turning the regulator handle fully in and out 3 times. Upon completion of the purge, back the regulator out until a minimal flow is observed exiting the gas line assembly.
23. Slightly open the GHe vent valve to allow for helium gas flow from the Dewar.
24. Attach the purged gas line assembly to the Dewar's helium gas pressurizing port. Fully back out the regulator adjusting handle.
25. Purge the transfer line and siphon. When the siphon is "plumed," continue with Step 21 of Section 9-5, Liquid Helium (LHe) Fill.
26. Prepare empty Dewar as follows:
    a. Liquid Helium Outlet Valve closed
    b. Helium Vent Valve closed
    c. Pressure Relief Valve open
    d. Replace all adapters on the liquid helium valve outlet
27. Proceed with Step 32 of Section 9-5, Liquid Helium (LHe) Fill.

9-7 CHANGING HELIUM GAS BOTTLES

DANGER!!

FATAL EXPLOSIVE HAZARD! TO PREVENT POSSIBLE FATAL EXPLOSIVE RELEASE OF GAS, OPEN MAIN VALVE ON GAS CYLINDER VERY SLOWLY. GAS IS AT 2400 PSI.

WARNING!

HELIUM GAS BOTTLE CAPS ARE FERROUS AND MUST BE REMOVED BEFORE BRINGING BOTTLE INTO MAGNET ROOM AS THEY CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.

Note

Change the helium gas bottle when bottle pressure drops below 5 psig. Approximately one 235 SCF helium gas bottle is needed to pressurize a 500 liter liquid helium Dewar.

1. Close vent valve on dewar and remove gas hose.
2. Close main valve and regulator on helium gas bottle.
3. Bring a full helium gas bottle, secured to a nonmagnetic cart, into the exam room and reconnect the regulator and hoses.
4. Purge the hose before reconnecting to the Dewar.

SECTION 10
PREPARATIONS FOR FIELD PLOTTING

10-1 FIELD PLOTTING RIG (2292542) SET-UP

Field Plotting Rig 2292542 requires one field service engineer and one assistant approximately 2 hours to set up.

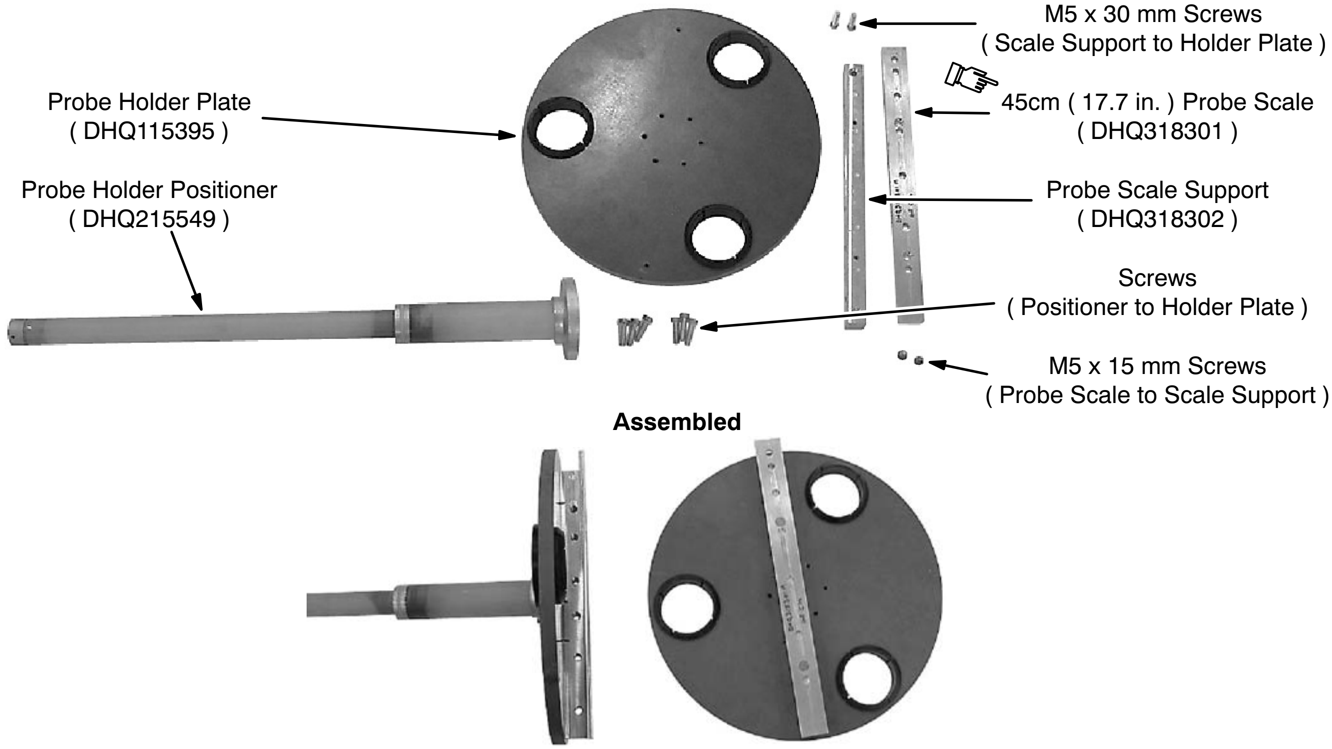
10-1-1 Tools / Equipment

- Field Plotting Rig (2292542). See RENEWAL PARTS, Section 4-14, Field Plotting Rig Assembly, Large Bore, for a list of component parts.
- MetroLab Teslameter (46-251865G4) or equivalent magnetic field measuring device
- No. 6 MetroLab Field Probe (2295525-5)
- 8 mm allen wrench
- Crescent wrench
- Flat blade screwdriver
- Two Foot (500 mm) Level

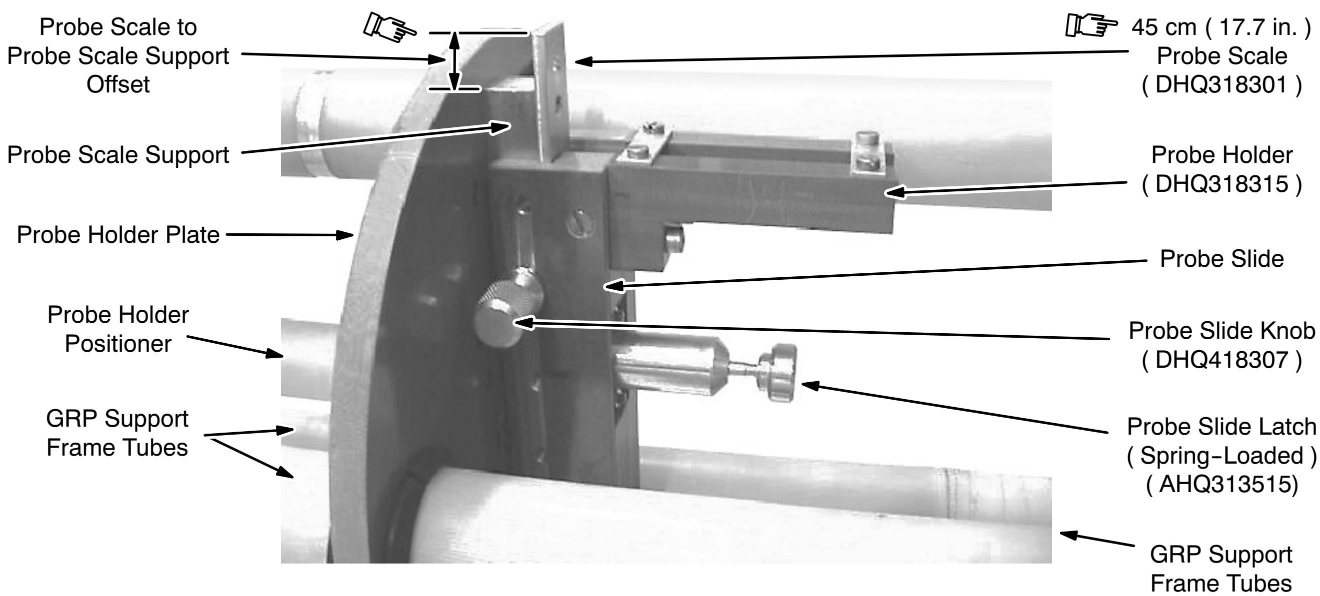
10-1-2 Probe Holder Assembly

1. Attach the Probe Scale Support (DHQ318302) to the Probe Holder Plate (DHQ115395). See Illustrations 10-1 and 10-2.
2. Attach the 45 cm (17.7 inch) Probe Scale (DHQ318301) to the Probe Scale Support so that the scale overhangs the support's top. See Illustrations 10-1 and 10-2.

10-1-2 Probe Holder Assembly (continued)



PROBE HOLDER PLATE ASSEMBLY
ILLUSTRATION 10-1



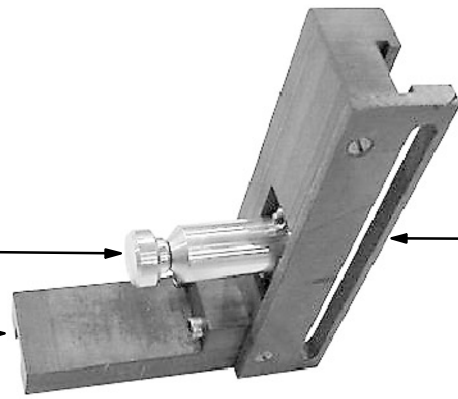
PROBE SCALE POSITIONING
ILLUSTRATION 10-2

10-1-2 Probe Holder Assembly (continued)

3. Pull the Probe Slide Latch out and slide the Probe Slide and Probe Holder (Illustration 10-3) onto the Probe Scale from the bottom of the Probe Scale and the Probe Holder facing upwards. See Illustration 10-2.

Note

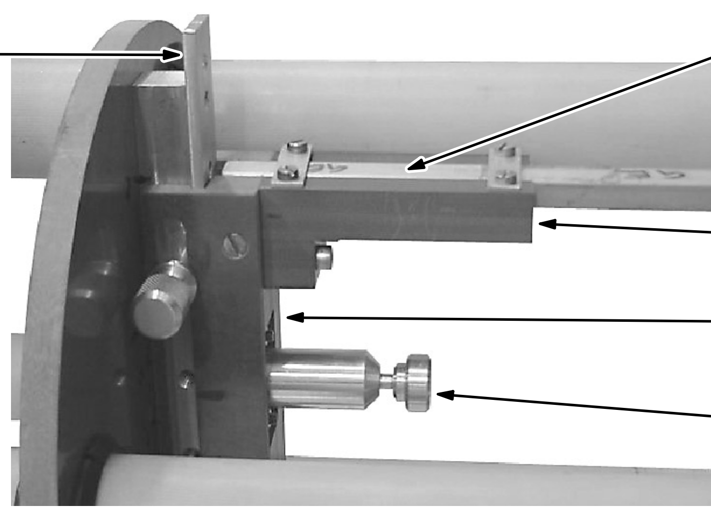
If the Probe Slide doesn't easily slide onto the Probe Scale, loosen or even temporarily remove the Probe Slide Knob. See Illustration 10-3.



PROBE HOLDER (FROM BELOW)
ILLUSTRATION 10-3

4. Position the Probe Slide with the Probe Slide Knob in the first hole in the 45 cm (17.7 inch) Probe Scale. That hole is 52 mm (2 inches) from the scale's bottom end and will place the probe at the center of the magnet's bore (R = 0) once the Field Plotting Rig is centered.

5. The Teslameter Probe is placed in the Probe holder as shown in Illustration 10-4.



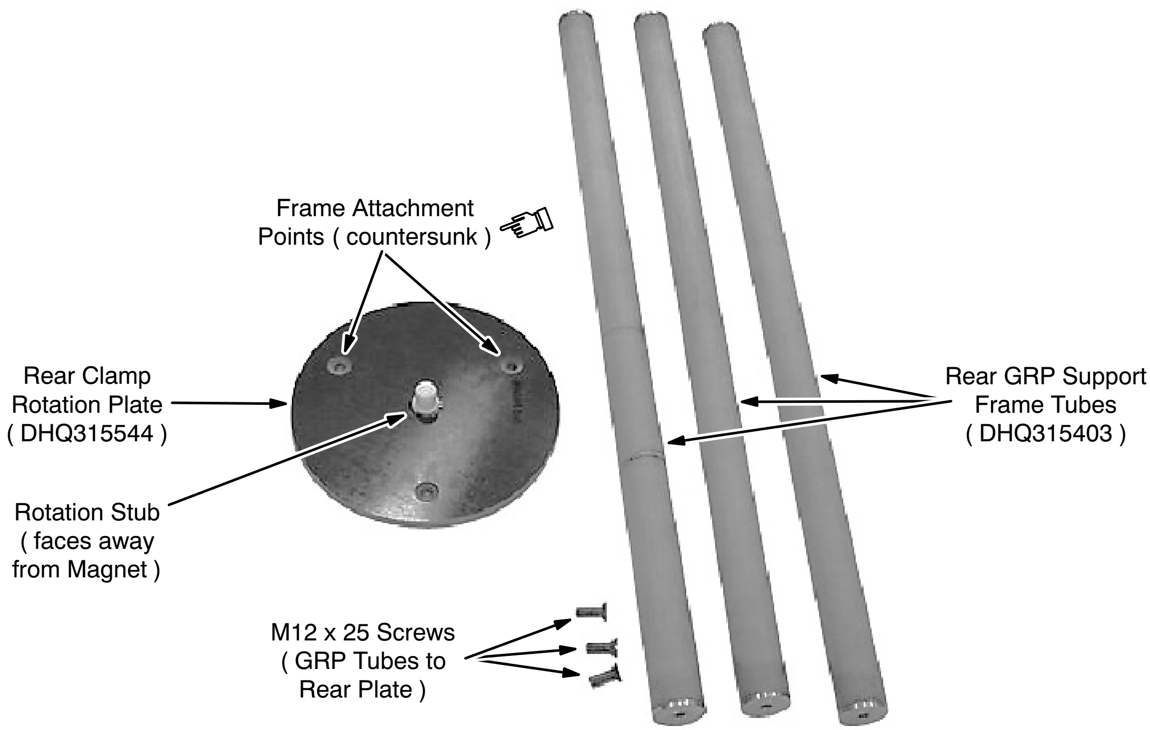
Note

Teslameter Probe is at Angle (α) = 0 when Probe Scale is vertical and Probe Holder faces up. See Illustration 10-16.

TESLAMETER PROBE MOUNTED IN PROBE HOLDER
ILLUSTRATION 10-4

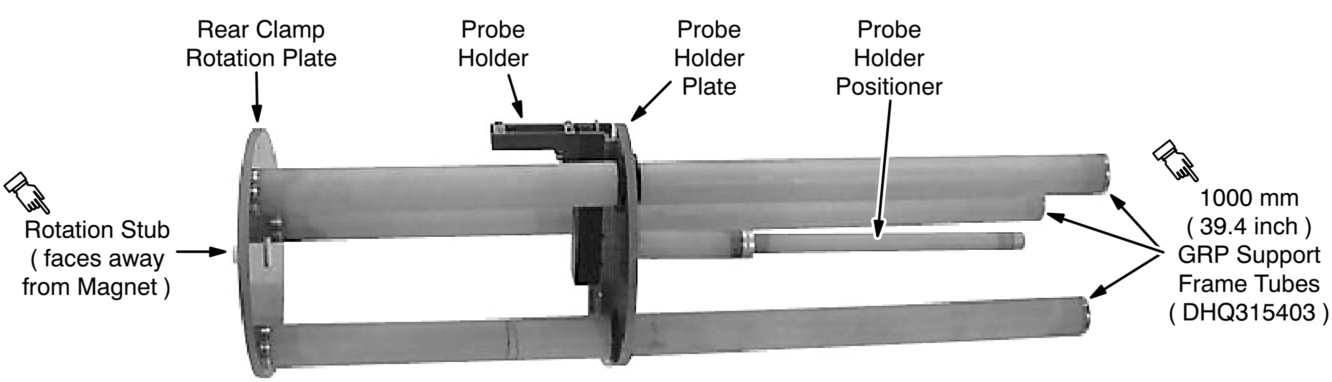
10-1-3 Plotting Rig Assembly and Mounting

1. Attach the 3 Rear GRP Support Frame Tubes to the face of the Rear Clamp Rotation Plate opposite the Rotation Stub. See Illustration 10-5.



REAR CLAMP ROTATION PLATE AND REAR SUPPORT FRAME TUBES
ILLUSTRATION 10-5

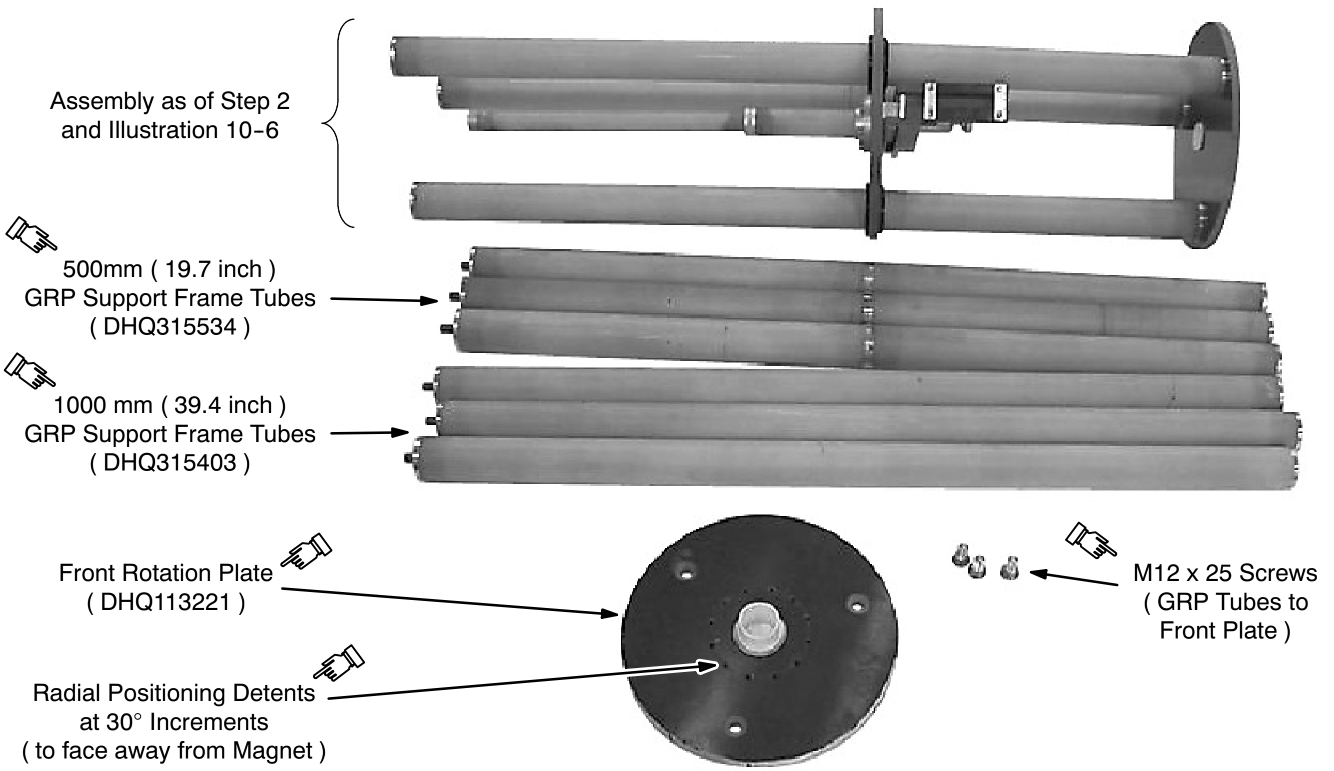
2. Slide the assembled Probe Holder and Plate onto the GRP Rear Support Frame Tubes. See Illustration 10-6.



PROBE HOLDER AND PROBE HOLDER PLATE ON REAR SUPPORT FRAME
(IN RADIAL 0° ORIENTATION)
ILLUSTRATION 10-6

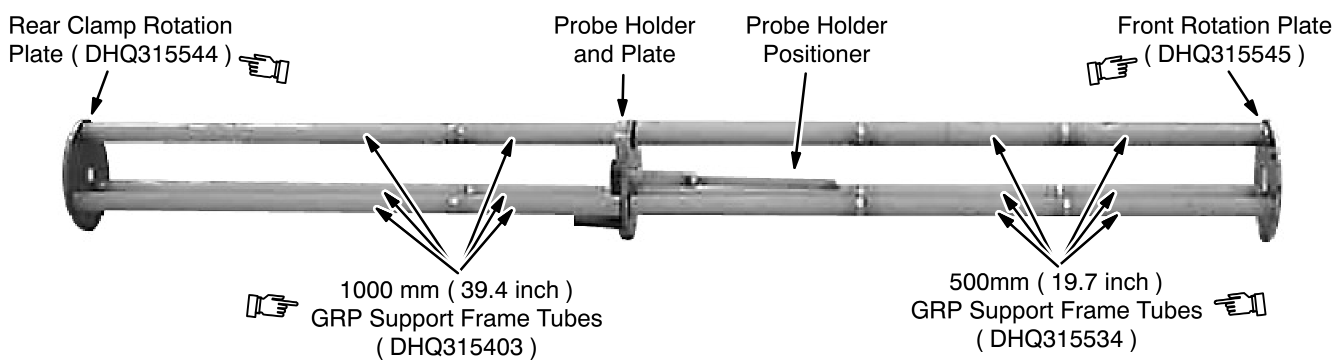
10-1-3 Plotting Rig Assembly and Mounting (continued)

3. Attach the remaining GRP Tubes to the existing GRP Tubes, then attach the Front Rotation Plate to the ends of those tubes. See Illustrations 10-7 and 10-8.



SUPPORT FRAME COMPONENTS
ILLUSTRATION 10-7

4. Insert the two Probe Holder Positioner Extensions through the Front Rotation Plate's center hole and attach them to the existing Probe Holder Positioner. See Illustration 10-8.



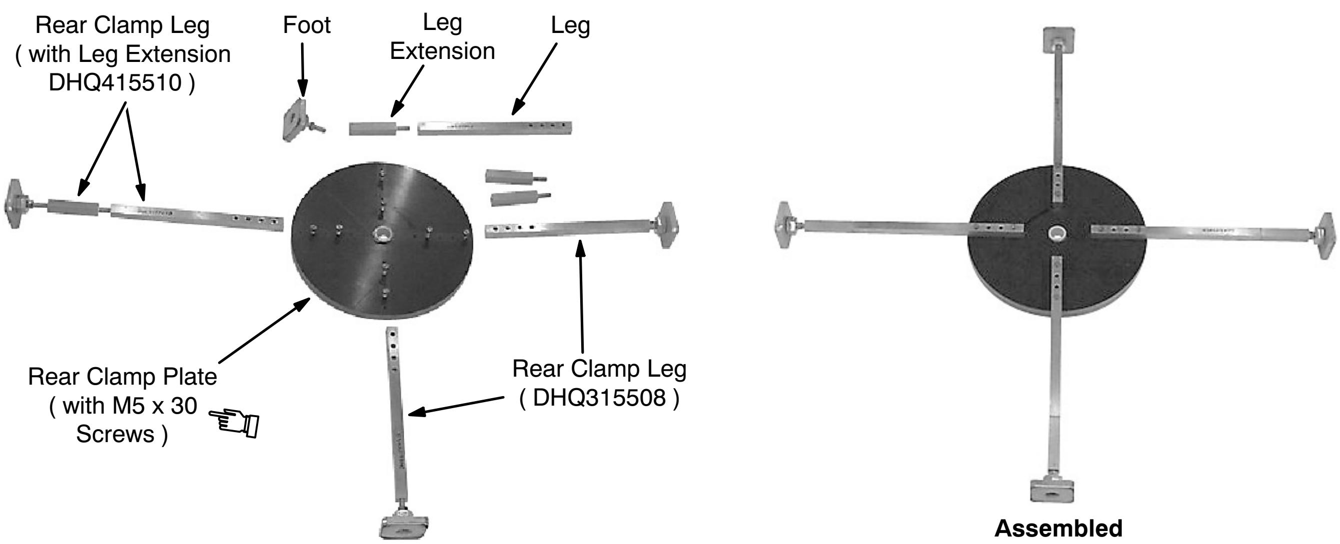
PROBE HOLDER AND SUPPORT FRAME
ILLUSTRATION 10-8

10-1-3 Plotting Rig Assembly and Mounting (continued)

CAUTION

Field Plotting Rig 2292542 is both bulky and heavy. Never move the assembled Support Frame or Plotting Rig without assistance.

5. Lay the assembly in the bore of the magnet. Have an assistant help.
6. Assemble the Rear Clamp Assembly by attaching the Rear Clamp Legs to the Rear Clamp Plate using M5 x 30 screws in the first and fourth holes from the edge of the Rear Clamp Plate. Attach the Leg Extensions to the Legs and then the Rear Clamp Feet to the Leg Extensions. See Illustrations 10-9.



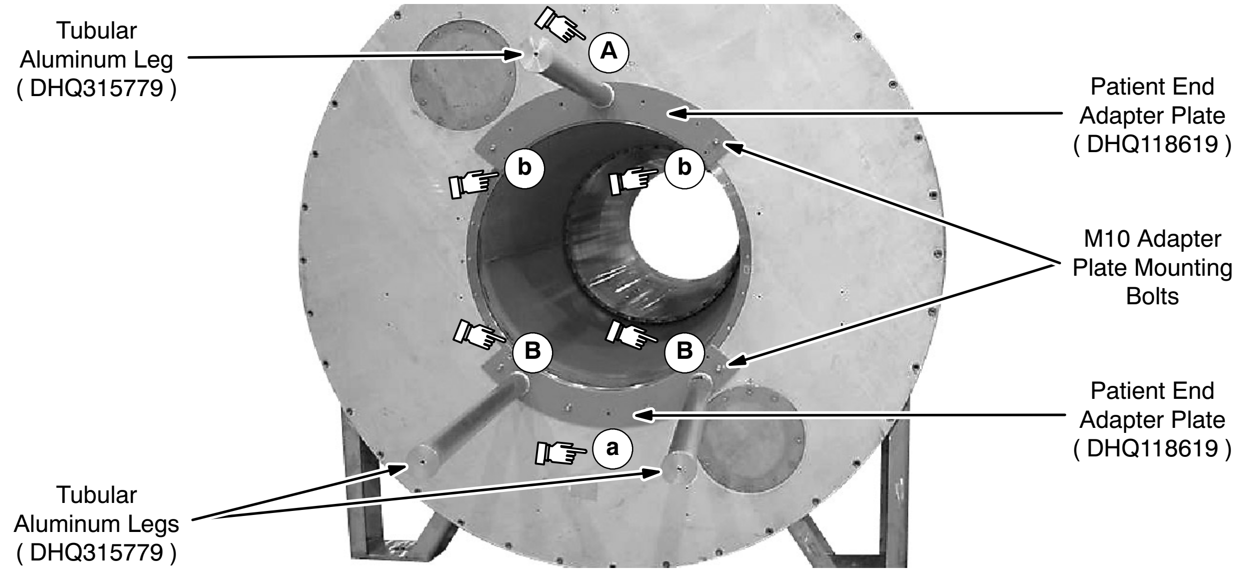
REAR CLAMP ASSEMBLY (AHQ115505) COMPONENTS
ILLUSTRATION 10-9

7. Attach one Tubular Aluminum Leg (DHQ315779) to one Patient End Adapter Plate (DHQ118619) at point “A” in Illustration 10-10. Do not attach legs to points “b” in Illustration 10-10 on this Patient End Adapter Plate.

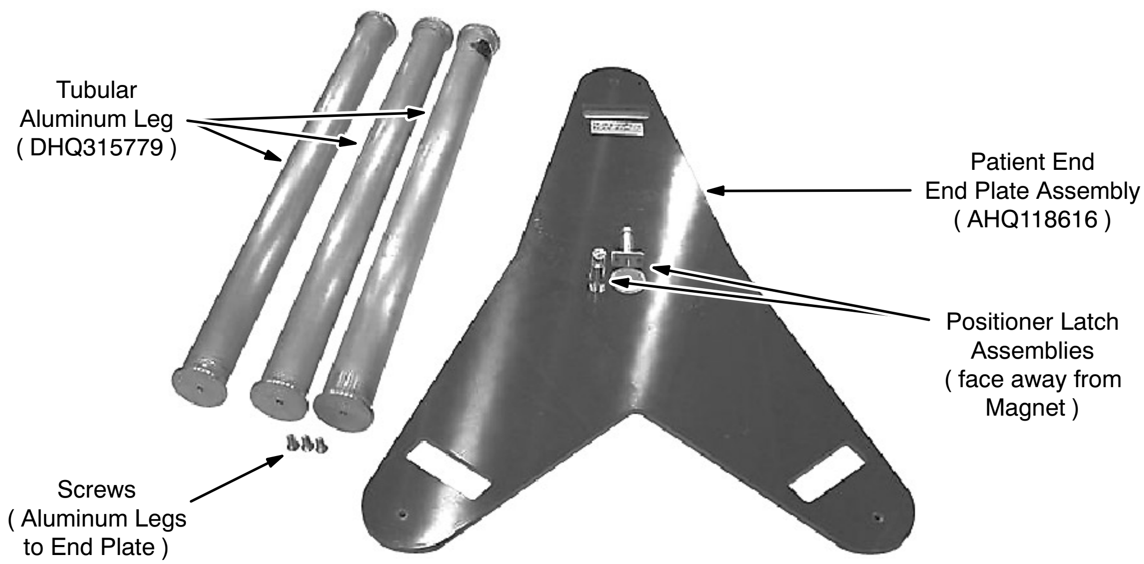
Note
The components of the Patient End End Plate Assembly (AHQ118616) and the Tubular Aluminum Legs (DHQ315779) used to attach the assembly to the Patient End Plate Adapters (DHQ118619) are shown in Illustration 10-11.

8. Attach two legs to the other Patient End Adapter Plate (DHQ118619) at points “B’ in Illustration 10-10. Do not attach legs to points “a” in Illustration 10-10 on this Patient End Adapter Plate.

10-1-3 Plotting Rig Assembly and Mounting (continued)



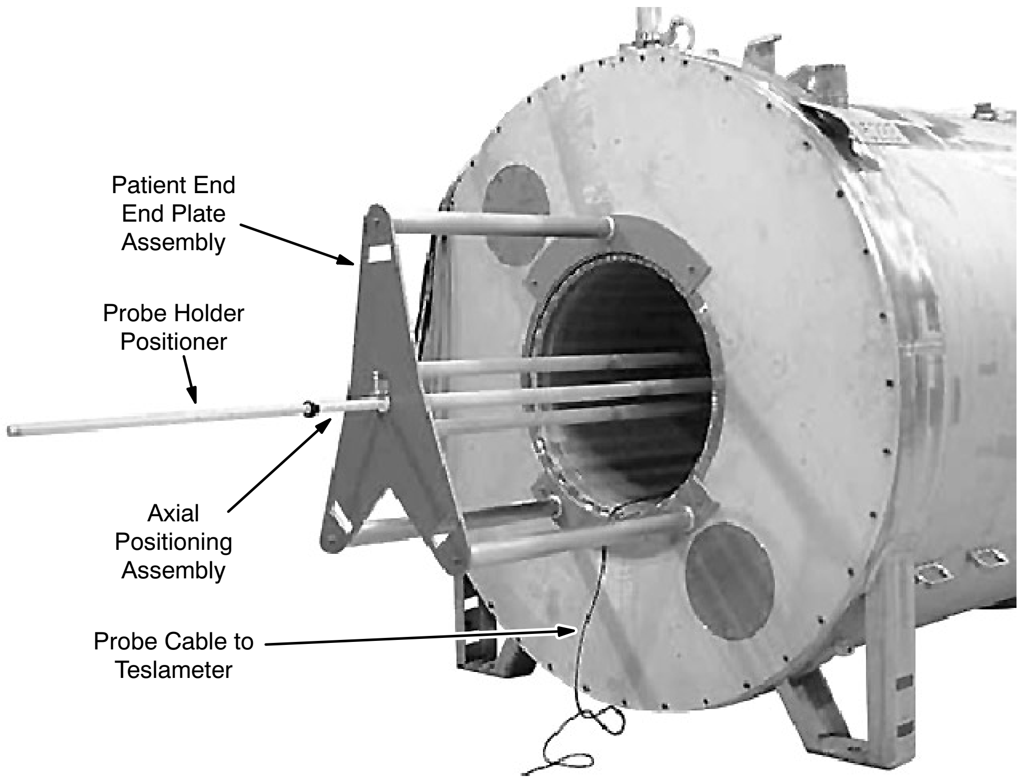
PATIENT END ADAPTER PLATE ATTACHMENT
ILLUSTRATION 10-10



PATIENT END END PLATE ASSEMBLY AHQ118616
ILLUSTRATION 10-11

10-1-3 Plotting Rig Assembly and Mounting (continued)

9. Attach the Patient End Adapter Plate with two legs below the bore on the patient end of the magnet. Attach the Patient End Adapter Plate with one leg above the bore. See Illustrations 10-10 and 10-12.



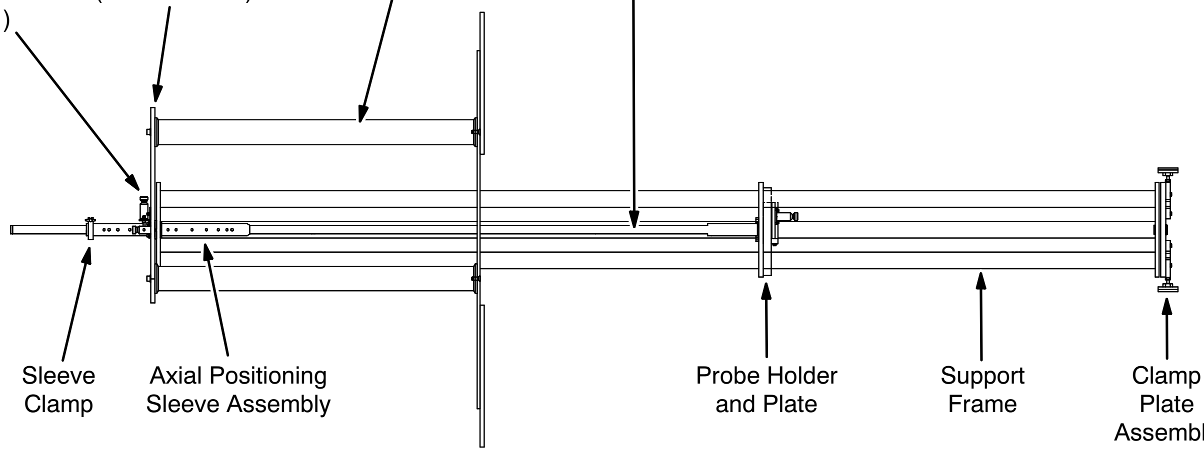
ASSEMBLED FIELD PLOTTING RIG, PATIENT END VIEW
ILLUSTRATION 10-12

10. Make sure the Positioner Latch Assemblies are attached to Patient End End Plate Assembly AHQ118616. See Illustration 10-11. If the Positioner Latch Assemblies (Illustration 10-11) are attached to the Service End End Plate Assembly (AHQ113317), relocate them to the Patient End End Plate Assembly (AHQ118616) and continue assembling the Field Plotting Rig as instructed.

Note

With the Service End Field Plotting Rig configuration shown in Illustration 10-13, passive shimming cannot be done. However, all components required for both the patient end and service end configurations are included in Field Plotting Rig 2292542.

10-1-3 Plotting Rig Assembly and Mounting (continued)



* Passive Shimming cannot be done with the Service End Field Plotting Rig configuration. Instead, assemble the Field Plotting Rig (2292542) using the Patient End End Plate Assembly (AHQ118616) as instructed.

FIELD PLOTTING RIG IN SERVICE END CONFIGURATION
ILLUSTRATION 10-13

11. Attach the Patient End End Plate Assembly (AHQ118616) to the Tubular Aluminum Legs (DHQ315779). See Illustartions 10-10, 10-11 and 10-12.

CAUTION

Field Plotting Rig 2292542 is both bulky and heavy. Never move the assembled Support Frame or Plotting Rig without assistance.

Note
Make sure the Probe Holder is oriented with the Probe facing up, as shown in Illustration 10-4, while performing Step 12.

12. With the aid of an assistant, lift the Probe Holder and Support Frame (shown in Illustration 10-8) that was placed in the magnet bore in Step 5. Push the Probe Holder Positioner Extensions through the center hole in the Patient End End Plate Assembly (shown in Illustration 10-11) until the bearing engages. See Illustration 10-12.
13. Temporarily support the rear of the Probe Holder and Support Frame.
14. Attach the Rear Clamp Assembly (shown in Illustration 10-9) to the Rear Clamp Rotation Plate (shown in Illustration 10-8). See Illustration 10-14.

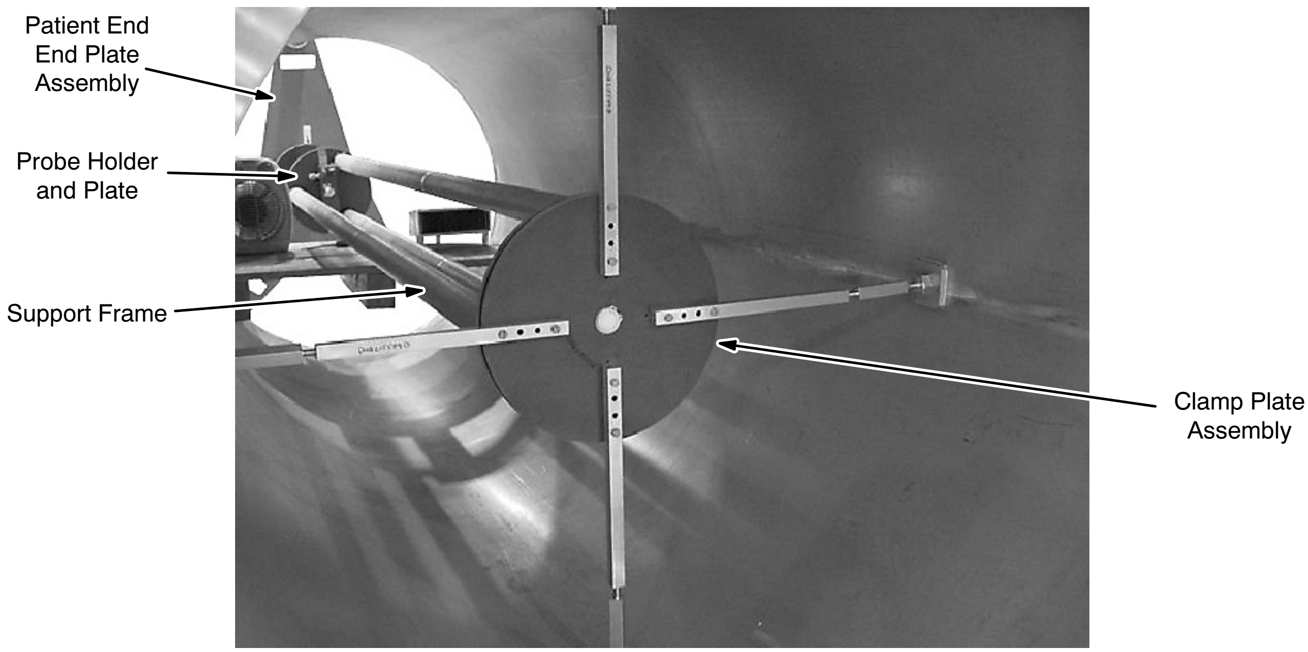
10-1-3 Plotting Rig Assembly and Mounting (continued)

15. Adjust the lengths of the Rear Clamp Legs until the Rear Clamp Rotation Plate measures as centrally located within the bore diameter, then tighten the lock nuts on each leg. Verify that Rear Clamp Rotation Plate remains centrally positioned. See Illustrations 10-14 and 10-15.

Note

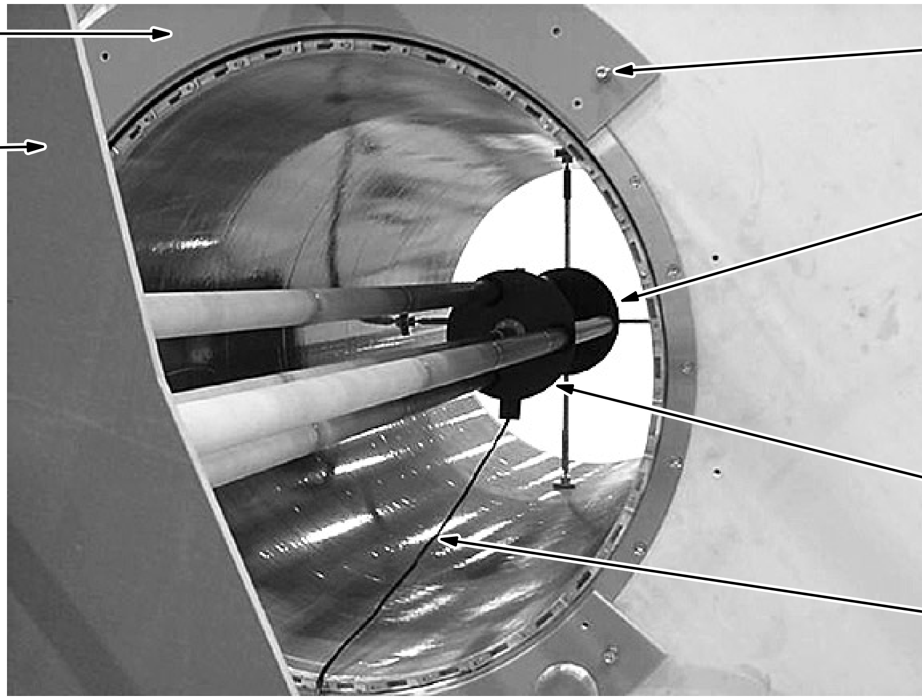
Plotting accuracy depends on the Field Plotting Rig being in the magnet's axial center.

16. Slide the Axial Positioning Sleeve Assembly over the Probe Holder Positioner and then through the bearing hole in the Patient End End Plate Assembly. See Illustrations 10-12, 10-11 and 10-13.



REAR CLAMP ASSEMBLY LOCATION, FROM REAR
ILLUSTRATION 10-14

10-1-3 Plotting Rig Assembly and Mounting (continued)



REAR CLAMP ASSEMBLY LOCATION, FROM FRONT
ILLUSTRATION 10-15

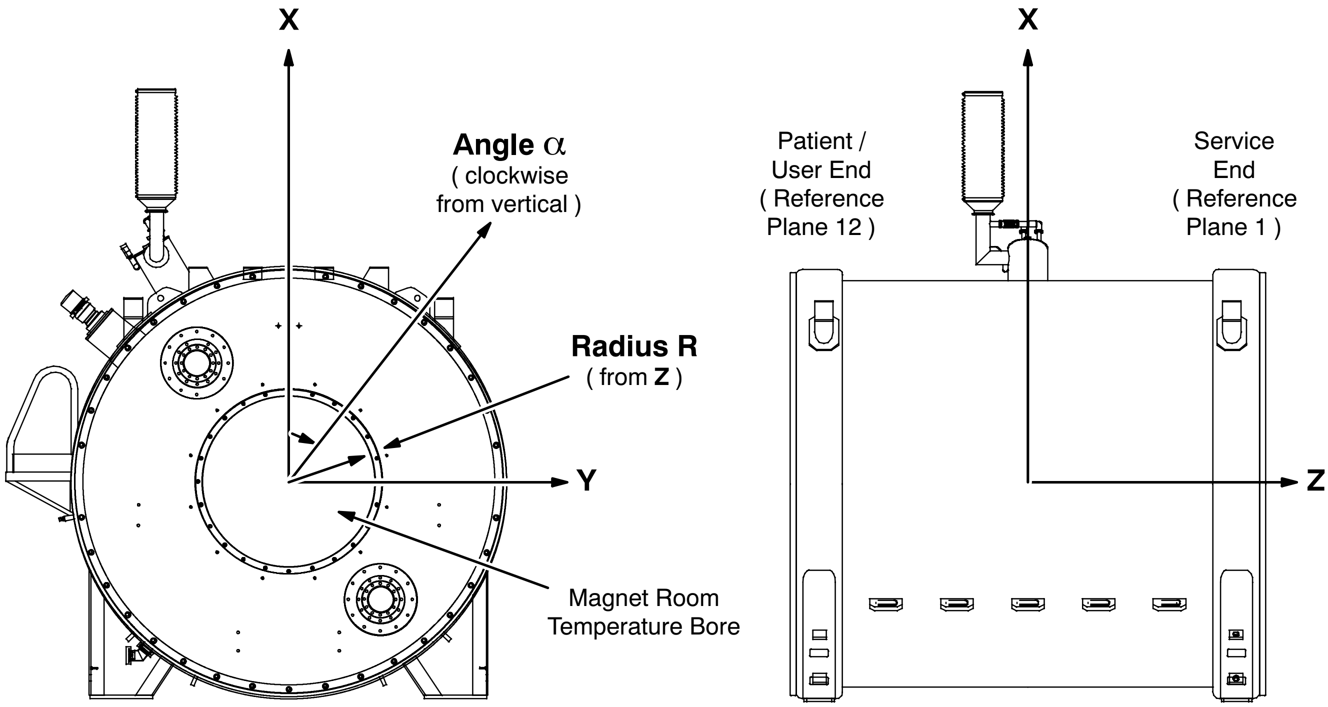
17. Place a two foot (500 mm) level on the Probe Holder while the Probe Holder is half way between the Clamp Plate Assembly and the Patient End End Plate Assembly.

18. Raise / lower and support the Patient End End Plate as required to make the Probe Holder and Support Frame level.

Note
Plotting accuracy depends on the Field Plotting Rig being in the magnet's axial center.

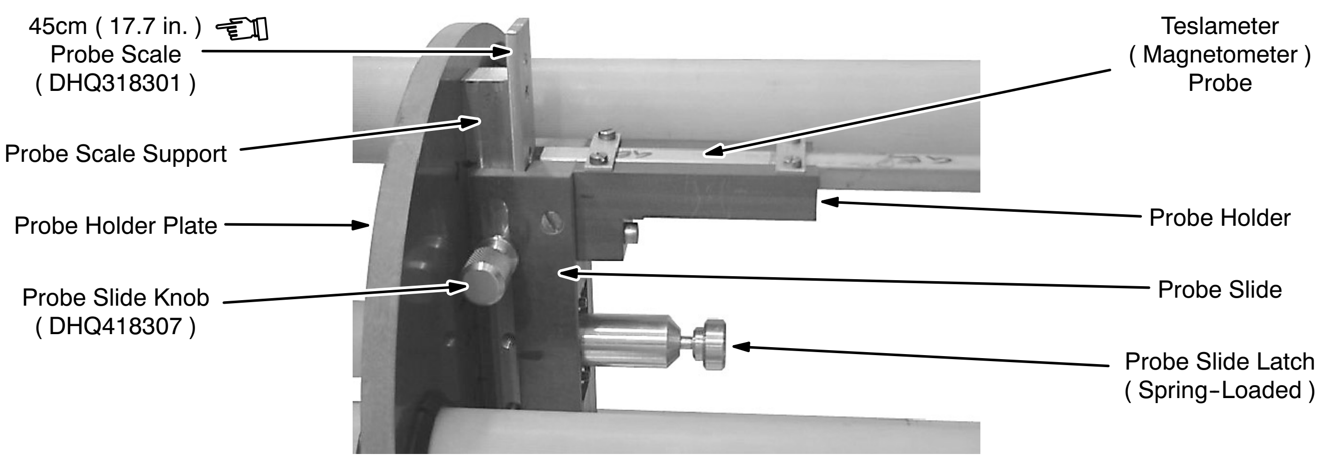
10-1-4 Plotting Rig Centering

The Field Plotting Rig must position the Teslameter (Magnetometer) Probe at the magnet's center for the field map to be accurate. Illustration 10-16 shows the magnet's coordinate system.



MAGNET COORDINATE SYSTEM
ILLUSTRATION 10-16

1. Mount the Teslameter (Magnetometer) Probe in the Probe Holder. See Illustration 10-17.
2. Measure the distance from the the Probe's sampling location (marked with an "X") and the closest surface of the Probe Holder Plate. Call this distance the "probe offset."



TESLAMETER PROBE MOUNTED IN PROBE HOLDER
ILLUSTRATION 10-17

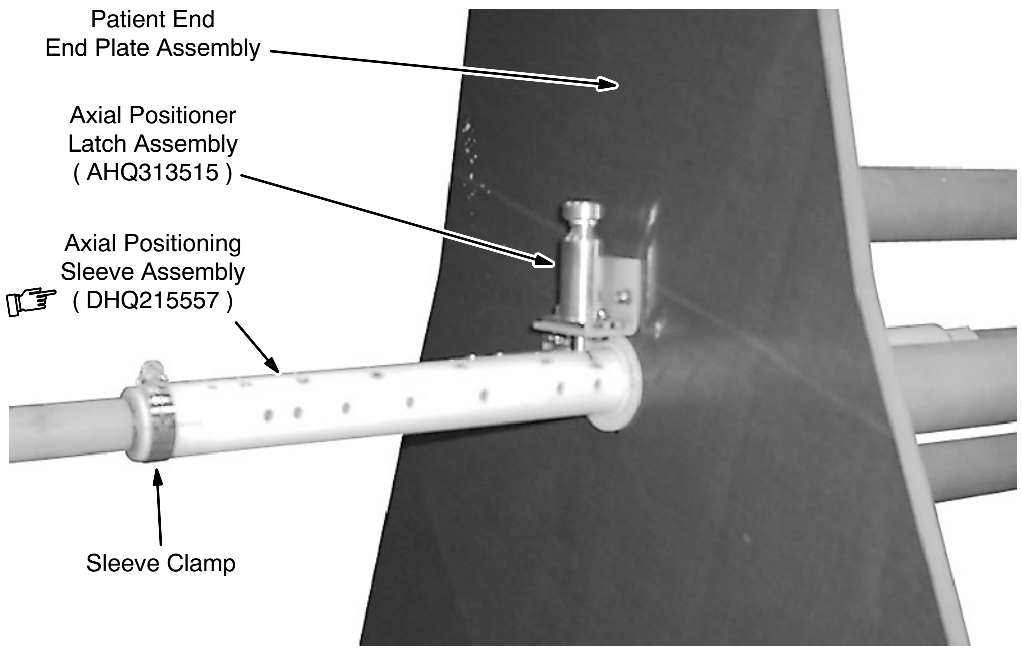
10-1-4 Plotting Rig Centering (continued)

3. Measure the distance between the magnet's end plates (nominally 2600 mm / 102.36 inches). Half this distance is the magnet's Z-axis center.

4. Move the Probe Holder Plate to the "probe offset" distance in front of the magnet's center. The Probe should now be at the magnet's Z-axis center.

5. Slide the white Axial Positioning Sleeve along the Probe Holder Positioner until the Axial Positioner Latch engages in the sleeve's center (zero) hole. See Illustration 10-18.

6. Verify that the Probe is still at the magnet's Z-axis center.



POSITIONING THE AXIAL POSITIONING SLEEVE
ILLUSTRATION 10-18

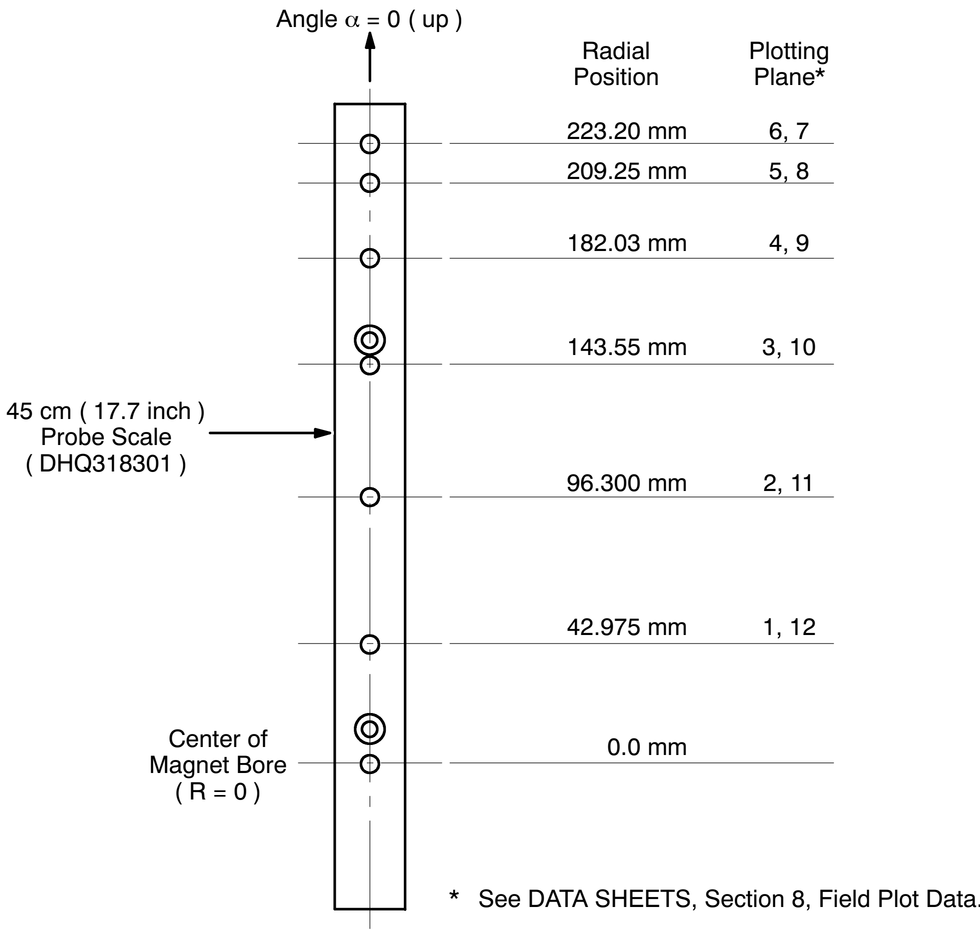
7. Lock the Sleeve Clamp (Illustration 10-13) to secure the Axial Positioning Sleeve to the Probe Holder Positioner.

8. Mark this location of the Axial Positioning Sleeve on the Probe Holder Positioner for future reference.

10-1-5 Probe Positioning (22.5 cm Plotting Radius)

Radial Positioning

1. Loosen the Probe Slide Knob. See Illustration 10-17.
2. Pull the Probe Slide Latch to disengage the Probe Slide from its current detent in the 45 cm (diameter) Probe Scale. See Illustration 10-17.
3. Move the Probe Slide to where the Probe Slide Latch will engage with the detent for the radial position to be sampled next. See Illustration 10-19. Release the Probe Slide Latch.
4. Tighten the Probe Slide Knob.



RADIAL PROBE POSITIONS ALONG 45 CM (17.7 INCH) PROBE SCALE (DHQ318301)
ILLUSTRATION 10-19

10-1-5 Probe Positioning (22.5 cm Plotting Radius) (continued)

Axial Positioning

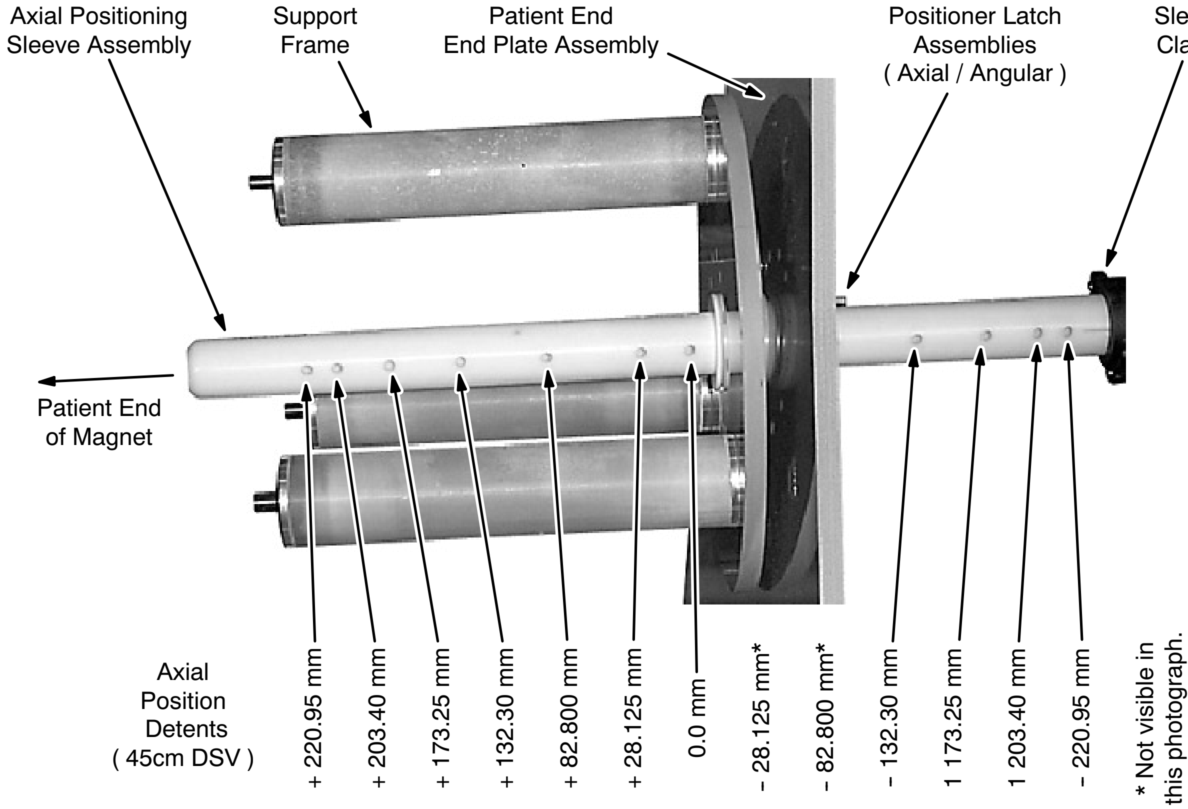
1. Pull the Axial Positioning Latch (Illustration 10-13) to disengage the latch from its current detent in the Axial Positioning Sleeve. See Illustration 10-18.
2. Move the Probe Holder along the Support Frame until the Axial Positioning Latch aligns with the desired Axial Position Detent. Table 10-1, Illustration 10-20 and DATA SHEETS, Section8, list the detents and resulting probe positions.
3. Release the Axial Positioning Latch into the detent.

TABLE 10-1
AXIAL POSITION DETENTS (WITH 22.5 CM PLOTTING RADIUS)

	Positive Detents (Towards Service End)							Negative Detents (Towards Patient End)					
	6	**5**	**4**	**3**	**2**	**1**	**0**	**- 1**	**- 2**	**- 3**	**- 4**	**- 5**	**- 6**
Axial Position (mm)	220.95	203.40	173.25	132.30	82.800	28.125	0.0	-28.125	-82.800	-132.30	-173.25	-203.40	-220.95
Plotting Plane*	1	2	3	4	5	6	—	7	8	9	10	11	12

* See DATA SHEETS, Section 8, Field Plot Data.

10-1-5 Probe Positioning (22.5 cm Plotting Radius) (continued)



AXIAL PROBE POSITIONING
ILLUSTRATION 10-20

Angular Positioning

1. Pull the Angular Positioning Latch (Illustration 10-13) to disengage the latch from its current position.

2. Rotate the Probe Holder until the Angular Positioning Latch aligns in the desired radial position.

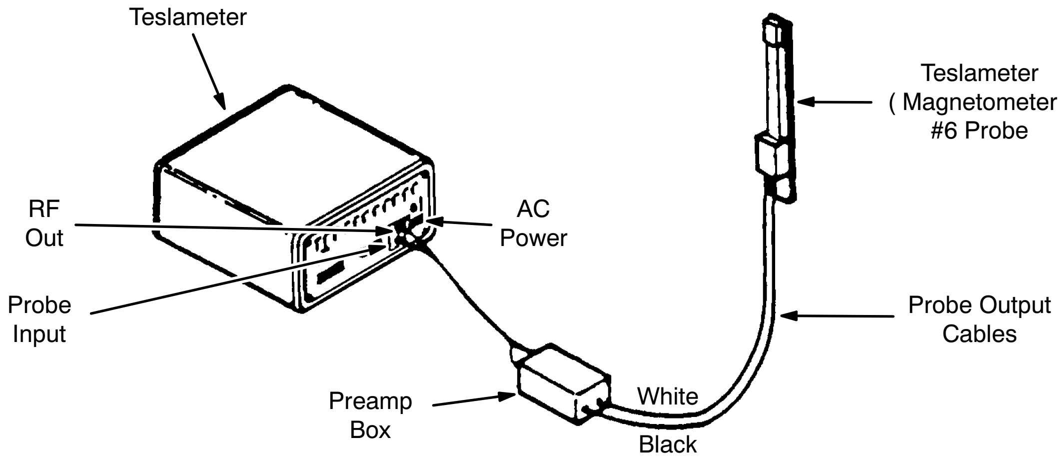
Note
The radial positions run from straight up (zero degrees) around clockwise (as viewed from the magnet's patient end) in 30 degree increments to 330 degrees. These positions match those in DATA SHEETS, Section 8.

3. Release the Angular Positioning Latch into the detent.

10-2 TESLAMETER ADJUSTMENT / RESYNCHRONIZATION

10-2-1 Teslameter Adjustment for Ramp Up

1. Connect the probe output cables to the Preamp Box. See Illustration 10-21.

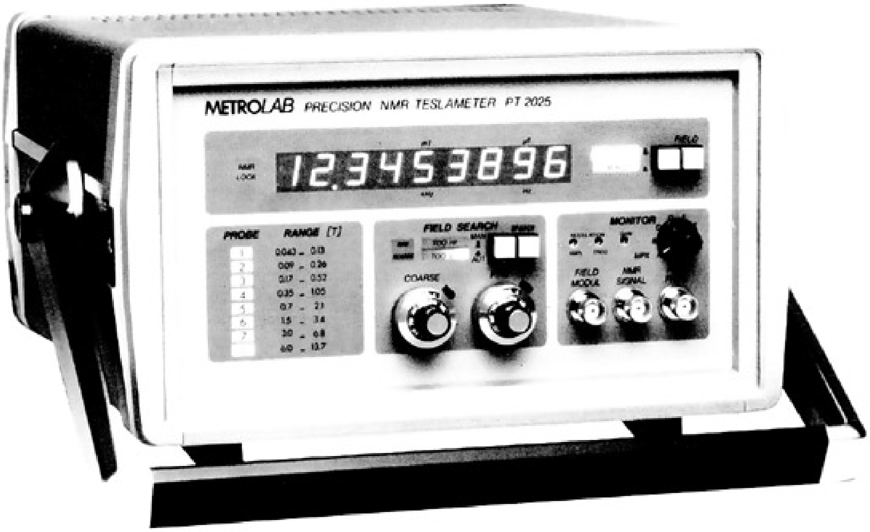


TESLAMETER INTERCONNECTIONS
ILLUSTRATION 10-21

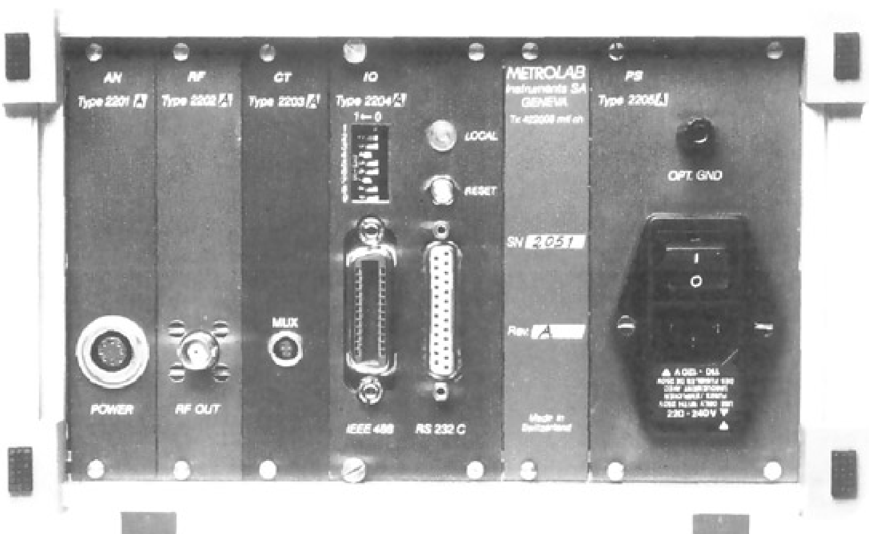
2. Connect the Preamp Box to the two-probe connection input plug on the Teslameter.
3. Position the Teslameter (Magnetometer) Probe at physical center of the bore (R = 0, Z = 0). See Illustration 10-16.

10-2-1 Teslameter Adjustment for Ramp Up (continued)

4. Turn on AC power to the Teslameter. See Illustration 10-22, Rear View.
5. Place the Lock / Manual Switch in the "Manual" position.



Front View



Rear View

TESLAMETER
ILLUSTRATION 10-22

10-2-1 Teslameter Adjustment for Ramp Up (continued)

CAUTION

If the Lock / Manual Switch is not in its "Manual" position when field search begins during ramping, the sweep will be in the ± 1 Tesla range, and the system could lock on to the mechanical oscillation harmonic of the Shield Cooler Coldhead, resulting in an erroneous reading.

6. Set the Coarse and Fine Field Search Control Knobs fully counterclockwise (CCW).
7. Set the NMR Freq. / Field Switch to the "Field" position.

Note
The Teslameter is now set up and prepared to monitor the field when ramping commences.

Note
The Teslameter will not lock in on the probe's signal until the magnetic field is > 1.0 Tesla, depending on the probe being used and the magnet field strength.

8. Once ramping has started monitor the Power Supply Current Meter until an indication of ≈ 85 amps is approached, then start monitoring the Teslameter.
9. As the developing field approaches the lower limit of the Teslameter probe (≈ 1.0 Tesla), the "No Strobe Signal" LED will start blinking. When this LED goes out, reposition the Lock / Manual Switch to the "Lock" position. See Illustration 10-22.

Note
If the signal does not lock in, reposition of the Probe Modulation Switch.

10. As the field continues to increase the "Hi" LED to the right of the Coarse Control Knob will light. Turn the Coarse Control Knob clockwise (towards the lighted LED) until the LED light is extinguished.
11. As the magnetic field increases, the resonant frequency of the probe sample will increase above the Teslameter's range setting. Increase the Coarse Control Knob setting periodically to keep the Teslameter locked on the probe sample.

Note
The Fine Control Knob does not function when the Lock / Manual Switch Is in the "Locked" position.

10-2-2 Teslameter Adjustment For Rampdown

1. Increase the Coarse Control Knob until a field reading of ≈ 3.0 Tesla is obtained. You should be near but slightly below the actual field. See Illustration 10-22.

2. Slowly start increasing the Fine Control Knob. When the "No Strobe Signal" LED stops blinking and remains out, reposition the Lock / Manual Switch to the "Lock" position.

Note
If the signal will not lock on, reposition of the Probe Modulation Switch.

3. Once the Teslameter is locked onto the field, either the "Lo" or "Hi" LED will be lit. Turn the Coarse Control Knob in the direction of whichever LED is lit. Slowly that LED's light will go out and stay out.

4. As the magnetic field decreases, the resonant frequency of the probe sample will decrease below the Teslameter's range setting. Decrease the Coarse Control Knob setting periodically to keep the Teslameter locked on the probe sample.

Note
The Fine Control Knob does not function when the Lock / Manual Switch Is in the "Locked" position.

10-2-3 Teslameter Resynchronization

If the Teslameter should go out of sync while ramping the magnet up (or down), it can easily be resynchronized by the following procedure.

Manual Resynchronization

1. Reposition the Lock / Manual Switch to "Manual".

Note
The "No Strobe" Signal will be on and both the "Lo" and "Hi" LEDs will be oscillating, indicating a search mode.

2. Read the Main Power Supply's Current Meter. Multiply meter's present current reading by 20 gauss (since there's ≈ 20 gauss / amp). Reset the Current Meter to this calculated value.

3. Slowly start Increasing (if ramping up) the Coarse and / or Fine Control Knob while monitoring the "No Strobe Signal" LED.

4. Once the "No Strobe Signal" LED light extinguishes, quickly place the Lock / Manual Switch in the "Lock" position. The meter will now be synchronized.

Note
If the "Hi" LED is lit, the Coarse Control Knob will have to be turned towards the "Hi" LED until the LED goes out. Repeat this adjustment as required until the parking field Is reached.

10-2-3 Teslameter Resynchronization (continued)

Manual Resynchronization (with Oscilloscope)

An oscilloscope can be set up near the Teslameter to display and trigger on the “Field Modulation” signal from a jack on the Teslameter’s front panel. Adjust the time base to display one or two ramp waveforms. On the second channel, display the “NMR Signal” from the Teslameter’s front panel.

1. Leave the Teslameter’s Lock / Manual Switch in the “Lock” position.
2. Slowly turn the Coarse Control Knob in the direction the field is going (i.e., if ramping up, turn the knob towards higher numbers; if ramping down, turn the knob towards lower numbers.)

Note

As the meter is approaching the actual field, the baseline of the “FID” display will start to wander. Once the meter is in range of the field, the “FID” will appear on the scope trace as the meter locks on.

3. When locked on, the “No Strobe Signal” LED will be out. Slightly readjust the Coarse Control Knob in the direction of the lighted “Lo” or “Hi” LED until that LED goes out.
4. Maintain tracking through end of ramp sequence.

SECTION 11 - RAMPING

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

11-1 INTRODUCTION

This procedure utilizes voltage control for ramping and current control for parking. Approximately 13.5 – 14 hours of ramp time are required to reach the 3T field.

WARNING!

THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RAMPING THE MAGNET:

- **HELIUM GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLORLESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:**
  - **A. MAKE SURE THE MAGNET ROOM DOOR IS PROPPED OPEN,**
  - **B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND**
  - **C. REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, INCLUDED WITH THIS MANUAL.**

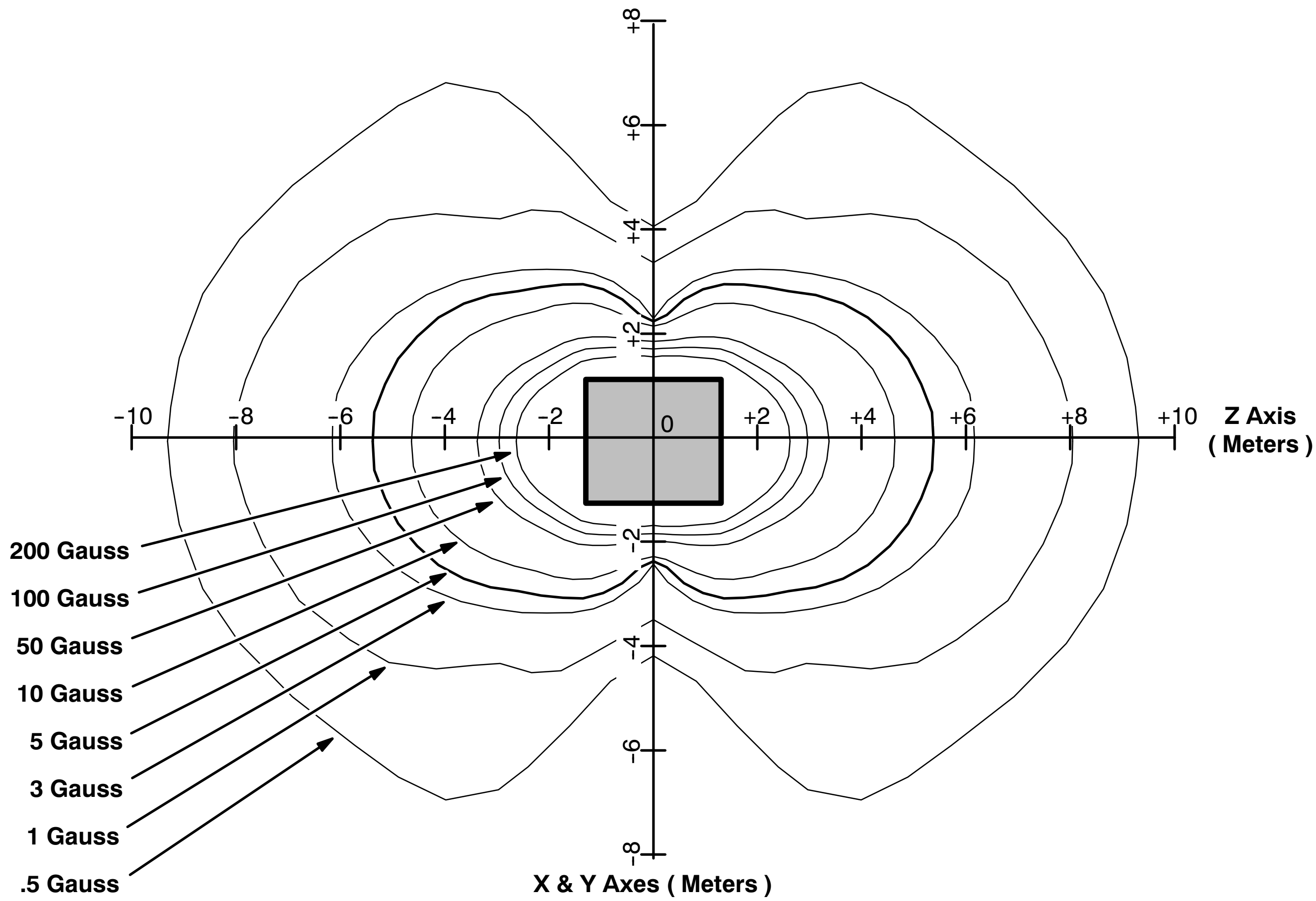
11-1 INTRODUCTION (continued)

WARNING!

(CONTINUED)

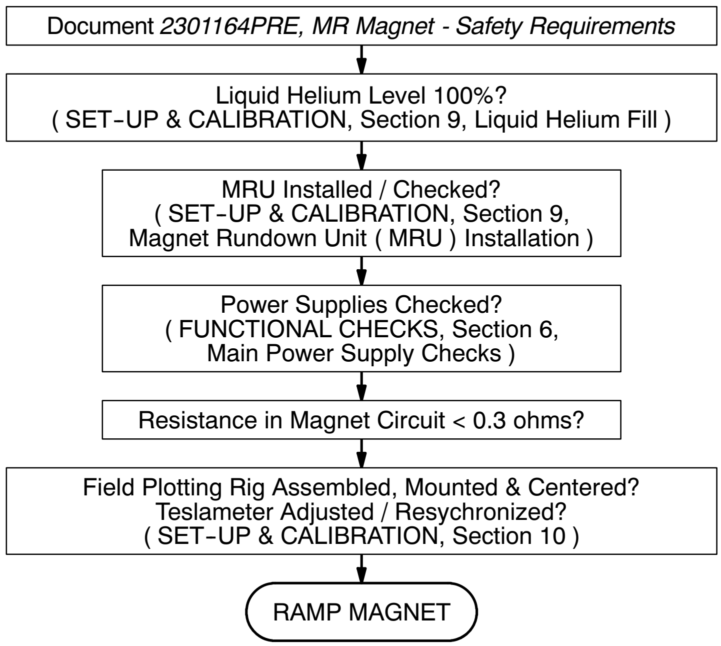
- **NOTIFY SITE ADMINISTRATION BEFORE RAMPING THE MAGNET THAT ALL MAGNETIC SAFETY PRECAUTIONS MUST BE TAKEN.**
- **POST WARNING SIGNS (46-255326P1) OUTSIDE THE 5 GAUSS ZONE TO ALERT PERSONNEL WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS AND OTHER BIOSTIMULATION DEVICES NOT TO PROCEED INTO THE DESIGNATED AREA. POST THESE SIGNS ON THE MAGNET ROOM LEVEL AS WELL AS AREAS BELOW THE MAGNET TO WHICH THE 5 GAUSS ZONE EXTENDS. SEE ILLUSTRATION 11-1 AND *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **POST "SECURITY ZONE" (46-255325P1) AND "AUTHORIZED PERSONNEL ONLY" (2289812) SIGNS AT THE MAGNET ROOM ENTRANCE AND MR SUITE ENTRANCE, RESPECTIVELY, PRIOR TO RAMPING THE MAGNET. THESE WARNINGS ARE TO ALERT PERSONNEL THAT NO FERROMAGNETIC MATERIAL OR INDIVIDUALS WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES ARE ALLOWED IN THE MAGNET ROOM WHEN THE MAGNET IS RAMPED. SEE *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. PULL THE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND SITE GEOMETRY ALLOW. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.**
- **MAKE SURE THE MAGNET RUNDOWN UNIT (MRU) IS INSTALLED AND OPERATIONAL TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE SET-UP AND CALIBRATION, SECTION 4.**
- **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT A PROPERLY FUNCTIONING OXYGEN MONITOR IS USED IN THE MAGNET AND/OR CRYOGEN STORAGE ROOM(S) BY SERVICE PERSONNEL.**
- **MAKE SURE THE MAGNET IS 100% (1200 MM) FULL OF LIQUID HELIUM TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING DURING RAMPING TO A POINT WHERE A QUENCH MAY OCCUR.**

11-1 INTRODUCTION (continued)



GEMSO 3T/940 FRINGE FIELD
ILLUSTRATION 11-1

11-1 INTRODUCTION (continued)



RAMPING FLOW DIAGRAM
ILLUSTRATION 11-2

11-2 TOOLS / EQUIPMENT

- Ramp / Energize Kit (2296056)
- Shimming Kit (2297304)
- Shim Power Supply (46-260777G3)
- Ramp Power Supply. The 1000 amp ESS7.5-1000-2-D-1236 (46-260776G4) is recommended because of its reduced ripple.
- Shim Cable Kit (2135558)
- Ramp Cable Kit (2135435)
- Metrolab Teslameter Kit (46-251865G4)
- No. 6 MetroLab Field Probe (2295525-5)
- Shim Adapter Box (2320885)
- Main Switch Adapter Box (2320946)

11-3 PREPARATION

1. Top off the magnet in conformance with REPLACEMENT / MAINTENANCE, Section 3, to bring the level of liquid helium inside the magnet to 100% (1200 mm).

Note

The helium level will drop during ramping, but must not fall below 90% (1040 mm) on "Probe 1." An initial top off decreases the chances of having to park the magnet and top off later.

2. Make sure the Magnet Rundown Unit (MRU) is installed and fully operational in accordance with SET-UP AND CALIBRATION, Section 4, Magnet Rundown Unit (MRU) Installation, and FUNCTIONAL CHECKS, Section 4, Magnet Rundown Unit (MRU) Checks.

3. Position the power supplies outside the magnet's 5 gauss zone (extending from the magnet's center approximately 11 feet or 3.3 meters on either side of the magnet and 19.5 feet or 6 meters at either end). See Illustration 11-1.

IMPORTANT !!!

DO NOT BLOCK the magnet room entrance or any other passageway.

4. Perform the power supply set-up checks detailed in FUNCTIONAL CHECKS, Section 6-2-4.

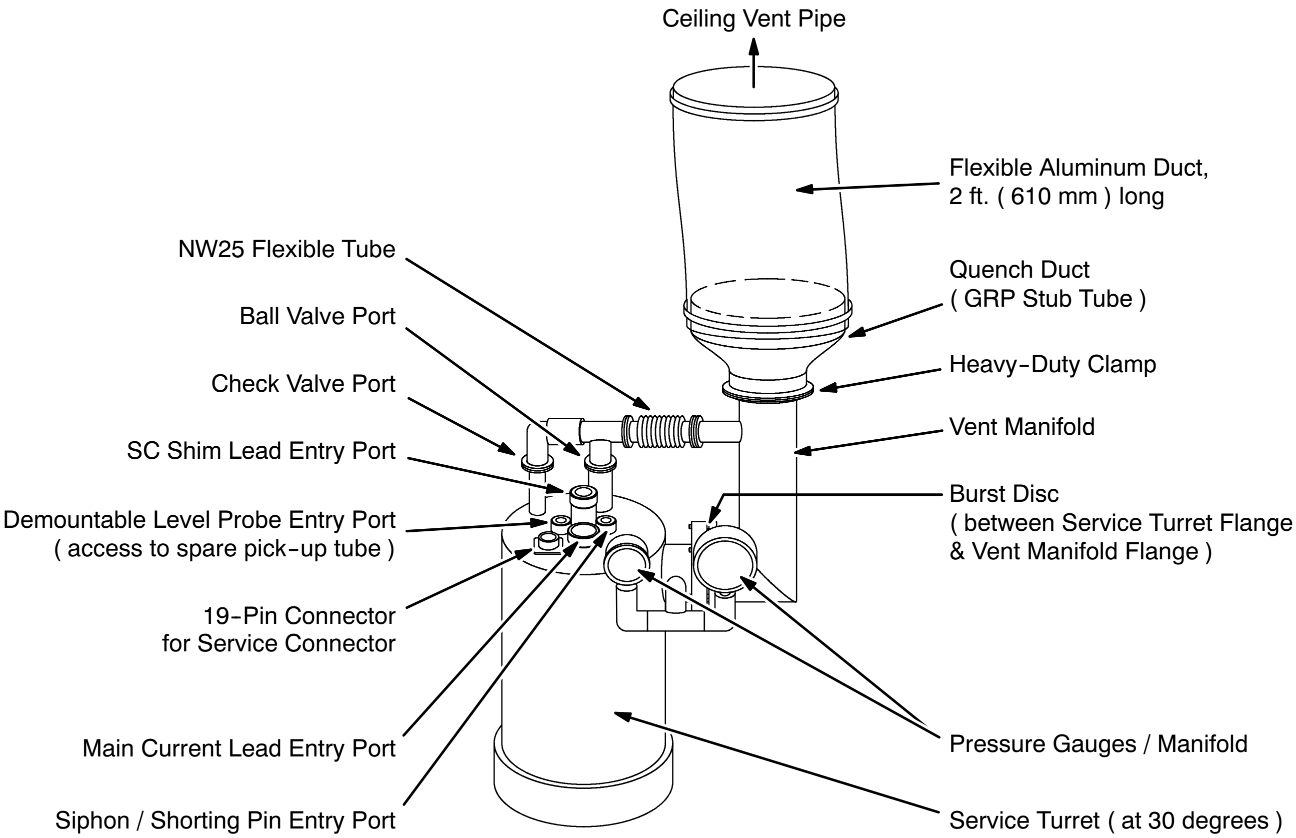
5. Assemble, mount and center the Field Plotting Rig (2292542) in conformance with SET-UP AND CALIBRATION, Section 10-1, Field Plotting Rig (2292542) Set-Up.

6. Open the Ball Valve Port to vent the magnet down to < 0.1 psig on the manifold gauge.

7. Close the Ball Valve and blank off the Check Valve Port.

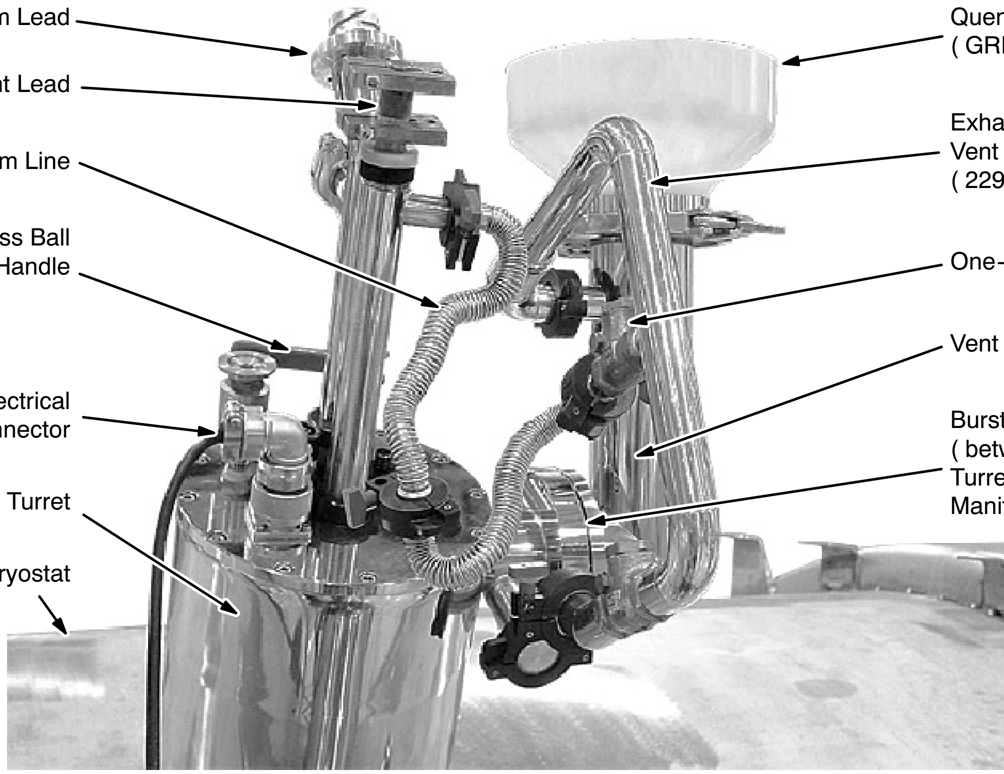
8. Remove the Pressure Gauge Manifold and install the Exhaust Gas Vent Assembly (2292536) in its place. See Illustration 11-3 for the factory-installed configuration and Illustration 11-4 for the Exhaust Gas Vent arrangement.

11-3 PREPARATION (continued)



TYPICAL FACTORY-INSTALLED 3T SERVICE TURRET PLUMBING
ILLUSTRATION 11-3

11-3 PREPARATION (continued)



EXHAUST GAS VENT CONNECTION
ILLUSTRATION 11-4

9. Test for ice / debris in the Service Turret by performing a trial fit of the Shorting Pin:
   a. Screw the Shorting Pin onto the Shorting Pin Insertion Tool (2255522).
   b. Remove the plug on the Siphon / Shorting Pin Entry. See Illustration 11-3.
   c. Slowly insert the Shorting Pin Insertion Tool into the Siphon / Shorting Pin Entry until the Shorting Pin makes contact with the Siphon Cone.
   d. Firmly push the Shorting Pin until it is seated in the cone.

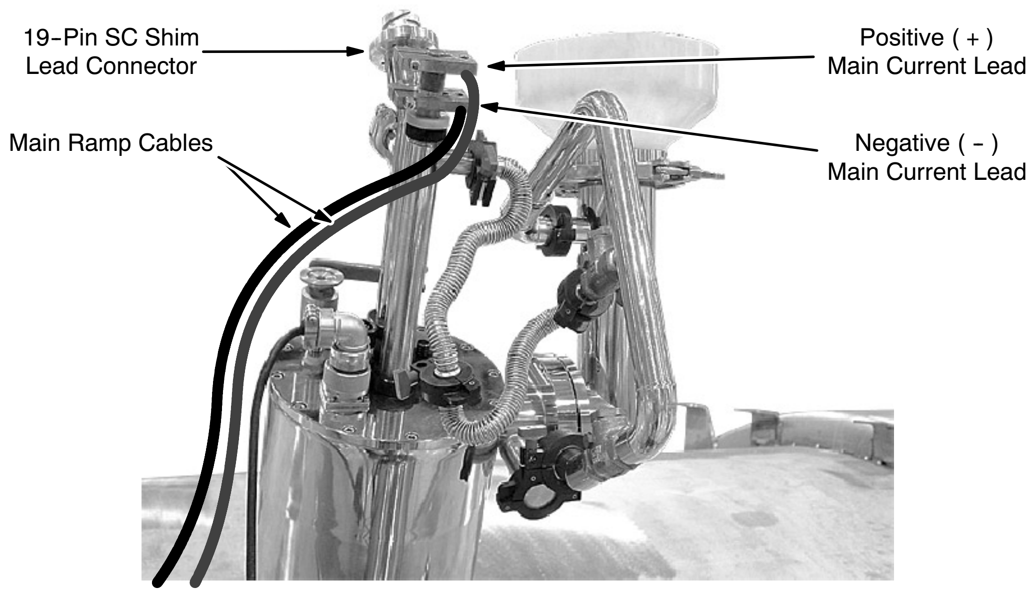
IMPORTANT !!!

If the Shorting Pin cannot be inserted and engaged, then remove ice / debris from the Service Turret in accordance with REPLACEMENT / MAINTENANCE, Section 6, before continuing to ramp the magnet.

   e. Slowly remove the Shorting Pin Insertion Tool with attached Shorting Pin from the Siphon / Shorting Pin Entry Port.
   f. Refit the plug covering the Siphon / Shorting Pin Entry.

11-3 PREPARATION (continued)

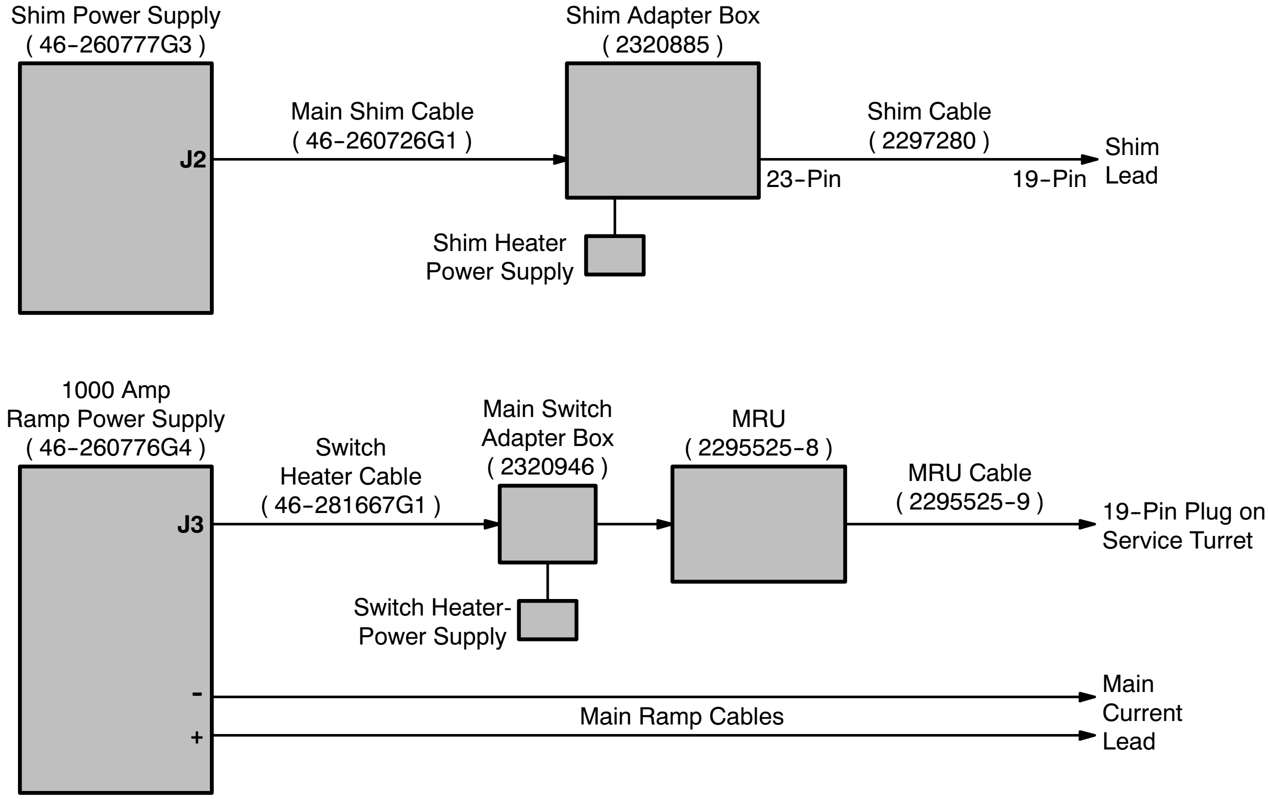
10. Connect the Main Ramp Cables to the copper lugs on the Main Current Lead. See Illustration 11-5. The positive current lead terminal is the center top block and has a M12 connection screw. The negative terminal is the lower block and has a M10 screw.



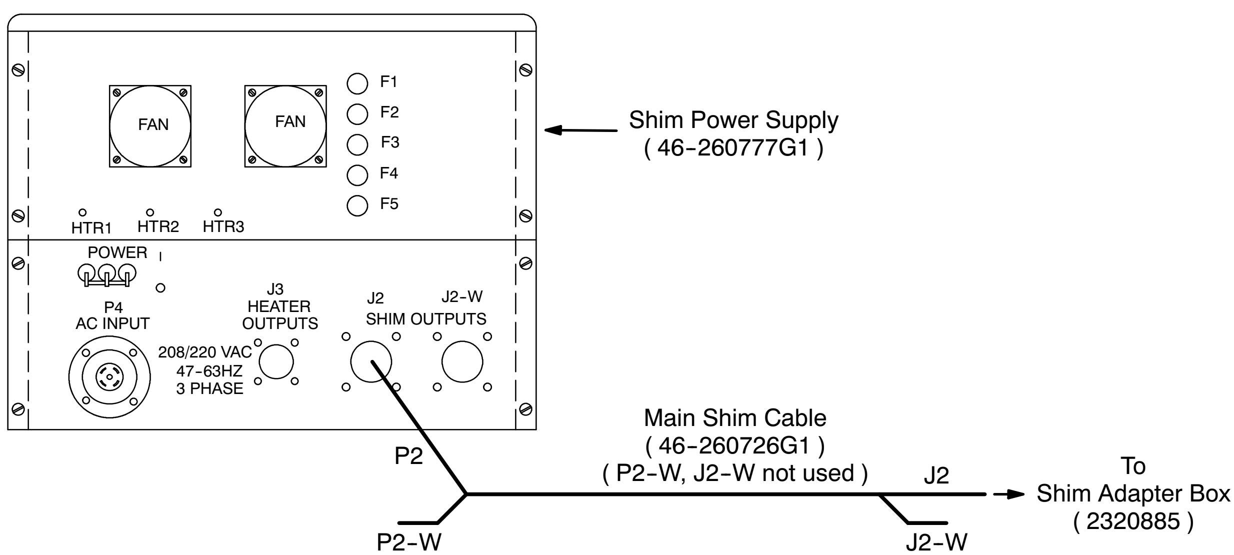
SHIM AND CURRENT LEAD CONNECTIONS
ILLUSTRATION 11-5

11. Connect the Shim Heater Power Supply to the Shim Adapter Box (2320885). See Illustrations 11-6 and 11-8.
12. Connect the Main Shim Cable (46-260726G1) between Shim Power Supply socket J2 and the input socket on the Shim Adapter Box. See Illustrations 11-6, 11-7 and 11-8.
13. Connect Shim Cable 2297280 to the Shim Adapter Box's 23-pin output and then to the Shim Lead. See Illustrations 11-6 and 11-8.

11-3 PREPARATION (continued)

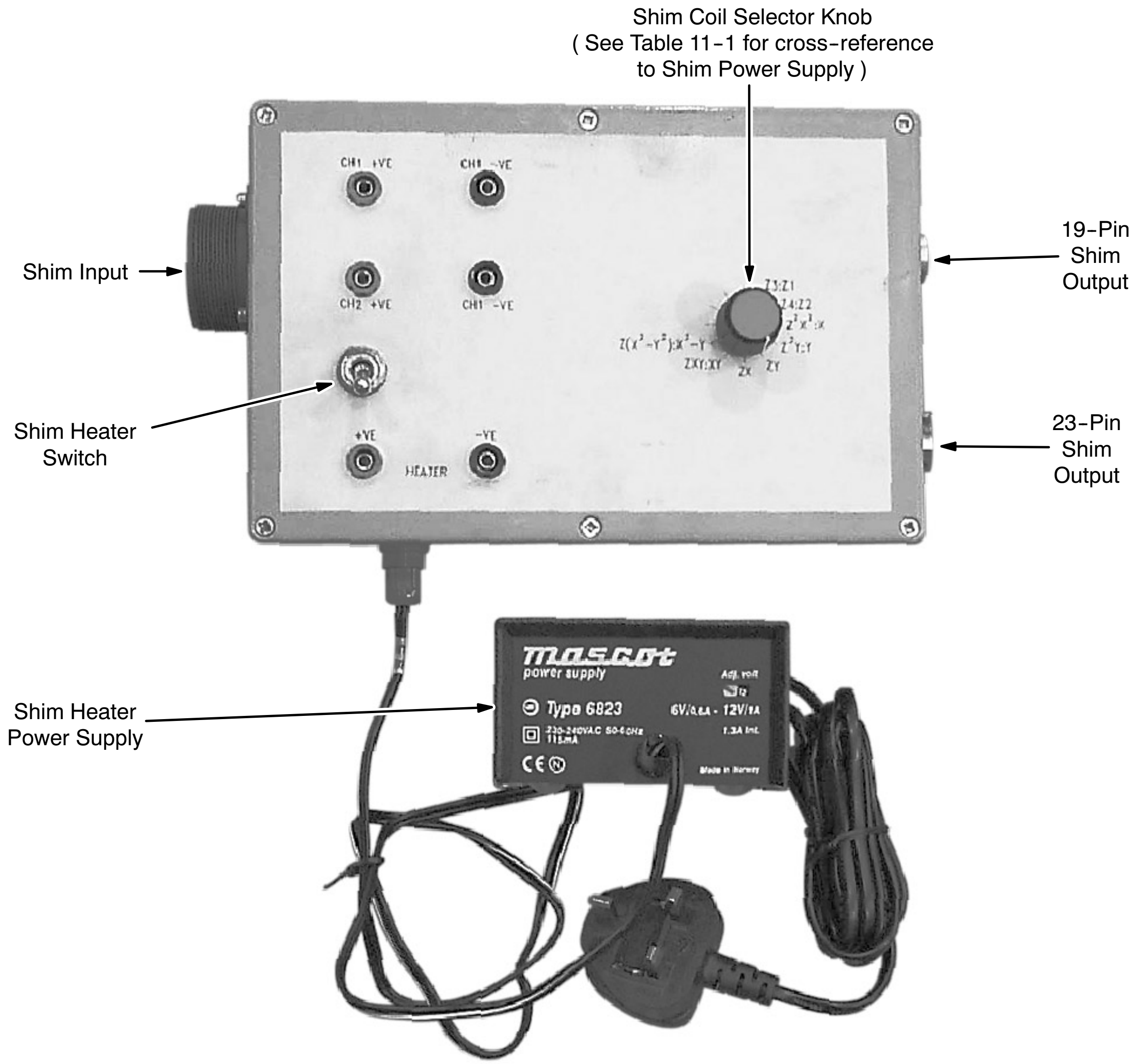


CABLE CONNECTIONS FOR RAMPING
ILLUSTRATION 11-6



SHIM POWER SUPPLY OUTPUT CONNECTIONS
ILLUSTRATION 11-7

11-3 PREPARATION (continued)



SHIM ADAPTER BOX (2320885)
ILLUSTRATION 11-8

TABLE 11-1
CROSS-REFERENCE, SHIM POWER SUPPLY CHANNELS TO SHIM ORDERS

Power Supply Channel (Shim Power Supply)	T1-1						T1-2							
Shim Order (Shim Adapter Box Label)	Z3	Z4	Z^2X	Z^2Y	ZXY	$Z(X^2-Y^2)$	Z1	Z2	X	Y	ZX	ZY	XY	X^2-Y^2

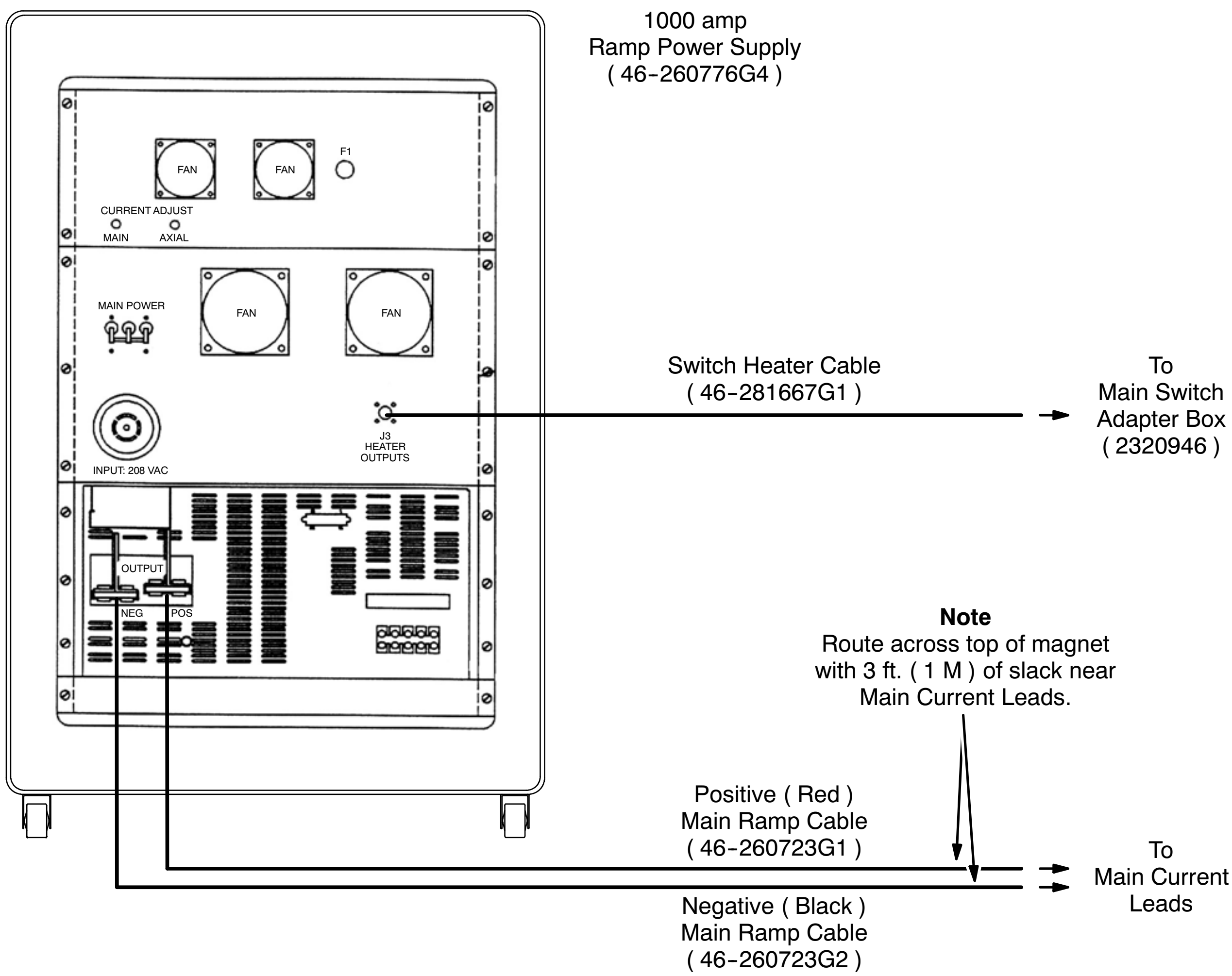
11-3 PREPARATION (continued)

14. Set the Main Axial Heater Current switch to “On” and the current to 825 mA through the dummy load in the Main Switch Adapter Box (2320946) to enable the power supply interlocks.

15. Connect the Switch Heater Cable (46-281667G1) to the Ramp Power Supply’s connector J3 and then to the Main Switch Adapter Box. See Illustrations 11-6, 11-9 and 11-10.

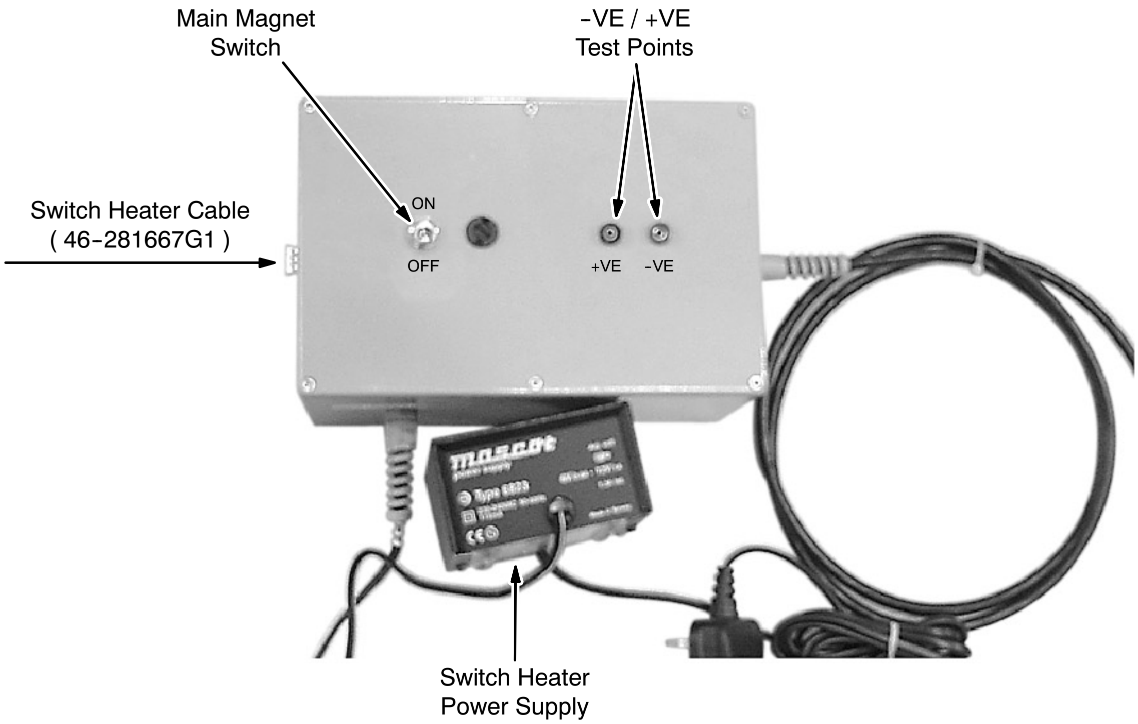
16. Connect a digital voltmeter (DVM), set to its volts range, to the test points on the Main Switch Adapter Box.

17. Plug in the Switch Heater Power Supply. Verify that the Main Magnet Switch on the Main Switch Adapter Box opens (on) and closes (off); leave the switch in the closed position. Unplug the Switch Heater Power Supply.



MAGNET POWER SUPPLY CONNECTIONS (1000 AMP)
ILLUSTRATION 11-9

11-3 PREPARATION (continued)



MAIN SWITCH ADAPTER BOX (2320946)
ILLUSTRATION 11-10

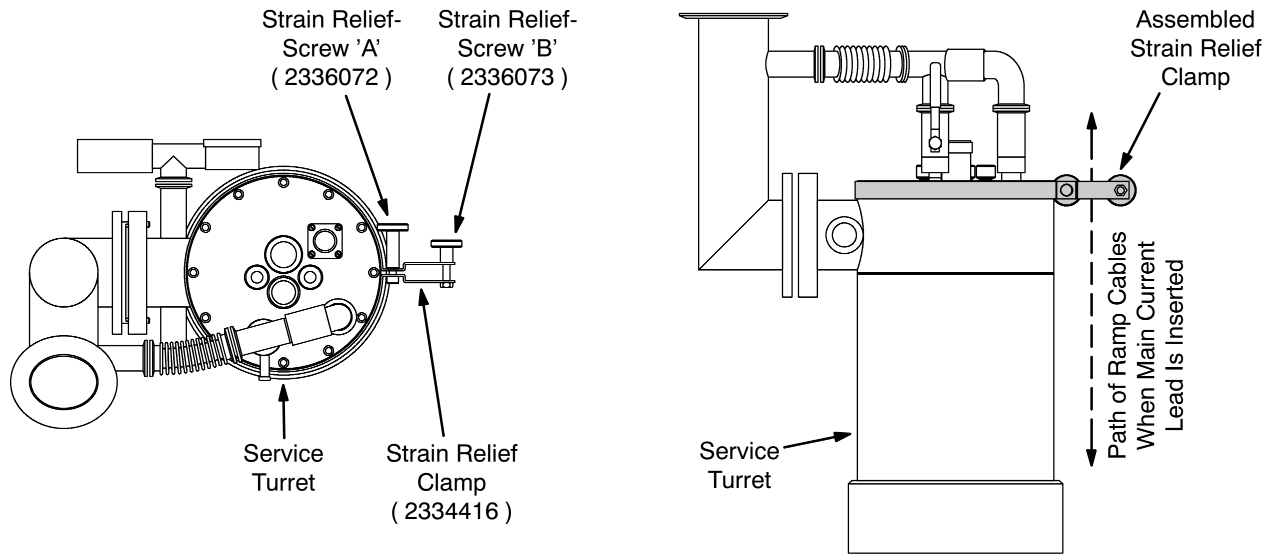
18. Connect the Main Switch Adapter Box to the MRU's 9-pin female connector labeled "PSU." See Illustrations 11-6 and 11-10.

Note

This arrangement gives access to the main magnet voltage and allows the magnet switch to be opened with the separate Switch Heater Power Supply.

19. Install the Ramp Cable Strain Relief Clamp (Clamp 22334416, Screw 'A' 2336072 and Screw 'B' 2336073) around the top edge of the Service Turret as shown in Illustration 11-11.

11-3 PREPARATION (continued)



Top View of Service Turret **Side View of Service Turret**

RAMP CABLE STRAIN RELIEF CLAMP
ILLUSTRATION 11-11

CAUTION

Do not leave the Main Current Lead Port open to the atmosphere for more than a few seconds since air will condense in the port, which will cause an ice blockage.

20. Remove the cap from the Main Current Entry Port, moving the nut, washers and o-ring to the Main Current Lead. See Illustration 11-3.

WARNING!

MAKE SURE THE MAIN CURRENT LEAD / SHIM LEAD EXHAUST PORTS ALWAYS FACE AWAY FROM THE SERVICE PLATFORM TO PREVENT COLD HELIUM GAS EXHAUSTING TOWARD THE SERVICE ENGINEER.

21. Insert the Main Current Lead into the Main Current Entry Port and push down gradually. Cold helium gas will emerge from the entry port. If the rush of gas becomes too great, hold the lead steady or retract it slightly until the rush decreases.

11-3 PREPARATION (continued)

22. Once contact is felt, push the Main Current Lead into full engagement with the male connector at the bottom of the entry port.

Note

Contact is necessary over the whole 2 inches (50 mm) of the male connector's pin to minimize contact resistance.

CAUTION

Do not force the Main Current Lead into full engagement.

23. Rotate the Main Current Lead gently to its fully engaged position.

IMPORTANT !!!

Do not allow the cables or exhaust port touch the Cryostat. Support the cables via the Ramp Cable Strain Relief Clamp shown in Illustration 11-11 to prevent the cables' weight from pulling on the Current Lead.

24. Tighten the nut to create a gas-tight seal.
25. Run the Ramp Cables through the Ramp Cable Strain Relief Clamp along the path shown in Illustration 11-11. Make sure the cables are not pulling on the Main Current Leads, then tighten the clamp.
26. Remove the cap from the SC Shim Lead Entry Port, moving the nut, washers and o-ring to the Shim Lead. See Illustration 11-3.

WARNING!

MAKE SURE THE MAIN CURRENT LEAD / SHIM LEAD EXHAUST PORTS ALWAYS FACE AWAY FROM THE SERVICE PLATFORM TO PREVENT COLD HELIUM GAS EXHAUSTING TOWARD THE SERVICE ENGINEER.

27. Insert the Shim Lead into the SC Shim Lead Entry Port and push down gradually. Cold helium gas will emerge from the entry port. If the rush of gas becomes too great, hold the lead steady or retract it slightly until the rush decreases.

11-3 PREPARATION (continued)

28. Once contact is felt, push the Shim Lead into full engagement.

Note

The Shim Lead is keyed, so it should only engage in one orientation: with the exhaust port facing directly away from the service platform. If it does not fit in the correct orientation, stop immediately; **DO NOT FORCE THE SHIM LEAD INTO ENGAGEMENT NOR ROTATE IT**. Instead, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715).

IMPORTANT !!!

If ice is encountered on the Shim Lead contacts, remove and warm the Shim Lead, then reinsert the warm lead to clear the ice.

29. Connect the 19-Pin Breakout Box (2295141) to the lead and measuring the resistances between pin "J" and the other pins as listed in Table 11-2. If all measurements are within the specified ranges, the Shim Lead has been engaged correctly. If any measurement is outside its expected range, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715).

TABLE 11-2
RESISTANCE CHECKING OF CORRECT SHIM LEAD INSERTION ORIENTATION

Pin Pair (at 19-Pin Breakout Box)	A – J	B – J	C – J	D – J	E – J	F – J	G – J	H – J
Expected Resistance Measurement (Ohms)*	200 ± 20	200 ± 20	200 ± 20	200 ± 20	100 ± 10	100 ± 10	200 ± 20	200 ± 20

30. Measure the resistance between the Main Current Lead's positive and negative terminals. Clean and reseat the Main Current Lead if resistance is not < 0.3 ohms.

31. Connect a NW16 vacuum line between the Main Current Lead exhaust and the one-way valve at the center of the Exhaust Gas Vent Assembly (2292536) to route exhaust gas safely out of the room through the quench duct. See Illustration 11-4.

32. Blank off the Shim Lead exhaust to make sure all exhaust gas is used to cool the Current Lead while energizing the magnet.

33. Connect the ramp cables to the power supply. Check the polarity.

34. Adjust / resynchronize the Teslameter in conformance with SET-UP AND CALIBRATION, Section 10-2.

11-4 RAMPING

TABLE 11-3
RAMPING SUMMARY

- Maximum Central Field: .... 3.0 Tesla
- Nominal Current for Maximum Central Field: .... 253.5 Amps
- Z2, Z4 and B0 Switches must be open during energization / de-energization
- Settle magnet once 186 amps is reached
- Overfield to 3.01 Tesla and persist with 0 Amps for 1 hour
- Re-persist at 3.0 Tesla
- Fit Shorting Pin after energization

TABLE 11-4
RAMP CURRENT DESIGNS

Magnet Serial Number	Ramp Current Design
Through 4402	250 Amp (Mk I)
4493, 4639	300 Amp (Mk II)
4492, 4494	300 Amp, then converted to 276 Amp
4694, 4775	276 Amp
Since 4776	250 Amp (Mk III)

11-4-1 Initial Ramp (to 186 Amps)

1. Make sure all current and voltage controls are set to zero (full CCW).
2. Connect input power to the Ramp Power Supply and then it turn on.
3. Turn on the Ramp Power Supply's Axial Heater Switch.

Note
The Axial Heater switch is shorted in the Adapter Cable / Box. However, the power supply interlocks require the Axial Heater switch be on.

11-4-1 Initial Ramp (to 186 Amps) (continued)

4. Perform resistance check:

   a. Set the current limit to 250 amps, then slowly adjust the voltage control to pass 250 amps through the Main Current Leads.

   b. Measure the voltage drop across the positive (+) (center top block) and negative (–) (lower block) terminals (copper lugs) on the Main Current Lead.

   Note
   The voltage drop should be ≤ 500 mV at 250 amps. If the voltage drop is > 500 mV at 250 amps, then remove, warm and reseat the Main Current Lead.

   c. Once voltage requirements are met, set all voltage and current controls to zero (full CCW) before proceeding.

5. Set the current limit to the Main Current Lead to 2 amps.

6. Turn the Shim Power Supply on.

7. Make sure all current potentiometers are set to zero (fully counterclockwise).

8. Select the “Z2:Z4” shim pair on the Shim Adapter Box. Plug in the Shim Heater Power Supply and turn on the Shim Heater Switch on the Shim Adapter Box. The Z2 and Z4 shims will now be held open while the magnet energizes.

9. Make sure the Main Magnet Switch on the Main Switch Adapter Box is set in the closed positon, then plug in the Switch Heater Power Supply. Turn on the Main Magnet Switch and monitor the Main Coil Voltage Indicator on the ramp supply for a positive voltage swing, which indicates the switch has opened.

10. Set the current limit to 5 amps and increase the current in the magnet at the energization rates given in Table 11-5.

11. When the current has reached 5 amps, allow the voltage to decay to ≈ 200 mV.

11-4-1 Initial Ramp (to 186 Amps) (continued)

WARNING!

THE MAXIMUM CURRENT LIMIT SETTING MUST NEVER EXCEED 275 AMPS TO PREVENT POTENTIAL MAIN COIL BURN-OUT.

TABLE 11-5
RECOMMENDED MAXIMUM ENERGIZATION RATES

Main Coil Current (Amps)	Current Limit (Amps)	Amps / Hour	Main Coil (Ramp) Voltage*	Duration
0.0 - 186.0	187	21.8	2.5	8.5 Hours
186.0 - 237.0	238	17.4	2.0	2.9 Hours
237.0 - 249.0	250	8.7	1.0	1.4 Hours
249.0 - (Value calculated in Section 11-4-2, Step 3b)	Value calculated in Section 11-4-2, Step 3b, but never > 275** Amps	4.4	0.5	1.0 Hours

* Adjust the Voltage on the Ramp Power Supply over the range given under "Main Coil Current (Amps)" in order to keep the corresponding "Main Coil (Ramp) Voltage" constant.
** Or the 5 Amps more than the Acceptance Test Report full field current amperage.

12. Switch on the B0 Coil at the MRU. The Main Coil Voltage Indicator should show an inductive voltage swing of 30 - 50 mV. If no swing is shown, turn off the B0 Coil, ramp the magnet to 10 amps and repeat. If no swing is shown this second time, persist (park) the magnet and contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715) before proceeding.

WARNING!

FAILURE TO HAVE THE B0 COIL ON DURING RAMP MAY RESULT IN A MAGNET QUENCH.

13. Switch on the B0 Coil and leave on throughout the remainder of the ramp.

14. Turn the Heater Switch back on and continue energizing the magnet at the ramp rates shown in Table 11-5. Make sure the current limit is reset for each current range to the correct "current limit" values.

11-4-1 Initial Ramp (to 186 Amps) (continued)

15. Check the helium level at 15 minute intervals. The Helium Level Monitor should remain > 900 mm on "Probe 1." If the reading drops below 900 mm:
    a. Record the Main Coil Current and park the magnet in conformance with Section 11-4-2.
    b. Top off the magnet with helium in accordance with REPLACEMENT / MAINTENANCE, Section 3, before continuing this ramp procedure.

11-4-2 "Parking" Overnight / 24 Hours

1. Once the current has reached 186 amps (approximately 2.2 Tesla), slowly turn the Current Control counterclockwise (CCW) until the Main Coil voltage just starts to decrease.
2. Adjust the Current Control as required to set and maintain 186 amps.

WARNING!

MAKE SURE THE CONNECTION POLARITY AND THE MAIN COIL CURRENT VALUE ARE RECORDED FOR FUTURE USE. THIS INFORMATION IS ESSENTIAL DURING FINAL RAMP. THE POWER SUPPLY MUST BE SET TO THE SAME POLARITY AND CURRENT TO AVOID A QUENCH WHEN TURNING ON THE MAIN MAGNET SWITCH.

3. Measure and record the following in DATA SHEETS, Section 7, Table 7-1, Magnet Ramping / Parking Current Log:
    a. Measure the total voltage drop across the current leads at the back of the power supply. Record this value under "Main Current Lead Voltage Drop (mV)".
    b. Record the stabilized Main Coil current from the power supply under "Stabilized Main Coil Current (A)". This value will be used to pre-set the power supply "current limit" in the final ramp phase. See Table 11-5.
    c. Record the stabilized magnet field reading from the Teslameter under "Stabilized Magnet Field (Gauss)".
    d. Calculate the magnet gauss / amp value from the recorded stabilized field and urrent readings. Record this value under "Gauss / Amp".

CAUTION

Make sure the Main Coil voltage is < 20 mV and the magnet's field is stable at the desired value before turning off the Main Magnet Switch to prevent the possibility of internal switch heating preventing the switch from going persistent.

4. Verify that the Main Coil voltage is stable at < 20 mV and that the magnet's field is stable at ≈ 2.2T, then turn off the Main Coil Switch on the Main Switch Adapter Box and allow the switch to cool for 5 minutes.

11-4-2 "Parking" Overnight / 24 Hours (continued)

CAUTION

Check during Step 5 that the Teslameter reading does not decrease. A decrease in field strength indicates the switch has NOT gone persistent; the Main Coil current will have to be slowly brought back to the recorded value and the park attempted again.

5. After the Main Magnet Switch has cooled for 5 minutes and the field gone persistent, slowly decrease (0.5 amps / second) the Current Control counterclockwise (CCW) to zero and then the Voltage Control (CCW) to zero.
6. Turn off the Shim Heater Switch and the input power to the Main Power Supply. Unplug the power cord.
7. Insert the Shorting Pin into the magnet:
   a. Screw the Shorting Pin onto the Shorting Pin Insertion Tool (2255522).
   b. Remove the plug on the Siphon / Shorting Pin Entry. See Illustration 3-1.

CAUTION

Be very careful not to contact the Main Lead Lugs in Step 7c. Little clearance is available.

   c. Slowly and carefully insert the Shorting Pin Insertion Tool into the Siphon / Shorting Pin Entry until the Shorting Pin makes contact with the Siphon Cone.
   d. Firmly push the Shorting Pin until it is seated in the cone.
   e. Slowly rotate the Shorting Pin Insertion Tool 10 turns counterclockwise (CCW) while keeping a firm downward pressure on the tool until it is separated from the pin.
   f. Remove the tool.
   g. Refit the plug covering the Siphon / Shorting Pin Entry.

11-4-2 "Parking" Overnight / 24 Hours (continued)

8. Blank off the Shim Lead exhaust.
9. Make sure the Ramp lead exhaust is connected to the Exhaust Gas Vent Assembly (2292536).
10. Make sure all seals are in place, warm, dry and tight.
11. Leave the B0 Coil Switch turned on during the overnight / 24 hour "park."

11-4-3 Final Ramp

1. After the magnet has settled overnight / 24 hours, top off the magnet in conformance with SET-UP AND CALIBRATION, Section 9-5, Liquid Helium (LHe) Fill.
2. Make sure the Current and Voltage controls are set to zero (full CCW), then connect the input power cords and turn on the Ramp and Shim Power supplies. Make sure the B0 Coil Switch is on.
3. Remove the Shorting Pin:
   a. Remove the plug from the Siphon / Shorting Pin Entry Port.

   CAUTION

   Be very careful not to contact the Main Lead Lugs in Step 3b. Little clearance is available.

   b. Slowly insert the Shorting Pin Insertion Tool until it contacts the Shorting Pin.
   c. Push the tool down and rotate it 10 turns clockwise (CW) to engage the Shorting Pin.
   d. Once the Shorting Pin is fully engaged, pull the Shorting Pin Insertion Tool out.
   e. Replug the Siphon / Shorting Pin Entry Port.
4. Reset the Main Coil Current in the power supply and external leads to the "Stabilized Main Coil Current (A)" value and polarity recorded in DATA SHEETS, Section 7, Table 7-1, Magnet Ramping / Parking Current Log, when the magnet was parked (in Section 11-4-2, Step 3b above).
5. Turn on the Shim Heater Switch on the Shim Adapter Box.
6. Rotate the Shim Coil Selector Knob on the Shim Adapter Box through all positions, pausing for 10 seconds at each position to quench any induced shim currents. Then select and leave the Shim Coil Selector Knob at "Z2:Z4" during the final ramp.

11-4-3 Final Ramp (continued)

WARNING!

MAKE SURE THE CONNECTION POLARITY AND THE MAIN COIL CURRENT VALUE ARE RECORDED FOR OVERNIGHT PARKING (SECTION 11-4-2, STEP 3B). THE POWER SUPPLY MUST BE SET TO THE SAME POLARITY AND CURRENT TO AVOID A QUENCH WHEN TURNING ON THE MAIN MAGNET SWITCH.

7. Verify that the Main Coil Current and polarity has been set to the overnight parking level, then turn on the Main Magnet Switch on the Main Switch Adapter Box. Wait 5 minutes for the switch to go resistive.

WARNING!

THE MAXIMUM CURRENT LIMIT SETTING MUST NEVER EXCEED 260 AMPS TO PREVENT POTENTIAL MAIN COIL BURN-OUT.

8. Adjust the Current Limit to 238.0 amps and continue energizing the magnet at a Main Coil Voltage of 2.0 volts, both as indicated in Table 11-5.

9. When the Main Coil Current reaches 237 amps, set the Current Limit to 250.0 amps and continue energizing the magnet at a Main Coil Voltage of 1.0 volts, as indicated in Table 11-5.

10. When the Main Coil Current reaches 249.0 amps, set the Current Limit to the value calculated according the formula below for 3.01 Tesla and continue energizing the magnet at a Main Coil Voltage of 0.5 volts, as indicated in Table 11-5.

$$\text{Amps at 3.01 Tesla} = \left(\begin{array}{c} \text{"Gauss / Amp" value calculated in Section 11-4-2,} \\ \text{Step 3d, and recorded in DATA SHEETS, Section 7,} \\ \text{Table 7-1, Magnet Ramping / Parking Current Log.} \end{array} \right) \times \text{ 30,100 Gauss}$$

Note

3.01 Tesla = 128.1534 MHz = approximately 254.34 amps.
3.00 Tesla = 127.7277 MHz = approximately 253.5 amps.
However, some magnets may require up to 273 amps ramp current for a full 3.0T field. Refer to Table 11-4 for the design current levels used on various magnets, by serial number.

11. When the field reaches 3.01T on the Teslameter, slowly turn the Current Control down (CCW) until the Main Coil Voltage just starts to decrease, then adjust the Current Control to maintain the field at a 3.01T steady state.

11-4-3 Final Ramp (continued)

WARNING!

MAKE SURE THE CONNECTION POLARITY AND THE MAIN COIL CURRENT VALUE ARE RECORDED FOR FUTURE USE. THIS INFORMATION IS ESSENTIAL FOR MAGNET FIELD ADJUSTUSTMENT AND / OR RAMP DOWN. THE POWER SUPPLY MUST BE SET TO THE SAME POLARITY AND CURRENT TO AVOID A QUENCH WHEN TURNING ON THE MAIN MAGNET SWITCH.

12. Measure and record the following in DATA SHEETS, Section 7, Table 7-1, Magnet Ramping / Parking Current Log:
    a. Measure the total voltage drop across the current leads at the back of the power supply. Record
    b. Record the total voltage drop across the current leads at the back of the power supply under "Main Current Lead Voltage Drop (mV)".
    c. Record the stabilized Main Coil Current from the power supply under "Stabilized Main Coil Current (A)".

CAUTION

Make sure the Main Coil voltage is < 20 mV and the magnet's field is stable at the desired value before turning off the Main Magnet Switch to prevent the possibility of internal switch heating preventing the switch from going persistent.

13. Verify that the Main Coil voltage is stable at < 20 mV and that the magnet's field is stable at 3.01T, then turn off the Main Magnet Switch on the Main Switch Adapter Box and allow the switch to cool for 5 minutes.

CAUTION

Check during Step 14 that the Teslameter reading does not decrease. A decrease in field strength indicates the switch has NOT gone persistent; the Main Coil current will have to be slowly brought back to the recorded value.

14. After the Main Magnet Switch has cooled for 5 minutes and the field has gone persistent, slowly decrease (0.5 amps / second) the Current Control counterclockwise (CCW) to zero and then the Voltage Control (CCW) to zero.

11-4-3 Final Ramp (continued)

15. Allow the magnet to "settle" for 1 hour with the Main Magnet Switch in persistent mode.
16. After 1 hour, slowly increase the current in the power supply and leads to the Stabilized Main Coil Current value recorded in Step 12.
17. Once the current in the power supply and leads is at the Stabilized Main Coil Current value, turn on the Main Magnet Switch on the Main Switch Adapter Box and wait 5 minutes for the switch to go resistive. If the field begins drifting upwards during this period, reduce the Current Control (CCW) slightly until the field starts reducing.
18. Decrease the Current Control (CCW) at the rate of 4.4 amps / hour (73 mA / min.; 1.2 mA / sec.) until the magnet's field reads 3.0000 Tesla on the Teslameter.

WARNING!

MAKE SURE THE CONNECTION POLARITY AND THE MAIN COIL CURRENT VALUE ARE RECORDED FOR FUTURE USE. THIS INFORMATION IS ESSENTIAL FOR MAGNET FIELD ADJUSTUSTMENT AND / OR RAMP DOWN. THE POWER SUPPLY MUST BE SET TO THE SAME POLARITY AND CURRENT TO AVOID A QUENCH WHEN TURNING ON THE MAIN MAGNET SWITCH.

19. Measure and record the following in DATA SHEETS, Section 7, Table 7-1, Magnet Ramping / Parking Current Log:
    - Measure the total voltage drop across the current leads at the back of the power supply. Record
    - Record the total voltage drop across the current leads at the back of the power supply under "Main Current Lead Voltage Drop (mV)".
    - Record the stabilized Main Coil Current from the power supply under "Stabilized Main Coil Current (A)".

CAUTION

Make sure the Main Coil voltage is < 20 mV and the magnet's field is stable at the desired value before turning off the Main Magnet Switch to prevent the possibility of internal switch heating preventing the switch from going persistent.

20. Verify that the Main Coil voltage is stable at < 20 mV and that the magnet's field is stable at 3.00T, then turn off the Main Coil Switch on the Main Switch Adapter Box and allow the switch to cool for 5 minutes.
21. After the Main Magnet Switch has cooled for 5 minutes and the field gone persistent, slowly decrease (0.5 amps / second) the Current Control counterclockwise (CCW) to zero and then the Voltage Control (CCW) to zero.

11-4-3 Final Ramp (continued)

22. When the power supply current is zero, turn all Shim Heater Switches off. This closes the Z2 and Z4 shims.
23. Turn off and unplug input power to the power supplies.
24. Reinsert the Shorting Pin into the magnet:
    a. Screw the Shorting Pin onto the Shorting Pin Insertion Tool (2255522).
    b. Remove the plug on the Siphon / Shorting Pin Entry. See Illustration 3-1.

CAUTION

Be very careful not to contact the Main Lead Lugs in Step 24c. Little clearance is available.

    c. Slowly insert the Shorting Pin Insertion Tool into the Siphon / Shorting Pin Entry until the Shorting Pin makes contact with the Siphon Cone.
    d. Firmly push the Shorting Pin until it is seated in the cone.
    e. Slowly rotate the Shorting Pin Insertion Tool 10 turns counterclockwise while keeping a firm down pressure on the tool.
    f. Remove the tool.
    g. Refit the plug covering the Siphon / Shorting Pin Entry.
25. Twelve hours after completing the magnet ramp process, blank off the Main Current Lead Port and connect the Shim Lead to the center port of the Exhaust Gas Vent Assembly (2292536).
26. Leave the B0 Coil Switch on for 72 hours after energizing the magnet.

Note
Wait at least 12 hours between ramping and shimming the magnet.

SECTION 12 – SHIMMING

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

12–1 INTRODUCTION

"Shimming" is the process of measuring the change in magnetic field over a fixed volume and then using the acquired data to improve the field homogeneity.

3T/940 Magnets ship with Passive Shim Trays fitted into the Room Temperature Shim Tube. These trays are loaded with shimming material according to a factory shim analysis. However, final magnet shimming must be done once the magnet is in position at the customer's site.

The magnet's field is plotted from the patient end which is always negative and reference plane 12. The magnet's service end is always positive and reference plane 1. The coordinate system used to map the 3T/940's magnetic field is shown in Illustration 12–1. The Shim Plotting Rig (2292542), covered in SET–UP AND CALIBRATION, Section 10, Preparation for Field Plotting, is rotated clockwise to the required field positions from the patient end of the magnet.

Fourteen orders of Superconducting (SC) Shim Coils are provided for on–site shimming: Z1, Z2, Z3, Z4, X, Y, ZX, ZY, XY, X^2–Y^2, Z^2X, Z^2Y, ZXY and $Z(X^2-Y^2)$.

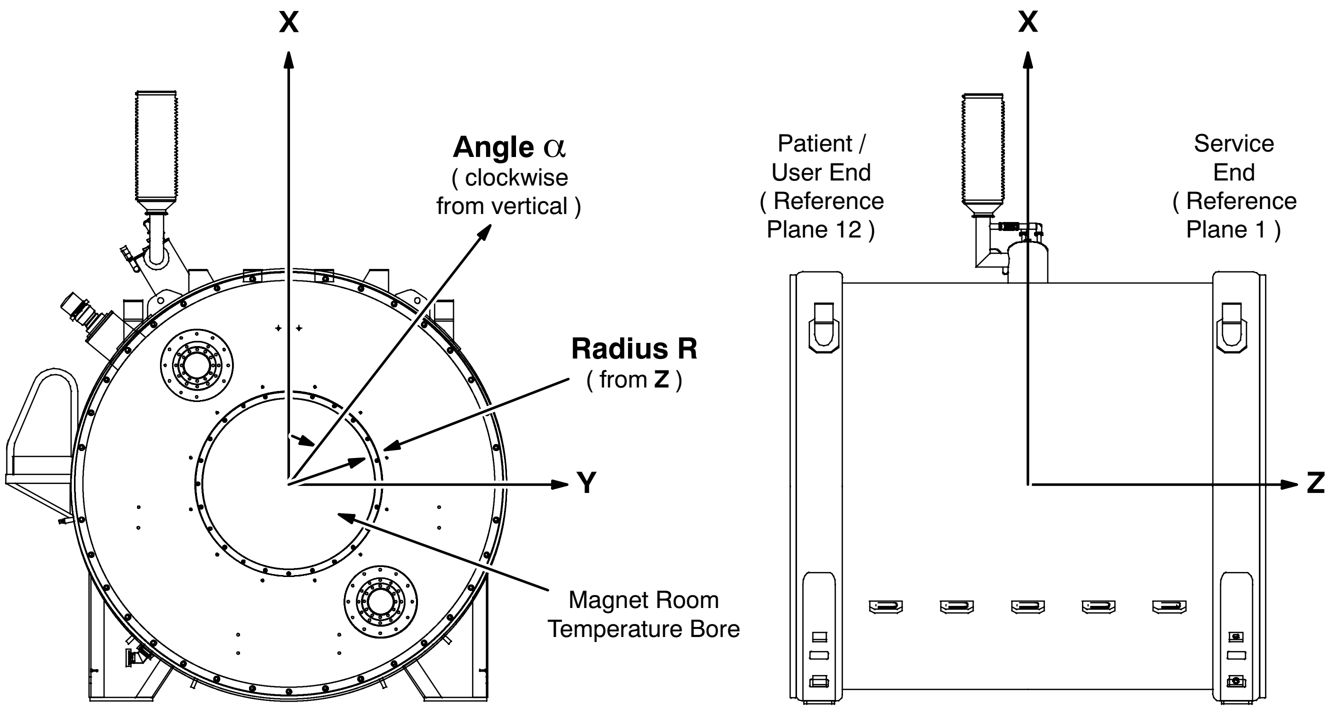
The goal in shimming is to maximize the homogeneity of the magnet's field. The specification is < 5 ppm (peak–to–peak) over a 45 cm DSV.

12-1 INTRODUCTION (continued)

Shimming may effect the magnet's field. On completion of shimming make sure the field is within the **2.9993 - 3.0007 Tesla** (30,000 ± 7 Gauss) range. Field adjustments may be required to meet this specification.

IMPORTANT !!!

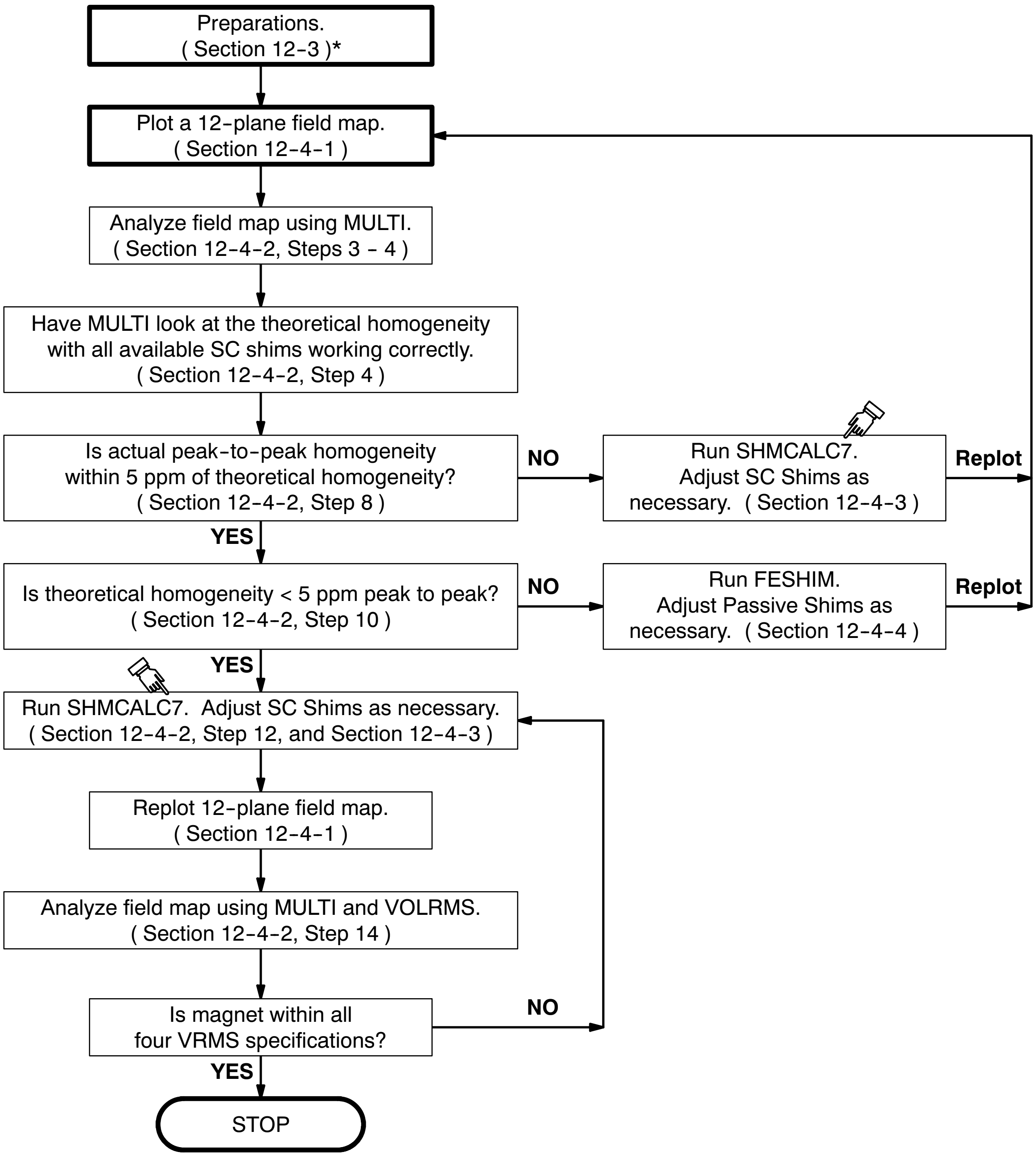
Make sure the dock motor is attached to the magnet before shimming. The ferromagnetic mass of the motor will effect shimming.



MAGNET COORDINATE SYSTEM
ILLUSTRATION 12-1

12-1 INTRODUCTION (continued)

An overview of the shimming procedure is shown in Illustration 12-2.



* Including set-up, mounting and centering of the Field Plotting Rig (2292542) in conformance with SET-UP AND CALIBRATION, Section 10-1, and adjustment / resynchronization of the Teslameter in conformance with SET-UP AND CALIBRATION, Section 10-2.

SHIM PROCEDURE OVERVIEW
ILLUSTRATION 12-2

12-2 EQUIPMENT

- Field Plotting Rig (2292542)
- MetroLab Teslameter (46-251865G4) or equivalent magnetic field measuring device
- No. 6 MetroLab Field Probe (2295525-5)
- Shimming Kit (2297304)
- Passive Shim Kit (2292493)
- Shim Power Supply (46-260777G3)
- Shim Adapter Box (2320885)
- Field analysis programs MULTI.EXE, FESHIM.EXE and VOLRMS.EXE; the spreadsheet file SHMCALC7.XLS; and the configuration file (3t940.fig). These are available online at "http://medfr-quickplace01.ge.com/quickplace/gemse_shimprog/Main.nsf?OpenDatabase" under "Library." If any of these files are not available on the web, contact GE Medical Systems Oxford at 8-011-44-1235-5490-00 before proceeding.
- Factory shimming information: For Oxford-shipped magnets, download the existing shim polarity information for the particular magnet serial number from "http://medfrquickplace01.ge.com/quickplace/gemse_shimprog/Main.nsf?OpenDatabase" under "Shim Settings." For Florence-shipped magnets, the same information is contained in the Acceptance Test Report (ATR) shipped with the magnet.

Note

If the shimming information for an Oxford-shipped magnet is missing from the web, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715) before proceeding.

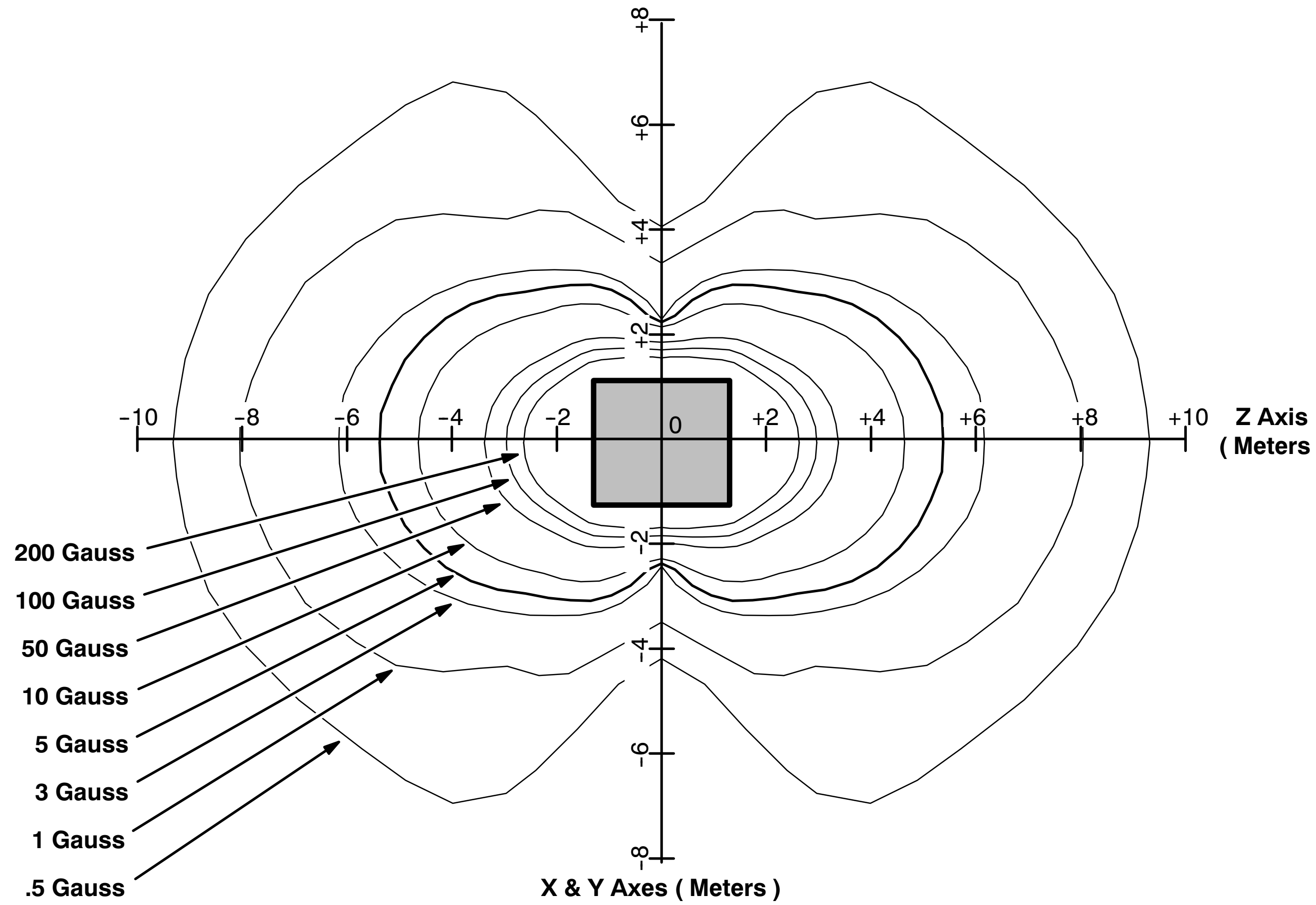
12-3 PREPARATION

1. Make sure the liquid helium level inside the magnet is ≥ 90% (1040 mm). Top off the magnet in conformance with REPLACEMENT / MAINTENANCE, Section 3, if not.
2. Position the power supplies outside the magnet's 5 gauss zone (extending from the magnet's center approximately 11 feet or 3.3 meters on either side, above and below the magnet and 19.5 feet or 6 meters at either end). See Illustration 12-3.

IMPORTANT !!!

DO NOT BLOCK the magnet room entrance or any other passageway.

12-3 PREPARATION (continued)



GEMSO 3T/940 FRINGE FIELD
ILLUSTRATION 12-3

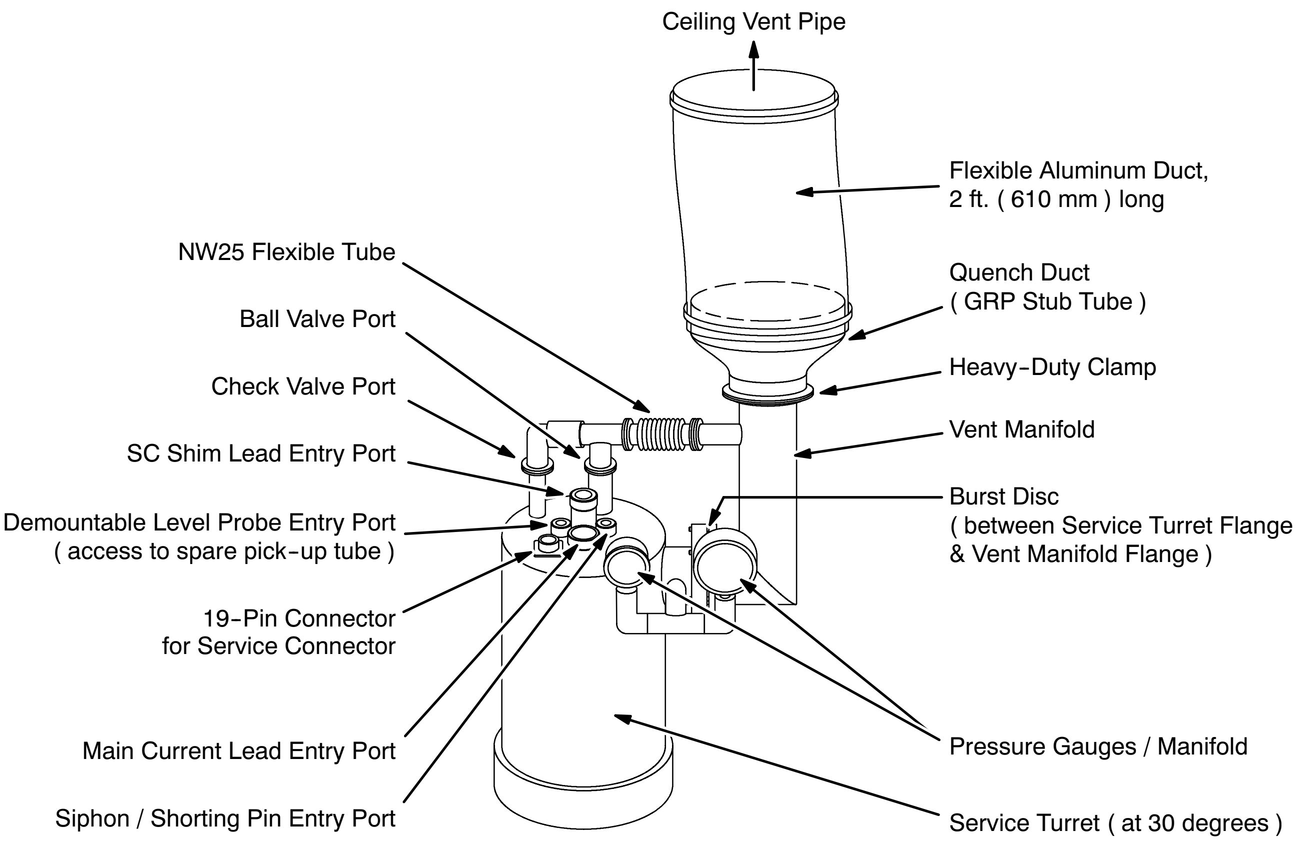
IMPORTANT !!!

If Shimming quickly follows Ramping, perform Step 3 and then skip Steps 4 - 16. However, if the magnet has been persisted for several days and both the Ramp and Shim Leads have been removed — making the vent arrangement shown in Illustration 12-4 — and Shim Cable 2297280 is not connected, then skip Step 3 and perform Steps 4 - 16.

3. If Shimming quickly follows Ramping (i.e., the magnet has not been persisted for several days, the Ramp and Shim Leads are engaged and Shim Cable 2297280 is connected):
   a. Remove the blanking plate from the Shim Lead exhaust.
   b. Disconnect the NW16 vacuum line from the Ramp Lead.
   c. Blank off the Ramp Lead.
   d. Continue at Step 17.

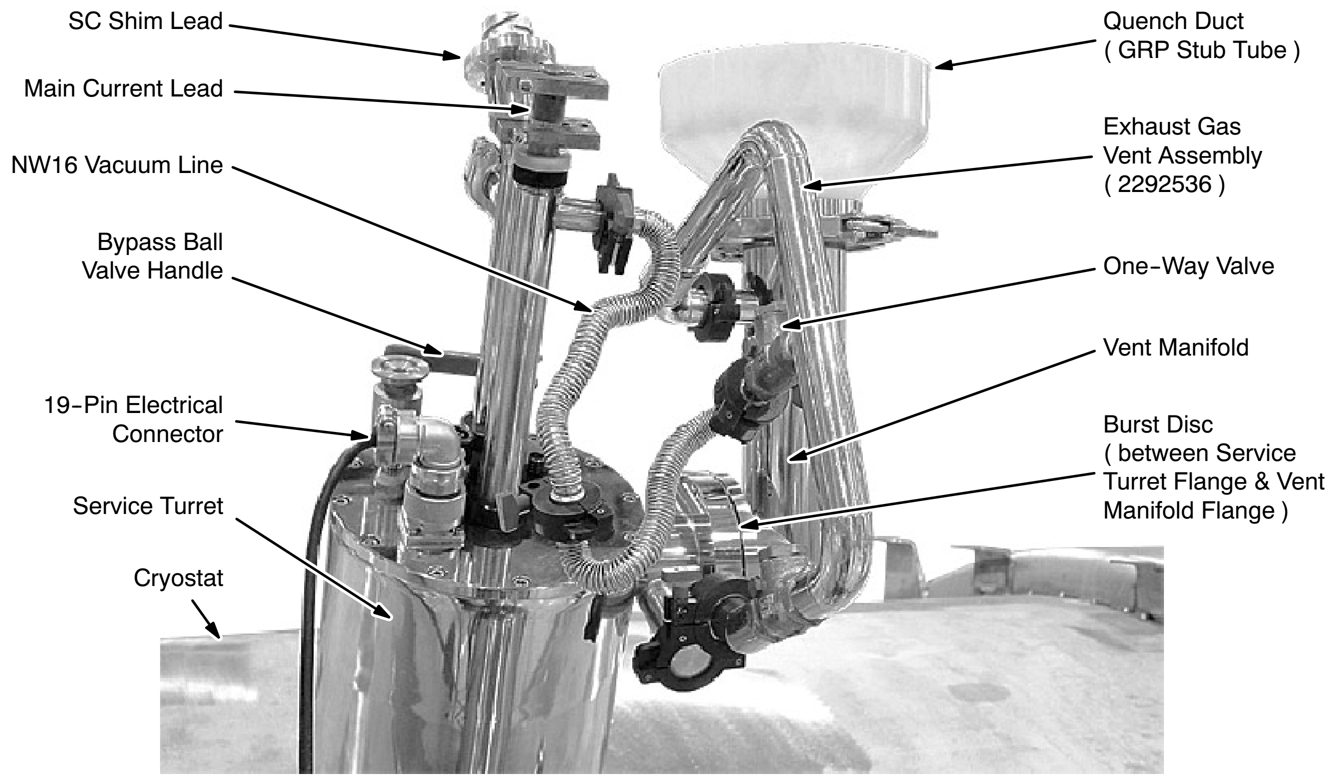
12-3 PREPARATION (continued)

4. Assemble, mount and center the Field Plotting Rig (2292542) in conformance with SET-UP AND CALIBRATION, Section 10-1, Field Plotting Rig (2292542) Set-Up.
5. Open the Ball Valve Port to vent the magnet down to < 0.1 psig on the manifold gauge.
6. Close the ball valve and blank off the check valve.
7. Remove the Pressure Gauge Manifold and install the Exhaust Gas Vent Assembly (2292536) in its place. See Illustration 12-4 for the factory-installed configuration and Illustration 12-5 for the Exhaust Gas Vent arrangement.



TYPICAL FACTORY-INSTALLED 3T SERVICE TURRET PLUMBING
ILLUSTRATION 12-4

12-3 PREPARATION (continued)

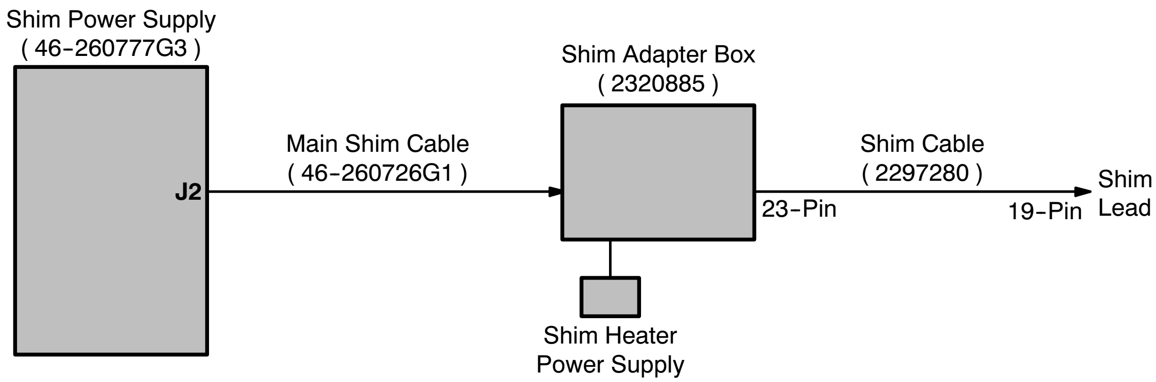


EXHAUST GAS VENT CONNECTION
ILLUSTRATION 12-5

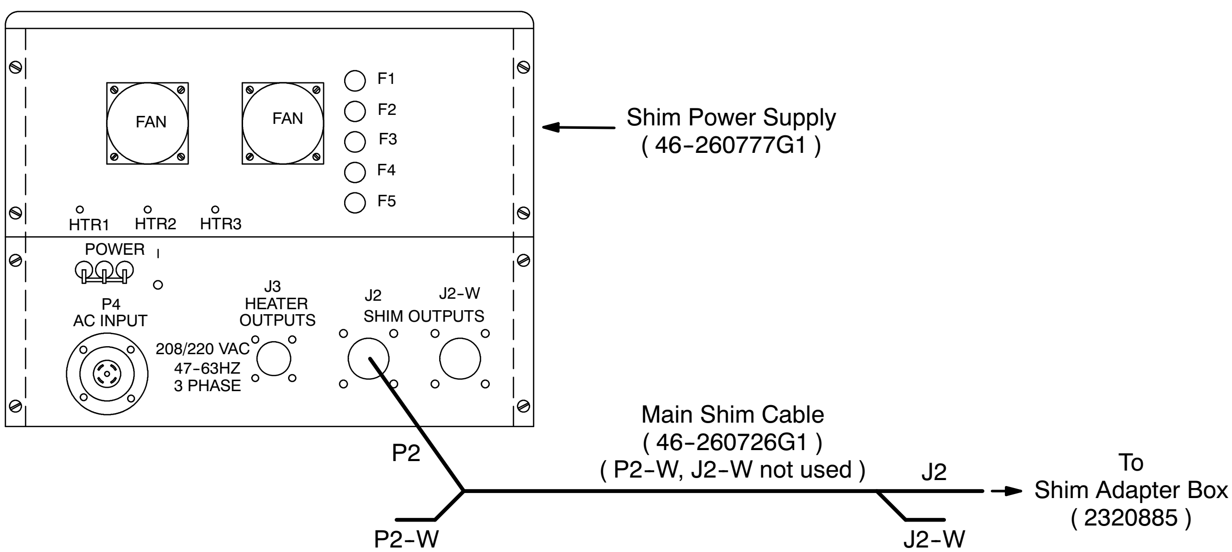
8. Connect the Main Shim Cable (46-260726G1) between Shim Power Supply socket J2 and the input socket on the Shim Adapter Box (2320885). See Illustrations 12-6, 12-7 and 12-8.

9. Connect Shim Cable 2297280 to the Shim Adapter Box's 23-pin output. See Illustrations 12-6 and 12-8.

12-3 PREPARATION (continued)

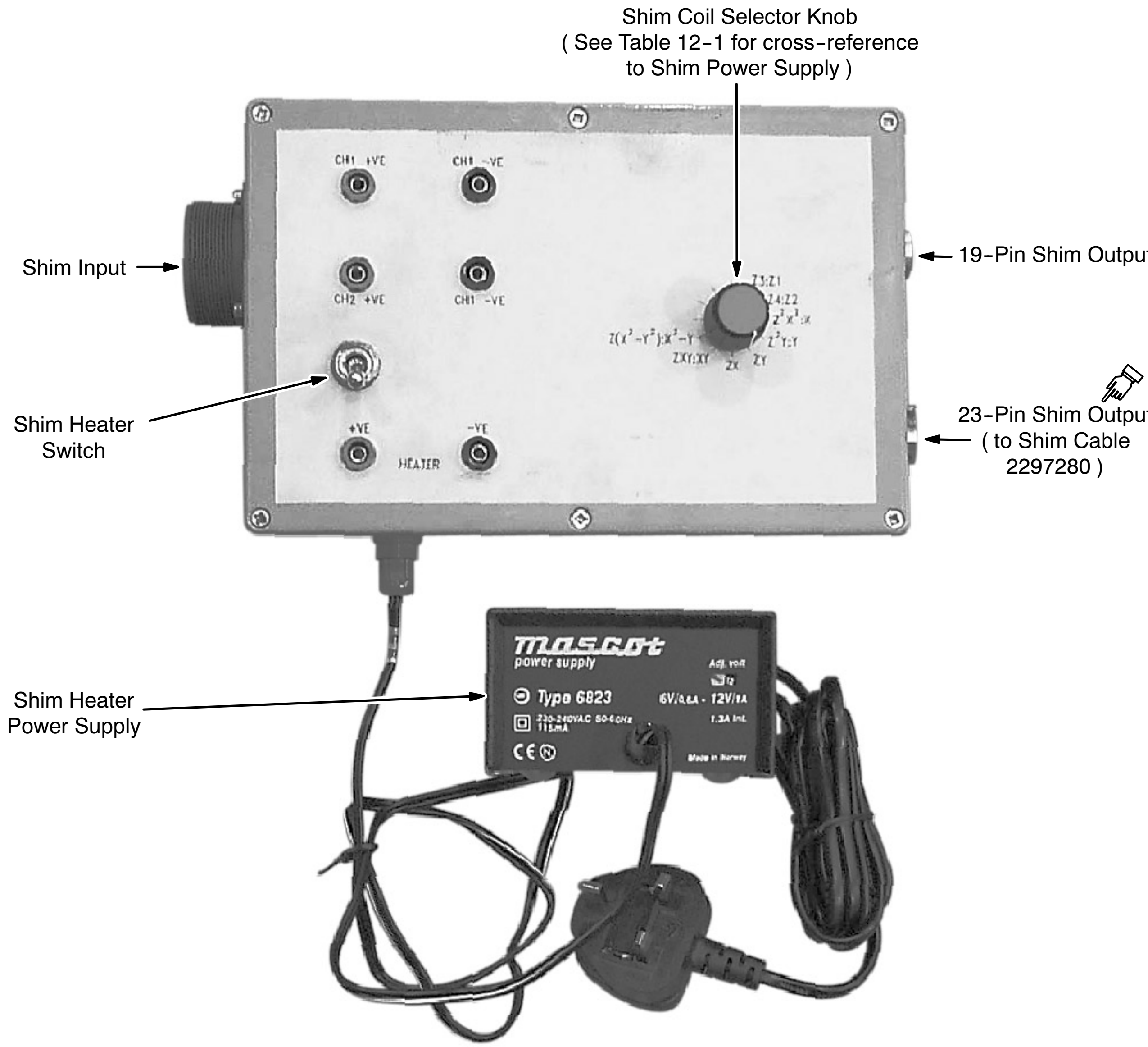


CABLE CONNECTIONS FOR SHIMMING
ILLUSTRATION 12-6



SHIM POWER SUPPLY OUTPUT CONNECTIONS
ILLUSTRATION 12-7

12-3 PREPARATION (continued)



SHIM ADAPTER BOX (2320885)
ILLUSTRATION 12-8

10. Turn on the Shim Power Supply.

11. Make sure all current potentiometers on the Shim Power Supply are set to zero (fully counterclockwise).

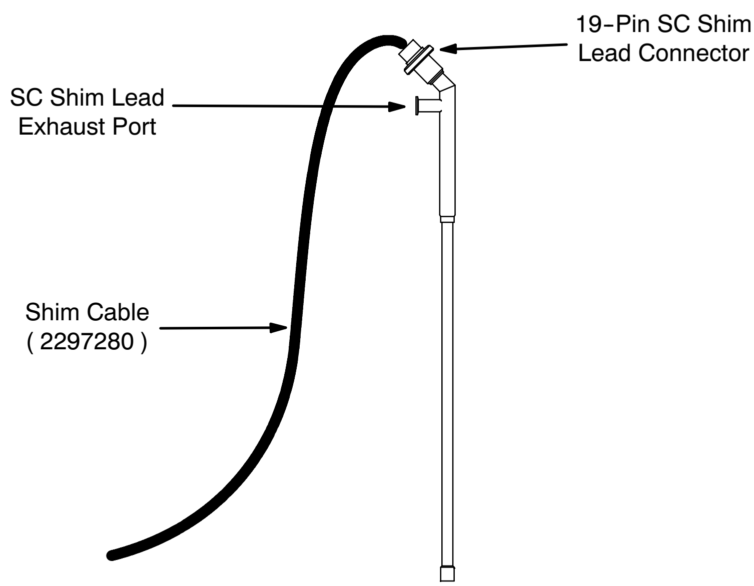
12. Attach the other end of the Shim Cable to the Shim Lead. See Illustration 12-9

12-3 PREPARATION (continued)

CAUTION

Do not leave the SC Shim Lead Entry Port open to the atmosphere for more than a few seconds since air will condense in the port, which will cause an ice blockage.

13. Remove the cap from the SC Shim Lead Entry Port, moving the nut, washers and o-ring to the Main Current Lead. See Illustration 12-4.



SC SHIM LEAD CONNECTIONS
ILLUSTRATION 12-9

12-3 PREPARATION (continued)

WARNING!

THE MAGNET MAY QUENCH IF A WARM SHIM LEAD IS NOT ALLOWED TO COOL SUFFICIENTLY DURING INSERTION.

WARNING!

MAKE SURE THE SHIM LEAD EXHAUST PORT ALWAYS FACES AWAY FROM THE SERVICE PLATFORM TO PREVENT COLD HELIUM GAS EXHAUSTING TOWARD THE SERVICE ENGINEER.

14. Insert the Shim Lead into the SC Shim Lead Entry Port with the exhaust port facing away from the service platform. Push down gradually, allowing the lead to cool. Cold helium gas will emerge from the entry port. If the rush of gas becomes too great, hold the lead steady or retract it slightly until the rush decreases.

15. Once contact is felt, push the Shim Lead into full engagement with the male connector at the bottom of the entry port. If it does not fit in the correct orientation, stop immediately; **DO NOT FORCE THE SHIM LEAD INTO ENGAGEMENT NOR ROTATE IT**. Instead, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715).

16. Tighten the nut to create a gas-tight seal.

17. Connect a NW16 vacuum line between the Shim Lead exhaust and the one-way valve at the center of the Exhaust Gas Vent Assembly (2292536) to route exhaust gas safely out of the room. See Illustration 12-5.

18. Verify that the Shim Heater Switch on the Shim Adapter Box is “Off,” then plug in the Shim Heater Power Supply.

19. Select the “Z1:Z3” shim pair via the Shim Coil Selector Knob on the Shim Adapter Box, then switch on the Shim Heaters on the Shim Adapter Box.

20. Rotate the Shim Coil Selector Knob through all positions, pausing for 10 seconds at each, to dump the current in all shim pairs.

21. Turn off the Shim Heater Switch.

22. Adjust / resynchronize the Teslameter in conformance with SET-UP AND CALIBRATION, Section 10-2.

12-4 SHIMMING

12-4-1 Plotting the Magnetic Field

IMPORTANT !!!

DO NOT dump the shim currents (as in Section 12-3, Steps 19 - 21) when plotting the field after the first iteration. Shimming is an iterative process; dumping clears the process to its beginning.

1. Position the Field Plotting Rig so the Teslameter can map Reference Plane 1.

   Note

   DATA SHEETS, Section 8, Field Plot Data, gives the axial and radial probe positions for all reference planes when using a 22.5 cm plotting radius (i.e., using the 45 cm DSV Probe Scale in the Field Plotting rig) and instructions on calculating positions for other plotting radius values.

2. Record the shim settings and potentiometer settings for the field's map on DATA SHEETS, Section 8, Field Plot Data.
3. Map the plane from 0° clockwise around a full 360° in 30° increments: 0°, 30°, 60°, 90°, 120°, 150°, 180°, 210°, 240°, 270°, 300°, 330°, and 360°.
4. Record each measurement on DATA SHEETS, Section 8, Field Plot Data.
5. Reposition the rig to the next plane in this sequence: 1, 12, 2, 11, 3, 10, 4, 9, 5, 8, 6, 7.
6. Repeat Steps 3 - 5 until the map documented in DATA SHEETS, Section 8, Field Plot Data, is complete.
7. Fill in the remaining portions of DATA SHEETS, Section 8, Field Plot Data.
8. Check the data for errors and retake any measurement outside these criteria:
   - All measurements should be within 20KHz (469.6 microTesla) of each other.
   - No measurement should deviate from adjoining measurements (above, below or to either side) by more than 1 KHz (23.48 microTesla).

12-4-2 Field Map Analysis

1. Download the field analysis program files MULTI.EXE, FESHIM.EXE and VOLRMS.EXE; the spreadsheet file SHMCALC5.XLS; and the configuration file (3t940.fig) from "http://medfrquickplace01.ge.com/quickplace/gemse_shimprog/Main.nsf?OpenDatabase" under "Library." If any of these files are not available on the web, contact GE Medical Systems Oxford at 8-011-44-1235-5490-00 before proceeding.

12-4-2 Field Map Analysis (continued)

2. For Oxford-shipped magnets, download the existing shim polarity information for the particular magnet serial number from "http://medfrquickplace01.ge.com/quickplace/gemse_shimprog/Main.nsf?OpenDatabase" under "Shim Settings." For Florence-shipped magnets, the same information is contained in the Acceptance Test Report (ATR) shipped with the magnet.

Note

If the shimming information for an Oxford-shipped magnet is missing from the web, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715) before proceeding.

Note

Make sure all of these files are stored in the same directory on your PC.

3. Run MULTI.EXE and enter / input data. Follow Steps 3a - 3f when typing new data into MULTI. Follow Steps 3g - 3i when inputting data from an existing file.

   When entering new data into MULTI:

   a. To type field map data into MULTI, choose "3" (12 plane plot) at the main menu. See Illustration 12-10.

Note

Always press the Enter Key after typing menu choices or information into MULTI.

```
***********************************************************
*                                                         *
*                       MULTI                             *
*                                                         *
*         Multi-plane field analysis programme            *
*                                                         *
*         Version 2.4                   8-12-00           *
*                                                         *
*                MAGNEX SCIENTIFIC LTD                    *
*                                                         *
*                21  Blacklands Way                       *
*               Abingdon Business Park                    *
*         Abingdon    Oxfordshire OX14 1DY    ENGLAND     *
*     TEL : 01235 534488         FAX : 01235 554917       *
*                                                         *
***********************************************************

type 1 to input from file
type 2 to type in a 7 plane plot
type 3 to type in a 12 plane plot
type 0 for all other cases
3

input centre field in Tesla (or MHz)  3.0000
```

Main Menu

Base Center Field Strength (Tesla)

MULTI.EXE - CHOOSING TO ENTER DATA
ILLUSTRATION 12-10

12-4-2 Field Map Analysis (continued)

3. (continued)

Note

’Base center field strength (Tesla)’ is the same concept as ’base frequency (MHz)’ used on other GE magnets: the value entered MUST be common to ALL field map data samples. Enter all data in Tesla / microTesla.

b. Enter the <u>base center field strength</u> in <u>Tesla</u>. For the 3T940 this is “3.00xx”. See Illustration 12-10.

c. Enter the smallest unit of microTesla containing significant change in the field map data. If the least significant digit of the Teslameter readings recorded was “ones” (e.g., “1 microTesla”), enter “1” for “Field Unit.” If the least significant digit was “tenths” (e.g., “0.1 microTesla”), enter “0.1”. See Illustration 12-11.

Note

MULTI’s acceptable choices are “0.1”, “1”, “10” or “100”.

d. Enter the plotting radius used and recorded in DATA SHEETS, Section 8, Field Plot Data. Enter “22.5” when using the standard 22.5 cm plotting radius. See Illustration 12-11.

```
      *         Multi-plane field analysis programme          *
      *                                                       *
      *         Version 2.4                  8-12-00          *
      *                                                       *
      *                MAGNEX SCIENTIFIC LTD                  *
      *                                                       *
      *                21  Blacklands Way                     *
      *               Abingdon Business Park                  *
      *         Abingdon    Oxfordshire OX14 1DY    ENGLAND   *
      *     TEL : 01235 534488         FAX : 01235 554917     *
      *                                                       *
      *********************************************************

type 1 to input from file
type 2 to type in a 7 plane plot
type 3 to type in a 12 plane plot
type 0 for all other cases
3

input centre field in Tesla (or MHz)  3.0000

input field unit in micro Tesla (or Hz)  .1


input plotting radius  22.5
```

Least Significant Digit of Teslameter Data (microTesla) → input field unit in micro Tesla (or Hz)

Plotting Radius (cm) → input plotting radius

MULTI.EXE - PREPARATION FOR DATA ENTRY
ILLUSTRATION 12-11

12-4-2 Field Map Analysis (continued)

3. (continued)

   e. Enter the field values found in Section 12-4-1, Plotting the Magnetic Field, and recorded in DATA SHEETS, Section 8, Field Plot Data. These values must be in microTesla since MULTI requires the same units be used thoroughout.

Note

Illustration 12-12 shows sample data entered for “Plane 1.”

   f. Continue with Step 4.

```
input centre field in Tesla (or MHz)  3.0000

input field unit in micro Tesla (or Hz)  .1


input plotting radius  22.5



   PLANE  1

integer field value for   0 degrees   4843
integer field value for  30 degrees   4880
integer field value for  60 degrees   4922
integer field value for  90 degrees   4969
integer field value for 120 degrees   4999
integer field value for 150 degrees   5016
integer field value for 180 degrees   5011
integer field value for 210 degrees   4986
integer field value for 240 degrees   4937
integer field value for 270 degrees   4888
integer field value for 300 degrees   4852
integer field value for 330 degrees   4841
integer field value for 360 degrees   4848
```

MULTI.EXE - DATA ENTRY, PLANE 1 OF 12
ILLUSTRATION 12-12

12-4-2 Field Map Analysis (continued)

3. (continued)

When inputting data from an existing file:

g. To input existing field map from a file, choose "1" (input from file) at MULTI's main menu. See Illustration 12-13.

h. Type the exact DOS path to the file. The example path in Illustration 12-13 will input the data stored in "`analysia.dat`" located in the root directory of drive "C".

i. Continue with Step 4.

```
      **********************************************************
      *                                                        *
      *                    MULTI                               *
      *                                                        *
      *          Multi-plane field analysis programme          *
      *                                                        *
      *          Version 2.4                  8-12-00          *
      *                                                        *
      *                 MAGNEX SCIENTIFIC LTD                  *
      *                                                        *
      *                 21  Blacklands Way                     *
      *                Abingdon Business Park                  *
      *          Abingdon    Oxfordshire OX14 1DY    ENGLAND   *
      *      TEL : 01235 534488         FAX : 01235 554917     *
      *                                                        *
      **********************************************************

type 1 to input from file
type 2 to type in a 7 plane plot
type 3 to type in a 12 plane plot
type 0 for all other cases
1

input file name
c:\analysia.dat_
```

Main Menu

DOS Path to File

MULTI.EXE - DATA ENTRY
ILLUSTRATION 12-13

12-4-2 Field Map Analysis (continued)

4. MULTI will display the data in two screens (Illustration 12-14). Make sure all data is correct before choosing either the “0 = continue analysis” or “-1 = write to file” options.

```
                 PLOTTED DATA                    MULTI  Version 2.3
                 ************
 Base field :   3.0000         Field units :     .1

PLANES     1       2       3       4       5       6       7

   Z    22.09   20.34   17.32   13.21    8.28    2.82   -2.82
   R     4.30    9.61   14.36   18.21   20.92   22.32   22.32

  0      4843    4753    4803    4832    4858    4798    4772
 30      4880    4814    4869    4918    5006    4981    4974
 60      4922    4911    4953    5035    5196    5185    5147
 90      4969    5006    5040    5140    5271    5205    5128
120      4999    5070    5096    5205    5297    5188    5062
150      5018    5167    5121    5246    5361    5242    5093
180      5011    5091    5096    5199    5326    5213    5107
210      4986    5023    5023    5074    5099    4996    4921
240      4937    4923    4936    4938    4878    4746    4672
270      4888    4836    4850    4848    4801    4651    4593
300      4852    4768    4807    4819    4847    4719    4637
330      4841    4736    4782    4809    4849    4762    4699
360      4848    4749    4806    4828    4849    4800    4771
HIT RETURN
```

```
PLANES     8       9      10      11      12

   Z    -8.28  -13.21  -17.32  -20.34  -22.09
   R    20.92   18.21   14.36    9.61    4.30

  0      4732    4803    4856    4885    5012
 30      4908    4892    4914    4962    5038
 60      5026    4953    4954    4999    5056
 90      5015    4969    4965    5004    5064
120      4910    4938    4955    4985    5052
150      4933    4914    4924    4945    5037
180      4956    4879    4891    4928    5018
210      4814    4798    4846    4874    4995
240      4585    4696    4783    4821    4974
270      4496    4640    4748    4795    4961
300      4536    4660    4758    4799    4970
330      4621    4725    4801    4947    4993
360      4732    4806    4859    4892    5014

Type plane number to edit plane   0 = continue analysis    -1 = write to file
0


Input radius of sphere for output
22.5_
```

Make sure all data is correct before continuing or writing to file.

MULTI.EXE - DATA REVIEW
ILLUSTRATION 12-14

12-4-2 Field Map Analysis (continued)

5. Record the list of measured harmonics next displayed by MULTI. (See Illustration 12-15.) Make sure to note the polarity of each. MULTI will also save this information in its “ANALYSIS.DAT” output file.

IMPORTANT !!!

MULTI always uses the same file name to save information. Rename each newly created ANALYSIS.DAT file using the format “ANALYSI” + sequential alphabetic character + “.dat” (8 characters + “.dat”) so each iteration is preserved.

Data shown was input from this file.

```
Evaluation of coeficients at a radius of  22.500     MULTI  Version 2.3
AXIAL GRADIENTS                                    Filename : c:\analysia.dat
Z 1    =     2.74 ppm
Z 2    =     -.23 ppm
Z 3    =    -5.53 ppm
Z 4    =     1.30 ppm
Z 5    =     1.92 ppm
Z 6    =      .74 ppm
Z 7    =     -.59 ppm
Z 8    =      .31 ppm
Z 9    =    -1.35 ppm
Z10    =      .29 ppm
Z11    =     1.56 ppm
TRANSVERSE GRADIENTS
X      =    -4.24 ppm          Y      =     7.67 ppm
ZX     =    -1.76 ppm          ZY     =     -.15 ppm
X2Y2   =      .41 ppm          XY     =      .13 ppm
Z2X    =     -.14 ppm          Z2Y    =     -.68 ppm
ZX2Y2  =     -.04 ppm          ZXY    =     -.14 ppm
X3     =     -.13 ppm          Y3     =     -.03 ppm
Z3X    =     -.08 ppm          Z3Y    =      .01 ppm
Z4X    =     -.13 ppm          Z4Y    =      .25 ppm
Z2X2Y2=      -.03 ppm          Z2XY   =     -.02 ppm    HIT RETURN
```

MULTI.EXE - GRADIENT COEFICIENTS
ILLUSTRATION 12-15

12-4-2 Field Map Analysis (continued)

6. Note the ppm values shown at the end of the "Measured Variation" display. See Illustration 12-16.

7. When MULTI gives the option of calculating the predicted homogeniety if all shims are used, choose "16 X3 & Y3" to mark the X3 and Y3 shims as unavailable. See Illustration 12-16.

Note

Illustration 12-2 gives an overview of the field map mapping / analysis and SC Shim adjustment process.

End of "Measured Variation" Display

```
210      -3.7    -4.3    -2.7    -1.7     2.3
240     -11.4    -7.7    -4.8    -3.5     1.6
270     -14.3    -9.5    -5.9    -4.4     1.2
300     -13.0    -8.9    -5.6    -4.2     1.5
330     -10.2    -6.7    -4.2      .7     2.2
360      -6.5    -4.0    -2.2    -1.1     2.9

   28.83 ppm peak to peak       Standard deviation   5.467 ppm      hit return




The following shim functions will be subtracted  1=yes 0=no
1   Z1 : 1                11     X & Y    : 1
2   Z2 : 1                12    ZX & ZY   : 1
3   Z3 : 1                13  X2Y2 &  XY  : 1
4   Z4 : 1                14   Z2X & Z2Y  : 1
5   Z5 : 0                15  ZX2Y2 & ZXY : 1
6   Z6 : 0                16    X3 & Y3   : 1
                          17   Z3X & Z3Y  : 0
                          18 Z2X2Y2 & Z2XY: 0

Input reference number of shim to be used/not used
Type 0 to continue with analysis (-1 to exit error analysis)
_
```

Available Shims (Choose "16" and reset "X3 & Y3" to "0")

MULTI.EXE - MARKING AVAILABLE SHIMS
ILLUSTRATION 12-16

12-4-2 Field Map Analysis (continued)

```
330        .5    -2.1     -.7     -.1     1.5      .0     -.1
360        .8    -1.6      .1      .1      .5     -.6     -.3
HIT RETURN



PLANES     8       9      10      11      12

  0       -.7     1.2      .6    -1.8     -.3
 30       -.3     -.1     -.2     -.6      .1
 60       -.6     -.8     -.4     -.1      .5
 90       -.4      .6      .7      .4      .8
120      -1.7     1.5     1.7      .4      .6
150      -1.3      .7      .9     -.6      .3
180       -.5     -.5      .0     -.8     -.1
210       -.2      .3      .4    -1.7     -.5
240      -1.1     1.7     1.2    -2.1     -.8
270      -1.7     1.8     1.4    -2.3    -1.0
300      -2.1     1.3     1.2    -2.4     -.8
330      -1.3     1.5     1.0     1.6     -.4
360       -.7     1.3      .7    -1.6     -.2

    5.84 ppm peak to peak       Standard deviation   1.146 ppm      hit return
```

End of "Theoretical Variation" Display

MULTI.EXE - THEORETICAL VARIATION (PPM)
ILLUSTRATION 12-17

8. Compare the ppm values at the end of the "Theoretical Variation" display (Illustration 12-17) with the "Measured Variation" values. If the field map plot is more than 5 ppm greater than the predicted theoretical value:

   a. Run SHMCALC7.XLS with the "default" variable set to "1" (one).

   Note
   SHMCALC7.XLS contains instructions on "sheet 2." Rename the copy of SHMCALC7.XLS used with a name in the format "SHMCALC" + magnet serial number + ".XLS".

   b. Enter the measured harmonics of the first field plot (from Step 4) in column "B."

   c. If the shim polarities from Step 2 differ from the default polarities in SHMCALC7, edit the polarities in column "C."

   d. Use SHMCALC7's shim current calculations to set the SC shims in accordance with Section 12-4-3.

9. Repeat Section 12-4-1, Plotting the Magnetic Field, and Steps 3 - 8 in this Section until the measured homogeneity is within 5 ppm of the predicted theoretical value.

12-4-2 Field Map Analysis (continued)

10. Once the field map plot is within 5 ppm of the predicted theoretical value, check passive shimming. See Section 12-4-4.
11. Repeat Section 12-4-1, Plotting the Magnetic Field, and Steps 3 - 10 in this Section until the theoretical homogeneity is < 5 ppm.
12. Run SHMCALC7.XLS (Step 8 in this Section) and adjust the SC shims in accordance with Section 12-4-3.
13. Replot the field map in conformance with Section 12-4-1, Plotting the Magnetic Field.
14. Run MULTI.EXE and VOLRMS.EXE to analyze the results.
15. Repeat Steps 12 - 14 until the Volume RMS calculated by VOLRMS is within specification:

    < 1.00 ppm on 45 cm
    < 0.50 ppm on 40 cm
    < 0.25 ppm on 30 cm
    < 0.05 ppm on 20 cm

IMPORTANT !!!

Save the copy of SHMCALC7.XLS used. Name the file "SHMCALC" + <u>magnet serial number</u> + ".XLS". Save in electronic and printed form and store in a secure place on-site. This information is important for future shimming on the same magnet.

12-4-3 Setting SC Shim Currents

CAUTION

Monitor the Main Coil Switch voltage during Shim current setting to make sure the switch does not go resistive. Voltage should be stable at approximately zero volts.

1. Prepare to set shim currents in accordance with Section 12-3. However, only dump the SC shim currents (Steps 19 - 21) prior to the initial plot, not subsequent ones.
2. If the B0 Shim Coil Switch is not "On," turn it on (open) at the MRU.
3. Make sure all current potentiometers are set to zero (fully counter clockwise).
4. Make sure the Switch Heater Switch on the Shim Adapter Box is off.
5. Turn on the Shim Power Supply.

12-4-3 Setting SC Shim Currents (continued)

TABLE 12-1
CROSS-REFERENCE, SHIM POWER SUPPLY CHANNELS TO SHIM ORDERS

Power Supply Channel (Shim Power Supply)	T1-1						T1-2							
Shim Order (Shim Adapter Box Label)	Z3	Z4	Z^2X	Z^2Y	ZXY	$Z(X^2-Y^2)$	Z1	Z2	X	Y	ZX	ZY	XY	X^2-Y^2

6. Select the "Transverse 1" bank of shims (channels T1-1, T1-2, etc.) by turning the selector knob on the Shim Power Supply to its central position. The Shim Adapter Box uses channels T1-1 and T1-2.
7. Position the MetroLab Probe to the location stated in Table 12-2 for the shim pair to be adjusted.
8. Select the shim pair to be adjusted using the Shim Coil Selector Knob on the Shim Adapter Box.

TABLE 12-2
OPTIMUM PROBE POSITIONS FOR MONITORING FIELD DRIFT

Shim	Axial (Z-Axis) Position*	Radial Position*	Angle (Degrees)
Z1	+ R	0	0
Z2	0	R	0
Z3	+ R	0	0
Z4	+ R	R	0
X	0	R	0
Y	0	R	90
ZX	+ R	R	0
ZY	+ R	R	90
XY	0	R	45
ZXY	+ R	R	45
$X^2 - Y^2$	0	R	0
Z^2X	+ R	R	0
Z^2Y	+ R	R	90
Z^2Y	+ R	R	45
$Z (X^2 - Y^2)$	+ R	R	0

* 0 = Axial Center of magnet.
R = Plotting Radius of 22.5 cm. Positive "R" is towards the magnet's service end; see Illustration 12-1.

12-4-3 Setting SC Shim Currents (continued)

9. Turn on the Shim Heater Switch at the Shim Adapter Box.

10. Adjust the T1-1 potentiometer to the shim current / polarity calculated in Section 12-4-2 for the shim being adjusted. Press the button below the T1-1 potentiometer to monitor its current.

Note
Do not change polarity until the current has been set to zero.

11. Monitor field change with the teslameter. When it has stabilized, repeat Step 10 for channel T1-2.

Note
Field increase with increasing shim current indicates a positive shim polarity for all Shim Orders except Z2 and Z4 and a negative shim polarity for Z2 and Z4.

Field decrease with increasing shim current indicates a negative shim polarity for all Shim Orders except Z2 and Z4 and a positive shim polarity for Z2 and Z4.

12. When each shim pair has stabilized, turn off the Switch Heater at the Shim Power Supply. Wait 3 minutes for the switch to become fully resistive, then set both channels to zero current.

13. Repeat Steps 7 - 12 for the next shim pair until all shim pairs have been set.

12-4-4 Passive Shimming

Introduction

Passive shimming may be required to fine tune the magnet's field to achieve the homogeneity specifications or to compensate for high-order gradients for for which there are no SC Shims.

Thirty-six Passive Shim Trays are inserted in the RT Shim Tube around its circumferance. The Passive Shim Tray channels are labeled on the patient end of the RT Shim Tube, from 0° (coordinate 'α' in Illustration 12-1) at the top every 10° clockwise to 350°. Each Shim Tray has 41 shim positions known as "pockets," 20 on either side of the magnet's center and marked from -30 at the patient end to +30 at the service end. Only the central 25 positions are used in shimming. See Illustration 12-18.

A record of passive shimming conducted at the factory on each Oxford-shipped magnet is online at "http://medfrquickplace01.ge.com/quickplace/gemse_shimprog/Main.nsf?OpenDatabase" under "Shim Settings." The file is named _____F__.dat, in the form of magnet serial number + "F" + factory iron iterations + ".dat". For Florence-shipped magnets, the same information is contained in the Acceptance Test Report (ATR) shipped with the magnet.

Note
If the shimming information for an Oxford-shipped magnet is missing from the web, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715) before proceeding.

12-4-4 Passive Shimming (continued)

WARNING!

PASSIVE SHIMS USE FERROMAGNETIC MATERIAL (STEEL) SUBJECT TO MAGNETIC FORCES PROPORTIONAL TO STEEL MASS AND FIELD STRENGTH:

- **ATTRACTING / REPELLING FORCES THAT VARIES IN STREGTH / DIRECTION DURING SHIM TRAY INSERTION / REMOVAL.**
- **A LOADING FORCE ON MOUNTED SHIM TRAYS THAT CAN PROPEL TRAYS OUT OF MAGNET WHEN RETAINING CLIPS ARE REMOVED.**

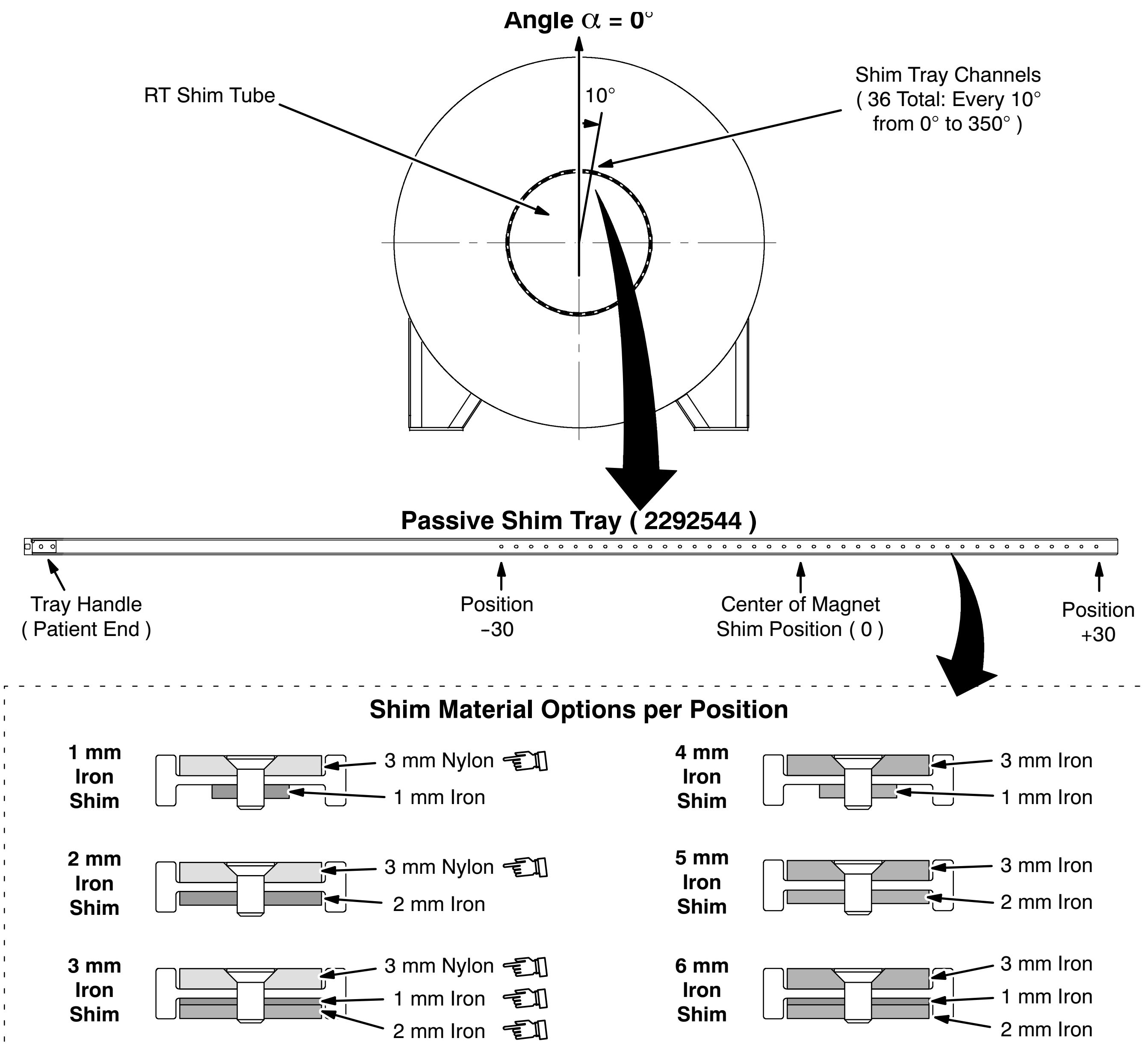
TAKE THE FOLLOWING STEPS TO PREVENT INJURY DUE TO THESE FORCES:

- **WEAR SAFETY GLASSES.**
- **WEAR GLOVES WHILE HANDLING SHIM TRAYS TO PROTECT FROM FIBERGLASS SPLINTERS / CUTS.**
- **DO NOT DIRECTLY FACE OR STAND IN THE FRONT OF A SHIM TRAY.**
- **ALWAYS USE NONMAGNETIC TOOLS.**
- **REMOVE / INSERT SHIM TRAYS CAREFULLY & GRADUALLY:**
    - **A. HOLD SHIM TRAY FIRMLY WHILE REMOVING RETAINING CLIP.**
    - **B. REMOVE SHIM TRAY WITH A GRADUAL MOTION WHILE USING YOUR PALM TO RESTRAIN ANY UNDESIRED INITIAL TRAY MOTION.**
    - **C. THEN GRIP THE TRAY WITH BOTH HANDS, MOVING YOUR GRIP ALONG THE TRAY AS IT GRADUALLY PULLS OUT.**
    - **D. HOLD EACH TRAY SECURELY WHILE MOVING AWAY FROM MAGNET.**
    - **E. REVERSE THESE STEPS INSERTING TRAY.**
- **MAKE SURE ALL PASSIVE SHIM MATERIAL IS SECURE IN THE TRAY BEFORE LEAVING THE TRAY LOADING AREA.**
- **NEVER LEAVE LOADED TRAYS OR LOOSE PASSIVE SHIM MATERIAL NEAR THE MAGNET.**

Note

The removal or insertion of passive shim trays may cause small changes in SC shim currents depending on the iron mass in each tray.

12-4-4 Passive Shimming (continued)



PASSIVE SHIM LAYOUT
ILLUSTRATION 12-18

Procedure

1. Run FESHIM.EXE and follow its instructions.
2. Open the most recent plot file generated by MULTI and the configuration file 3t940.fig downloaded from the web site.

12-4-4 Passive Shimming (continued)

3. Use the default list of available SC Shims. However, if shims are known to not be working properly or have excessive currents, mark these specific shims as unavailable and compensate with more iron.

4. Open the most recent file recording previous shims. Record the opened file's name in DATA SHEETS, Section 9, Shim Plot Data, beside "Input File Name" for the plot iteration.

Note

For the initial on-site shim iteration, this is the factory's _____F__.dat file (magnet serial number + "F" + factory iron iterations + ".dat") downloaded from the web site. For subsequent iterations, use the output file from the immediately previous iteration.

5. Enter an output file name for FESHIM's results. Record this file name in DATA SHEETS, Section 9, Shim Plot Data, beside "Output File Name" for the plot iteration.

Note

Use this file naming convention: _____F__.dat (magnet serial number + "F" + sequential iron iteration number + ".dat"). For instance, if the last factory shim iteration produced 12222F005.dat, the first iteration of the installed magnet would be named 12222F06.dat, the second 12222F007.dat, etc.

6. Enter the maximum number of shim pieces to use. This is initially 200. If the predicted homogeneity is < 4 ppm, consider re-running FESHIM with fewer shim pieces. If the predicted homogeneity is > 5 ppm, consider re-running FESHIM with more shim pieces.

7. When given the option to reduce the number of shim positions, answer "No" unless homogeneity already is low.

8. Exit FESHIM by typing "Q".

Note

FESHIM produces a table of standard iron mass units (1 to 6) required at each shim position, with the Shim Trays numbered in the table rows and the positions in separate columns.

9. For each Shim Tray containing positions identified by FESHIM's output table as needing adjustment:

   a. Remove the Shim Tray and take to a shim loading area outside the magnet room.

   b. Add /delete iron from each position needing adjustment until the count of standard iron units at each position matches FESHIM's recommendation.

   c. Return the Shim Tray to the magnet and reinsert in its Shim Tray Channel.

10. Repeat Section 12-4-1, Plotting the Magnetic Field, and Section 12-4-2, Field Map Analysis.

FUNCTIONAL CHECKS

TABLE OF CONTENTS

TABLE OF CONTENTS (continued)

SECTION 1 - MAGNET SPECIFICATIONS

1-1 MAGNET DATA

TABLE 1-1
MAGNET DATA

Maximum Central Field		3.0 Tesla (127.7277 MHz; 30,000 Gauss)
Magnet / System Bandwidth for a Shimmed Magnet		29,993 - 30,007 Gauss
Nominal Current for Maximum Field		253.5 Amps*
Field Stability		Drift < 0.05 ppm / hour
Magnet Inductance		413.0 Henries
S/C Switch	Normal Resistance	5 ohms
	Heater Resistance	100 ohms
	Heater Energization Current	60 - 80 mA
Field map	Fringe Field	See SET-UP AND CALIBRATION, Section 10, Ramping, Illustration 10-1
	5 Gauss	5.4 M (17.7 feet) Axially 3.1 M (10.2 feet) Radially
Fully Shimmed, plotted over 12 planes	45 cm (17.7 inches) DSV	1 ppm RMS Deviation
	40 cm (15.7 inch) DSV	0.5 ppm RMS Deviation
	30 cm (11.8 inches) DSV	0.25 ppm RMS Deviation
	20 cm (7.87 inch) DSV	0.05 ppm RMS Deviation
* Some magnets may require up to 273 amps ramp current for a full 3.0T field. Refer to SET-UP AND CALIBRATION, Table 11-4, for the design levels used on various magnets, by serial number.		

1–2 CRYOSTAT DATA

TABLE 1–2
CRYOSTAT DATA

Helium Vessel Volume		3066 litres
Installation Requirements, Warm Shipment	Liquid Helium	8500 liters (4000 liters extra if “training” quench occurs)
	Liquid Nitrogen	6000 liters
Liquid Helium Refill Volume		1200 liters**
Minimum Operating Level, Liquid Helium*	Normal Operation	500 mm (1876 liters), probe 1
	During Ramp Up	920 mm (2466 liters), probe 1
Nominal Dimensions	Clear Bore Diameter	940 ± 3 mm (37.0 ± 0.1 inches)
	Bore Diameter, RT Shims Installed	900 ± 3 mm (35.4 ± 0.1 inches)
	Cryostat Length	2600 ± 5 mm (102.4 ± 0.2 inches)
	Cryostat Diameter	2380 ± 5 mm (93.7 ± 0.2 inches)
	Center Line Height Above Floor	1205 ± 5 mm (47.4 ± 0.2 inches)
	Cryostat Weight	13 metric tons (28,665 pounds)

* Measuring LHe level is covered in FUNCTIONAL CHECKS, Section 9, Liquid Helium Level Measurement.
** Exact liquid helium volume required to refill depends upon level when refill is begun and efficiency of liquid helium transfer. See FUNCTIONAL CHECKS, Section 9, Liquid Helium Level Measurement.

SECTION 2 – MAGNET FIELD STABILITY

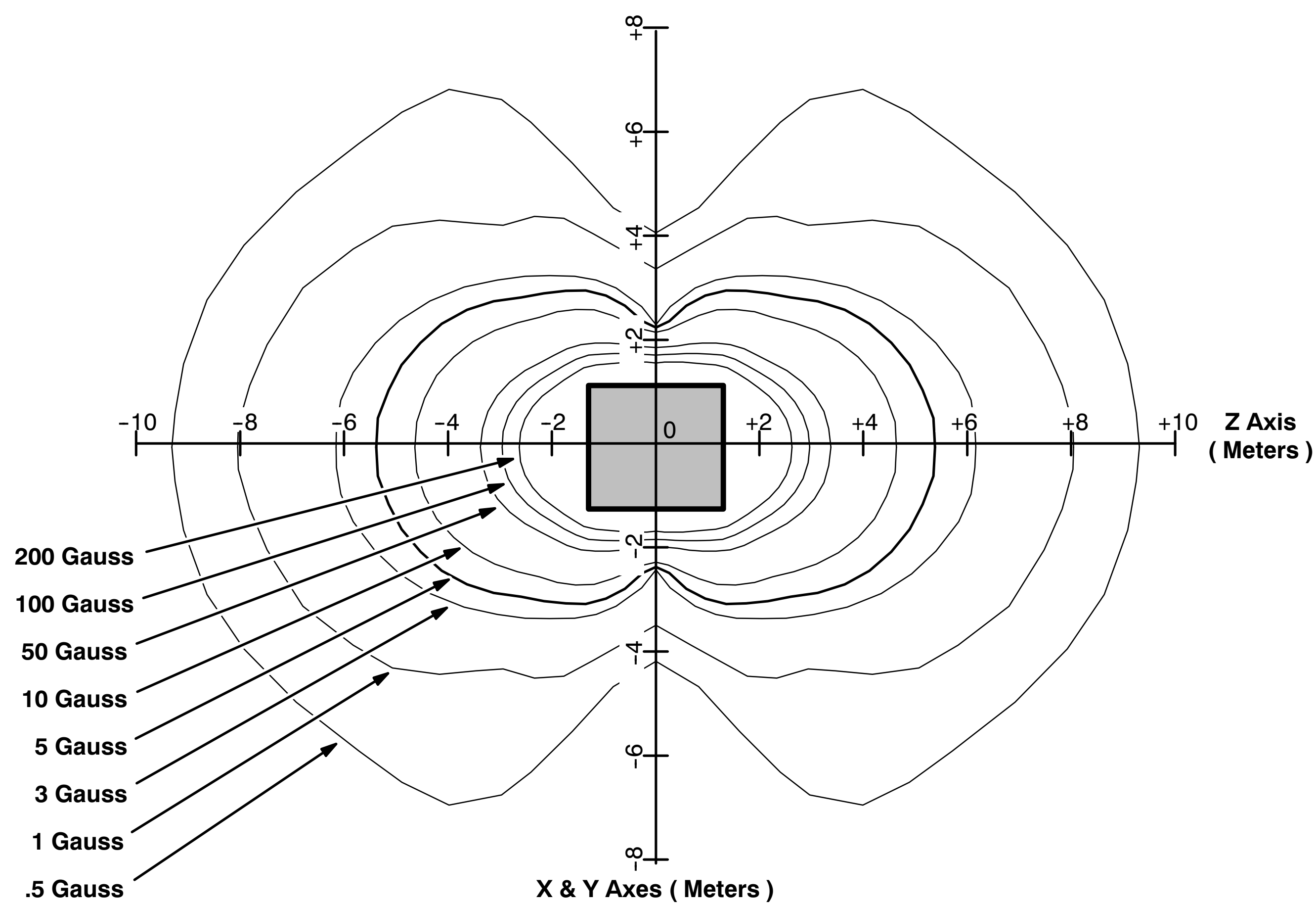
Note

This procedure determines the *uncompensated* main field drift of the magnet.

CAUTION

Moving equipment may effect the field readings.

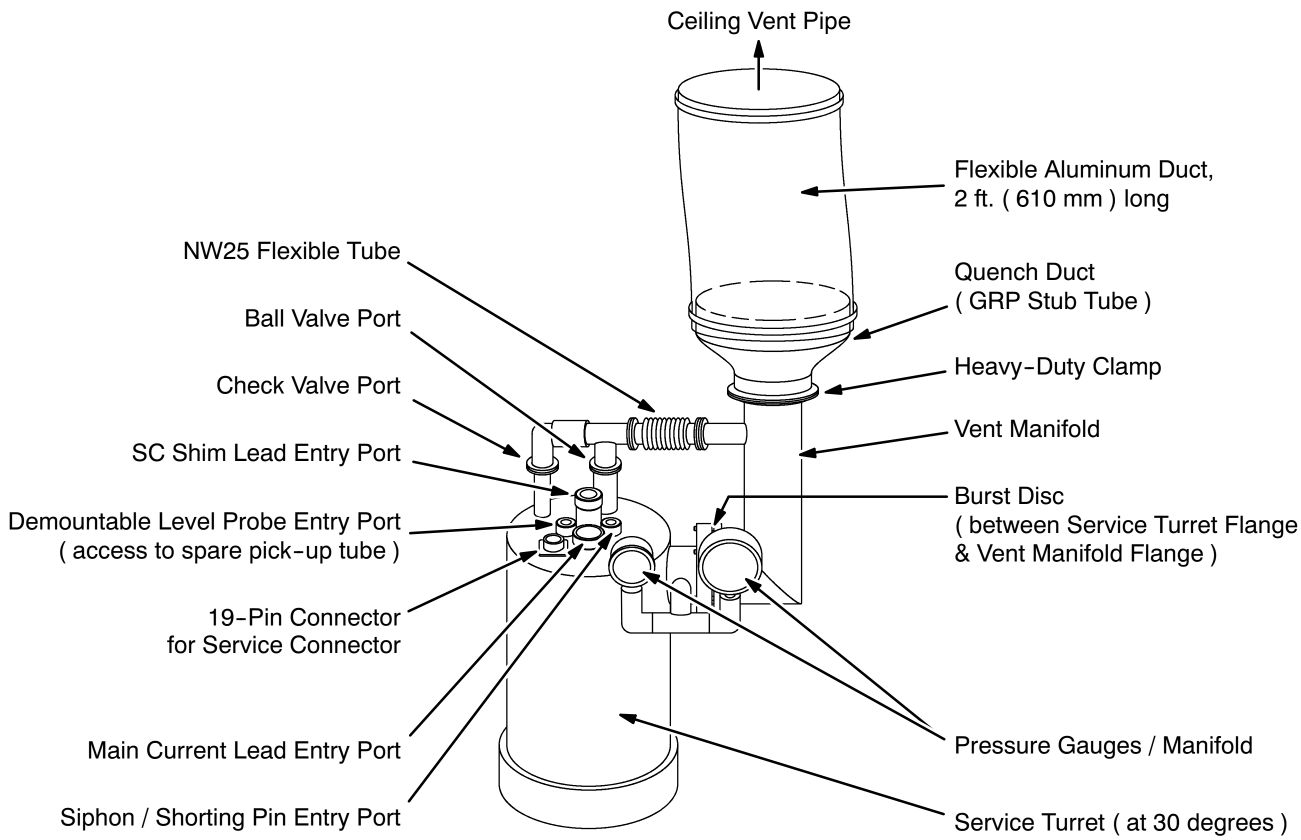
1. Post signs outside the 0.5 Gauss line (extending from the magnet's center approximately 23 feet or 7.0 meters on either side, above and below the magnet and 30.7 feet or 9.4 meters at either end) indicating a magnetic drift test is in progress. See Illustration 2–1. Do not move or rearrange any ferromagnetic articles or equipment inside the 0.5 Gauss line during the test.



GEMSO 3T/940 FRINGE FIELD
ILLUSTRATION 2–1

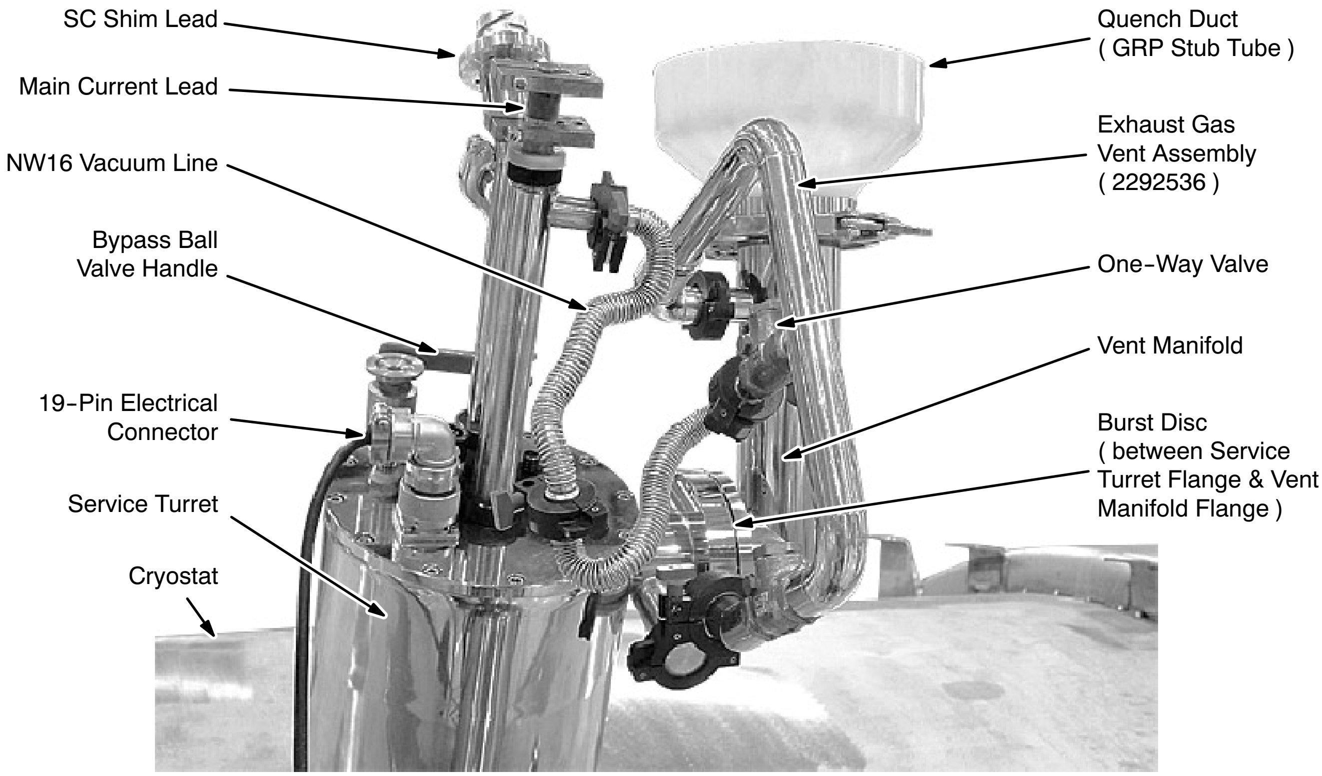
SECTION 2 - MAGNETIC FIELD STABILITY (continued)

2. Assemble, mount and center the Field Plotting Rig (2292542) in conformance with SET-UP AND CALIBRATION, Section 10-1, Field Plotting Rig (2292542) Set-Up.
3. Open the Ball Valve Port to vent the magnet down to < 0.1 psig on the manifold gauge.
4. Close the ball valve and blank off the check valve.
5. Remove the Pressure Gauge Manifold and install the Exhaust Gas Vent Assembly (2292536) in its place. See Illustration 2-2 for the factory-installed configuration and Illustration 2-3 for the Exhaust Gas Vent arrangement.



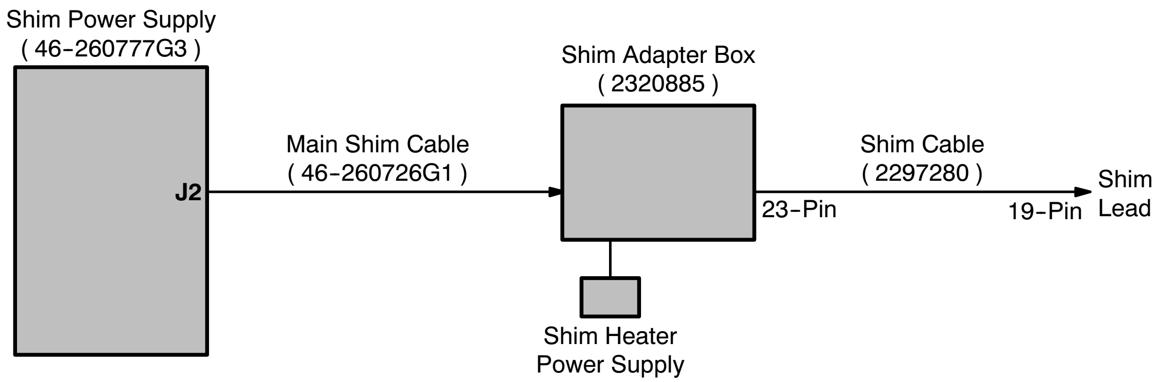
TYPICAL FACTORY-INSTALLED 3T SERVICE TURRET PLUMBING
ILLUSTRATION 2-2

SECTION 2 - MAGNETIC FIELD STABILITY (continued)



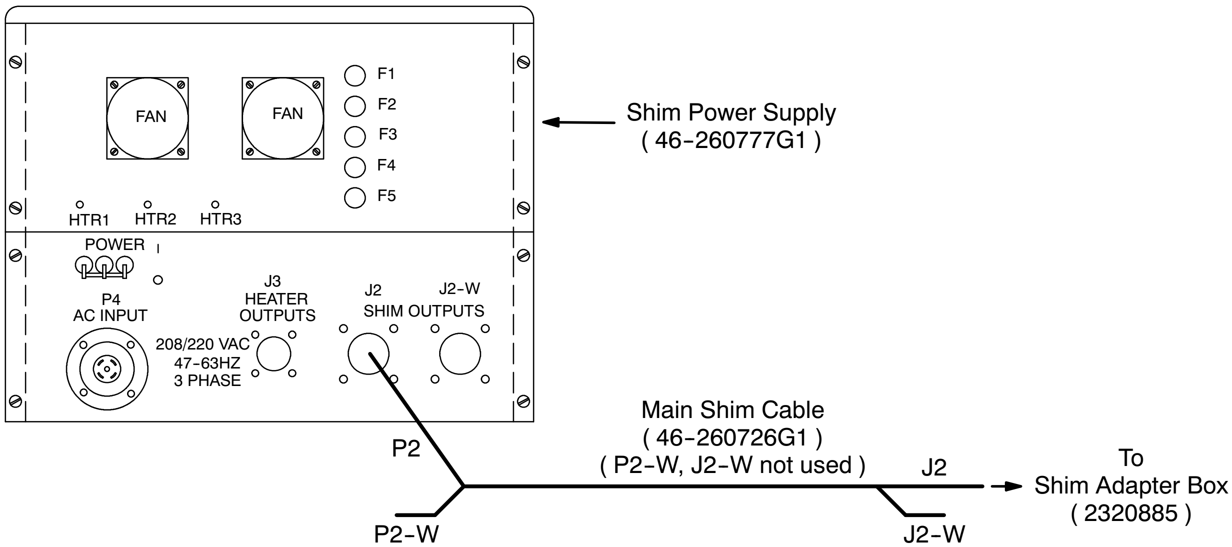
EXHAUST GAS VENT CONNECTION
ILLUSTRATION 2-3

6. Connect the Main Shim Cable (46-260726G1) between Shim Power Supply socket J2 and the input socket on the Shim Adapter Box (2320885). See Illustrations 2-4, 2-5 and 2-6.
7. Connect Shim Cable 2297280 to the Shim Adapter Box's 23-pin output. See Illustrations 2-4 and 2-6.

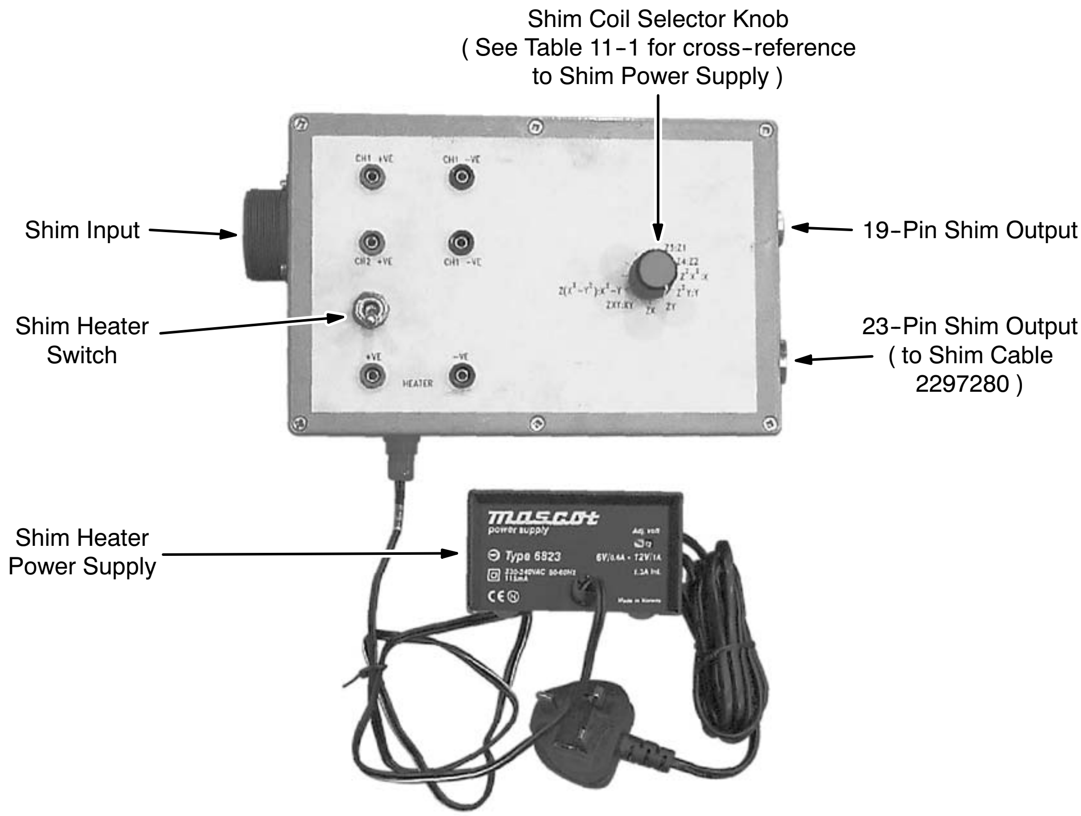


CABLE CONNECTIONS FOR SHIMMING
ILLUSTRATION 2-4

SECTION 2 - MAGNETIC FIELD STABILITY (continued)



SHIM POWER SUPPLY OUTPUT CONNECTIONS
ILLUSTRATION 2-5



SHIM ADAPTER BOX (2320885)
ILLUSTRATION 2-6

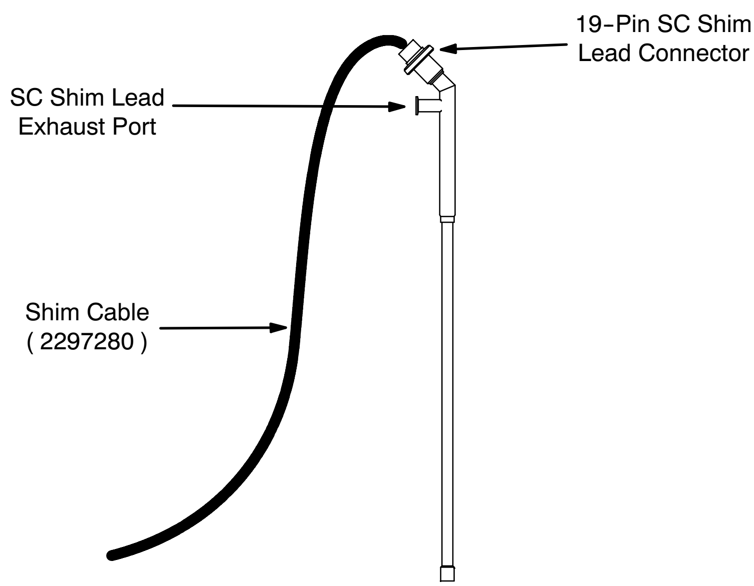
SECTION 2 – MAGNETIC FIELD STABILITY (continued)

8. Turn on the Shim Power Supply.
9. Make sure all current potentiometers on the Shim Power Supply are set to zero (fully counterclock-wise).
10. Attach the other end of the Shim Cable to the Shim Lead. See Illustration 2–7

CAUTION

Do not leave the SC Shim Lead Entry Port open to the atmosphere for more than a few seconds since air will condense in the port, which will cause an ice blockage.

11. Remove the cap from the SC Shim Lead Entry Port, moving the nut, washers and o–ring to the Main Current Lead. See Illustration 2–2.



SC SHIM LEAD CONNECTIONS
ILLUSTRATION 2–7

SECTION 2 – MAGNETIC FIELD STABILITY (continued)

WARNING!

THE MAGNET MAY QUENCH IF A WARM SHIM LEAD IS NOT ALLOWED TO COOL SUFFICIENTLY DURING INSERTION.

WARNING!

MAKE SURE THE SHIM LEAD EXHAUST PORT ALWAYS FACES AWAY FROM THE SERVICE PLATFORM TO PREVENT COLD HELIUM GAS EXHAUSTING TOWARD THE SERVICE ENGINEER.

12. Insert the Shim Lead into the SC Shim Lead Entry Port with the exhaust port facing away from the service platform. Push down gradually, allowing the lead to cool. Cold helium gas will emerge from the entry port. If the rush of gas becomes too great, hold the lead steady or retract it slightly until the rush decreases.
13. Once contact is felt, push the Shim Lead into full engagement with the male connector at the bottom of the entry port. If it does not fit in the correct orientation, stop immediately; **DO NOT FORCE THE SHIM LEAD INTO ENGAGEMENT NOR ROTATE IT**. Instead, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715).
14. Tighten the nut to create a gas-tight seal.
15. Connect a NW16 vacuum line between the Shim Lead exhaust and the one-way valve at the center of the Exhaust Gas Vent Assembly (2292536) to route exhaust gas safely out of the room. See Illustration 2-3.
16. Verify that the Shim Heater Switch on the Shim Adapter Box is “Off,” then plug in the Shim Heater Power Supply.
17. Select the “Z1:Z3” shim pair via the Shim Coil Selector Knob on the Shim Adapter Box.
18. Dial into the Shim Power supply the shim currents and polarities last recorded in DATA SHEETS, Section 9, Shim Plot Data.
19. Switch on the Shim Heaters on the Shim Adapter Box.
20. Wait for the Switch Heaters to open, then reduce the current to zero.
21. Turn off the Shim Heater Switch, then rotate the Shim Coil Selector Knob to the next shim pair and repeat Steps 17 – 20.
22. Make sure the Shim Heater Switch is off.

SECTION 2 – MAGNETIC FIELD STABILITY (continued)

23. Adjust / resynchronize the Teslameter in conformance with SET-UP AND CALIBRATION, Section 10-2.

24. Set the Teslameter switch to “NMR Frequency (Hz)” and allow the Teslameter to stabilize within a 10 Hz band.

25. Rotate the Shim Coil Selector Knob on the Shim Adapter Box to the “Z4:Z2” position and then turn on the Shim Heater Switch on the Shim Adapter Box. See Illustration 2-6.

26. After 3 minutes record the frequency as “Frequency 1” in the DATA SHEETS, Table 6-2.

27. Repeat Steps 1 through 26 after 24 hours. Record this frequency as “Frequency 2” in DATA SHEETS, Table 6-2.

28. Calculate the Main Field Drift Rate by using the following formula:

$$\text{Drift Rate (ppm/hr)} = \frac{(\text{Freq 1} - \text{Freq 2}) \times 10^6}{(\text{Freq 1}) \times 24}$$

For the Initial Drift Rate use: Freq 1 = initial reading
Freq 2 = reading after 24 hours

29. If the Drift Rate is greater than 6.3 Hz/hr, the Drift Rate is outside guidelines. Contact the MAC Team Representative or the Regional Service Engineer. High Drift Rates will require frequent field adjustment and reshimming.

Note

The Teslameter has a resolution of ± 5 Hz; therefore, a month or more may be required to establish a significant frequency difference (drift rate).

SECTION 3 – MAGNET ELECTRICAL CHECKS

Note

This section provides "go" / "no go" tests for internal magnet circuitry.

WARNING!

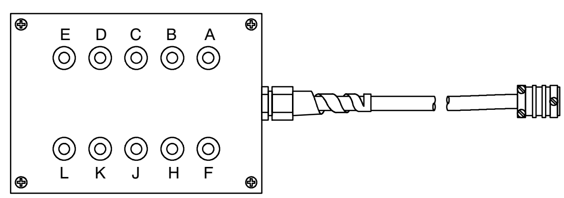
ELECTRICAL CHECKS CAN ONLY BE PERFORMED WITH THE MAIN & SHIM COILS RAMPED DOWN (0 AMPS). DO NOT MAKE CONTACT WITH THE 19-PIN SERVICE LEAD CONNECTOR WHILE ARE COILS RAMPED UP.

1. Locate the connector pins using Tables 3-1 through 3-6.

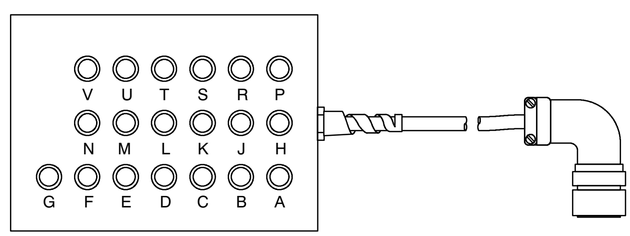
Note

Use 10-Pin Breakout Box 2295140 to access the left and right Pump-Out Port 10-pin connectors, the Shield Cooler Coldhead 10-pin connector and the Demountable Level Probe 10-pin connector. Use 19-Pin Breakout Box 2295141 to access the Service Turret 19-pin connector. See Illustration 3-1.

10-Pin Breakout Box 2295140
(For 10-pin connectors
shown in Illustration 7-3.)



19-Pin Breakout Box 2295141
(For the 19-Pin Service Turret Connector
shown in Illustration 7-3.)



Note

Breakout box socket lettering directly correlates to connector pin identification.

10-PIN (2295140) AND 19-PIN (2295141) BREAKOUT BOXES

ILLUSTRATION 3-1

SECTION 3 – MAGNET ELECTRICAL CHECKS (continued)

2. Use a digital meter to measure the resistance across the identified connector pins.

3. Measure the resistances of the pin pairs shown.

   - If readings deviate more than ± 10% from the typical values given, contact your MAC Team representative before installing the magnet.
   - If the reading across pins A and B of the Service Turret 19–pin connector shows an open cicuit, the **cable side** of the 19–pin connector can be rewired to connect to the spare Level Probe on pins D and E:

     a Access the wiring to the cable side of the Service Turret 19–pin connector.

     b Relocate the wire connecting to pin A to connect to pin D.

     c Relocate the wire connecting to pin B to connect to pin E.

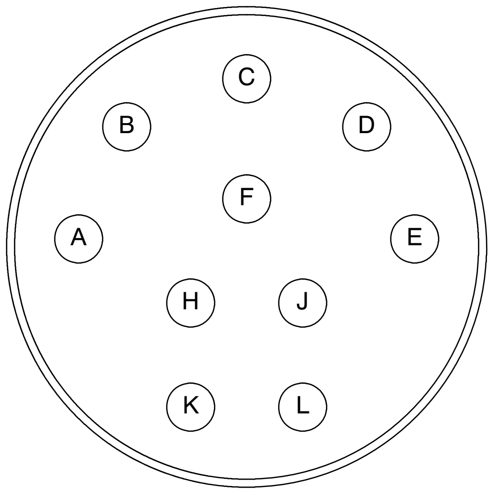
     Note
     Do not touch or in any way change any other wiring to the Service Turret 19–pin connector.

SECTION 3 - MAGNET ELECTRICAL CHECKS (continued)

TABLE 3-1
WIRING CHECK SHEET,
LEFT-HAND TEN-PIN CONNECTOR ON PUMP-OUT PORT

Magnet Model No.________________ Service Engineer:________________
Magnet Serial No.________________ Date:________________

Check box if ALL actual measurements match typical values: ☐



Pin	A	B	C	D	E	F	H	J	K	L	GND
A	—										(∞)
B	(110)	—									(∞)
C	(∞)	(∞)	—								(∞)
D	(400)	(400)	(∞)	—							(∞)
E	(400)	(400)	(∞)	(110)	—						(∞)
F	(∞)	(∞)	(∞)	(∞)	(∞)	—					(∞)
H	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	—				(∞)
J	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(100)	—			(∞)
K	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(215)	(215)	—		(∞)
L	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(215)	(215)	(100)	—	(∞)

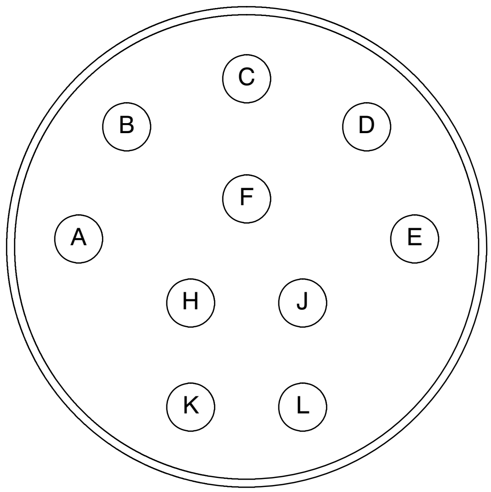
Note: Typical values in Ohms are shown in parentheses. Polarity is not relevant.
If readings deviate more than ± 10% from the typical values given, contact your MAC Team representative before installing the magnet.

SECTION 3 – MAGNET ELECTRICAL CHECKS (continued)

TABLE 3-2
WIRING TEST,
RIGHT-HAND TEN-PIN CONNECTOR ON PUMP-OUT PORT

Magnet Model No.______________________
Magnet Serial No.______________________
Service Engineer:______________________
Date:______________________

Check box if ALL actual measurements match typical values: ☐



Pin	A	B	C	D	E	F	H	J	K	L	GND
A	—										(∞)
B	(110)	—									(∞)
C	(∞)	(∞)	—								(∞)
D	(390)	(390)	(∞)	—							(∞)
E	(∞)	(∞)	(∞)	(110)	—						(∞)
F	(∞)	(∞)	(∞)	(∞)	(∞)	—					(∞)
H	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	—				(∞)
J	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(100)	—			(∞)
K	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(220)	(220)	—		(∞)
L	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(220)	(220)	(100)	—	(∞)

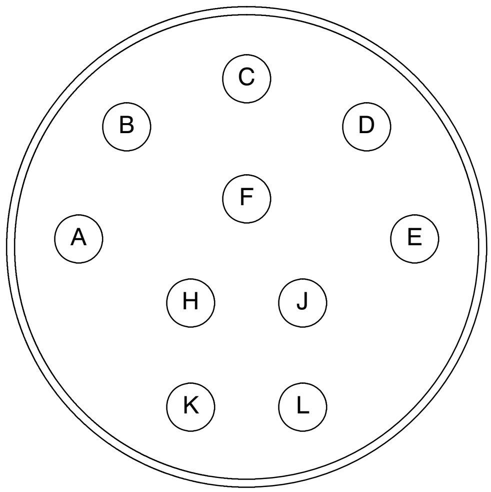
Note: Typical values in Ohms are shown in parentheses. Polarity is not relevant.
If readings deviate more than ± 10% from the typical values given, contact your MAC Team representative before installing the magnet.

SECTION 3 – MAGNET ELECTRICAL CHECKS (continued)

TABLE 3–3
WIRING TEST,
SHIELD COOLER TEN–PIN CONNECTOR ON COLDHEAD

Magnet Model No.__________________ Service Engineer:__________________
Magnet Serial No.__________________ Date:__________________

Check box if ALL actual measurements match typical values: ☐



Pin	A	B	C	D	E	F	H	J	K	L	GND
A	—										(∞)
B	(50)	—									(∞)
C	(∞)	(∞)	—								(∞)
D	(330)	(330)	(∞)	—							(∞)
E	(330)	(330)	(∞)	(50)	—						(∞)
F	(∞)	(∞)	(∞)	(∞)	(∞)	—					(∞)
H	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	—				(∞)
J	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(45)	—			(∞)
K	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(150)	(150)	—		(∞)
L	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(150)	(150)	(45)	—	(∞)

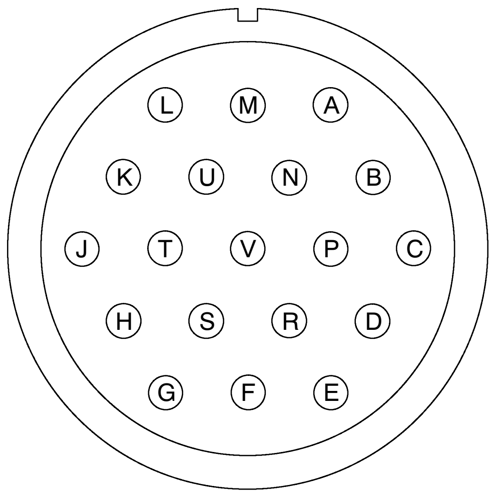
Note: Typical values in Ohms are shown in parentheses. Polarity is not relevant.
If readings deviate more than ± 10% from the typical values given, contact your MAC Team representative before installing the magnet.

SECTION 3 – MAGNET ELECTRICAL CHECKS (continued)

TABLE 3–4
WIRING TEST,
19–WAY SERVICE LEAD CONNECTOR

Magnet Model No.______________________ Service Engineer:______________________
Magnet Serial No.______________________ Date:______________________

Check box if ALL actual measurements match typical values: ☐



Level Probes							
Pin	A	B	C	D	E	F	GND
A	—						(∞)
B	(50)	—					(∞)
C	(322)	(306)	—				(∞)
D	(580)	(563)	(322)	—			(∞)
E	(563)	(550)	(305)	(50)	—		(∞)
F	(296)	(280)	(39)	(295)	(279)	—	(∞)

Quench Heat		Voltage Sense		Switch Heat			
G–H	H–GND	J–K	J–GND	L–M	L–N	M–N	M–GND
(12)	(∞)	(34)	(16)	(114)	(216)	(114)	(∞)

Note: Typical values in Ohms are shown in parentheses. Polarity is not relevant.
If readings deviate more than ± 10% from the typical values given, contact your MAC Team representative before installing the magnet.

SECTION 3 – MAGNET ELECTRICAL CHECKS (continued)

TABLE 3-4 (CONTINUED)
WIRING TEST,
19-WAY SERVICE LEAD CONNECTOR

Temperature Sensors / B0 Coil							
Pin	P	R	S	T	U	V	GND
P	—						(∞)
R	(172)	—					(∞)
S	(173)	(64)	—				(∞)
T	(470)	(365)	(365)	—			(∞)
U	(∞)	(∞)	(∞)	(∞)	—		(∞)
V	(∞)	(∞)	(∞)	(∞)	(113)	—	(∞)

Pin	A	G	J	L	P
A	—				
G	(∞)	—			
J	(∞)	(∞)	—		
L	(∞)	(∞)	(∞)	—	
P	(∞)	(∞)	(∞)	(∞)	—

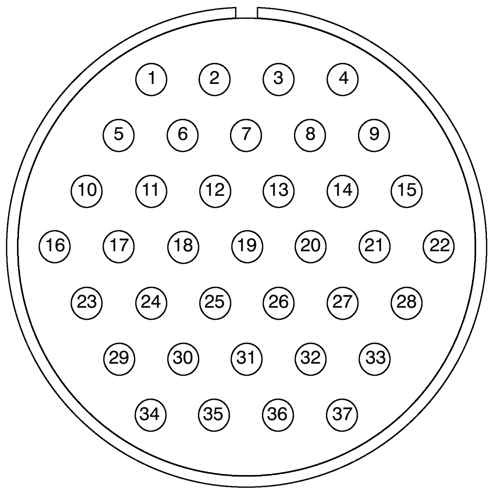
Note: Typical values in Ohms are shown in parentheses. Polarity is not relevant.
If readings deviate more than ± 10% from the typical values given, contact your MAC Team representative before installing the magnet.

SECTION 3 – MAGNET ELECTRICAL CHECKS (continued)

TABLE 3–5
WIRING TEST,
ROOM TEMPERATURE SHIM STRENGTHS & RESISTANCES (37–PIN PIGTAIL)

Magnet Model No.__________________ Service Engineer:__________________
Magnet Serial No.__________________ Date:__________________

Check box if ALL actual measurements match typical values: ☐



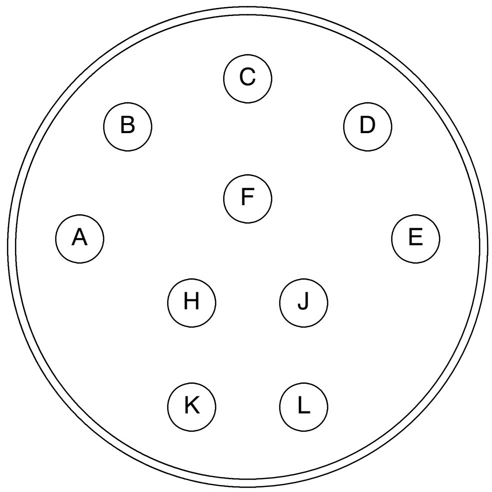
Pin	Shim	Function	Resistance (Ohms)	Strength at 22.5 cm Radius (ppm / amp)
5	Z2	+ve	4.8	5.1
6		–ve		
7	Z3	+ve	4.6	2.9
8		–ve		
17	ZX	+ve	1.9	3.2
18		–ve		
19	ZY	+ve	1.9	3.2
20		–ve		
27	XY	+ve	4.5	1.3
28		–ve		
25	X^2–Y^2	+ve	4.5	1.3
26		–ve		

SECTION 3 – MAGNET ELECTRICAL CHECKS (continued)

TABLE 3-6
WIRING TEST,
TEN-PIN CONNECTOR ON DEMOUNTABLE LEVEL PROBE

Magnet Model No.____________________ Service Engineer:____________________
Magnet Serial No.____________________ Date:____________________

Check box if ALL actual measurements match typical values: ☐



Pin	A	B	C	D	E	F	H	J	K	L	GND
A	—										(∞)
B	(27)	—									(∞)
C	(92)	(93)	—								(∞)
D	(83)	(84)	(17)	—							(∞)
E	(∞)	(∞)	(∞)	(∞)	—						(∞)
F	(∞)	(∞)	(∞)	(∞)	(∞)	—					(∞)
H	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	—				(∞)
J	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	—			(∞)
K	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(220)	—		(∞)
L	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	(∞)	—	(∞)

Note: Typical values in Ohms are shown in parentheses. Polarity is not relevant.
If readings deviate more than ± 10% from the typical values given, contact your MAC Team representative before installing the magnet.

SECTION 4
MAGNET RUNDOWN UNIT (MRU) CHECKS

WARNING!

DO NOT LIFT THE MRU COVER AND PRESS THE “EMERGENCY DISCHARGE” BUTTON UNLESS THERE IS A LIFE-THREATENING CRISIS WHICH WOULD BENEFIT FROM AN IMMEDIATE MAGNET DISCHARGE.

4-1 INTRODUCTION

The Magnet Rundown Unit (MRU) is a small power supply with battery back-up. Should an emergency occur, pressing the MRU’s red Emergency Discharge Button will cause approximately 1 amp to be delivered to a quench heater embedded in the magnet coil windings for 30 seconds (± 20%). This will warm the magnet and initiate a quench to discharge the magnet in about 1 minute.

The MRU’s power supply is backed-up by lead acid batteries. These batteries are constantly being trickle-charged while the main power is connected. **Check the battery level weekly**. If both the battery and heater lights illuminate when the battery’s green “Test” button is pressed, the battery is charged and the heater connected.

The MRU also houses timed (the B0 Dump Clock) and manual controls to heat and release the field in the magnet’s B0 shield coil. The B0 coil picks up current from the magnet’s main coil over time. The B0 coil should be manually discharged during ramping and shimming procedures and at regular intervals via the clock thereafter. **Check the B0 Dump Clock settings and battery level whenever the MRU has been without main power for more than 72 hours.**

4-2 ROUTINE MRU CHECKS

WARNING!

PERFORM THE FOLLOWING CHECK AT WEEKLY INTERVALS. IF THE CHECK FAILS, IMMEDIATELY CONTACT YOUR GENERAL ELECTRIC SERVICE REPRESENTATIVE.

At each Signa System PM:

1. Check that the “Power” LED is lit. See Illustration 4-1. If not, turn the power on.

4-2 ROUTINE MRU CHECKS (continued)

2. Press the green "Test" button.

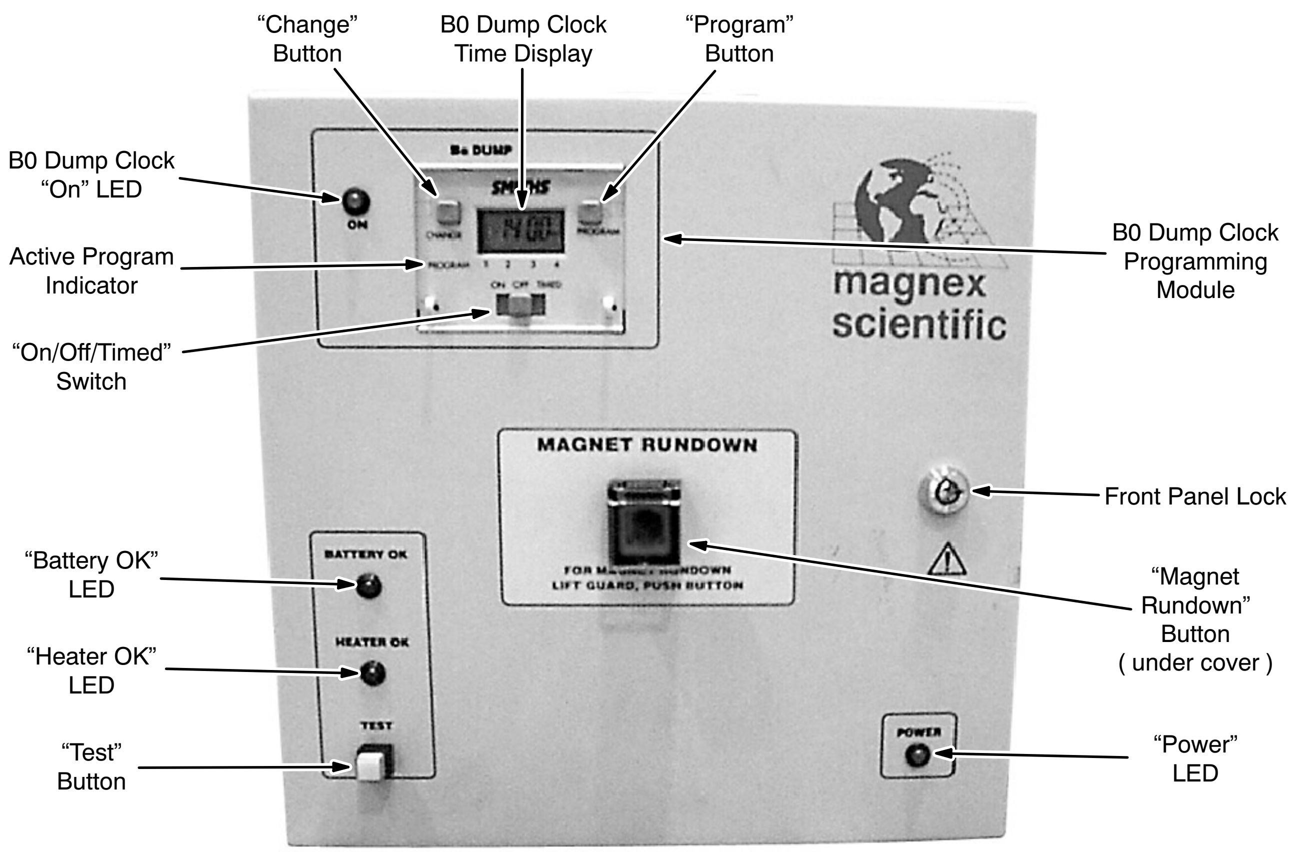
3. Check to see if both the "Heater OK" and the "Battery OK" LED's light up. If either LED fails to light, immediately call your General Electric Service Representative.

4. Make sure the B0 dump clock "ON" LED is lit, indicating the B0 Dump Clock has been set. If the LED is not lit, perform Section 4-3, Setting the B0 Dump Clock.

5. Make sure the "On/Off/Timed" slider switch on the B0 Dump Clock Programming Module is set to "Timed" and that the display shows the correct dump time. See Illustration 4-1. Reset the B0 Dump Clock in conformance with Section 4-3 as required.

Note

The B0 Coils are permanently dumped while the "On/Off/Timed" switch is in the "On" position, regardless of timer setting. The B0 Coils are never dumped while the switch is in the "Off" position. Only the "Timed" position dumps the B0 Coils according to the B0 Dump Clock.



E7006 MAGNET RUNDOWN UNIT FRONT PANEL

ILLUSTRATION 4-1

4–3 SETTING THE B0 DUMP CLOCK

CAUTION

Failure to set the B0 Dump Clock may cause the B0 coil to be permanently energized.

CAUTION

Check the B0 Dump Clock settings and the MRU battery whenever the MRU has been without main power for more than 72 hours.

1. Set the "On/Off/Timed" slider switch on the B0 Dump Clock Programming Module to the "Timed" position. See Illustration 4–1.
2. Press the "Change" and "Program" buttons simultaneousness until the timer display goes blank. Release both buttons and the display will begin flashing the hours digits.
3. Enter the local time.

   a. Press the "Change" button to advance the hour display.

   Note
   When setting current time "00:00" represents midnight on a 24 hour clock.

   b. Press the "Program" button to store the hours setting. The minutes digits will begin flashing.

   c. Press the "Change" button to advance the minutes display.

   d. Press the "Program" button to store the minutes setting.

   Note
   Once the local time is stored in the B0 Dump Clock Programming Module the Timer Display will show "– –:– –" and flash the word "ON". The module is ready to receive "Program 1."

4-3 SETTING THE B0 DUMP CLOCK (continued)

4. Set B0 Dump Clock for the “Program 1” B0 Coil dump.

Note

Read and set the B0 Dump Clock as:
“ON (begin dump) at Hour (24-hour clock)” : “OFF (stop dump) at Minute past that hour.”

a. Press the “Change” button to advance the ON hour of scheduled B0 Coil dumps. For example, “00:xx” begins scheduled B0 Coil dumps at midnight.

Note

The B0 Coil Dump should be scheduled for a time when the scanner is not in use.

b. Press the “Program” button to store the dump ON hour setting.

c. Press the “Change” button to advance the OFF minute of scheduled dumps. For example, “xx:01” stops scheduled B0 coil dumps one minute after the ON hour.

d. Press the “Program” button to store the dump OFF minute setting.

e. Press the “Change” button to advance to the next program.

5. Repeat Step 4, setting “Program 2,” “Program 3” and “Program 4” to the same time as “Program 1.” This will make sure the B0 Coil is properly dumped when “On/Off/Timed” slider switch is set to “Timed.”

Note

Once all programs have been set, pressing the “Change” button will cause the time display to show the present local time again.

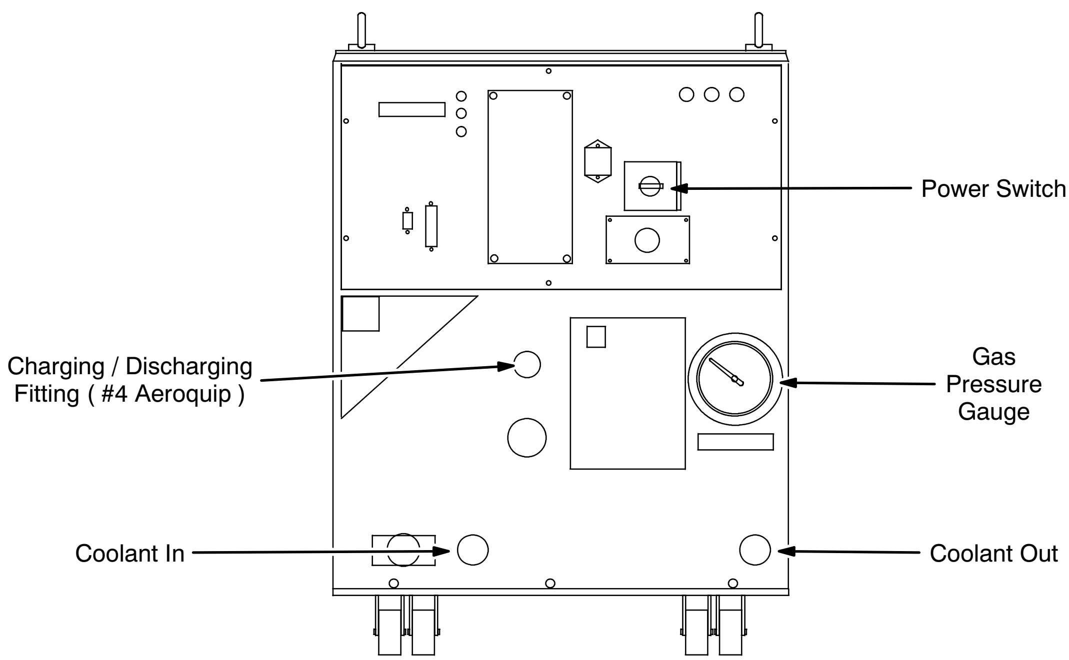
SECTION 5
SHIELD COOLER COMPRESSOR CHECKS

Introduction

The Shield Cooler system consists of a Shield Cooler Compressor, located in the equipment room, and a Coldhead thermally attached to the heat shields of the cryostat. A reduction in helium pressure in the Compressor results in reduced performance. Check the pressure level weekly.

Procedure

1. Turn the Shield Cooler Compressor off and allow the static gas pressure to equalize on the front panel pressure gauge. See Illustration 5-1.



SHIELD COOLER COMPRESSOR PRESSURE GAUGE
ILLUSTRATION 5-1

SECTION 5 – SHIELD COOLER COMPRESSOR CHECKS (continued)

2. Read and record the gauge–equalized pressure in DATA SHEETS, Section 1, Magnet Commissioning Log, under “Static Pressure Reading.” The static pressure should be between 218 psig (15 bar) and 232 psig (16 bar).

Note

If pressure is outside specification range in Step 2, refer to the vendor’s manual for troubleshooting instructions.

3. When static gas pressure is within the specification range, turn on the Shield Cooler Compressor.

4. Check the water flow to the Shield Cooler Compressor. Make sure there is a minimum flow rate of 1.0 gallons per minute (gpm) available at the supply and that the water temperature at the supply is between 40°F (5°C) and 80°F (27°C).

SECTION 6 – MAIN POWER SUPPLY TEST

6-1 INTRODUCTION

The power supply checkout procedure should always be done before ramp up, ramp down, or readjusting the magnet field after shimming. Heater Current checkout only needs to be performed when the Heater 1 Main or Heater 2 Shim Axial currents are outside the specified range (810 mA ± 10 mA). If any of the checks in this section fail, the Main Power Supply must be repaired before attempting to ramp up, ramp down or readjust the magnet field after shimming.

The 1000 amp ESS7.5-1000-2-D-1236 (46-260776G4) is now normally used since it has less ripple than the 750 amp TCR 7.5T750 (46-260776G3). Section 6-2 covers the check-out procedures for the ESS 7.5-1000-2-D-1236, and Section 6-3 covers them for the TCR 7.5T750.

6-2 ESS7.5-1000-2-D-1236 MAIN POWER SUPPLY TESTS

All ESS7.5-1000-2-D-1236 Power Supplies should be checked in conformance with this procedure at least once a year. In addition, it could be used when a ESS7.5-1000-2-D-1236 Power Supply is suspect of faulty operation and the supply cannot be sent to an approved calibration / repair facility before it is needed. This document is intended only for MAC team members who have been trained by a member of the “Power Supply Team”.

6-2-1 Equipment

- 4.5-Digit True RMS Voltmeter (Fluke 87 or equivalent)
- Oscilloscope
- Set of 3 red and 3 black Ramping Cables (for 1000 amp power supply)
- Power Supply Checkout Kit (2101360)
- Nonmetallic Tuning Tool

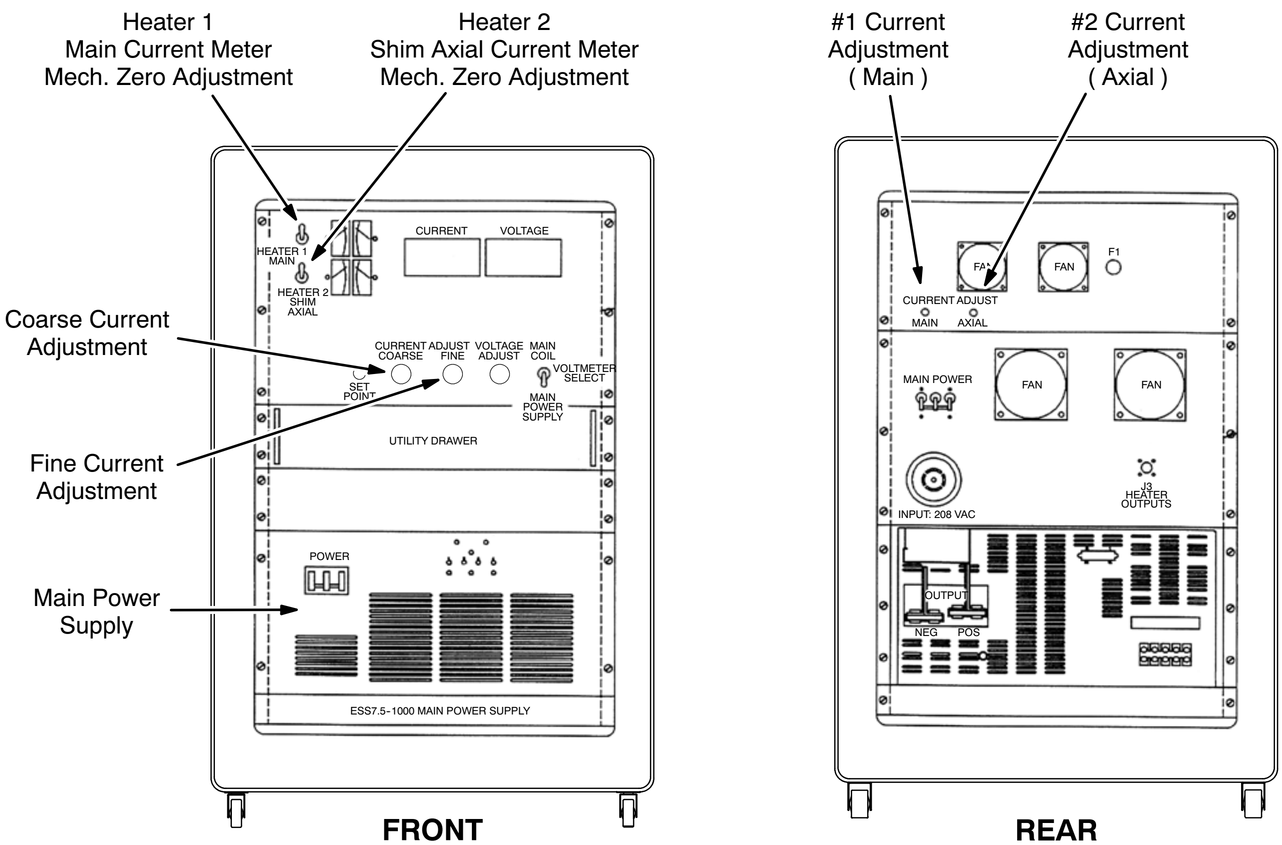
6-2-2 Heater Current Meter Zero Calibration

1. Disconnect the input power cable from the power supply to be tested.
2. Remove the cover plates to access the meter’s mechanical Zero Adjustment Screw (located adjacent to each meter). See Illustration 6-1.
3. Set the Heater 1 Main and Heater 2 Shim Axial Switches to “O” (off).

6-2-2 Heater Current Meter Zero Calibration (continued)

4. Adjust each meter's mechanical Zero Adjustment Screw as required to position the meter indicator needle at "0" (zero).

5. Replace the cover plates.



SUPERCONDUCTING MAIN COIL SERVICE POWER SUPPLY CABINET
ILLUSTRATION 6-1

6-2-3 Heater Calibration

1. Disconnect the input power cable from the power supply to be tested.

2. Connect the P3 test connector to J3, located on the rear of the power supply.

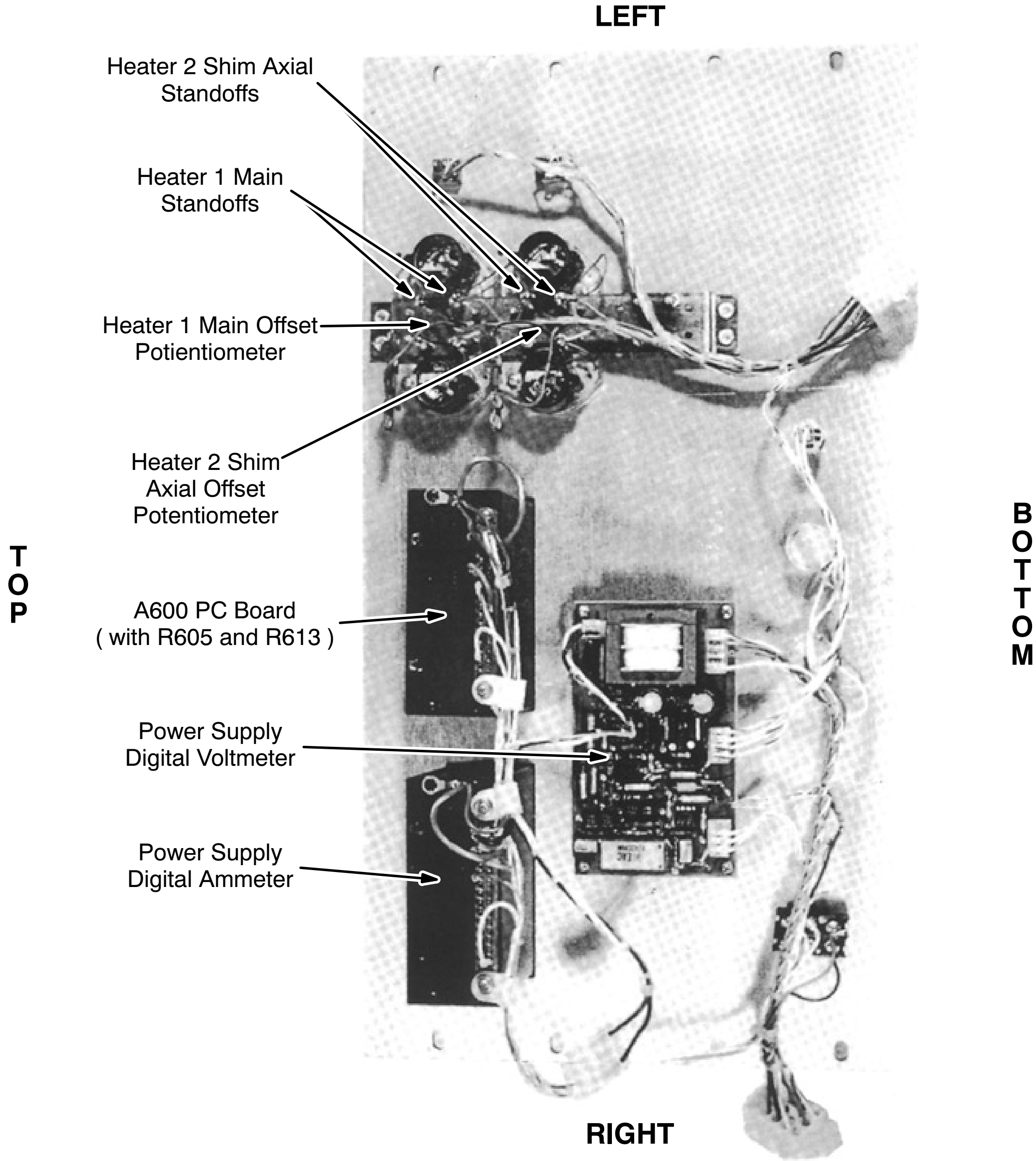
3. Remove the eight mounting screws holding the Control Panel to the power supply. See Illustration 6-1.

4. Pull out the Utility Drawer, then place the Control Panel face down on the drawer to expose the component side of the panel.

5. Connect an external DVM (set to "DC Volts") to the Heater 1 Main Standoffs, located on the component side of the Control Panel. Use the set of standoffs nearest the heater voltmeter. See Illustration 6-2.

6-2-3 Heater Calibration (continued)

6. Connect 208 VAC to AC input connector. Set the Main Power and Power On Switches to “ON.”
7. Set the Heater 1 Main Switch to “I” (on).
8. Adjust the #1 Current Adjustment Potentiometer, located on the rear of the power supply, until the external voltmeter reads 24.3 VDC. See Illustration 6-1.
9. Adjust the Heater 1 Main Meter Offset Potentiometer, located on the component side of the Control Panel, until the Heater 1 Main Current Meter reads 810 mA DC. See Illustration 6-2.



REAR VIEW CONTROL PANEL
ILLUSTRATION 6-2

6-2-3 Heater Calibration (continued)

10. Set the Heater 2 Shim Axial Switch to “I” (on).
11. Connect an external DVM (set to “DC Volts”) to the Heater 2 Shim Axial standoffs, located on the component side of the Control Panel. See Illustration 6-2.
12. Adjust the #2 Current Adjustment Potentiometer, located on the rear of the power supply, until the external voltmeter reads 24.3 VDC.
13. Adjust the Heater 2 Shim Axial Meter Offset Potentiometer, located on the component side of the Control Panel, until the Heater 2 Shim Axial Current Meter reads 810 mA DC. See Illustration 6-2.
14. Set both heater switches to “O” (off).
15. Set the Main Power and Power On Switches to “OFF.”

6-2-4 Voltage Calibration

Description

Voltage Calibration is checked under “no load” condition. Make sure the Ramp Leads are disconnected for this test.

Procedure

1. Turn off the power supply, then disconnect the power supply input power cable.
2. Remove the TB2 Access Cover Plate. See Illustration 6-3.
3. Make sure the P3 test connector is connected to J3, located on the rear of the power supply.
4. Set the DVM to the “DC Volt” scale.
5. Connect the DVM to TB2 pin 1 (positive) and to TB2 pin 7 (negative).

Note
TB2 is numbered from left to right.

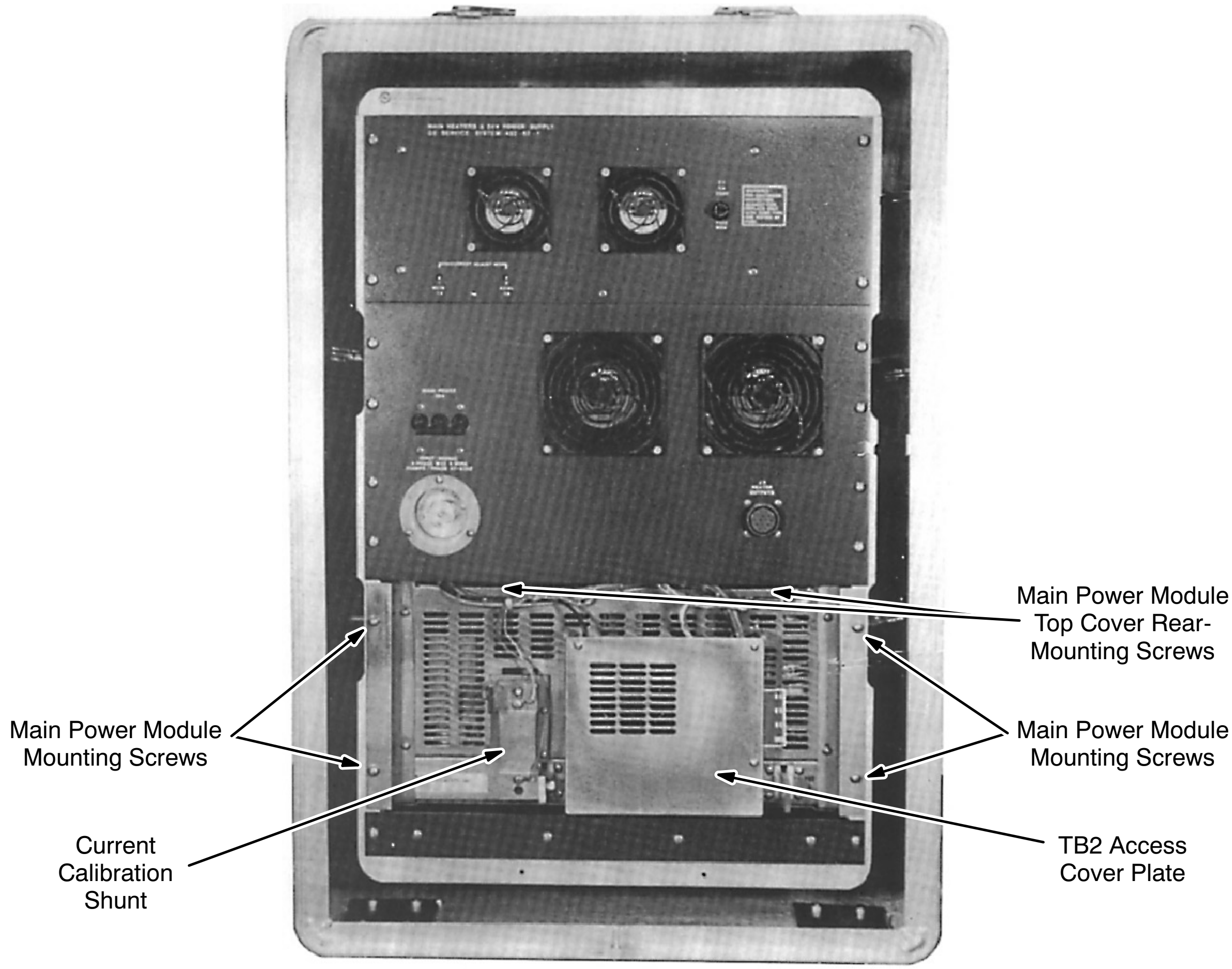
6. Adjust the Voltage Adjustment Potentiometer to minimum (full CCW) and both Current Adjustment Potentiometers to maximum (full CW).

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! WITH THE TB2 ACCESS COVER PLATE REMOVED, 208 VAC 3-PHASE ON TB1 IS EXPOSED. BE EXTREMELY CAREFUL NOT TO COME INTO CONTACT WITH TB1.

7. Connect the power supply input power cable to the power supply.

6-2-4 Voltage Calibration (continued)



MAIN POWER SUPPLY REAR VIEW
ILLUSTRATION 6-3

8. Set the Main Power and Power On Switches to “ON.”

9. Set the Heater 1 Main and Heater 2 Shim Axial Switches to “I” (on).

10. Adjust the Voltage Adjustment Potentiometer to maximum (full CW). The DVM should display within the range of 7.0 VDC and 8.0 VDC.

Note

The ideal DVM reading in Step 10 is 7.5 Volts. Voltages lower than 7.5 volts could increase ramp up time. Voltages greater than 8.0 Volts could damage output capacitors. If the voltage is outside the range indicated in Step 10, return the power supply for repair.

11. Adjust the Current Coarse Adjustment Potentiometer to minimum (full CCW).

6-2-4 Voltage Calibration (continued)

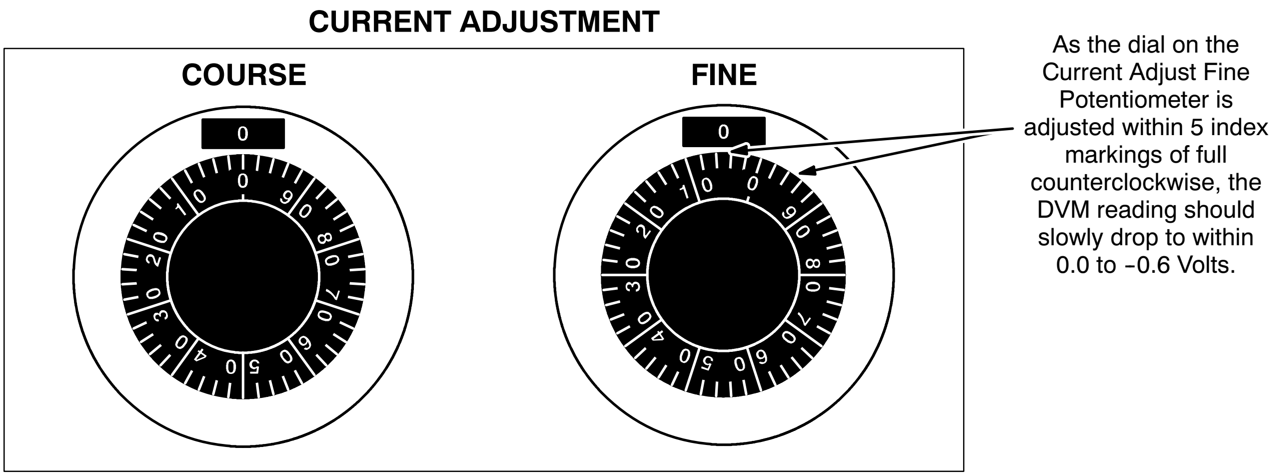
Note

The adjustment in Step 12, below, is a step function. The DVM reading will **not** vary continuously as R605 is adjusted. The function of R605 is to set the threshold at which no voltage is available at the power supply output when the Current Adjustment Potentiometers are set to minimum. If R605 is adjusted much beyond the threshold point, there will be a delay between the point at which the Current Adjustment Potentiometers are adjusted, from minimum, until there is a noticeable current output of power supply.

Note

It takes approximately 2 minutes for the DVM reading, in Step 12, to decay to a value between 0.0 and -0.6 Volts. A few trials of adjusting the Current Adjust Fine Potentiometer must be done to insure the power supply output voltage drops and rises as the Current Fine Adjust Potentiometer is adjusted in and out of the range shown in Illustration 6-4.

12. Adjust the Current Adjust Fine Potentiometer to minimum (full CCW). As the dial on the Current Adjust Fine Potentiometer is adjusted to within 5 index markings of full counterclockwise, as shown in Illustration 6-4, the DVM reading should drop to within the range of 0.0 VDC and -0.6 VDC. If necessary, adjust the R605 potentiometer, located on the A600 PC board behind the power supply Control Panel, to the point at which the DVM drop occurs. See Illustration 6-2.



CURRENT ADJUST POTENTIOMETERS
ILLUSTRATION 6-4

13. Adjust the Current Adjust Fine Potentiometer in and out of the range shown in Illustration 6-3 to insure proper adjustment of R605.

Note

The Current Fine Adjustment Potentiometer is very sensitive.

14. Reinstall the Control Panel if previously removed.

Note

The Control Panel will have to be removed again later this procedure. It is only necessary to replace a couple of the Control Panel mounting screws at this time.

6-2-5 Voltage Adjustment Potentiometer Noise Test

Description

The Voltage Adjustment Potentiometer is checked under "no load" condition. Make sure the Ramp Leads and test plug J3 are disconnected for this test.

Procedure

1. Disconnect the input power cable from the Main Coil Power Supply.
2. Measure the resistance, using an ohmmeter, from the Negative Buss Bar to any point on the power supply chassis. The resistance will increase to well within the Megohm range as the output capacitor is being charged by the ohmmeter. Send the power supply to a repair facility if the resistance check indicates a shorted or leaky output capacitor.
3. Connect the X1 Probe to the Oscilloscope.
4. Connect the X1 Probe's ground lead to the Output Bus Bar's negattive terminal and the signal lead to the positive terminal.
5. Set the Oscilloscope's "Time Base" Control to 5 msec/div (X1 Probe).
6. Set the Oscilloscope's "VOLT / DIV" Control to 100 mV / div.
7. Set the input coupling to "AC."
8. Plug the Oscilloscope power cable into an AC outlet and turn the Oscilloscope power switch on.
9. Connect input power to the Main Coil Power Supply and turn all power supply breaker switches on.

Note

The Oscilloscope beam will move upward and downward as the Voltage Adjustment Control is adjusted.

10. Adjust the Voltage Adjustment Potentiometer through its full range of operation while observing the Oscilloscope. The signal should be free from spikes. A bad potentiometer will intermittently display large spikes as the potentiometer is adjusted through its full range. See Illustration 6-5 for an example of a bad pot. Do not confuse spikes or noise seen while the pot is **not** being adjusted with potentiometer noise.

Note

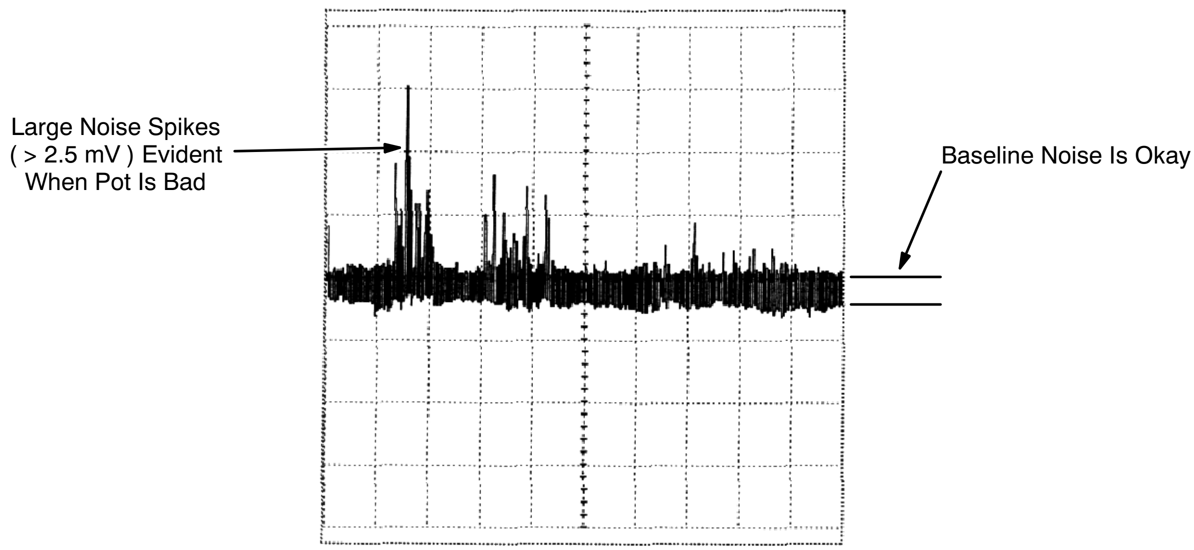
A storage type oscilloscope, if properly used, will greatly aid in determining if a pot is excessively noisy.

Note

Spikes seen riding on the baseline at regular intervals (baseline noise) are caused by switching circuits in the main power module. Do not confuse these spikes as potentiometer noise. To isolate spikes of less than 1 Volt, make the probe and ground lengths as short as possible.

6-2-5 Voltage Adjustment Potentiometer Noise Test (continued)

11. If the potentiometer fails this test, turn off and disconnect input power to the power supply and replace the pot. The Voltage Adjustment Potentiometer (46-281468P12) is 5K 10-turn.
12. Adjust the Voltage Adjustment Potentiometer to minimum (CCW).
13. Set the power supply Main Power and Power On Switches to "OFF."



POTENTIOMETER NOISE
ILLUSTRATION 6-5

6-2-6 Current Calibration and Current Potentiometer Noise Check

Description

The external Current Shunt (2101358) should be replaced or calibrated at least once a year. If dropped, it could be forced out of tolerance and should be replaced. The resistance tolerance of the shunt is ± 0.1%. This means that the power supply output could be 750 ± 0.75 amps at rated current. The DVM used in the following section should be a Fluke 87 4.5-digit Digital Voltmeter (DVM) or equivalent.

Procedure

1. Plug the Oscilloscope power cable into an AC outlet and turn the Oscilloscope power switch on.
2. Connect the P3 test connector to J3, located on the rear of the power supply.
3. Set the Main Power and Power On Switches to "ON."
4. Set the Heater 1 Main and Heater 2 Shim Axial Switches to "I" (on).

6-2-6 Current Calibration and Current Potentiometer Noise Check (continued)

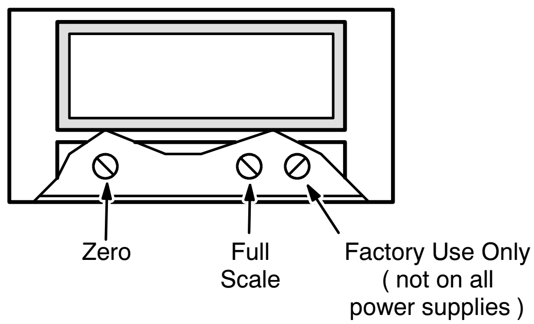
5. Adjust the Current Adjustment and Voltage Adjustment potentiometers to minimum (full CCW)

6. Remove the snap-on cover plate from the power supply's Current Meter.

Note
Step 7 must be performed with no load (i.e., ramp cables disconnected).

7. Adjust the Current Meter Zero "trim pot" on the lower left corner of the meter to 000.0 Amps. This potentiometer is located behind the snap-on cover plate removed in Step 6. See Illustration 6-6.

CURRENT METER



POWER SUPPLY CURRENT METER ADJUSTMENT POTS
ILLUSTRATION 6-6

8. Set both heater switches to "O" (off).

9. Set the Main Power and Power On Switches to "OFF."

10. Connect the positive and negative Ramp Cables to the output buss bars on the magnet power supply.

CAUTION

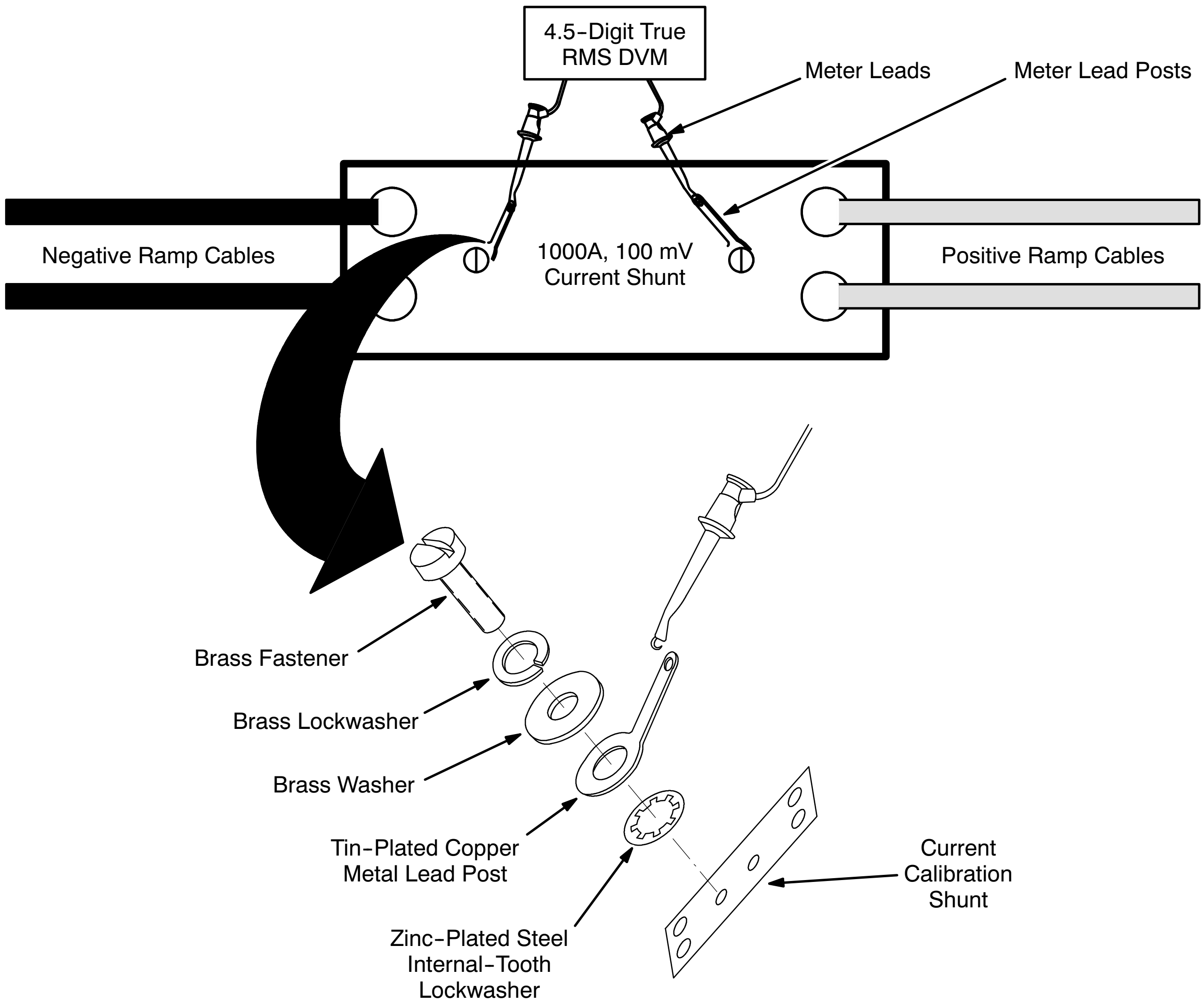
Heat in excess of 60 watts could be present at the junction between the Ramp Cable and the 1000 amp 100mV Current Shunt if the cables in Steps 11 and 12 are not tightly connected to the Current Shunt.

Because of the high current capacity of the power supply, jewelry, such as rings, should not be worn during this section of the checkout procedure as a potential burn hazard exists.

Place the Current Shunt on brick or ceramic material to avoid heat damage to tile or carpet.

11. Tightly bolt the other end of the Positive Ramp Cables to one side of the 1000 amp 100mV Current Shunt. See Illustration 6-7.

6-2-6 Current Calibration and Current Potentiometer Noise Check (continued)



CURRENT SHUNT CONNECTIONS
ILLUSTRATION 6-7

12. Tightly bolt the other end of the Negative Ramp Cables to the other side of the 1000 amp 100mV Current Shunt.
13. Set the Heater 1 Main and Heater 2 Shim Axial Switches to “O” (off).
14. Adjust the Voltage Control Potentiometer to maximum (full CW). Make sure the Current Adjustment Potentiometers are set to minimum (full CCW).

6-2-6 Current Calibration and Current Potentiometer Noise Check (continued)

15. Install the Meter Lead Posts, fasteners and washers to external Current Shunt as shown in Illustration 6-7. Make sure Meter Lead Post fasteners are tight.

Note

Erroneous readings will result in Steps 24 through 26 below if the Meter Lead Post fasteners are not tight. All materials used for the Meter Lead Posts, fasteners, and washers were chosen to avoid the possibility of a thermocouple action.

16. Connect a 4.5-digit true RMS Digital Voltmeter (DVM) to the Meter Lead Posts on the Current Shunt. See Illustration 6-7.
17. Set the DVM to read DC millivolts.
18. Connect the X1 Probe to the Oscilloscope, then connect the X1 Probe's ground lead to the Output Bus Bar's negative terminal and the signal lead to the positive terminal.
19. Set the Oscilloscope "Time Base" Control to 5 msec/div.
20. Set the Oscilloscope "VOLT / DIV" Control to 100 mV / div.
21. Set the input coupling to AC.
22. Set the power supply Main Power and Power On Switches to "ON."
23. Set the Heater 1 Main and Heater 2 Shim Axial Switches to "I" (on).

Note

Each millivolt increment on the DVM corresponds to 10 amps. Make sure the meter used in this procedure is capable of reading to three decimal places.

24. Adjust the Current Adjust Fine / Coarse Controls to maximum (full CW). The external DVM should indicate a minimum of 75.0 mVDC. If this voltage is low, maximum rated current (750 amps) will not be achievable, and the power supply should be returned for repair. Include a written description of the problem when returning power supplies.
25. Adjust the Current Adjustment Controls to obtain a reading of 75.0 mVDC on the external DVM. This corresponds to a power supply output of 750 amps. Adjust R613, located on the A600 board, to display a reading of 750 amps on the power supply's digital Current Meter. If a reading of 750 amps cannot be achieved, perform Step 26 below.

Note

Perform Step 26 only if a reading of 750 amps on the power supply's digital Current Meter cannot be achieved by adjusting R613.

26. Adjust the power supply's Current Meter's "Full Scale" Trim Potentiometer for 750 Amps. See Illustration 6-6.

6-2-6 Current Calibration and Current Potentiometer Noise Check (continued)

27. Adjust the Current Adjustment Potentiometers through the full range of operation while observing the Oscilloscope. The signal should be free from spikes. See Illustration 6-5 for an example of a bad pot.

28. If either potentiometer fails this test, turn off and disconnect the power supply input power and replace the potentiometer. The Current Adjustment Potentiometer is a 5K 10 turn pot. The Current Adjustment Fine Potentiometer is a 100 ohm 10 turn pot. Extra potentiometers are included in the Power Supply Checkout Kit (2101360).

29. Adjust all Current and Voltage Adjustment Potentiometers to minimum (full CCW).

30. Set the Heater 1 Main and Heater 2 Shim Axial Switches to “O” (off).

31. Set the Power On and Main Power Switches to “OFF.”

32. Reinstall the power supply Control Panel.

6-3 TCR 7.5T750 MAIN POWER SUPPLY TESTS

All TRC 7.5T750 Power Supplies should be checked in conformance with this procedure at least once a year. In addition, use this procedure when a TRC 7.5T750 Power Supply is suspected of faulty operation and the supply cannot be sent to an approved calibration / repair facility before it is needed. This document is intended only for MAC team members who have been trained by a member of the “Power Supply Team.”

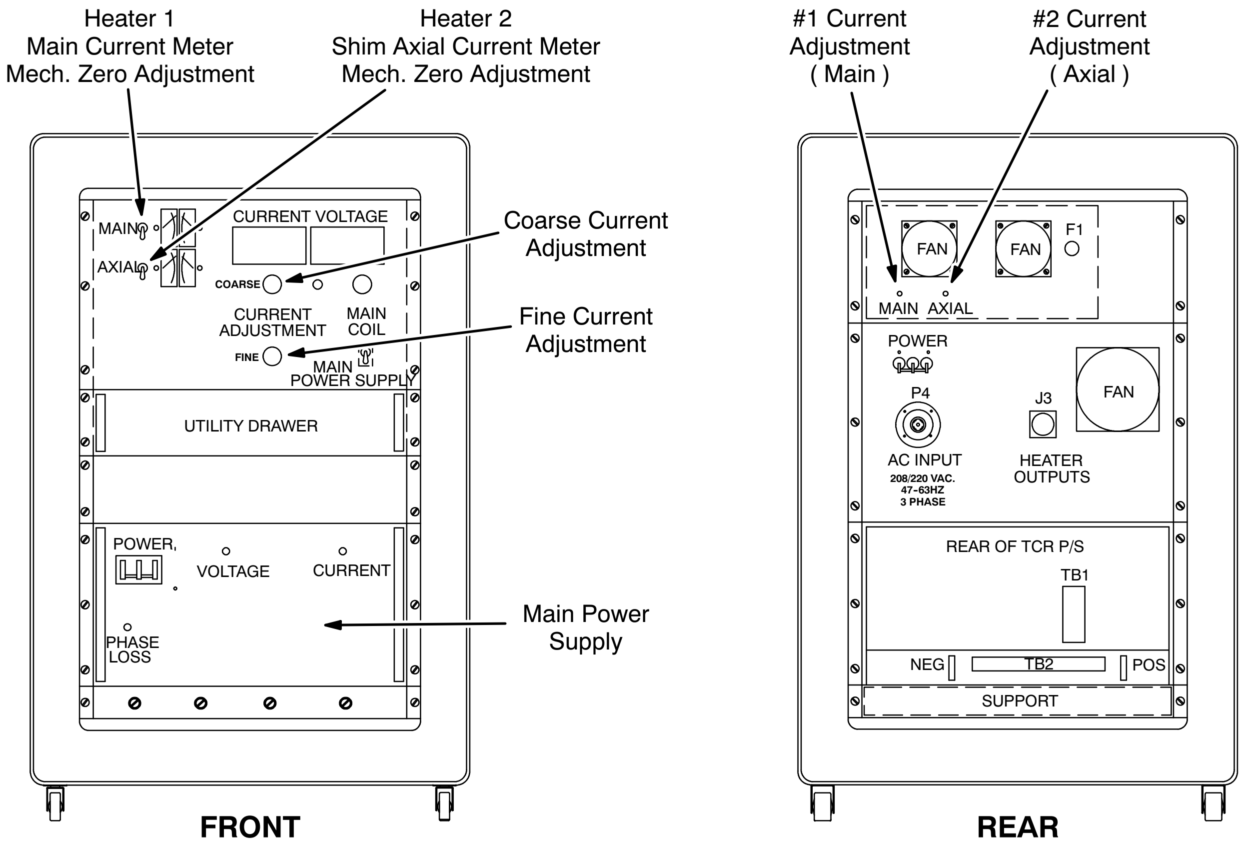
6-3-1 Equipment

- 4.5-Digit True RMS Voltmeter (Fluke 87 or equivalent)
- Oscilloscope
- Set of 2 red and 2 black Ramping Cables (for 750 amp power supply)
- Power Supply Checkout Kit (2101360)
- Nonmetallic Tuning Tool

6-3-2 Heater Current Meter Zero Calibration

1. Disconnect the input power cable from the power supply to be tested.
2. Remove the cover plates to access the meter’s mechanical Zero Adjustment Screw (located adjacent to each meter). See Illustration 6-8.
3. Set the Heater 1 Main and Heater 2 Shim Axial Switches to “O” (off).
4. Adjust each meter’s mechanical Zero Adjustment Screw as required to position the meter indicator needle at “0” (zero).
5. Replace the cover plates.

6–3–2 Heater Current Meter Zero Calibration (continued)



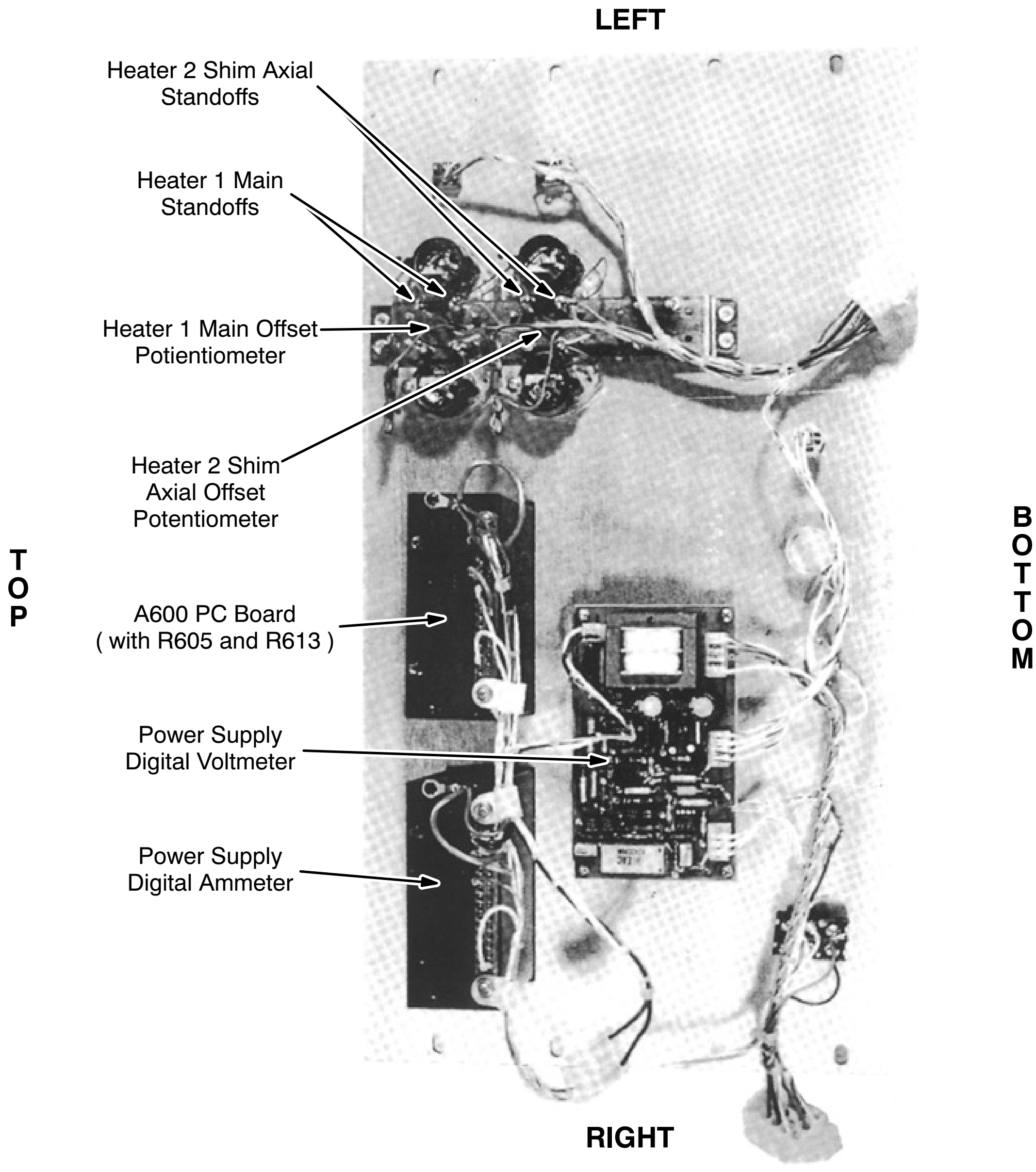
750 AMP TCR 7.75T750 SUPERCONDUCTING MAIN COIL SERVICE POWER SUPPLY CABINET
ILLUSTRATION 6–8

6–3–3 Heater Calibration

1. Disconnect the input power cable from the power supply to be tested.
2. Connect the P3 test connector to J3, located on the rear of the power supply.
3. Remove the eight mounting screws holding the Control Panel to the power supply. See Illustration 6–8.
4. Pull out the Utility Drawer, then place the Control Panel face down on the drawer to expose the component side of the panel.
5. Connect an external DVM (set to “DC Volts”) to the Heater 1 Main Standoffs, located on the component side of the Control Panel. Use the set of standoffs nearest the heater voltmeter. See Illustration 6–9.
6. Connect 208 VAC to the AC input connector. Set the Main Power and Power On Switches to “ON.”
7. Set the Heater 1 Main Switch to “I” (on).

6-3-3 Heater Calibration (continued)

8. Adjust the #1 Current Adjustment Potentiometer, located on the rear of the power supply, until the external voltmeter reads 24.3 VDC. See Illustration 6-8.
9. Adjust the Heater 1 Main Meter Offset Potentiometer, located on the component side of the Control Panel, until the Heater 1 Main Current Meter reads 810 mA DC. See Illustration 6-9.
10. Set the Heater 2 Shim Axial Switch to "I" (on).
11. Connect an external DVM (set to "DC Volts") to the Heater 2 Shim Axial standoffs, located on the component side of the Control Panel. See Illustration 6-9.



REAR VIEW CONTROL PANEL
ILLUSTRATION 6-9

6-3-3 Heater Calibration (continued)

12. Adjust the #2 Current Adjustment Potentiometer, located on the rear of the power supply, until the external voltmeter reads 24.3 VDC.
13. Adjust the Heater 2 Shim Axial Meter Offset Potentiometer, located on the component side of the Control Panel, until the Heater 2 Shim Axial Current Meter reads 810 mA DC. See Illustration 6-9.
14. Set both heater switches to “O” (off).
15. Set the Main Power and Power On Switches to “OFF.”

6-3-4 Voltage Calibration

Description

Voltage Calibration is checked under “no load” condition. Make sure the Ramp Leads are disconnected for this test.

Procedure

1. Turn off the power supply, then disconnect the power supply input power cable.
2. Remove the TB2 Access Cover Plate. See Illustration 6-10.
3. Make sure the P3 test connector is connected to J3, located on the rear of the power supply.
4. Set the DVM to the “DC Volt” scale.
5. Connect the DVM to TB2 pin 1 (positive) and to TB2 pin 7 (negative).

Note
TB2 is numbered from left to right.

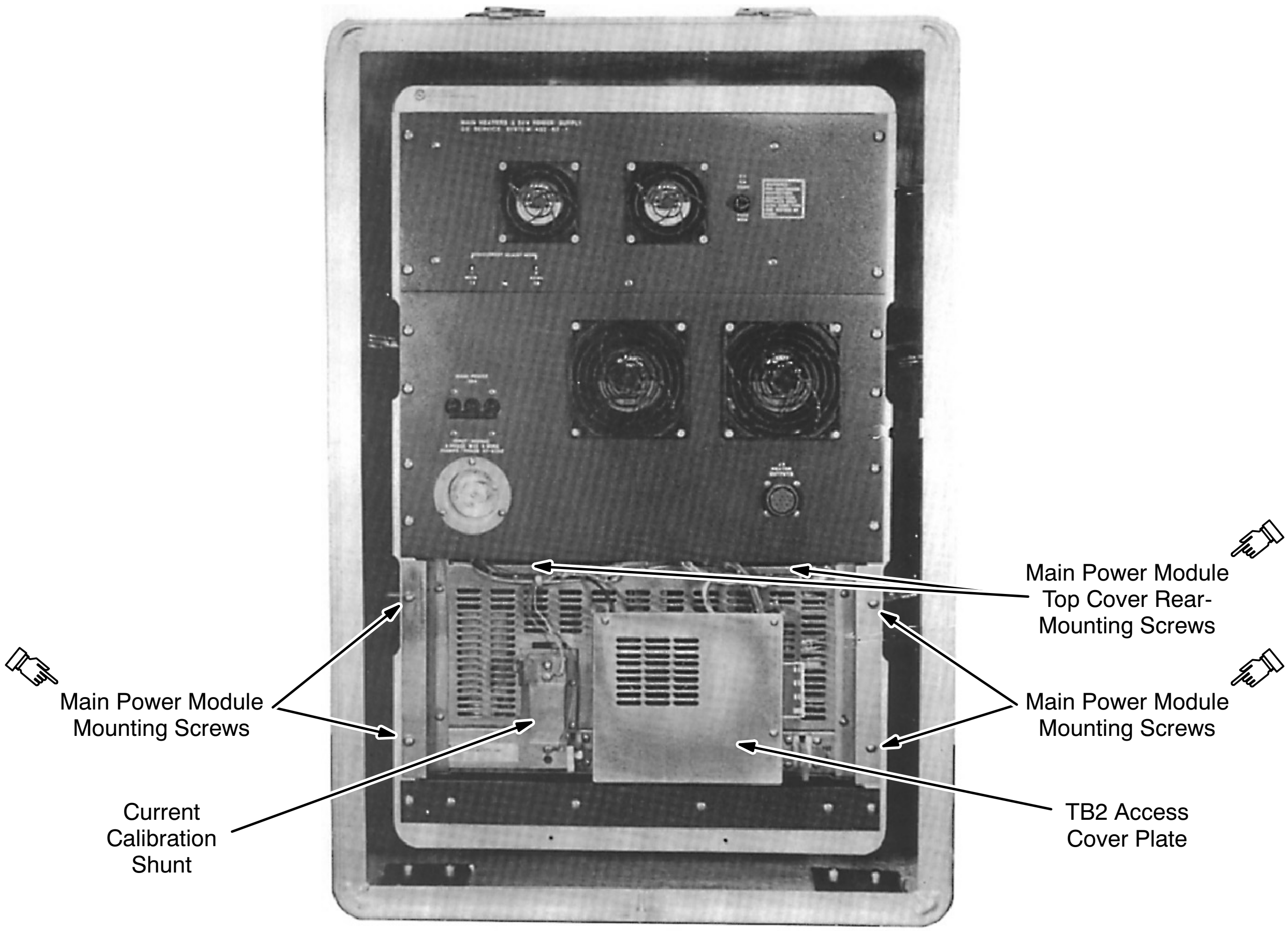
6. Adjust the Voltage Adjustment Potentiometer to minimum (full CCW) and both Current Adjustment Potentiometers to maximum (full CW).

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! WITH THE TB2 ACCESS COVER PLATE REMOVED, 208 VAC 3-PHASE ON TB1 IS EXPOSED. BE EXTREMELY CAREFUL NOT TO COME INTO CONTACT WITH TB1.

7. Connect the power supply input power cable to the power supply.
8. Set the Main Power and Power On Switches to “ON.”

6–3–4 Voltage Calibration (continued)



MAIN POWER SUPPLY REAR VIEW
ILLUSTRATION 6–10

9. Set the Heater 1 Main and Heater 2 Shim Axial Switches to “I” (on).

10. Adjust the Voltage Adjustment Potentiometer to maximum (full CW). The DVM should display within the range of 7.0 to 8.0 VDC.

Note

The ideal DVM reading in Step 10 is 7.5 Volts. Voltages lower than 7.5 volts could increase ramp up time. Voltages greater than 8.0 Volts could damage output capacitors. If the voltage is outside the range indicated in Step 10, return the power supply for repair.

11. Adjust the Current Coarse Adjustment Potentiometer to minimum (full CCW).

6-3-4 Voltage Calibration (continued)

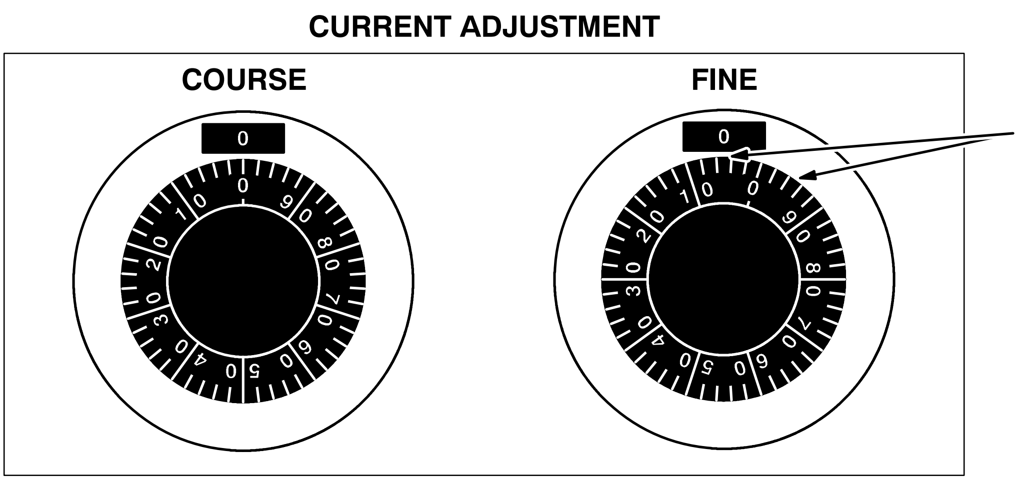
Note

The adjustment in Step 12, below, is a step function. The DVM reading will **not** vary continuously as R605 is adjusted. The function of R605 is to set the threshold at which no voltage is available at the power supply output when the Current Adjustment Potentiometers are set to minimum. If R605 is adjusted much beyond the threshold point, there will be a delay between the point at which the Current Adjustment Potentiometers are adjusted, from minimum, until there is a noticeable current output of power supply.

Note

It takes approximately 2 minutes for the DVM reading, in Step 12, to decay to a value between 0.0 and −0.6 Volts. A few trials of adjusting the Current Adjust Fine Potentiometer must be done to insure the power supply output voltage drops and rises as the Current Fine Adjust Potentiometer is adjusted in and out of the range shown in Illustration 6-11.

12. Adjust the Current Adjust Fine Potentiometer to minimum (full CCW). As the dial on the Current Adjust Fine Potentiometer is adjusted to within 5 index markings of full counterclockwise, as shown in Illustration 6-11, the DVM reading should drop to within the range of 0.0 VDC and −0.6 VDC. If necessary, adjust the R605 potentiometer, located on the A600 PC board behind the power supply Control Panel, to the point at which the DVM drop occurs. See Illustration 6-9.



CURRENT ADJUST POTENTIOMETERS
ILLUSTRATION 6-11

13. Adjust the Current Adjust Fine Potentiometer in and out of the range shown in Illustration 6-11 to insure proper adjustment of R605.

Note

The Current Fine Adjustment Potentiometer is very sensitive.

14. Reinstall the Control Panel if previously removed.

Note

The Control Panel will have to be removed again later this procedure. It is only necessary to replace a couple of the Control Panel mounting screws at this time.

6-3-5 Voltage Adjustment Potentiometer Noise Test

Description

The Voltage Adjustment Potentiometer is checked under "no load" condition. Make sure the Ramp Leads are disconnected for this test.

Procedure

1. Disconnect the input power cable from the Main Coil Power Supply.
2. Measure the resistance, using an ohmmeter, from the Negative Buss Bar to any point on the power supply chassis. The resistance will increase to well within the Megohm range as the output capacitor is being charged by the ohmmeter. Send the power supply to a repair facility if the resistance check indicates a shorted or leaky output capacitor.
3. Remove the mounting screws securing the TB2 Access Cover Plate. The plate is located between the output buss bars on the bottom rear of the power supply and protects a fan and terminal board TB2. See Illustration 6-10.
4. Connect the X1 Probe to the Oscilloscope.
5. Connect the X1 Probe ground lead to TB2 pin 7 and the signal lead on TB2 pin 3.

Note
TB2 is numbered from left to right.

6. Set the Oscilloscope's "Time Base" Control to 5 msec/div (X1 Probe).
7. Set the Oscilloscope's "VOLT / DIV" Control to 5 mV / div.
8. Set the input coupling to "AC."
9. Plug the Oscilloscope power cable into an AC outlet and turn the Oscilloscope power switch on.
10. Connect input power to the Main Coil Power Supply and turn all power supply breaker switches on.

Note
The Oscilloscope beam will move upward and downward as the Voltage Adjustment Control is adjusted.

6-3-5 Voltage Adjustment Potentiometer Noise Test (continued)

11. Adjust the Voltage Adjustment Potentiometer through its full range of operation while observing the Oscilloscope. The signal should be free from spikes. A bad potentiometer will intermittently display large spikes as the potentiometer is adjusted through its full range. See Illustration 6-12 for an example of a bad pot. Do not confuse spikes or noise seen while the pot is **not** being adjusted with potentiometer noise.

Note

A storage type oscilloscope, if properly used, will greatly aid in determining if a pot is excessively noisy.

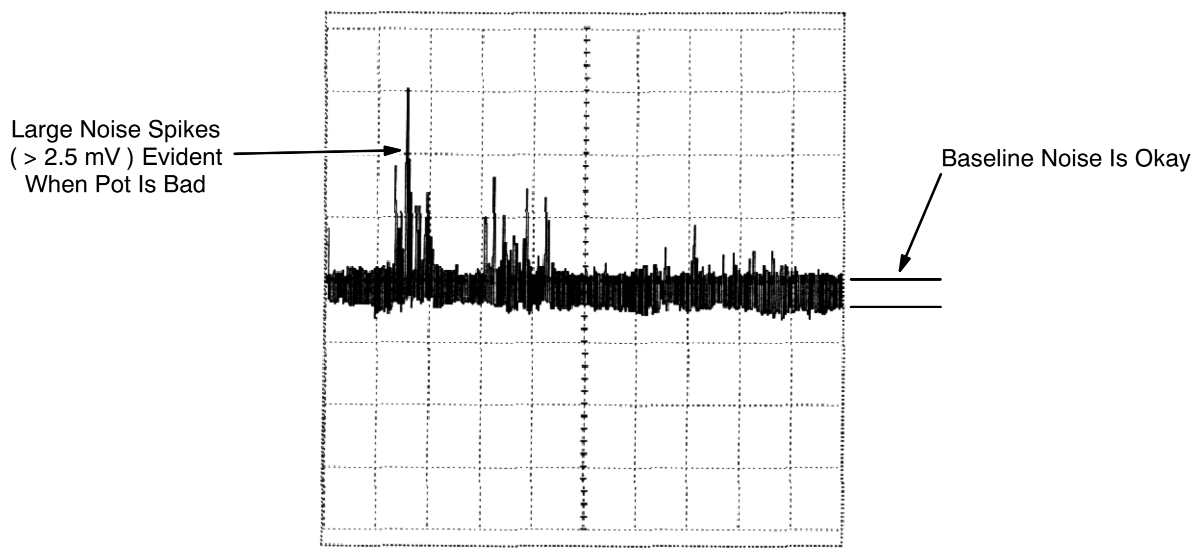
Note

Spikes seen riding on the baseline at regular intervals (baseline noise) are caused by switching circuits in the main power module. Do not confuse these spikes as potentiometer noise.

12. If the potentiometer fails this test, turn off and disconnect input power to the power supply and replace the pot. The Voltage Adjustment Potentiometer (46-281468P12) is 5K 10-turn.

13. Adjust the Voltage Adjustment Potentiometer to minimum (CCW).

14. Set the power supply Main Power and Power On Switches to “OFF.”



POTENTIOMETER NOISE
ILLUSTRATION 6-12

6-3-6 Current Calibration and Current Potentiometer Noise Check

Description

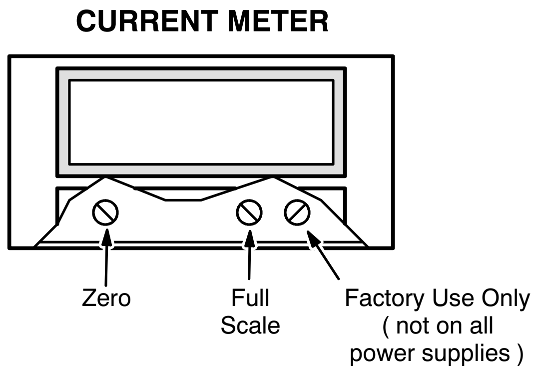
The external Current Shunt (2101358) should be replaced or calibrated at least once a year. If dropped, it could be forced out of tolerance and should be replaced. The resistance tolerance of the shunt is ± 0.1%. This means that the power supply output could be 750 ± 0.75 amps at rated current. The DVM used in the following section should be a Fluke 87 4.5-digit Digital Voltmeter (DVM) or equivalent.

Procedure

1. Plug the Oscilloscope power cable into an AC outlet and turn the Oscilloscope power switch on.
2. Connect the P3 test connector to J3, located on the rear of the power supply.
3. Set the Main Power and Power On Switches to “ON.”
4. Set the Heater 1 Main and Heater 2 Shim Axial Switches to “I” (on).
5. Adjust the Current Adjustment and Voltage Adjustment potentiometers to minimum (full CCW)
6. Remove the snap-on cover plate from the power supply’s Current Meter.

Note
Step 7 must be performed with no load (i.e., ramp cables disconnected).

7. Adjust the Current Meter Zero “trim pot” on the lower left corner of the meter to 000.0 Amps. This potentiometer is located behind the snap-on cover plate removed in Step 6. See Illustration 6-13.



POWER SUPPLY CURRENT METER ADJUSTMENT POTS
ILLUSTRATION 6-13

8. Set both heater switches to “O” (off).
9. Set the Main Power and Power On Switches to “OFF.”
10. Connect the positive and negative Ramp Cables to the output buss bars on the magnet power supply.

6-3-6 Current Calibration and Current Potentiometer Noise Check (continued)

CAUTION

Heat in excess of 60 watts could be present at the junction between the Ramp Cable and the 1000 amp 100mV Current Shunt if the cables in Steps 11 and 12 are not tightly connected to the Current Shunt.

Because of the high current capacity of the power supply, jewelry, such as rings, should not be worn during this section of the checkout procedure as a potential burn hazard exists.

Place the Current Shunt on brick or ceramic material to avoid heat damage to tile or carpet.

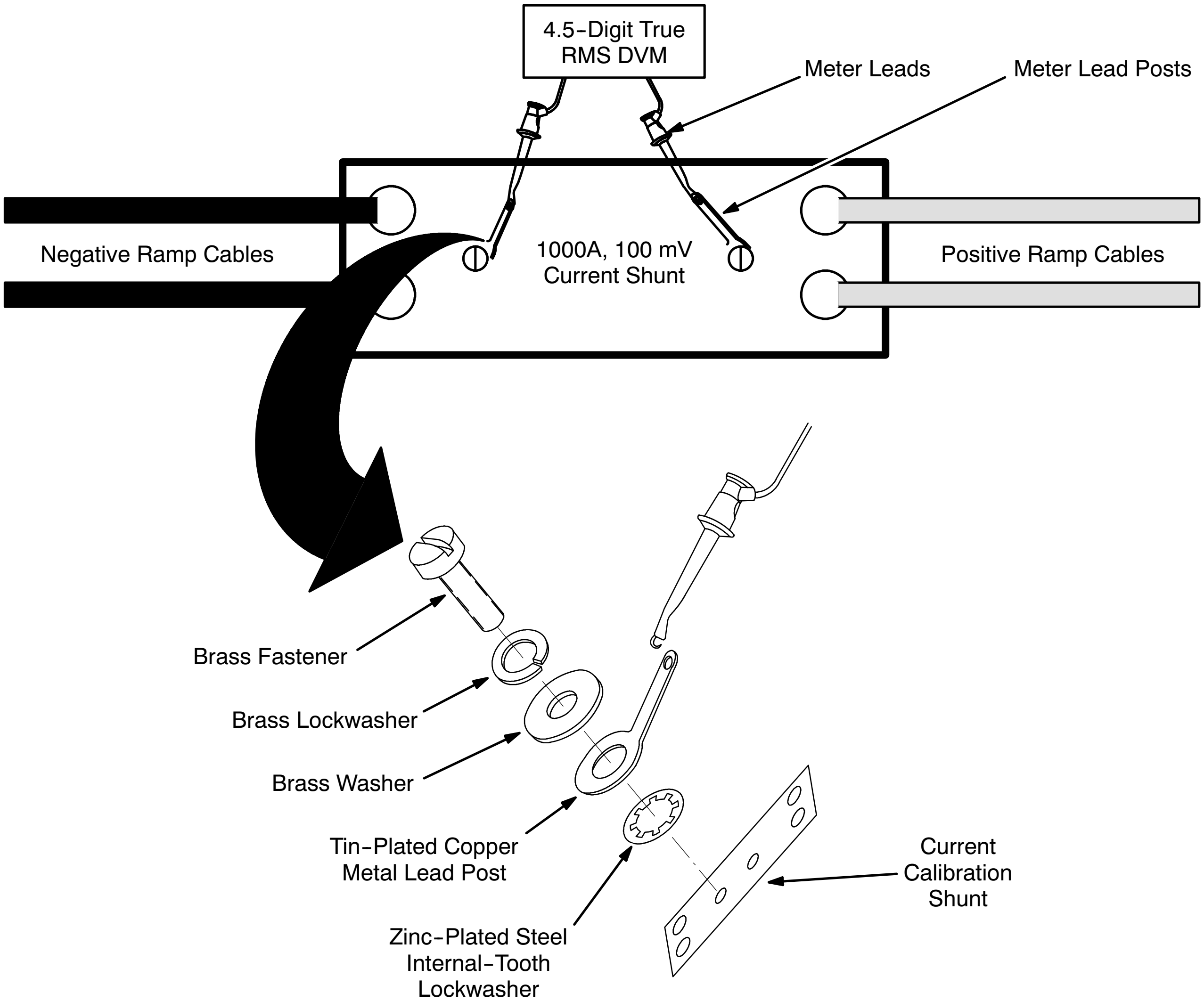
11. Tightly bolt the other end of the Positive Ramp Cables to one side of the 1000 amp 100mV Current Shunt. See Illustration 6-14.
12. Tightly bolt the other end of the Negative Ramp Cables to the other side of the 1000 amp 100mV Current Shunt.
13. Set the Heater 1 Main and Heater 2 Shim Axial Switches to "O" (off).
14. Adjust the Voltage Control Potentiometer to maximum (full CW). Make sure the Current Adjustment Potentiometers are set to minimum (full CCW).
15. Install the Meter Lead Posts, fasteners and washers to external Current Shunt as shown in Illustration 6-14. Make sure Meter Lead Post fasteners are tight.

Note

Erroneous readings will result in Steps 24 through 26 below if the Meter Lead Post fasteners are not tight. All materials used for the Meter Lead Posts, fasteners, and washers were chosen to avoid the possibility of a thermocouple action.

16. Connect a 4.5-digit true RMS Digital Voltmeter (DVM) to the Meter Lead Posts on the Current Shunt. See Illustration 6-14.
17. Set the DVM to read DC millivolts.
18. Connect an Oscilloscope to TB2 pins 9 (+) and 7 (-).
19. Set the Oscilloscope "Time Base" Control to 5 msec/div.
20. Set the Oscilloscope "VOLT / DIV" Control to 5 mV / div.
21. Set the input coupling to AC.

6-3-6 Current Calibration and Current Potentiometer Noise Check (continued)



CURRENT SHUNT CONNECTIONS
ILLUSTRATION 6-14

22. Set the power supply Main Power and Power On Switches to “ON.”
23. Set the Heater 1 Main and Heater 2 Shim Axial Switches to “I” (on).

Note

Each millivolt increment on the DVM corresponds to 10 amps. Make sure the meter used in this procedure is capable of reading to three decimal places.

24. Adjust the Current Adjust Fine / Coarse Controls to maximum (full CW). The external DVM should indicate a minimum of 75.0 mVDC. If this voltage is low, maximum rated current (750 amps) will not be achievable, and the power supply should be returned for repair. Include a written description of the problem when returning power supplies.

6-3-6 Current Calibration and Current Potentiometer Noise Check (continued)

25. Adjust the Current Adjustment Controls to obtain a reading of 75.0 mVDC on the external DVM. This corresponds to a power supply output of 750 amps. Adjust R613, located on the A600 board, to display a reading of 750 amps on the power supply's digital Current Meter. If a reading of 750 amps cannot be achieved, perform Step 26 below.

Note

Perform Step 26 only if a reading of 750 amps on the power supply's digital Current Meter cannot be achieved by adjusting R613.

26. Adjust the power supply's Current Meter's "Full Scale" Trim Potentiometer for 750 Amps. See Illustration 6-13.
27. Adjust the Current Adjustment Potentiometers through the full range of operation while observing the Oscilloscope. The signal should be free from spikes. See Illustration 6-12 for an example of a bad pot.
28. If either potentiometer fails this test, turn off and disconnect the power supply input power and replace the potentiometer. The Current Adjustment Potentiometer is a 5K 10 turn pot. The Current Adjustment Fine Potentiometer is a 100 ohm 10 turn pot. Extra potentiometers are included in the Power Supply Checkout Kit (2101360).
29. Adjust all Current and Voltage Adjustment Potentiometers to minimum (full CCW).
30. Set the Heater 1 Main and Heater 2 Shim Axial Switches to "O" (off).
31. Set the Power On and Main Power Switches to "OFF."
32. Reinstall the power supply Control Panel.

6-3-7 Voltage Ripple Check

Description

Typical voltage ripple values, using this procedure, are ≤ 30mV RMS. To achieve the lowest voltage ripple reading the power supply being tested must be warmed up for at least 15 minutes before checking and / or adjustment. This test is to be performed using a True RMS Voltmeter.

Procedure

1. Connect the positive and negative Ramp Cables to the main power supply Output Buss Bars.
2. Connect the P3 test connector to J3, located on the rear of the power supply.

6–3–7 Voltage Ripple Check (continued)

WARNING!

THE RAMP CABLES COULD CAUSE A SLIGHT BURN AT THE POSITIVE TO NEGATIVE CABLE JUNCTION (WHERE THE TWO CABLES ARE BOLTED TOGETHER). MAKE SURE THE CONNECTION IS TIGHT TO AVOID RESISTANCE, WHICH CAN CAUSE EXCESSIVE HEAT.

3. Bolt the other ends of the positive and negative Ramp Cables together.
4. Set a True RMS Voltmeter to its AC millivolt scale.
5. Connect the Voltmeter to the main power supply Buss Bars.
6. Set the Main Power and Power On Switches to “ON.”
7. Set the Heater 2 Shim Axial Switches to “I” (on).
8. Adjust the Current Adjustment and Voltage Adjustment Potentiometers to maximum (full CW).
9. Monitor the Voltmeter, after at least 15 minutes to allow power supply to sufficiently warm up, for a “voltage ripple” indication.
10. If the voltage ripple is > 35 millivolts, send the main power supply for repair.
11. Adjust the Voltage Adjustment and Current Adjustment Potentiometers to minimum (CCW).
12. Set the Heater 1 Main and Heater 2 Shim Axial Switches to “I” (off).
13. Set the power supply Main Power and Power On Switches to “OFF.”

SECTION 7
TEMPERATURE SENSOR MEASUREMENT

7-1 INTRODUCTION

Log the Cryostat Shield temperature in DATA SHEETS, Section 6, Temperature Sensor Log. Any increasing temperature trend indicates Shield Cooler performance is degenerating.

The resistance of each temperature sensor is found by taking two 2-wire resistance measurements with a Digital Volt-Ohm Meter and subtracting to eliminate lead resistance.

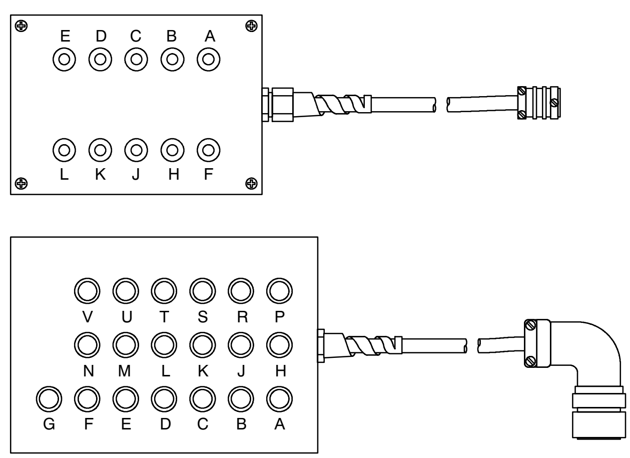
Note
Resistance vs. temperature tables are included here as Tables 7-2 through 7-4.

7-2 EQUIPMENT

- Digital Volt-Ohm Meter (DVOM) with output ≤ 10mA.
- 10-Pin Breakout Box 2295140 (See Illustration 7-1.)
- 19-Pin Breakout Box 2295141 (See Illustration 7-1.)

10-Pin Breakout Box 2295140
(For 10-pin connectors
shown in Illustration 7-3.)

19-Pin Breakout Box 2295141
(For the 19-Pin Service Turret Connector
shown in Illustration 7-3.)

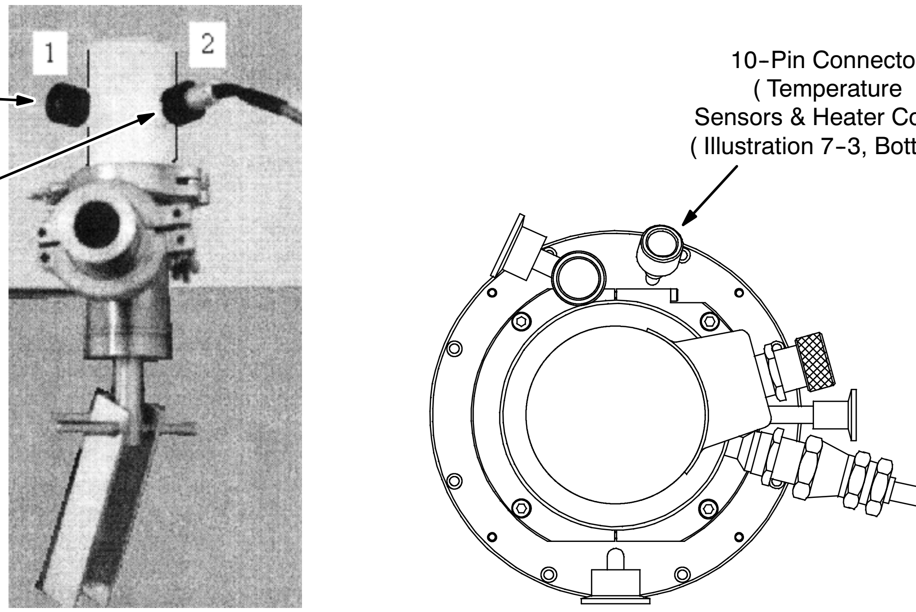


Note
Breakout box socket lettering directly correlates to connector pin identification.

10-PIN (2295140) AND 19-PIN (2295141) BREAKOUT BOXES
ILLUSTRATION 7-1

7-3 PROCEDURE

1. Access the Pump-Out Port. See Illustration 7-2.



Pump-Out Port Sensor Connections

Coldhead Turret Sensor Connections

TEMPERATURE SENSOR PLUG LOCATIONS
ILLUSTRATION 7-2

2. To find the 20K Shield temperature:

   a. Attach a Digital Volt-Ohm Meter (DVOM) to pins A and E (refer to Illustration 7-3 and Table 7-1) on the right-hand Pump-Out Port connector (see Illustration 7-2) via 10-Pin Breakout Box 2295140 (Illustration 7-1) and measure the resistance.

   b. Attach the DVOM to the plug's A and B pins and measure the resistance.

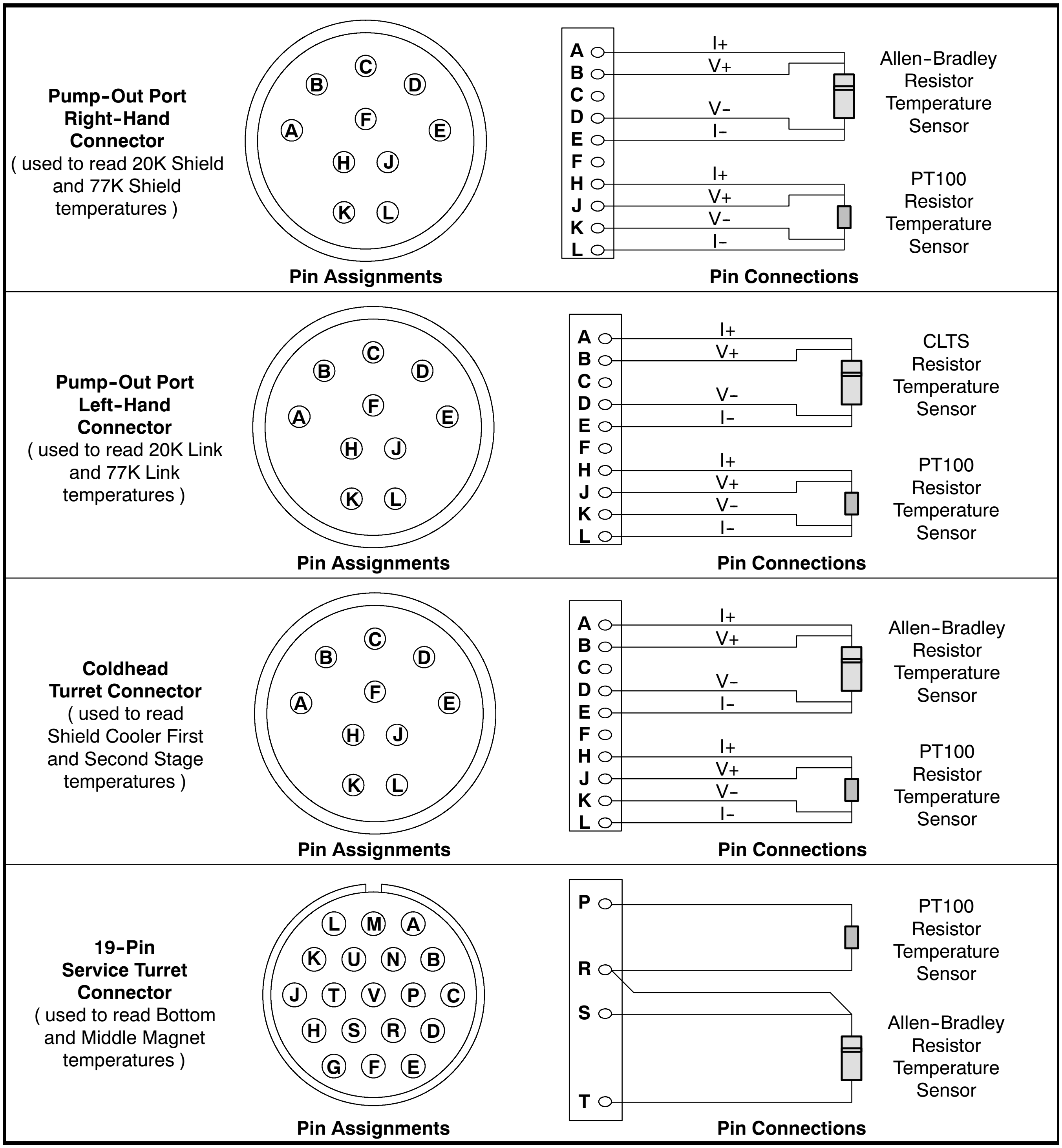
   c. Subtract the measurement taken in Step 2b from that taken in Step 2a to obtain the 20K Shield Sensor resistance:

   20K Shield Sensor Resistance = (Pins A and E) - (Pins A and B)

   d. Locate the value calculated in Step 2c within the "Resistance (Ohm)" column of Table 7-2, Allen-Bradley Resistor Calibration, and read the "Temperature (K)" value beside it.

   e. Log the temperature value found in Step 2d for the "20K Shield" in DATA SHEETS, Section 6, Temperature Sensor Log.

7-3 PROCEDURE (continued)



TEMPERATURE SENSOR PIN CONNECTIONS

ILLUSTRATION 7-3

7-3 PROCEDURE (continued)

TABLE 7-1
TEMPERATURE SENSOR MEASUREMENT WORKSHEET
(SUMMARIZING CONNECTORS, RESISTORS, AND TEMPERATURES MEASURED)

Component Temperature Measured*	Sensor Plug Location (Illustration 7-2)	Read DVOM Across Plug Pins		Calculated Resistance*	Table to Consult
		Reading 1	Reading 2		
		Reading 1 - Reading 2 = Resistance			
20K Shield	Pump-Out Port Right-Hand Sensor 10-Pin Connector	A and E: _____ Ohms	A and B: - _____ Ohms	= _____ Ohms	Table 7-2, Allen-Bradley
		(A and E) - (A and B) = Resistance			
77K Shield	Pump-Out Port Right-Hand Sensor 10-Pin Connector	J and K: _____ Ohms	H and J: - _____ Ohms	= _____ Ohms	Table 7-3, PT100
		(J and K) - (H and J) = Resistance			
20K Link	Pump-Out Port Left-Hand Sensor 10-Pin Connector	A and E: _____ Ohms	A and B: - _____ Ohms	= _____ Ohms	Table 7-4, CLTS
		(A and E) - (A and B) = Resistance			
77K Link	Pump-Out Port Left-Hand Sensor 10-Pin Connector	J and K: _____ Ohms	H and J: - _____ Ohms	= _____ Ohms	Table 7-3, PT100
		(J and K) - (H and J) = Resistance			
Shield Cooler First Stage	Coldhead Turret 10-Pin Connector	J and K: _____ Ohms	H and J: - _____ Ohms	= _____ Ohms	Table 7-3, PT100
		(J and K) - (H and J) = Resistance			
Shield Cooler Second Stage	Coldhead Turret 10-Pin Connector	B and D: _____ Ohms	A and B: - _____ Ohms	= _____ Ohms	Table 7-2, Allen-Bradley
		(B and D) - (A and B) = Resistance			
Magnet Bottom	Service Turret 19-Pin Connector	S and T: _____ Ohms	S and R: - _____ Ohms	= _____ Ohms	Table 7-2, Allen-Bradley
		(S and T) - (S and R) = Resistance			
Magnet Middle	Service Turret 19-Pin Connector	P and R: _____ Ohms	S and R: - _____ Ohms	= _____ Ohms	Table 7-3, PT100
		(P and R) - (S and R) = Resistance			

* Log in DATA SHEETS, Section 6, Temperature Sensor Log, the calculated resistances and the corresponding resistor temperatures.

7-3 PROCEDURE (continued)

3. To find the 77K Shield temperature:

   a. Attach a DVOM to pins J and K (refer to Illustration 7-3 and Table 7-1) of the right-hand Pump-Out Port connector (see Illustration 7-2) via 10-Pin Breakout Box 2295140 (Illustration 7-1) and measure the resistance.

   b. Attach the DVOM to the plug's H and J pins and measure the resistance.

   c. Subtract the measurement taken in Step 3b from that taken in Step 3a to obtain the 77K Shield Sensor resistance:

   77K Shield Sensor Resistance = (Pins J and K) - (Pins H and J)

   d. Locate value calculated in Step 3c within the "Resistance (Ohm)" column of Table 7-3, PT100 Resistor Calibration, and read the "Temperature (K)" value beside it.

   e. Log the the temperature value found in Step 3d for the "77K Shield" in DATA SHEETS, Section 6, Temperature Sensor Log.

4. To find the 20K Link temperature:

   a. Attach a DVOM to pins A and E (refer to Illustration 7-3 and Table 7-1) of the left-hand Pump-Out Port connector (see Illustration 7-2) via 10-Pin Breakout Box 2295140 (Illustration 7-1) and measure the resistance.

   b. Attach the DVOM to the plug's A and B pins and measure the resistance.

   c. Subtract the measurement taken in Step 4b from that taken in Step 4a to obtain the 20K Link Sensor resistance:

   20K Linkd Sensor Resistance = (Pins A and E) - (Pins A and B)

   d. Locate value calculated in Step 4c within the "Resistance (Ohm)" column of Table 7-4, CLTS Resistor Calibration, and read the "Temperature (K)" value beside it.

   e. Log the temperature value found in Step 4d for the "20K Link" in DATA SHEETS, Section 6, Temperature Sensor Log.

7-3 PROCEDURE (continued)

5. To find the 77K Link temperature:

   a. Attach a DVOM to pins H and L (refer to Illustration 7-3 and Table 7-1) of the left-hand Pump-Out Port connector (see Illustration 7-2) via 10-Pin Breakout Box 2295140 (Illustration 7-1) and measure the resistance.

   b. Attach the DVOM to the plug's H and J pins and measure the resistance.

   c. Subtract the measurement taken in Step 5b from that taken in Step 5a to obtain the 77K Link Sensor resistance:

      77K Link Sensor Resistance = (Pins H and L) – (Pins H and J)

   d. Locate the value calculated in Step 5c within the "Resistance (Ohm)" column of Table 7-3, PT100 Resistor Calibration, and read the "Temperature (K)" value beside it.

   e. Log the temperature value found in Step 5d for the "77K Link" in DATA SHEETS, Section 6, Temperature Sensor Log.

6. Access the temperature sensor plug inside the Coldhead Turret. See Illustration 7-2.

7. To find the Shield Cooler First Stage temperature:

   a. Attach a DVOM to pins H and L (refer to Illustration 7-3 and Table 7-1) of the Coldhead Turret connector (see Illustration 7-2) via 10-Pin Breakout Box 2295140 (Illustration 7-1) and measure the resistance.

   b. Attach the DVOM to the plug's H and J pins and measure the resistance.

   c. Subtract the measurement taken in Step 7b from that taken in Step 7a to obtain the Shield Cooler First Stage Sensor resistance:

      Shield Cooler First Stage Sensor Resistance = (Pins H and L) – (Pins H and J)

   d. Locate the value calculated in Step 7c within the "Resistance (Ohm)" column of Table 7-3, PT100 Resistor Calibration, and read the "Temperature (K)" value beside it.

   e. Log the temperature value found in Step 7d for the "Shield Cooler First Stage" in DATA SHEETS, Section 6, Temperature Sensor Log.

8. To find the Shield Cooler Second Stage temperature:

   a. Attach a DVOM to pins A and E (refer to Illustration 7-3 and Table 7-1) of the Coldhead Turret connector (see Illustration 7-2) via 10-Pin Breakout Box 2295140 (Illustration 7-1) and measure the resistance.

   b. Attach the DVOM to the plug's A and B pins and measure the resistance.

   c. Subtract the measurement taken in Step 8b from that taken in Step 8a to obtain the Shield Cooler Second Stage Sensor resistance:

      Shield Cooler Second Stage Sensor Resistance = (Pins A and E) – (Pins A and B)

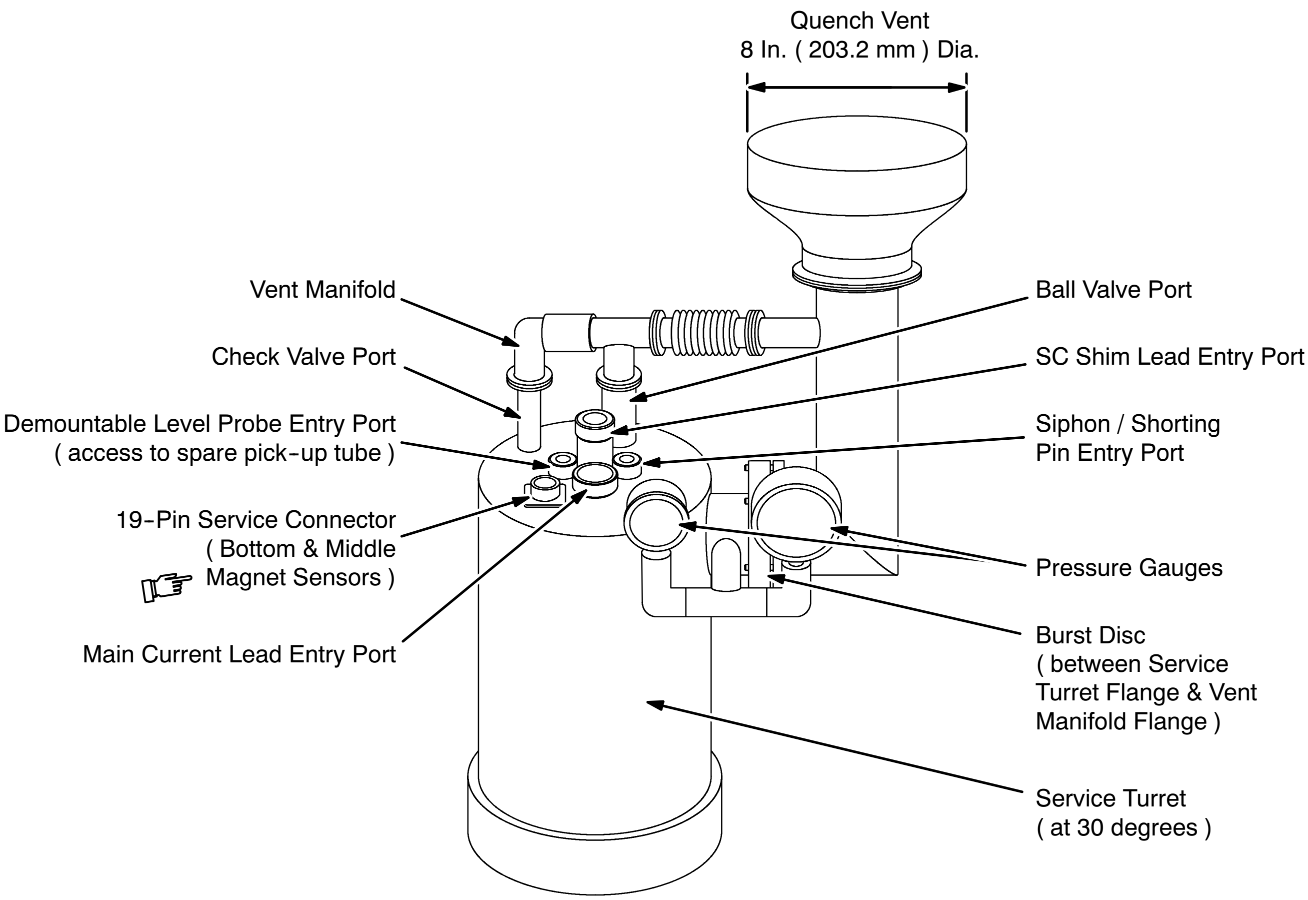
7-3 PROCEDURE (continued)

8. (continued)

   d. Locate the value calculated in Step 8c within the “Resistance (Ohm)” column of Table 7-2, Allen-Bradley Resistor Calibration, and read the “Temperature (K)” value beside it.

   e. Log the temperature value found in Step 8d for the “Shield Cooler Second Stage” in DATA SHEETS, Section 6, Temperature Sensor Log.

9. Connect 19-Pin Breakout Box 2295141 (Illustration 7-1) to the Service Turret’s 19-pin service connector (Illustration 7-4). Attach a DVOM to the Breakout Box.



TYPICAL FACTORY-INSTALLED 3T SERVICE TURRET PLUMBING
ILLUSTRATION 7-4

7-3 PROCEDURE (continued)

10. To find the Bottom Magnet temperature:

    a. Measure the resistance between pins S and T (refer to Illustration 7-3 and Table 7-1) of the Service Turret's 19-pin service connector (see Illustraion 7-4) via 19-Pin Breakout Box 2295141 (Illustration 7-1).

    b. Measure the resistance between pins S and R.

    c. Subtract the measurement taken in Step 8b from that taken in Step 8a to obtain the bottom Cryostat resistance:

    Bottom Magnet Resistance = (Pins S and T) – (Pins S and R)

    d. Locate the value calculated in Step 8c within the "Resistance (Ohm)" column of Table 7-2, Allen-Bradley Resistor Calibration, and read the "Temperature (K)" value beside it.

    e. Log the temperature value found in Step 8d for the "Magnet Bottom" in DATA SHEETS, Section 6, Temperature Sensor Log.

11. To find the Middle Magnet temperature:

    a. Measure the resistance between pins P and R (refer to Illustration 7-3 and Table 7-1) of the Service Turret's 19-pin service connector (see Illustraion 7-4) via 19-Pin Breakout Box 2295141 (Illustration 7-1).

    b. Measure the resistance between pins S and R.

    c. Subtract the measurement taken in Step 8b from that taken in Step 8a to obtain the bottom Cryostat resistance:

    Middle Magnet Resistance = (Pins P and R) – (Pins S and R)

    d. Locate the value calculated in Step 8c within the "Resistance (Ohm)" column of Table 7-3, PT-100 Resistor Calibration, and read the "Temperature (K)" value beside it.

    e. Log the temperature value found in Step 8d for the "Magnet Middle" in DATA SHEETS, Section 6, Temperature Sensor Log.

7-3 PROCEDURE (continued)

TABLE 7-2
ALLEN-BRADLEY RESISTOR CALIBRATION
(20K SHIELD, SHIELD COOLER SECOND STAGE AND MAGNET BOTTOM TEMPERATURES)

Temperature (K)	Resistance (Ohm)	Temperature (K)	Resistance (Ohm)	Temperature (K)	Resistance (Ohm)	Temperature (K)	Resistance (Ohm)
11	689.34	34	431.09	57	377.83	80	352.45
12	655.48	35	427.52	58	376.37	81	351.63
13	627.56	36	424.11	59	374.95	82	350.83
14	604.08	37	420.86	60	373.57	83	350.05
15	584.04	38	417.75	61	372.24	84	349.29
16	566.69	39	414.82	62	370.94	85	348.54
17	551.53	40	412.02	63	369.67	86	347.81
18	538.15	41	409.33	64	368.44	87	347.09
19	526.26	42	406.75	65	367.24	88	346.39
20	515.57	43	404.28	66	366.07	89	345.70
21	505.93	44	401.90	67	364.94	90	345.02
22	497.36	45	399.62	68	363.84	91	344.36
23	489.42	46	397.44	69	362.76	92	343.71
24	482.04	47	395.33	70	361.70	93	343.08
25	475.16	48	393.30	71	360.67	94	342.45
26	468.74	49	391.34	72	359.67	95	341.84
27	462.89	50	389.45	73	358.69	96	341.24
28	457.50	51	387.62	74	357.74	97	340.65
29	452.43	52	385.85	75	356.81	98	339.97
30	447.64	53	384.14	76	355.89	99	339.26
31	443.11	54	382.49	77	355.00	100	338.57
32	438.81	55	380.88	78	354.13		
33	434.85	56	379.33	79	353.28		

7-3 PROCEDURE (continued)

TABLE 7-3
PT100 RESISTOR CALIBRATION
(77K SHIELD, 77K LINK, SHIELD COOLER FIRST STAGE AND MAGNET MIDDLE TEMPERATURES)

Temperature (K)	Resistance (Ohm)	Temperature (K)	Resistance (Ohm)	Temperature (K)	Resistance (Ohm)	Temperature (K)	Resistance (Ohm)
77	20.00	133	42.85	189	65.70	245	88.54
79	20.82	135	43.66	191	66.51	247	89.36
81	21.63	137	44.48	193	67.33	249	90.18
83	22.45	139	45.30	195	68.14	251	90.99
85	23.26	141	46.11	197	68.96	253	91.81
87	24.08	143	46.93	199	69.78	255	92.62
89	24.90	145	47.74	201	70.59	257	93.44
91	25.71	147	48.56	203	71.41	259	94.26
93	26.53	149	49.38	205	72.22	261	95.07
95	27.34	151	50.19	207	73.04	263	95.89
97	28.16	153	51.01	209	73.86	265	96.70
99	28.98	155	51.82	211	74.67	267	97.52
101	29.79	157	52.64	213	75.49	269	98.34
103	30.61	159	52.46	215	76.30	271	99.15
105	31.42	161	54.27	217	77.12	273	99.97
107	32.24	163	55.09	219	77.94	275	100.78
109	33.06	165	55.90	221	78.75	277	101.60
111	33.87	167	56.72	223	79.57	279	102.42
113	34.69	169	57.54	225	80.38	281	103.23
115	35.50	171	58.35	227	81.20	283	104.05
117	36.32	173	59.17	229	82.02	285	104.86
119	37.14	175	59.98	231	82.83	287	105.68
121	37.95	177	60.80	233	83.65	289	106.50
123	38.77	179	61.62	235	84.46	291	107.31
125	39.58	181	62.43	237	85.28	293	108.13
127	40.40	183	63.25	239	86.10	295	108.94
129	41.22	185	64.06	241	86.91	297	109.76
131	42.03	187	64.88	243	87.73	299	110.58

7-3 PROCEDURE (continued)

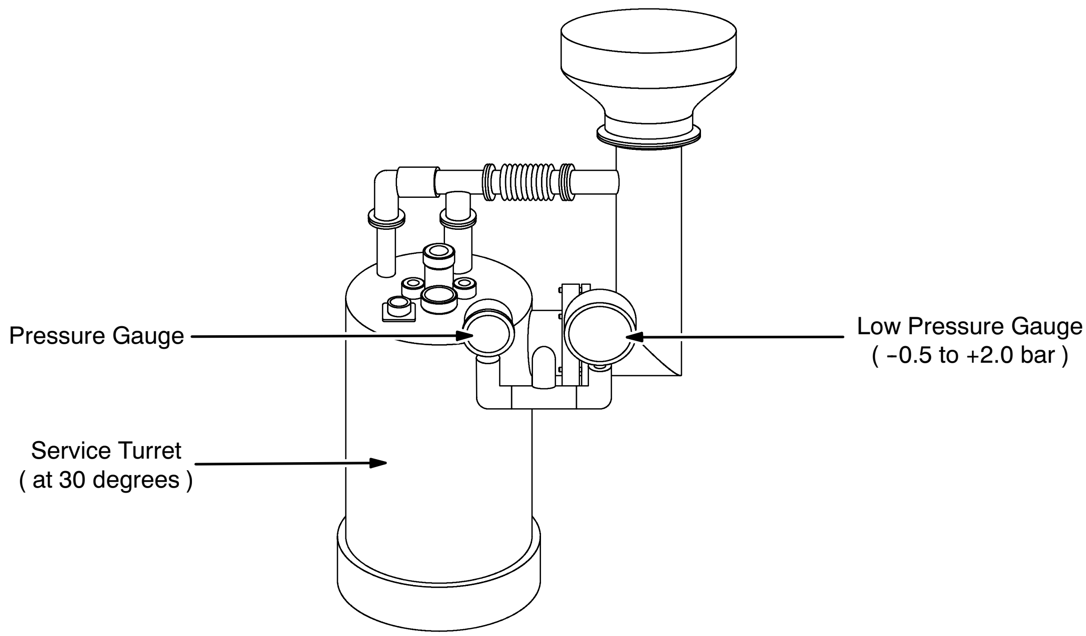
TABLE 7-4
CLTS RESISTOR CALIBRATION
(20K LINK TEMPERATURE)

Temperature (K)	Resistance (Ohm)	Temperature (K)	Resistance (Ohm)	Temperature (K)	Resistance (Ohm)	Temperature (K)	Resistance (Ohm)
10	221.4	74	236.7	138	252.2	202	267.3
12	221.9	76	237.2	140	252.5	204	267.8
14	222.3	78	237.6	142	252.9	206	268.2
16	222.8	80	236.1	144	253.4	208	268.7
18	223.3	82	238.6	146	253.9	210	269.2
20	223.8	84	239.1	148	254.4	212	269.7
22	224.3	86	239.6	150	254.8	214	270.1
24	224.7	88	240.0	152	255.3	216	270.6
26	225.2	90	240.5	154	255.8	218	271.1
28	225.7	92	241.0	156	256.3	220	271.6
30	226.2	94	241.5	158	256.8	222	272.1
32	226.6	96	241.9	160	257.2	224	272.5
34	227.1	98	242.4	162	257.7	226	273.0
36	227.6	100	242.9	164	258.2	228	273.5
38	228.1	102	243.4	166	258.7	230	274.0
40	228.6	104	243.9	168	259.1	232	274.4
42	229.0	106	244.3	170	259.6	234	274.9
44	229.5	108	244.8	172	260.1	236	275.4
46	230.0	110	245.3	174	260.6	238	275.9
48	230.5	112	245.8	176	261.1	240	276.4
50	230.9	114	246.2	178	261.5	242	276.8
52	231.4	116	246.7	180	262.0	244	277.3
54	231.9	118	247.2	182	262.5	246	277.8
56	232.4	120	247.7	184	263.0	248	278.3
58	232.9	122	248.2	186	263.2	250	278.7
60	233.3	124	248.6	188	263.9	252	279.2
62	233.8	126	249.1	190	264.4	254	279.7
64	234.3	128	249.6	192	264.9	256	280.2
66	234.8	130	250.1	194	265.4	258	280.7
68	235.2	132	250.5	196	265.8	260	281.1
70	235.7	134	251.0	198	266.3		
72	236.2	136	251.5	200	266.8		

SECTION 8
CRYOSTAT PRESSURE MEASUREMENT

Introduction

To maintain safe operation the magnet helium vessel must remain completely sealed to air entry and helium loss. Normally this vessel operates slightly above atmospheric pressure. This pressure is checked by reading the large low pressure gauge on the Service Turret. See Illustration 8-1.



SERVICE TURRET PRESSURE GAUGES
ILLUSTRATION 8-1

Procedure

1. Read the large low pressure gauge on the magnet Service Turret. See Illustration 8-1.
2. If the gauge reads in the green "Normal" zone, the helium vessel is operating with normal pressure.
3. If the gauge reads in the "Low" or "Vacuum" zones, a potential for icing exists. Immediately call your General Electric Service Representative.

SECTION 9
LIQUID HELIUM LEVEL MEASUREMENT

9-1 INTRODUCTION

The Helium Level Monitor (2255532) measures the amount of helium in the helium vessel, displaying the results in millimeters, although the monitor can also trigger a low helium warning. The volume of helium remaining is a non-linear function of depth because of the annular nature of the helium vessel and the monitor probe's arc shape inside the vessel.

Each Helium Level Monitor probe inside the magnet is a fine superconducting wire with a resistive matrix. During the instrument's read cycle a pulse of current is passed along the wire. This heats the top of the wire and creates a "normal zone" on the wire. The normal zone propagates along the wire. If the current pulse is chosen correctly, the normal zone propagates only to where the wire meets the liquid helium's surface. Thus, the resistance of the wire is proportional to the wire's length above the liquid level. This is measured and converted to a reading in millimeters.

The readings obtained from the longer, fixed probe, Probe 1, are used during normal operating conditions (Section 9-2), while those from the shorter Probe 2 are used during ramp up.

TABLE 9-1
HELIUM LEVELS BY CONDITION

Condition	Monitor Reading (Probe 1)	Volume (liters)
Helium Vessel Capacity ("Full")	1200 mm (47.24 inches)	3066 liters
Minimum Operating Level, Normal Operation	490 mm (19.3 inches)	1864 liters
Minimum Level During Ramping	1040 mm (89.37 inches)	2740 liters

9-2 HELIUM LEVEL MONITOR CHECKS

Note
Check the Helium Level Monitor daily.

1. Make sure the "Power" LED illuminates, showing the monitor is on.
2. Check the monitor's alarm light. If it is flashing, the helium level is getting extremely low and should be refilled as soon as possible.

9-2 HELIUM LEVEL MONITOR CHECKS (continued)

3. Check the monitor's "Read" light and helium level display. The monitor or a monitor probe could be malfunctioning if either the "Read" light is always lit or the display constantly reads full (1200 mm or 47.24 inches). Call your General Electric Service Representative.
4. Press the monitor's update button while the monitor is set to Probe 1. Read the resulting display.
5. Use Table 9-2 to convert the monitor's millimeter display to liters of helium remaining in the vessel.
6. Record helium level in DATA SHEETS, Section 3, Table 3-1, Cryogen Log.
7. Calculate the Helium Boil-Off Rate according to FUNCTIONAL CHECKS, Section 10.
8. Examine the Data Sheets log for trends and contact your General Electric Service Representative as required:
   a. Is the helium level decreasing to near the refill point? Schedule a refill.
   b. Is the helium consumption rate increasing? The coldhead may need replacing, or the vacuum space within the magnet need re-pumping.

9-2 HELIUM LEVEL MONITOR CHECKS (continued)

TABLE 9-2
HELIUM MONITOR PROBE 1 READINGS AND HELIUM LEVELS
(PROBE 1 WIRE RESISTANCE = 193.17 OHMS / METER)

Monitor Reading (mm)	Percent Full of Helium	Remaining Helium (liters)	Comment	Monitor Reading (mm)	Percent Full of Helium	Remaining Helium (liters)	Comment
1200	**100.00**	**3066**	**Full**	580	64.45	1976	
1180	98.86	3031		560	63.63	1951	
1160	97.68	2995		540	62.82	1926	
1140	96.38	2955		520	62.00	1901	
1120	95.04	2914		**500**	**61.19**	**1876**	**Minimum Operating Level**
1100	93.64	2871		480	60.40	1852	
1080	92.17	2826	1 Dewar to Fill*	460	59.59	1827	
1060	90.67	2780		440	58.81	1803	6 Dewars to Fill*
1040	**89.37**	**2740**	**Minimum Level During Ramping**	420	57.99	1778	
1020	87.90	2695		400	57.18	1753	
1000	86.33	2647	2 Dewars to Fill*	380	56.39	1729	
980	84.74	2598		360	55.58	1704	
960	83.11	2548		340	54.76	1679	
940	81.70	2505		320	53.91	1653	
920	80.43	2466		300	53.10	1628	
900	79.22	2429	3 Dewars to Fill*	280	52.25	1602	
880	78.08	2394		260	51.40	1576	7 Dewars to Fill*
860	77.01	2361		240	50.95	1562	
840	75.96	2329		220	50.10	1536	
820	74.95	2298		200	49.22	1509	
800	73.97	2268		180	48.30	1481	
780	72.99	2238		160	47.42	1454	
760	72.05	2209	4 Dewars to Fill*	140	46.48	1425	
740	71.14	2181		120	45.53	1396	
720	70.22	2153		100	44.55	1366	8 Dewars to Fill*
700	69.34	2126		80	43.57	1336	
680	68.46	2099		60	42.99	1318	
660	67.61	2073		40	41.98	1287	
640	66.76	2047		20	40.90	1254	
620	66.11	2027		0	39.76	1220	
600	65.36	2001	5 Dewars to Fill*				

* Dewars to fill assumes 250 liter Dewars and 85 percent transfill efficiency.

9-2 HELIUM LEVEL MONITOR CHECKS (continued)

TABLE 9-3
HELIUM MONITOR PROBE 2 READINGS AND HELIUM LEVELS
(PROBE 2 WIRE RESISTANCE = 193.17 OHMS / METER)

Monitor Reading (mm)	Remaining Helium (liters)	Monitor Reading (mm)	Remaining Helium (liters)	Monitor Reading (mm)	Remaining Helium (liters)	Monitor Reading (mm)	Remaining Helium (liters)
270	**3066 (Full)**	200	2939	130	2789	60	2642
260	3049	190	2919	120	2768	50	2617
250	3032	180	2898	110	2749	40	2591
240	3015	170	2877	100	2728	30	2563
230	2997	160	2856	90	2712	20	2536
220	2978	150	2834	80	2690	10	2510
210	2959	140	2812	70	2666	0	2506

TABLE 9-4
CALIBRATION OF PROBE 1 VS. PROBE 2
(PROBE CURRENT = 95 mA)

Probe 1		Probe 2	
Resistance (Ohm)	Monitor Reading (mm)	Resistance (Ohm)	Monitor Reading (mm)
0	1250	0	320
9.7	1200 (Full)	9.7	270 (Full)
241.5	0	61.8	0

SECTION 10
CRYOGEN BOIL-OFF RATE CALCULATION

Helium in the magnet gradually "boils off" and needs to be replaced. The rate at which helium is dissipating can be calculated from a pair of Helium Level Monitor by this formula:

$$\underset{\text{(approximate)}}{\textbf{Helium Boil-Off Rate}} = \frac{\underset{\text{(DATA SHEETS, Section 3, Table 3-1)}}{\textbf{Current LHe Level}} - \underset{\text{(DATA SHEETS, Section 3, Table 3-1)}}{\textbf{Past LHe Level}}}{\underset{\text{(DATA SHEETS, Section 3, Table 3-1)}}{\textbf{Time @ Current Reading}} - \underset{\text{(DATA SHEETS, Section 3, Table 3-1)}}{\textbf{Time @ Past Reading}}}$$

Make this calculation daily, entering the results in DATA SHEETS, Section 3, Cryogen Log.

REPLACEMENT / MAINTENANCE

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SECTION 1 – RAMPDOWN TO ZERO FIELD

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

WARNING!

THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RAMPING THE MAGNET:

- **HELIUM GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLORLESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:**
  - **A. MAKE SURE THAT THE MAGNET ROOM DOOR IS PROPPED OPEN,**
  - **B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND**
  - **C. REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, INCLUDED WITH THIS MANUAL.**

SECTION 1 – RAMPDOWN TO ZERO FIELD (continued)

WARNING!

(CONTINUED)

- **NOTIFY SITE ADMINISTRATION BEFORE RAMPING THE MAGNET THAT ALL MAGNETIC SAFETY PRECAUTIONS MUST BE TAKEN.**
- **POST WARNING SIGNS (46–255326P1) OUTSIDE THE 5 GAUSS ZONE TO ALERT PERSONNEL WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS AND OTHER BIOSTIMULATION DEVICES NOT TO PROCEED INTO THE DESIGNATED AREA. POST THESE SIGNS ON THE MAGNET ROOM LEVEL AS WELL AS AREAS BELOW THE MAGNET TO WHICH THE 5 GAUSS ZONE EXTENDS. SEE *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **POST "SECURITY ZONE" (46–255325P1) AND "AUTHORIZED PERSONNEL ONLY" (2289812) SIGNS AT THE MAGNET ROOM ENTRANCE AND MR SUITE ENTRANCE, RESPECTIVELY, PRIOR TO RAMPING THE MAGNET. THESE WARNINGS ARE TO ALERT PERSONNEL THAT NO FERROMAGNETIC MATERIAL OR INDIVIDUALS WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES ARE ALLOWED IN THE MAGNET ROOM WHEN THE MAGNET IS RAMPED. SEE *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. PULL THE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND SITE GEOMETRY ALLOW. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.**
- **MAKE SURE THE MAGNET RUNDOWN UNIT (MRU) IS INSTALLED AND OPERATIONAL TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE SET–UP AND CALIBRATION, SECTION 4.**
- **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT A PROPERLY FUNCTIONING OXYGEN MONITOR IS USED IN THE MAGNET AND/OR CRYOGEN STORAGE ROOM(S) BY SERVICE PERSONNEL.**
- **MAKE SURE THE MAGNET IS AT LEAST 90% (≥ 1040 mm) FULL OF LIQUID HELIUM TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING DURING RAMPING TO A POINT WHERE A QUENCH MAY OCCUR.**

SECTION 1 – RAMPDOWN TO ZERO FIELD (continued)

Document *2301164PRE, MR Magnet - Safety Requirements*

↓

Liquid Helium Level 100%?
(SET-UP & CALIBRATION, Section 9, Liquid Helium Fill)

↓

MRU Installed / Checked?
(SET-UP & CALIBRATION, Section 9, Magnet Rundown Unit (MRU) Installation)

↓

Power Supplies Checked?
(FUNCTIONAL CHECKS, Section 6, Main Power Supply Checks)

↓

Resistance in Magnet Circuit < 0.3 ohms?

↓

Field Plotting Rig Assembled, Mounted & Centered?
Teslameter Adjusted / Resychronized?
(SET-UP & CALIBRATION, Section 10)

↓

RAMP MAGNET

RAMPING FLOW DIAGRAM
ILLUSTRATION 1-1

1-1 TOOLS / EQUIPMENT

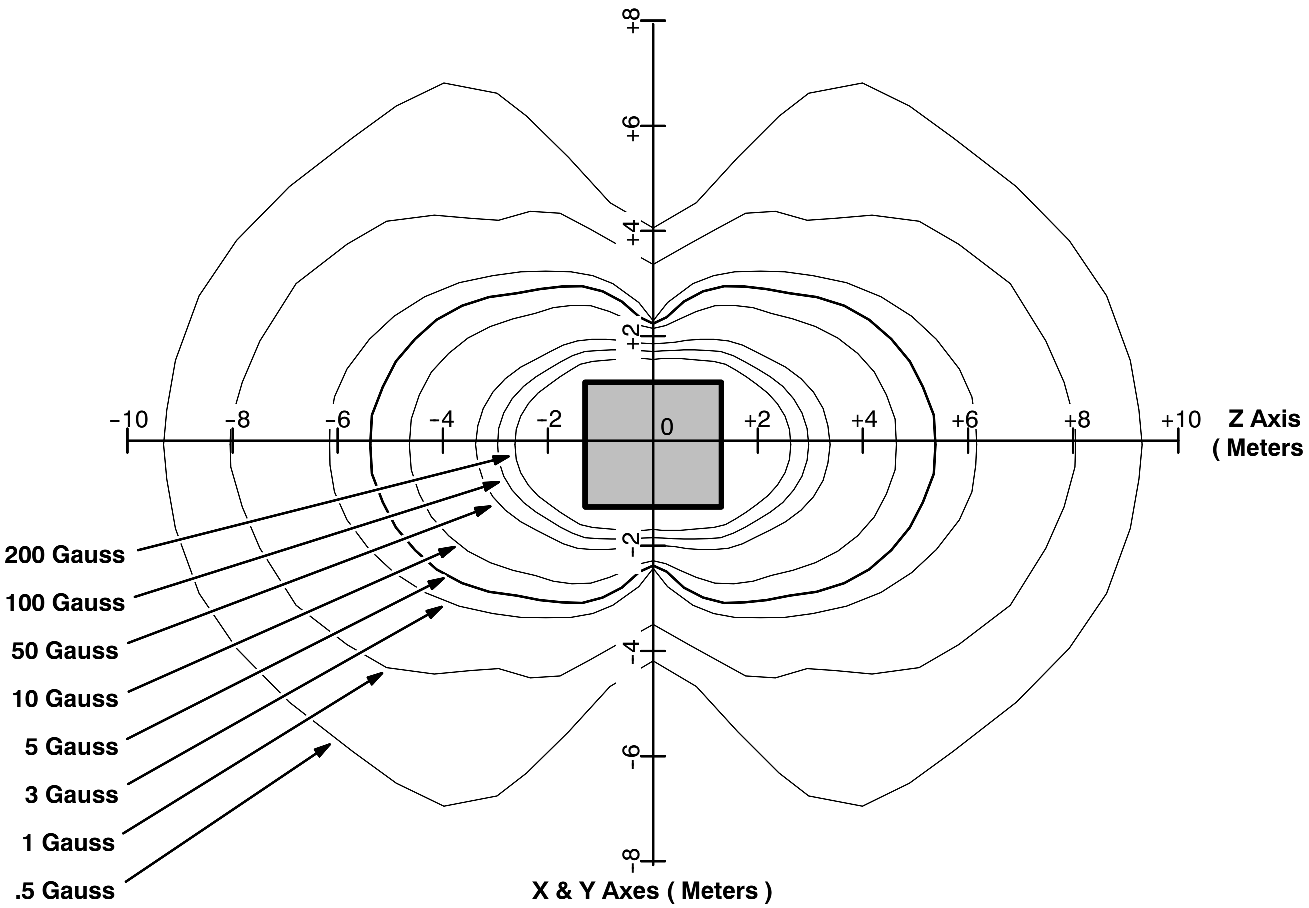
- Ramp / Energize Kit (2296056)
- Shimming Kit (2297304)
- Shim Power Supply (46-260777G3)
- Ramp Power Supply. The 1000 amp ESS7.5-1000-2-D-1236 (46-260776G4) is recommended because of its reduced ripple.
- Shim Cable Kit (2135558)
- Ramp Cable Kit (2135435)
- Metrolab Teslameter Kit (46-251865G4)
- No. 6 MetroLab Field Probe (2295525-5)
- Main Switch Adapter Box (2320946)

1-2 PREPARATION

1. Make sure the liquid helium level inside the magnet is ≥ 90% (1040 mm). Top off the magnet in accordance with REPLACEMENT / MAINTENANCE, Section 3, if not.

2. Position the power supplies outside the magnet's 5 gauss zone (extending from the magnet's center approximately 11 feet or 3.3 meters on either side of the magnet and 19.5 feet or 6 meters at either end). See Illustration 1-2.

IMPORTANT !!!
DO NOT BLOCK the magnet room entrance or any other passageway.



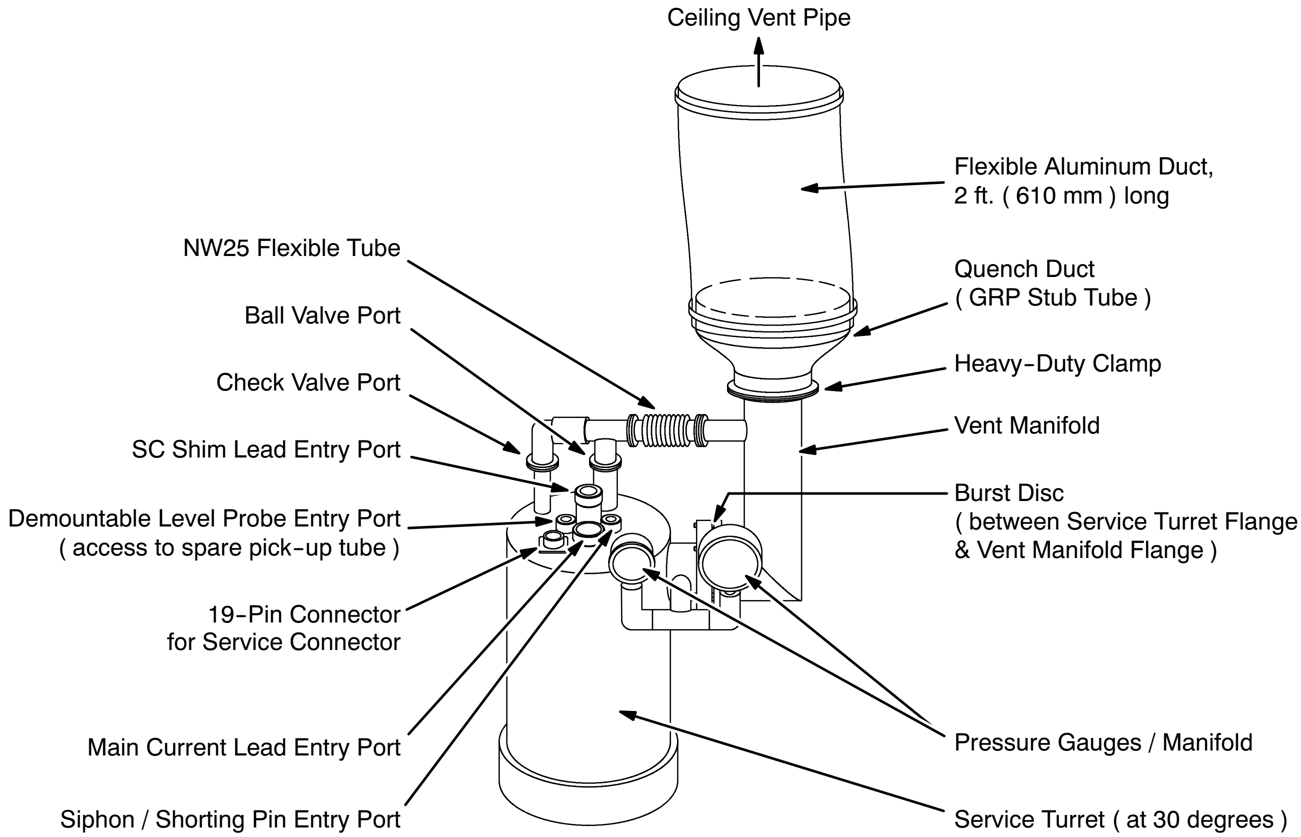
GEMSO 3T/940 FRINGE FIELD
ILLUSTRATION 1-2

3. Perform the power supply set-up checks detailed in FUNCTIONAL CHECKS, Section 6-2-4.

4. Assemble, mount and center the Field Plotting Rig (2292542) in conformance with SET-UP AND CALIBRATION, Section 10-1, Field Plotting Rig (2292542) Set-Up.

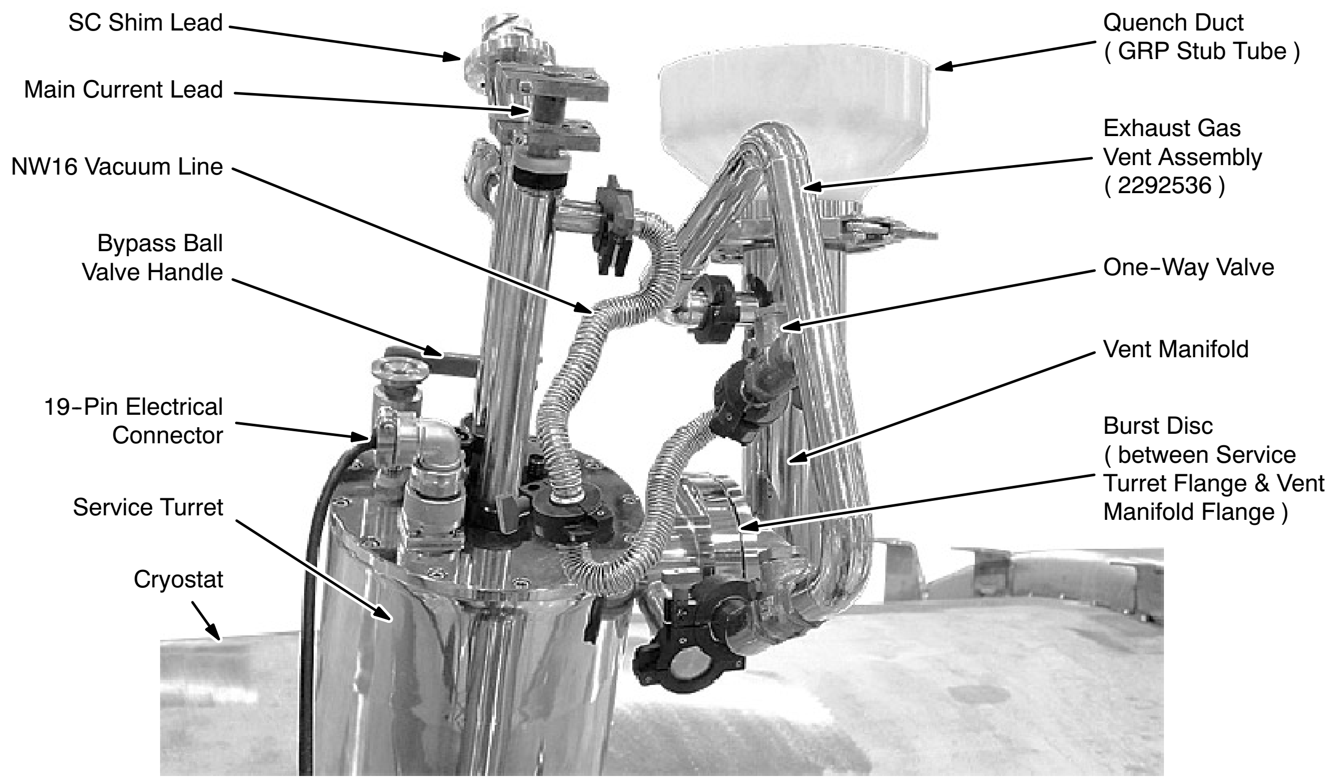
1-2 PREPARATION (continued)

5. Remove the Pressure Gauge Manifold and install the Exhaust Gas Vent Assembly (2292536) in its place. See Illustration 1-3 for the factory-installed configuration and Illustration 1-4 for the Exhaust Gas Vent arrangement.



TYPICAL FACTORY-INSTALLED 3T SERVICE TURRET PLUMBING
ILLUSTRATION 1-3

1-2 PREPARATION (continued)

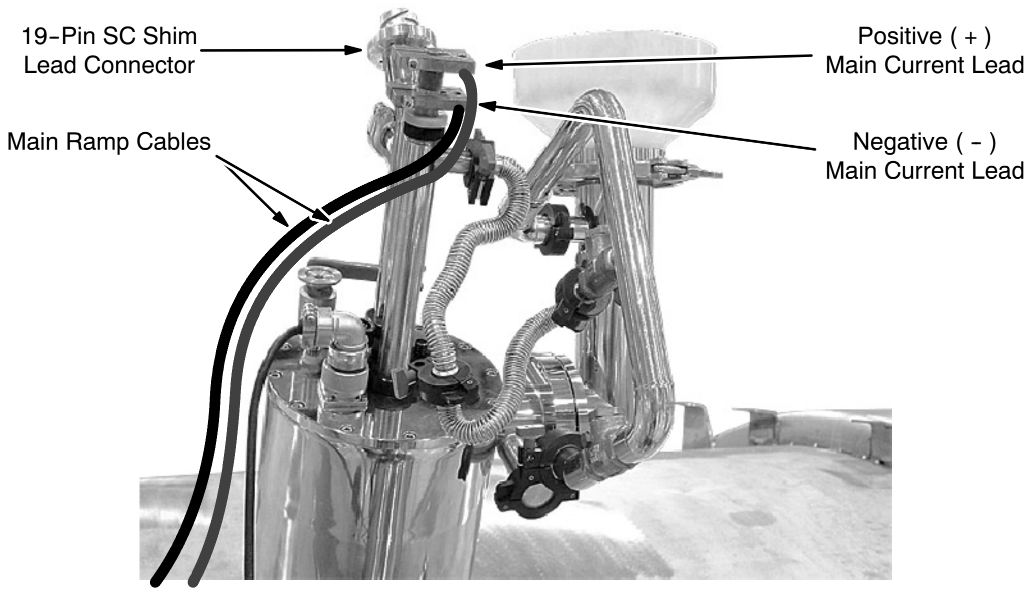


EXHAUST GAS VENT CONNECTION
ILLUSTRATION 1-4

6. Make sure the terminals on the Main Current Lead are not shorting to each other or the other end of the current lead.

7. Connect the Main Ramp Cables to the copper lugs on the Main Current Lead. See Illustration 1-5. The positive current lead terminal is the center top block and has a M12 connection screw. The negative terminal is the lower block and has a M10 screw.

1-2 PREPARATION (continued)



SHIM AND CURRENT LEAD CONNECTIONS
ILLUSTRATION 1-5

8. Connect the Shim Heater Power Supply to the Shim Adapter Box (2320885). See Illustrations 1-6 and 1-8.

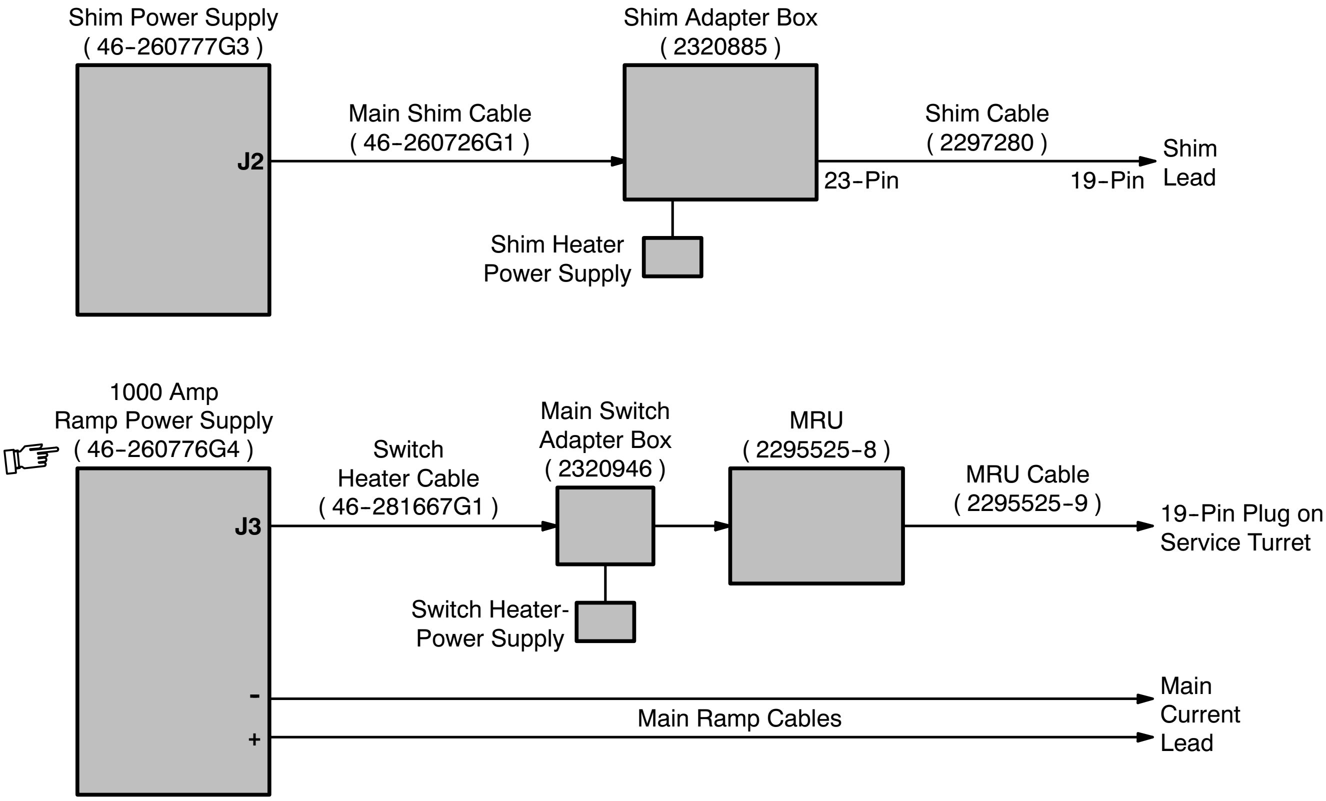
9. Connect the Main Shim Cable (46-260726G1) between Shim Power Supply socket J2 and the input socket on the Shim Adapter Box. See Illustrations 1-6, 1-7 and 1-8.

CAUTION

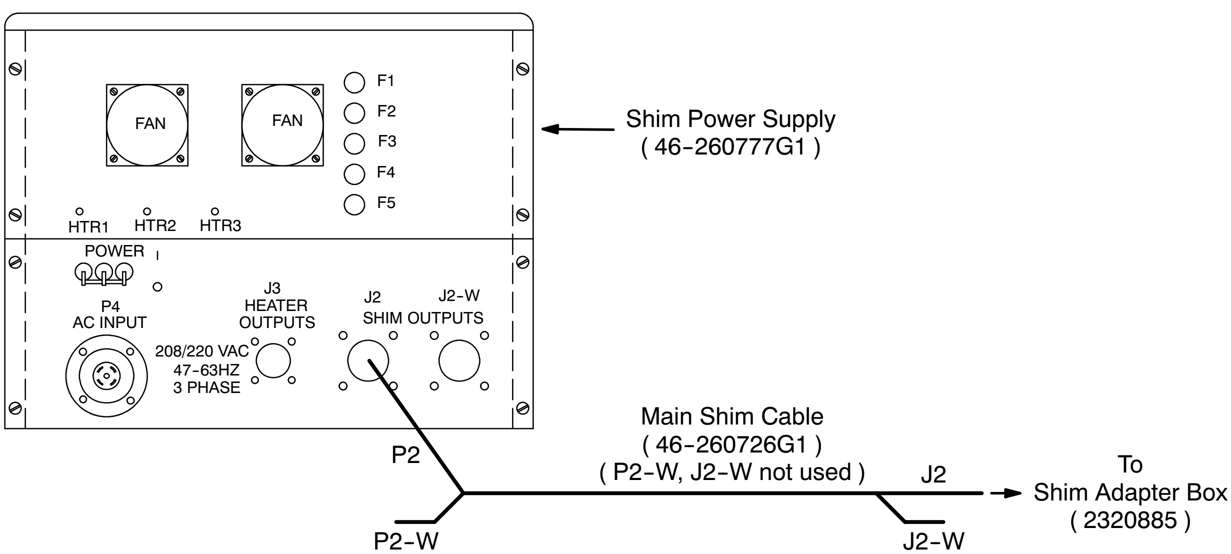
To prevent Shim Heater Coil burnout, DO NOT connect the heater supply from the Shim Power Supply. Only use the separate Shim Heater Power Supply.

10. Connect Shim Cable 2297280 to the Shim Adapter Box's 23-pin output and then to the Shim Lead. See Illustrations 1-6 and 1-8.

1-2 PREPARATION (continued)

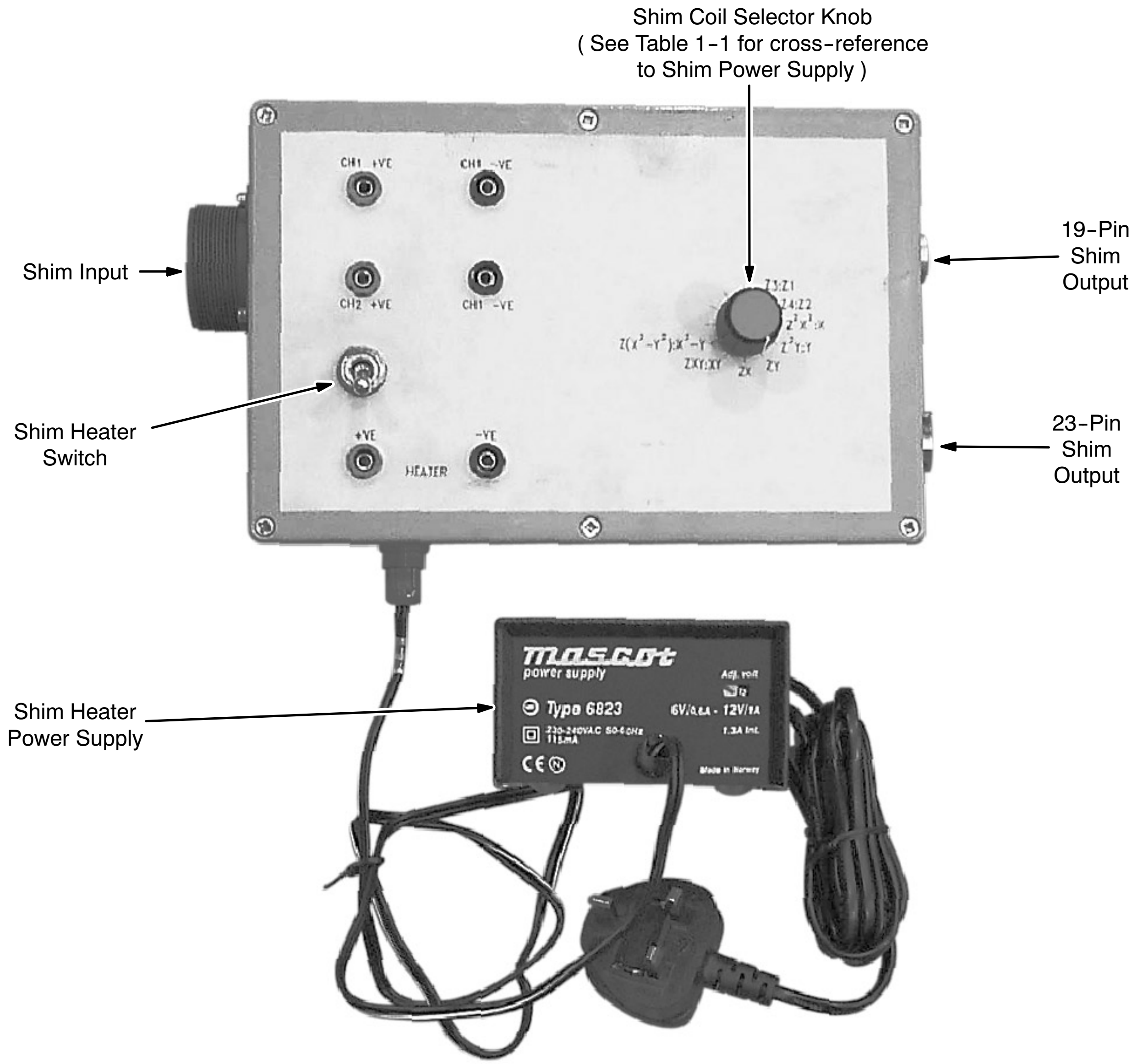


CABLE CONNECTIONS FOR RAMPDOWN
ILLUSTRATION 1-6



SHIM POWER SUPPLY OUTPUT CONNECTIONS
ILLUSTRATION 1-7

1-2 PREPARATION (continued)



SHIM ADAPTER BOX (2320885)
ILLUSTRATION 1-8

TABLE 1-1
CROSS-REFERENCE, SHIM POWER SUPPLY CHANNELS TO SHIM ORDERS

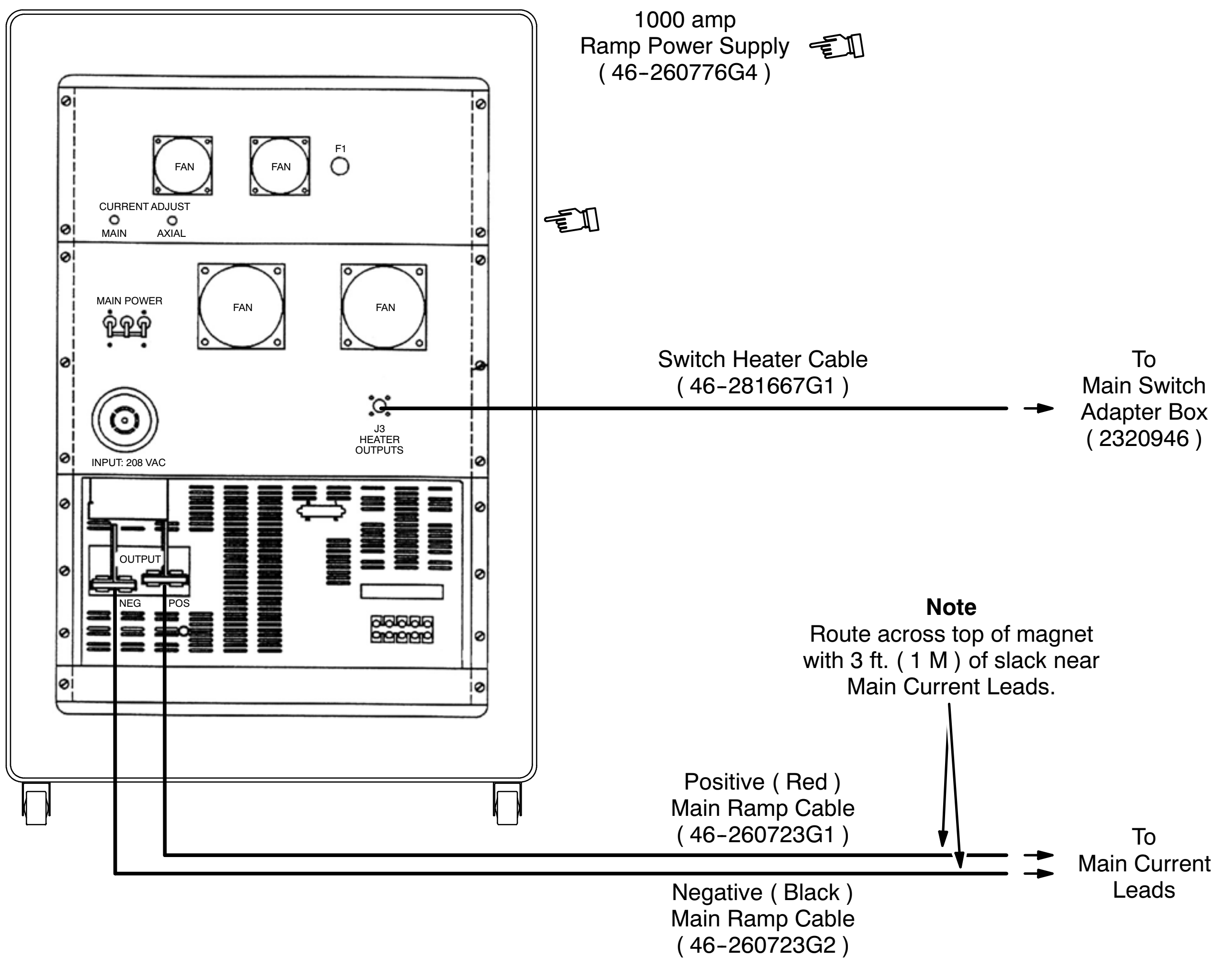
Power Supply Channel (Shim Power Supply)	T1-1						T1-2							
Shim Order (Shim Adapter Box Label)	Z3	Z4	Z^2X	Z^2Y	ZXY	$Z(X^2-Y^2)$	Z1	Z2	X	Y	ZX	ZY	XY	X^2-Y^2

1-2 PREPARATION (continued)

11. Connect the Switch Heater Cable (46-281667G1) to the Ramp Power Supply's connector J3 and then to the Main Switch Adapter Box (2320946). See Illustrations 1-6, 1-9 and 1-10.

12. Connect a digital voltmeter (DVM), set to its volts range, to the test points on the Main Switch Adapter Box.

13. Plug in the Switch Heater Power Supply. Verify that the Main Magnet Switch on the Main Switch Adapter Box opens (on) and closes (off); leave the switch in the closed position. Unplug the Switch Heater Power Supply.



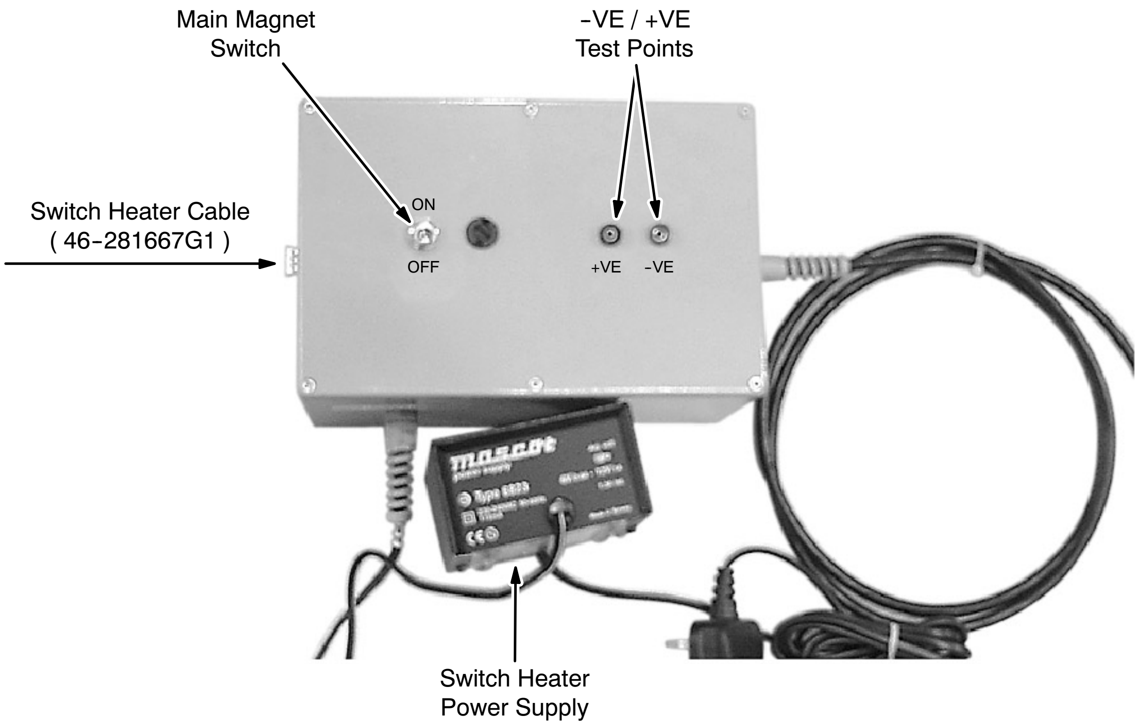
MAGNET POWER SUPPLY CONNECTIONS (1000 AMP)
ILLUSTRATION 1-9

1-2 PREPARATION (continued)

14. Connect the Main Switch Adapter Box to the MRU's 9-pin female connector labeled "PSU." See Illustrations 1-6 and 1-10.

Note

This arrangement gives access to the main magnet voltage and allows the magnet switch to be opened with the separate Switch Heater Power Supply.



MAIN SWITCH ADAPTER BOX (2320946)

ILLUSTRATION 1-10

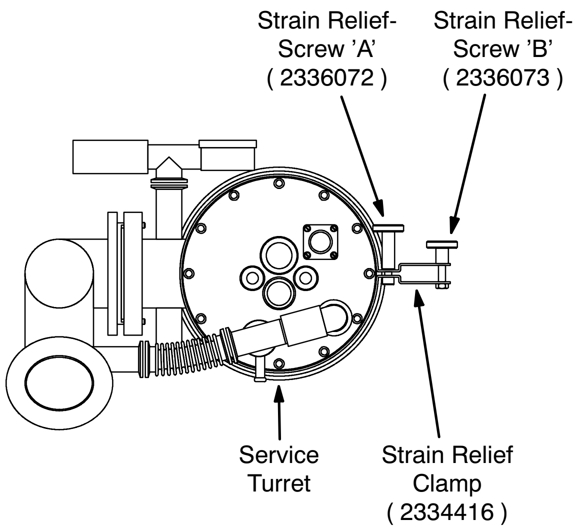
15. Install the Ramp Cable Strain Relief Clamp (Clamp 22334416, Screw 'A' 2336072 and Screw 'B' 2336073) around the top edge of the Service Turret as shown in Illustration 1-11, if not already installed.

CAUTION

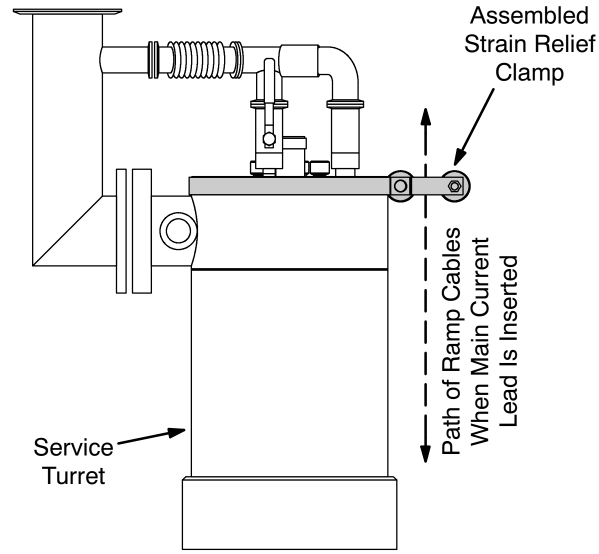
Do not leave the Main Current Lead Port open to the atmosphere for more than a few seconds since air will condense in the port, which will cause an ice blockage.

16. Remove the cap from the Main Current Entry Port, moving the nut, washers and o-ring to the Main Current Lead. See Illustration 1-3.

1–2 PREPARATION (continued)



Top View of Service Turret



Side View of Service Turret

RAMP CABLE STRAIN RELIEF CLAMP
ILLUSTRATION 1–11

WARNING!

MAKE SURE THE MAIN CURRENT LEAD / SHIM LEAD EXHAUST PORTS ALWAYS FACE AWAY FROM THE SERVICE PLATFORM TO PREVENT COLD HELIUM GAS EXHAUSTING TOWARD THE SERVICE ENGINEER.

17. Insert the Main Current Lead into the Main Current Entry Port and push down gradually. Cold helium gas will emerge from the entry port. If the rush of gas becomes too great, hold the lead steady or retract it slightly until the rush decreases.

18. Once contact is felt, push the Main Current Lead into full engagement with the male connector at the bottom of the entry port.

Note

Contact is necessary over the whole 2 inches (50 mm) of the male connector's pin to minimize contact resistance.

1-2 PREPARATION (continued)

CAUTION

Do not force the Main Current Lead into full engagement.

19. Rotate the Main Current Lead gently to its fully engaged position.

IMPORTANT !!!

Do not allow the cables or exhaust port touch the Cryostat. Support the cables via a strain relief to prevent the cables' weight from pulling on the Current Lead.

20. Tighten the nut to create a gas-tight seal.
21. Run the Ramp Cables through the Ramp Cable Strain Relief Clamp along the path shown in Illustration 1-11. Make sure the cables are not pulling on the Main Current Leads, then tighten the clamp.
22. Remove the cap from the SC Shim Lead Entry Port, moving the nut, washers and o-ring to the Shim Lead. See Illustration 11-3.

WARNING!

MAKE SURE THE MAIN CURRENT LEAD / SHIM LEAD EXHAUST PORTS ALWAYS FACE AWAY FROM THE SERVICE PLATFORM TO PREVENT COLD HELIUM GAS EXHAUSTING TOWARD THE SERVICE ENGINEER.

23. Insert the Shim Lead into the SC Shim Lead Entry Port and push down gradually. Cold helium gas will emerge from the entry port. If the rush of gas becomes too great, hold the lead steady or retract it slightly until the rush decreases.

1-2 PREPARATION (continued)

24. Once contact is felt, push the Shim Lead into full engagement.

Note

The Shim Lead is keyed, so it should only engage in one orientation: with the exhaust port facing directly away from the service platform. If it does not fit in the correct orientation, stop immediately; **DO NOT FORCE THE SHIM LEAD INTO ENGAGEMENT NOR ROTATE IT**. Instead, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715).

IMPORTANT !!!

If ice is encountered on the Shim Lead contacts, remove and warm the Shim Lead, then reinsert the warm lead to clear the ice.

25. Connecting the 19-Pin Breakout Box (2295141) to the lead and measuring the resistances between pin "J" and the other pins as listed in Table 1-2. If all measurements are within the specified ranges, the Shim Lead has been engaged correctly. If any measurement is outside its expected range, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715).

TABLE 1-2
RESISTANCE CHECKING OF CORRECT SHIM LEAD INSERTION ORIENTATION

Pin Pair (at 19-Pin Breakout Box)	A – J	B – J	C – J	D – J	E – J	F – J	G – J	H – J
Expected Resistance Measurement (Ohms)*	200 ± 20	200 ± 20	200 ± 20	200 ± 20	100 ± 10	100 ± 10	200 ± 20	200 ± 20

26. Measure the resistance between the Main Current Lead's positive and negative terminals. Clean and reseat the Main Current Lead if resistance is not < 0.3 ohms.

27. Connect a NW16 vacuum line between the Main Current Lead exhaust and the one-way valve at the center of the Exhaust Gas Vent Assembly (2292536) to route exhaust gas safely out of the room through the quench duct. See Illustration 1-4.

28. Blank off the Shim Lead exhaust to make sure all exhaust gas is used to cool the Current Lead while ramping down the magnet.

29. Connect the ramp cables to the power supply. Check the polarity.

30. Adjust / resynchronize the Teslameter in conformance with SET-UP AND CALIBRATION, Section 10-2.

1-3 RAMPING DOWN TO ZERO FIELD

1. Turn on the Ramp Power Supply.
2. On the MRU turn on the B0 Heater.
3. Turn on the Ramp Power Supply's Axial Switch Heater.
4. Turn on the Shim Power Supply.
5. Remove the Shorting Pin:
   a. Remove the plug from the Siphon / Shorting Pin Entry Port.
   b. Slowly insert the Shorting Pin Insertion Tool until it contacts the Shorting Pin.
   c. Push the tool down and rotate it 10 turns clockwise.
   d. Once the Shorting Pin is fully engaged, pull the Shorting Pin Insertion Tool out.
   e. Replug the Siphon / Shorting Pin Entry Port.
6. Make sure all current potentiometers on the Shim Power Supply are set to zero (fully counterclockwise).
7. Set all shims to zero current:
   a. Select the "Z1:Z3" shim pair via the Shim Coil Selector Knob on the Shim Adapter Box.
   b. If the magnet has been shimmed previously, set the current and polarity in the T1-1 and T1-2 channels equal to that in the shims. Refer to SET-UP AND CALIBRATION, Section 12-4-3, Steps 2 through 13, although it is not necessary to position the probe or wait for the field to stabilize.
   c. Turn on the Shim Heater Switch, then wait 10 seconds for the switch to go resistive.
   d. Reduce the currents to zero, then wait one minute before turning off the Shim Heater Switch.
   e. Select the next shim pair. Repeat Steps 7b - 7d until all shims have been set to zero.
8. Select the "Z2:Z4" shim pair and turn on the Shim Heater Switch to hold these open during the ramp down.

1-3 RAMPING DOWN TO ZERO FIELD (continued)

WARNING!

THE MAXIMUM CURRENT LIMIT SETTING MUST NEVER EXCEED 260 AMPS TO PREVENT POTENTIAL MAIN COIL BURN-OUT.

WARNING!

THE POWER SUPPLY MUST BE SET TO THE SAME POLARITY AND CURRENT AS THE LAST VALUES RECORDED IN DATA SHEETS, SECTION 7, TO PREVENT A MAGNET QUENCH WHEN TURNING ON THE MAIN MAGNET SWITCH.

9. Set the Current Limit on the Ramp Power Supply to one amp more than the persistence current value recorded under “Final Persistence Current (A)” in DATA SHEETS, Section 7, Table 7-1.
10. Set the magnet voltage to zero and then increase the voltage until the magnet current matches the recorded persistence current value.
11. Plug in the Switch Heater Power Supply on the Main Switch Adapter Box and wait 5 minutes for it to go resistive. If the field begins drifting upwards during this period, slightly reduce the voltage until the field starts reducing.
12. Once the switch is resistive, decrease the voltage until the field begins to reduce.
13. Continue periodically decreasing the voltage in the stages shown in Table 1-3 in order to keep the field running down. Keep the voltage within 100 mV of the “Voltage” value in Table 1-3 (e.g., 0.4 - 0.5 volts for full field to 249.0 amps).

TABLE 1-3
RECOMMENDED MAXIMUM ENERGIZATION RATES

Main Coil Current (Amps)	Current Limit (Amps)	Amps / Hour	Main Coil Voltage*	Duration
Final Persistence Value* - 249.0	Final Persistance Value* + 1.0	4.4	0.5	1.0 hours
249.0 - 237.0	250	8.7	1.0	1.4 Hours
237.0 - 186.0	238	17.4	2.0	2.9 Hours
186.0 - 0.0	187	21.8	2.5	8.5 Hours
* “Final Persistence Current (A)” value from DATA SHEETS, Section 7, Table 7-1.				

1-3 RAMPING DOWN TO ZERO FIELD (continued)

14. When the current reaches zero, adjust both voltage and current to zero and wait for magnet voltage to decay to zero (≈ 10 minutes).
15. Turn off the Main Magnet Switch on the Main Switch Adapter Box.
16. Turn off the Shim Heater Switch and the B0 Coil Heater Swtich.
17. Remove the Shim Lead and the Main Current Lead.
18. Remove the Exhaust Gas Vent Assembly and reinstall the Pressure Gauge Manifold.

SECTION 2 – BURST DISC REPLACEMENT

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

2–1 INTRODUCTION

Defective Burst Discs and those that rupture during a magnet quench must be replaced to prevent cryopumping and ice blockages forming in the system. Quenches release cryogens, so cryogen refill will be required. If cryogen refill cannot take place during the Burst Disc replacement service call, apply helium gas to the Helium Vessel to prevent continued cryopumping and ice block formation in the Exhaust Lines and Vertical Stack.

2–2 TOOLS / EQUIPMENT

- Replacement Burst Disc (2255516)
- Portable Oxygen Monitor (2106236 or 2106237) if Oxygen Monitor Kit 2107184 is not installed and functional in room.
- Multimeter
- 5 mm Allen Key

2-3 VERIFYING BURST DISC REPLACEMENT IS REQUIRED

1. Check the Service Turret Pressure Gauge. A broken Burst Disc doesn't hold pressure in the Cryostat.
2. Connect common and Ohms leads from a Multimeter to the Burst Disc's "Tell" leads; the meter should read "O.L." Ohms if the disc is broken.

Note

Remove and visually inspect the Burst Disc if not absolutely certain its condition is good.

2-4 BURST DISC REMOVAL

WARNING!

MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, INCLUDED WITH THIS MANUAL.

WEAR PROTECTIVE CLOTHING (LONG-SLEEVE SHIRT, LONG PANTS AND A PROTECTIVE APRON / JACKET), INSULATED NONABSORBENT GLOVES AND GOGGLES / SAFETY GLASSES WHEN REPLACING THE BURST DISC ON COLD VENT SYSTEM.

MAKE SURE THE MAGNET IS RAMPED DOWN TO ZERO FIELD TO PREVENT ANY POSSIBILITY OF A QUENCH.

VENT OFF CRYOSTAT PRESSURE BEFORE COMMENCING THIS PROCEDURE.

CAUTION

Replace the Burst Disc immediately after rupturing to avoid cryopumping. Ice blocks can form inside the exhaust lines and Vertical Stack if the Burst Disc is not replaced promptly.

Make sure Coldhead / Shield Cooler power is turned off before starting this procedure to prevent ice formation in Recondenser.

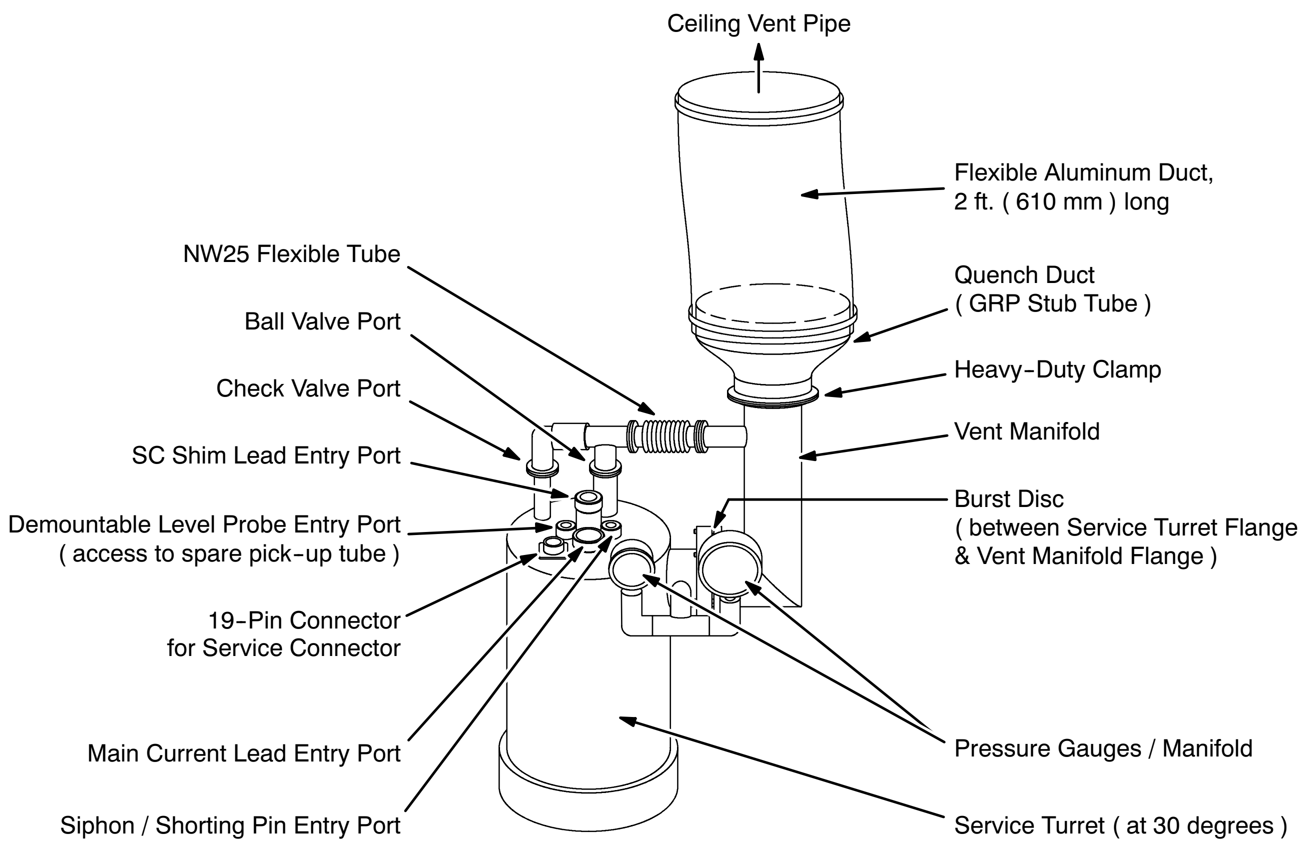
1. Turn off Coldhead / Shield Cooler power.

2-4 BURST DISC REMOVAL (continued)

2. Slowly open the ball valve on the Service Turret to allow any pressure within the Helium Vessel to excape. Do not continue with this procedure until the Pressure Gauges read less than 0.5 psig.

3. Remove the heavy-duty clamp securing the 8 inch (203 mm) Quench Duct (GRP Stub Tube) to the Vent Manifold. See Illustration 2-1.

4. The gasket between the Quench Duct and Vent Manifold is now accessible. The gasket needs to be removed; however, if removal now might be damage the gasket, remove it in Step 8.



EXHAUST VENT TO SERVICE TURRET CONNECTIONS
ILLUSTRATION 2-1

5. Remove the NW25 Clamp securing the Vent Manifold to the NW25 Flexible Tube. See Illustration 2-1.

6. The Centering Ring at the end of theNW25 Flexible Tube is now accessible. The ring needs to be removed; however, if removal now might damage the ring, remove it in Step 8.

2-4 BURST DISC REMOVAL (continued)

7. Using a M5 Allen key, loosen the six M6 cap screws securing the Vent Manifold Flange to the Service Turret Flange. See Illustration 2-1. Do not remove the screws nor allow the Vent Manifold to drop off the Service Turret Flange at this time.

8. Remove the Quench Duct gasket and NW25 Centering Ring if these could not be removed without damage earlier (Steps 3 and 5, respectively).

9. Carefully lower the Vent Manifold to a resting position against the O.V.C.

10. Remove the lowest five M6 cap screws loosened in Step 7, then securely hold the Vent Manifold while removing the remaining M6 cap screw.

11. Take the Vent Manifold off the Service Turret Flange and place the Manifold where it will not fall or get damaged.

Note
The Burst Disc is now accessible for inspection / removal.

2-5 BURST DISC REPLACEMENT

1. Clean any ruptured Burst Disc debris in the exhaust vent system. Make sure the sealing surfaces on the Service Turret Flange and the Vent Manifold Flange are completely clean.

2. Fit a replacement Burst Disc (2255516) onto the Service Turret Flange, with the white Teflon seal in full contact with the flange. See Illustration 2-1. Do not let the Burst Disc fall out.

3. Carefully position the Vent Manifold Flange against the Service Turret Flange while resting the top of the Vent Manifold against the O.V.C.

4. Loosely install six M6 cap screws between the Vent Manifold Flange and the Service Turret Flange. Do not tighten yet.

5. Swing the Vent Manifold into position with the NW25 Flexible Tube. See Illustration 2-1.

6. Install the NW25 Centering Ring between the Vent Manifold and the Flexible Tube, then install the NW25 Clamp.

7. Place the Quench Duct Gasket between the Vent Manifold and the Quench Duct. See Illustration 2-1.

8. Install and tighten the heavy-duty clamp to secure the Quench Duct to the Vent manifold. See Illustration 2-1.

9. Using a M5 Allen key, evenly tighten the six M6 cap screws securing the Vent Manifold Flange to the Service Turret Flange.

10. Close the ball valve on the Service Turret. See Illustration 2-1.

SECTION 3 – LIQUID HELIUM TOP–OFFS

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

During normal magnet operation the helium level in the magnet needs to be maintained above 1864 liters (60% of the vessel's 3066 liter capacity). Consequently, the Helium Level Monitor must be checked regularly. See FUNCTIONAL CHECKS, Section 9, Liquid Helium Level Monitor. A refill should be scheduled whenever the monitor reading from Probe 1 approaches 490 mm. If the monitor's alarm LED is flashing, a refill should be scheduled as soon as possible.

IMPORTANT !!!

Liquid helium is used in this procedure. Make sure of the following prior to starting the precool:

- **The procedure will be performed by trained and authorized personnel.**
- **All required equipment is in place and functional.**

Perform the preparations and follow the procedure found SET–UP AND CALIBRATION Section 9, Liquid Helium Fill / Top–Off, to refill the 3T/940 magnet.

SECTION 4 – RE–EVACUATION OF VACUUM SPACE

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

Equipment

- Helium mass–spectrometer leak detector
- Appropriate vacuum hoses

Note
Only use hoses that have not been previously used with gaseous helium.

Procedure

1. Ramp the magnet down to zero field in accordance with REPLACEMENT / MAINTENANCE, Section 1.
2. Connect the leak detector to the vacuum port of ______________________, but do not open the valve.
3. Evacuate the vacuum hoses to < 10 milliTorr.
4. Set the leak detector to register background level.

Procedure (continued)

5. Slowly open the cryostat valve.

6. Monitor pressure and helium background level.

Note

Helium level may temporarily increase for a minute or so as outgassing causes the valve's o-ring to momentarily unseat.

SECTION 5
SHIELD COOLER COLDHEAD REPLACEMENT

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

WARNING!

THE FOLLOWING SAFETY PRECAUTIONS MUST BE TAKEN BEFORE STARTING TO CHANGE OUT A COLDHEAD OF A MAGNET AT FIELD. THESE PRECAUTIONS ARE REQUIRED TO PREVENT FERROMAGNETIC MATERIAL FROM BECOMING DANGEROUS PROJECTILES IN THE MAGNETIC FIELD OR THE COLDHEAD FROM BEING ATTRACTED INTO THE BORE OR ONTO THE OUTER VESSEL OF THE MAGNET.

- **THIS PROCEDURE MUST BE PERFORMED BY PERSONNEL TRAINED AND QUALIFIED IN COLDHEAD REPLACEMENT WHILE THE MAGNET IS AT FIELD.**
- **NEVER BRING COLDHEAD NEAR THE MAGNET BORE OR IN CONTACT WITH THE OUTER VESSEL DURING REMOVAL / REPLACEMENT.**
- **DO NOT BRING ANY FERROMAGNETIC TOOLS OR EQUIPMENT INTO MAGNET ROOM. FERROMAGNETIC MATERIAL CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.**
- **MAKE SURE THE MAGNET RUNDOWN UNIT IS FUNCTIONING PROPERLY TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE FUNCTIONAL CHECKS, SECTION 4.**

SECTION 5 – SHIELD COOLER REPLACEMENT (continued)

WARNING!

(CONTINUED)

- **BOTH OLD AND NEW COLDHEADS MUST BE CARRIED FROM / TO SIDE OF MAGNET IN A DIRECT PATH FROM / TO WALL OF THE ROOM. NEVER BRING THE COLDHEAD NEAR THE END FLANGES OF THE MAGNET WHERE IT CAN BE DRAWN IN THE BORE.**
- **A SECOND PERSON IS REQUIRED TO ASSIST CHANGING OUT A COLDHEAD IN A MOBILE VAN. THIS ENABLES THE PERSON REMOVING THE COLDHEAD TO HAND IT OFF TO THE SECOND PERSON.**
- **IF ALL OF THE ABOVE CONDITIONS ARE NOT MET, THE MAGNET MUST BE RAMPED DOWN BEFORE CHANGING THE COLDHEAD.**

5-1 EQUIPMENT

- 30V 2A Power Supply
- Full helium gas cylinder (99.9995%)
- Regulator Kit (46-306734G1)
- Gas Cylinder Cart (46-258150P1)
- RGD-5100 Coldhead Assembly (2255528)
- High-Vac Pump
- Leak Detector
- 10-Pin Breakout Box (2295140)
- Digital Volt-Ohm Meter (DVOM)
- Various Allen keys, spanners, etc.
- Leather gloves (for opening and closing coldhead motor shield)
- Portable Oxygen Monitor (2106236 or 2106237)
- Flashlight

5-2 REMOVAL

1. Turn the Compressor off.
2. Disconnect the power cable from the Coldhead to the Compressor.

CAUTION

Keep strain relief ty-wraps on gas lines. Do not put a downward force on the Aeroquip fittings during removal of gas line connections to prevent a bending moment on the Aeroquip stem and increase friction force on the Aeroquip fitting threads.

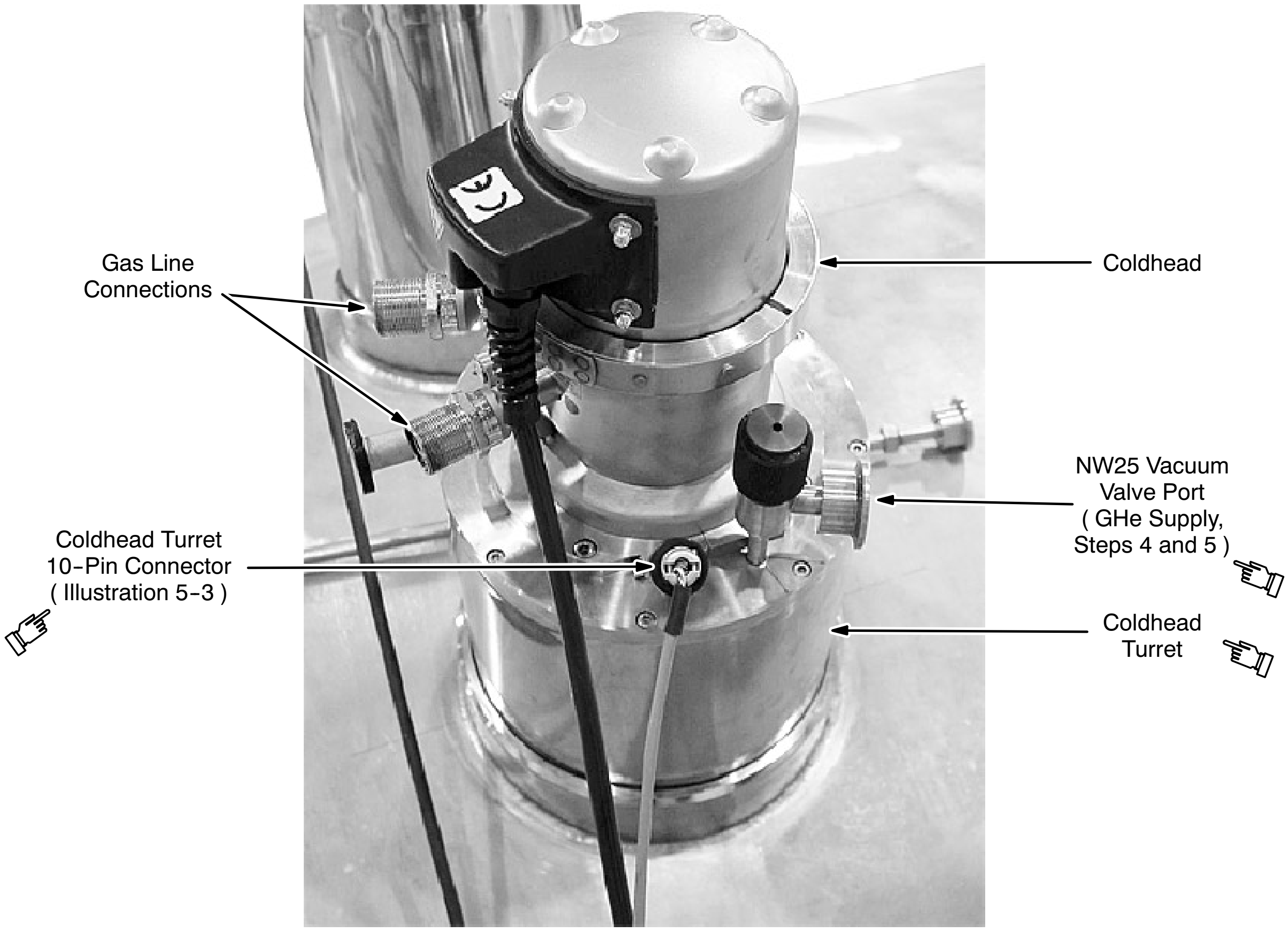
3. Disconnect both Coldhead gas lines at the Coldhead. See REPLACEMENT / MAINTENANCE, Section 11, Helium Flexlines Connections / Replacement.
4. Connect the helium gas supply line to the NW25 Vacuum Valve Port on the Coldhead Turret. Start the gas flowing at 1 - 2 psig.

CAUTION

Do not fully open the knob on the NW25 Vacuum Valve Port as a leak can occur while the valve is fully open. Make sure the valve is at least a half turn less than fully open.

5. Vent the vacuum space by allowing helium gas to flow through the NW16 Vacuum Valve on the Coldhead Turret. See Illustraion 5-1.

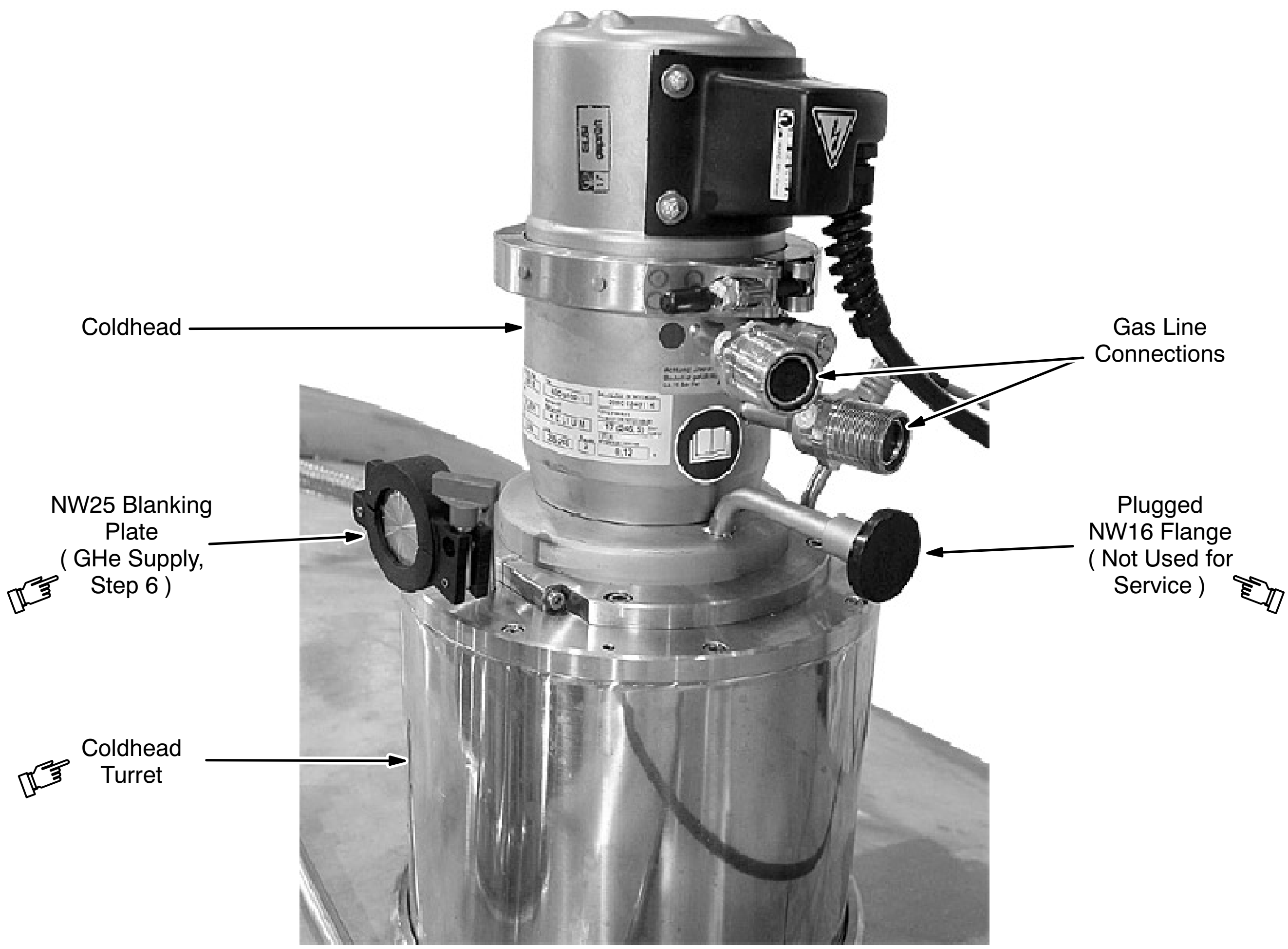
5-2 REMOVAL (continued)



COLDHEAD TURRET AND COLDHEAD
ILLUSTRATION 5-1

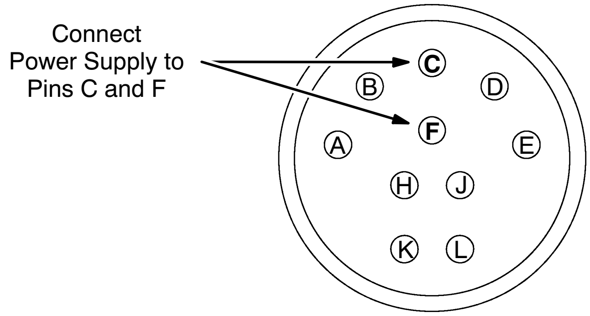
6. Remove the NW25 Blanking Plate on the Coldhead Turret. See Illustration 5-2. Allow helium gas to flow through at 1 - 2 psig, measured at the regulator. Make sure this helium gas flow is continuous to keep air / water from entering the turret.

5-2 REMOVAL (continued)



COLDHEAD TURRET AND COLDHEAD
ILLUSTRATION 5-2

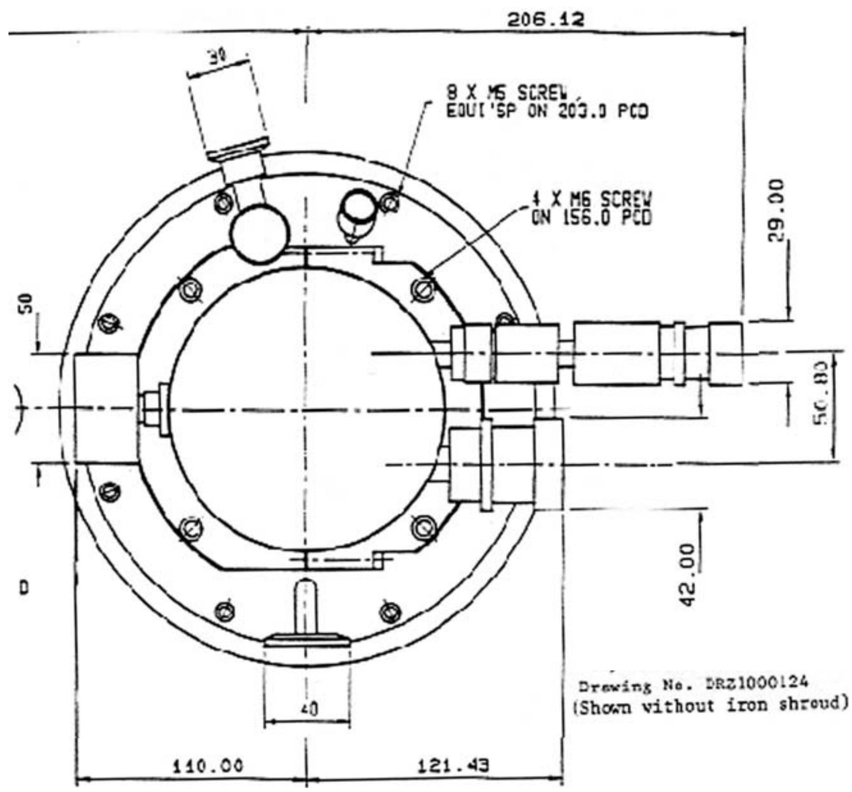
7. Connect the power supply to pins C and F of the Coldhead Turret 10-Pin Connector (Illustration 5-3).



COLDHEAD TURRET 10-PIN CONNECTOR
ILLUSTRATION 5-3

5-2 REMOVAL (continued)

8. Turn the power supply on. Make sure 1 to 1.5 Amps flows to the heater.
9. Monitor the Coldhead first and second stage temperature sensors for warming. The temperature sensors should approach room temperature in approximately 30 minutes. See FUNCTIONAL CHECKS, Section 7, Temperature Sensor Measurement.
10. Once the Coldhead is near room temperature remove the M5 x 8 mm screws from the Coldhead top flange. See Illustration 5-4.



SHIELD COOLER COLDHEAD, TOP VIEW
ILLUSTRATION 5-4

5-2 REMOVAL (continued)

WARNING!

THE FOLLOWING PRECAUTIONS MUST BE TAKEN WHEN REMOVING THE COLDHEAD TO PREVENT THE IT FROM BEING ATTRACTED BY THE MAGNETIC FIELD AND AVOID CRYOGEN BURNS:

- **MAKE SURE ALL REQUIRED EQUIPMENT IS ACCESSIBLE WITHIN YOUR REACH.**
- **MAKE SURE TO WEAR NONABSORBENT LEATHER GLOVES WHEN REMOVING THE COLDHEAD. THE FIRST AND SECOND STAGES OF THE COLDHEAD ARE COLD AND CAN CAUSE CRYOGENIC BURNS IF CONTACT IS MADE WITH THE SKIN.**
- **MAKE SURE AN ASSISTANT IS AT THE CRYOSTAT'S SIDE BEFORE DISLODGING COLDHEAD. THE ASSISTANT MUST BE CERTIFIED IN ACCORDANCE WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

11. Try removing the Coldhead by pulling vertically along the Coldhead's axis. If it does not dislodge, wait another 15 minutes before trying again.

WARNING!

DO NOT LOWER OR CARRY COLDHEAD DIRECTLY PAST END OF MAGNET TO PREVENT COLDHEAD BEING DRAWN INTO THE BORE.

12. Immediately pass the dislodged Coldhead to the assistant, who should safely carry the removed Coldhead directly to the wall and then away from the magnet room.

Note

Handle the Coldhead carefully. Do not drop nor allow objects to strike the Coldhead as the lower portion is constructed with thin tubes which can easily be damaged.

13. Quickly cover the turret openning with a blanking flange fully seated on the Coldhead o-ring to prevent air from entering the turret.
14. Complete Shield Cooler warming to room temperature.

5-2 REMOVAL (continued)

15. Make sure the Coldhead's copper flanges were not bent during removal.
16. Turn off the heater to pins C – F.
17. Disconnect the power supply.

5-3 REPLACEMENT

WARNING!

THE FOLLOWING PRECAUTIONS MUST BE TAKEN WHEN REPLACING A COLDHEAD TO PREVENT THE IT FROM BEING ATTRACTED BY THE MAGNETIC FIELD AND AVOID CRYOGEN BURNS:

- **MAKE SURE ALL REQUIRED EQUIPMENT IS ACCESSIBLE WITHIN YOUR REACH.**
- **MAKE SURE TO WEAR NONABSORBENT LEATHER GLOVES.**
- **USE AN ASSISTANT CERTIFIED IN ACCORDANCE WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

1. Loosen the clamp securing the Shield Cooler to the top flange.
2. Make sure the linkages inside the turret are free of ice. De-ice as needed.

WARNING!

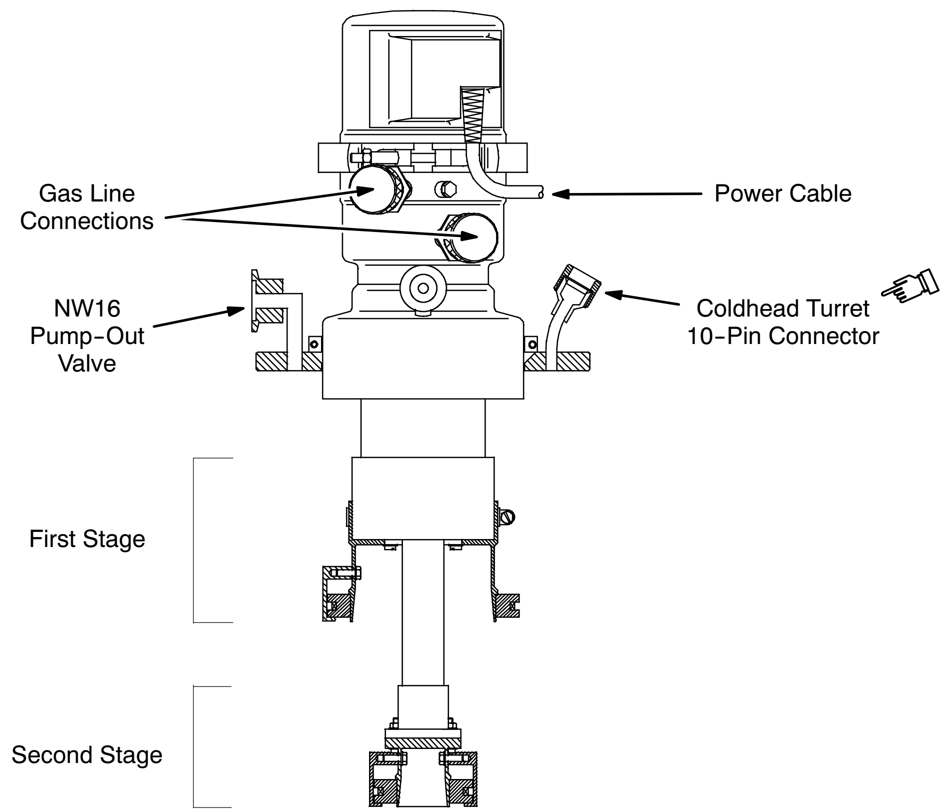
DO NOT CARRY OR RAISE COLDHEAD DIRECTLY PAST END OF MAGNET TO PREVENT COLDHEAD BEING DRAWN INTO THE BORE.

3. Have an assistant carefully bring the replacement Coldhead along the magnet room walls to the magnet's side.

Note
Handle Coldhead carefully. Do not drop nor allow objects to strike Coldhead as lower portion is constructed with thin tubes which can easily be damaged.

4. Insert replacement Coldhead into position in the Cryocooler turret. See Illustration 5-5. Make sure the Coldhead o-ring is fully seated, and the Coldhead is firmly in place.

5-3 REPLACEMENT (continued)



SHIELD COOLER COLDHEAD, SIDE SECTION
ILLUSTRATION 5-5

5. Tighten all Coldhead mounting screws.
6. Replace the NW25 blanking flange on the Coldhead Turret.
7. Repump the vacuum space through the NW16 valve.
8. Tighten the clamp securing the Shield Cooler to the top flange.
9. Check for leaks around the Coldhead o-ring, etc.

Note

The pressure gauge of the Compressor should read 20 –22 bar when running 14 – 16 bas static. A lower reading indicates a leak, which must be corrected by carefully reseating the Coldhead and checking for leaks again.

10. Turn the Shield Cooler on.
11. Monitor the temperature sensors in conformance with FUNCTIONAL CHECKS, Section 7, Temperature Sensor Measurement..

SECTION 6 – DE–ICING MAGNET

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2–5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

6–1 INTRODUCTION

Air leaking into the magnet can cause a build–up of ice in the Service Turret's neck that can eventually completely block the exhaust plumbing for helium boil–off. The resulting increased helium vessel pressure can rupture the thin–walled neck tube.

Two gauges on the Service Turret report helium vessel pressure, with the normal pressure range marked in yellow on the right–hand gauge. However, these gauges measure pressure at the top of the neck. They are effected by changes in atmospheric pressure, but will not indicate helium vessel pressure once a complete ice block has developed.

An ice block will be diagnosed when the siphon can not be inserted during helium fill. However, the ice could have been building for some time without obvious symptoms. Once symptoms of ice build–up are apparent, the magnet will requiring de–icing.

WARNING!

THE MAGNET MUST BE RAMPED DOWN BEFORE BEGINNING DE–ICING PROCEDURE IN ORDER TO PREVENT ANY QUENCH POSSIBILITY. A QUENCH WOULD SUDDENLY RELEASE CRYOGENIC HELIUM GAS INTO THE MAGNET ROOM.

6-1 INTRODUCTION (continued)

IMPORTANT !!!

Liquid cryogens are released in this procedure. Make sure of the following prior to starting the precool:

- **The procedure will be performed by trained and authorized personnel.**
- **All required equipment is in place and functional.**

IMPORTANT !!!

Any ice preventing the insertion / connection of the Ramp Lead must be cleared first to enable the magnet to be ramped down to complete ice clearing. This initial ice clearing will be performed with the magnet at field. The following warnings MUST be adhered to prevent exposure to the rapid expulsion of cryogenically cold helium gas and the blow-out tube should the magnet quench during this ice clearing.

WARNING!

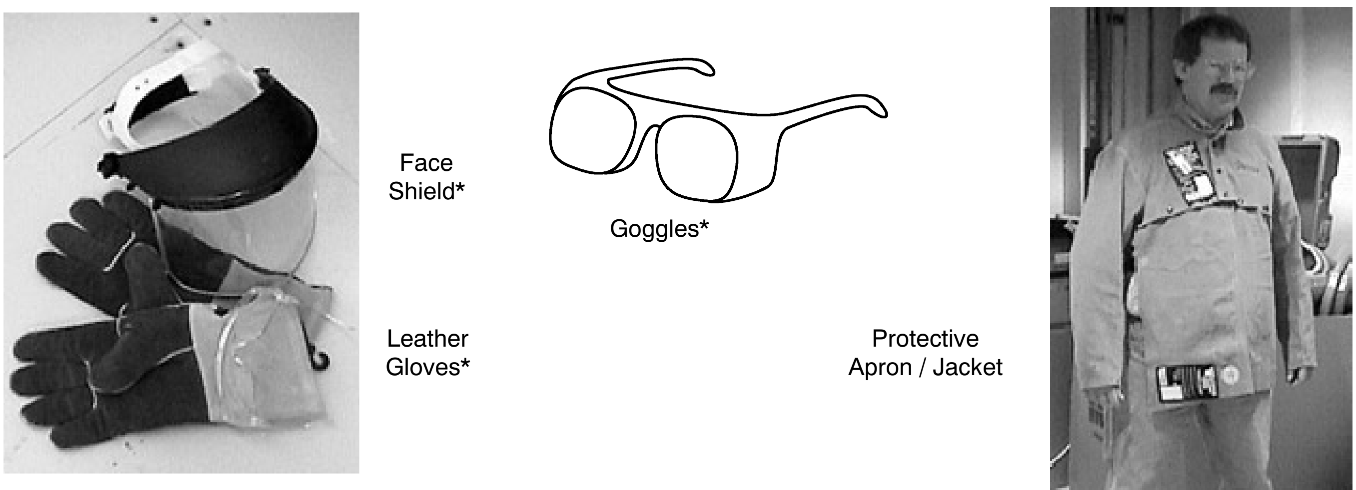
THE FOLLOWING SAFETY MEASURES MUST BE TAKEN DURING THIS PROCEDURE TO PREVENT EXPOSURE TO THE RAPID EXPULSION OF CRYOGENIC HELIUM GAS AND BLOW-OUT TUBE SHOULD A MAGNET QUENCH OCCUR:

- **WEAR ALL PERSONNAL PROTECTIVE EQUIPMENT REQUIRED FOR CRYOGEN SAFETY (NONABSORBANT GLOVES, JACKET AND PANTS; PROTECTIVE APRON / JACKET; GOGGLES / FACESHIELD).**
- **MAKE SURE THE MAGNET ROOM DOOR IS SECURED IN THE OPEN POSITION AND THAT THE EXHAUST VENT SYSTEM AND FANS ARE OPERATING PROPERLY.**
- **DO NOT PUT YOUR FACE ABOVE OR NEAR THE SERVICE TURRET.**
- **MAKE SURE THIS PROCEDURE IS PERFORMED CAREFULLY BY A PROPERLY TRAINED PERSON.**
- **MAKE SURE THE SERVICE LADDER PLATFORM AT THE MAGNET IS SECURE AND ALLOWS RAPID ESCAPE.**
- **MAKE SURE AN EXIT PATH AT LEAST 28 INCHES (711 MM) WIDE IS MAINTAINED AT ALL TIMES.**
- **MAKE SURE A SECOND PERSON TRAINED IN MAGNET / CRYOGEN SAFETY PRACTICES AND FAMILIAR WITH THE SITE (EXITS, OUTSIDE PHONE LOCATION, ETC.) IS PRESENT.**
- **UNDERSTAND AND COMPLY WITH ALL MAGNET SAFETY REQUIREMENTS AND SERVICE SAFETY REQUIREMENTS ITEMS PUBLISHED FOR GE SUPERCONDUCTING MAGNETS.**
- **IF A QUENCH STARTS TO OCCUR, ABANDON ALL WORK UNDERWAY AND EXIT THE ROOM IMMEDIATELY.**

6-2 TOOLS / EQUIPMENT

6-2-1 Personal Protection Equipment

- Goggles, face shield and gloves from the Cryogen Safety Kit 46-271137G1. See Illustration 6-1.
- Protective Apron / Jacket.
- Portable Oxygen Monitor (2106236 or 2106237) if Oxygen Monitor Kit 2107184 is not installed and functional in room.
- Cryogen Warning Signs specified in 2301164PRE*, MR Magnet - Safety Requirements***.**



* Contained in Cryogen Safety Kit (46-271137G1).

SAFETY EQUIPMENT
ILLUSTRATION 6-1

6-2-2 Tools

- Blow-Out Tube (2255509; for 2.9M ceiling: 2294703) with Phase Separator (2294434)
- Source of pure helium gas (Dewar or gas bottle)
- Connecting hose. If a ferromagnetic helium gas bottle is used, use a 15 meter (50 foot) nonferrous extension hose so the bottle may be located outside the magnet's 10 gauss zone.

6-3 PREPARATION

WARNING!

- **HELIUM GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLORLESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:**

  A. MAKE SURE THAT THE MAGNET ROOM DOOR IS PROPPED OPEN,

  B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND

  C. THE PORTABLE FLOOR FAN IS INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THIS PROCEDURE.

- **MAKE SURE ALL REQUIRED PERSONNEL PROTECTIVE EQUIPMENT IS WORN TO PREVENT CRYOGEN EXPOSURE / BURNS: GOGGLES / FACE SHIELD, NONABSORBENT GLOVES, LONG-SLEEVE SHIRT AND LONG PANTS, PROTECTIVE APRON / JACKET.**

- **SMOKING IS PROHIBITED AROUND CRYOGENS. CRYOGENS CAN CONDENSE ATMOSPHERIC OXYGEN PRODUCING A HIGHLY COMBUSTION SUPPORTING FLUID. CLOTHING WHICH HAS ABSORBED LIQUID OXYGEN CAN REMAIN FLAMMABLE FOR LONG PERIODS OF TIME.**

- **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT PROPERLY FUNCTIONING OXYGEN MONITORS ARE WORN ON ALL PERSONNEL.**

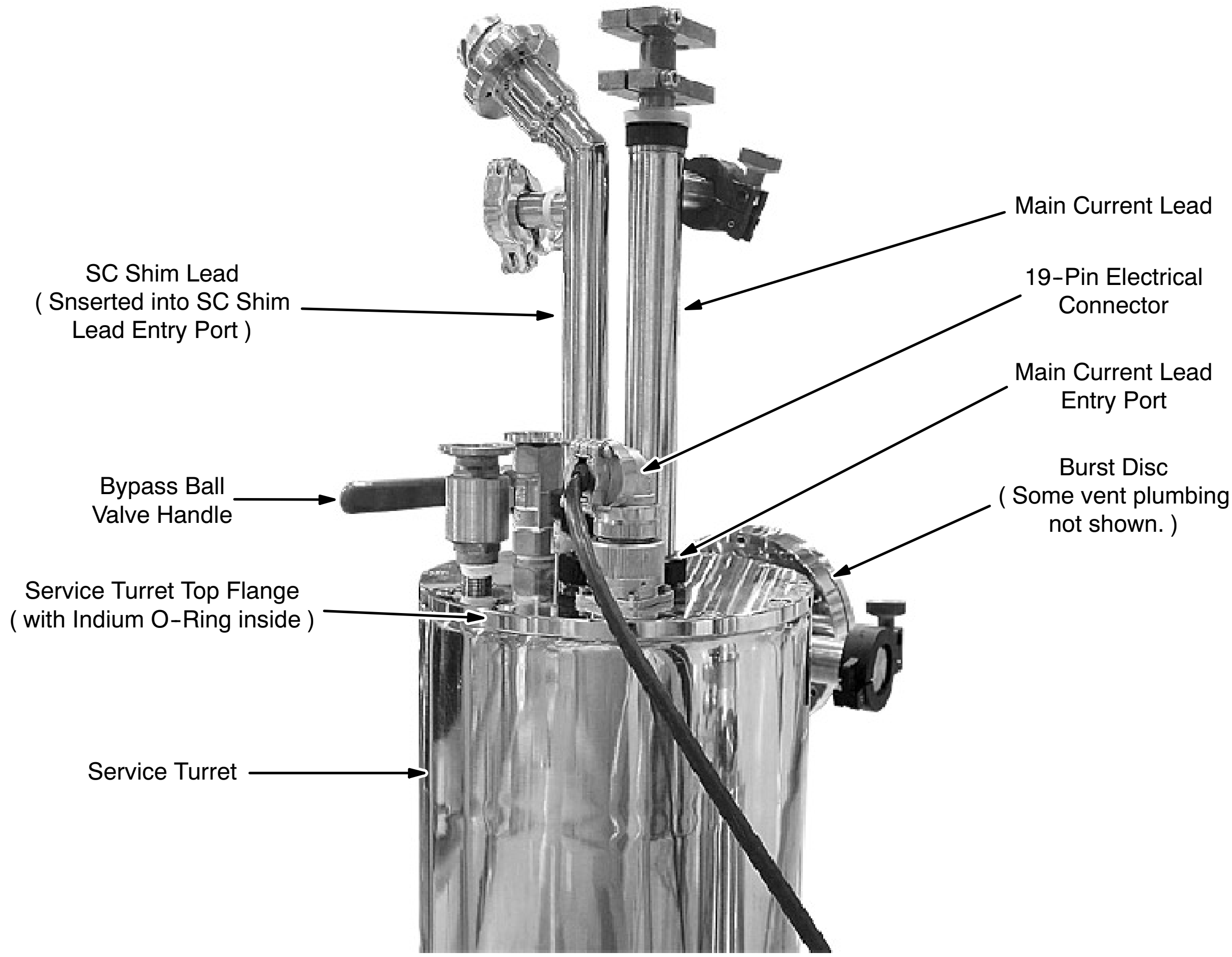
1. Complete all items marked “X” in the “Magnet / Recondenser De-Icing” column of SAFETY REQUIREMENTS, *2301164PRE, MR Magnet - Safety Requirements*, Table 1, Service Procedure Safety Requirements.

IMPORTANT !!!

Make sure all identified De-Icing defects are corrected before continuing with magnet service.

6-4 PROCEDURE

1. Open the blue-handled Bypass Valve on the Service Turret by moving the handle to the vertical position. See Illustration 6-1.



SERVICE TURRET
ILLUSTRATION 6-1

2. Continue venting until the sound of excaping gas is no longer heard, indicating the upper neck is at atmospheric pressure.

Note
No sound idicates EITHER a) the helium vessel is already at atmospheric pressure or b) the exhaust passage is completely blocked by ice.

3. Connect the long hose from the helium gas bottle to the end of the Blow-Out Tube. Make sure the Phase Separator is in place on the Blow-Out Tube's end.

6-4 PROCEDURE (continued)

4. Start the helium gas flowing at 1.0 psig.

5. Carefully unscrew the Siphon Entry Port's cap and lift the plug.

WARNING!

WHEN THE BLOW-OUT TUBE BREAKS THROUGH AN ICE BLOCKAGE A SUDDEN EXPULSION OF CRYOGENIC HELIUM GAS MAY OCCUR.

6. Slowly insert the end of the Blow-Out Tube into the Siphon / Shorting Pin Entry Port until resistance from ice is felt.

7. Continue pushing the Blow-Out Tube into the ice with a small force while the helium gas begins melting the ice. If after 3 minutes the Blow-Out Tube has not pushed further into the ice, increase the helium flow to 2.0 psig.

8. If the ice block can not be cleared in 30 minutes, contact / call your "MAC" Team Leader.

CAUTION

Do not allow large amounts of warm helium gas into the lower neck as a magnet quench may result.

9. When the Blow-Out Tube has broken through the ice, retract it 0.5 inch (10 mm) or so.

10. Open the Ball Valve Port to allow excess gas to vent from the helium vessel into the magnet exhaust vent.

11. Once pressure has been dissipated, direct the Blow-Out Tube to flow warm helium gas on the remaining ice.

12. Make sure ice is clear from the Shorting Pin connector:

    a. When ice has cleared from the Shorting Pin, remove the pin with the Shorting Pin Insertion Tool (2255522).

    b. Screw the Blow-Out Tube into the Shorting Pin connector at least two turns.

    c. Flow 2 psig helium gas until the bottom connector is clear of ice.

    d. Remove the Blow-Out Tube.

6-4 PROCEDURE (continued)

13. Replace the Siphon Port plug and nut once the port is clear of ice.
14. Stop the helium gas flow.
15. Repeat Steps 4 – 14 to clear ice from the SC Shim Lead Entry Port and the Main Current Lead Entry Port.

IMPORTANT !!!

At Step 12 make sure to test insert the SC Shim Lead in its connector and the Main Current Lead in its connector. If connection isn't made, flow 2 psig helium gas on the connector until obstructing ice is cleared.

16. Diagnose the source of the ice formation:
    a. Verify that the Burst Disc has not ruptured and is seated correctly. Replace or reseat the Burst Disc as necessary.
    b. Check the o–rings in the Siphon Entry Port, the SC Shim Lead Entry Port and the Main Current Lead Entry Port for breakage. Check the nuts for the ability to tighten. Replace o–rings / nuts and reseat the port plugs as necessary.
    c. Make sure the Top Flange is seated and that the indium seal is not leaking. Tighten screws as necessary.
    d. Make sure the Service Turret Check Valve and Ball Valve seals do not leak. Replace the seals using Teflon or PTFE tape as necessary.
    e. Verify that all Service Turret and Vent Kit fittings are correctly assembled and installed.

SECTION 7 - MAGNET WARM-UP

DANGER!!

MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, *2301164PRE, MR MAGNET - SAFETY REQUIREMENTS*, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

7-1 INTRODUCTION

Very few situations will require warming up the magnet in the field (for example, removing an internal ice block or field repairing a vacuum leak that requires reevacuation). If a magnet warm-up is required, contact the Regional MAC Team Representative before proceeding.

IMPORTANT !!!

Liquid cryogens are released in this procedure. Make sure of the following prior to starting the precool:

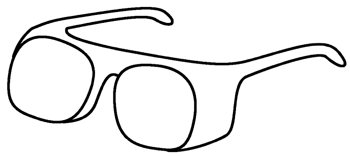
- **The procedure will be performed by trained and authorized personnel.**
- **All required equipment is in place and functional.**

7-2 TOOLS / EQUIPMENT

7-2-1 Personal Protection Equipment

- Goggles, face shield and gloves from the Cryogen Safety Kit 46-271137G1. See Illustration 7-1.
- Protective Apron / Jacket.
- Portable Oxygen Monitor (2106236 or 2106237) if Oxygen Monitor Kit 2107184 is not installed and functional in room.
- Floor exhaust fan. Fan to be specified by Installation Specialist and obtained locally.
- Cryogen Warning Signs specified in *2301164PRE, MR Magnet - Safety Requirements*.

Face Shield*



Goggles*

Leather Gloves*

Protective Apron / Jacket

* Contained in Cryogen Safety Kit (46-271137G1).

SAFETY EQUIPMENT
ILLUSTRATION 7-1

7-2-2 Tools

- Empty 1500 liter helium Dewars to hold recovered liquid helium
- Full helium gas cylinder with pressure regulator
- Two full dry nitrogen gas cylinders
- 0.5 inch (12.7 mm) diameter Helium Transfer Siphon (2255866; for 2.9M ceilings: 2292330)
- Burst Disc Blank
- Pump-Out Tool (See Illustraion 7-2.)
- Pressure gauge
- Vacuum fittings (o-rings and clamps)
- NW40 to NW25 reducer
- Diaphram valve
- Non-return check valve
- NW25 tee

7-3 PREPARATION

WARNING!

- **HELIUM GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLORLESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:**

  A. MAKE SURE THE MAGNET ROOM DOOR IS PROPPED OPEN,

  B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND

  C. THE PORTABLE FLOOR FAN IS INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THIS PROCEDURE.

- **MAKE SURE ALL REQUIRED PERSONAL PROTECTIVE EQUIPMENT IS WORN TO PREVENT CRYOGEN EXPOSURE / BURNS: GOGGLES / SAFETY GLASSES AND FACE SHIELD; INSULATED NONABSORBENT GLOVES, LONG-SLEEVE SHIRT AND LONG PANTS; AND PROTECTIVE APRON / JACKET.**

- **SMOKING IS PROHIBITED AROUND CRYOGENS. CRYOGENS CAN CONDENSE ATMOSPHERIC OXYGEN PRODUCING A HIGHLY COMBUSTION SUPPORTING FLUID. CLOTHING WHICH HAS ABSORBED LIQUID OXYGEN CAN REMAIN FLAMMABLE FOR LONG PERIODS OF TIME.**

- **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT PROPERLY FUNCTIONING OXYGEN MONITORS ARE WORN ON ALL PERSONNEL.**

- **METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD. DO NOT BRING ANY FERROMAGNETIC OBJECTS, EQUIPMENT OR TOOLS INTO THE MAGNET ROOM.**

1. Complete all items marked “X” in the “Helium Precool / Warm-Up” column of SAFETY REQUIREMENTS, *2301164PRE, MR Magnet - Safety Requirements*, Table 1, Service Procedure Safety Requirements.

IMPORTANT !!!

Make sure all identified Helium Precool / Warm-Up defects are corrected before continuing with magnet service.

7-4 PROCEDURE

WARNING!

DO NOT BRING A FERROMAGNETIC HELIUM GAS BOTTLE INSIDE MAGNET FIELD'S 10 GAUSS ZONE. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.

1. Ramp the magnet down in accordance with REPLACEMENT / MAINTENANCE Section 1, Rampdown to Zero Field.

WARNING!

Do not activate the Magnet Rundown Unit for this rampdown.

2. Remove liquid helium from the cryostat:
   a. Fully screw the M10 thread of the Helium Transfer Siphon into the magnet's siphon cone.
   b. Exchange the Burst Disc with a Burst Disc Blank.
   c. Pressurize the helium vessel to 4.5 psig.
   d. Insert other end of transfer siphon in empty helium Dewar when liquid helium is seen emerging from siphon end.
3. When helium has been removed from cryostat, vent helium gas through Cryocooler "sock."

Note

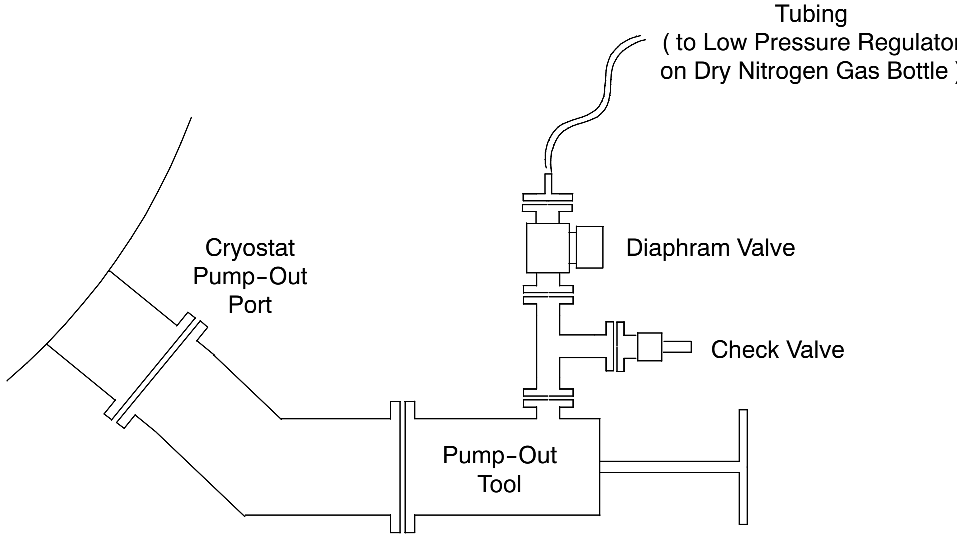
Vent the Cryocooler "sock" before venting the main vacuum to prevent possible "sock" damage.

7-4 PROCEDURE (continued)

CAUTION

Remove the Room Temperature Shim Tube before venting the Cryostat vacuum in Step 4 to prevent the tube being crushed.

4. Vent the cryostat vacuum vessel with dry nitrogen gas:
   a. Connect the Pump-Out Tool to the cryostat's Pump-Out Port. See Illustration 7-2.
   b. Attach a check valve, diaphram valve, tubing and bottle of dry nitrogen gas with a low pressure regulator to the Pump-Out Tool as shown in Illustration 7-2.



PLUMBING ARRANGEMENT FOR CRYOSTAT VENTING
ILLUSTRATION 7-2

   c. Make sure the low pressure regulator is working by operating it and then back its valve fully out.
   d. Screw the Pump-Out Tool plunger into the tapped hole in the brass transit bung.
   e. Pull the Pump-Out Tool plunger out very slowly.
   f. Open the valve on the bottle of dry nitrogen gas and slowly twist in the regulator valve to start gas flow through this plumbing and into the cryostat.
   g. Continue the gas flow until pressure within the vacuum space is slightly above atmospheric pressure. This may take one or two hours.
   h. Use the gas bottle's regulator to balance pressure against the check valve to keep pressure within the vacuum space at 0.0 - 0.5 psig.
5. Leave cryostat in this condition for at least seven days without leakage.

SECTION 8 – TRANSITING THE MAGNET

8-1 INTRODUCTION

The magnet should not be moved without the Transit Bungs installed to secure the internal cryostat vessel.

IMPORTANT !!!

The magnet must first be warmed up in accordance with REPLACEMENT / MAINTENANCE Section 7 before commencing this procedure.

8-2 PROCEDURE

CAUTION

The outer cover plates are heavy. To avoid injury:

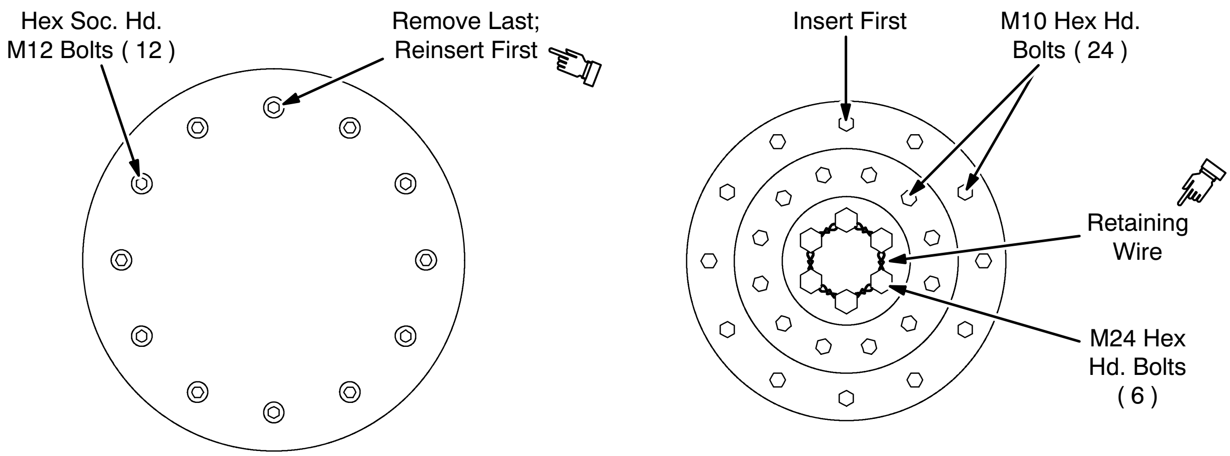
- **Wear leather gloves.**
- **Use each cover plate's top bolt for support while removing the other eleven bolts, then hold the plate with a "free" hand while removing the top bolt. Use both hands when lowering each plate.**

IMPORTANT !!!

Remove one pair of outer and inner cryostat cover plates and reinstall one transit bung assembly at a time.

1. Remove the eleven lowest M12 bolts securing one outer cover plate in place. Only slightly loosen the top bolt. See Illustration 7-3.
2. Remove the top bolt while securely holding the cover plate with a "free" hand. Carefully lower the plate with both hands. Rest the palte with the o-ring surface facing up.

8-2 PROCEDURE (continued)



OUTER COVER PLATE **INSIDE TRANSIT BUNG**

BUNG BOLT ARRANGEMENTS
ILLUSTRATION 7-3

3. Remove the larger inner thermal vessel cover plate. Set plate and fasteners aside.

IMPORTANT !!!

Make sure no component part or tool is dropped inside the cryostat as they can not be retreived.

4. Remove the mylar superinsulation patches from the plate.

5. Remove the smaller inner thermal vessel cover plate. Place plate and fasteners with the larger cover plate.

CAUTION

The transit bung weighs approximately 35 pounds (15.88 kg). To avoid injury:

- **Wear leather gloves.**
- **Have a second technician use both hands to lift each bung and hold in place while the top bolt is inserted and secured.**

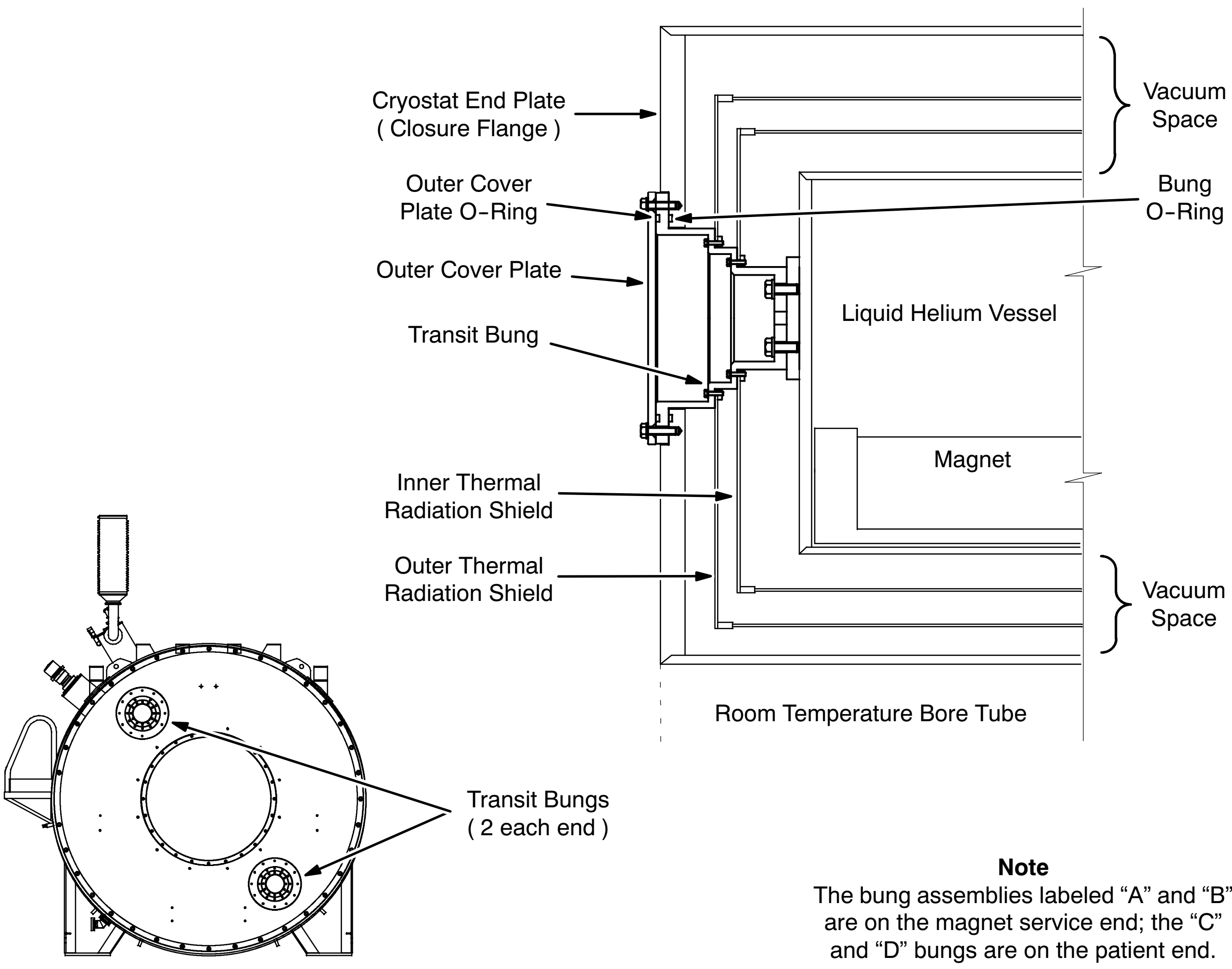
6. Clean with isopropyl alcohol and lightly grease the Transit Bung o-ring.

7. Fit the Transit Bung o-ring into the o-ring groove on the cryostat end plate.

8-2 PROCEDURE (continued)

8. Have a second technician wearing leather gloves carefully lift the magnet's Transit Bung Assembly into the bung hole and align the bung bolt holes with those of the vessel. See Illustration 7-4.
9. Insert the top M10 bolt and secure sufficiently to hold the bung's weight. See Illustration 7-3.
10. Insert the six M24 bolts into the innermost holes in the bung. Only loosely tighten. See Illustration 7-3.
11. Insert the other 23 M10 bolts into the Transit Bung Assembly. Only loosely tighten. See Illustration 7-3.
12. Temporarily insert four M12 bolts in opposing holes to locate the Bung Assembly with the cryostat end plate. Only loosely tighten. See Illustration 7-3.

Note
At this point the bung should be fully located.



Note
The bung assemblies labeled "A" and "B" are on the magnet service end; the "C" and "D" bungs are on the patient end.

TRANSIT BUNGS
ILLUSTRATION 7-4

8-2 PROCEDURE (continued)

13. Fully tighten all M10 and M24 bolts in a cross-pattern sequence. Do not tighten the M12 bolts. See Illustration 7-3.
14. Remove the four M12 bolts loosely inserted in Step 12.
15. Clean the original outer cover plate o-ring with isopropyl alcohol and lightly grease.
16. Mount the outer cover plate o-ring in the o-ring groove of the bung. See Illustration 7-4.

CAUTION

The outer cover plates are heavy. To avoid injury:

- **Wear leather gloves.**
- **Use each cover plate's top bolt for support while removing the other eleven bolts, then hold the plate with a "free" hand while removing the top bolt. Use both hands when lowering each plate.**

17. Replace the outer cover plate on the cryostat end plate and secure with all twelve M12 bolts.
18. Evacuate the cryostat vacuum vessel to 0.1 mb in accordance with REPLACEMENT / MAINTENANCE, Section 4, Re-Evacuation Of Vacuum Space.

The magnet is now ready to be shipped.

SECTION 9 – QUENCH RECOVERY

Introduction

During a magnet quench extremely cold helium gas ruptures the Burst Disc to excape through the Exhaust Vent System, making the Service Turret very cold and allowing air into the helium vessel. The helium vessel can begin cryopumping, and enough ice can form in the Service Turret to make Main Current Lead and SC Shim Lead insertion impossible. The magnet must be returned to its operating condition quickly to minimize cryopumping and icing. All debris and ice must be cleared prior to helium fill.

Procedure

DANGER!!

DO NOT ENTER THE MAGNET ROOM UNTIL THE ROOM / PERSONAL OXYGEN MONITOR INDICATEDS ROOM CONDITIONS ARE SAFE.

WARNING!

THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RECOVERING FROM A MAGNET QUENCH:

- **MAKE SURE THE MAGNET ROOM DOOR IS PROPPED OPEN.**
- **THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS.**
- **MAKE SURE A PROPERLY FUNCTIONING OXYGEN MONITOR IS USED IN THE MAGNET ROOM BY SERVICE PERSONNEL.**
- **REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN _2301164PRE, MR MAGNET - SAFETY REQUIREMENTS_, INCLUDED WITH THIS MANUAL.**

IMPORTANT !!!

If a magnet quench occurs, perform the following procedure immediately in order to minimize “cryopumping” and icing in the magnet.

Report all magnet quenches via The Online Quench Report.

Procedure (continued)

1. Verify with a functioning Oxygen Monitor that the magnet room is safe to enter.
2. Allow the helium exhaust from the magnet to subside. This typically takes at least 30 minutes.
3. Replace the Burst Disc in conformance with REPLACEMENT / MAINTENANCE, Section 2.
4. Deice the Service Turret in conformance with REPLACEMENT / MAINTENANCE, Section 6.

IMPORTANT !!!

Make sure the Shorting Pin, SC Shim Lead and Main Current Lead connectors are free of ice.

5. Order liquid helium as quickly as possible and replace in conformance with SET-UP AND CALIBRATION, Section 9, Liquid Helium Fill, as soon as liquid helium is on-site.

Note

If liquid helium is not immediately available, make sure that a positive cryostat pressure is maintained. If necessary, pressurize the cryostat with 99.999% helium gas until liquid helium arrives.

6. Inspect the Vent System for damage and debris. Clear out debris and repair damage as required.
7. Check the plumbing and pressure gauges for leaks and / or damage. Repair and replace components as necessary.
8. Order a replacement Burst Disc (2255516) to keep on-site as a spare.
9. Make sure the magnet is filled to the 100% level with liquid helium. Check magnet pressure.
10. Ramp and shim the magnet in conformance with SET-UP AND CALIBRATION, Sections 10 and 11, respectively.

SECTION 10 – PREPARATION FOR SITE REMOVAL

1. Perform the following procedures from this manual (Direction 2295400) in sequence:

   a. Ramp down the magnet in conformance with REPLACEMENT / MAINTENANCE, Section 1, Ramp-down to Zero Field.

   b. Warm up the magnet in conformance with REPLACEMENT / MAINTENANCE, Section 7, Magnet Warm–Up.

   c. Prepare the magnet for transit in conformance with REPLACEMENT / MAINTENANCE, Section 8, Transiting the Magnet.

2. Move the magnet in conformance with Direction 2341078, GEMSO 3.0T/940 Active Shield Magnet Delivery and Installation, Section 5, Moving Magnet to MR Suite.

SECTION 11
HELIUM FLEXLINE CONNECTIONS / REPLACEMENT

11-1 INTRODUCTION

Follow the procedures in this section to prevent helium loss and damage to the Coldhead and Aeroquip fittings during gas line replacement. Make sure the gas line insulation is reinstalled on the new gas lines to prevent “spike” noise and image problems.

CAUTION

Order and technique are important when connecting helium flexlines to prevent inoperative Coldheads and system contamination.

WARNING!

DO NOT PUT ANY BENDING FORCE ON AEROQUIP FITTINGS WHILE CONNECTING / DISCONNECTING HELIUM FLEXLINES. A BENDING FORCE WILL CREATE DIFFICULTY IN THE RAPID THREAD ENGAGEMENT / DISENGAGEMENT REQUIRED TO PREVENT HELIUM LOSS AND CONTAMINATION. IT CAN ALSO RESULT IN BENDING OR LEAKING OF THE AEROQUIP STEMS. SUPPORT THE GAS LINES TO PREVENT A BENDING FORCE WHEN CONNECTING / DISCONNECTING TO THE COLDHEAD AT THE TOP OF THE MAGNET OR TO THE COMPRESSOR.

11-2 TOOLS / EQUIPMENT

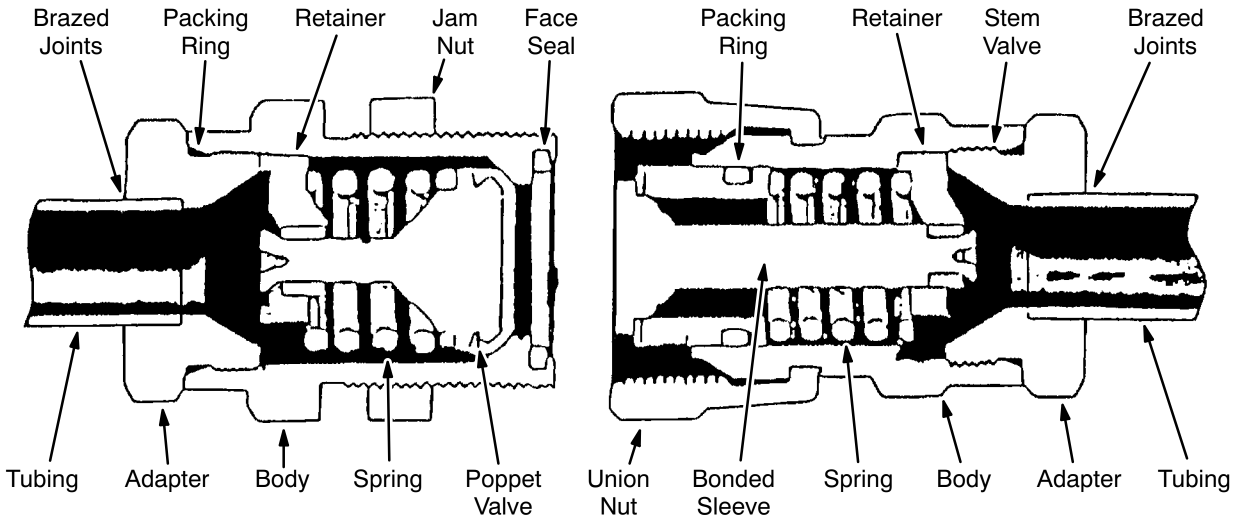
- Aeroquip fitting wrenches (included in Cold Head / Compressor Installation & Maintenance Kit 46-281088G3)
- Flexible Corregated Plastic Raceway Tubing 2251611
- Clean, lint-free nonabrasive cloth

11-3 CONNECTING AEROQUIP COUPLINGS

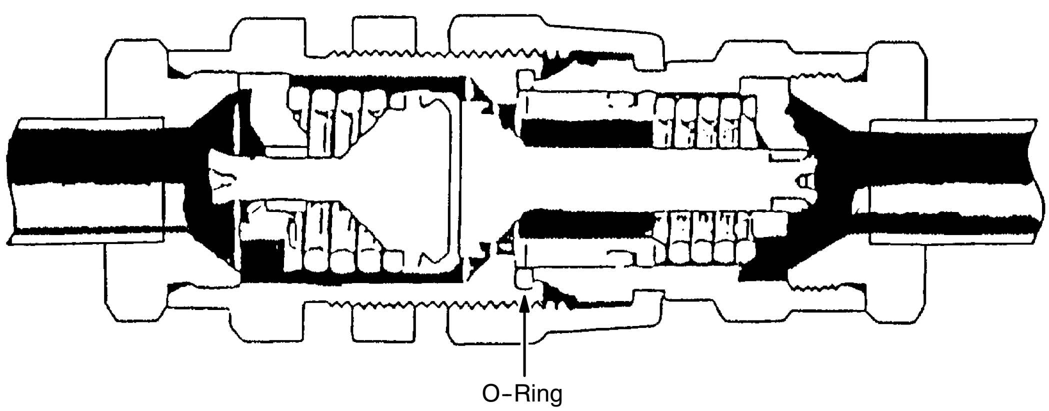
Note

Refer to Illustrations 11-1 through 11-3 for this procedure. Make sure the Aeroquip o-ring is present and in good condition on female fittings on the Coldhead side.

Coupling Halves, Disconnected



Coupling Halves, Connected



VIEW OF DISCONNECTED AND CONNECTED SELF-SEALING (AEROQUIP) COUPLINGS
ILLUSTRATION 11-1

11-3 CONNECTING AEROQUIP COUPLINGS (continued)

1. The system gas connections are shipped with caps and plugs to keep the fittings clean and free from damage. Remove the caps and plugs and thread them together for storage.

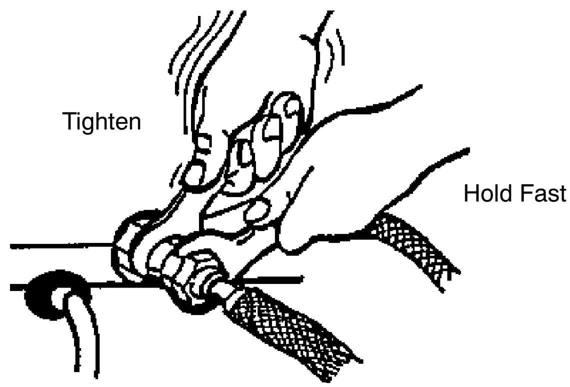
2. Wipe the faces of the couplings with a lint-free cloth to insure they are clean and free of chips and dust.

3. Make sure the face seal is in place on the inside periphery of the male coupling and is not damaged.

4. To make the connection, start the hose-side union nut onto the male connector by hand. Then, with the wrenches supplied, hold the stationary part of the female coupling while turning the union nut with the other wrench.

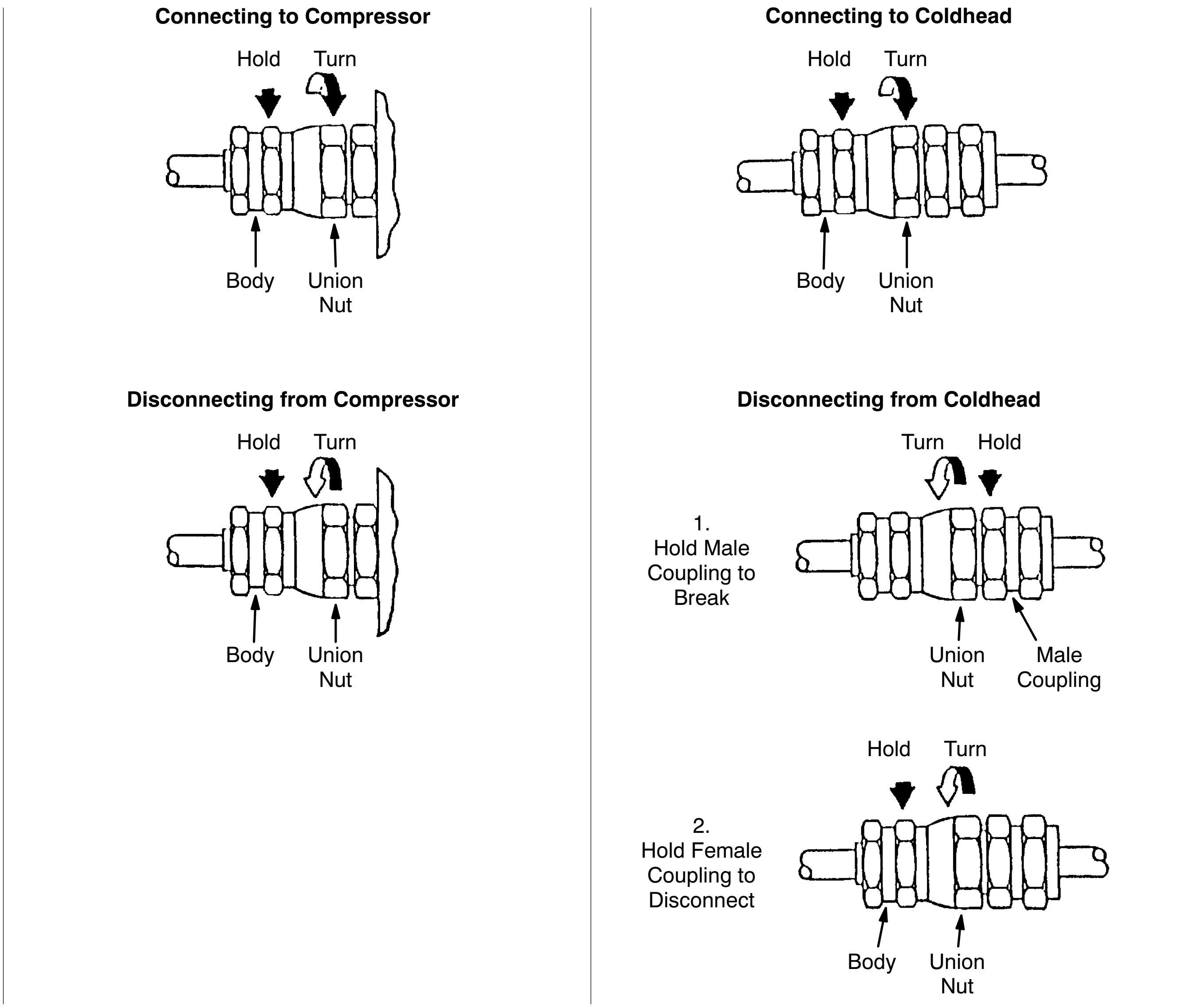
5. As the poppet begins to open there will be a slight venting of gas from the fitting. Continue to tighten the connection until the female coupling is firmly seated against the face seal on the male coupling. The required torques are:

   - 35 ft-lbs (47.5 N-m) for a 1/2 inch connection
   - 45 ft-lbs (61.0 N-m) for a 3/4 inch connection



PROPER 2-WRENCH CONNECTION TECHNIQUE FOR AEROQUIP COUPLINGS
ILLUSTRATION 11-2

11-3 CONNECTING AEROQUIP COUPLINGS (continued)



CONNECTING AND DISCONNECTING GAS LINES
ILLUSTRATION 11-3

11-4 DISCONNECTING AEROQUIP COUPLINGS

IMPORTANT !!!

Make sure the male couplings at the Compressor and Coldhead DO NOT ROTATE when disconnecting lines. Avoid torsional forces on the flex sections.

Note

Excessive gas will escape if the fittings are not alligned properly during connection or disconnection.

1. To disconnect the gas line at the Coldhead, first use one wrench to turn the female coupling union nut about 1/8 turn, while holding the male coupling with the other wrench. This will overcome the initial torque required to break the connection without loosening the male connector from its adapter.
2. Make sure the hose does not rotate to avoid a torsional force on the hose.
3. Place the second wrench on the stationary part of the female coupling and continue to unthread the union nut. Make sure the male connector does not rotate when disconnecting.
4. Make sure the bulkhead jam nut is secure and the male coupling does not rotate when removing the gas line.
5. To disconnect the gas line at the Compressor, turn the union nut on the female coupling while holding the stationary part of female coupling with a second wrench. Since the male coupling is bulkheaded to the Compressor front panel with a lock washer, the male coupling should not rotate from its adapter while removing.
6. When the hoses are disconnected, check each coupling to make sure the face seal is in place.

Note

While the hose is venting during disconnection, the face seal is often blown out of its gland and into the female coupling. Failure to remove the seal from the female coupling will cause the connection to leak when reconnected, with or without another face seal installed.

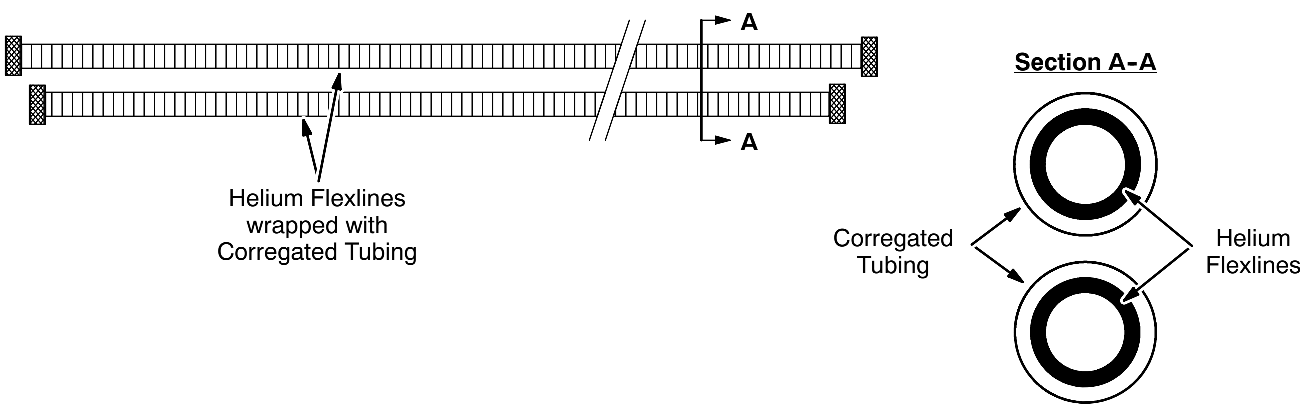
7. If all seals are in place, replace the dust caps and plugs to the coupling halves.

11-5 REPLACING GAS LINE INSULATION CASING

1. Slide the old insulated tubing off the gas line. See Illustration 11-4.
2. Cover both lines completely with Corregated Plastic Tubing (2251611) from the Penetration Panel where the lines exit the magnet room to the helium line fittings at the Coldhead. See Illustration 11-4.

Note

Corregated plastic tubing is used to isolate helium gas lines from each other and from ground to prevent spike noise.



HELIUM LINE CASING INSULATION
ILLUSTRATION 11-4

SECTION 12 – PERIODIC MAINTENANCE

Maintenance procedures should be performed at the intervals specified in Table 10–1. Some procedures require specially trained personnel.

TABLE 10–1
MAINTENANCE INTERVALS

Item	Perform Once Every				Trained Service Personnel Required?
	Work Day	Two Months	Year	Two Years	
1. Check for presence of spare parts kit and service ladder for magnet.	X				
2. Inspect magnet for ice (REPLACEMENT / MAINTENANCE Section 6, De–Icing Magnet)	X				
3. Inspect for damage at vertical stack and plumbing joints.	X				
4. Measure cryogen level (FUNCTIONAL CHECKS Section 9, Liquid Helium Level Measurement)	X				
5. Calculate helium boil–off rate (FUNCTIONAL CHECKS Section 10, Cryogen Boil–Off Rate Calculation)	X				
6. Check helium vessel operating pressure (FUNCTIONAL CHECKS Section 8, Cryostat Pressure Measurement)	X				
7. Check Coldhead operation (46–294439P4, Leybold Coldhead vendor manual)	X				
8. Check Magnet Rundown Unit (FUNCTIONAL CHECKS Section 4, Magnet Rundown Unit)	X				
9. Check Shield Cooler Compressor operation (FUNCTIONAL CHECKS Section 5, Shield Cooler Compressor Checks)		X			
10. Measure Cryostat Shield temperature (FUNCTIONAL CHECKS Section 7, Temperature Sensor Calibration & Measurement)		X			
11. Check Compressor cooling water flow rate (46–294439P4, Leybold Coldhead vendor manual)		X			
12. Replace Coldhead Displacer (46–294439P4, Leybold Coldhead vendor manual)			X		Yes
13. Replace Burst Disc (REPLACEMENT / MAINTENANCE Section 2, Burst Disc Replacement)			X		Yes
14. Replace Shield Cooler Compressor Absorber (46–294439P4, Leybold Coldhead vendor manual)				X	Yes

SCHEMATICS / INTERCONNECTS

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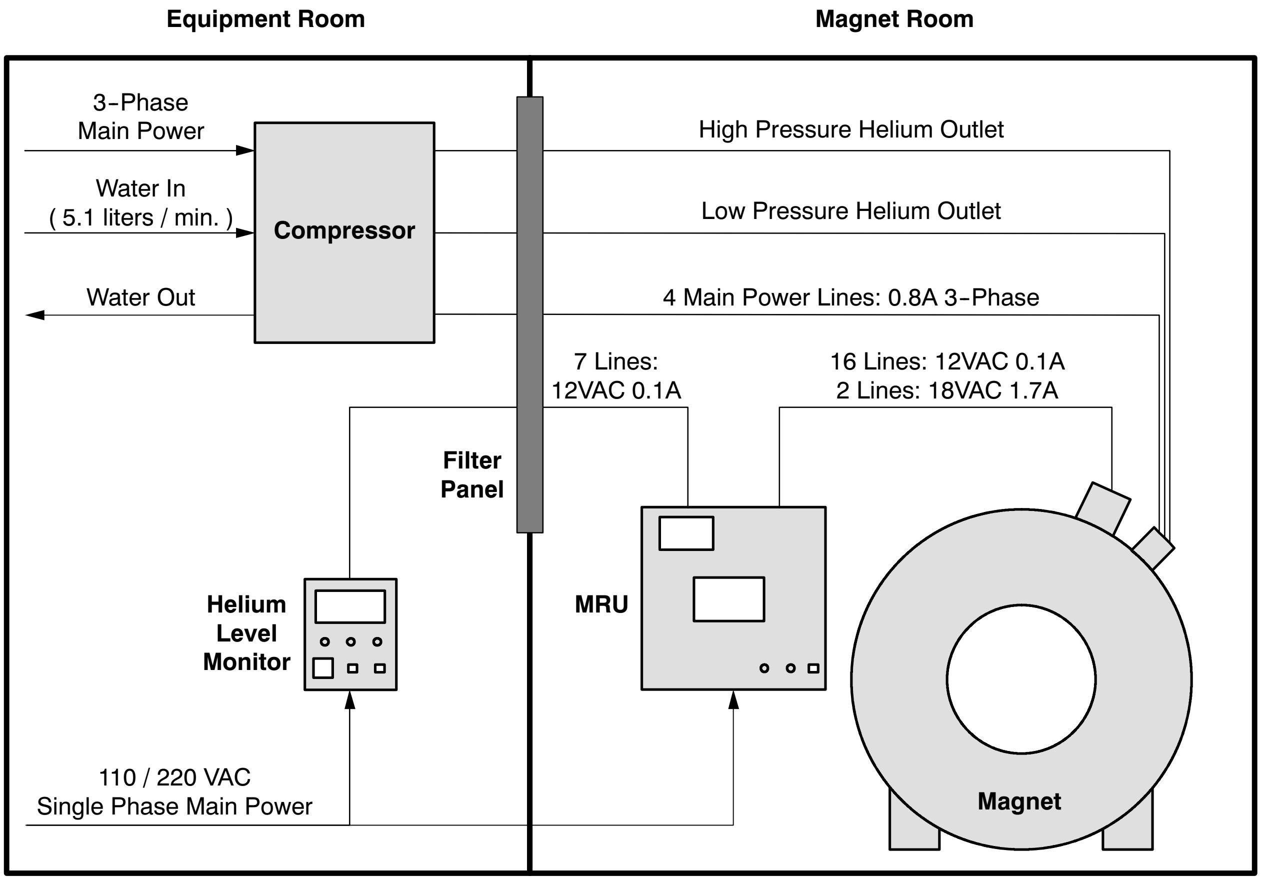
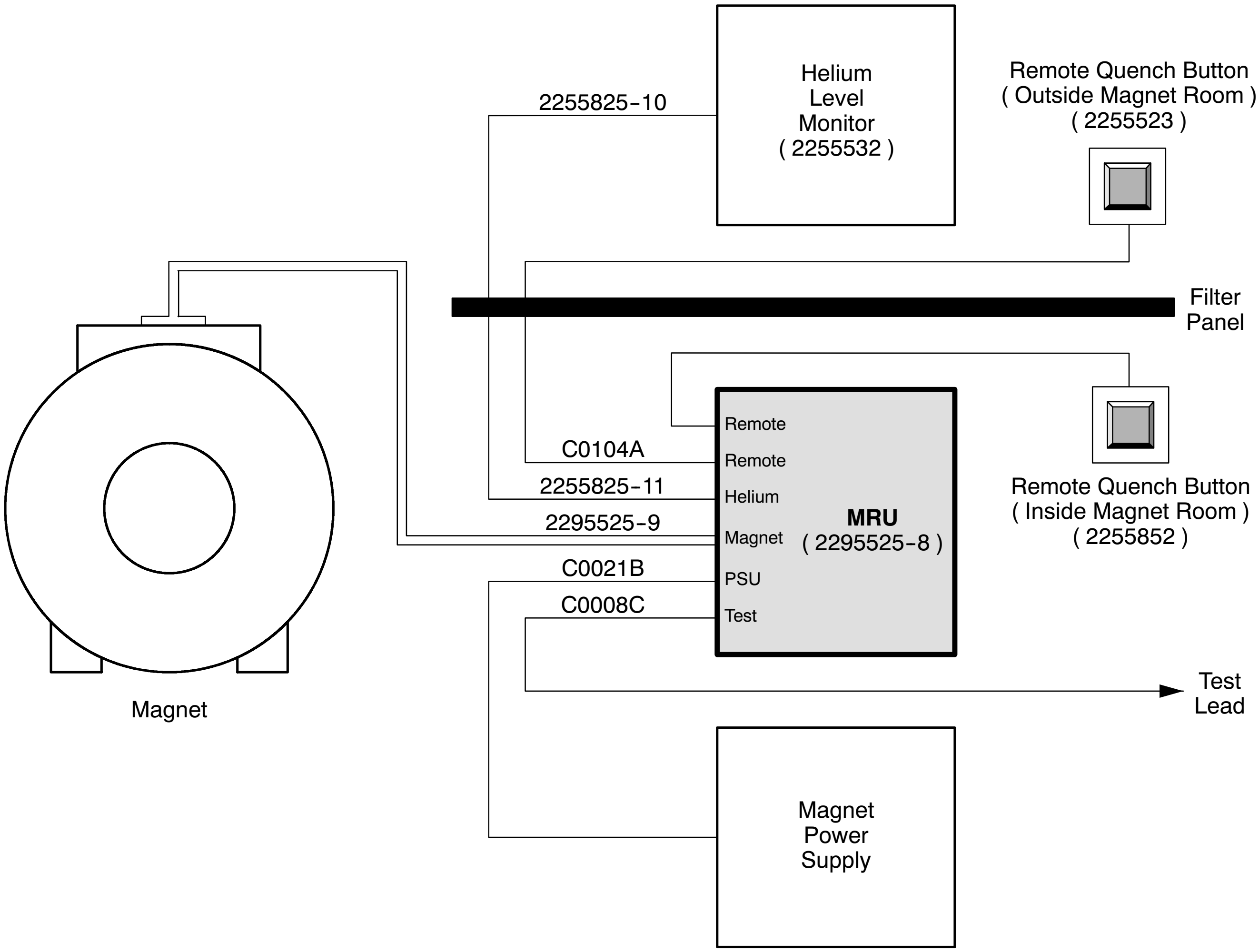
SECTION 1 – SYSTEM BLOCK DIAGRAMS



SYSTEM BLOCK DIAGRAM

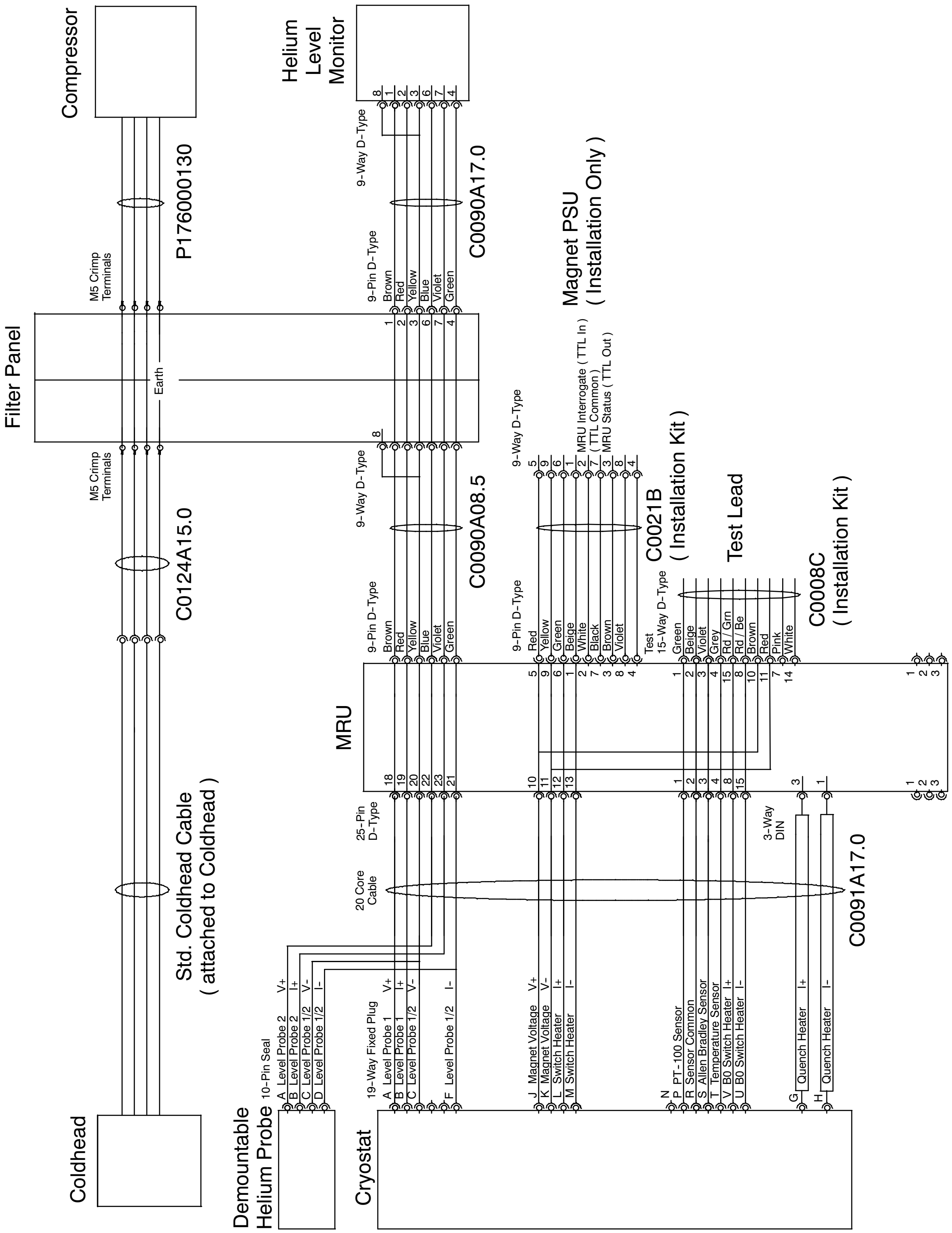
ILLUSTRATION 1-1

SECTION 1 – SYSTEM BLOCK DIAGRAMS (continue)



MRU CABLE CONNECTIONS
ILLUSTRATION 1–2

SECTION 1 – SYSTEM BLOCK DIAGRAMS (continue)



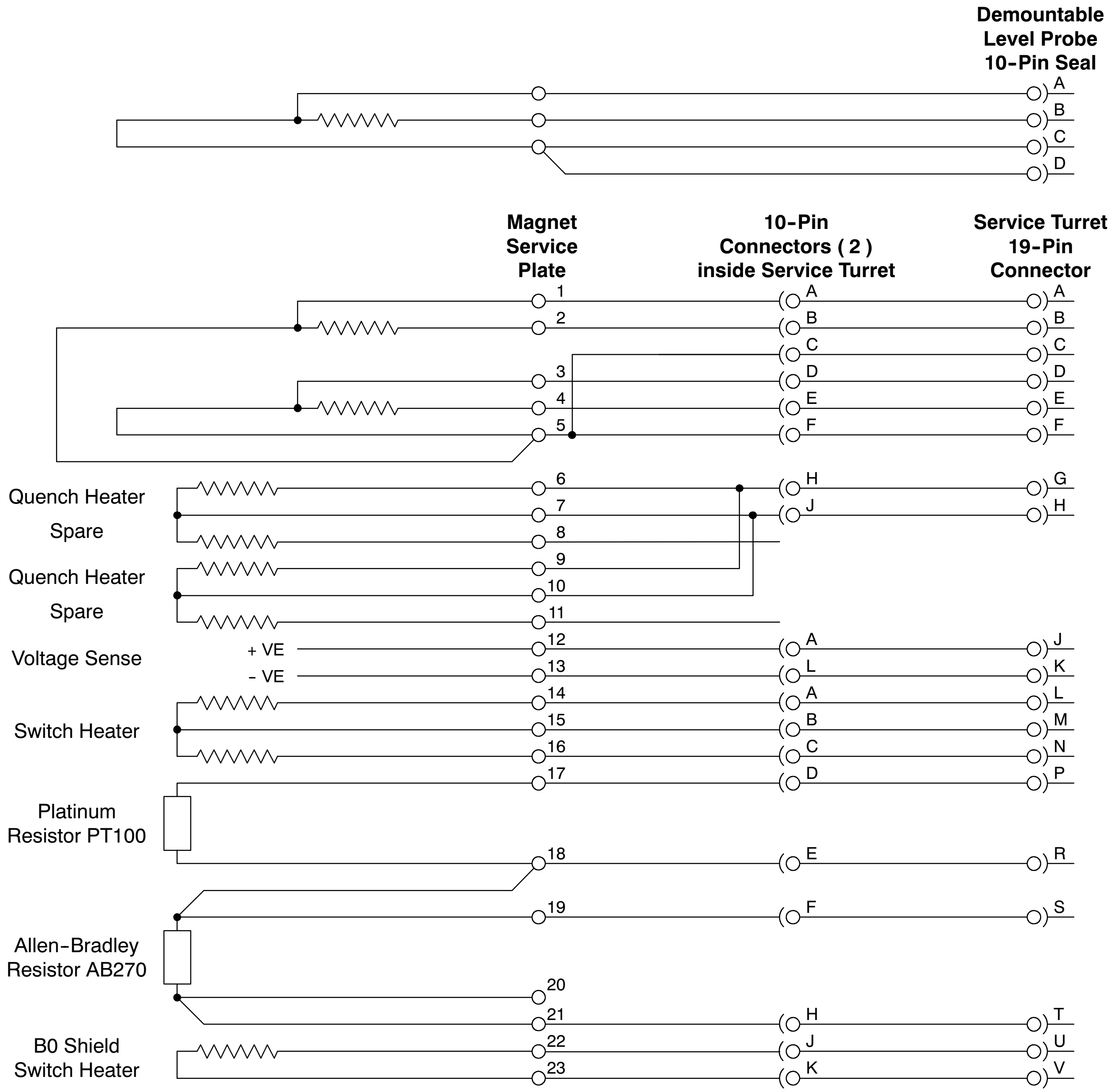
SYSTEM WIRING DIAGRAM

ILLUSTRATION 1–3

SECTION 2 – SHIELD COOLER

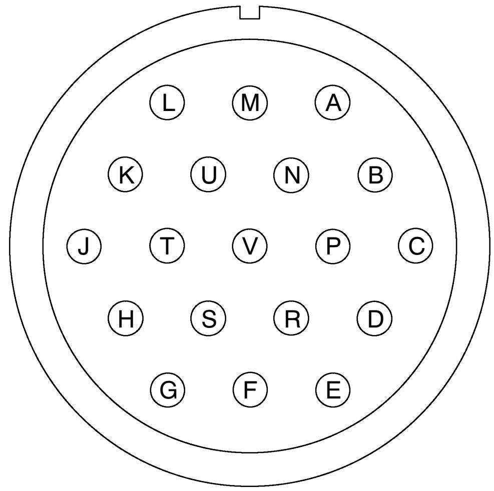
SECTION 3 – MAGNET

3-1 MAGNET SERVICE WIRING



MAGNET SERVICE WIRING SCHEMATIC
ILLUSTRATION 3-1

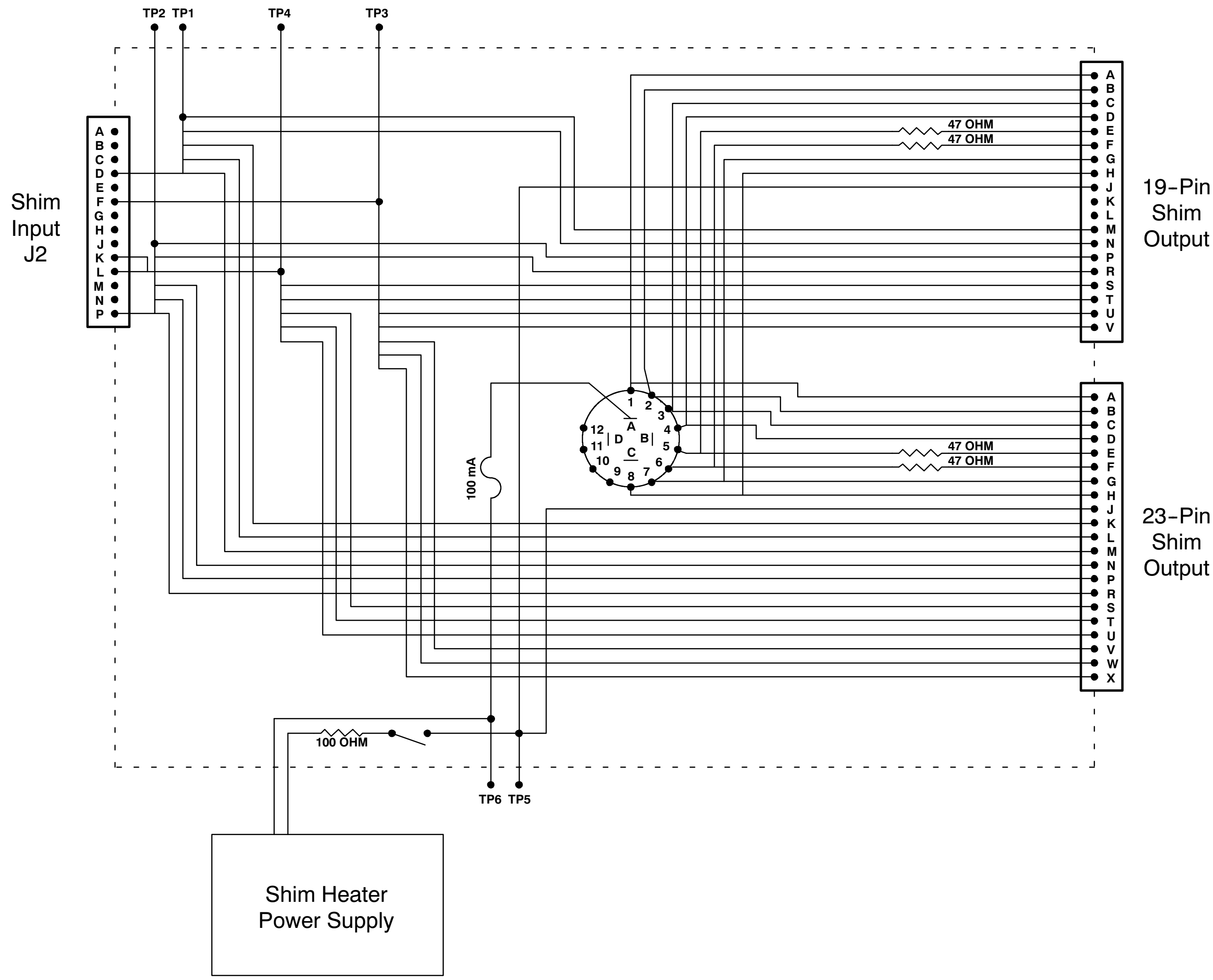
3-1 MAGNET SERVICES WIRING (continued)



Pin	Function		Pin	Function	
A	Level Probe 1	V +	L	Main Switch Heater	I +
B		I +	M		I −
C	Level Probe Common	V −	N	Spare Switch Heater	I +
D	Spare Level Probe	V +	P	PT100 Sensor	
E		I +	R	Temperature Sensor Common	
F	Level Probe Common	I −	S	Allen-Bradley Sensor	
G	Quench Heater	I +	T	Temperature Sensor	
H		I −	U	B0 Shield Switch Heater	I +
J	Magnet Voltage Sense	V +	V		I −
K		V −			

SERVICE TURRET 19-PIN CONNECTOR PINOUTS
ILLUSTRATION 3-2

SECTION 4 – SHIM ADAPTER BOX (2320885)



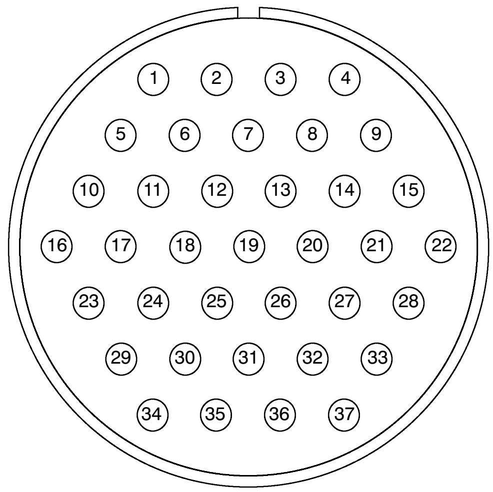
SHIM ADAPTER BOX SCHEMATIC
ILLUSTRATION 4–3

SECTION 4 – SHIM ADAPTER BOX (2320885) (continue)

TABLE 4–1
SHIM ADAPTER BOX PIN OUTS

Shim Adapter Box Connection		GE Shim Power Supply Channel	Test Point		19–Pin Shim Output Pin	23–Pin Shim Output Pin
Shim Cable (Shim Input J2)	D	T1–1	TP1	—	M, N	K, L, M
	P	T1–1 Return	TP2	—	P, R	N, P, R
	F	T1–2	TP3	—	U, V	V, W, X
	K, L	T1–2 Return	TP4	—	S, T	S, T, U
Shim Adapter Box Connection		**Heater**	**Test Point**	**Multipole Switch Position**	**19–Pin Shim Output Pin**	**23–Pin Shim Output Pin**
Heater Cable (Heater Input J3)	E	Transverse 1 +	TP5	Pole A	—	—
	F	Transverse 1 –	TP6	—	J	J
	Switch Position	**Shim Pair**	**Wire Color**		**19–Pin Shim Output Pin**	**23–Pin Shim Output Pin**
Shim Coil Selector Switch	1	Z1, Z3	Black	—	A	A
	2	Z2, Z3	Brown	—	B	B
	3	X, Z^2X	Orange	—	C	C
	4	Y, Z^2Y	Yellow	—	D	D
	5	ZX	Green	—	E	E
	6	ZY	Blue	—	F	F
	7	ZXY, ZY	Purple	—	G	G
	8	$Z(X^2-Y^2)$, X^2-Y^2	Pink	—	H	H
	9	—	—	—	—	—
	10	—	—	—	—	—
	11	—	—	—	—	—
	12	—	—	—	—	—

SECTION 4 – SHIM ADAPTER BOX (2320885) (continue)



PIN	WIRE COLOR	FUNCTION		PIN	WIRE COLOR	FUNCTION	
1	—	—	—	19	Grey	ZY	+ve
2	—	—	—	20	White / Red	ZY	–ve
3	—	—	—	21	—	—	—
4	—	—	—	22	—	—	—
5	Green / Red	Z2	+ve	23	—	—	—
6	Yellow / Red	Z2	–ve	24	—	—	—
7	Red / Black	Z3	+ve	25	White	X^2–Y^2	+ve
8	Turquoise	Z3	–ve	26	Black	X^2–Y^2	–ve
9	—	—	—	27	Green	XY	+ve
10	—	—	—	28	Yellow	XY	–ve
11	Blue	PT100 Temperature Sensor (I+)		29	—	—	—
12	Violet	PT100 Temperature Sensor (I–)		30	—	—	—
13	—	—	—	31	—	—	—
14	—	—	—	32	—	—	—
15	—	—	—	33	—	—	—
16	—	—	—	34	—	—	—
17	Red	ZX	+ve	35	—	—	—
18	Red / Blue	ZY	–ve	36	—	—	—
				37	—	—	—

37-PIN RESISTIVE SHIM COIL PIGTAIL PINOUTS
ILLUSTRATION 4–2

RENEWAL PARTS

TABLE OF CONTENTS

TABLE OF CONTENTS (continued)

SECTION 1 – MAGNET SYSTEM

1-1 DOCUMENTATION (Box 1)

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
		MAN11009CV01		1	Magnex 3T/940 AS Manuals (includes the E5011 Liquid Helium Level Monitor Manual and the E7006 Emergency Discharge Unit Manual)
	2295400			1	3T/940 Magnet & Cryogens Subsystem manual, on CD
		LOGBOOK		1	Log Book
				1	ATR w/ Final Test Data, 3.5 In. Floppy Disc (Florence shipments)
				1	ICD Card for Magnet / Collector
				1	ICD Card for Compressor

1-2 SAFETY SIGNAGE KIT (Box 2) 46-258770G4 REV. 1

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
	46-255326P1			10	5-Gauss Exclusion Zone Warning Sign
	46-255325P1			2	Security Zone Warning Sign
	46-255813P1			50	Magnetic Item Warning Label, 5 x 3 In. (127 x 76 mm)
	2289812			2	Warning Sign: Authorized Personnel Only
	46-255814P1			50	Magnetic Warning Label, 2.5 x 1.5 In. (64 x 38 mm)
	46-282214P1			2	Coolant Caution Label
	46-301671P1			1	Warning Placard

1-3 DE-TRANSIT KIT (Box 3)

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
		DHC412690		4	CCS Cat Flap
		DHC412691		4	GCS Cat Flap
				4	Insulation Pads (from unit)
	2324428	FB19050012		96	M6 Screws
	2324795	FS13120025		48	M12 x 25 mm CSk. Hd. Screw, SS (transit blanking plate)
	2212419-5			1 roll	Aluminized Mylar Tape, 1 In. (25 mm) wide

1-4 EXHAUST DUCTNG (Box 4)

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
	2292545	AHC318635		1	8 In. (203 mm) Quench Duct Assembly
	2325534	P194000013		1	8 In. (203 mm) Flexible Flue Duct, 24 In. (610 mm) long
	2292512	DHC118913		2	8 In. (203 mm) Duct Clamp (2 sections; 4 total pieces)
	46-318764P6	FS10060025		4	M6 x 25 mm Soc. Hd. Cap Screw, Stl.
	2109878-3	FS40000006		8	M6 Flat Washer, Stl.
	2324385	FS50000006		4	M6 Nut, SS
	2293036	P208000051		1	Heavy Duty Clamp
	2293038	P208000053		1	Gasket

1-5 VENT AND BAFFLES (Box 5)

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
	2255519	AHC114786		1	Copper Berilium Baffle Assembly
	2255520	AHC315883		1	Shim Lead Baffle Assembly
	2292514	AHC309787		1	Elbow Tube Assembly
	2292513	DHC409800		1	Bypass Manifold Tube, Short
	2292541	AHC315227		1	Tee Manifold Assembly
	2255516	P200000019		2	Burst Disc / Burst-Tel
	2294711	DVU318651		1	NW25 - NW25 Flex Hose, 110 mm Long
	46-241180P3	P171022530		2	O-Ring, 22.5 mm I.D. x 3.0 mm
	2294083-16	P208000038		3	KF25 Clamp
	2294083-17	P208000039		3	KF25 O-Ring with Center

1-6 CABLES (Box 6)

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
	2295525-11	C0090085		1	Cable, EDU to He Monitor, 8.5 M (27.9 Ft.)
	2295525-10	C0090170		1	Cable, EDU to He Monitor, 17 M (55.8 Ft.)
	2295525-9	C0091170		1	Service Cable
	2255533	P180000002		2	Main Power Lead, IEC American 3-Pin

1-7 MAGNET LEVELING KIT (Box 7)

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
	2343055			1	3T/940 Magnet Leveling Kit

1–8 3T/940 PASSIVE SHIM INSTALLATION KIT (Box 8) 2292493 REV. 0

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
		FB14040007		1 bag	M4 x 7 mm Screw, Brass (~1000)
		DRZ415933		1 bag	2 mm Tapped Alum. Square (~500)
		DRZ415932		1 bag	3 mm Alum. Shim, Sq. (~500)
		DRZ415930		1 bag	1 mm Tapped Iron Shim, Rect. (~500)
		DRZ415929		1 bag	2 mm Tapped Iron Shim, Sq. (~500)
		DRZ418785		1 bag	1 mm Iron Shim, Sq. (~500)
		DRZ415928		1 bag	3 mm Iron Shim, Sq. (~500)
				1 bag	2 mm Iron Shim, (~1000) – Not Required

1–9 ITEMS PACKED SEPARATELY

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
	46–317271G1			1	Oxygen Monitor
	2295525–8	E7006		1	E7006 EDU (Magnet Rundown Unit)
		E5011		1	E5011 He Monitor mounted in rack frame:
	2255532	P214000006		1	E5011 He Monitor
		P214000044		1	19 In. Subrack Frame (84F)
	2255515	P222000019		1	Ladder
	2321775	DHC115301		1	Adjustable Service Platform
	2255522	AUM117523		1	Syphonic Shorting Pin & Tool
	2255856	AUE116445		1	Helium Level Probe

1–10 CRYOCOOLER AND COMPRESSOR (Pallet 2)

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
	2255524	P196000025		1	CoolPak 6000 Compressor
	2255527	P196000008		1 box	Semi–Flexible Gas Lines, 11 M (36 Ft.) (2 per box)
	46–294150P56	P196000030		1	CP6000 Compressor Power Cable, U.S., 3.5 M (11.5 Ft.)
	46–294144P2	P176000130 ?		1	Cable, Compressor to Filter Panel, 50 Ft. (15.24 M)
	2251611			2	Flexible Corregated Tubing, Orange

SECTION 2 – MAGNET COMPONENTS

2–1 MAGNET COMPONENTS

GEMSO 3T/940 MAGNET

Item	Part Number	FRU	Name	Quantity	Description
	2255510	1	Lead	1	Main Current Lead
	2294705	1	Lead	1	Main Current Lead, 3–Section (2.9M / 9.5 ft. Ceilings)
	2255539	1	Probe	1	Demountable Level Probe*
	2255856	1	Probe	1	Demountable Level Probe
	2255519	2	Assembly	1	Baffle & Bung Assembly, Main Current Lead
	2255520	2	Assembly	1	Baffle & Bung Assembly, Shim Lead
	2294434	1	Fitting	1	Phase Separator
	2292496	1	Fitting	1	Venting Manifold (Quench Vent Elbow)
	2292545	1	Funnel	1	8 Inch (203.2 mm) Duct Assembly
	2293038	1	Gasket	1	Clamp Seal, Duct Assembly to Vent Elbow
	2293036	1	Clamp	1	Clamp, 8 Inch (203.2 mm)
	2292512	1	Clips	2	8 Inch (203.2 mm) Ducting Clamp
	2292513	1	Tube	1	Bypass Manifold Tube, Short
	2294711	1	Tube	1	Bypass Manifold Tube, Flexible
	2292541	1	Manifold	1	Tee Manifold Assembly
	2292514	1	Fitting	1	Elbow Tube Assembly
	2255521	1	Manifold	1	Pressure Gauge Manifold
	46–294150P55	1	Coupling	1	Gas Line Coupling, Set of 2, Leybold 891 71
	2255862	1	Absorber	1	Magnet Absorber (Compressor)
	2255516	1	Burst Disc	1	Burst Disc, 80 mm (3.1 inch)
	2255524	1	Compressor	1	CP6000 Compressor
	2255528	1	Coldhead Asm.	1	RGD–5100 Coldhead Assembly
	2255527	1	Gas Lines	1	Semi–Flexible Gas Lines, 11M (36 foot), box of 2
	2295525–8	1	Module	1	E7006 Magnet Rundown Unit (MRU)
	2255867	1	Module	1	E7001 Magnet Rundown Unit (MRU)
	2255532	1	Module	1	E5011 Helium Lever Monitor
	2255523	1	Switch	1	Remote Quench Switch (and Cable), Outside**
	2255852	1	Switch	1	Remote Quench Switch (and Cable), Inside
	2255531	1	Cable	1	Cable, Coldhead to O/S Room
	2295525–9	1	Cable	1	Cable, MRU to Magnet
	2295525–11	1	Cable	1	Cable, MRU to Filter Panel
	2295525–10	1	Cable	1	Cable, Filter Panel to He Monitor
	2255530	1	Cable	1	Coldhead Extension Cable
	46–294150P56	1	Cable	1	Compressor Main Power Cable, U.S., 3.5M (11.5 foot)

* Magnet 4108 only.

** Magnets 4108 – 4402 only.

2-1 MAGNET COMPONENTS (continued)

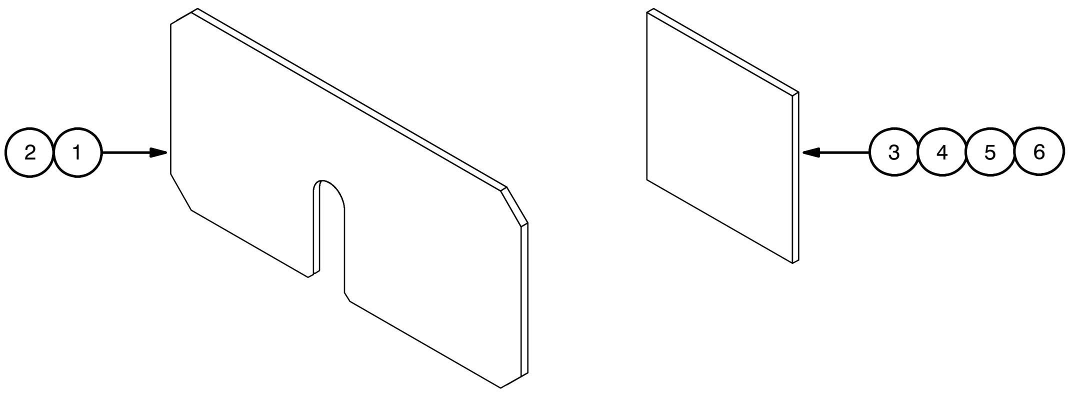
GEMSO 3T/940 MAGNET (continued)

Item	Part Number	FRU	Name	Quantity	Description
	2255534	1	Cable	1	Magnet Service Cable, Inside**
	2255535	1	Cable	1	Magnet Service Cable, Outside**
	2255854	1	Cable	1	Quench filter - MRU Interconnect
	2255851	1	Cable	1	MRU - He Monitor Interconnect**
	2255857	1	Connector	1	9-Way D Male-Male Gender Changer
	2255533	1	Cable	1	He Monitor - MRU Main Power Cable
	46-294142P2	1	Gas Line	1	Helium Supply Line, 22M (72.2 foot), Leybold 893-62†
	46-294143P2	1	Gas Line	1	Helium Return Line, 22M (72.2 foot), Leybold 893-63†
			Adapter	2	Flexible Gas Line Adapter, .5 inch female to .75 inch male, Leybold 892-90†
	2336061		Kit		3T/940 Check Valve / Service Turret Retrofit Plumbing Kit
	2335308		Fitting	1	NW25-.375 NPT Flange††
	2335148		Vent Asm	1	3T/940 Warm Vent Assembly††
	2335441		Check Valve	1	Check Valve, .375 NPT Brass††
	2335040		Fitting	2	NW25-.5 NPT 14 TPI Fitting††
	2335336		Tee	1	NW25 Tee††
	2335732		Vent Asm	1	Small Copper Vent Tube Assembly††
	2325754		Centering Ring	6	NW25 Centering Ring, Polymer††
	2325755		Clamp	6	NW25 Clamp, Polymer††

** Magnets 4108 - 4402 only.
† 22M (72.2 foot) special order alternative to 2295525-8.
†† Included in 3T/940 Check Valve / Service Turret Retrofit Plumbing Kit 2336061

SECTION 3 - MAGNET ACCESSORIES & KITS

3-1 MAGNET LEVELING KIT 2343055 REV. A



Item	Part Number	FRU	Name	Quantity	Description
1	2343040	N	Shim Plate	4	Shim Plate, Steel, 1 mm (.039 in.) thick
2	2343044	N	Shim Plate	4	Shim Plate, Steel, 3 mm (.118 in.) thick
3	2180016	2	Shim Plate	24	Shim, Aluminum, 6 x 6 x .020 in.
4	2180016-2	2	Shim Plate	16	Shim, Aluminum, 6 x 6 x .032 in.
5	2180016-3	2	Shim Plate	8	Shim, Aluminum, 6 x 6 x .040 in.
6	2180016-4	2	Shim Plate	4	Shim, Aluminum, 6 x 6 x .063 in.

SECTION 4 – SERVICE TOOLS & EQUIPMENT

4–1 Miscellaneous FRUs

Item	Part Number	FRU	Name	Quantity	Description
	2321737	1	Kit	1	3T/940 Passive Shim Kit*
	2292544	1	Assembly	36	Shim Trays
	2321736	N	Assembly	1	RT Passive Shim Assembly
	2321775	2	Platform	1	Adjustable Service Platform
	2255515	2	Ladder	1	Ladder
	2255529	1	Cable	1	Coldhead Diagnostic Cable
	2295525–5	1	Probe	1	Miniture Probe, MetroLab #1062/6/10M
	2320885	1	Adapter Box	1	Shim Adapter Box
	2320946	1	Adapter Box	1	Main Switch Adapter Box
	2294434	1	Fitting	1	Phase Separator

* See Section 4–13, 3T/940 Passive Shim Installation Kit, for kit contents.

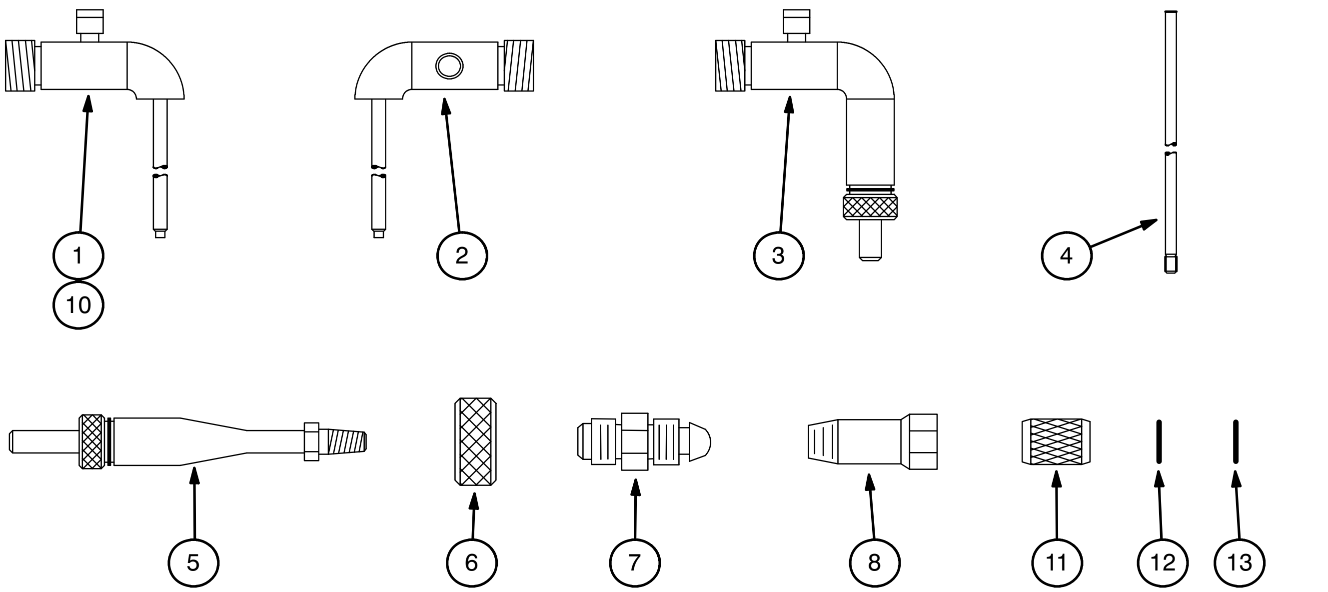
4–2 HELIUM FILL KIT, 3T — 2295507 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2294075–2	1	Regulator	1	Cylinder Regulator
2	2294083–6	1	Nozzle	1	Nozzle
3	2294083–14	1	Clamp	3	Swing Clamp, NW50, aluminum
4	2294083–17	1	Centering Ring	3	Centering Ring & O–Ring
5	46–252065P138	1	Spray	1	Leak Detector Spray, 300mL aerosol
6	2294083–7	1	Adapter	1	Gauge Tube (Siphon Entry) Adapter & O–Ring, NW25
7	2292330	1	Transfer Line	1	LHe Transfer Line (low ceiling height)
8	2255866	1	Siphon	1	He Transfer Siphon w/ Extension
9*	2294434	1	Fitting	1	Phase Separator

* Not included in kit, but required for REPLACEMENT / MAINTENANCE, Section 3, Liquid Helium Top–Offs.

4-3 UNIVERSAL FILL LINE KIT

46-294705G1 REV. 4



Item	Part Number	Name	Quantity	Description
1	46-294512P3	7 In. Stinger	1	Cryostat Stinger, 7 in., Air Products #CSA-7
2	46-294512P4	14.75 In. Stinger	1	Cryostat Stinger, 14.75 in., Air Products #CSA-14
3	46-294512P5	3 In. Reducer	1	Reducer, .50 in. x 3.00 in. male bayonet, Air Products #MBR-6-3
4	46-294512P12	16 In. Stinger Ext. Tip	1	Cryostat Stinger Extension, 16 in. with teflon tip, Air Products #CSE-16
5	46-294512P13	Adapter	1	Purge / Precool Adapter, Air Products #PPA-1
6	46-294512P15	Blanking Cap	1	Blanking Cap, 1.5 in. OD knurled brass, Air Products #BBC-1.5
7	46-294512P16	Nitrogen Adapter	1	Nitrogen Fill Line Adapter, Air Products #FLA-1
8	46-294512P17	Helium Adapter	1	Helium Fill Line Adapter, Air Products #GLA-1
9	46-294512P14	Case	1	Black Carrying Case, Air Products #PCC-27
10	46-294512P25	17 In. Stinger	1	Cryostat Stinger Assembly, 17 in., Air Products #CSE-17
11	46-318619P1	Nut	1	Extended Coupling Nut, .75-20UNEF knurled brass
12	46-260272P1	Retaining Ring	1	Retaining Ring, .687 in. OD stainless steel
13	46-260342P9	O-Ring	1	O-Ring, BUNA-N, .489 in. ID x .625 in. OD

4-4 OTHER HELIUM TRANSFER SERVICE ITEMS

Item	Part Number	Name	Quantity	Description
1*	46-271137G1	Safety Kit	1	Safety Wear Kit, Air Products #295-A-MRK500
1A*	46-271135P10	Head Gear	1	Head Gear without Visor
1B*	46-271135P11	Vision Shield	1	Vision Shield for 46-271135P10
1C*	46-271135P12	Safety Glasses	1	Safety Glasses
1D*	46-258217P2	Cryogen Gloves	1	Cryogen Handling Gloves
2	46-265286G1	He Resistance Box	1	GE Magnet Helium Resistance Box
3	46-306734G1	Regulator Kit	1	High Pressure Helium Regulator And Hose Kit
4	46-258150P1	Helium Cart	1	Nonmagnetic Cylinder Cart

* Item 1 includes items 1a, 1b, 1c and 1d, which can be ordered separately.

4-5 POWER SUPPLIES

Item	Part Number	Name	Quantity	Description
1	46-260776G4	Main P/S	1	1000 Amp Main Coil Power Supply
1A	46-260776G3	Main P/S	1	750 Amp Main Coil Power Supply
2	46-260777G3	Power Supply	1	Shimming Power Supply

4-6 SERVICE POWER SUPPLY RENEWAL PARTS FOR 46-260776G3, G4, 46-260777G3 (Part Number Cross Reference, Supplier P/N* to GE P/N)

Item	Vendor Part Number*	Description	GE Part Number
1	12-452-028	Panel, Control Assembly	46-281468P1
2	12-452-026	Heater, P/S, Heater & 24 VDC PS	46-281468P2
3	25-611-000	Panel, Input / Output Connect	46-281468P3
4	00-467-498	TCR 7.5T750, Main Coil Power Supply	46-281468P4
5	51-001-001	Fan, 3.12 in., 115V	46-260219P12
6	58-005-010	Fuse, MDA 4A, 250V	46-281468P5
7	20-292-002	Printed Circuit Board Assembly	46-281468P6
8	20-230-000	Printed Circuit Board, +24 & ±15 Power Supply	46-281468P7
9	71-024-000	10-Turn Locking Indicator Dial	46-260219P27
10	66-065-006	Analog Voltmeter, 0-36 VDC	46-281468P8
11	62-062-009	Analog Ammeter, 0-1 ADC	46-281468P9
12	66-082-007	Digital Volmeter, 3.5 Digits	46-281468P10
13	66-082-008	Digital Ammeter, 5.5 Digits	46-281468P11
14	67-055-007	10-Turn Potentiometer, 5K, WW	46-281468P12
15	68-012-005	Pushbutton Switch, MOM, White	46-260219P33
16	68-004-001	Toggle Switch, DPDT	46-281468P13
17	68-008-003	Toggle Switch, DPDT, Locking	46-281468P14
18	20-354-000	PWP, Printed Circuit Board P-Set Amplifier	46-281468P15
19	58-006-010	Fuse, MDV 0.125A, 250V	46-281468P16
20	58-001-008	Fuse, AGC 2A, 250V	46-281468P17

* Supplier parts shown in supplier service manual.

4-6 SERVICE POWER SUPPLY RENEWAL PARTS FOR 46-260776G3, G4, 46-260777G3 (continued) (Part Number Cross Reference, Supplier P/N* to GE P/N)

Item	Vendor Part Number*	Description	GE Part Number
21	20-137-087	Control Printed Circuit Board, A100	46-281468P18
22	54-072-002	Capacitor, 350 KMF/10V	46-281468P19
23	61-011-001	SCR, Dual Pack	46-281468P20
24	56-069-004	Breaker, 3-Pole, 30A	46-281468P21
25	51-002-002	Fan, 4.68 in. sq., 220V	46-260219P2
26	67-023-005	Resistor, 3 Ohm, 25W	46-281468P22
27	63-004-001	LED Indicator, Red	46-281468P23
28	51-009-001	Biscuit Fan	46-281468P24
29	51-002-001	Fan, 4.68 in. sq., 115V	46-260219P20
30	56-001-002	Breaker, 3-Pole, 30A	46-281468P25
31	12-452-027	Control Assembly Panel	46-281469P1
32	58-005-013	Fuse, MDA 6.25A, 250A	46-281469P2
33	12-452-025	Switch Heater & Internal Power Supply	46-281469P
34	25-612-000	Input / Output Panel	46-281469P4
35	00-452-084	Shim Power Supply Module, #1 thru #6	46-281469P5
36	58-005-006	Fuse, MDA 1A, 250V	46-281469P11
37	60-010-001	Diode, Dual Pak, 600V, 15A	46-281469P12
38	20-292-001	PWB, A700, Printed Circuit Board Assembly	46-281469P13
39	65-047-001	Relay, DPDT, 24VDC	46-281469P14
40	66-082-012	Digital Voltmeter, 4.5 Digits	46-281469P15
41	66-082-011	Digital Ammeter, 4.75 Digits	46-281469P16
42	68-037-007	Selector Switch, 8PL, 6STN	46-281469P17
43	67-055-011	10-Turn Potentiometer, 2000 Ohm	46-281469P18
44	65-024-007	Relay, 4PDT, 25A	46-281469P19
45	68-008-001	Thermostat N/O 195° F	46-281469P20
46	62-005-020	Transistor, 2N5685	46-281469P21
47	62-005-014	Transistor, MJ2955	46-281469P22
48	68-002-002	Thermostat, N/C, 210° F	46-281469P23
49	20-350-001	PWB, A100 Assembly	46-281469P24
50	67-055-005	10-Turn Potentiometer, 20K, 2W	46-281469P25
51	00-481-1305	Main Coil Power Supply, ESS7.5-1000-2-D-1236	46-260219P113
52	67-136-001	10-Turn Potentiometer, 100 OHM WW	46-260219P114
53	66-062-009	Analog Ammeter, 0-1 ADC	46-260219P115
54	66-082-014	Digital Voltmeter / Ammeter, 5.5 DIGIT	46-260219P116
55	80-001-013	PS1 MTR, P.S. +/- 12, +5	46-260219P117
56	67-136-002	10-Turn Potentiometer, 5K, WW, Oil-Filled	46-260219P118

* Supplier parts shown in supplier service manual.

4-7 ELECTRICAL TEST KIT — 2296053 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2295140	1	Box	1	10-Pin Breakout Box
2	2295141	1	Box	1	19-Pin Breakout Box
3	46-294167P50	1	Cleaner	1	Contact Cleaner
4	46-294167P51	1	Cord	4	Patch Cord
5	46-294167P52	1	Test Lead	1	Modular Test Lead
6	46-294167P53	1	Handle	1	Modular Test Lead Handle, pointed banana plugs
7	46-294167P54	1	Handle	1	Modular Test Lead Handle, sheathed alligator clips
8	46-294167P55	1	Handle	1	Modular Test Lead Handle, retractable pincers
9	46-294167P56	1	Brush	1	Coating Brush
10	46-294167P57	1	Case	1	Storage Case

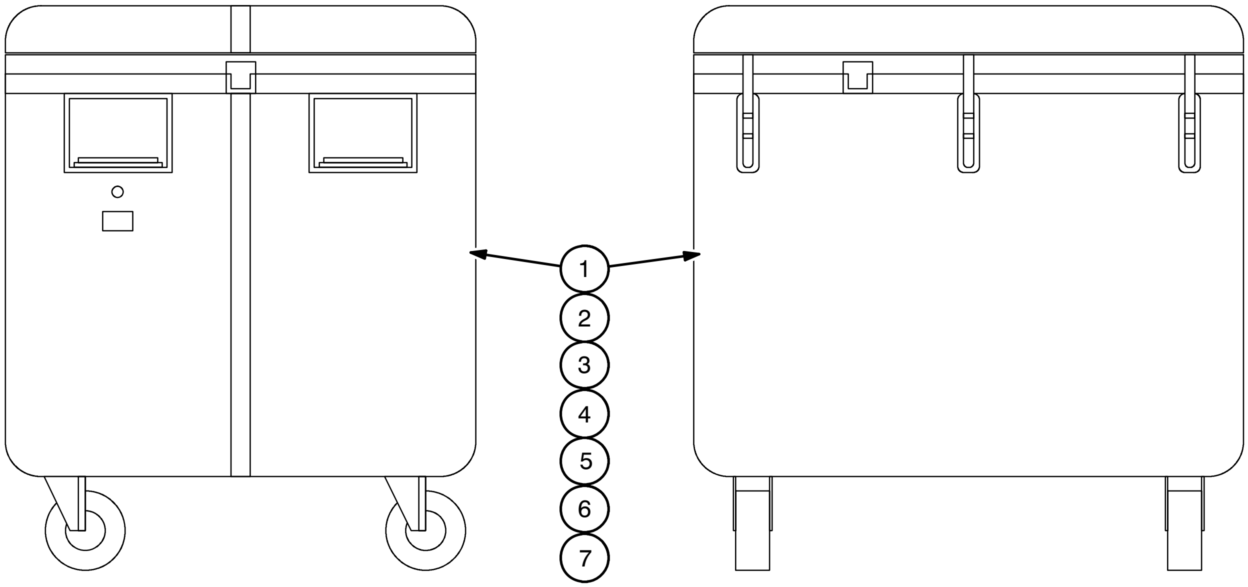
4-8 RAMP / ENERGIZE KIT — 2296056 REV. 1+

Item	Part Number	FRU	Name	Quantity	Description
2	2135435	1	Ramp Cables	1	750 Amp Ramp Cart / Cable Kit
3	2292536	1	Vent Assembly	1	Exhaust Gas / Supply Vent Assembly
4	2294083-29	1	Flex Pipe	2	Flexible Pipeline, NW16 x 500 mm
5	2294083-21	1	Swing Clamp	2	Swing Clamp, NW10/16
6	2294083-22	1	Centering Ring	2	Centering Ring & O-Ring
7	2294083-16	1	Swing Clamp	1	Swing Clamp, NW20/25
8	2294083-17	1	Centering Ring	1	Centering Ring & O-Ring
9	2255510	1	Current Lead	1	Main Current Lead
10	2294705	1	Current Lead	1	Main Current Lead, 3-Section (2.9M / 9.5 ft. Ceilings)
11	2255522	1	Pin / Tool	1	Shorting Pin and Insertion Tool
12	2294710	1	Tool	1	Hinged Syphonic Shorting Plug
13	2293772	1	Lead Clamp	1	Negative Current Lead Clamp
14	2293858	1	Lead Clamp	1	Positive Current Lead Clamp
15	46-294167P64	1	Case	1	Storage Case
16	46-294167P67	1	Padlock	1	Padlock
17	2294083-27	1	Flange	1	Blank Flange, NW16
18	2334416	N	Clamp	1	Current Cable Hold Tool (Strain Relief Clamp)
19	2336072	N	Screw	1	Hold Tool Screw 'A'
20	2336073	N	Screw	1	Hold Tool Screw 'B'

4-9 RAMP / SHIM SUPPLY CABLE (5-#8 Cond. 30A Cable, 50 Feet / 15.24 M) — 46-320460G1 REV. 1

Item	Part Number	FRU	Name	Quantity	Description
1	2192535-3	2	Cable	50 Ft.	5-Conductor #8 AWG Type SO Cable
2	46-278445P1	2	Plug	1	4-Pole 5-Wire Plug, 30A 208V, Male
3	46-208781P2	2	Label	2	Write-On Label, .75 x 1.90 inch (19 x 48.3 mm)
4	46-221917P2	2	Plug	1	3-Phase 5-Wire Twist-Lock Plug, 30A 120/208V, Female

4–10 RAMP CART / CABLE KITS



4–10–1 1000 AMP RAMP CART / CABLE KIT 2180589 REV. 0

Item	Part Number	Name	Quantity	Description
1	2135434	Cart	1	Ramp Cable Cart
2	46–260723G1	Cable	3	Power Cable Assembly, Positive 4/0
3	46–260723G2	Cable	3	Power Cable Assembly, Negative 4/0
4	46–260724G1	Cable	1	Switch Heater Cable Assembly
5	46–281667G1	Cable	1	Switch Heater Cable, RJC & RJD
6	46–281235P1	Cable	1	Voltage Sense Lead, 40 feet (12.2 meters)
7	46–294167P20	Tape	1	Black / yellow checkered tape, roll

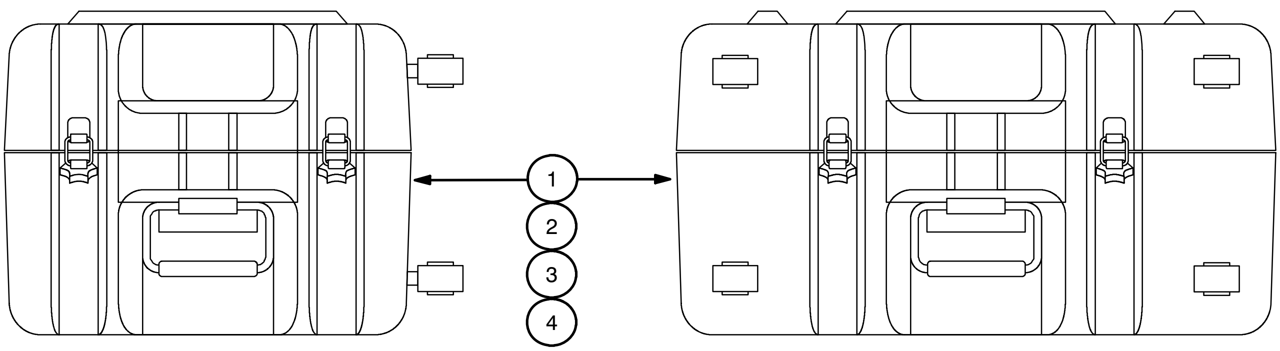
4–10–2 750 AMP RAMP CART / CABLE KIT 2135435 REV. 0

Item	Part Number	Name	Quantity	Description
1	2135434	Cart	1	Ramp Cable Cart
2	46–260723G1	Cable	2	Power Cable Assembly, Positive 4/0
3	46–260723G2	Cable	2	Power Cable Assembly, Negative 4/0
4	46–260724G1	Cable	1	Switch Heaters Cable Assembly
5	46–281667G1	Cable	1	Switch Heaters Cable, RJC & RJD
6	46–281235P1	Cable	1	Voltage Sense Lead, 40 feet (12.2 meters)

4-11 SHIMMING KIT 2297304 REV. 0+

Item	Part Number	FRU	Name	Quantity	Description
2	2297280	1	Shim Cable	1	Shim Cable, 15M
3	2255512	1	Shim Lead	1	SC Shim Lead, Long
4	2255511	1	Shim Lead	1	SC Shim Lead, Short
5	2292542	1	Plotting Rig	1	Field Plotting Rig Assembly, Large Bore
8	2294083-21	1	Clamp	1	Swing Clamp, NW10/16
9	2294083-22	1	Centering Ring	2	Centering Ring & O-Ring
10	2294083-27	1	Flange	2	Blank Flange, NW16
11	2294083-16	1	Clamp	2	Swing Clamp, NW20/25
12	2294083-17	1	Centering Ring	1	Centering Ring & O-Ring
13	2292536	1	Vent Assembly	1	Exhaust Gas / Supply Vent Assembly
15	2295525-6	1	Box	1	Transport Box

4-12 SHIM CASE / CABLE KIT 2135558 REV. 0



Item	Part Number	Name	Quantity	Description
1	2135557	Case	1	Shim Cable Case
2	46-260726G1	Cable	1	Shim Coil Cable Assembly
3	46-260724G1	Cable	1	Switch Heaters Cable
4	2135362	Cable	1	Shim Lead Adapter Cable, SV

4–13 3T/940 PASSIVE SHIM KIT 2321737 REV. A

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
1	2322018	DRZ415928	N	500	3 mm Iron Upper Shim, Sq.
2	2322025		N	500	1 mm Spacer, Sq.
3	2322019	DRZ415929	N	500	2 mm Tapped Iron Lower Shim, Sq.
4	2322020	DRZ415930	N	500	1 mm Tapped Iron Lower Shim, Rect.
5	2322022	DHW360683	N	500	3 mm Nylon Upper Shim
6	2322021		N	1000	0.2 mm Iron Lower Shim
7	2322023		N	500	2 mm Nylon Lower Shim
8	2324435	FB14040007	N	1000	M4 x 7 mm Csk. Hd. Screw, Brass

4–14 FIELD PLOTTING RIG ASSEMBLY, LARGE BORE 2292542 REV. 0

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
		—		1	Plotting Rig End Plate Assembly
		AHQ313515		2	. Probe Latch Body Assembly
		DHQ313230		1	. . Plotting Rig Latch Body
		DHQ417838		1	. . Plotting Rig Probe Latch Pin
		DHQ417839		1	. . Plotting Rig Latch Knob
		P209000071		1	. . M10 x 1.5 x 29.25 mm Cross, Ph. Br.
		P209000072		1	. . M10 x 1.5 x 19.25 mm Cross, Ph. Br.
		DHQ113221		1	. Plotting Rig Frame End Plate
		DHQ313228		1	. Plotting Rig Frame Bearing, Outer
		DHQ313229		1	. Plotting Rig Latch Mounting Bracket
		FS10050016		2	. M5 x 16 mm Soc. Hd. Cap Screw, St.
		FS13050025		4	. M5 x 25 mm Soc. Hd. Csk. Screw, St.
		FS42000005		6	. M5 Spring Washer, St.
		FS52000005		6	. M5 NyLock Nut
		P202000015		1	. 48 mm O.D. CirClip Ext., St.
		AHQ115408		1	Plotting Rig Rotating Assembly
		AHQ318330		1	. Plotting Rig Probe Plate Asm. (Probe Holder & Plate)
		AHQ318329		1	. . Plotting Rig Probe Carrier (Probe Holder Asm.)
		AHQ313515		1	. . . Probe Latch Body Assembly
		DHQ313230		1	. . . . Plotting Rig Latch Body
		DHQ417838		1	. . . . Plotting Rig Probe Latch Pin
		DHQ417839		1	. . . . Plotting Rig Latch Knob
		P209000071		1	. . . . M10 x 1.5 x 29.25 mm Cross, Ph. Br.
		P209000072		1	. . . . M10 x 1.5 x 19.25 mm Cross, Ph. Br.
		DHQ318315		1	. . . Plotting Rig Probe Carrier (Probe Holder)
		DHQ318316		1	. . . Probe Carrier Side Plate
		DHQ322115		1	. . . Probe Support
		DHQ413299		2	. . . Plotting Rig Probe Retainer
		FB12050012		4	. . . M5 x 12 mm Slotted Ch. Hd. Screw
		FB12050016		2	. . . M5 x 16 mm Slotted Ch. Hd. Screw
		FB14050016		2	. . . M5 x 16 mm Csk. Slotted Hd. Screw
		FB40000005		2	. . . M5 Flat Washer
		FB50000005		2	. . . M5 Nut
		FB10060030		2	. . . M6 x 30 mm Soc. Hd. Cap Screw
		DHQ115395		1	. . Plotting Rig Probe Plate (Probe Holder Plate)

4-14 FIELD PLOTTING RIG ASSEMBLY, LARGE BORE (continued) 2292542 REV. 0

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
		—		1	. . 40 cm Probe Scale Plate
		DHQ318301		1	. . 45 cm Probe Scale Plate
		DHQ318302		1	. . Probe Plate Adapter (Probe Holder Support)
		—		1	. . Probe Plate Scale
		—		1	. . Probe Plate Scale
		—		1	. . Probe Plate Scale
		DHQ418307		1	. . Probe Carrier Knob (Probe Slide Knob)
		FB14050016		2	. . M5 x 16 mm Csk. Slotted Hd. Screw
		FB14050030		2	. . M5 x 30 mm Csk. Slotted Hd. Screw
		DHQ313227		1	. Plotting Rig Frame Bearing, Inner
		DHQ315403		6	. 1000 mm Support Tube Assembly
		CONS000599		.001	. . Adhesive, Scotweld DP-190
		DHQ315410		1	. . 1000 mm Support Tube
		DHQ415409		2	. . Plotting Rig Tube Bung (End Cap)
		DHQ315534		6	. 500 mm Support Tube Assembly
		CONS000599		.001	. . Adhesive, Scotweld DP-190
		DHQ315535		1	. . 500 mm Support Tube
		DHQ415409		2	. . Plotting Rig Tube Bung (End Cap)
		DHQ315536		3	. 250 mm Support Tube Assembly
		CONS000599		.001	. . Adhesive, Scotweld DP-190
		DHQ415409		2	. . Plotting Rig Tube Bung (End Cap)
		DHQ415537		1	. . 250 mm Port. Plotting Rig Support Leg
		DHQ315544		1	. Plotting Rig Rear Rotating Plate (Rear Clamp Rotation Plate)
		DHQ315545		1	. Plotting Rig Front Rotating Plate
		DHQ315767		1	. Plotting Rig Bearing, Inner
		DHQ415542		15	. Plotting Rig M12 Stud
		DHQ415546		3	. Plotting Rig Plate Guide
		FB14120035		6	. M12 x 35 mm Csk. Slotted Hd. Screw
		P202000015		1	. 48 mm O.D. CirClip Ext., St.
		P202000017		1	. 26 mm O.D. CirClip Ext., St.
		AHQ115505		1	Plotting Rig Clamp
		DHQ115507		1	. Plotting Rig Clamp Plate
		DHQ215509		4	. Plotting Rig Long Leg
		DHQ315508		4	. Plotting Rig Clamp Leg
		DHQ315530		1	. Plotting Rig Clamp End Bearing
		DHQ413243		4	. Plotting Rig Clamp
		DHQ413307		4	. Plotting Rig Clamp Bolt
		DHQ413308		4	. Plotting Rig Clamp Retainer
		DHQ415510		4	. Plotting Rig Extension Leg
		FB11050010		4	. M5 x 10 mm Hex Hd. Screw
		FB11050030		8	. M5 x 30 mm Hex Hd. Screw
		FB50000008		4	. M8 Nut
		P201000020		.02	. Rubber, 1/8 Inch (3.2 mm) Thick, White
		P202000017		1	. 26 mm O.D. CirClip Ext., St.
		—		2	Plotting Rig OVC Quadrant*

4–14 FIELD PLOTTING RIG ASSEMBLY, LARGE BORE (continued) 2292542 REV. 0

Item	GE Part Number	Magnex P/N	FRU	Quantity	Description
		DHQ118615		1	Patient End Mounting Plate
		DHQ118619		2	Plotting Rig Patient End Adapter Plate
		—		1	Plotting Rig OVC End Ring*
		DHQ215549		1	Plotting Rig Axial Positioner (Probe Holder Positioner)
		CONS000599		.001	. Adhesive, Scotweld DP-190
		DHQ215557		1	. Axial Index Sleeve, 40 and 45 cm
		DHQ315555		1	. Plotting Rig Handle Extension Tube
		DHQ415554		1	. Plotting Rig Handle, End
		DHQ415558		1	. Plotting Rig Handle, End
		P193000004		.03	. M8 Studding, Grade 2 Titanium
		P208000039		1	. NW25 Clamp, Polymer
		DHQ315411		1	Plotting Rig Rotator Handle (Axial Positioning Assembly)
		CONS000599		.001	. Adhesive, Scotweld DP-190
		DHQ315785		1	. Plotting Rig Rotator Tube (Axial Positioning Sleeve)
		DHQ315786		1	. Plotting Rig Rotator Bushing
		DHQ315787		1	. Plotting Rig Handle Bushing
		DHQ315792		1	. Plotting Rig Tube
		DHQ315793		1	. Plotting Rig Plate
		DHQ415553		1	. Plotting Rig Handle, End Fitting
		DHQ415790		1	. Plotting Rig Handle Washer
		DHQ415791		1	. Plotting Rig Bearing
		FB40000010		1	. M10 Flat Washer
		FB50000010		2	. M10 Nut
		DHQ315551		2	. Plotting Rig Handle Extension
		CONS000599		.001	. Adhesive, Scotweld DP-190
		DHQ315555		1	. Plotting Rig Handle Extension Tube
		DHQ415553		1	. Plotting Rig Handle, End Fitting
		DHQ415554		1	. Plotting Rig Handle, End
		DHQ315779		3	Plotting Rig Frame Tube Assembly
		DHQ313264		2	. Plotting Rig Frame Tube Cap
		DHQ315784		1	. Plotting Rig Frame Tube
		FB12060020		6	M6 x 20 mm Slotted Ch. Hd. Screw
		FB17030006		3	M3 x 6 mm Grub [Hex Soc. Headless] Screw
		FS10060025		8	M6 x 25 mm Soc. Hd. Cap Screw
		FS10080025		8	M8 x 25 mm Soc. Hd. Cap Screw
		FS10100025		11	M10 x 25 mm Soc. Hd. Cap Screw
		FS13060020		20	M6 x 20 mm Csk. Soc. Hd. Screw
		FS13100025		3	M10x 25 mm Csk. Soc. Hd. Screw
		FS40000010		3	M10 Flat Washer
		P193000004		.12	M8 Studding, Grade 2 Titanium

* For Service End mounting only.

4-15 PUMP / PRECOOL KIT, U.S. VERSION 2295481 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2297067	1	Kit	1	Fittings and Electronics Kit, U.S. Version
2	2296794	1	Kit	1	Vacuum Pump Kit, U.S. Version
3	2296796	1	Kit	1	Leak Detector Kit, U.S. Version

4-15-1 VACUUM PUMP KIT, U.S. VERSION 2296794 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2296610	1	Pump	1	Rotary Vucuum Pump, 115V 60Hz, Leybold
2	46-294150P50	1	Filter	2	Exhaust Filter, AF 16-25
3	46-294150P38	1	Valve	1	Partial Flow Valve, 100V with D16B (Leybold 140-26)
4	2295525	1	Case	1	Heavy-Duty Flight Case

4-15-2 LEAK DETECTOR KIT, U.S. VERSION 2296796 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	46-294150P48	1	Leak Detector	1	Leak Detector, UL-200, 115V
2	46-294150P42	1	Case	1	Transport Case

4-15-3 FITTINGS AND ELECTRONICS KIT, U.S. VERSION 2297067 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2294075	1	Regulator	1	Cylinder Regulator, Air Products E75
2	2294083-14	1	Clamp	8	Swing Clamp, NW50
3	2294083-19	1	Centering Ring	8	Centering Ring & O-Ring
4	2294083-15	1	Reducer	1	Reducer, NW50 to NW25
5	2294083-16	1	Clamp	35	Swing Clamp, NW20 / NW25
6	2294083-17	1	Centering Ring	35	Centering Ring & O-Ring
7	2294083-18	1	Fitting	3	Tee Fitting, NW25
8	2294083-2	1	Valve	6	SpeediValve SP25K
9	2294083-6	1	Nozzle	5	Nozzle
10	46-294167P59	1	Gauge	1	Vacuum Gauge, 4 inch dia., .25 NPTM
11	2294083-10	1	Adapter	2	Pipe Adapter, NW25 to .25 NPTF
12	2294585	1	Socket	1	Socket, .5 inch (12.7 mm) square drive
13	2294585-2	1	Handle	1	Ratchet Handle, .5 inch (12.7 mm) square drive
14	2294585-3	1	Extension Bar	1	Extension Bar, .5 inch (12.7 mm) ratchet
15	46-294167P58	1	Tool	1	Heavy-Duty Wire Cutters
16	2294584	1	Pump	1	Turbomolecular Pump, Leybold
17	46-294150P40	1	Air Cooler	1	Turbo Pump Air Cooler, 115V
18	46-294150P47	1	Converter	1	Main Power Frequency Convertor, 120V, U.S. plug
19	46-294150P43	1	Line	1	Pump Converter Connecting Line, 5 mm
20	2293688	1	Adapter	1	Adapter, ISO 100 / NW50

4-15-3 FITTINGS AND ELECTRONICS KIT, U.S. VERSION (continued) 2297067 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
21	2293780	1	Trapped O-Ring	1	ISO 100 Trapped O-Ring
22	2293010	1	Clamp	8	Claw Clamp
23	2294083-13	1	Pipe	1	Flexible Exhaust Pipe, NW50
24	2294083-20	1	Fitting	1	Reducing Cross Fitting, NW50 to NW25
25	2293621-2	1	Valve	1	Right-Angle Solenoid Valve, NW25
26	2294083-11	1	Pipe	4	Flexible Exhaust Pipe, NW25 x 1000 mm (39.4 inches)
27	2292510	1	Actuator	1	Pump-Out Port Actuator
28	2294083-23	1	Reducer	1	Reducer, NW25 / NW16
29	2294083-21	1	Clamp	4	Swing Clamp, NW10 / NW16
30	2294083-22	1	Centering Ring	3	Centering Ring & O-Ring
31	2293317	1	Gauge	1	Pressure Gauge, Leybold CombiVac
32	2293303	1	Gauge	1	Pressure Gauge, Leybold ThermoVac
33	2293309	1	Gauge	1	Pressure Gauge, Leybold PenningVac
34	46-294150P53	1	Cable	1	TR Sensor Cable, 5M
35	46-294150P54	1	Cable	1	PR Sensor Cable, 5M
36	46-294150P35	1	Sniffer Line	1	Helium Sniffer Line, SL 200G x 4M
37	46-294150P36	1	Spray Gun	1	Search Gas Spray Gun
38	2294075-2	1	Regulator	1	Cylinder Regulator, Air Products E69
39	2294083-9	1	Grease	1	Silicon High-Vacuum Grease
40	46-252065P138	1	Spray	1	Leak Detector Spray, 300 mL aerosol
41	46-294167P60	1	Platform	1	Light-Duty Service Platform
42	2294083	1	Fitting	1	Reducing Cross Fitting, NW 40 / NW25
43	2294083-3	1	Valve	1	SpeediValve SP40K
44	2294083-24	1	Clamp	10	Swing Clamp, NW 32 / NW40
45	2294083-25	1	Centering Ring	10	Centering Ring & O-Ring
46	2294083-12	1	Pipe	3	Flexible Exhaust Pipe, NW40
47	2294083-26	1	Reducer	2	Reducer, NW40 / NW25
48	2255509	1	Tube	1	Nitrogen Blow-Out Tube
49	2294703	1	Tube Assembly	1	Split Blow-Out Tube Assembly
50	2293227	1	Gasket	4	Centering Gasket, NW25, PTFE
51	46-294167P62	1	Tape	1	Teflon Sealant Tape, .0025 inch (.064 mm) thick
52	2295525-3	1	Hose	1	LN_2 Transfer Line
53	46-294167P66	1	Heat Gun	1	Heavy-Duty Heat Gun, 1500w
54	2292536	1	Vent Assembly	1	Exhaust Gas / Supply Vent Assembly
55	2294083-27	1	Flange	1	Blank Flange, NW16
56	46-294167P63	1	Gauge	1	Gauge, 4.5 inch dia., .25 NPTM
57	2294083-28	1	Pipe	1	Flexible Pipe, NW25 x 500 mm
58	2294083-4	1	Diaphragm	2	SP25K Diaphragm, spare
59	2294083-5	1	Diaphragm	2	SP40K Diaphragm, spare
60	46-294150P41	1	Oil	5	Rotary Vacuum Pump Oil, N62
61	46-294167P65	1	Funnel	1	Plastic Funnel
62	2295525-2	1	Chest	1	Locking Tool Chest on Casters

4-16 PUMP / PRECOOL KIT, JAPAN VERSION 2296782 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2296797	1	Kit	1	Leak Detector Kit, Japan Version
2	2296798	1	Kit	1	Vacuum Pump Kit, Japan Version
3	2297071	1	Kit	1	Fittings and Electronics Kit, Japan Version

4-16-1 VACUUM PUMP KIT, JAPAN VERSION 2296798 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2296760	1	Pump	1	Rotary Vucuum Pump, 100V 50Hz, Leybold
2	46-294150P50	1	Filter	2	Exhaust Filter, AF 16-25
3	46-294150P52	1	Valve	1	Partial Flow Valve, 100V with 25B (Leybold 140-27)
4	2295525	1	Case	1	Heavy-Duty Flight Case

4-16-2 LEAK DETECTOR KIT, JAPAN VERSION 2296797 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	46-294150P51	1	Leak Detector	1	Leak Detector, UL-200, 100V
2	46-294150P42	1	Case	1	Transport Case

4-16-3 FITTINGS AND ELECTRONICS KIT, JAPAN VERSION 2297071 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2294075	1	Regulator	1	Cylinder Regulator, Air Products E75
2	2294083-14	1	Clamp	8	Swing Clamp, NW50
3	2294083-19	1	Centering Ring	8	Centering Ring & O-Ring
4	2294083-15	1	Reducer	1	Reducer, NW50 to NW25
5	2294083-16	1	Clamp	35	Swing Clamp, NW20 / NW25
6	2294083-17	1	Centering Ring	35	Centering Ring & O-Ring
7	2294083-18	1	Fitting	3	Tee Fitting, NW25
8	2294083-2	1	Valve	6	SpeediValve SP25K
9	2294083-6	1	Nozzle	5	Nozzle
10	46-294167P59	1	Gauge	1	Vacuum Gauge, 4 inch dia., .25 NPTM
11	2294083-10	1	Adapter	2	Pipe Adapter, NW25 to .25 NPTF
12	2294585	1	Socket	1	Socket, .5 inch (12.7 mm) square drive
13	2294585-2	1	Handle	1	Ratchet Handle, .5 inch (12.7 mm) square drive
14	2294585-3	1	Extension Bar	1	Extension Bar, .5 inch (12.7 mm) ratchet
15	46-294167P58	1	Tool	1	Heavy-Duty Wire Cutters
16	2295480	1	Pump	1	Turbomolecular Pump, Leybold
17	46-294150P40	1	Air Cooler	1	Turbo Pump Air Cooler, 115V
18	46-294150P46	1	Converter	1	Main Power Frequency Convertor, 110V, U.S. plug
19	46-294150P43	1	Line	1	Pump Converter Connecting Line, 5 mm
20	2293688	1	Adapter	1	Adapter, ISO 100 / NW50

4-16-3 FITTINGS AND ELECTRONICS KIT, JAPAN VERSION (continued) 2297071 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
21	2293780	1	Trapped O-Ring	1	ISO 100 Trapped O-Ring
22	2293010	1	Clamp	8	Claw Clamp
23	2294083-13	1	Pipe	1	Flexible Exhaust Pipe, NW50
24	2294083-20	1	Fitting	1	Reducing Cross Fitting, NW50 to NW 25
25	2293621-2	1	Valve	1	Right-Angle Solenoid Valve, NW25
26	2294083-11	1	Pipe	4	Flexible Exhaust Pipe, NW25 x 1000 mm (39.4 inches)
27	2292510	1	Actuator	1	Pump-Out Port Actuator
28	2294083-23	1	Reducer	1	Reducer, NW25 / NW16
29	2294083-21	1	Clamp	4	Swing Clamp, NW10 / NW16
30	2294083-22	1	Centering Ring	3	Centering Ring & O-Ring
31	2293317	1	Gauge	1	Pressure Gauge, Leybold CombiVac
32	2293303	1	Gauge	1	Pressure Gauge, Leybold ThermoVac
33	2293309	1	Gauge	1	Pressure Gauge, Leybold PenningVac
34	46-294150P53	1	Cable	1	TR Sensor Cable, 5M
35	46-294150P54	1	Cable	1	PR Sensor Cable, 5M
36	46-294150P35	1	Sniffer Line	1	Helium Sniffer Line, SL 200G x 4M
37	46-294150P36	1	Spray Gun	1	Search Gas Spray Gun
38	2294075-2	1	Regulator	1	Cylinder Regulator, Air Products E69
39	2294083-9	1	Grease	1	Silicon High-Vacuum Grease
40	46-252065P138	1	Spray	1	Leak Detector Spray, 300 mL aerosol
41	46-294167P60	1	Platform	1	Light-Duty Service Platform
42	2294083	1	Fitting	1	Reducing Cross Fitting, NW40 / NW25
43	2294083-3	1	Valve	1	SpeediValve SP40K
44	2294083-24	1	Clamp	10	Swing Clamp, NW32 / NW40
45	2294083-25	1	Centering Ring	10	Centering Ring & O-Ring
46	2294083-12	1	Pipe	3	Flexible Exhaust Pipe, NW40
47	2294083-26	1	Reducer	2	Reducer, NW40 / NW25
48	2255509	1	Tube	1	Nitrogen Blow-Out Tube
49	2294703	1	Tube Assembly	1	Split Blow-Out Tube Assembly
50	2293227	1	Gasket	4	Centering Gasket, NW25, PTFE
51	46-294167P62	1	Tape	1	Teflon Sealant Tape, .0025 inch (.064 mm) thick
52	2295525-3	1	Hose	1	LN_2 Transfer Line
53	46-294167P66	1	Heat Gun	1	Heavy-Duty Heat Gun, 1500w
54	2292536	1	Vent Assembly	1	Exhaust Gas / Supply Vent Assembly
55	2294083-27	1	Flange	1	Blank Flange, NW16
56	46-294167P63	1	Gauge	1	Gauge, 4.5 inch dia., .25 NPTM
57	2294083-28	1	Pipe	1	Flexible Pipe, NW25 x 500 mm
58	2294083-4	1	Diaphragm	2	SP25K Diaphragm, spare
59	2294083-5	1	Diaphragm	2	SP40K Diaphragm, spare
60	46-294150P41	1	Oil	5	Rotary Vacuum Pump Oil, N62
61	46-294167P65	1	Funnel	1	Plastic Funnel
62	2295525-2	1	Chest	1	Locking Tool Chest on Casters

4-17 PUMP / PRECOOL KIT, EURO VERSION — 2296909 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2296920	1	Kit	1	Leak Detector Kit, Euro Version
2	2296912	1	Kit	1	Vacuum Pump Kit, Euro Version
3	2297712	1	Kit	1	Fittings and Electronics Kit, Euro Version

4-17-1 VACUUM PUMP KIT, EURO VERSION — 2296912 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2296760-2	1	Pump	1	Rotary Vucuum Pump, 230V 50Hz, Leybold
2	46-294150P50	1	Filter	2	Exhaust Filter, AF 16-25
3	46-294150P37	1	Valve	1	Partial Flow Valve, 200V with D25B (Leybold 140-25)
4	2295525	1	Case	1	Heavy-Duty Flight Case

4-17-2 LEAK DETECTOR KIT, EURO VERSION — 2296920 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	46-294150P49	1	Leak Detector	1	Leak Detector, UL-200, 220V
2	46-294150P42	1	Case	1	Transport Case

4-17-3 FITTINGS AND ELECTRONICS KIT, EURO VERSION — 2297712 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
1	2294075	1	Regulator	1	Cylinder Regulator, Air Products E75
2	2294083-14	1	Clamp	8	Swing Clamp, NW50
3	2294083-19	1	Centering Ring	8	Centering Ring & O-Ring
4	2294083-15	1	Reducer	1	Reducer, NW50 to NW25
5	2294083-16	1	Clamp	35	Swing Clamp, NW20 / NW25
6	2294083-17	1	Centering Ring	35	Centering Ring & O-Ring
7	2294083-18	1	Fitting	3	Tee Fitting, NW25
8	2294083-2	1	Valve	6	SpeediValve SP25K
9	2294083-6	1	Nozzle	5	Nozzle
10	46-294167P59	1	Gauge	1	Vacuum Gauge, 4 inch dia., .25 NPTM
11	2294083-10	1	Adapter	2	Pipe Adapter, NW25 to .25 NPTF
12	2294585	1	Socket	1	Socket, .5 inch (12.7 mm) square drive
13	2294585-2	1	Handle	1	Ratchet Handle, .5 inch (12.7 mm) square drive
14	2294585-3	1	Extension Bar	1	Extension Bar, .5 inch (12.7 mm) ratchet
15	46-294167P58	1	Tool	1	Heavy-Duty Wire Cutters
16	2295480	1	Pump	1	Turbomolecular Pump, Leybold TurboVac 361
17	46-294150P39	1	Air Cooler	1	Turbo Pump Air Cooler, 220V
18	46-294150P45	1	Converter	1	Main Power Frequency Convertor, 230V, Euro plug
19	46-294150P43	1	Line	1	Pump Converter Connecting Line, 5 mm
20	2293688	1	Adapter	1	Adapter, ISO 100 / NW50

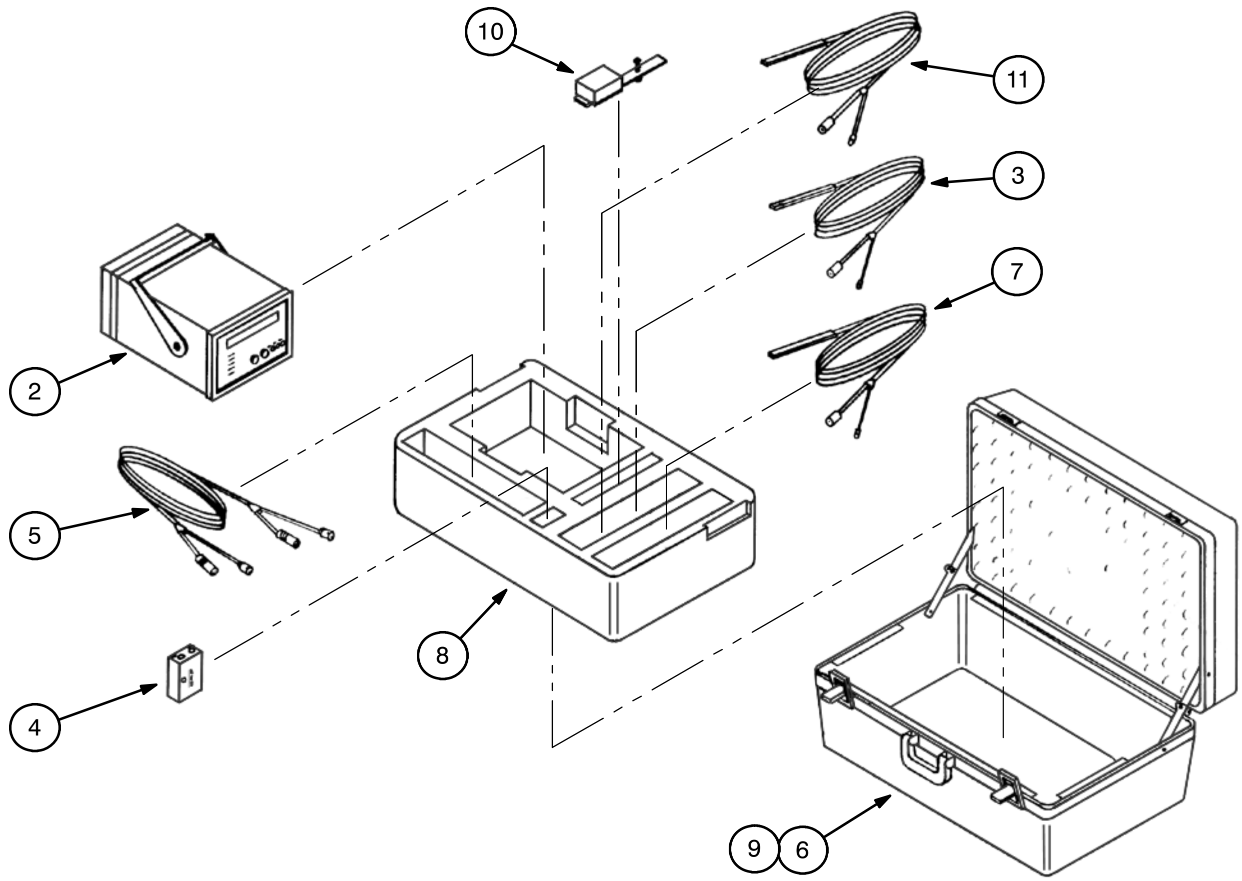
4-17-3 FITTINGS AND ELECTRONICS KIT, EURO VERSION (continued) 2297712 REV. 0

Item	Part Number	FRU	Name	Quantity	Description
21	2293780	1	Trapped O-Ring	1	ISO 100 Trapped O-Ring
22	2293010	1	Clamp	8	Claw Clamp
23	2294083-13	1	Pipe	1	Flexible Exhaust Pipe, NW50
24	2294083-20	1	Fitting	1	Reducing Cross Fitting, NW50 to NW25
25	2293621	1	Valve	1	Right-Angle Solenoid Valve, NW25
26	2294083-11	1	Pipe	4	Flexible Exhaust Pipe, NW25 x 1000 mm (39.4 inches)
27	2292510	1	Actuator	1	Pump-Out Port Actuator
28	2294083-23	1	Reducer	1	Reducer, NW25 / NW16
29	2294083-21	1	Clamp	4	Swing Clamp, NW10 / NW16
30	2294083-22	1	Centering Ring	3	Centering Ring & O-Ring
31	2293317	1	Gauge	1	Pressure Gauge, Leybold CombiVac
32	2293303	1	Gauge	1	Pressure Gauge, Leybold ThermoVac
33	2293309	1	Gauge	1	Pressure Gauge, Leybold PenningVac
34	46-294150P53	1	Cable	1	TR Sensor Cable, 5M
35	46-294150P54	1	Cable	1	PR Sensor Cable, 5M
36	46-294150P35	1	Sniffer Line	1	Helium Sniffer Line, SL 200G x 4M
37	46-294150P36	1	Spray Gun	1	Search Gas Spray Gun
38	2294075-2	1	Regulator	1	Cylinder Regulator, Air Products E69
39	2294083-9	1	Grease	1	Silicon High-Vacuum Grease
40	46-252065P138	1	Spray	1	Leak Detector Spray, 300 mL aerosol
41	46-294167P60	1	Platform	1	Light-Duty Service Platform
42	2294083	1	Fitting	1	Reducing Cross Fitting, NW40 / NW25
43	2294083-3	1	Valve	1	SpeediValve SP40K
44	2294083-24	1	Clamp	10	Swing Clamp, NW32 / NW40
45	2294083-25	1	Centering Ring	10	Centering Ring & O-Ring
46	2294083-12	1	Pipe	3	Flexible Exhaust Pipe, NW40
47	2294083-26	1	Reducer	2	Reducer, NW40 / NW25
48	2255509	1	Tube	1	Nitrogen Blow-Out Tube
49	2294703	1	Tube Assembly	1	Split Blow-Out Tube Assembly
50	2293227	1	Gasket	4	Centering Gasket, NW25, PTFE
51	46-294167P62	1	Tape	1	Teflon Sealant Tape, .0025 inch (.064 mm) thick
52	2295525-3	1	Hose	1	LN_2 Transfer Line
53	46-294167P61	1	Heat Gun	1	Heavy-Duty Heat Gun, 2000w
54	2292536	1	Vent Assembly	1	Exhaust Gas / Supply Vent Assembly
55	2294083-27	1	Flange	1	Blank Flange, NW16
56	46-294167P63	1	Gauge	1	Gauge, 4.5 inch dia., .25 NPTM
57	2294083-28	1	Pipe	1	Flexible Pipe, NW25 x 500 mm
58	2294083-4	1	Diaphragm	2	SP25K Diaphragm, spare
59	2294083-5	1	Diaphragm	2	SP40K Diaphragm, spare
60	46-294150P41	1	Oil	5	Rotary Vacuum Pump Oil, N62
61	46-294167P65	1	Funnel	1	Plastic Funnel
62	2295525-2	1	Chest	1	Locking Tool Chest on Casters

4-18 3T/940 O-RING KIT — 2347635 REV. A

Item	Part Number	FRU	Name	Quantity	Description
1	2324860		O-Ring	4	O-Ring, BS 112, 0.487 I.D. x 0.103 inch (12.4 x 2.6 mm)
2	2324862		O-Ring	2	O-Ring, BS 122, 1.112 I.D. x 0.103 inch (28.2 x 2.6 mm)
3	2324861		O-Ring	2	O-Ring, BS 120, 0.987 I.D. x 0.103 inch (25.1 x 2.6 mm)
4	2324883		O-Ring	1	O-Ring, 7.26 I.D. x 0.118 inch (184.5 x 3.0 mm)
5	2324879		O-Ring	1	O-Ring, 5.09 I.D. x 0.224 inch (129.3 x 5.7 mm)

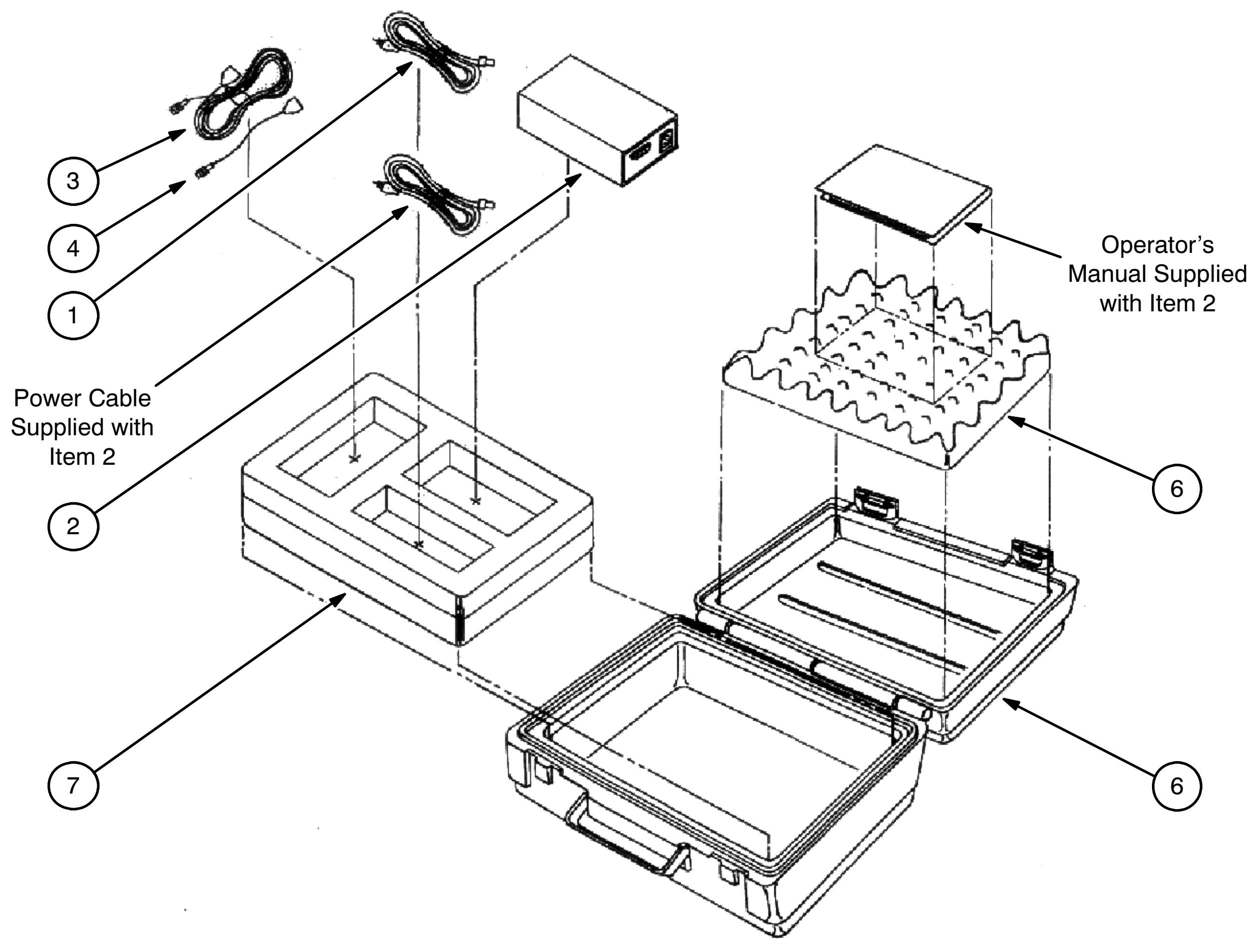
4-19 METROLAB TESLAMETER KIT 46-251865G4 REV. 0



Item	Part Number	FRU	Name	Quantity	Description
2	46-255839P2	2	Teslameter	1	Teslameter, MetroLab 2025
3	46-255839P104	2	Probe	1	#5 Probe, without amplifier
4*					Battery
5	46-255939P121	2	Cable	1	Cable, Teslameter to Preamp
6**	46-255839P200	N	Case	1	Carrying Case
7	46-255839P103	2	Probe	1	#4 Probe, without amplifier
8**	46-306832P1	N	Insert	1	Foam Insert
9**	46-306813P3	N	Label	1	Kit Identification Label
10	46-306836G1	2	Holder	1	Probe Holder
11	46-255839P102	2	Probe	1	#3 Probe, without amplifier
12*	2295525-5	1	Probe	1	Miniture Probe, MetroLab #1062/6/10M (required for GEMSO 3T/940)

* Not included in kit.
** Can be ordered assembled as 46-251865G50.

4–20 LAKESHORE 208 THERMOMETER KIT 46–301477G2 REV. A



Item	Part Number	Name	Quantity	Description
1	46–301453P6	Case	1	Blow–Molded Case
2	46–301478P1	Thermometer	1	Thermometer, Digital, 3W, 4–digit
3	46–301619P1	Cable	1	Interconnecting Cable to Balzer Diodes
4	46–301620P1	Cable	1	Interface Cable to GE Magnet
5	2129919	Cable	1	Interface Cable to GE Magnet w / DIN Connector
6	46–301618P1	Foam	1	Convoluted Top Foam
7	46–301618P2	Foam	1	Two–Layered Bottom Foam
8	2139962	Label	1	Kit Label

DATA SHEETS

TABLE OF CONTENTS

SECTION 1 – MAGNET COMMISSIONING LOG

TABLE 1–1
MAGNET COMMISSIONING LOG

Date	Cryostat Pressure	Static Pressure Reading (Spec 218 psig – 232 psig)	Cold Head Temperatures	
			First Stage	Second Stage

SECTION 1 – MAGNET COMMISSIONING LOG (continued)

TABLE 1–1 (CONTINUED)
MAGNET COMMISSIONING LOG

Date	Cryostat Pressure	Cold Head Temperatures	
		First Stage	Second Stage

SECTION 2 – HELIUM REFILL LOG

TABLE 2–1
LIQUID HELIUM REFILL CHART

PREPARED BY:___________________________________ DATE:____

Dewar Serial Number	Time		Dewar Weight (Pounds)		Liters Helium Used (L_U) (Δ lbs x 3.63)	Helium Level (L_I)				Transfer Efficiency (Δ L_I / L_U)
						Start		Finish		
	Start	Finish	Start	Finish		Meter %	Amt (L)	Meter %	Amt (L)	

SECTION 2 – HELIUM REFILL LOG (continued)

TABLE 2–1 (CONTINUED)
LIQUID HELIUM REFILL CHART

PREPARED BY:______________________ DATE:______________________

Dewar Serial Number	Time		Dewar Weight (Pounds)		Liters Helium Used (L_U) (Δ lbs x 3.63)	Helium Level (L_I)				Transfer Efficiency ($\Delta L_I / L_U$)
						Start		Finish		
	Start	Finish	Start	Finish		Meter %	Amt (L)	Meter %	Amt (L)	

SECTION 2 – HELIUM REFILL LOG (continued)

TABLE 2–1 (CONTINUED)
LIQUID HELIUM REFILL CHART

PREPARED BY:______________________________ DATE:______________________________

Dewar Serial Number	Time		Dewar Weight (Pounds)		Liters Helium Used (L_U) (Δ lbs x 3.63)	Helium Level (L_I) Start		Helium Level (L_I) Finish		Transfer Efficiency (Δ L_I / L_U)
	Start	Finish	Start	Finish		Meter %	Amt (L)	Meter %	Amt (L)	

SECTION 2 – HELIUM REFILL LOG (continued)

TABLE 2–1 (CONTINUED)
LIQUID HELIUM REFILL CHART

PREPARED BY:________________________ DATE:________________________

Dewar Serial Number	Time		Dewar Weight (Pounds)		Liters Helium Used (L_U) (Δ lbs x 3.63)	Helium Level (L_I)				Transfer Efficiency ($\Delta L_I / L_U$)
						Start		Finish		
	Start	Finish	Start	Finish		Meter %	Amt (L)	Meter %	Amt (L)	

SECTION 3 – CRYOGEN LOG

TABLE 3–1
CRYOGEN LOG

Date	Time	Monitor Reading (mm)	He Level	Change from Previous Reading		Boil–Off Rate	Date	Time	Monitor Reading (mm)	He Level	Change from Previous Reading		Boil–Off Rate
	t		L	Δt	ΔL	$\Delta L / \Delta t$		t		L	Δt	ΔL	$\Delta L / \Delta t$

SECTION 3 – CRYOGEN LOG (continued)

TABLE 3–1 (continued)
CRYOGEN LOG

Date	Time	Monitor Reading (mm)	He Level	Change from Previous Reading		Boil–Off Rate	Date	Time	Monitor Reading (mm)	He Level	Change from Previous Reading		Boil–Off Rate
	t		L	Δt	ΔL	$\Delta t - \Delta L$		t		L	Δt	ΔL	$\Delta t - \Delta L$

SECTION 4 – HELIUM FILL DATA

TABLE 4–1
HELIUM FILL DATA SHEET

1	Date	
2	Filled By	
3	Fill Line Number	
4	Dewar Serial Number	
5	Dewar Tare Weight	
6	Dewar Weight Full	
7	Dewar Weight Empty	
8	Helium Volume Lbs. (Line 6 – Line 7)	
9	Helium Volume Liters (Line 8 X 3.63)	
10	End % Volume (From Table Below)	
11	Start % Volume (From Table Below)	
12	Actual Helium Transfer Volume (Line 10 – Line 11)	
13	Transfer Efficiency (Line 12 / Line 9)	
14	Fill Time from Fill History (T! – T1)	
NOTE: Volumetric conversion of liquid helium level is provided in Tables 5–1 and 5–2 and Graph 5–1.		

FILL HISTORY

	START TIME	START MEASUREMENT	GAS PSI	MAGNET PSI
T1				
T2				
T3				
T4				
T5				
T6				
T7				
T8				
T9				
END T!				

FOR YOUR INFORMATION: Predicted Monitor Reading*: __________
Volume of Helium Gas Used**: __________

* Use starting monitor reading. Look up helium liter level on table. Subtract Line 5 from Line 6. Multiply level. Record monitor measurement that reflects this level as your predicted monitor reading.

** Record and note the amount of helium gas used by the pressure level gauge on the helium gas tanks.

SECTION 4 – HELIUM FILL DATA (continued)

TABLE 4–1
HELIUM FILL DATA SHEET

1	Date	
2	Filled By	
3	Fill Line Number	
4	Dewar Serial Number	
5	Dewar Tare Weight	
6	Dewar Weight Full	
7	Dewar Weight Empty	
8	Helium Volume Lbs. (Line 6 – Line 7)	
9	Helium Volume Liters (Line 8 X 3.63)	
10	End % Volume (From Table Below)	
11	Start % Volume (From Table Below)	
12	Actual Helium Transfer Volume (Line 10 – Line 11)	
13	Transfer Efficiency (Line 12 / Line 9)	
14	Fill Time from Fill History (T! – T1)	
NOTE: Volumetric conversion of liquid helium level is provided in Tables 5–1 and 5–2 and Graph 5–1.		

FILL HISTORY

	START TIME	START MEASUREMENT	GAS PSI	MAGNET PSI
T1				
T2				
T3				
T4				
T5				
T6				
T7				
T8				
T9				
END T!				

FOR YOUR INFORMATION: Predicted Monitor Reading*: ______________________
Volume of Helium Gas Used**: ______________________

* Use starting monitor reading. Look up helium liter level on table. Subtract Line 5 from Line 6. Multiply level. Record monitor measurement that reflects this level as your predicted monitor reading.

** Record and note the amount of helium gas used by the pressure level gauge on the helium gas tanks.

SECTION 4 – HELIUM FILL DATA (continued)

TABLE 4–1
HELIUM FILL DATA SHEET

1	Date	
2	Filled By	
3	Fill Line Number	
4	Dewar Serial Number	
5	Dewar Tare Weight	
6	Dewar Weight Full	
7	Dewar Weight Empty	
8	Helium Volume Lbs. (Line 6 – Line 7)	
9	Helium Volume Liters (Line 8 X 3.63)	
10	End % Volume (From Table Below)	
11	Start % Volume (From Table Below)	
12	Actual Helium Transfer Volume (Line 10 – Line 11)	
13	Transfer Efficiency (Line 12 / Line 9)	
14	Fill Time from Fill History (T! – T1)	
NOTE: Volumetric conversion of liquid helium level is provided in Tables 5–1 and 5–2 and Graph 5–1.		

FILL HISTORY

	START TIME	START MEASUREMENT	GAS PSI	MAGNET PSI
T1				
T2				
T3				
T4				
T5				
T6				
T7				
T8				
T9				
END T!				

FOR YOUR INFORMATION: Predicted Monitor Reading*: ______________________
Volume of Helium Gas Used**: ______________________

* Use starting monitor reading. Look up helium liter level on table. Subtract Line 5 from Line 6. Multiply level. Record monitor measurement that reflects this level as your predicted monitor reading.

** Record and note the amount of helium gas used by the pressure level gauge on the helium gas tanks.

SECTION 4 – HELIUM FILL DATA (continued)

TABLE 4–1
HELIUM FILL DATA SHEET

1	Date	
2	Filled By	
3	Fill Line Number	
4	Dewar Serial Number	
5	Dewar Tare Weight	
6	Dewar Weight Full	
7	Dewar Weight Empty	
8	Helium Volume Lbs. (Line 6 – Line 7)	
9	Helium Volume Liters (Line 8 X 3.63)	
10	End % Volume (From Table Below)	
11	Start % Volume (From Table Below)	
12	Actual Helium Transfer Volume (Line 10 – Line 11)	
13	Transfer Efficiency (Line 12 / Line 9)	
14	Fill Time from Fill History (T! – T1)	
NOTE: Volumetric conversion of liquid helium level is provided in Tables 5–1 and 5–2 and Graph 5–1.		

FILL HISTORY

	START TIME	START MEASUREMENT	GAS PSI	MAGNET PSI
T1				
T2				
T3				
T4				
T5				
T6				
T7				
T8				
T9				
END T!				

FOR YOUR INFORMATION: Predicted Monitor Reading*: ______________________
Volume of Helium Gas Used**: ______________________

* Use starting monitor reading. Look up helium liter level on table. Subtract Line 5 from Line 6. Multiply level. Record monitor measurement that reflects this level as your predicted monitor reading.

** Record and note the amount of helium gas used by the pressure level gauge on the helium gas tanks.

SECTION 5 – VOLUMETRIC CONVERSION

TABLE 5-1
HELIUM MONITOR PROBE 1 READINGS AND HELIUM LEVELS
(PROBE 1 WIRE RESISTANCE = 193.17 OHMS / METER)

Monitor Reading (mm)	Percent Full of Helium	Remaining Helium (liters)	Comment	Monitor Reading (mm)	Percent Full of Helium	Remaining Helium (liters)	Comment
1200	**100.00**	**3066**	**Full**	580	64.45	1976	
1180	98.86	3031		560	63.63	1951	
1160	97.68	2995		540	62.82	1926	
1140	96.38	2955		520	62.00	1901	
1120	95.04	2914		**500**	**61.19**	**1876**	**Minimum Operating Level**
1100	93.64	2871		480	60.40	1852	
1080	92.17	2826	1 Dewar to Fill*	460	59.59	1827	
1060	90.67	2780		440	58.81	1803	6 Dewars to Fill*
1040	89.37	2740		420	57.99	1778	
1020	87.90	2695		400	57.18	1753	
1000	86.33	2647	2 Dewars to Fill*	380	56.39	1729	
980	84.74	2598		360	55.58	1704	
960	83.11	2548		340	54.76	1679	
940	81.70	2505		320	53.91	1653	
920	**80.43**	**2466**	**Minimum Ramping Level**	300	53.10	1628	
900	79.22	2429	3 Dewars to Fill*	280	52.25	1602	
880	78.08	2394		260	51.40	1576	7 Dewars to Fill*
860	77.01	2361		240	50.95	1562	
840	75.96	2329		220	50.10	1536	
820	74.95	2298		200	49.22	1509	
800	73.97	2268		180	48.30	1481	
780	72.99	2238		160	47.42	1454	
760	72.05	2209	4 Dewars to Fill*	140	46.48	1425	
740	71.14	2181		120	45.53	1396	
720	70.22	2153		100	44.55	1366	8 Dewars to Fill*
700	69.34	2126		80	43.57	1336	
680	68.46	2099		60	42.99	1318	
660	67.61	2073		40	41.98	1287	
640	66.76	2047		20	40.90	1254	
620	66.11	2027		0	39.76	1220	
600	65.36	2001	5 Dewars to Fill*				

* Dewars to fill assumes 250 litre Dewars and 85 percent transfill efficiency.

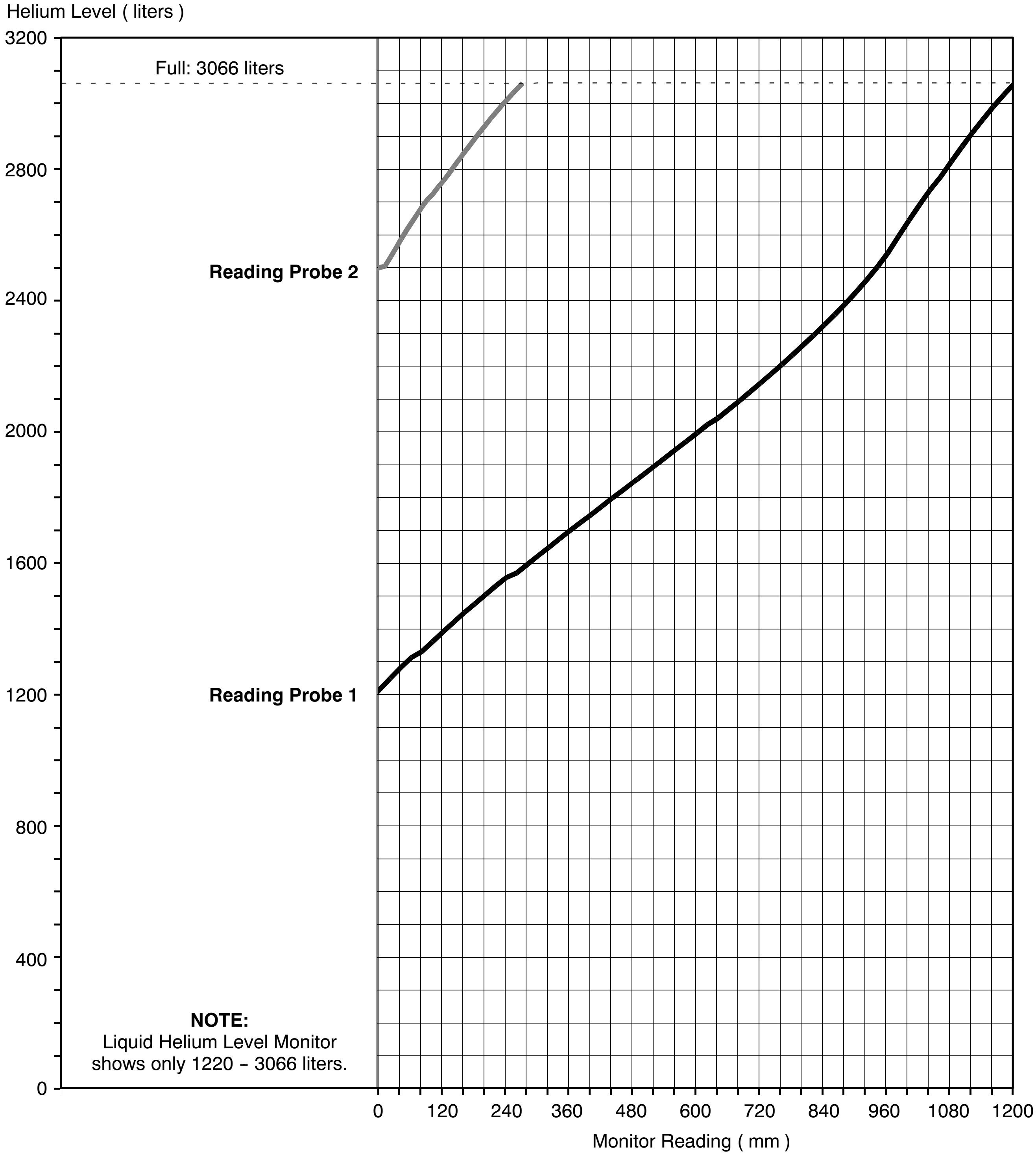
SECTION 5 - VOLUMETRIC CONVERSION (continued)

TABLE 5-2
HELIUM MONITOR PROBE 2 READINGS AND HELIUM LEVELS
(PROBE 2 WIRE RESISTANCE = 193.17 OHMS / METER)

Monitor Reading (mm)	Remaining Helium (liters)	Monitor Reading (mm)	Remaining Helium (liters)	Monitor Reading (mm)	Remaining Helium (liters)	Monitor Reading (mm)	Remaining Helium (liters)
270	**3066 (Full)**	200	2939	130	2789	60	2642
260	3049	190	2919	120	2768	50	2617
250	3032	180	2898	110	2749	40	2591
240	3015	170	2877	100	2728	30	2563
230	2997	160	2856	90	2712	20	2536
220	2978	150	2834	80	2690	10	2510
210	2959	140	2812	70	2666	0	2506

SECTION 5 – VOLUMETRIC CONVERSION (continued)

GRAPH 5–1
VOLUMETRIC CONVERSION OF LIQUID HELIUM LEVEL



SECTION 6 – TEMPERATURE SENSOR LOG

TABLE 6–1
TEMPERATURE SENSOR LOG

Date	Time	20K Shield (A–B)		77K Shield (PT100)		20K Link (CLTS)		77K Link (PT100)		Shield Cooler First Stage (PT100)		Shield Cooler Second Stage (A–B)		Magnet Bottom (PT100)		Magnet Middle (A–B)	
		Ohm	K	Ohm	K	Ohm	K	Ohm	K	Ohm	K	Ohm	K	Ohm	K	Ohm	K

SECTION 6 – CRYOGEN LOG (continued)

TABLE 6–1 (continued)
TEMPERATURE SENSOR LOG

Date	Time	20K Shield (A–B)		77K Shield (PT100)		20K Link (CLTS)		77K Link (PT100)		Shield Cooler First Stage (PT100)		Shield Cooler Second Stage (A–B)		Magnet Bottom (PT100)		Magnet Middle (A–B)	
		Ohm	K	Ohm	K	Ohm	K	Ohm	K	Ohm	K	Ohm	K	Ohm	K	Ohm	K

SECTION 7 – RAMP CURRENT LOG

TABLE 7–1
MAGNET RAMPING / PARKING CURRENT LOG

Magnet Model No.______________________________
Magnet Serial No.______________________________

Service Engineer:____________________________________
CHECK POLARITY BEFORE RAMPING!

Date	Time	Main Current Lead Voltage Drop (mV)	Stabilized Main Coil Current (A)	Stabilized Magnet Field (Gauss)	Gauss / Amp	3.06T Persistence Current (A)	Final Persistence Current (A)

SECTION 7 – MAGNET CURRENT & DRIFT LOGS (continued)

TABLE 7–1 (continued)
MAGNET RAMPING / PARKING CURRENT LOG

Magnet Model No. ______________________________
Magnet Serial No. ______________________________

Service Engineer: ______________________________
CHECK POLARITY BEFORE RAMPING!

Date	Time	Main Current Lead Voltage Drop (mV)	Stabilized Main Coil Current (A)	Stabilized Magnet Field (Gauss)	Gauss / Amp	3.06T Persistence Current (A)	Final Persistence Current (A)

SECTION 8 – FIELD PLOT DATA

TABLE 8–1–1
12–PLANE FIELD MAP

Magnet Model No.__________ Plot No.__________ Date__________
Magnet Serial No.__________ Plotting Radius: 22.5 cm Base Field__________
Service Engineer:__________

	Z1	Z2	Z3	Z4	X	Y	ZX	ZY	XY	X^2-Y^2	Z^2X	Z^2Y	ZXY	$Z(X^2-Y^2)$
Shim Setting (Amps)														
Pot. Setting														

PLANE	1	2	3	4	5	6	7	8	9	10	11	12
Plot Sequence	1	3	5	7	9	11	12	10	8	6	4	2
PROBE POSITION (WITH 22.5 CM PLOTTING RADIUS)												
Axial (cm)	22.095	20.340	17.325	13.230	8.2800	2.8125	–2.8125	–8.2800	–13.230	–17.325	–20.340	–22.095
Radial (cm)	4.2975	9.6300	14.355	18.203	20.925	22.320	22.320	20.925	18.203	14.355	9.6300	4.2975
ANGLE	MEASURED DATA											
0°												
30°												
60°												
90°												
120°												
150°												
180°												
210°												
240°												
270°												
300°												
330°												
360°												

* The positions given above are for a 22.5 cm Plotting Radius. If using a different Plotting Radius, enter the radius used at the top of this page and calculate the axial and radial positions by multipling the Plotting Radius times each of the constants below.

Axial Constant	0.982	0.904	0.770	0.588	0.368	0.125	–0.125	–0.368	–0.558	–0.770	–0.904	–0.982
Radial Constant	0.191	0.428	0.638	0.809	0.930	0.992	0.992	0.930	0.809	0.638	0.428	0.191

SECTION 8 – FIELD PLOT DATA (continued)

TABLE 8–1–2
12–PLANE FIELD MAP

Magnet Model No.________________ Plot No.________________ Date________________
Magnet Serial No.________________ Plotting Radius: 22.5 cm Base Field________________
Service Engineer:________________

	Z1	Z2	Z3	Z4	X	Y	ZX	ZY	XY	X^2-Y^2	Z^2X	Z^2Y	ZXY	$Z(X^2-Y^2)$
Shim Setting (Amps)														
Pot. Setting														

PLANE	1	2	3	4	5	6	7	8	9	10	11	12
Plot Sequence	1	3	5	7	9	11	12	10	8	6	4	2
PROBE POSITION (WITH 22.5 CM PLOTTING RADIUS)												
Axial (cm)	22.095	20.340	17.325	13.230	8.2800	2.8125	–2.8125	–8.2800	–13.230	–17.325	–20.340	–22.095
Radial (cm)	4.2975	9.6300	14.355	18.203	20.925	22.320	22.320	20.925	18.203	14.355	9.6300	4.2975
ANGLE	**MEASURED DATA**											
0°												
30°												
60°												
90°												
120°												
150°												
180°												
210°												
240°												
270°												
300°												
330°												
360°												

* The positions given above are for a 22.5 cm Plotting Radius. If using a different Plotting Radius, enter the radius used at the top of this page and calculate the axial and radial positions by multipling the Plotting Radius times each of the constants below.

Axial Constant	0.982	0.904	0.770	0.588	0.368	0.125	–0.125	–0.368	–0.558	–0.770	–0.904	–0.982
Radial Constant	0.191	0.428	0.638	0.809	0.930	0.992	0.992	0.930	0.809	0.638	0.428	0.191

SECTION 8 – FIELD PLOT DATA (continued)

TABLE 8–1–3
12–PLANE FIELD MAP

Magnet Model No.____________ Plot No.____________ Date____________
Magnet Serial No.____________ Plotting Radius: 22.5 cm Base Field____________
Service Engineer:____________

	Z1	Z2	Z3	Z4	X	Y	ZX	ZY	XY	X^2–Y^2	Z^2X	Z^2Y	ZXY	Z(X^2–Y^2)
Shim Setting (Amps)														
Pot. Setting														

PLANE	1	2	3	4	5	6	7	8	9	10	11	12
Plot Sequence	1	3	5	7	9	11	12	10	8	6	4	2
PROBE POSITION (WITH 22.5 CM PLOTTING RADIUS)												
Axial (cm)	22.095	20.340	17.325	13.230	8.2800	2.8125	–2.8125	–8.2800	–13.230	–17.325	–20.340	–22.095
Radial (cm)	4.2975	9.6300	14.355	18.203	20.925	22.320	22.320	20.925	18.203	14.355	9.6300	4.2975
ANGLE	**MEASURED DATA**											
0°												
30°												
60°												
90°												
120°												
150°												
180°												
210°												
240°												
270°												
300°												
330°												
360°												

* The positions given above are for a 22.5 cm Plotting Radius. If using a different Plotting Radius, enter the radius used at the top of this page and calculate the axial and radial positions by multipling the Plotting Radius times each of the constants below.

Axial Constant	0.982	0.904	0.770	0.588	0.368	0.125	–0.125	–0.368	–0.558	–0.770	–0.904	–0.982
Radial Constant	0.191	0.428	0.638	0.809	0.930	0.992	0.992	0.930	0.809	0.638	0.428	0.191

SECTION 8 – FIELD PLOT DATA (continued)

TABLE 8–1–4
12–PLANE FIELD MAP

Magnet Model No.______________ Plot No.______________ Date______________
Magnet Serial No.______________ Plotting Radius: 22.5 cm Base Field______________
Service Engineer:______________

	Z1	Z2	Z3	Z4	X	Y	ZX	ZY	XY	X^2-Y^2	Z^2X	Z^2Y	ZXY	$Z(X^2-Y^2)$
Shim Setting (Amps)														
Pot. Setting														

PLANE	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Plot Sequence	1	3	5	7	9	11	12	10	8	6	4	2
PROBE POSITION (WITH 22.5 CM PLOTTING RADIUS)												
Axial (cm)	22.095	20.340	17.325	13.230	8.2800	2.8125	–2.8125	–8.2800	–13.230	–17.325	–20.340	–22.095
Radial (cm)	4.2975	9.6300	14.355	18.203	20.925	22.320	22.320	20.925	18.203	14.355	9.6300	4.2975
ANGLE	**MEASURED DATA**											
0°												
30°												
60°												
90°												
120°												
150°												
180°												
210°												
240°												
270°												
300°												
330°												
360°												

* The positions given above are for a 22.5 cm Plotting Radius. If using a different Plotting Radius, enter the radius used at the top of this page and calculate the axial and radial positions by multipling the Plotting Radius times each of the constants below.

Axial Constant	0.982	0.904	0.770	0.588	0.368	0.125	–0.125	–0.368	–0.558	–0.770	–0.904	–0.982
Radial Constant	0.191	0.428	0.638	0.809	0.930	0.992	0.992	0.930	0.809	0.638	0.428	0.191

SECTION 8 - FIELD PLOT DATA (continued)

TABLE 8-1-5
12-PLANE FIELD MAP

Magnet Model No.______________________ Plot No.____________________ Date__________________
Magnet Serial No.______________________ Plotting Radius: 22.5 cm__________ Base Field______________
Service Engineer:______________________

	Z1	Z2	Z3	Z4	X	Y	ZX	ZY	XY	X^2-Y^2	Z^2X	Z^2Y	ZXY	$Z(X^2-Y^2)$
Shim Setting (Amps)														
Pot. Setting														

PLANE	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Plot Sequence	1	3	5	7	9	11	12	10	8	6	4	2
PROBE POSITION (WITH 22.5 CM PLOTTING RADIUS)												
Axial (cm)	22.095	20.340	17.325	13.230	8.2800	2.8125	-2.8125	-8.2800	-13.230	-17.325	-20.340	-22.095
Radial (cm)	4.2975	9.6300	14.355	18.203	20.925	22.320	22.320	20.925	18.203	14.355	9.6300	4.2975
ANGLE	**MEASURED DATA**											
0°												
30°												
60°												
90°												
120°												
150°												
180°												
210°												
240°												
270°												
300°												
330°												
360°												

* The positions given above are for a 22.5 cm Plotting Radius. If using a different Plotting Radius, enter the radius used at the top of this page and calculate the axial and radial positions by multipling the Plotting Radius times each of the constants below.

Axial Constant	0.982	0.904	0.770	0.588	0.368	0.125	-0.125	-0.368	-0.558	-0.770	-0.904	-0.982
Radial Constant	0.191	0.428	0.638	0.809	0.930	0.992	0.992	0.930	0.809	0.638	0.428	0.191

SECTION 8 – FIELD PLOT DATA (continued)

TABLE 8–1–6
12–PLANE FIELD MAP

Magnet Model No.____________ Plot No.____________ Date____________

Magnet Serial No.____________ Plotting Radius: 22.5 cm Base Field____________

Service Engineer:____________

	Z1	Z2	Z3	Z4	X	Y	ZX	ZY	XY	X^2-Y^2	Z^2X	Z^2Y	ZXY	Z(X^2-Y^2)
Shim Setting (Amps)														
Pot. Setting														

PLANE	1	2	3	4	5	6	7	8	9	10	11	12
Plot Sequence	1	3	5	7	9	11	12	10	8	6	4	2
PROBE POSITION (WITH 22.5 CM PLOTTING RADIUS)												
Axial (cm)	22.095	20.340	17.325	13.230	8.2800	2.8125	–2.8125	–8.2800	–13.230	–17.325	–20.340	–22.095
Radial (cm)	4.2975	9.6300	14.355	18.203	20.925	22.320	22.320	20.925	18.203	14.355	9.6300	4.2975
ANGLE	**MEASURED DATA**											
0°												
30°												
60°												
90°												
120°												
150°												
180°												
210°												
240°												
270°												
300°												
330°												
360°												

* The positions given above are for a 22.5 cm Plotting Radius. If using a different Plotting Radius, enter the radius used at the top of this page and calculate the axial and radial positions by multipling the Plotting Radius times each of the constants below.

Axial Constant	0.982	0.904	0.770	0.588	0.368	0.125	–0.125	–0.368	–0.558	–0.770	–0.904	–0.982
Radial Constant	0.191	0.428	0.638	0.809	0.930	0.992	0.992	0.930	0.809	0.638	0.428	0.191

SECTION 9 – SHIM PLOT DATA

TABLE 9–1–1
SHIM PLOT DATA

Magnet Model No.________________ Field Before Shim:________________ Service Engineer:________________
Magnet Serial No.________________ Field After Shim:________________ Date________________

Shim		Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___
Input File Name							
Z1	PPM						
	Current						
Z2	PPM						
	Current						
Z3	PPM						
	Current						
Z4	PPM						
	Current						
X	PPM						
	Current						
Y	PPM						
	Current						
ZX	PPM						
	Current						
ZY	PPM						
	Current						
XY	PPM						
	Current						
X^2-Y^2	PPM						
	Current						
Z^2X	PPM						
	Current						
Z^2Y	PPM						
	Current						
ZXY	PPM						
	Current						
$Z(X^2-Y^2)$	PPM						
	Current						
Output File Name							

SECTION 9 – SHIM PLOT DATA (continued)

TABLE 9–1–2
SHIM PLOT DATA (CONTINUED)

Magnet Model No.________________ Field Before Shim:________________ Service Engineer:________________
Magnet Serial No.________________ Field After Shim:________________ Date________________

Shim		Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___
Input File Name							
Z1	PPM						
	Current						
Z2	PPM						
	Current						
Z3	PPM						
	Current						
Z4	PPM						
	Current						
X	PPM						
	Current						
Y	PPM						
	Current						
ZX	PPM						
	Current						
ZY	PPM						
	Current						
XY	PPM						
	Current						
X^2-Y^2	PPM						
	Current						
Z^2X	PPM						
	Current						
Z^2Y	PPM						
	Current						
ZXY	PPM						
	Current						
$Z(X^2-Y^2)$	PPM						
	Current						
Output File Name							

SECTION 9 – SHIM PLOT DATA (continued)

TABLE 9–1–3
SHIM PLOT DATA (CONTINUED)

Magnet Model No.________________ Field Before Shim:________________ Service Engineer:________________
Magnet Serial No.________________ Field After Shim:________________ Date________________

Shim		Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___
Input File Name							
Z1	PPM						
	Current						
Z2	PPM						
	Current						
Z3	PPM						
	Current						
Z4	PPM						
	Current						
X	PPM						
	Current						
Y	PPM						
	Current						
ZX	PPM						
	Current						
ZY	PPM						
	Current						
XY	PPM						
	Current						
X^2-Y^2	PPM						
	Current						
Z^2X	PPM						
	Current						
Z^2Y	PPM						
	Current						
ZXY	PPM						
	Current						
$Z(X^2-Y^2)$	PPM						
	Current						
Output File Name							

SECTION 9 – SHIM PLOT DATA (continued)

TABLE 9–1–4
SHIM PLOT DATA (CONTINUED)

Magnet Model No.________________ Field Before Shim:________________ Service Engineer:________________
Magnet Serial No.________________ Field After Shim:________________ Date________________

Shim		Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___
Input File Name							
Z1	PPM						
	Current						
Z2	PPM						
	Current						
Z3	PPM						
	Current						
Z4	PPM						
	Current						
X	PPM						
	Current						
Y	PPM						
	Current						
ZX	PPM						
	Current						
ZY	PPM						
	Current						
XY	PPM						
	Current						
X^2-Y^2	PPM						
	Current						
Z^2X	PPM						
	Current						
Z^2Y	PPM						
	Current						
ZXY	PPM						
	Current						
$Z(X^2-Y^2)$	PPM						
	Current						
Output File Name							

SECTION 9 – SHIM PLOT DATA (continued)

TABLE 9–1–5
SHIM PLOT DATA (CONTINUED)

Magnet Model No.____________ Field Before Shim:____________ Service Engineer:____________
Magnet Serial No.____________ Field After Shim:____________ Date____________

Shim		Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___
Input File Name							
Z1	PPM						
	Current						
Z2	PPM						
	Current						
Z3	PPM						
	Current						
Z4	PPM						
	Current						
X	PPM						
	Current						
Y	PPM						
	Current						
ZX	PPM						
	Current						
ZY	PPM						
	Current						
XY	PPM						
	Current						
X^2-Y^2	PPM						
	Current						
Z^2X	PPM						
	Current						
Z^2Y	PPM						
	Current						
ZXY	PPM						
	Current						
$Z(X^2-Y^2)$	PPM						
	Current						
Output File Name							

SECTION 9 – SHIM PLOT DATA (continued)

TABLE 9–1–6
SHIM PLOT DATA (CONTINUED)

Magnet Model No.__________ Field Before Shim:__________ Service Engineer:__________
Magnet Serial No.__________ Field After Shim:__________ Date__________

Shim		Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___
Input File Name							
Z1	PPM						
	Current						
Z2	PPM						
	Current						
Z3	PPM						
	Current						
Z4	PPM						
	Current						
X	PPM						
	Current						
Y	PPM						
	Current						
ZX	PPM						
	Current						
ZY	PPM						
	Current						
XY	PPM						
	Current						
X^2–Y^2	PPM						
	Current						
Z^2X	PPM						
	Current						
Z^2Y	PPM						
	Current						
ZXY	PPM						
	Current						
$Z(X^2$–$Y^2)$	PPM						
	Current						
Output File Name							

SECTION 9 – SHIM PLOT DATA (continued)

TABLE 9–1–7
SHIM PLOT DATA (CONTINUED)

Magnet Model No.________________ Field Before Shim:________________ Service Engineer:________________
Magnet Serial No.________________ Field After Shim:________________ Date________________

Shim		Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___
Input File Name							
Z1	PPM						
	Current						
Z2	PPM						
	Current						
Z3	PPM						
	Current						
Z4	PPM						
	Current						
X	PPM						
	Current						
Y	PPM						
	Current						
ZX	PPM						
	Current						
ZY	PPM						
	Current						
XY	PPM						
	Current						
X^2-Y^2	PPM						
	Current						
Z^2X	PPM						
	Current						
Z^2Y	PPM						
	Current						
ZXY	PPM						
	Current						
$Z(X^2-Y^2)$	PPM						
	Current						
Output File Name							

SECTION 9 – SHIM PLOT DATA (continued)

TABLE 9–1–8
SHIM PLOT DATA (CONTINUED)

Magnet Model No.________________ Field Before Shim:________________ Service Engineer:________________
Magnet Serial No.________________ Field After Shim:________________ Date________________

Shim		Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___	Plot No. ___
Input File Name							
Z1	PPM						
	Current						
Z2	PPM						
	Current						
Z3	PPM						
	Current						
Z4	PPM						
	Current						
X	PPM						
	Current						
Y	PPM						
	Current						
ZX	PPM						
	Current						
ZY	PPM						
	Current						
XY	PPM						
	Current						
X^2–Y^2	PPM						
	Current						
Z^2X	PPM						
	Current						
Z^2Y	PPM						
	Current						
ZXY	PPM						
	Current						
$Z(X^2$–$Y^2)$	PPM						
	Current						
Output File Name							